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MRM: Wireless Sensor Network for Online Milling Roller Monitoring in Harsh Industrial Environment

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Abstract: Roller is a key component of milling machine in iron and steel industry, with the development of industrial wireless technology. The online predictive maintenance of the roller plays an important role in effective production. But the harsh working environment – spray of cooling liquid, moving, high temperature and heavy shelter – cause low quality of wireless communication and need high reliability. The paper proposed a novel roller monitoring system based on star/mesh topology, which use 2.4 GHz in mesh subnet and 433 MHz in star subnet to avoid interference between nodes in large scale networks. The paper designed a self-organized multi-channel link layer protocol optimized network flexibility and power consumption under high reliability. The system was deployed in the real iron and steel plant to monitor support roller of strip mill with one years, which shown a high reliability and low power consumption, and improved the efficiency of the roller maintenance. *Copyright © 2013 IFSA.*

Keywords: Wireless sensor network, Milling roller monitoring, MAC protocol, Predictive maintenance.

1. Introduction

Wireless sensor networks has been deployed in many areas [1], especially, industrial wireless sensor networks have been rapid developed in recent years [2-4]. Because of the harsh industrial environments and the blocking led by the layout of the factory, communication is easily interfered. Reliability has been the first requirement in industrial wireless application like equipment monitoring, feedback control etc. In process industrial like steel or chemical production, which equipped with expensive equipment and the quality control has big relationship with the working condition of those equipment. The monitoring, maintenance and repair cost huge investments each year. Compared with artificial point inspection, wireless equipment monitoring is an accurate, real-time and low cost way to ensure the safety of the machinery and equipment, improve product quality, and reduce labor working intensity, especially for the moving, rotating points of the production line. Strip mill is the key equipment in the cold rolling production. A typical six high cold mill has six rollers including two support rollers. The

temperature of the bearings will go high when there are faults occurred on rollers. On-line temperature monitoring of the bearings is an efficient way to monitor the working condition of the rollers, guide the maintenance and prevent major accidents. Industrial wireless sensor network provides a low-cost, flexible and scalable industrial automation technology, which can eliminate wires, save the time of installation, and provide convenience for maintenance.

The challenge of the roller monitoring in strip mill include:

1) The worst environment of the factory wireless communication. The whole strip mill is covered by steel wall and rollers are under spray of the cooling liquid. Thus, the application scenario needs a low complex but high reliable communication protocol.

2) Mobility of rollers. Since rollers need to be changed periodically, sensor nodes need a self-organized protocol to form topology as soon as possible.

The remainder of this paper is organized as follows. Section II describes the related work. Section III presents the design requirement of the application. Section IV shows the system architecture and the protocol design. Section V, we introduced a real field deployment of the system. We conclude our work in Section VI and future work is also discussed.

2. Related Work

There are increasing deployments in industrial wireless sensor networks especially in equipment pre-maintenance [5]. Suratsavdee Korkua [6] proposed a health monitoring system of electric machine based on wireless sensor network using ZigBee, to obtain higher accuracy in diagnostic for the predictive maintenance of induction motors by vibration detection. Faria SPS [7] proposed a predictive maintenance industrial system to monitor rotating machinery, which integrated two temperature and vibration sensors and two kind networks: a wireless ZigBee network and a wired Controller Area Network (CAN). Paper [8] introduced wireless sensor network to the area of monitoring and protection of oil and gas operations. In addition, the paper proposed a model to provide new tools for research in predictive maintenance and condition-based monitoring of factory machinery. Paper [9] studied MEMS accelerometer based monitoring of machine tool vibrations which shown the relationship of the faults and vibration data, and proposed a MEMS-accelerometer based WSN platform. Paper [10] introduced predictive or condition-based maintenance results on selected rotating equipment at High Flux Isotope Reactor (HFIR), which demonstrated that wireless sensors offer an effective method for monitoring key process conditions continuously and remotely. Albe CLF [11] proposed an IEEE 802.15.4 based WSN to monitor the torque and efficiency in induction motors, which acquires electrical signals from the motor in a noninvasive manner, and then performs local processing for torque and efficiency estimation. Pedro Cheong [12] developed a ZigBee based wireless sensor network node for the ultraviolet (UV) detection of flame. The sensor node was verified when the luminous flame was imaged onto the sensor node with different angles ranging from -30° to 30° and distances of 0.1, 0.2, and 0.3 m enabling effective fire safety applications.

Industrial researches formed three standards: WirelessHART [13], ISA SP100.11a [14] and WIA-PA [15]. WirelessHART is the first wireless standard for industrial applications released by the HART Communication Foundation in 2007, as an extension of HART bus [16]. It uses TDMA schedule in MAC layer based on IEEE802.15.4 and supports mesh topology, which targets the process monitoring and control applications. And WirelessHart has been voted to be the IEC standards in 2011. ISA-100.11a was started by ISA in 2007, which targets for one wireless standards supporting different industrial field bus, not only the HART bus. WIA-PA is the second IEC standards published in 2011. It uses mesh/star topology and two layers network management. All of those standards use TDMA schedule and central network management by Network Manager and support security communication.

Zigbee Protocol used in many literatures is not suite for industrial application because of its low reliability and low guarantee of delay. The industrial standards, which have high reliability and security technologies, are too complex for the application in harsh cool strip mill environment. For the special requirements of the roller monitoring in cool strip mill, we proposed a simple but reliable protocol using two frequency bands and two layer network architecture, which achieves the low power consumption, mobility of the roller and the reliability requirements.

3. System Design

3.1. Design Goals

The roller of the cold strip mill works in relative harsh environment. Steel walls cover the whole strip mill and rollers work under high temperature and spray of the cooling liquid. Sensor nodes are fixed on the end shield of the support rollers, so that the wireless communications of sensor nodes are weak and easy interfered. Sensor node cannot be artificial maintained when the strip mill is working. The support roller will be replaced periodically, which causes online and offline automatically switching of the sensor nodes. According to the environmental characteristics and application requirements, the design goals are as follows:

1) High reliability.

Reliability is the first requirement in industrial applications. Because of the harsh environment of the

cold strip mill, the monitoring system needs higher reliability than common industrial wireless applications. Also, it is impossible to maintain the sensor nodes when the sensor nodes are working online. Additional, to quick trace the temperature variation, when the temperature of a sensor node beyond its alarm temperature, it should automatically short the sample period and alarm the manager by sound, light and SMS.

2) Self-organization.

Support roller works in two states: online working and offline maintenance. Sensor nodes mounted on rollers will leave the original network and join the new network when the roller changes its states. Online sensors form a monitoring network, and offline sensors form a maintenance network with a slow communication period to avoid searching network. When a roller goes online, the sensors mounted on this roller will change to online monitoring network quickly. And also, the online sensor nodes maybe change its router to another for the bad communication caused by the dynamic harsh environment. Therefore, compared with general quasi-static network in industrial applications, cold strip mill roller monitoring network has more dynamic changes and needs self-organization protocol.

3) Low power consumption.

Because of the hard replacement and maintenance of the sensor node, the system need low power consumption to reach a long battery life. Typical work life is 12 months under 1 minute sampling period, or 20 months under 5-minute sampling period, or 30 months under 10-minute sampling period.

4) Controllability and reconfiguration.

Different from the common sensor networks, industrial sensor networks, have a special demand for the network controllability and reconfiguration, which make the network management easily. In the application of support roller monitoring, sample period should be adjusted, in a range of 1 minute to 20 minutes with the minimum step of 1 minute. And also, the alarm level of the sensor node should be configured by the network manager. For the system management, information such as battery capacity, communication quality and work states of each node should be displayed in real-time.

3.2. System Architecture

Based on the above requirements, the architecture of the network is designed as shown in Fig. 1.

As shown in Fig. 1, the system is consisted by two parts: mesh subnet and star subnet. Mesh subnet includes cluster heads and sink node, responsible for transferring data form sensor to sink. Star subnet is consisted by sensor nodes and cluster head, which is responsible for the sensor data collection. Mesh and star using different frequency bands and protocols. Mesh subnet uses 2.4 GHz IEEE 802.15.4 [17] compacted device and DJmesh [18] protocol. Star subnet uses 433 MHz ISM frequency band and a new designed simple TDMA MAC protocol. The mesh/star topology divided the network into small area, which is flexible for control and expansion, and avoids the disturbing between sensor nodes and cluster heads by using two separated frequency band.

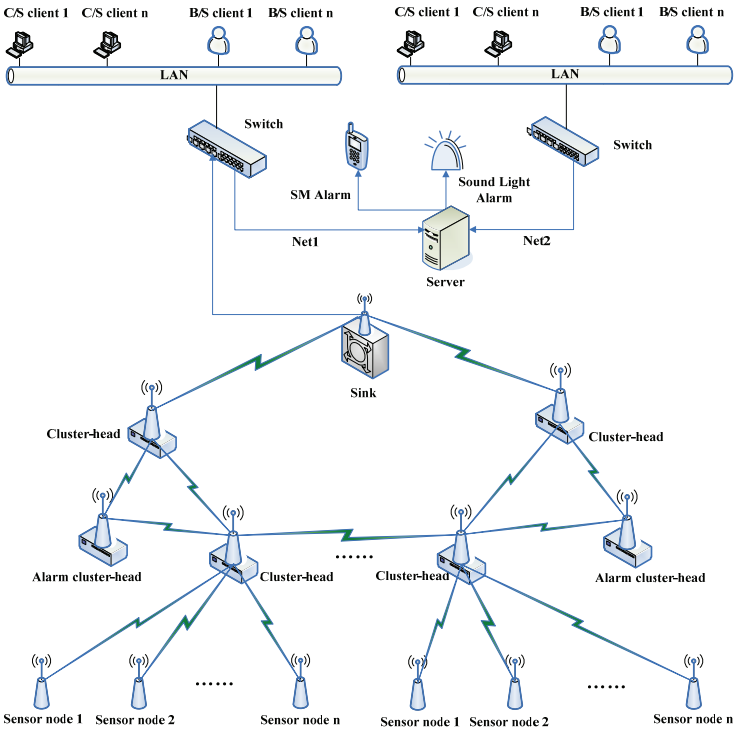


Fig. 1. Architecture of the roller-monitoring network.

Network is consisted by three different kind devices. Sensor node is responsible for the period temperature sampling and forwarding data to the cluster head. When the sampled temperature exceeds the alarm level, it will speed up the sampling period. Sensor node also takes star network transmission performance statistics and responses for parameter configuration commands from cluster head. Cluster head is responsible for the maintenance of the star network, which has two RF transceivers. It will collect sensor data of the whole cluster and performance statistic results of star subnet, and forward the data to sink through mesh subnet. Sink node is connected to a PC server and transfer data of wireless sensor network to Ethernet.

3.3. Hardware Design

Sensor node uses Atmega64 MCU and CC1100 RF transceiver with a PA of 15dbm output. DS18B20 digital sensor is used for temperature sampling. Sensor node is packed in an IP68 protect case to prevent from the spray of the cooling liquid. Cluster head uses ATmega128L MCU and has two RF modules. The CC1100 RF is used for star subnet and the Xbee module is connect to MCU through UART bus. Xbee module uses DJmesh protocol for multi-hop transmitting data between clusters. Sensor node and cluster head are shown in Fig. 2.

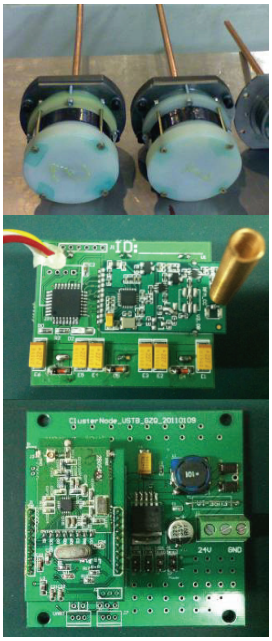


Fig. 2. Architecture of sensor node and cluster head.

4. MAC Protocol Design

In star subnet, each star is a cluster and we dispatch each cluster different frequency channel to avoid interference from neighbor cluster. And the channel is calculated by cluster ID. In a star subnet, each sensor node has a physic address PA, which is a two byte unique number in the whole network. And each sensor node has a logical address *LA*, *LA* is a one byte number start from 1, and the sink node identifies a node by combine cluster ID and *LA* together. For cluster head, *LA* is zero.

We defined a NPT structure to describe the relationship of *LA* and *PA*. NPT also stores the parameters of each node. A NPT table includes *PA* address, *LA* address and alarm level, link quality etc.

4.1. Network Joining and Leaving

4.1.1. Channel Scanning

The process of the channel scanning is shown in Fig. 3. First, when sensor nodes start to work or waking up from sleep mode, sensor nodes will judgment whether the network state flag *bInNet* is true. If sensor node is already in network, skip scanning process. If *bInNet* is false, node will start a channel scanning from channel 1 to channel n. Sensor node listens beacon packet on each channel for *T_BEACON_LISTEN* seconds. If beacon packets received, set current channel *channel[i]* to 1, and record the information of cluster head contained in beacon, then, change to next channel. If no beacon received in *T_BEACON_LISTEN* seconds, switch to the next channel to continue listen until all the channels are scanned. When sensor node scanned all channels, if no beacon received in any of the channels, in other words, element of array *channel[n]* is zero, sensor node will start sleep algorithm. If at least one beacon is received, sensor node begins cluster head association steps described in the cluster head association.

Sensor sleep algorithm is a periodic detection process to avoid power consumption in the situation of no cluster heads. Sensor node will sleep for *T_SCAN_SLEEP* seconds and then starts a new channel scanning.

4.1.2. Cluster Head Association

As shown in Fig. 4, sensor nodes scan the array *channel[n]* to search available cluster heads, and check the numbers of sensor nodes in each cluster. If the number in cluster using *channel[i]* equals *MAX_NODES*, set *channel[i]* to zero. If no cluster head found, sensor node will start sleep algorithm. If only one cluster head is available, join the network through this cluster head. If more than one cluster head exist, sensor node selects cluster head with high link quality by function *maxLQI()*, and if several cluster heads have same link quality, choose the cluster head with minimum sensor nodes in the cluster by function *minNODE()*.

When sensor node finished cluster head selection, sends a join request command to selected cluster head, start network join process described in network joining.

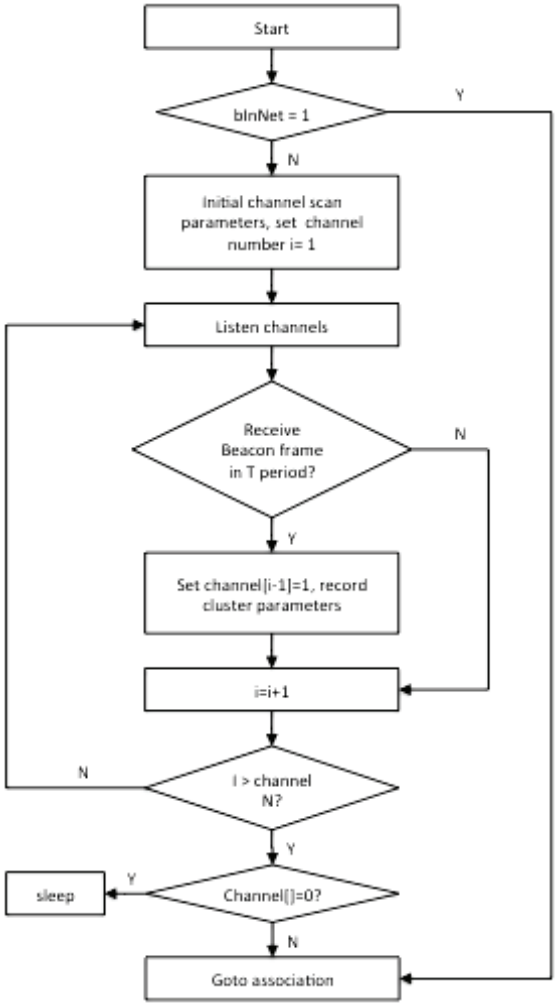


Fig. 3. Process of network scanning.

```
FOR each element channel[i] in array channel[]
    IF channel[i] !=0
       cluster_head[j] = cluster head work in channel[i]
            IF number of sensor nodes of
          cluster_head[j]                    >=
          MAX_NODES_IN_CLUSTER
              channel[i]=0
        ELSE
       candidate[k]=cluster_head[j], j++,k++
        ENDIF
     ENDIF
     i++
ENDFOR
IF  k==0
   sleep
ENDIF
IF k>1
    FOR each element of array candidate[]
          selected[i]=maxLQI(candidate[]),i++
    ENDFOR
    IF  i>1
      FOR  each element of array selected[]
           selected_head = minNODE(selected[])
          ENDFOR
    ELSE
      selected_head = maxLQI(candidate[])
    ENDIF
ENDIF
join network
```

Fig. 4. Cluster head association algorithm.

4.1.3. Network Joining Process

The sensor node joining process block diagram is shown in Fig. 5. After finished cluster head selection, sensor node will send a JoinRequestFrame with its physical address *PA* to cluster head using CSMA, if no ACK received, retransmits JoinRequestFrame at max *MAX_SEND_NUM* times. When cluster head received the JoinRequestFrame, it will assign a logic address *LA* and corresponding TDMA slots for the new joining device through an ACK frame. The allocation method is as following:

1) Search NPT table to find a *LA* where *PA* is zero, assign this *LA* to the new sensor node as the logic address, and then, fill *LA* address in ACK frame.

2) Set *accept_flag* in ACK frame to one, if no available *LA* is found, set *accept_flag* to zero to reject the join request of the new sensor node.

3) Transmit the ACK frame to sensor node.

Sensor node will check *accept_flag* when ACK frame is received. The *accept_flag* equal one indicates that join network successfully, and then sensor node will set *LA* in ACK as its logical address and calculate the TDMA slots. Otherwise if *accept_flag* equals zero, join request is reject by cluster head. Sensor node will mark current cluster head invalid (set *channel[i]* to zero) and select a new cluster head using algorithm in cluster head association to join the network.

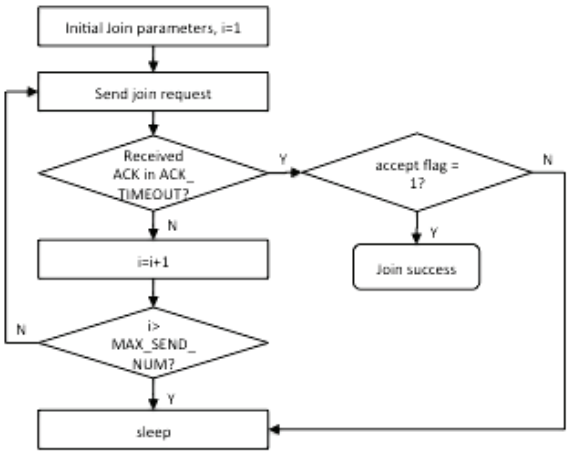


Fig. 5. Sensor node joining process.

Cluster head will send a join notification to sink node after successfully transmitted ACK, join notification packet includes updated NPT table and cluster ID.

4.1.4. Network Leaving

Network leaving happens in two scenarios. The first is communication interruption caused by environment interferences; sensor node cannot communicate with its original cluster head and then

rejoin the network. The second is the support roller has been replaced with another roller, which causes moving of the sensor nodes, thus sensor node will change the state of online and offline and search a new cluster to join.

Sensor node must judge whether leaving one cluster in a limited time when it leaved the network. When *bInNet* flag is true, if sensor node continuously lost ACK in *PACKAGE_LOST_MAX* times sample periods, sensor node sets *bInNet* flag to false and starts join process.

Cluster head can detect leaving of sensor node. For cluster head, in each sample period, PRR (packet receive rate) minus one, when sensor data is received, the corresponding parameter PRR in NPT table is set to value *MaxPRR*. If PRR equals zero, corresponding sensor node has left the network, cluster head will set PA of this sensor node to zero. Network leaving process is shown in Fig. 6.

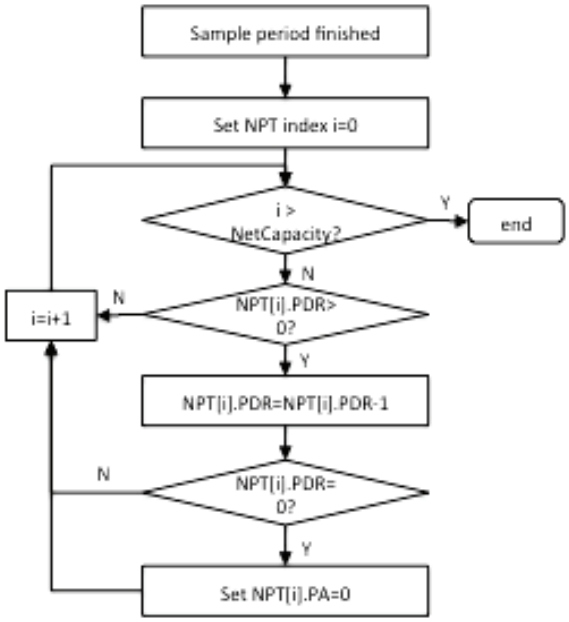


Fig. 6. Sensor node leaving process.

4.1.5. Slot Dispatch

Slot allocation in the MAC protocol using a fixed dispatching method. Because each sensor node has a one byte *LA* address in a cluster, thus, we use *LA* address as the input of the slot allocation algorithm. The whole slots are divided into two parts, the front slots are allocated for sensor node sending data to cluster head and the rest slots are allocated for cluster head send packets to sink. In sensor and cluster communication slots, each sensor node has a base slot, which is the same of its *LA* address number, and the rest slots are for alarm packets transmission. Thus, we do not need a slot request or slot allocation command, which reduced the communication complexity and increased the reliability.

4.2. Network Running

4.2.1. Sensor Running Process

As the flowchart shown in Fig. 7, sensor node samples temperature and battery voltage after it wakes up. If the temperature is higher than the alarm level, set *bIsAlarm* to true and send the data using alarm slots, then change sample period to speed monitoring. Otherwise, when temperature is lower than alarm level, make sure *bIsAlarm* is false and resume normal sample period and transmitting slots.

Sensor node will create a DataFrame filled with temperature, voltage, link RSSI and LQI, and send it to cluster head. When received the ACK, sensor node will compare the parameters (such as alarm level, sample period etc.) in ACK with local parameters to decide whether to modify the parameters. If ACK is not received in *ACK_TIMEOUT* time, go to sleep.

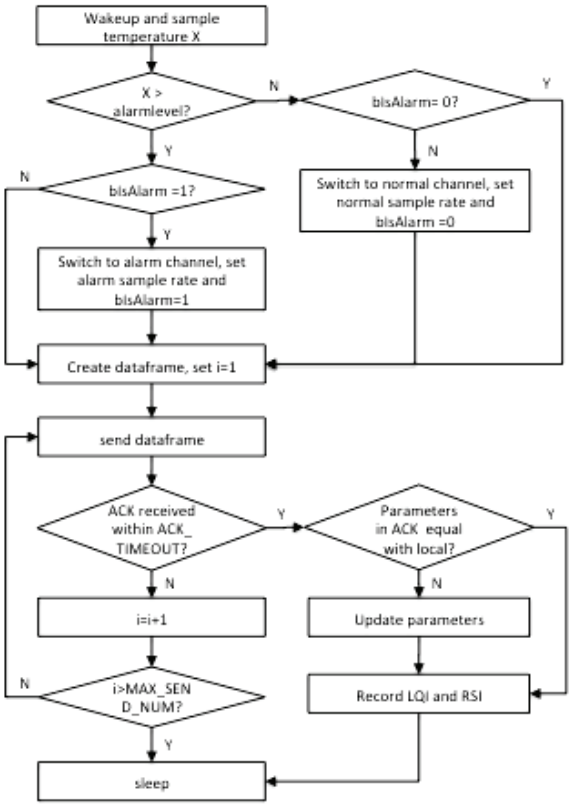


Fig. 7. Sensor running process.

4.2.2. Cluster Workflow

Cluster head connects the sensor node and sink, which is responsible for two functions: communication with sink and communication with sensor nodes.

Cluster head uses mesh topology and multi-hop DJmesh protocol, when server change a parameter of sensor node, sink will send a command packet to cluster head. When cluster head received the

command packet, it will store parameter (such as sample period, alarm period etc.) in a local variable. When cluster head received the data frame from sensor node, it will fill the parameters into ACK frame and send to sensor node. Parameters will be sent with ACK frame in each sample period, so if an accidental disconnection between sensor node and cluster head happened, parameters can be updated in the next several sample periods, thus ensure the reliability of parameter setting.

As the role of sensor nodes management, when received data from sensor node, cluster head firstly searches parameters of sensor node using *LA*, stores temperature and voltage values, and sends an ACK to sensor node. Cluster head will periodically send data fame including temperature and voltage of all sensor nodes in the cluster to sink through mesh network. Also, a link quality frame will be sent to sink including the LQI of uplinks and downlinks.

4.3. Reliability and Power Analysis

This part we will analysis the reliability and power consumption of sensor nodes. For cluster head, we often use active power supply. Consider the situation of intra-cluster communication and assume the packet transmitting successful rate on different channel is *pi* (i =1, 2, ... m), for a sensor node *Si*, if it has at max *r* retransmissions, the transmitting successful rate between sensor and cluster head is:

$$P = P_i(1+(1-P_i)+(1-P_i)^2+\cdots+(1-P_i)^{r-1}) = P_i\sum_{x=0}^{x=r}(1-P_i)^x$$

Low power consumption is one of the targets of the system design, which is high related to hardware. Assume the system transmitting reliability requirement is P, and sensor node has at max r retransmissions, the reliability of the channel is p, then, average retransmission times is: $\alpha = log_{1-p}(1-P)$ For sensor nodes, if the sample period is T_{sample} then

$$T_{sample} = \alpha T_{w\text{-}fifo} + T_{r\text{-}fifo} + T_{sensor} + T_{protocol} = \alpha T_{w\text{-}fifo} + T_{r\text{-}fifo} + \Delta$$

where sensor data process time T_{sensor} and packet process time $T_{protocol}$ are almost fixed , $T_{w\text{-}fifo}$ and $T_{r\text{-}fifo}$ are writing packet and reading packet time form RF FIFO, which are related to packet length. The RF working time includes sending, receiving ACK and waiting ACK timeout, as shown in bellowing formula:

$$T_{RF} = \alpha T_{tx} + T_{rx} + (\alpha - 1) T_{ack}$$

In the worst situation, if all transmission are failed, $T_{RF} = r T_{tx} + r T_{ack}$. Thus, in a sample period, sensor nodes power consumption is:

$$\begin{aligned} E_{sensor} &= (\alpha T_{w-fifo} + T_{r-fifo} + \Delta) I_{cpu} + \alpha T_{tx} I_{tx} + \\ &(T_{rx} + (\alpha - 1) T_{ack}) I_{rx} + (T_{cycle} - T_{cpu}) I_{sleep} \\ &= (\alpha\sigma L_{tx} + \sigma L_{ack} + \Delta)(I_{cpu} - I_{sleep}) + \alpha\omega L_{tx} I_{tx} \\ &+ \omega L_{ack} I_{rx} + (\alpha - 1) T_{ack} I_{rx} + T_{cycle} I_{sleep} \end{aligned}$$

where σ is the speed of SPI bus and ω is the RF transmitting rate, L_{tx} and L_{ack} are the length of data frame and ACK frame separately.

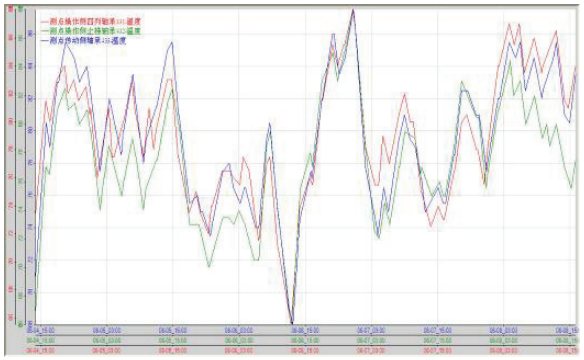
5. Deployment and Validation

We deployed a cold strip mill support roller monitoring system in Ansteel of China in 2010, which works reliably and power efficiently. The cold strip mill has five stands and 6 roller each stand, in which, supporting roller is the biggest roller and with the longest working period. According to the mechanical structure of the supporting rollers, the operating-side of each supporting rollers installed two temperature sensor nodes, and the electrical side installed one node. There are ten supporting rollers for online production, and also ten supporting rollers for offline maintenance, thus the whole network includes 60 sensor nodes. For a reliable communication, we use 5 clusters for online sensor nodes and 2 clusters for offline sensor nodes. The environment of cold strip mill is shown in Fig. 8.

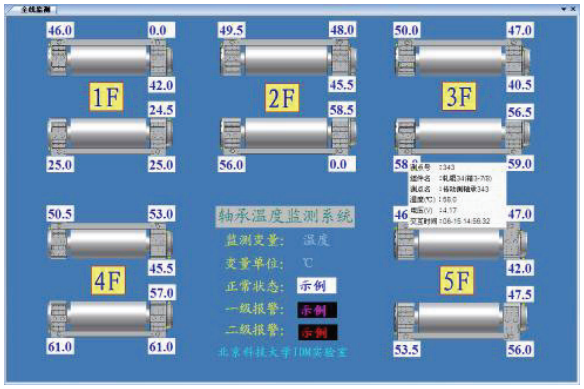
Fig. 8. Cold strip mill application environment.

The monitoring results of the system are shown in Fig. 9(a) and (b). As shown in Fig. 9(a), three sensor nodes on one supporting roller has the same tendency, and we can see form the figure that when a roller offline, the temperature goes down and when changed with a new roller, the temperature goes high. Fig. 9(b) is the display interface for system manager, when temperature beyond the alarm level, the color will be changed from blue to red to notify system manager.

Fig. 10 shows the relationship of temperature and battery voltage. The voltage will go high when temperature goes high which is the characteristic of the battery. Fig. 11 is the variation of the RSS. When the temperature is low, which means roller is not working and store on the maintained ground. In such situation, link reliability is not good as roller working because of the heavy obstacles and long distance between sensor nodes and cluster head.



(a)



(b)

Fig. 9. Temperature monitoring results and system interface.

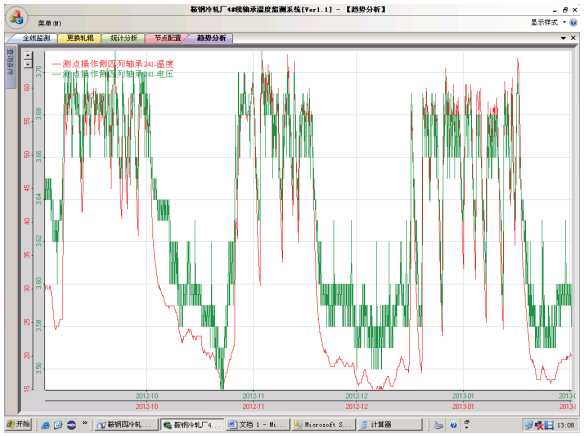


Fig. 10. Voltage variation with temperature.

6. Conclusions

Equipment pre-maintenance is one of the key factors for high quality production in industry. In a cold strip mill factory, the health of the roller direct affect the products, but rollers in mill factory work in a very harsh environment and also impossible for human repair when they are working. The paper proposed a novel wireless system for online roller temperature monitoring. The system uses mesh/star topology and two different frequency bands to avoid the frequency interference. Mesh subnet uses DJmesh protocol and in star subnet, we proposed a simple but reliable TDMA MAC protocol. Each star subnet forms a cluster and uses a unique frequency channel. Sensor node in a cluster will choose the best cluster head to join the network and have a self-organized strategy with low power consumption. The real deployment in factory shows that the protocol has high reliability and low power consumption, which satisfied the requirement in harsh industrial environment.

Fig. 11. RSS variation with roller change.

Acknowledgements

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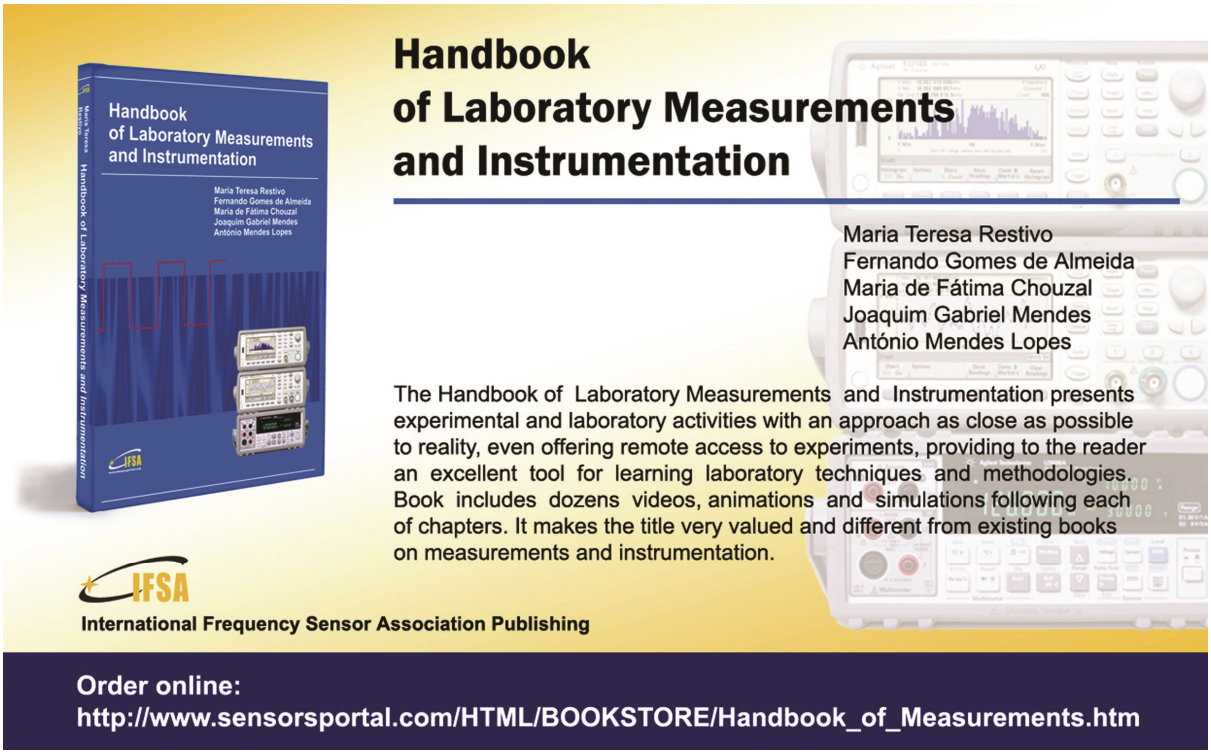

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Low-Power Design and Critical Event-Triggered Transmission Scheme for Wireless Strain Gauge Sensor Node in Aircraft Structural Health Monitoring

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Abstract: The development of wireless sensor network integrated into aircraft structural health monitoring (SHM) can help to achieve the goal of deploying more sensors for the significant benefits in flexibility, interoperability, mass reduction, system redundancy and enhanced robustness. The objective of this paper is research and development concerning wireless strain gauge sensor node mounted on aircraft, towards prolonging the battery lifetime and balancing the power-performance tradeoff. This wireless sensor node can acquire the strain in a structure, convert, conditioning, process and finally transmit data by wireless connection. This wireless sensor node included mainly a nonlinear bridge strain transducer, bridge sensor conditioner, lookup table, low pass filter, ADC, MCU, low-power RF transceiver, external memory, RTC, energy harvesting, energy storage and management. The functions, considerations and specifications of these main parts were described. A low-power design and a critical event-triggered transmission scheme were illustrated and applied to reduce power consumption and conserve network bandwidth. The experimental results show that this wireless strain gauge sensor node is reasonable potential for aircraft SHM. *Copyright © 2013 IFSA.*

Keywords: Aircraft structural health monitoring, Wireless strain gauge sensor, ZigBee, Energy management.

1. Introduction

Structural health monitoring (SHM) is the process of detecting and interpreting any "changes" in a structure, especially for adverse phenomenon. The goal of SHM is to improve the safety and reliability, and reduce life-cycle costs in civil and military infrastructures by detecting damage before they reach a critical status. To achieve this goal, the relevant technology is being developed to transit from qualitative visual inspections and schedule-based maintenance procedures to more quantifiable and automated damage assessment processes. These processes are implemented by using both hardware and software with the intent of achieving more effective condition-based maintenance (CBM). The extensive technology based on SHM has been developed over the last 20 years [1]. A living example of this technology is embedding sensors in engineering structures such as bridge [2,3], where the distributed sensors acquire various real-time structural changes like stress and strain, and then transmit them to a remote data acquisition centre for monitoring and evaluation. As a result, more information needs to be moved somewhere, so that the only hope for the traditional hard-wired method may not be an optimum result, especially for some special situations.

The wireless sensor network (WSN) will be a most promising technology to replace bulky cables and fixed wiring. Today, WSN is used in many industrial and consumer applications, such as

structural health monitoring, industrial process monitoring and control [4], consumer electronics [5] and so on [6, 7]. Moreover, the development of WSN is motivated by military applications such as battlefield surveillance [8, 9]. Wireless communication can remove the cabling problem of the traditional monitoring system and considerably reduce the maintenance cost. Compared to a hard-wired sensor network, WSN has double or even quadruple redundancy in the size of sensor hardware and wire harnesses [10]. With the aid of modern technology, the real-time monitoring based on WSN becomes possible. As a result, WSN can bring many special favors for SHM.

To select a suitable communication protocol standard is a key to the interoperable issue of WSN. ZigBee is a suite of high level wireless communication protocols based on an IEEE 802.15.4 standard. ZigBee is targeted at the applications that require a low data rate, long battery lifetime, secure networking and supporting up to 255 nodes per network, therefore it is particularly suitable for WSN [11, 12]. ZigBee Devices can self-organize into a short-range wireless communication network to support relatively low throughput applications such as distributed sensing and monitoring [13, 14]. ZigBee network ranges from simple single-hop star topologies to more complex multi-hop mesh networks. Although ZigBee technology is currently still developing and costly in a certain extent, the emergence of ZigBee standardization is appealing because its potential is closely related to the objectives for relatively fast, low cost and simplified implementations compared to more traditional wired network installations applied in WSN.

2. Aircraft Structural Health Monitoring

Several studies have been conducted to develop SHM which use the data collected from distributed wireless sensors embedded into the structure [15, 16]. Although the WSN used in SHM has been developed for many years, the most examples are for the fixed building or workplace. The researches on vehicle and aircraft are only just beginning, whereas many problems still need to be addressed. With the increasing use of novel materials and sophisticated equipments in aircraft structures, it is necessary to develop an innovative method for structural safety assessment. For example, composite structures and metal structures constitute almost the bodies of the aircraft. Most failures of the laminated composite structures originate in the repeated cyclic stresses and impacts to form delamination. Most failures of the fabricated metal structures originate in the development of corrosion and fatigue with the passage of time. For these cases, visual inspection is not a reliable method for failure detection. Moreover, the widely used schedule-based maintenance procedures have a high cost [10]. The development of an integrated SHM and reporting system for pitch links on a helicopter had been described in [17].

Our previous works are to develop and evaluate a versatile aircraft SHM system installed for a certain unmanned aerial vehicle (UAV). SHM system has capable of synchronized monitoring data from a range of wireless sensors cooperated with some hard-wired sensors. The wireless sensors typically are low-power level or passive devices, including strain gauges, accelerometers and thermocouples, etc. In contrast, the hard-wired sensors typically are relatively high power level or high sampling rate devices, including gyroscopes, magnetometers and GPS, etc.

Fig. 1 shows the approximate deployment of several representative devices required for aircraft SHM system. These wireless devices create a mesh network based on ZigBee protocol. The wireless sensor nodes are located spatially on engine, wing, rudder and other critical structural elements to acquire information about the aircraft status. Combining a single personal area network (PAN) coordinator, some ZigBee routers with various wireless sensor nodes are a typical WSN. The sensor node (generally reduced dunction device, RFD) may be tiny, simple device with a very limited power consumption, which has only sufficient functionality to talk with its parent device but cannot relay. ZigBee Router (full function device, FFD) acts as an intermediate node to relay data from other devices, which can gain an additional communication link to extend distance and reliability. PAN coordinator known as central gateway can originally establish and configure the network, which is the most capable of the three node types but exactly one coordinator in each network [13]. Once the aircraft status happens to change, the vibration, pressure, temperature and other characteristic signals will carry a wealth of information. These distributed wireless sensor nodes have access to the response to changes in physical parameters on critical area, then process and transmit the data from child nodes via their parent routers to only PAN coordinator. PAN coordinator provides a wired connection to the data aggregator which implements collect, process, analyze, storage and eventually present available information to the operator.

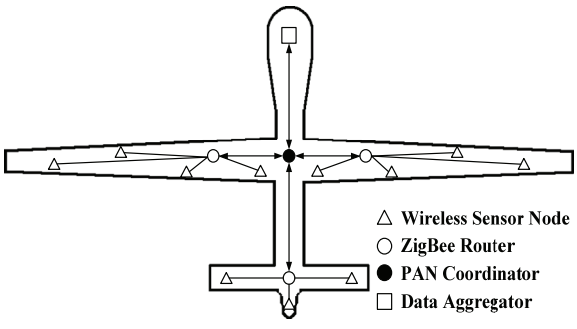

Fig. 1. The WSN Approximate Deployment for Aircraft SHM System.

3. Hardware Descriptions

Generally speaking, a wireless sensor node includes the following main parts: signal acquisition, signal conditioning, MCU, external memory, RF transceiver and power source. Power source contains energy storage, power supply and an option of energy harvesting. Fig. 2 shows a modular design approach that provides a flexible and versatile platform to address the needs of various applications. Signal acquisition is the processes of sampling signals that measure and convert physical parameters into electrical signals. Signal conditioning regulates and filters the resulting samples into a form that can be converted to digital values. The MCU performs tasks, processes data and controls the functionality of other components in the node, such as executing communication protocol and running local computation algorithms. External memory uses for storing application related and programming the device. RF transceiver is an essential component of wireless communication to transmit and receive RF signals. Power supply provides electronic components with stable and sustained electric powers extracting from energy storage including the usual form of either a battery or a capacitor. Energy harvesting generates electric powers from ambient energy source and renews energy storage. The functions, considerations and specifications of these main parts used in our demonstration are described as follows.

Strain gauge sensors are the most common devices for the electrical measurement of structural deformation in aircraft. Strain gauge sensors can be used to measure the fatigue loading experienced under various conditions of speed, weight and mission configuration more accurately far better than the older, less accurate counting accelerometer methods. These strain gauge sensors can be embedded into the composite structures or attached to the metal surfaces to more accurately monitor for potential fatigue damage induced by thousands of flight hours, high stress maneuvers and landings.

3.1. Signal Acquisition and Conditioning

3.1.1. Nonlinear Bridge Strain Transducer

A strain gauge (also known as strain gage) is a component used to measure the strain of an object. The bonded metal foil strain gauges are a mature technology used for many situations. The typical foil resistive-type gauge consists of a wire grid network embedded on an elastic layer, which is bonded with an epoxy layer onto the object of interest. When the object occurs to deform under load, the deformation is transferred to the wire network, causing resistive changes in the material. Since the changes in the resistive characteristics of metallic foil gauges to applied strain are very small, so signal processing is required to convert these changes into voltages that are detectable and able to be further processed. One of the most popular detection circuits is the Wheatstone bridge. The Wheatstone bridge can be either a quarter bridge or a half bridge, a full bridge. The quarter bridge has only one active arm, i.e. strain gauge, and it is the simplest form of Wheatstone bridge configuration. The quarter bridge is shown in Fig. 3 and used in our demonstration.

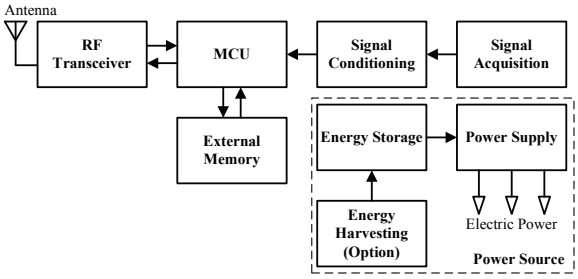

Fig. 2. The Typical Architecture of Wireless Sensor Node with Single-Channel Sampling.

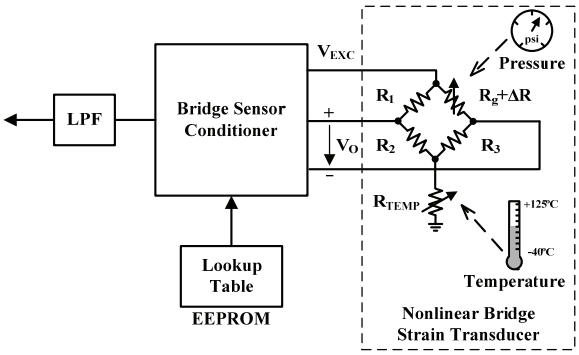

Fig. 3. Signal Acquisition and Signal Conditioning.

A fundamental parameter of the strain gauge is its sensitivity to strain, expressed quantitatively as the gauge factor (GF). Gauge factor is defined as the ratio of fractional change in electrical resistance to the fractional change in strain length:

$$GF = \frac{\Delta R/R_g}{\Delta L/L} = \frac{\Delta R/R_g}{\varepsilon}, \quad (1)$$

where ε is the axial strain, L is the nominal length and R_g is the equivalent resistance of the strain gauge. ΔL and ΔR are respectively the change in the length and resistance due to applied strain. A detailed treatment on stress-strain relationship of strain gauge can be found in [18]. When an excitation voltage, V_{EXC}, is applied across the bridge, the output voltage of the bridge, V_O will be equal to:

$$V_O = \left(\frac{R_2}{R_1 + R_2} - \frac{R_3}{R_3 + R_g} \right) V_{EXC} \quad (2)$$

In the absence of strain, it is apparent that when $R_1 = R_2$ and $R_3 = R_g$, the bridge is said to be balanced and its output is zero. When strain is applied to the strain gauge, R_g is changed by a small amount, ΔR, this will result in an unbalance of bridge and produce a nonzero output voltage. The bridge equation above can be rewritten to express V_O/V_{EXC} as a function of strain:

$$\frac{V_O}{V_{EXC}} = \left(\frac{R_2}{R_1 + R_2} - \frac{R_3}{R_3 + R_g + \Delta R} \right) = \frac{GF \cdot \varepsilon}{4} \left(\frac{1}{1 + GF \cdot \frac{\varepsilon}{2}} \right) \quad (3)$$

The bridge excitation voltage, $V_{EXC,}$ affects both the output sensitivity and the gauge self-heating. From a measurement standpoint, a higher excitation voltage level is desirable, but a lower voltage level can reduce gauge self-heating. The electric power in the gauge is dissipated as heat which must be transferred from the gauge to the surroundings. The gauge temperature is therefore a function of the ambient temperature and the temperature rises due to high power dissipation [19]. Assuming that all nominal resistances exactly equal, I_g is the gauge current, P_g is the gauge power dissipation and given by the following equation:

$$P_g = \frac{1}{2} \times \frac{V_{EXC}^{\ 2}}{R_3 + R_g} = \frac{V_{EXC}^{\ 2}}{4R_g} = I_g^{\ 2} R_g \quad (4)$$

Table 1 illustrates the relationship between gauge resistance and power dissipation under different excitation voltages. It is worth noting that reducing the excitation voltage or increasing the gauge resistance will lower power dissipation, whereas an over-adjustment will cause an insensitive detection. Moreover, increasing some the non-gauge resistance values of R_1 and R_2 is also a good way to further lower the total power consumption.

Table 1. Gauge Power Dissipation in a Quarter Wheatstone Bridge.

Power Dissipation (mW) / Resistance (Ω) / Excitation Voltage (V)	1000	500	350	120
0.1	0.0025	0.005	0.007	0.021
0.2	0.010	0.020	0.029	0.083
0.5	0.0625	0.125	0.179	0.521
1.0	0.250	0.500	0.714	2.083
2.0	1.000	2.000	2.857	8.333
3.0	2.250	4.500	6.429	18.750
4.0	4.000	8.000	11.429	33.333
5.0	6.250	12.500	17.857	52.083
10.0	25.000	50.000	71.400	208.300

In our demonstration, the nominal values are: $R_1 = R_2 = 3000\ \Omega$; $R_3 = R_g = 350\ \Omega$; $V_{EXC} = 2.0$ V. The total equivalent resistance of this Wheatstone bridge is about 626.866 Ω. Since the change value of R_g (ΔR) is very small (only single-digit Ohm), the power consumption between full load and empty load is not significantly different. Hence the gauge power dissipation is about 2.857 mW and the total bridge power consumption is about 6.381 mW.

3.1.2. Bridge Sensor Conditioner

As illustrated in the above section, the output voltage from the Wheatstone bridge is very small and cannot be increased more by some changes in the circuit configuration. Hence, the amplification of the bridge output voltage is required so that it can be further processed. Since the presence of the $1/(1+GF \bullet \varepsilon/2)$ term in equation (3) that indicates the nonlinearity of the quarter-bridge output with respect to applied strain, in addition, temperature factor also makes a serious impact on the output response [20]. Therefore, the appropriate calibration and compensation are required to perform. Analog processing or digital processing is the common way to convert and linearize the sensor sampling signal, but a highly integrated solution is often more fast, precise and even energy-efficient. Compared to a simple operational amplifier, a programmable gain amplifier (PGA) has more advantages, including adjustable Programmable Gain, Programmable Offset, Update of Calibration, Temperature Compensation, Over-Scale and Under-Scale Limits. These feathers are more suitable for connection to nonlinear bridge strain transducer. The PGA309 is a programmable analog signal conditioner designed especially for nonlinear bridge strain transducer, it may perform necessary functions to calibrate, compensate for temperature variance, scale and linearize the sensor signal [21]. Up to 17 temperature coefficients are stored in external EEPROM to allow a linear interpolation algorithm for a piecewise linear temperature compensation. These temperature coefficients are measured in our laboratory over −40 °C to 125 °C.

The PGA309 draws a quiescent current including from analog voltage supply (SA) and digital voltage supply (SD). Its typical value ($I_{SA} + I_{SD}$) is 1.2 mA and the maximum value is 1.6 mA at $V_{SA} = V_{SD} = 5$V without the bridge load. Thus the maximum quiescent power consumption is 8 mW. The dynamic power consumption actually equals to bridge strain transducer power consumption, about 6.381 mW mentioned in above section.

3.1.3. Lookup Table

Lookup table allows multi-slope compensation over temperature for nonlinear bridge strain transducer. Lookup table has access to 17 calibration points consisting of three adjustment values (Temperature, Zero DAC, Gain DAC) that are stored in 16-bit data format (17×3×16 = 816 bit). A 1k-bit EEPROM is enough when using all 17 temperature

coefficients. Once power-up, the PGA309 resets the internal registers and then reads the configuration data from the external EEPROM device [22].

In our demonstration, 24LC01B is a 1 Kbit EEPROM from Microchip Technology Inc. This chip is organized as one block of 128 × 8-bit memory with a 2-wire serial interface. 24LC01B can operate at low voltages down to 1.7 V with low active (1 mA) and standby (1 μA) current, thus suitable for battery energy or restricted power applications. It has a two-wire serial interface which is used to communicate with the PGA309, and can also be used to interface with the MCU on the node. It is designed for more than 1 million erase/write cycles, and is also available in the 5-lead SOT-23 package. More details can be found in the datasheet [23].

3.1.4. Low Pass Filter

Since the aircraft structural element, such as the fuselage in flight, is also subjected to high frequency vibration, however, a relatively slowly varying quantity has been more interests in measuring strain. Thus the output of the PGA309 is fed to a low pass filter (LPF) to eliminate high frequency noise. Moreover, LPF may eliminate radio frequency interference (RFI) presented in the environment, it is especially important for a high-precision measurement device.

3.2. RF Transceiver and MCU

3.2.1. Low-Power RF Transceiver

An integrated RF transceiver solution (Fig. 4) may minimize device size and shorten the interrupt handling time. The CC2530 is a true system-on-chip (SOC) solution for IEEE 802.15.4 and ZigBee applications. The CC2530 combines inside the excellent performance of a RF transceiver with an industry-standard enhanced 8051 MCU, in-system programmable flash memory, 8-kB RAM, and many other powerful features. It provides extensive hardware support for frame handling, data buffering, burst transmissions, data encryption, data authentication, clear channel assessment, link quality indication and frame timing information, which can reduce largely the overhead of the processor [24]. Due to the low active power consumption, the standby power consumption and wide supply range, the CC2530 prolongs battery lifetime and reduces battery maintenance.

3.2.2. MCU

Wireless sensor nodes belong to embedded devices. Since a general-purpose microprocessor generally has higher power consumption than a microcontroller, it is often not considered a suitable choice for an embedded device. Digital signal processor (DSP) may be chosen for broadband wireless communication applications, but the WSN communication is often modest: simpler protocol, easier to handle modulation and the signal processing tasks on nodes are less complicated. Therefore the advantages of DSP are not usually of much importance to wireless sensor node. Field programmable gate array (FPGA) can be reprogrammed and reconfigured according to requirements, but this takes up more time and energy than desired. Therefore, the above processor architectures are almost infeasible for battery-powered wireless devices.

A microcontroller is often used in many embedded devices, because of its low cost, flexibility to connect, ease of programming, and low-power consumption. In our demonstration, the architecture option inclines to MSP430 microcontrollers, which is designed specifically for low-power operation. Depending on MSP430F2274 from the 2 Series MSP family, the current consumption of 0.7 μA during standby mode (LPM3) and 4.32 mA during active mode (AM) that make it an excellent choice for battery-powered applications. The power consumptions of main operating modes in MSP430F2274 will be shown in Fig. 9. MSP430 features a powerful 16-bit RISC CPU, 16-bit registers and constant generators that contribute to maximum code efficiency, from being the main applications and protocol process to coordinating communications [25]. In addition, an integrated 10-bit analog-to-digital converter supports conversion rates of up to 200 ksps that enough to complete sampling signal digitization for our demonstration.

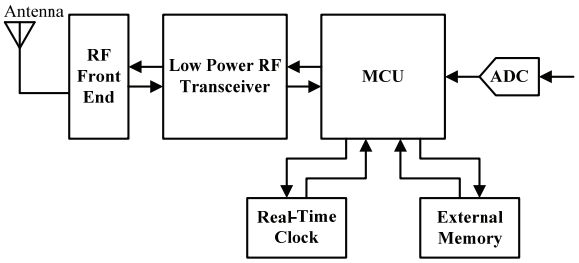

Fig. 4. Low-Power RF Transceiver and MCU.

3.3. Power Source

As shown in Fig. 5, power source module consists of an energy harvesting, a boost converter, energy storage and a smart switch in our demonstration. Energy harvesting resorts efforts to capturing ambient energy sources and transforms into electric power. Boost converter regulates and converts ultra-low level electric power into more usable level like a charger. Energy storage stores power generated from energy harvesting for later use. Energy storage

may be an efficient quick charging capacitor or one of the newly developed thin film batteries. Generally, capacitor is used when the application needs to provide huge energy spikes. Battery is due to less current leakage and therefore used when the application needs to provide steady energy flows. The stored energy is triggered at the required condition to connect up power supply by a smart switch. All of these devices must be sufficiently robust to endure long-term exposure to hostile environments and have a broad range of dynamic sensitivity to exploit the entire energy spectrum. Moreover, extracting maximum power from energy harvestings places unique demands on the energy storage management architecture.

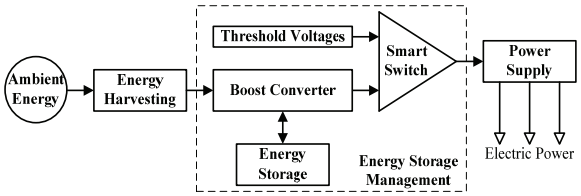

Fig. 5. Switch Mode Power Supply with Energy Storage Management and Energy Harvesting.

3.3.1. Energy Harvesting

Since the most wireless sensor nodes are powered from chemical batteries in the past, the schedule-based maintenance is needed. The battery lifetime is inherently limited so as to cause a maintenance problem [26]. Especially for a large-scale sensor networks, battery maintenances are inconvenient to implement, and also the battery reinforcements are infeasible to realize. Moreover, the growing demands of distributed computations and coordinating communications, such as the preprocessing and packaging of sensor data with the intent of minimizing wireless overhead to reduce power consumption and conserve network bandwidth, will result in such a rapid increase in the power usage of a wireless sensor. Energy harvesting can effectively provide a virtually perpetual energy source so that the battery does not require frequent maintenance [27].

Fig. 6 is a schematic diagram of the flexure element and proof mass. The piezoelectric (PZT) materials are utilized to convert vibration energy into electric power. It is comprised of a tapered flexure element with PZT mounted on the top and bottom of a cantilever beam. The tapered cantilever structure is designed to create a nearly uniform strain field on the PZT elements. A proof mass is affixed to the end of the beam, which resonated at a specific frequency. Such equipments capture vibration energy for powering wireless sensor nodes that have been reported [28]. Other researchers have been active in this area as well, and have described various strategies for harvesting, or scavenging energy from the environment, such as thermoelectric harvesting, photovoltaic harvesting, RF harvesting, etc [27].

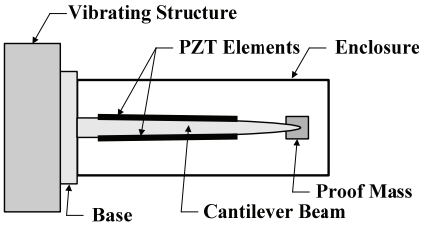

Fig. 6. Piezoelectric Vibration Energy Harvesting Cantilever Flexure Element with Proof Mass.

3.3.2. Energy Storage Management

Although the current energy harvesting technology offers the significant advantages over WSN, the availability of the energy source is often sporadic or time-varying so that an energy storage element is typically needed. The energy storage element can provide certain constant power and allow the device to handle any peak currents that can not directly come from energy harvesting. To prevent energy storage element from damaging, the output voltage of boost converter is compared against the pre-set battery threshold by a smart nanoampere comparator switch (see Fig. 5). This switch features very low closing resistance and very high switching frequency. Such a configuration is very critical for aircraft flight condition where the ambient energy levels may be low or intermittent. Without this switch, the device may never successfully start up, because stored energy levels may always remain insufficient for the task at hand.

Fig. 7 shows the key threshold voltages and operation states that are programmed independently for successfully harvesting ambient energy source and protecting energy storage element. When the input voltage coming from energy harvesting exceeds the minimum input voltage with sufficient power, the cold start state turns on the device to initiate boost converter. After the output voltage continues to increase and keep in pre-set time, cold start state is turned off and the energy storage element starts to charge up. To prevent rechargeable batteries from being deeply discharged and damaged, or to prevent a capacitive storage element from being completely depleting charge, the under voltage threshold must be set to disconnect battery powered. To prevent rechargeable batteries from being exposed to excessive charging voltages, or to prevent a capacitive storage element from being over charging, the over voltage threshold must be set to step down the charging voltage or disable battery charging. An acceptable battery good threshold must be set to instruct work under normal conditions. To avoid misuse and erroneous act due to ripple voltage, the respective hysteresis voltage thresholds should also be reasonable to set.

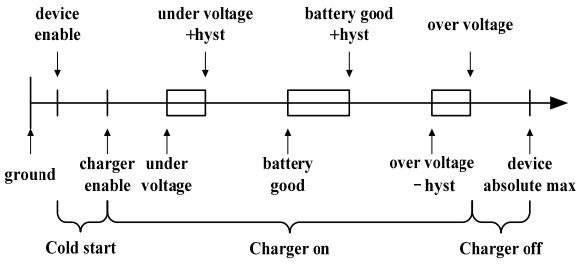

Fig. 7. The Relevant Location of Key Threshold Voltages and Operation States.

The bq25504 and its corresponding external circuit can be designed to efficiently acquire and manage power generated from energy harvesting. The bq25504 integrates a highly efficient boost converter that requires only microwatts of power to begin operating, especially for WSN which has stringent power and operating demands [29].

4. System-level Energy Management

The trade-off between the low-power consumption and high processing speed is a key design for embedded devices. On the one hand, the battery capacity has improved very slowly. But on the other hand, the wireless communication and local computation have drastically increased over the same time frame. Therefore, the development of wireless sensor nodes has stressed even more the needs of the energy minimization. Due to the fact that the wireless nodes are energy constrained even if joining with an energy harvesting, power consumption must be minimized reasonably and managed effectively. A better low-power hardware design technique has helped to lower the power consumption. On the other hand, managing power consumption at higher system-levels can considerably decrease energy requirements, and thus prolong battery lifetime, lower packaging and cooling costs [30]. Two different approaches for lowering the power consumption at the system level have been proposed: dynamic voltage scaling (DVS) and system-level dynamic power management (DPM).

4.1. Dynamic Voltage Scaling

One approach of lowering the power consumption is to slow down the execution or disable some functions and features, when appropriate, also lower the operating voltage of the component. Fig. 8 shows that MSP430 MCU operates at the workable processor frequency corresponding to the supply voltage. Processor frequency is defined by supply voltage. Flash program or erase operations require a minimum V_{CC} of 2.2 V [25].

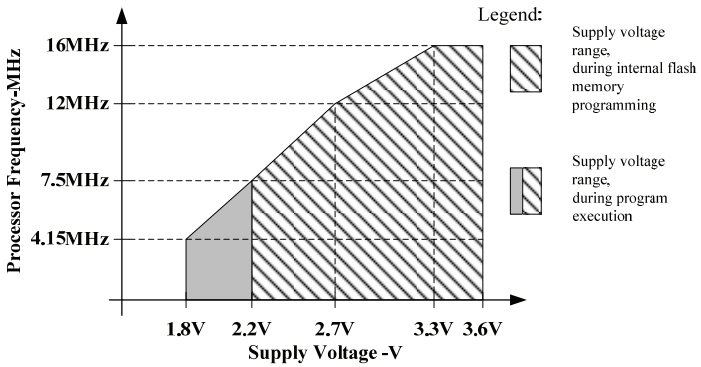

Fig. 8. Operating Area of MSP430 MCU.

4.2. Dynamic Power Management

Embedded sensor devices are designed to be able to deliver peak performance when needed, but their components execute at utilizations less than 100% over the most time. In contrast to DVS, DPM decreases the energy consumption by selectively placing idle components into lower power states. In fact, the wireless communications consume a large fraction of the overall energy consumption in most current wireless sensor nodes. Fig 9 shows a power consumption breakdown for our demonstration. In the same power supply rail, the wireless communication takes up above 90 % of the overall power budget and the second power hungry device is the MCU. Therefore, the optimization of both computing and communication energy consumption is critically important to achieve long battery lifetime.

DVS can only be applied to MCU, while DPM can be applied to RF transceiver, MCU, and all other components that have low-power states. The two approaches are combined effectively in our demonstration.

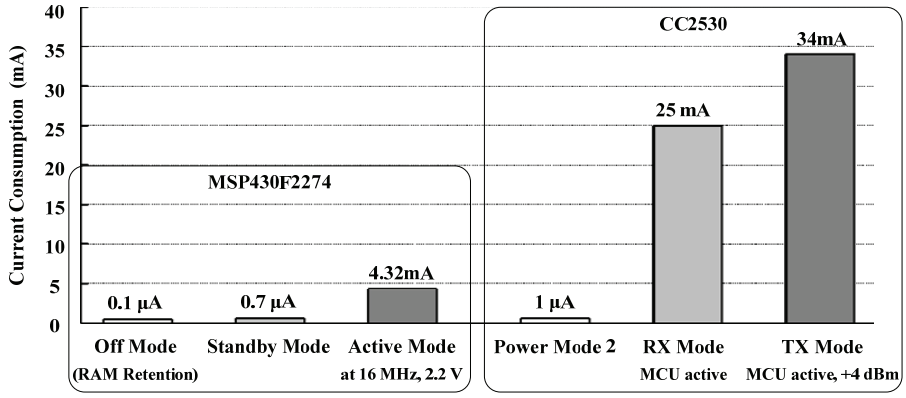

Fig. 9. Power consumption of the two main power hungry components in our sensor node demonstration without counting the external circuit consumption.

4.3. Critical Event-Triggered Transmission

Besides the selection of the low-power components in hardware, system-level energy management design is also needed to consider further. In most cases, the sensor signal swings within a range and then restores stability in the new status phase for some time. The real-time transmission of sensor data needs to consume energy and occupy bandwidth. An autonomous wireless sensor node can adequately observe the dynamics of monitored object and also perform suitable signal processing and feature extraction. Bandwidth is an important resource as well as energy in the scale WSN. A universal solution towards decreasing power consumption is to employ periodic wake-up scheduling, where nodes stay in low-power or sleep modes for the most time that has been described in many literatures [31, 32], however, this method will possibly cause the loss of critical data and signal delay resulting in a deteriorate online monitoring.

Fig. 10 shows the conversion workflow of critical event-triggered transmission scheme between the two working modes applied in our demonstration. In contrast with the periodic wake-up scheduling, our scheme is a good way to reasonably reduce power consumption and conserve network bandwidth. When the variation range of acquired sensor data does not exceed the comparison threshold based on a predetermined initial value, normal mode turns on the device to storage temporarily sensor data into the external flash memory, set sampling rate at 32 Hz, perform timing synchronization and upload data packet to data aggregator once every 60 s. Hence, the maximum value of data delay is theoretically 60 s for remote monitoring side.

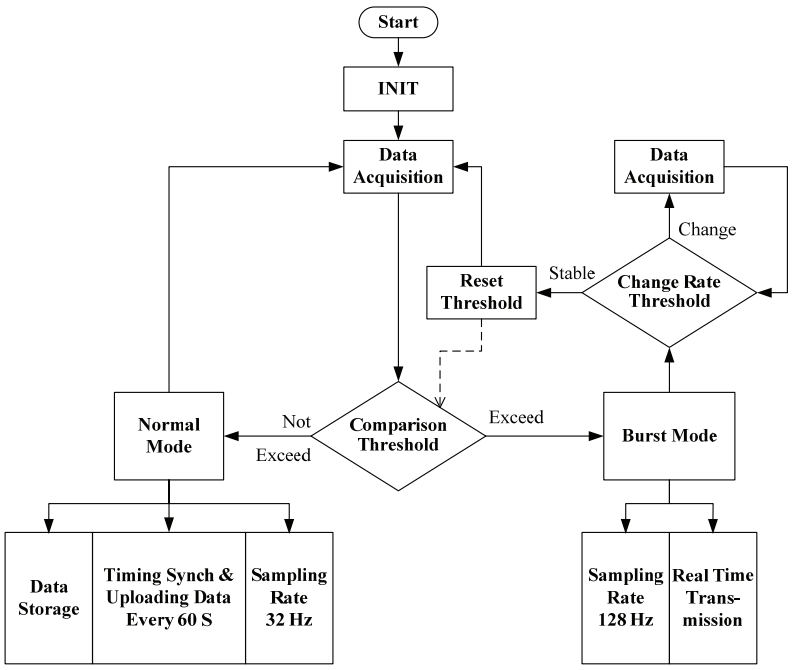

Fig. 10. Critical Event-Triggered Transmission Scheme Workflow Chart.

Additionally, the RF part (CC2530) can even sleep for most of the time, ignore most broadcasts from the network and only needs to periodically wake up to perform timing synchronization to conserve energy. It is worth noting that timing synchronization is a critical piece of infrastructure for any distributed system, especially for WSN. Although each node contains a real-time clock, environmental temperature differences, parameter error and cumulative error will lead to timing drift among network, which can be corrected by a centrally broadcast synchronization beacon. A good algorithm for the compensation of the timing drift is inevitable to achieve high precision and low communication overhead.

When the variation range of sensor data exceeds the comparison threshold and keeps in pre-set time, the node performs the transition from normal mode to burst mode, the sampling rate is set up to 128 Hz and the real-time transmission is perform to track the monitored object. If the change rate of sensor data is greater than zero, the node still sticks in burst mode. Once the change rate is stable within a permissible minimal range for some time, the node returns to normal mode and resets a new comparison threshold based on current stable value.

Fig. 11 shows that our aircraft SHM system can monitor synchronously the structures on GPS, inertia, pressure, acceleration, temperature, humidity, and illumination data via a range of wireless sensors cooperated with some hard-wired sensors.

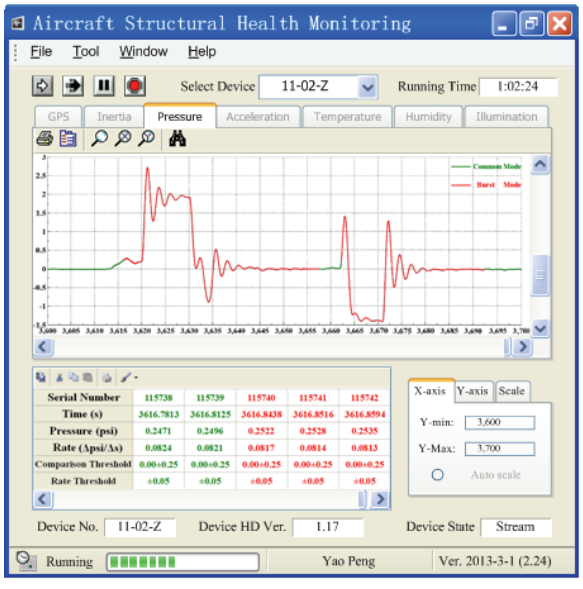

Fig. 11. Human-Machine Interactive Interface of Aircraft Structural Health Monitoring System.

As shown in Table 2, the sensor data of a critical event example can be obtained to illustrate the transition from normal mode to burst mode. Before the 3616.8125 s, the green mark is used to indicate that the node works in the normal mode, i.e. demands for data acquisition at the interval of 1/32 s. At the 3616.8438 s, since the pressure data is 0.2522 psi that exceeds the current given comparison threshold of ± 0.25, the node transits to the burst mode, i.e. demands for data acquisition at the interval of 1/128 s and upload data in real-time. The red mark is used to indicate that the node works in the burst mode.

Table 2. Transition from Normal Mode to Burst Mode (screenshot from aircraft SHM system).

Serial Number	115738	115739	115740	115741	115742
Time (s)	3616.7813	3616.8125	3616.8438	3616.8516	3616.8594
Pressure (psi)	0.2471	0.2496	0.2522	0.2528	0.2535
Rate (Δpsi/Δs)	0.0824	0.0821	0.0817	0.0814	0.0813
Comparison Threshold	0.00±0.25	0.00±0.25	0.00±0.25	0.00±0.25	0.00±0.25
Rate Threshold	±0.05	±0.05	±0.05	±0.05	±0.05

As shown in Table 3, after the 3683 s (not shown in the Table 3), since the change rate consistently lowers than the rate threshold value of ± 0.05, and the preset stable time is set for 10 s, therefore, at the 3693 s, the node transits back to the normal mode. Meanwhile, according to the approximation of current data value, the node resets the comparison threshold value from 0.01 ± 0.25 to − 0.01± 0.25.

Table 3. Transition from Burst Mode to Normal Mode (screenshot from aircraft SHM system).

Serial Number	125897	125898	125899	125900
Time (s)	3692.9922	3693	3693.0313	3693.0625
Pressure (psi)	-0.0096	-0.0095	-0.0090	-0.0085
Rate (Δpsi/Δs)	0.0176	0.0173	0.0165	0.0151
Comparison Threshold	0.01±0.25	0.01±0.25	-0.01±0.25	-0.01±0.25
Rate Threshold	±0.05	±0.05	±0.05	±0.05

5. Conclusions

Despite the limitations of the current prototype, the overall results presented by the experimental tests are satisfactory. Nevertheless, further tests are needed to perform in order to assess the behaviors of the system under more harsh conditions. In real scenario of onboard aircraft, the system design will be much more complicated and sophisticated because of the enhanced efficiency and reliability by a greater number of monitoring points to be measured. Furthermore, The SHM of a full-scale aircraft requires the higher demands of transmission rate, timing synchronization, data processing capacity and so on. Therefore, there are some significant technical challenges for the successful implementation of aircraft SHM.

The WSN represents an exciting development, with the potential to integrate into an aircraft environment for SHM applications. The combination of advanced technologies including WSN, low-power design, system-level energy management, energy

harvesting, signal acquisition and data processing, etc is a powerful solution for SHM. These capabilities coupled with the appropriate methods will enable critical structure to be monitored remotely, securely and automatically. Our future research and work should be directed in addressing the below technical challenges: The SHM system is capable of transmitting monitoring results via satellite. The area of information fusion between wireless network and hard-wired network (or bus) is conducted. A new wireless aircraft certification regulation needs to be developed to prevent the various security and safety threats.

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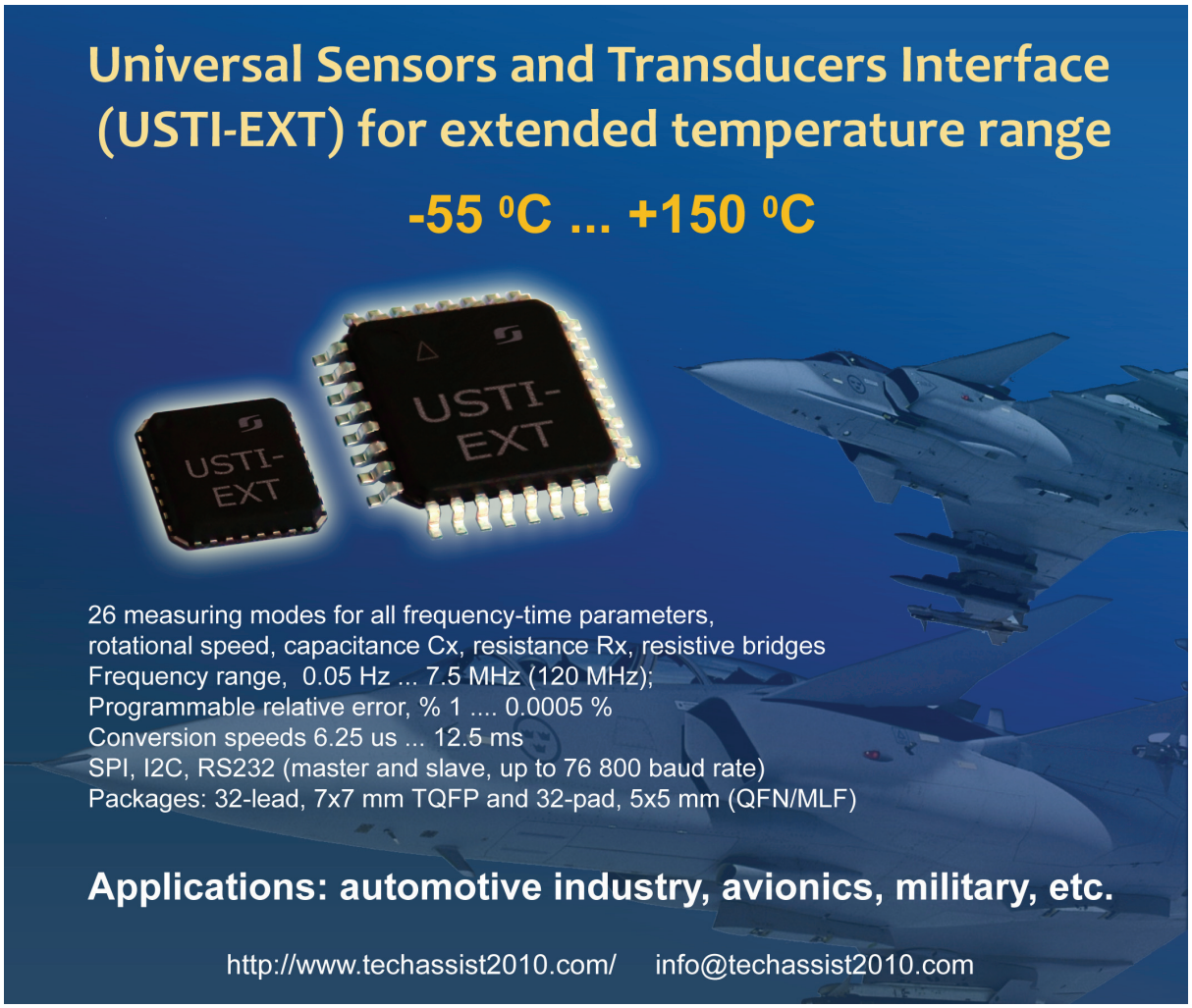

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Imaging of a Metallic Object Using Magnetic Induction Tomography

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Abstract: This paper describes the metallic imaging results from magnetic induction tomography (MIT) hardware. Copper phantom, representing a high conductivity material, was placed into the measurement region of interest. The initial 16-channel MIT consists of eight excitation coils and eight receiving coils stacked alternately. The measurement was undertaken on single and dual arrangements of copper phantom aligned in random coordinates so that the sensitivity of the excitation and receiving coil sensors could be experimentally observed. The results of the imaging results show that the hardware and algorithm used were capable of processing the data captured by the receiver and are discussed at the end of the paper. *Copyright © 2012 IFSA.*

Keywords: Magnetic induction tomography, Excitation coil, Reconstruction algorithm, Conductivity, Copper phantom.

1. Introduction

Magnetic induction tomography (MIT) is a non-invasive contactless tomography modality which works the on passive electrical properties of tissue; conductivity (σ), permeability (μ) and permittivity (ε) to map the object under investigation [1-3]. MIT applies a magnetic field from an excited coil to induce eddy currents in the material; the induced eddy current is then detected by sensing coils [1]. This technique has been variously named as mutual inductance tomography (also MIT) and electromagnetic tomography (EMT). The MIT modality system has three main components: sensors, conditioning electronics and the host computer. The sensors consist of excitation coils, transmitters at one end and receivers at the other, for data measurement. Typically, the operating frequency for an MIT system is from several tens of kHz up to several tens of MHz. Unlike any other electrode-based tomography, MIT may appear better as the method is contactless; it does not suffer from poor electrode contact [17].

2. Image Reconstruction Algorithm: Linear Back Projection (LBP)

The image reconstruction algorithm plays a vital role in converting the sensor-end data into appropriate output values in the form of images. In the LBP algorithm, the concentration profile is generated by combining the projection data from each sensor with its computed sensitivity maps [4]. The algorithm offers fast processing time due to its simplicity and it can be expressed mathematically as below [5]:

$$V_{LBP}(x,y) = \sum_{Tx=0}^{n} \sum_{Rx=0}^{n} S_{Tx,Rx} \times \overline{M}_{Tx,Rx}(x,y), \quad (1)$$

where $V_{LBP}(x,y)$ is the voltage distribution obtained using the LBP algorithm in an n x n matrix where n is equal to the dimension of the sensitivity maps; $S_{Tx,Rx}$ is the signal loss value of receiver *Rx-th* for projection *Tx-th* in volts and $\overline{M}_{Tx,Rx}(x,y)$ is the normalized sensitivity matrix for the view of *Tx-Rx*. In this system, the sensor loss value can be obtained by using the following formula:

$$V_{in} = \frac{D_{out} \times 5}{1023}, \quad (2)$$

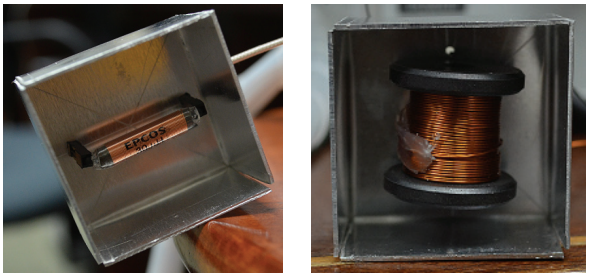
where D_{out} represents the voltage loss (in digital value) in the sensitivity matrices and 5 V is the input voltage to the excitation coil. Compared to the general LBP equation, V_{in} now represents the signal loss value of receiver *Rx-th* for projection *Tx-th*. Therefore, the LBP equation for the system can be expressed as below:

$$V_{LBF}(x_{th}, y_{th}) = \sum_{Tx=th}^{n=th} \sum_{Rx=th}^{n=th} V_{in} \times \bar{M}_{Tx,Rx}(x_{th}, y_{th}) \quad (3)$$

3. Experimental Setup

3.1 Sensors: Excitation and Receiving Coil

Most of the previous work on MIT has usually focused on two main designs of coil sensor: the air core and the ferromagnetic core. Relatively, the low sensitivity of an air coil sensor can be partially overcome by the incorporation of a ferromagnetic core, which acts as a flux concentrator inside the coil [6]. The coils with ferrite core provide a more stable and concentrated field while the coils without ferrite a core produce less concentrated field [7-10]. In this experiment, the excitation coil receives an alternating sine wave at a 500 kHz frequency to generate a magnetic field, the so called primary field. Fig. 1 (a) shows an excitation coil and (b) a receiving coil, both with the square-type electromagnetic field screen.



(a) (b)

Fig. 1. (a) Excitation coil, (b) hand wound ferrite core receiving coil with n=200. Aluminium screen with outer dimension of 42 mm × 42 mm × 22 mm and 1 mm thickness applied to both coils.

The square-type electromagnetic screen was applied to the sensors to minimize the effect of noise from the capacitive coupling and other external magnetic fields [11]. This is an important precaution in protecting the detected signal since it is relatively weak [12]. The magnetic field produced by the excitation coil of this architecture is frequency dependent; therefore, a GFB-8255A function generator has been used for this particular purpose.

3.2. Sensor Fixture

Fig. 2 shows a novel 16-channel single layer sensor jig, made from aluminium, introduced into the system. The perturbation performance in the primary and secondary fields of the system can be detected within a 26.5 cm range. As such perturbation was performed in simulation, it is not discussed further in this paper. Therefore, the measurement area, the so called region of interest (ROI), designed with a 26.5 cm diameter, is the area where the measurements take place and is fabricated from acrylic plotted with a simple x and y-axes ranging from -12 to 12 as shown on Fig. 3. The sensors lie towards imaginary z-axis which sensed the conductivity profile of the materials.

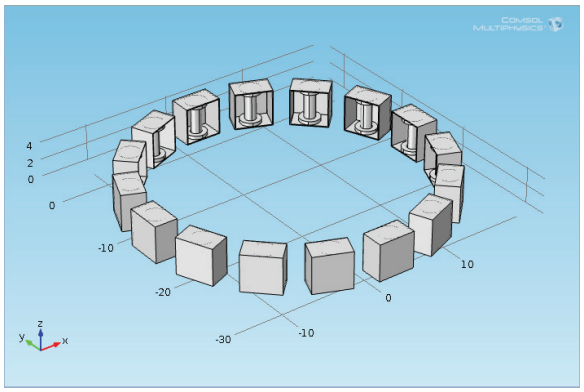


Fig. 2. 16-channel (8Tx x 8Rx) single plane sensor jig.

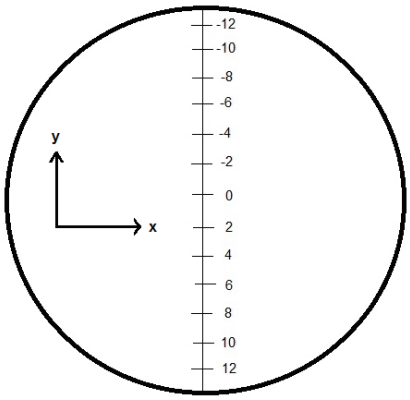


Fig. 3. Region of interest (ROI).

The measurements begin with zero phantom calibration to obtain the maximum voltage value: this means that there is no copper; the copper phantoms were placed in the ROI. Then follow full phantom calibration to obtain the minimum voltage value: this means that a copper phantom approximately the size of the ROI is placed in the measurement area. The uncovered adjacent space between the sensor jig and the phantom during full phantom calibration is negligible. The system is ready to take some measurements once the zero and full phantom calibrations are complete.

3.3. Excitation and Receiving Circuit

AD817 has been chosen to be used as the amplifier in the excitation circuit due to its high output current drive capability and the ability to drive an unlimited capacitive load [14]. Fig. 4 shows the amplifier circuit using AD817. The receiving circuit of the hardware is shown in Fig. 5. The variable amplifier and peak detector architecture have been designed to reduce the effect of phase shift on the receiving signals.

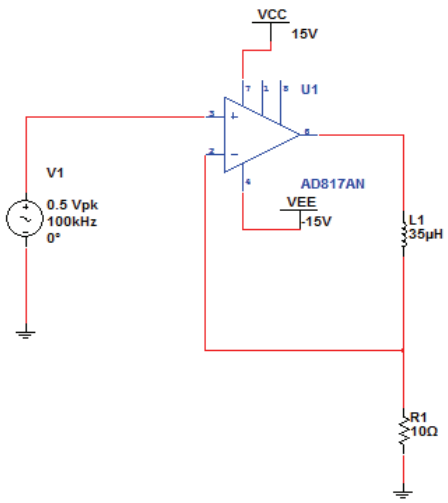


Fig. 4. Excitation circuit [13].

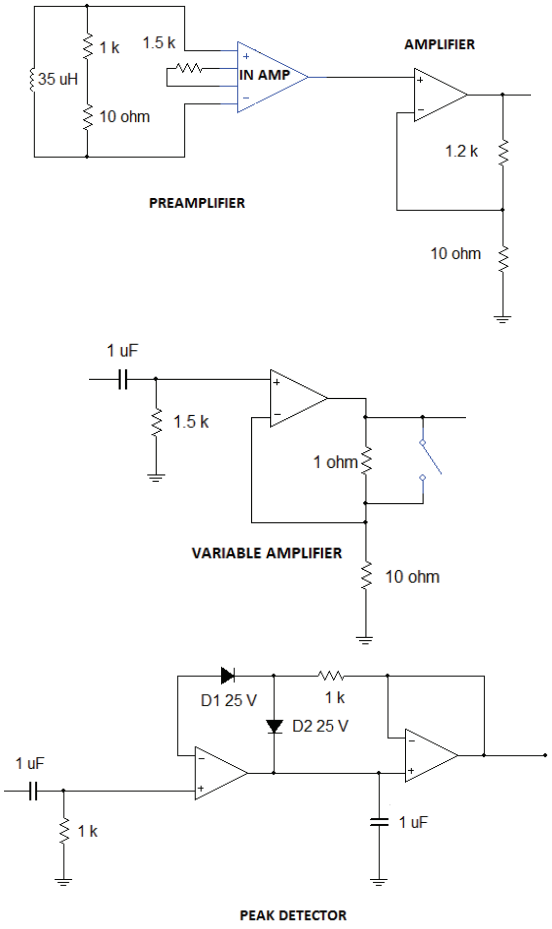


Fig. 5. Receiving circuit.

Fig. 6 shows an excitation circuit suited to an 8-channel excitation coil. The function of the transmitter circuit is to amplify and stabilize the pulse generated by the microcontroller so that the excitation coil only receives the desired time-varying sine wave. The analogue switch ADG333A has been chosen because it provides fast switching times of $t_{on} < 175\ ns$ and $t_{off} < 145\ ns$ [15]. This is important to avoid the overlapping of excitation signals from the microcontroller.

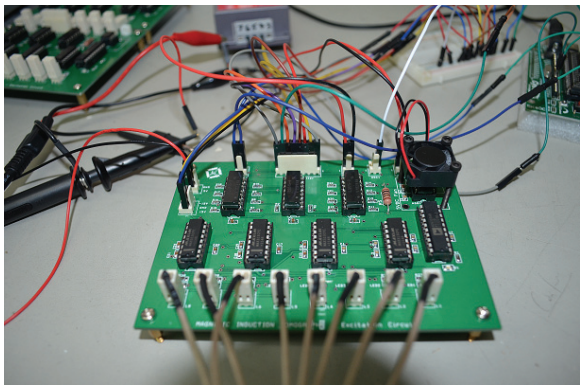


Fig. 6. 8-channel transmitter circuit board.

3.4. Sensor Switching

The excitation signal should be software controlled so that the timing of the sine wave or pulse can be easily tuned respectively. For that reason, a microcontroller, as shown on Fig. 7, is used to provide the excitation signal to the transmitter circuit which will amplify the signal and consequently pass the amplified signal to the excitation coil. The 16-channel sensors, comprising 8Tx and 8Rx mounted to the jig, are controlled by the PIC18F4550 microcontroller. When t_s = 1, Tx1 is ON while the rest are OFF and all Rx are ON. The cycle continues to Tx2 when t_s = 2 until Tx8 when t_s = 8, respectively, and the cycles are separated by 100 ms delays.

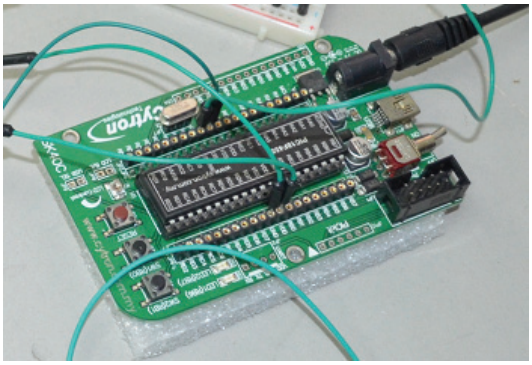


Fig. 7. PIC18F4550.

4. Imaging Results

4.1. Single Phantom

From the imaging tomogram provided, the sensors are initially able to map the conductivity profile of the simple copper phantom model. The smearing effect can be seen clearly on (d), (e), (f), (g), (i), (j), (k) and (l) in Fig. 8. In much of the advanced research into the image reconstruction algorithm in MIT, in common with such methods as the Newton-Raphson method [16], a non-linear algorithm has been chosen to eliminate this effect.

To better explain how the smearing effect occurs, it is necessary to return to the LBP expression itself, where the algorithm is the result of multiplication between sensor loss value and sensitivity matrices. Every sensitivity matrix has its own projection path; every single multiplication of these two factors will produce a straight line, but in the case of MIT, it is a curved line. These curved lines will overlap with each other. The most overlapped line produces a high intensity pixel, while the least overlapped produces a low intensity pixel and the object detected by the LBP is the object with the high intensity pixel. The balance of the low intensity pixel, which is not detected by the LBP, will produce the smearing effect in the tomogram.

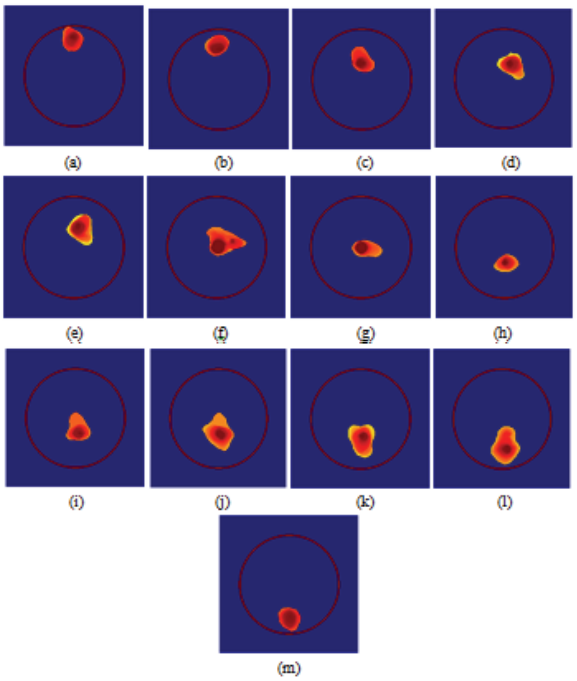


Fig. 8. The tomograms for a single copper phantom model on the y-axis position as (a) -12, (b) -10, (c) -8, (d) -6, (e) -4, (f) -2, (g) 0, (h) 2, (i) 4, (j) 6, (k) 8, (l) 10, (m) 12.

4.2. Dual Phantom

Copper 1 and Copper 2 of 18 mm and 27 mm diameter, respectively, were positioned towards the y-axis. In all three positions as shown on Fig. 9, Copper 1 has been moved towards the ROI centre, while Copper 2 has remained in the same position.

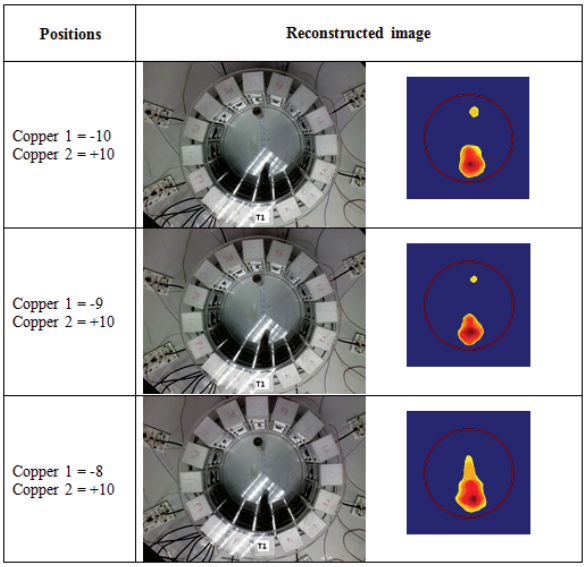


Fig. 9. Dual phantom tomogram.

When using a linear reconstructed algorithm, the smearing effect appears as Copper 1 approaches Copper 2 due to the non-linearity of the MIT. The

lack of colour gradient is due to less sensors being used and it is agreed that the resolution of the images can be enhanced by using an increased number of sensors.

5. Conclusion

This paper presents the preliminary metallic imaging result for MIT hardware. From the results, the MIT system developed is capable at least of mapping the copper-based object. For future development, it is recommended that the following factors are taken into consideration; increase the number of sensors used to enhance the resolution, implement the reconstruction algorithm to reduce the smearing effect and apply a semi-conical aluminium electromagnetic field screen to better concentrate and shield the field.

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Belief and Reputation Based Recommended Trust Computation in Wireless Sensor Networks

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Abstract: Due to limited capacity of individual node, cooperation is a feasible solution to cope with complex task in wireless sensor networks (WSNs). However, node cooperation might be launched with an unreliable collaborative group in large-scaled WSN. To solve this challenge, a node can be organized through recommendation. In this paper, we propose a belief and reputation based recommended trust computation mechanism, which is used to build up creditable relationships among strange nodes. Firstly, we define subjective trustable relationship among nodes, named belief, and objective common trust view, named reputation, for nodes in WSNs. Then, we present a recommended trust computation based on aforementioned two notions. The recommended trust computation is divided into three aspects, i.e., recommended trust computation with different route composition, objective coincidence and optional degree. Furthermore, a recommended trust computation framework (RTCF) and its related proofs are proposed. Finally, we give example and examinations to further explain the efficiency and feasibility of our mechanism. *Copyright © 2013 IFSA.*

Keywords: Wireless sensor networks (WSN), Recommended trust, Belief, Reputation.

1. Introduction

Due to the ability of changing the world significantly, wireless sensor networks (WSNs) obtains more and more attentions in recent years [1-2]. Consisting of large number of tiny and cheap nodes, WSNs would meet the complicated application requirements through the cooperation of nodes. Different from the cooperation in wired network environments (e.g., Internet and Intranet), the cooperation in WSNs faces with some limitations because nodes are unreliable devices with features of weakness, temporary access, low power, and limited capacity. In such a network, node cooperation is vulnerable to threats and risks. Therefore the security of node selection is an unavoidable challenge of node cooperation. Once selecting an unreliable node, its weakness would definitely endanger the cooperation to achieve goals. In our consideration, the problem of securing node cooperation has been a roadblock to large scale application of WSNs. Research field of securing wireless sensor networks is still in its infancy.

In an open and dynamic environment, it is impossible for the node to recognize all the other thousands of nodes and to select appropriate nodes for cooperation, which means that nodes would be selected without any objective and creditable

standards for measurement. Such embarrassing situation brings more unreliability to cooperation in WSNs. In recent years, trust has been suggested as an effective security mechanism for inspiring the cooperation organizing in network environment [2]. Through subjective or objective impressions, i.e., belief or reputation, one can make decisions for node selection depending on their trust evaluation. Some research work has shown that rating nodes' trust and reputation is an effective approach to improve security or promote node cooperation in WSNs [3, 6-10].

In this paper, we introduce a belief and reputation based recommended trust computation mechanism for node cooperation in WSNs. The rest of this paper is organized as follows. In Section 2, we propose the related works of this paper. In Section 3, we address the subjective trust relationship named belief between nodes and objective trust degree named reputation respectively. Then, we present the definition of recommended trust and the computation mechanism which is composed by three aspects: recommended trust computation with different route composition, recommendation objective coincidence and recommendation optional degree in section 4. And further, we address a recommended trust computation framework (RTCF) in this section. Moreover, we present some lemmas and theorems in order to prove the correctness for our mechanism in section 5. Finally, example was given to explain the efficiency and feasibility in section 6.

2. Related Works

The key problem, which lacks of enough attentions, is that traditional trust establishment mechanisms focus on providing trust relationship among nodes, ranking reputation for nodes or managing trust for nodes [11]. Such mechanisms are not efficient because there are a large number of sensor nodes and there are limited communications and unfamiliar relationships between nodes in WSNs. As aforementioned, it is impossible for a node to be familiar with all other ones so that it is indispensable for WSNs to help the node selecting strange ones in order to ensure the performance of cooperation based on most competent nodes. Moreover, wireless sensors networks are deployed in open, unsupervised and insecure environment which is prone to increase the risks of unsuccessful cooperation. And also the absence of any centralized authority in WSNs makes it challenging to monitor every node's reputation accurately. Hence, building cooperative relationships just depending on nodes' trust relationships or reputation is not a viable solution since evaluating trust relationship between every two nodes or scanning reputation would lead to large amounts of cost by flooding the network with request message [2].

Some existing researches aim to address recommended trust for WSNs, such as the ones given in [2, 4, 14, 16]. However, the recommended trust mechanisms in these researches are calculated based on subjective or objective views and ignored the comprehensive views. And also, some researches pay less attention to the recommendation route composition and only calculate recommended trust though single route or one intermediate recommender. In our consideration, recommendation is a feasible and effective solution. A Node can evaluate the trust degree of one who is recommended for cooperation. Recommendation builds up an initial trust relationships between strange nodes. Therefore, our motivation of this paper is to build up a mechanism of recommend trust evaluation for WSNs and make the recommendation fitting the truth of nodes.

3. Belief and Reputation

3.1. Belief Among Nodes

Trust, as an inherent willingness of human beings, shows the emotional and logic confidence relationships between peoples. It is derived from judging trustworthiness by evaluating various facts which can leads to the confidence or distrust. Firstly, we address the subjective trust notion which is named as belief in this paper. Belief demonstrates one's subjective willingness about others. As mentioned in reference [12], subjective trust is based on the ones the past experience and observation. We can define belief between nodes as following,

Definition 1. Belief is a node's subjective opinion towards another one based on the past interactions, judgments and result observations between nodes.

Let there be node d_i and d_j in WSNs, and d_j do n times of interaction with d_i in past. Each time of interaction generated a judgment value $jud_t(d_j)\left(jud_t(d_j)\in[0,1]\right)$ which was given by d_i for interaction with d_j. Assume that there were m times of malicious judgments from d_i to d_j. Let there be p nodes, $type(d_j)$, which are same type of node d_j and did direct interactions with d_i in past. $belief(d_i, type(d_j))$ indicates the belief from d_i to each same type nodes of d_j. Then, belief of past judgments can be calculated as following,

$$belief(d_i,d_j)=\begin{cases}\dfrac{\sum_{t=1}^{n} jud_t(d_j)}{n}\times\left(\dfrac{n-m}{n}\right)^{\frac{1}{n-m}} & n\neq 0\\ \alpha\times\dfrac{\sum belief(d_i,type(d_j))}{p} & n=0\end{cases} \quad (1)$$

In formula 1, belief computation is based on the overall past judgments given by d_i. The number p

denotes the parameter $\alpha \in [0,1]$ reflects the impact to the initial belief from the type of target node. Contrast with traditional average computation of past judgments, we address an impact factor according to the result observations, i.e. total interaction quantity and malicious interaction quantity. In the impact factor calculation, malicious interaction would lead to belief reducing, which corresponds to the facts that people should execute more trustable actions to maintain their trust relationships while even a little of malicious actions would damage their trust relationships in human society.

3.2. Reputation

Reputation is a different complex notion from belief because reputation spans across multiple disciplines and is easily confused with trust [2]. Reputation denotes a public and authoritative trust view from an adiaphorous community [11].

In addition, due to the distributed nature of WSN, it is impossible to set a centralized reputation center to publish and renew each node's reputation. To reducing computation complexity and storage, distributed scheme is a feasible solution for reputation management. In this paper, we proposed a distributed reputation calculation method for nodes in WSN. Our method allows nodes to request local reputation records from creditable nodes and then combine the records with node's own reputation into a comprehensive value. Here, we proposed the reputation in WSN as follows,

Definition 2. Reputation is a global trustable view which combines the nodes' local reputation views in a working community.

Let there be a community D in WSNs, and $d_j \in D$. Assume that the set, CD, includes nodes which have reputation value larger than a threshold. For a node d_i in D, it received h times of judgment, $jud(d_j, d_i)$, in past. Hence, its reputation can be calculated as follows,

$$reputation(d_i) = \frac{\sum_{d_j \in CD} \left[jud(d_j, d_i) \times reputation(d_j) \right]}{h} \quad (2)$$

Similar to formula 1, reputation computation is not a simple average of past judgment. We appoint that only those nodes which have a good reputation can vote judgments to calculate reputation. Further, since the judgment given from various nodes in community, we add their own reputations as impact factor to enhance or weaken their judgments.

4. Recommended Trust

4.1. Single Route Recommended Trust

In large network environment, it is difficult to request for inquiring the reputation of every node, let alone obtain belief relationship between all nodes [4]. In WSNs, node can recognize strange ones through familiar nodes' recommendation. Through the recommendation, node of WSNs will evaluate the subjective belief impression from recommender, and also will measure the objective reputation of both the recommenders and the presentee. It is an integrated consideration for node. Recommended trust is built up through intermediate nodes that have beliefs with both recommendation demander and the recommended strange node. We divide recommendation into three cases according to different route compositions.

We first now present the simplest case of recommended trust computation method. Let the recommendation demander be $\Re$, the intermediate nodes as recommenders be $d_1, d_2, \ldots$, and the presentee strange node be $\Im$. Assume that the recommendation is a single-track and single route through $route = [\Re, d_1, d_2, \ldots, \Im]$ between $\Re$ and $\Im$.

Definition 3. Single route recommended trust is an integrated and single-track trust which consists of subjective beliefs of recommenders by transmitting methodology.

Single route recommended trust can be calculated as follows,

$$\begin{aligned} sr(\Re, \Im) = & \\ & \frac{1}{dep(\Im)} \times \{ [belief(d_i, \Im) \times reputation(\Im)] \\ & + \sum_{i=1}^{dep(\Im)-1} \left[belief(d_{i-1}, d_i) \times reputation(d_i) \right] \\ & + \left[belief(\Re, d_1) \times reputation(d_1) \right] \} \end{aligned}, \quad (3)$$

where function $dep(\Im)$ indicates the route depth from $\Re$ to $\Im$ and $dep(\Im) = |route| - 1$.

In traditional researches, recommended trust is usually treated as a static transmitting value [4, 13, 14]. Nevertheless, recommendation is a sensitive and dynamic process. It builds up on subjective relationships between ones. In fact, recommended trust usually damps with the quantity of intermediate recommender rising. For example, it is possible to bring mistakes in recommendation if there were numerous intermediate links in a single-track recommendation in human society, what is called passing mouth to mouth leading to gossip. Likewise, in WSNs, recommended trust computation definitely should pay attention to such problem. In order to identify such evolution, we propose the damping effect of recommended trust. We modify above formula as following,

$$sr(\Re,\Im)=\frac{1}{dep(\Im)}\times\{[belief(d_i,\Im)\times reputation(\Im)]^{dep(\Im)}+\sum_{i=1}^{dep(\Im)-1}[belief(d_{i-1},d_i)\times reputation(d_i)]^{dep(\Im)-i+1}+[belief(\Re,d_1)\times reputation(d_1)]\} \quad (4)$$

The difference between formula 3 and 4 is that we add the route depth as impact exponent. The net result is that each node's recommendation would be weakened according to its depth in recommending route. That means distant recommendations are deemed less untrustworthy.

4.2. Multiple Routes Recommended Trust

In WSNs, demander node usually has various candidate nodes who can recommend the competent node for its recommendation request. In such case, the presentee node would be recommended to demander node through multiple routes. Therefore, recommended trust integrates with presentee node' reputation and the subjective believes of intermediate nodes.

As aforementioned, single route recommended trust can be calculated through the single-track belief transmitting among intermediate nodes, we now need to find out an effective and feasible mechanism which has capacity of combining multiple single route recommended trusts.

Different from single route recommended trust calculation, multiple routes recommended trust have two kinds of route composition in recommendation, i.e., independent routes and overlapping routes. Fig. 1 shows two kinds of route composition for multiple routes recommended trust.

Definition 4. Multiple routes recommended trust is an integrated trust which combines with single route recommended trusts in independent or overlapping routes from presentee node to demander node.

We first address the multiple routes recommended trust computation with independent routes. Let there be $m(m\geq 2)$ independent routes of recommendation between $\Im$ and $\Re$, each independent route can be identified as $route_k$, and each independent single route recommended trust be $sr_k(\Im)$. Then, multiple routes recommended trust can be calculated as follows,

$$mr(\Re,\Im)_=\min\left[\frac{\sum_{k=1}^{m}sr_k(\Im)}{m}+0.1\times\left(\frac{1}{m+1}\right)^{\frac{1}{m+1}},1\right] \quad (5)$$

In formula 5, multiple routes recommended trust is calculated based on the average of summary single route recommended trust. Moreover, we add a weight parameter as $0.1\times\left(\frac{1}{m+1}\right)^{\frac{1}{m+1}}$ whose value is between 0 and 0.1 to emphasize the importance of recommendation route quantity. While value of m increasing, the weight parameter would increase with it. In like manner, a person would win more trust if he/she gets more recommendations in human society.

Then, we propose the multiple routes recommended trust computation with overlapping routes. Our calculation method is to treat the whole overlapping route as a single route and calculate the multiple parts following the independent route mechanism. We assume that there are two kinds of node in an overlapping route: joint node jd and disjoint node dd. We divide the computation into two steps: calculating the independent multiple routes and calculating the whole overlapping route. The first step aims to remove the disjoint nodes in whole overlapping route and the second step aims to calculate the whole recommended trust as a single route.

Let the overlapping route be expressed by $route=[\Re,jd_1,...,\Im]$, and the disjoint nodes be dd_q^s which indicates that the disjoint nodes are following joint node jd_q in overlapping route. Then, belief between joint node jd_q and jd_{q+1} can be calculate as following,

$$belief(jd_q,jd_{q+1})=\begin{cases}belief(jd_q,jd_{q+1}) & no\ dd_q^s\ following\ jd_q\\ mr(jd_q,jd_{q+1}) & else\end{cases} \quad (6)$$

Through formula 6, there would be no disjoint nodes, and overlapping route can be regarded as a single route which is consisted through joint nodes. Therefore, overall recommended trust of overlapping can be calculated as follows,

$$or(\Re,\Im)=sr(\Re,\Im) \quad (7)$$

Here, $dep(\Im)=|route|-1$ and there is no disjoint nodes in the route.

4.3. Objective Coincidence of Recommended Trust

In human society, people would not make decision for potential selections only depending on other's recommendation information. As a rational decision, a person usually tends to evaluate potential selections with the others' opinions facts. In common sense, if a recommendation obviously departs from the members' trustworthiness realities, the recommendation would be unworthy to believe.

Likewise, it is important and necessary for node to evaluate coincidence between others' impressions and the recommended trust in WSNs.

Let there be demander node $\Re$ and presentee node $\Im$ in recommendation, and there be p different routes for the recommendations. For $\Re$, the objective fact of $\Im$ is the reputation information. Therefore, $\Re$ needs to measure whether the final recommended trust, $rt(\Re,\Im)$, is coincident with its reputation or not in order to realize the trustworthiness of the recommendation.

Assume that there is a neighbor recommender node set, Nd, which includes the neighbor nodes of $\Re$ and each neighbor recommender node nd_i obtains recommended trust, $rt(nd_i,\Im)$ and. Then the objective coincidence of recommended trust can be calculated as follows,

$$coin(\Re,\Im)=\sqrt{\frac{\sum\left[\left(rt(nd_i,\Im)-rt(\Re,\Im)\right)/rt(nd_i,\Im)\right]^2}{|Nd|}}, \quad (8)$$

where $|Nd|$ denotes the total number of neighbor recommender node set. Obviously, the smaller value of $coin(\Re,\Im)$ implies that the recommendation is more trustworthy.

4.4. Optional Degree for Recommendation

By computing the parameters of recommended trust and objective coincidence, demander node have the first-hand impression of presentee node. But these parameters are not sufficient for making decision to select. In human society, people would decide to select a recommended candidate with more rational facts in additional to first-hand information. Consequently, we propose an optional degree for recommendation based on parameters of recommended trust, objective coincidence and other facts.

Let the recommended trust be $rt(d_i,\Im)$ and the objective coincidence of recommended trust be $coin(\Re,\Im)$. Assume that there are p intermediate recommenders d_i in the recommender node set Rd and the presentee node $\Im$ was accepted ξ times and was rejected ω times in past recommendations. Then, the optional degree of presentee node can be calculated as follows,

$$op(\Re,\Im)=rt(\Re,\Im)\times reputation(\Im)^{\frac{\omega}{\xi}}\times\left[\frac{\sum_{d_i\in Rd}reputation(d_i)}{|Rd|}\right]^{coin(\Re,\Im)} \quad (9)$$

In above formula, $|Rd|$ denotes the total number of recommender node set. The value of $reputation(\Im)$ is impacted by accepted time ξ and rejected time ω. The more accepted times and the less rejected times occurred for $\Im$, the value $reputation(\Im)^{\frac{\omega}{\xi}}$ would be larger. Likewise, in formula 9, $coin(\Re,\Im)$ indicates that whether recommender nodes' recommendation is honest and accords with objective fact or not, while the average of recommender's reputation indicates the whole credit of recommenders. Therefore, the factor is more optional while average of recommenders' reputation is larger and $coin(\Re,\Im)$ is smaller.

4.5. Proposed Framework

Generally, people would not trust strange one depending on just one recommender. In order to ensure the presentee's reliability, people may inquire the trustable facts from multiple recommendation sources. Likewise, node should consider trust degree of $\Im$ from multiple recommendations in WSNs. Therefore, we propose a recommendation trust computation framework (RTCF) for recommended trust evaluation.

Definition 5. A recommendation trust computation framework for WSNs can be defined formally as following,

$$\Theta=(\Re,\Im,D,Route,Acceptability), \quad (10)$$

where, $\Re$ and $\Im$ denote the recommendation receiver and the presentee respectively, $D=(d_1,...,d_n)$ denotes the recommenders, $Route$ is a set of vector as $route_k=[d_1^k...]$ which denotes the recommendation routes from $\Im$ to $\Re$, and $Acceptability$ denotes the acceptable degree of $\Im$ which consists of three parameters, $rt(\Re,\Im)$, $coin(\Re,\Im)$ and $op(\Re,\Im)$. $rt(\Re,\Im)$ is the recommended trust from $\Im$ to $\Re$ through single route sr or multiple routes mr. $coin(\Re,\Im)$ is the matching degree between recommendation and reputation of $\Im$ to measure whether the recommended trust of $\Im$ is coincident with its reputation. $op(\Re,\Im)$ denotes a qualification degree whether the presentee can be trusted or not.

5. Proof of Correctness

In this section, we present a set of lemmas and theorems for our recommended trust computation in order to prove the correctness.

Lemma 1. In WSNs, trust can be conditional transmitted between nodes in recommendation.

Proof. Generally, trust does not have the character of transitivity [15]. But in recommendation, trust can be transmitted in some conditions, which was addressed in [4].

1) Intermediate recommenders explicitly communicate their trusts in presentee to demander, as a 'recommendation'.

2) Demander trusts recommenders, i.e. recommender trust exists in the system.

3) Demander is allowed to make judgments about the 'quality' of recommendation (based on demander's policies).

4) Trust is not absolute, i.e. demander may trust presentee less than recommender does, based on recommender's recommendation.

In our paper, nodes can explicitly communicate their trust as recommendation. And intermediate nodes do exist in WSNs. Moreover, demander can judge the recommendation based on its willingness, and the recommended trust obviously dumps in recommendation in our proposed formulas. Hence, trust can be transmitted conditionally in recommendation.

Lemma 2. In WSNs, all nodes can hold the recommended trust information from different intermediate recommenders about presentees so that the case of multiple intermediate recommenders is available.

Proof. For each node in WSNs, it can hold past transactions, including recommendations because it can generate and renew subjective belief with other nodes. Therefore, while a node receives a recommended trust, it would hold such recommendation as first impression about a strange node. Thus, every node has potential to deliver its recommendation to others and become an intermediate recommender. Lemma 2 holds its correctness.

Theorem 1. Recommended trust in RTCF computation reflects both objective and subjective judgments about pressentee.

Proof. According to lemma 1, demander node is allowed to make judgment about the recommendation based on its own policies. Then, the recommended trust indicates the subjective judgment of demander node. At the same time, formula 3, 4 and 5 shows that such judgment of demander node is based on trust information of belief and reputation. Hence, recommended trust is calculated from both subjective and objective aspects.

Theorem 2. Spurious recommendation from malicious recommender nodes is identifiable for demander node in RTCF.

Proof. According to formula 2, it is impossible for a malicious node to win a high reputation since the reputation reflects the whole members' opinions in WSNs. In case a malicious node is a recommender in recommendation, demander node would get a lower $sr(\Im)$ or $mr(\Im)$ through formula 4 and 5 because of low reputations of $reputation(d_i)$. On the other hand, if a spurious presentee node is recommended by others, the value of $coin(\Re,\Im)$ or $op(\Re,\Im)$ would be low through formula 8 and 9 because of the low reputation of $reputation(\Im)$. Moreover, theorem 1 shows that recommended trust is evaluated from both subjective and objective aspects. Then, demander node would not get high recommended trust value based on low reputations of malicious nodes.

6. Example and Examination for RTCF

6.1 Example

In this section, we proposed an example to explain the efficiency of our recommended trust framework. In our scenario of example, we set a recommendation between demander node $\Re$ and presentee node $\Im$. There were nine intermediate recommender nodes which composed three recommending routes between $\Re$ and $\Im$. Route I was an overlapping multiple routes composed by recommender node 1, 4, 5 and 8. Route II and III was single routes composed by recommender node 2, 6, 9 and node 3, 7 respectively. The recommendation is shown in Fig. 1 as following.

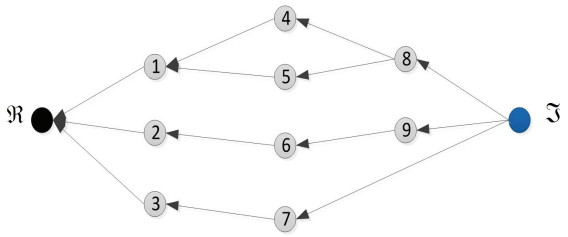

Fig. 1. Example of recommended trust.

We set the trust information of belief and reputation about the nodes in Table 1 and Table 2.

Based on computation method of RTCF, parameters of recommended trust, objective coincidence and optional degree were calculated as following (Table 3).

Table 1. Belief between Nodes.

Node	Belief	Node	Belief
$\Im \rightarrow 9$	0.75	$6 \rightarrow 2$	0.95
$\Im \rightarrow 8$	0.95	$5 \rightarrow 1$	1.0
$\Im \rightarrow 7$	0.8	$4 \rightarrow 1$	0.9
$9 \rightarrow 6$	0.9	$3 \rightarrow \Re$	0.75
$8 \rightarrow 5$	0.95	$2 \rightarrow \Re$	1.0
$8 \rightarrow 4$	0.65	$1 \rightarrow \Re$	0.8
$7 \rightarrow 3$	0.8		

Table 2. Reputation of Nodes.

Node	Reputation	Node	Reputation
$\Im$	1.0	5	1.0
1	1.0	6	0.65
2	0.9	7	0.9
3	0.95	8	0.75
4	0.7	9	1.0

Table 3. Parameters of RTCF.

	Parameters
8→1	$belief(1,8) = 0.663$
Route I	$or_I(\Re, \Im) = 0.52$
Route II	$sr_{II}(\Re, \Im) = 0.582$
Route III	$sr_{III}(\Re, \Im) = 0.581$
Recommended trust	$rt(\Re, \Im) = 0.632$
Objective coincidence	$coin(\Re, \Im) = 0.07$
Optional degree	$op(\Re, \Im) = 0.626$

Compared with each route of recommendation, the whole recommended trust is larger because more recommenders imply more trustable recommendation. And also, while recommended trust is quite deviating from the presentee node's reputation, the coincidence value would be larger and optional degree value would be less.

6.2. Examination

In this section, we examine our proposed trust management method through a series of simulations. We performed our simulations on the NS-3 simulator [17] which allows the simulation of wireless networks. For initial setting, we deployed 300 nodes in simulator. As initial setting, reputation values of honest nodes followed a normal distribution with mean 0.8 and variance 0.1 and reputation values of malicious nodes were set between 0.2 and 0.8 at random. The network topology was generated at random and average out-degree of a node was 4. Nodes which had a direct link between them could execute transactions by default. Then, we set initial past working data, including past judgments and working capacities, for nodes according to their honest and malicious identities and allowed nodes to query the past data from their neighbor nodes at any time.

In first examination, we aimed to testify the efficiency of recommended trust computation. We assumed that there were not malicious nodes in our simulation. For all users who received recommendation, they must accept and transmit it to their neighbors while recommended trust value satisfied $recommend(\Re, \Im) \geq 0.75$. We set three groups as follows: Baseline1 used the EigenRep method, which was given in reference 12, to calculate the trust relationships among users and decided whether transmitting the information or not according to the direct recommender. Baseline2 used the belief and reputation computation methods in this paper. All users decided whether transmitting information or not based on the average of belief and reputation. Baseline3 used the recommended trust computation in SRTCF to measure whether information could be transmitted. We repeated 100 times of recommendation dissemination and recorded the average dissemination depth of above three groups, which was shown in Fig. 2 as follows.

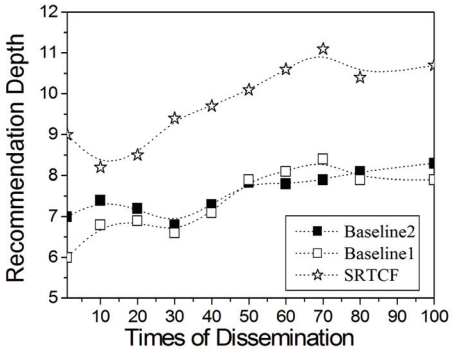

Fig. 2. Average dissemination depth of recommendation.

In second examination, we introduced malicious nodes whose reputation were lower than 0.2 and we appointed them to release fake recommendation. We defined that a malicious recommendation was the one which satisfied the condition $\left(recommend(\Re, \Im) \geq \alpha\right) \wedge \left(optional(\Theta) \geq \beta\right)$. We set three groups in this examination and the thresholds of $<\alpha, \beta>$ were set as $<0.7, 0.8>$, $<0.8, 0.85>$ and $<0.8, 0.9>$ respectively in group 1 to 3. We repeated 50 times of recommendation dissemination in three groups under different percent of malicious users. With the increasing of malicious users, the percent of malicious recommendations which were identified accurately were shown in Fig. 3. As shown in Fig. 3, we can see that percent of identified malicious recommendation increased with the enlargement of thresholds in three groups. That means our method can avoid the malicious recommendation effective.

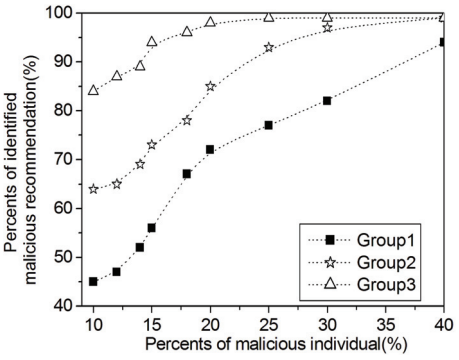

Fig. 3. Malicious recommendation identification with RTCF.

7. Conclusion

Security is very important for WSNs because of the relax-coupled network environment. Existing researches proved that trust is a feasible to cope with this challenge. In this paper, we introduced subjective and objective trust to build up a recommended trust computation for strange nodes so that they could get indirect trust through recommenders. We proposed notions of belief and reputation to evaluate the subjective trust relationships and the objective trust view for nodes. Then, we addressed the recommended trust notion based on above two trust notions. In our paper, recommended trust is calculated through single route or multiple routes between recommendation demander and presentee. Moreover, we presented two parameters of objective coincidence and optional degree for recommendation. Further, we proposed a recommended trust computation framework (RTCF) for WSNs nodes. Finally, lemmas and theorems were presented in order to prove the correctness RTCF. We plan to testify the efficiency of our work in several simulations or applications in near future.

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Research on Dynamic Calibration of Piezo-two-dimensional Force Sensor

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Abstract: Based on the piezoelectric effect of piezoelectric quartz crystal, a two-dimensional force sensor with good dynamic characteristics is designed, whose sensitive component is single-cut piezoelectric quartz wafers, and it realizes the measurement of force by measuring the charge which is generated by piezoelectric quartz crystal. This paper analyzes measurement principle of the sensor based on the mechanics of materials and gives structure of the sensor. Dynamic calibration system of the sensor is designed, and the curves between sensor output and input are obtained by dynamic calibration experiment and data analysis. This article provides a new way of thinking for in-depth study in dynamic calibration of the sensor and accurate measurement.
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Keywords: Sensor, Dynamic Calibration, Dynamic force, Dynamic torque.

1. Introduction

With the development of modern high-precision test equipment, the means and methods of static measurements of force are more and more matured, and the measurement accuracy is also rising. With the development of science and technology, people are setting foot in more fields, and new demands on the measurement technique have also been put forward. Dynamic force measurement has increasingly attracted attention [1].

Gu Baodong et. al. [2] analyzed precision of calibration system using interval arithmetic, and deduced specific expression of calibration accuracy, and studied Frequency-change characteristics of the system with additional mass. Austrian Qi [1] presented dynamic calibration device for force sensor, and it could realize dynamic calibration with dynamic loading. Yusaku Fujii [3] discussed the way of establishing dynamic calibration systems, and described three methods of measuring dynamic response of the sensor with different forces. Rolf Kumme [4] calibrated force sensor by using comparative law, that is to say, using the standard sensor to conduct dynamic calibration. D. J. Mee [5] analyzed different technologies of calibrating force balance, and supersonic instrument with short duration was used. J. P. Damion [6] described different methods of determining transfer function of sensors, and introduced the equipment which was used in dynamic calibration of sensor. A. Fisher, S. Watkins and J. Watmuff [7] introduced a portable calibration system that can quickly and accurately calibrate pressure measurement system. David J. Mee

[8] described different techniques of calibrating stress wave force balance.

With the rapid development of testing technology, it has been sufficiently attracted attention. The dynamic characteristic is a key parameter of the force sensor, and dynamic calibration is essential for sensor to measure dynamic force accurately and reliably. The requirements of measurement system measuring dynamic signal are that it could measure the signal quickly and accurately and reproduce changed signal waveform truly [9]. This paper proposes a two-dimensional force sensor based on single-cut piezoelectric quartz, and analyzes its measuring principle. The dynamic calibration of the sensor is also conducted, and the dynamic follow-up characteristic of the sensor is studied.

2. Measuring Principle

The measuring principle of measuring moment is analyzed based on Mechanics of Materials. Fig. 1 shows longitudinal section moment diagram of quartz group. According to relation between moment and distributed load in Mechanics of Materials, distributed force of piezoelectric quartz group which is caused by the effect of torque is linear distribution. Wherein, M is the torque acted on the piezoelectric quartz group ,while R and r are the radius of outer and inner circles of piezoelectric quartz group, respectively, and q represents the uniform load. F1 and F2 mean left integrated force and right integrated force, respectively, and l1 and l2 indicate distance from integrated forces to center of crystal group.

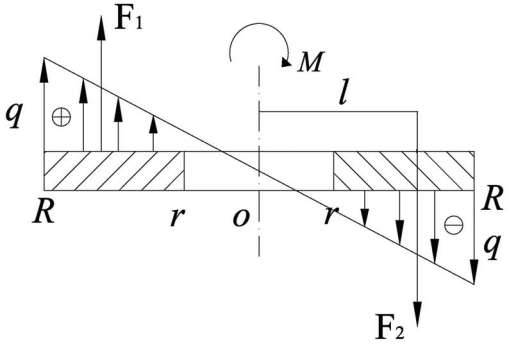


Fig. 1. Torque measurement schematic.

According to stress analysis，moment equation can be listed as formula (1):

$$\frac{1}{2}q\cdot R\cdot\frac{2}{3}\cdot R-\frac{1}{2}\left(\frac{r}{R}q\right)r\cdot\frac{2}{3}r=\frac{M}{2}, \tag{1}$$

q could be obtained by calculating formula (1).

$$q=\frac{3RM}{2\left(R^3-r^3\right)}$$

Therefore, the left integrated force F_1 of piezoelectric quartz group can be calculated.

$$F_1=\frac{3M}{4}\left(\frac{R+r}{R^2+Rr+r^2}\right)$$

Similarly, the right integrated force F_2 of piezoelectric quartz group can also be obtained. Therefore, according to the formula (2), moment M applied to the piezoelectric crystal can be obtained.

$$F_1l_1+F_2l_2=M, \tag{2}$$

where l_1 and l_2 are determined by calibration experiments. The equation (2) shows that it is linear relationship between the input torque and output voltage of sensor.

3. Sensor Structure

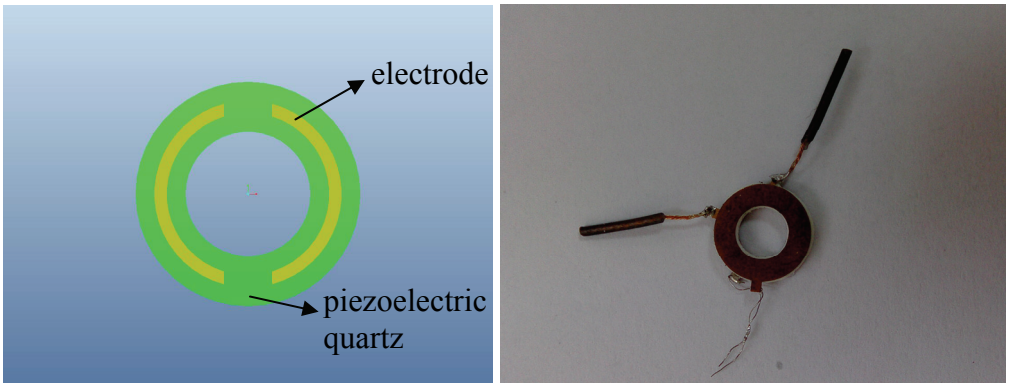
On the basis of measuring principle, this paper uses the single-cut piezoelectric quartz crystal, namely, piezoelectric quartz group is made of x0 cut piezoelectric quartz crystal. When the force combined with moment are applied to piezoelectric quartz crystal, the potential distribution of left and right parts of the crystal is asymmetric, so this article proposes piezoelectric quartz group with dual-electrode structure that could measure two-dimensional force.

Compared with single-electrode structure, dual-electrode structure could reduce the mutual interference between two parts of single-electrode and ensure minimal loss of the charge generated by piezoelectric quartz crystal. In this paper, the sensor uses the piezoelectric quartz crystal whose thickness is 1mm and outer diameter is 9 mm. The structure and in-kind of crystal group are shown as Fig. 2 (a) and (b), respectively.

4. Dynamic Calibration of the Sensor

According to calibration principle and method of the sensor, dynamic calibration system is designed as shown in Fig. 3, and the block diagram of the calibration system is showed as Fig. 4. The experiment is designed to calibrate sensor when the dynamic loading is applied to the force sensor. Output section of dynamic load and data collection section compose calibration system. The dynamic load is outputted by exciter in the calibration process, and the input signal of the exciter is outputted by real-time controller that is controlled by program of software controldesk, and program is compiled from the algorithm that is edited in the Simulink module of software Matlab. Dynamic force could be loaded continuously to the sensor by editing different algorithms, and the value and frequency of force

could also be changed. The experiment is done by increasing the frequency of the input signal of exciter to adjust the output amplitude of the exciter. Real-time control system could output the real-time signal and collect real-time data, and display graphics of real-time signal simultaneously. In the role of dynamic loading, the output end of the sensor is connected to the charge amplifier firstly, and the signal is amplified, and then the charge amplifier is connected to the data collection port of real-time controller. The purpose of this experiment is to study the output change of sensor with the change of dynamic load, and the change of dynamic load that is applied to the sensor is obtained by using laser displacement sensor to measure the change of vibration displacement of mass block, which is connected to the sensor.



(a) (b)

Fig. 2. The structure (a), and in-kind (b) of piezoelectric quartz crystal group

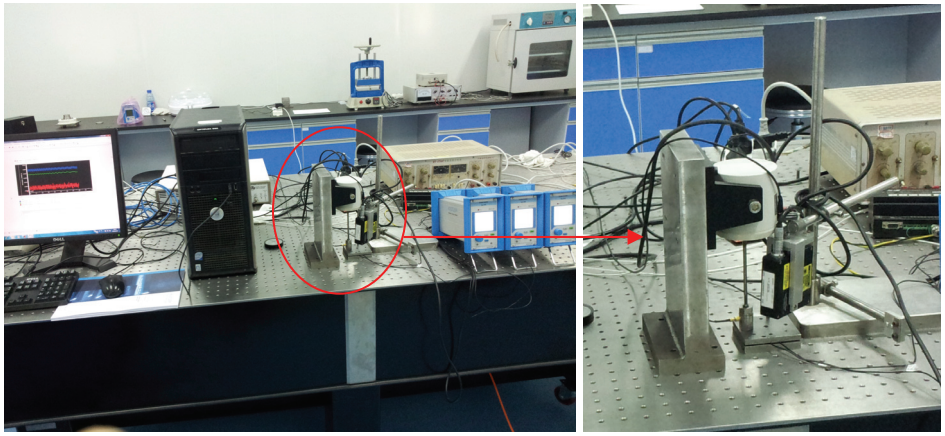


Fig. 3. Dynamic calibration system of the sensor.

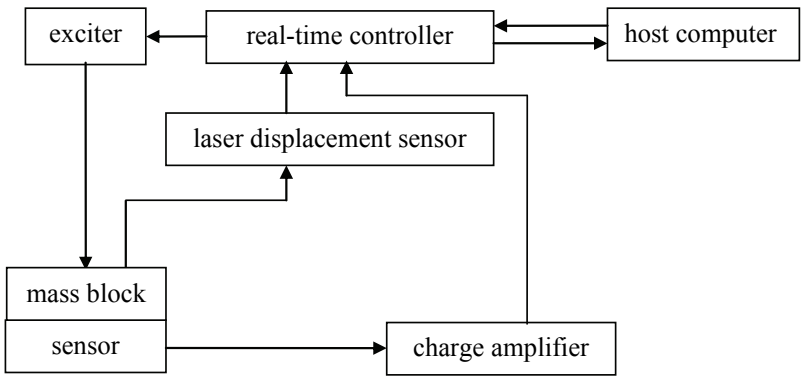


Fig. 4. Block diagram of the calibration system.

4.1. Dynamic Force Calibration Experiment

In the static calibration experiment of the sensor, usually, the calibration curve is obtained by the method of applying normal load to the sensor. The linearity of this piezoelectric sensor is excellent in the static calibration. In the dynamic calibration experiment, the output signal of the sensor could be monitored in real time when the dynamic force is applied to the sensor. The relationship between output signal and the input signal of the sensor is expressed by the relationship between output voltage and the vibration displacement of mass block that is connected to the sensor, and the curve is shown as Fig. 5.Wherein, the solid line with the node is practical Curve, and another curve is fitting curve with the method of square law. It can be seen from the Fig. 5, output voltage signal of the sensor increases linearly with the increase of the vibration displacement of the mass block, namely, the linearity of the sensor under the effect of dynamic force is good.

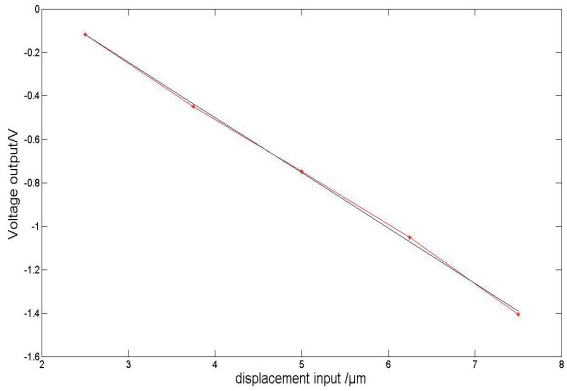


Fig. 5. The curve of dynamic force calibration.

4.2. Dynamic Torque Calibration Experiment

Static moment calibration of the sensor was conducted in the static calibration experiment of the sensor, and the linearity of calibration curve is excellent. Dynamic torque calibration experiment is designed to study the linearity between output voltage of the sensor and the input torque that is continuously changed over time. Fig. 5 shows that the curve between the output voltage of the sensor and the vibration displacement of mass block, namely, it is equivalent to the curve between output voltage of the sensor and input torque that is applied to the sensor. Wherein, the solid line with the node is practical curve, and another curve is fitting curve with the method of square law. It can be seen from the Fig. 6, the linear relation between output voltage and input torque is perfect.

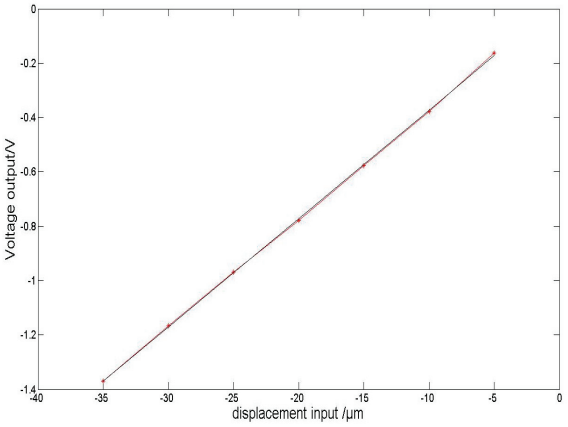


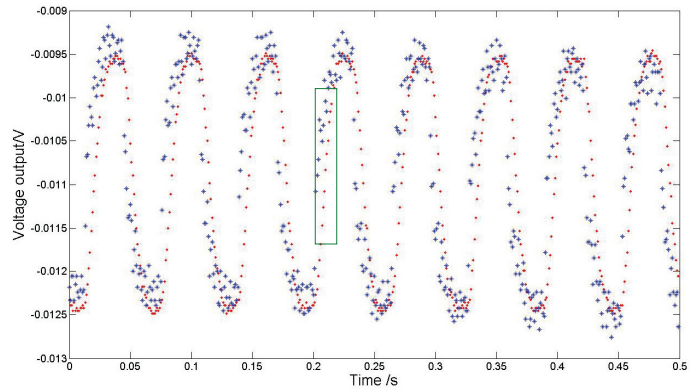
Fig. 6. The curve of dynamic torque calibration.

4.3. Dynamic Following Feature Experiment

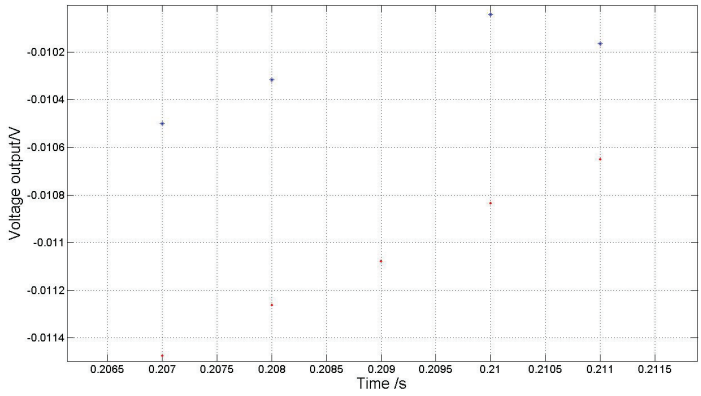
In order to test the dynamic performance of the sensor, dynamic following feature experiment is designed. Output voltage signal of the sensor is measured When the angular velocity of input dynamic force is100rad/s. Fig. 7 (a) shows comparison between output voltage and the input force of the sensor, and the smaller scatter indicates the force signal while the larger represents voltage signal of the sensor. Fig. 7 (b) is the partially enlarged view of the box area from Fig. 7 (a), and what can be seen from Fig. 7 (b) is that there is no time difference between voltage signal and the corresponding force signal, therefore, the dynamic following feature of this sensor is good.

5. Conclusion

In this paper, measurement principle based on Mechanics of Materials of two-dimensional piezoelectric force sensor is analyzed, and the sensitive element with special Dual-electrode structure made of single-cut piezoelectric quartz crystal is proposed. The dynamic calibration system including the real-time control system, the charge amplifier, and the host computer etc. is designed. Calibration experiments of dynamic force and dynamic torque and dynamic following feature experiment are designed and completed. According to data achieved from the dynamic calibration experiments, the curves between output of the sensor and dynamic force and dynamic torque are obtained, and the dynamic following characteristic curve of the sensor is also plotted. This sensor that is loaded with the dynamic loads, not only has good linearity, but also has excellent dynamic follow feature. The research results lay a new foundation for the dynamic calibration technology of the force sensor.



(a)



(b)

Fig. 7. The comparison between output and input of sensor (a), and Partial enlarged view (b).

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A Novel Identification Method of Obstacles Based on Multi-sensor Data Fusion in Forest

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Abstract: In this paper an obstacle identification system was set up and an artificial neural network (ANN) was established to identify obstacles in forest based on multi-sensor fusion, aiming to improve the efficiency of automated operations of forestry harvester in the complicated environment of forest areas. Firstly, a 2D laser scanner and an infrared thermal imager were used to collect the information of obstacles in forest areas. Based on the collected infrared photo, visible pictures and laser scan data, image fusion and data association were conducted to obtain features of obstacles. Second, a BP neural network was established for forest obstacles identification by training samples. Then, to improve the performance of BP model, several common used functions were adopted to construct BP model by permutations and combinations. The test results showed that the model with trainlm training function had a best performance in forest obstacles identification. Finally, three best models were selected to be evaluated by testing all samples and the results illustrated that the three selected BP neural network models can identify trees, people, stones at a high correct rate of 93.3 % or more.
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Keywords: Laser scanner, Infrared image, Visible image, Multi-sensor data fusion, Obstacle identification.

1. Introduction

In order to meet the needs of the rapid development of forestry, automation and high efficiency for forestry equipment are desired. However, various obstacles and the complicated environment in forest will lead to intermittent operations and driving dangers for forestry equipment [1]. Therefore, it is necessary to develop a highly efficient obstacle identification method in forest areas.

Regarding the obstacles identification, various research works have been done, in which different kinds of sensors are used as tools. L. Y. Chang designed an optical system of visible light detection camera to realize the detection of dark and weak target in space [2]. Mendes, A. used a laser range finder to detect and track moving objects. It cannot only classify several kinds of objects but also detect new ones [3]. Navarro-Serment L. E. employed three-dimensional LADAR measurement on a moving vehicle to detect and track pedestrians over time [4]. Sume, A. utilized a stepped-frequency radar to detect moving objects based on Doppler-based techniques [5].

Based on the above, camera, laser scanner, and radar are commonly used tools in obstacles detection. But one single sensor cannot collect full information of object. And multi-sensor fusion technology can solve this problem. Sensors can complement each other to make use of information effectively and obtain accurate results. How to fuse the information derived from different sensors is the focus of our study.

So far, multi-sensor fusion technology has been used in many areas including: image fusion [6-7], intelligent robots [8-9], remote sensing [10], fault diagnosis [11], and pattern recognition [12-13]. In obstacles identification, it is mainly used in vehicle system and robot movement. Raphaël Labayrade achieved real time-detection of multi-obstacles in the automotive context based on the fusion between stereovision and laser scanner [14]. Broggi, A. developed a pedestrian-detection system with fusion of a laser scanner and a vision system to localize pedestrians in front of the vehicle [15]. X. Z. Zhang and R.C. Luo both used sensor fusion strategy to realize simultaneous localization and mapping for robots [16-17]. Baltzakis H. employed laser and visual data for robot motion planning and collision avoidance [18].

However, the sensors fusion technology has not been used in forest obstacles identification. In the study on the forest environment, researchers have used terrestrial laser scanning (TLS) and airborne laser scanning (ALS) as main tools to acquire detailed parameters of standing trees and differentiate tree species from air. Based on using TLS and ALS, many researchers have proposed various algorithms to determine the tree parameters like position and diameter at breast height (DBH). X. Liang et al [19] developed a fully automatic stem-mapping algorithm by using single-scan TLS data to collect individual tree information from forest plots. Jari Vauhkonen et al [20] used six different algorithms to extract treetop position and estimate tree height. The performance of all algorithms were compared and analyzed. Eva lindberg et al [21] proposed a new method to measure tree stems automatically by terrestrial laser scanning (TLS), and to further link these ground measurements to ALS data analyzed at the single tree level. Leeuwen M. et al [22] presented an efficient and robust method for stem detection from terrestrial laser scanning based on the Medial Axis Transformation. Brolly G. et al [23] introduced a method to extract individual tree parameters including tree location, DBH, and tree height by constructing a digital terrain model and a digital crown surface model.

Although researchers have used laser scanner to measure detailed parameters of standing trees, one single sensor cannot satisfy the requirement of obstacles identification due to complex environment and insufficient light in forest. Therefore, an obstacle identification method based on multi-sensor fusion is proposed in this paper.

First, a 2D laser scanner and an infrared thermal imager were used to collect the information of obstacles in forest areas. Based on the collected infrared photo, visible pictures and Laser scan data, image fusion and data association were done to obtain 6 features of obstacles: R, G, B, aspect ratio, rectangularity, and temperature. Second, a BP neural network was established for forest obstacles identification by training samples and it had a correct rate at 90.00 % for one test. Next, to improve the performance of BP model, several common used functions were adopted to construct BP model by permutations and combinations. The test results showed that the model with trainlm training function had a best performance in forest obstacles identification. Finally, three best models were selected to be evaluated by testing all samples and the results illustrated that the three selected BP neural network models can identify trees, people, stones at a high correct rate of 93.3 % or more.

2. Obstacle Identification System

2.1. Flowchart of Proposed Method

Fig. 1 illustrates the technical flowchart of the obstacle identification system developed in this research. The system functions in the following procedure:

1) Collection and Pre-processing of the data by a laser scanner and an infrared thermal imager;
2) Extract features of obstacles from the laser data and images;
3) Obtain the region of interest (ROI) by data association of the laser points and the images;
4) ANN model construction by training samples;
5) Select the transfer function and the learning function to construct ANN model with high identification correct rate and low errors.
6) Finally, test all samples with selected model.

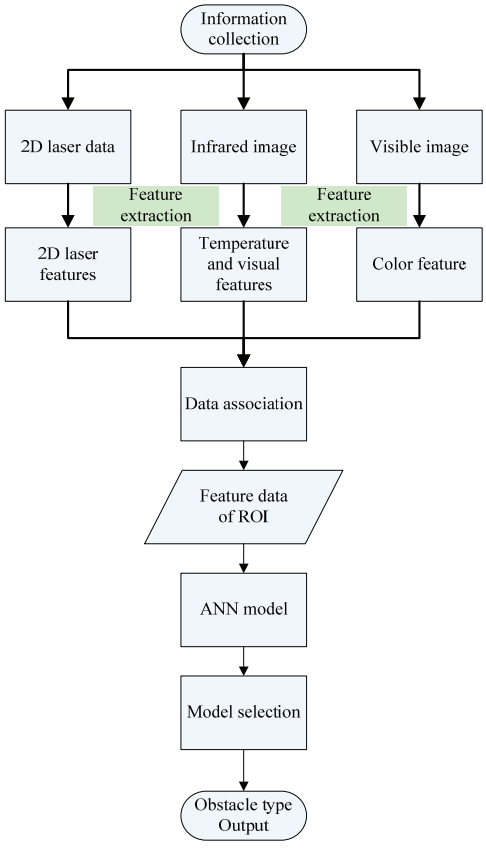

Fig. 1. Flowchart of the proposed method.

2.2. Test Equipment and Placement

Obstacles information is collected by a 2D laser scanner and an infrared thermal imager. As shown in Fig. 2, both the devices were placed on the same horizontal plane in this experiment conducted in Beijing Bajia Contry Park.

Fig. 2. Sensors placement in this experiment.

2D laser scanner LMS291 from SICK Inc. within the scan range of 32 m with a field of view (FOV) of 100° and angular resolution of 0.25°, is a non-contact measurement system, which detect two-dimensional space around as laser radar. Its operating temperature is between 0 ^{0}C-50 ^{0}C, and is an active scanning system that does not require external passive components. This makes it particularly suitable for the transport system and vehicle control.

The infrared thermal imager FLUKE Ti55 is used to capture the image of both visible light spectrum and infrared light simultaneously. Its image resolution is 240×320. It adopts a Fluke IR-Fusion technology to put the visible and infrared images fused. So it can speed up and facilitate the subsequent data acquisition process.

3. Obstacle Identification Approach

3.1. Information Extraction from Laser Point Data

The 2D laser scanner used reflected echo of the scanning laser to detect obstacle. The scanning data was a series of points that represented the distances between the obstacles to the scanner. How to distinguish and recognize these points from different objects was very important. Therefore, data filtering and obstacle data extraction were performed to classify the laser points into different segments for the corresponding objects. An edge detection method was adopted to calculate laser point clusters, which was described as follows: Calculated the distance of two adjacent laser points in the y-axis. If the distance value was smaller than a threshold, the two points were considered to belong to the same object. If not, they were separated. Additionally, a laser point that is far from its adjacent points was considered as a noise variable and excluded for further analysis. Only the clusters with a certain size were retained.

In this way, a large number of laser points were excluded and a group of candidate obstacles were obtained. Then, the information of each object including the depth, width, and the location in the laser coordinate system can be obtained from laser data.

3.2. Information Extraction by Visible-infrared Image Fusion

3.2.1. Image Fusion by PCNN

Both visible and infrared images were used in obstacle observation and obstacle feature extraction, but both of them were subject to the influence of different environmental conditions and had their own disadvantages. To improve obstacle detection in the forest environment, a pixel-level image fusion algorithm for visible-infrared images was used. It enhances clarity and analyticity of pictures and emphasizes the feature information of the target.

In this paper, PCNN (Pulse Coupled Neural Network) was adopted to complete the fusion of visible-infrared images. PCNN is a mathematical model inspired by the study of the cat nervous system, so it has a biological basis when used for image processing. The neurons represent the pixels in the image, and the luminance pixel value corresponds to an external stimulus signal. Each pixel is connected to the surrounding pixels with range of 3 × 3 or 5 × 5. PCNN model has properties of synchronous pulse excitation and dynamic pulse distribution. When used in image processing, the sync pulse excitation allows the area information, edge, texture with the similar brightness in the image excited simultaneously; on the other hand, dynamic impulses can obtain pulsed images at different times, forming time series excitation diagram. The characteristics of the original image can be extracted with the invariance of displacement, scale, rotation and distortion.

By merging visible-infrared images by PCNN, sharper edges of the obstacles in the images can be obtained, which provided useful information for feature extraction.

3.2.2. Features Selection from Visible-Infrared Images

To identify the obstacles in the images, the features of the objects were extracted based on image fusion (Fig. 3). The most common obstacles in the forest environment are trees, rocks and human, and

the characteristics need to be found to represent each of the three categories with independence and discrimination. Therefore, a number of features such as the temperature, color, aspect ratio, and rectangularity were chosen for feature extraction from images.

Fig. 3. (a) Infrared Image.

Fig. 3. (b) Visible image.

Fig. 3. (c) Fusion image.

1) Temperature is an important feature for distinguishing obstacles because the texture of the objects in each category is different. Therefore, the obstacle temperatures were calculated according to the infrared temperatures using a comparison table.
2) Colors were obtained from visible images and expressed in red-green-blue (RGB), and a 3-vector was used to match each color feature.
3) Aspect ratio is the ratio of width and height of the obstacles. The aspect ratio describes the obstacle shapes and can distinguish between slender objects and round or square objects.
4) Rectangularity is the ratio of obstacle areas and its minimum enclosing rectangle (MER) areas. The rectangularity reflects the proximity degree of an object with its MER and it can be used as a standard for distinguishing trees and other objects.

3.3. Data Association between Laser Points and Images

In order to make full use of the information acquired by each sensor, laser data need to be associated with images so that it can describe which object the features belonging to. In this research our goal is to study the static information fusion algorithm, so a simple data association is adopted to get the region of interest [24].

Software settings make laser scanner and infrared thermal imager synchronous to ensure the consistency of the time. Ignore the error caused by the sensors center position deviation, then, data association can be achieved as shown in Fig. 4.

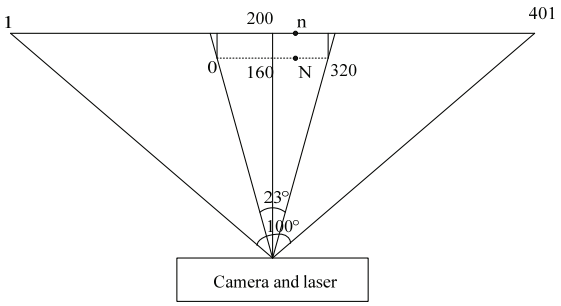

Fig. 4. Illustration of the data association.

In Fig. 4, laser scanner and infrared thermal imager are on the same center position with the same height. The FOV of the laser scanner and infrared thermal imager are 100" and 23" respectively. *N* represents the pixel number of the row in a picture and *n* is the number of laser points in one scan. Then the relationship of laser points and image can be described by calculating the laser points and the pixel number corresponding to 1 degree. After that, objects in laser and image could be associated to find the ROI.

3.4. ANN for Obstacle Identification

After the ROI in an image was obtained, an ANN model was used for obstacle identification based on 6 features derived from laser points and visible-infrared images.

ANN is a mathematical model applying structures that similar to the brain synaptic couplings to achieve information processing. BP (Back Propagation) neural network is a common neural network learning algorithm, which is a hierarchical neural network, composed of an input layer, an intermediate layer and an output layer. The intermediate layer can be extended to multiple layers. The neurons between adjacent layers are fully connected, and that in each layer have no connection.

From the Fig. 5, in the three layers, the input vector is $\mathrm{X}=\left(x_1,x_2,...x_i,...x_n\right)^T$; the output vector of hidden layer is $\mathrm{Y}=\left(y_1,y_2,...y_j,...y_m\right)^T$; the output vectors of output layer is $\mathrm{O}=\left(o_1,o_2,...o_k,...o_l\right)^T$; and the desired output vector is $\mathrm{d}=\left(d_1,d_2,...d_k,...d_l\right)^T$. The weights matrix between the input layer to the hidden layer is $\mathrm{V}=\left(v_1,v_2,...v_j,...v_m\right)^T$, where the column vector $\mathrm{V_j}$ is the weight vector corresponding to the j-th neurons in hidden layer; and the weights matrix between hidden layers to output layer is $\mathrm{W}=\left(w_1,w_2,...w_k,...w_l\right)^T$, where the column vector $\mathrm{W_j}$ is the weight vector corresponding to the k-th neurons in output layer.

When the output signal does not match with the teaching signal, there is output error E:

$$\mathrm{E}=\frac{1}{2}(\mathbf{d}-\mathbf{o})^2=\frac{1}{2}\sum_{k=1}^{l}\left(d_k-o_k\right)^2 \tag{1}$$

Expand the above formula to the hidden layer

$$\begin{aligned}\mathrm{E}&=\frac{1}{2}\sum_{k=1}^{l}\left[d_k-f\left(net_k\right)\right]^2\\&=\frac{1}{2}\sum_{k=1}^{l}\left[d_k-f\left(\sum_{j=0}^{m}w_{jk}y_i\right)\right]^2\end{aligned} \tag{2}$$

Expand the above formula to the input layer

$$\begin{aligned}\mathrm{E}&=\frac{1}{2}\sum_{k=1}^{l}\left\{d_k-f\left[\Sigma_{j=0}^{m}w_{jk}f\left(net_j\right)\right]\right\}^2\\&=\frac{1}{2}\sum_{k=1}^{l}\left\{d_k-f\left[\Sigma_{j=0}^{m}w_{jk}f\left(\sum_{j=0}^{n}v_{jk}x_i\right)\right]\right\}^2\end{aligned} \tag{3}$$

As can be seen from the above equation, the output error E is the function of network weights w_{jk} , v_{ij} of each layer, so the weights can be changed to adjust the error E. The purpose is to adjust the weights to decreases the error continuously, so the weight adjustment amount should be proportional to the error gradient descent. Therefore, BP neural network is called error gradient descent algorithm.

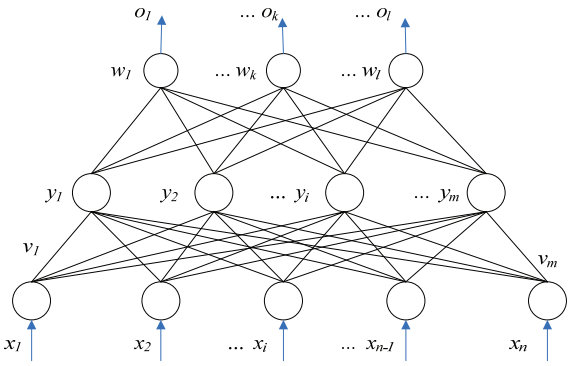

Fig. 5. BP neural network model.

4. Experiments and Results

The aim of this study is to identify the obstacles in the forest according to the acquired obstacle characteristic information. Applying MATLAB software to carry on simulation of BP algorithm, we can find a suitable BP network model to distinguish between the types of obstacles. There are six kinds of obstacle features including RGB values, temperature, aspect ratio, rectangularity, which are considered as inputs. And three types of obstacles that people, trees, and stone are outputs.

Then the BP network can be established as follow:

1) Input layer has 6 neurons.
2) Nodes in the hidden layer is set to 12 according to the number of training samples and the dimension of the input and output layer
3) The dimension of output layer number is equal to the number of the obstacle types 3. Construction principle of the output layer is that the output in the position corresponding to different types is 1, and the rest position is 0, such as trees [1,0,0], people [0,1,0], stone [0,0,1].
4) Tansig-tansig is selected as the network error transfer function.
5) Powell-Beale connection gradient BP training function traincgb is selected as learning function.
6) The number of learning is set to 500 times; the learning rate is set to 0.0005; and the training target is set to 0.001.

In this research, 150 samples including 50 trees, 50 peoples and 50 stones have been collected. Randomly use 120 samples to train the network and test the remaining 30 samples. Evaluate the obstacle classification capabilities of BP Network based on the correct rate and mean square error. One test result is shown in Table 1.

Table 1. One test result of the BP network for obstacles identification in forest.

yT =									
0.9684	0.9712	0.9577	0.9663	0.9722	0.9721	0.9721	0.9721	0.9687	0.9676
0.0885	0.0245	0.2767	0.1314	-0.0027	0.0033	0.0039	0.0045	0.0279	0.0058
-0.0611	-0.061	-0.0569	-0.0602	-0.0581	-0.061	-0.0613	-0.0615	-0.0219	0.0106
Tree	Tree	**Invalid**	**Invalid**	Tree	Tree	Tree	Tree	Tree	Tree
0.0009	0.0038	0.0007	0.0008	0.0289	0.0304	0.0008	0.0008	0.0007	0.0027
0.9926	0.9925	0.9926	0.9926	0.992	0.992	0.9926	0.9926	0.9926	0.9925
-0.0028	-0.0028	-0.0027	-0.0027	-0.0032	-0.0026	-0.0027	-0.0028	-0.0026	-0.0027
People	People	People	People	People	People	People	People	People	People
-0.0208	-0.021	0.0113	0.0163	0.3197	-0.02	-0.0199	-0.0189	-0.0201	-0.0201
0.0011	0.0005	-0.0002	-0.0002	-0.0007	-0.0002	-0.0002	-0.0002	-0.0002	-0.0002
0.9678	0.9674	0.9648	0.9645	0.9353	0.9669	0.9669	0.9668	0.9669	0.9669
Stone	Stone	Stone	Stone	**Invalid**	Stone	Stone	Stone	Stone	Stone

Mean square error of the testing results is 0.0939. From Table 1 it can be seen that there are three invalid outputs, so obstacle classification accuracy of BP neural network model is 90.00 %. But the mean square error of the testing results is large, so it is need to adjust the parameters, learning function and the error transfer function to improve the BP model.

Aiming to obstacle classification problem, there are three error transfer functions: S-type logarithmic function logsig, S-type tangent tansig, pure linear function purelin can be used in BP model construction. Furthermore, common used learning functions are Levenberg-Marquardt BP training function trainlm, adaptive learning rate gradient descent BP network training function traingda, momentum and adaptive learning rate BP gradient descent training function traingdx, Powell-Beale continuous gradient BP training function traincgb.

By permutations and combinations of these functions, it can form a variety of BP network models. Evaluate the performance of the BP neural network according to the training times, the training mean square error, the test mean square error and the identification correct rate four indicators. By calculation and contrast, it is found that the model with the trainlm training function performed best. The contrasts of all models with trainlm training function are shown in Table 2.

Table 2. The contrasts of all models with trainlm training function.

Indicators / **Transfer function**		**Model number**	**Training times**	**Training mean square error**	**Test mean square error**	**Identification correct rate (%)**
logsig	logsig	①	46	5.5e-2	0.7307	73.33 %
tansig	logsig	②	10	6.2e-5	0.4108	83.30 %
purelin	logsig	③	—	—	—	—
logsig	tansig	④	39	6.5e-5	0.0250	96.67 %
logsig	purelin	⑤	42	6.7e-5	0.0853	96.67 %
tansig	tansig	⑥	32	4.9e-6	0.1990	93.33 %
tansig	purelin	⑦	54	9.6e-5	0.0584	96.67 %
purelin	tansig	⑧	—	—	—	—
purelin	purelin	⑨	—	—	—	—

In the above table 2, model ③, ⑧, ⑨ cannot have a correct output for BP network; model ①, ② have a relatively low correct rate; the correct rate of model ⑥ is relatively high, but the test error is also high. In model ④, ⑤, ⑦, the mean square error is within 10% of the requirements and the identification correct rate is relatively high, so model ④, ⑤, ⑦ can be adopted as BP model to identify obstacles in forest.

To sum up, select logsig or tansig as the transfer function from output layer to the hidden layer, tansig or purelin as the transfer function from hidden layer to the output layer, and trainlm as the learning function for BP network model, which is suitable for obstacles classification and identification in forest.

Next, BP model ④, ⑤, ⑦ are considered as model I, II, III, then they are evaluated by testing all the 150 samples data. The training performance and the test indicators are shown in Fig. 6 and Table 3 respectively.

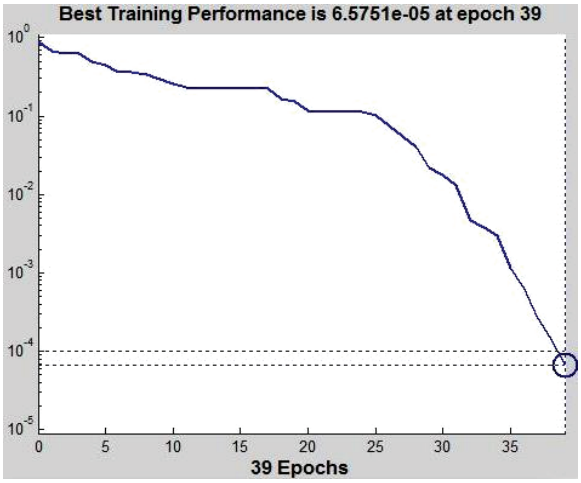

Fig. 6. (a) The training performance of model I.

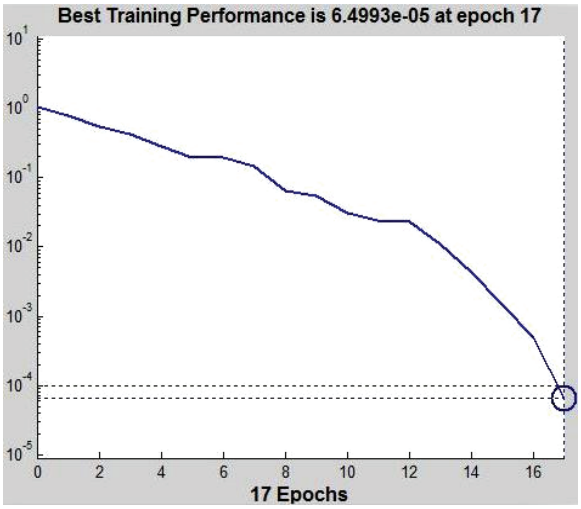

Fig. 6. (b) The training performance of model II.

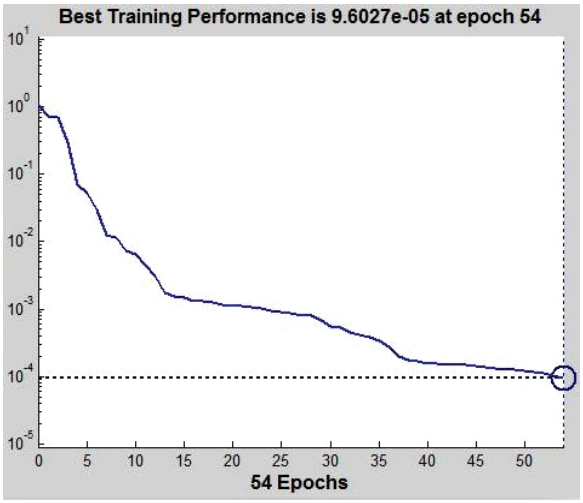

Fig. 6. (c) The training performance of model III.

Fig. 6 showed that the three models can reach the predetermined target value in a short training time, so they had a good learning performance.

Table 3. The contrasts of three BP model with testing all samples.

BP Model	Training times	Training mean square error	Test mean square error	Correct rate (%)
Model I	39	6.6e-5	0.0166	96.6 %
Model II	17	6.5e-5	0.0396	93.3 %
Model III	54	9.6e-5	0.0296	96.6 %

It can be seen that the three BP neural network models can identify trees, people, stones at a high correct rate based on the 6 features of the three common obstacles in forest. The correct identification rate can be up to 93.00 % or more, and the mean square error of the test less than 5 %. By testing and selecting, three BP models that suitable for obstacles identification in forest are obtained. It shows that the structure of the BP neural network model is reasonable, and the overall performance is also good.

5. Conclusions

In this paper, firstly, a 2D laser scanner and an infrared thermal imager were used to collect the information of obstacles in forest areas. Based on the collected infrared photo, visible pictures and Laser scan data, image fusion and data association were conducted to obtain 6 features of obstacles: R, G, B, aspect ratio, rectangularity, and temperature. Second, a BP neural network was established for forest obstacles identification by training samples and it had a correct rate at 90.00 % for one test. Next, to improve the performance of BP model, several common used functions were adopted to construct BP model by permutations and combinations. The test results showed that the model with trainlm training function had a best performance in forest obstacles identification. Finally, three best models were selected to be evaluated by testing all samples and the results illustrated that the three selected BP neural network models can identify trees, people, stones at a high correct rate of 93.3 % or more.

The current obstacle detection method was proposed on the basis of a static system. In the future, dynamic obstacle detection will be studied in order to achieve real-time and rapid detection of obstacles in the forest environment.

Acknowledgements

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The Design of Tension Test Sensor and its Finite Element Analysis and Test Research

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Abstract: The tension test method of strip steel on stretching winding straightener tension roll group was elaborated. The material and paster form of resistance strain foil in tension test sensor were designed. The measure circuit of tension test sensor was designed. The material and structural form of elastic body of tension test sensor were designed. Strength computation was made on elastic body of tension test sensor. Three-dimensional entity model of elastic body of tension test sensor was built by SolidWorks software. Finite element analysis was made on relative model by ANSYS software. Relative stress figure and displacement figure of elastic body of tension test sensor were obtained. Tension test sensor was manufactured based on design. Static state compression rigidity test of elastic body of tension test sensor was made. Relative test system constitution, test method and test process were elaborated. Relative test data were disposed. This research provides reliable base for the design of tension test sensor. *Copyright © 2013 IFSA.*

Keywords: Tension test sensor, Strength computation, Finite element analysis, Test research.

1. Introduction

Many industries such as spaceflight, national defence, domestic appliance, printing, and so on were developed rapidly at present. The need quantity of cold rolling strip steel increases rapidly. The quality requirement of cold rolling strip steel is stricter. During the continuous production process of cold rolling strip steel, stretching winding straightener was adopted used widely to realize relative techniques purpose. So, more and more attentions were paid to stretching winding straightener in various cold rolling strip steel production line [1, 2].

The important way of eliminate strip steel defect and improve strip steel plate shape quality is adopting stretching winding straightening technology. The accurate control of stretching winding straightening process and the improvement of strip steel quality can be realized by effective strip steel tension test. Tension test sensor is the key of strip steel tension test. Tension test sensor is the window of perceiving and obtaining test information in tension test system. The information obtained by tension test sensor decides the capability of tension test system directly. Automation control system research of strip tension leveller was made by Zhiguo Jia and Zhiyong Chen in cold rolling mill of Benxi Iron and Steel Company in 2002 [3]. Fault analyzing of elongation tension control system of tension leveller stretcher was made by Qin Wen and Zhiyong Chen in Benxi Iron and

Steel Company in 2003 [4]. The application of microcomputer control electron universal testing machine in materials mechanics teaching experiment was made by Wenlong Zheng in National University of Defense Technology in 2008 [5]. Study on the static rigidity of damper rubber product tested with computer controlled amsler tupe tester was made by Junqing Ren and Guiyi Wang in Zhongnan Rubber Group Company in 2008 [6]. The application of weigh sensor on tension test and control of twenty rollers rolling mill was made by Wenhong Yang, Baowen Li and Dongqing Wang in Baoji Titanium Group Limited Company in 2010 [7]. Automatic control system of tension stretching straightener in scouring and rolling union assemblies was made by Zhaogang Lu, Junxiang Dong and Yungang Huang in Liuzhou cold rolling strip mill in 2010 [8]. Study and application of tension leveller of PL-TCM at Meigang was made by Yingjie Zhang, Qingling Yang and Zhigang Li in cold rolling plant of Meishan Iron and Steel Company in 2010 [9]. Intelligent detection system tension was made by Xin Zhang, Kun Yang and Cheng Zhang in Tianjin Polytechnic University in 2010 [10]. Design and analysis on intelligent tension detection system based on bluetooth was made by Hongjun Liu and Shengkun Sun in Tianjin Military Affairs Deputy Bureau of Navy Equipment Ministry in 2012 [11].

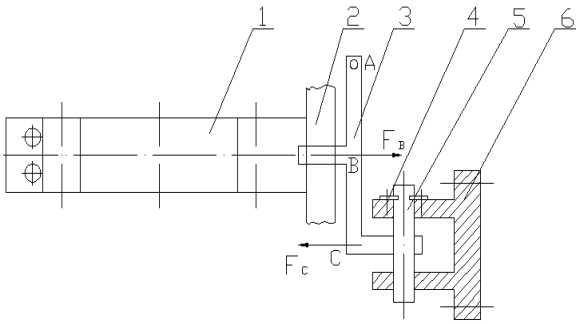
In a word, now, some researches have been made on the test and control of strip steel tension of stretching winding straightener. While, there are also some deficiencies. The sensor used in certain cold rolling mill stretching winding straightener tension test is imported. Its price is very high. Its work condition is very bad. There is not blueprint in mill. This causes great influence on the test precision and working life of sensor. This also causes many inconveniences in sensor servicing. So, the sensor was designed and manufactured in this paper. Relative finite element analysis and test research were made to insure the normal operation of straightener.

2. Tension Test Method

Tension is an important parameter in stretching winding straightener. It relates to the regular work state of stretching winding straightener. Tension test method is to transform the inlet tension and outlet tension of every tension roll into horizontal resultant. The resultant was transmitted to lever connected with sensor by support arm. The resultant can be solved by lever principle. Tension test picture is shown as Fig. 1.

In Fig. 1, A point is fixed hinge point between lever and framework. B point is the horizontal force pass point acted on lever by bearing block. C point is the force pass point acted on sensor by lever. F_B is the horizontal resultant transformed from the inlet tension and outlet tension of every tension roll. F_c is the force tested by sensor. The distance between A point and C point is two times as long as the distance between A point and B point. Strip steel tension can be solved by lever principle, which is shown as Formula (1).

$$F_C = F_B \frac{AB}{AC}, \tag{1}$$



1- Bearing Block, 2- Framework,
3- Lever, 4- Baffle, 5- Sensor, 6- Base
Fig. 1. Tension Test Picture.

3. Tension Test Sensor Design

3.1. Tension Test Sensor Constitution

Resistance strain sensor designed independently was used as tension test sensor in this paper. The three parts of this sensor are resistance strain foil, test circuit and elastic body. Strain foil material is nicochrome. Strain foil standard resistance value is 120 Ω. Test circuits are all full bridge equiarm constant current bridge, which is shown as Fig. 2. Elastic body structure is solid pin roll form, which is shown as Fig. 3.

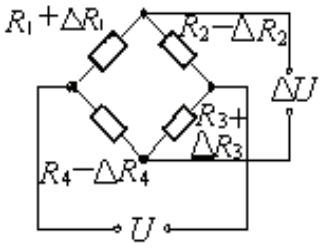


Fig. 2. Test circuit picture.

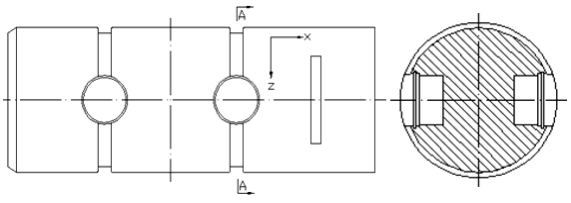


Fig. 3. Elastic body structure picture.

After loading on elastic body, shear stress caused by shear force need to be tested. So, four strain foils were stuck on both sides of I beam sterna. They are placed on 45° angle relative to central axis. The strain foils paster place is shown as Fig. 4. The four strain foils constitute full bridge. The symmetric output engendered on bridge diagonal is proportional to load with resistance value increase of strain foil R_1 and R_3 and resistance value decrease of strain foil R_2 and R_4. Strip steel tension can be tested by this principle.

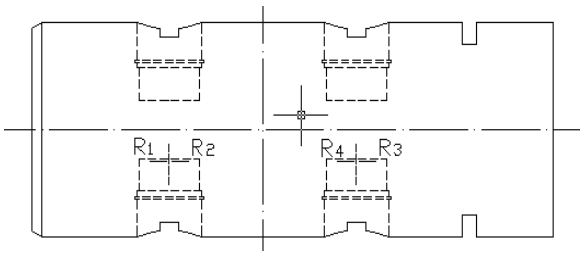


Fig. 4. The strain foils paster place picture.

3.2. Strength Computation of Elastic Body of Tension Test Sensor

The section passed through elastic body blind hole centre can be regarded as part I beam section. The section contains upper flange, bottom flange and middle plate. The section and relative shear stress and bend stress are shown as Fig. 5.

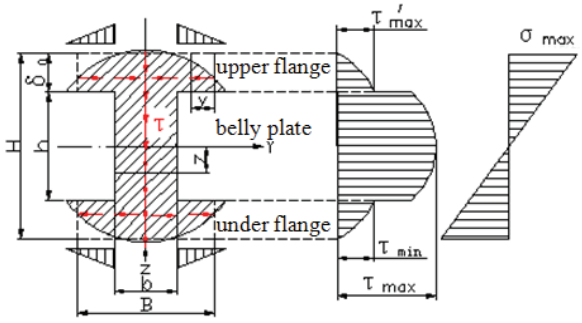


Fig. 5. Beam section and relative shear stress and bend stress.

The maximum bend stress $\sigma_{\max}$ and shear stress $\tau_{\max}$ of sensor elastic body can be solved by formula (2) and (3) respectively.

$$\sigma_{\max} = \frac{6QeH}{BH^3 - Bh^3 + bh^3}, \tag{2}$$

$$\tau_{\max} = \frac{3Q}{2b} \cdot \frac{BH^2 - Bh^2 + bh^2}{BH^3 - Bh^3 + bh^3}, \tag{3}$$

The yield stress of elastic body σ_s is 930 MPa. The allowable bend stress $[\sigma]$ of elastic body is 279 MPa. The allowable shear stress $[\tau]$ of elastic body is 186 MPa. The distance from section poid to section fringe e is 40 mm. The load on the fourth tension roll is 85.843 kN. This load was loaded on elastic body of sensor. I beam section height h is 30 mm. I beam section width B is 65 mm. The plate width b is 22 mm. After substituting these parameters into formula (2) and (3) respectively, $\sigma_{\max}$ and $\tau_{\max}$ were computered respectively with relevant value 60.93 MPa and 74.21 MPa. $\sigma_{\max}$ and $\tau_{\max}$ are all less than relevant allowable stress. So, the tension test sensor designed in this paper can satisfy spot strength requirement.

4. Finite Element Analysis of Elastic Body of Tension Test Sensor

4.1. The Construction of Elastic Body Finite Element Model of Sensor

Three-dimensional entity model of elastic body of sensor was constructed by SolidWorks software in this paper. It was saved as parasolid document form. Then, the relevant model was led into and analyzed by ANSYS software. The relevant model was shown as Fig. 6.

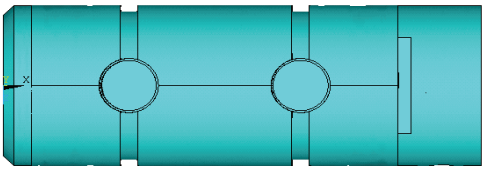


Fig. 6. Three-dimensional entity model of elastic body of sensor.

4.2. Property Definition, Grid Division, Constraintion and Load of Elastic Body of Sensor

The material property of elastic body E is 2.06×10^5 N/mm^2. Poisson ratio μ is 0.26. Density ρ is 7.81 kg/mm^3. SOLID187 element and free grid division method were adopted on grid division of elastic body. The displacements on Y axis and Z axis of elastic body circle face supported by base were set as zero constraintion. The displacements on X axis of two elastic body groove sides were set as zero constraintion. The displacements on Y axis of elastic body underside were set as zero constraintion. The connected position between elastic body and support arm was loaded as equispaced load. The tension on the fourth roll outlet is 85.843 kN. It was set as the load during analysis process. The analysis of this

research is finite element statics analysis. So, equispaced load can be transformed to concentrate load according to equivalent principle. The relevant concentrate load was loaded on five nodes. The relevant grid division, constraintion and load of elastic body are shown as Fig. 7.

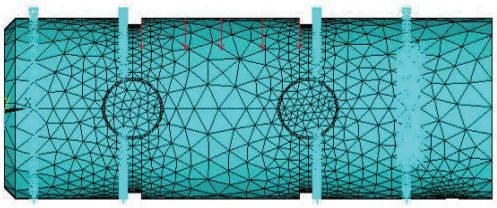


Fig. 7. The relevant grid division, constraintion and load of elastic body.

4.3. The Finite Element Analysis Result of Elastic Body of Sensor

After ANSYS computation, relative stress figure and displacement figure of elastic body of sensor were obtained. They are shown as Fig. 8 and Fig. 9 respectively.

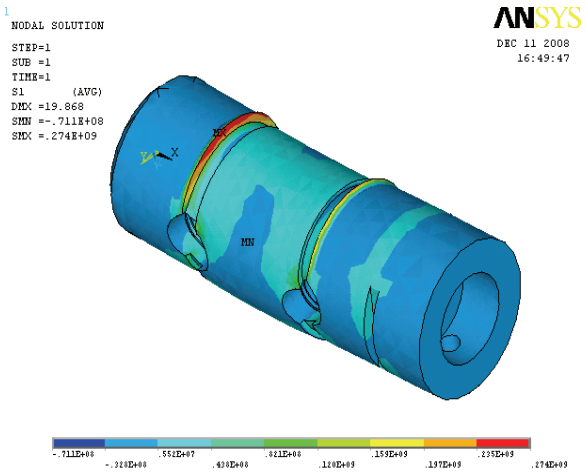


Fig. 8. Stress figure of elastic body of sensor.

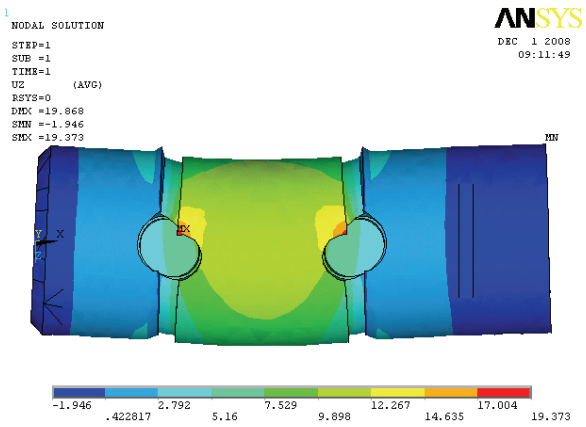


Fig. 9. Displacement figure of elastic body of sensor.

It can be seen from Fig. 8 that, the maximum stress $\sigma_{\max}$ of elastic body of sensor is 274 MPa. Its place is on two elastic body groove sides. It was caused mainly by part stress concentration. There is an equal shear stress area with small stress change gradient. The material of elastic body of sensor is 35CrMoV spring steel. Relevant yield strength σ_s is 930 MPa. Allowable safety factor [s] is 2. Relevant allowable stress $[\sigma]$ is the ratio of σ_s and $[\sigma]$. Its value is 465 MPa. The maximum stress $\sigma_{\max}$ is less than allowable stress $[\sigma]$. So, the tension test sensor designed in this paper can satisfy spot strength requirement. It can be seen from Fig. 9 that, the main load bearing of elastic body is on Z axis direction. So, the displacement deformation is mainly on elastic body Z axis direction. The displacement deformation form is parabola. The maximum displacement deformation value of elastic body is 0.019373 mm. The value is very small. So, the tension test sensor designed in this paper can satisfy spot rigidity requirement.

5. Static State Compression Rigidity Test of Elastic Body of Tension Test Sensor

5.1. Test System Constitution and Test Method

Static state compression rigidity test system of elastic body of tension test sensor contains universal test machine, sensor, junction box, strain gage, A/D and computer. Relevant principle diagram is shown as Fig. 10.

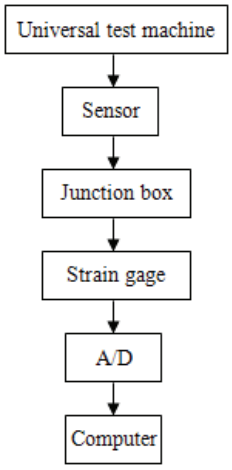


Fig. 10. Test system principle diagram.

WDW3100 electronic universal test machine is a new type test machine. The test machine mainly contains host computer, whole digital test control system, user software package and function

attachment, and so on. It is shown as Fig. 11. The host computer has the characteristics such as function stability, reliability and high test control precision. The tension test sensor designed and manufactured independently is shown as Fig. 12.

Fig. 11. WDW3100 electronic universal test machine.

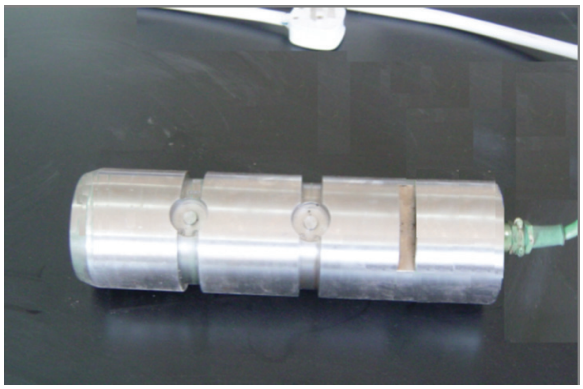


Fig. 12. The tension test sensor designed and manufactured independently.

The strain foils paster place of elastic body of sensor is shown as Fig. 13. In Fig. 13, the strains of R1 and R3 are stretching strains. The strains of R2 and R4 are compress strains. Bridge arm working condition coefficient C is 4. The strain foil resistance values are all 120 Ω .The strain foil sensitive coefficient K_s is 2.20. The calibrated strain of strain instrument ε_s is 1000 $\mu\varepsilon$. The join pattern between strain foil and junction box is shown as Fig. 14.

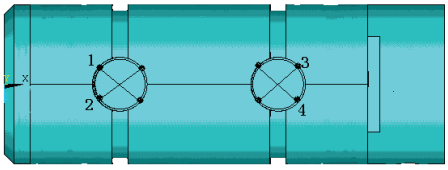


Fig. 13. The strain foils paster picture

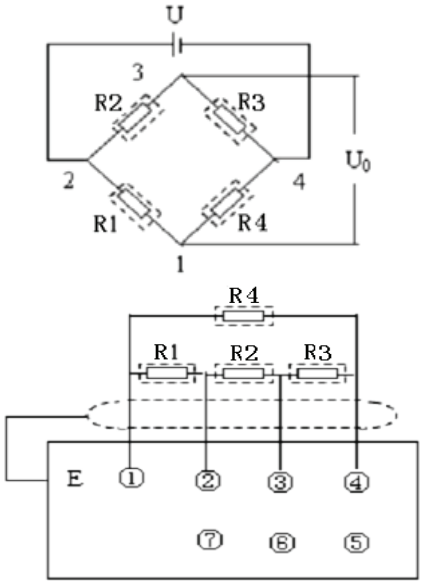


Fig. 14. The join pattern between strain foil and junction box.

BZ2201 dynamic strain gauge is an automatic balance strain instrument. Relevant alternating power supply voltage is 220 voltages. Relevant constant current power supply voltage is 12 voltages. It is shown as Fig. 15. The relevant test method contains following steps. Firstly, strain foils were stuck on blind holes underside of elastic body of sensor. Secondly, the strain of blind holes underside can be transformed as strain foil resistance change. Then, the strain foil resistance change can be transformed as voltage change. Finally, the voltage change can be shown by strain gauge.

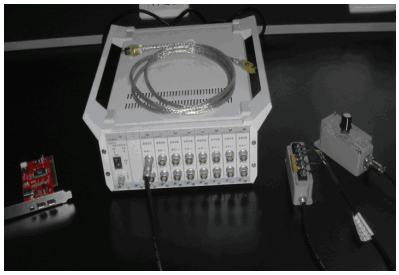


Fig. 15. BZ2201 dynamic strain gauge.

5.2. Test Process and Data Processing

Static state compression rigidity test of elastic body of tension test sensor was made by universal test machine. Relevant load signals and displacement signals come from load sensor and displacement sensor placed on universal test machine were sent into controller. After amplifying and A/D transforming, relevant signals were sent into computer. The disposed signals were shown on screen synchronously. So, load-displacement curve was formed. It is shown in Fig. 16.

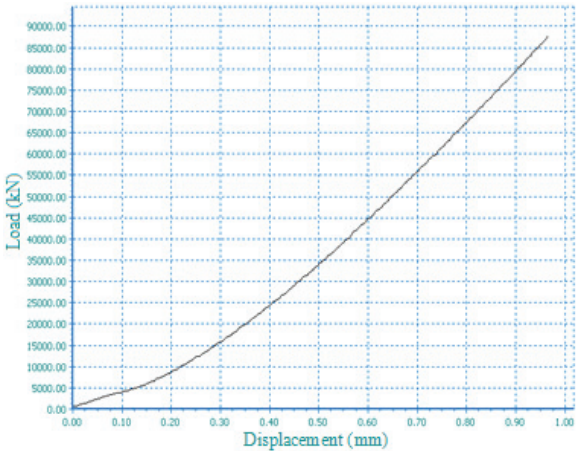


Fig. 16. Load-displacement curve.

It can be seen from Fig. 16 that, the initial short section of load-displacement curve is a curve. Hinder long section of load-displacement curve is an approximative slant. Relevant load scope is 0 kN to 90 kN. So, under this load, elastic body of tension test sensor is on elastic stage. The load on the fourth tension roll is 85.843 kN. This load is in the load scope of universal test machine. So, under the normal work of tension test sensor, relevant elastic body is also on elastic stage. Relevant deformation is also elastic deformation. So, the tension test sensor designed in this paper can satisfy spot rigidity and strength requirement. Three load tests were made on tension test sensor designed in this paper. Relevant output wave shapes of strain gauge are shown as (a), (b) and (c) of Fig. 17 respectively.

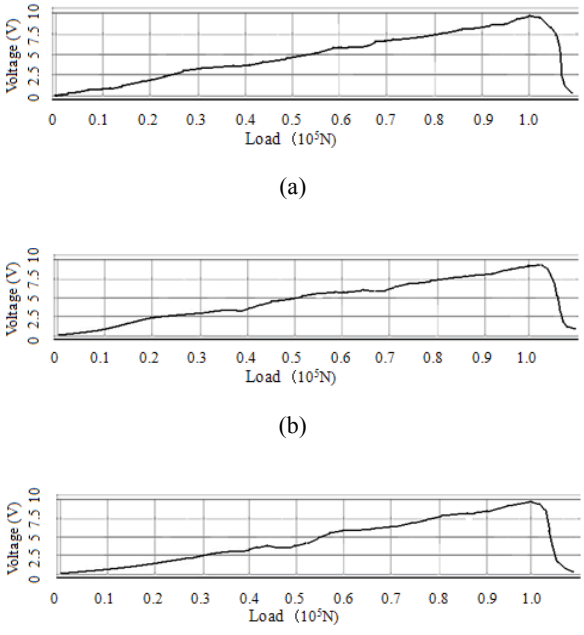


Fig. 17. Output wave shapes of strain gauge under three load tests.

It can be seen from Fig. 17 that, the relevant load scope is 0 kN to 90 kN. The output voltage is regarded as linear increase with the load increase approximatively. So, under the normal work of tension test sensor, relevant elastic body is also on elastic stage. Relevant deformation is also elastic deformation. So, the tension test sensor designed in this paper can satisfy spot rigidity and strength requirement.

6. Conclusions

The tension test sensor designed in this paper can satisfy spot strength requirement, which was validated by the strength computation of elastic body of tension test sensor. The tension test sensor designed in this paper can satisfy spot rigidity and strength requirement, which was validated by the finite element analysis of elastic body of tension test sensor by ANSYS software. The tension test sensor designed in this paper can satisfy spot rigidity and strength requirement, which was validated by the static state compression rigidity test of elastic body of tension test sensor. The strength computation, finite element analysis and static state compression rigidity test of elastic body of tension test sensor provide reliable base for the design of tension test sensor.

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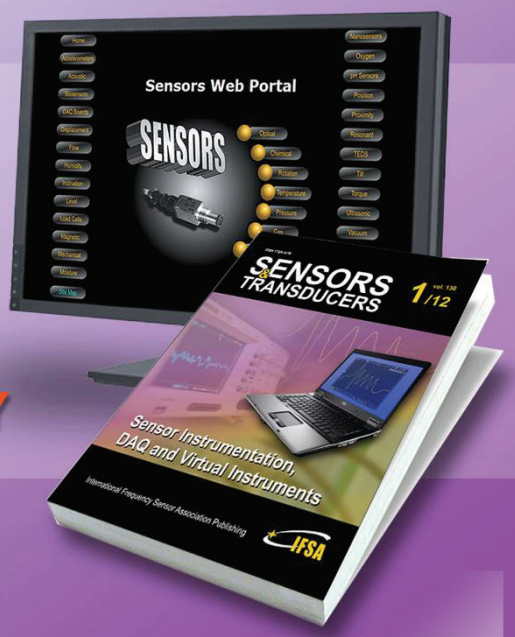

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The Optimization Algorithm of Wireless Sensor Network Node Based on Improved Ant Colony

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Abstract: Ant colony optimization algorithm is a way to be used to find the optimal solutions is enormous, the number of ants on the path, which will lead to reduced the network node on the path energy drastically. The wireless sensor network is a large number of tiny sensor nodes with communication and computing power. Using wireless sensor networks collaboration to perception, the perception of the object information acquisition and processing in the network coverage area, and send to the observer. This paper describes the basic principles and characteristics of the algorithm and the improved ant colony algorithm, ant colony algorithm in wireless sensor network node. The paper presents the optimization algorithm of wireless sensor network node based on improved ant colony. In order to verify the performance of the algorithm, the paper establishes a wireless sensor network simulation environment in Matlab. *Copyright © 2013 IFSA.*

Keywords: Ant colony, Wireless sensor network, Optimization.

1. Introduction

Ant colony algorithm (ant colony optimization, ACO), also known as ant algorithm, the probability type technology is a way to be used to find the optimal solution. Inspired by the behavior of ants found in the process of looking is in order for food path. When ants move forward on the path to choose their path to go based secretions the ants traveled by the front left. The probability of selecting a path is proportional to the intensity of secretions in the path. Therefore, the collective behavior of the group consisting of large number of ants' positive are feedback phenomenon in fact constitute a learning information: a path through ants more, greater ants behind the possibility to select the path. Individual ants seek the shortest path to the food through the exchange of such information.

Wireless sensor network node processor / radio frequency communication board and sensor board. The processor/radio frequency communication board with a MSP430F149 microcontroller and a variety of operating modes of the ultra-low-power radio frequency (RF) chip nRF905, sensor board with a sound, acceleration and magnetic properties of three sensors for environmental data collection and target the detection of the information, buzzer, microphone, and the reception of the audio signal amplification and decoding circuitry at the same time to achieve ranging between nodes. The system uses two battery-powered. Aggregation node is using the same structural design and ordinary sensor nodes.

Ant colony algorithm ants through pheromone path to reach the destination, so if the path is the optimal path to reach the destination, the number of ants in the path would be great [1]. This will lead to the path on the network node energy drastically reduced to shorten the life cycle of the entire network, and is likely to lead to the path congestion ant colony algorithm easy to fall into local optimal

solution, the performance of the routing information table stagnation, so when network topology changes, the artificial ants soon found new better path. In order to solve these problems, it is necessary to improve the ant colony algorithm, network data flow through different possible delay countdown.

Ant colony optimization algorithm is a simulated ants foraging behavior of analog optimization algorithm, it is first proposed in 1991 by the Italian scholar Dorigo M, et al, after study the basic principles and mathematical models of the ant colony algorithm, combined optimization problems with genetic algorithms, taboo search algorithm, simulated annealing algorithm, hill climbing simulation experiments laid the foundation for the development of the ant colony algorithm, and attracted the attention of scholars and research around the world.

Wireless sensor network (Wireless Sensor Network, WSN) is deployed in the monitoring area of a large number of cheap micro sensor nodes, formed a multi-hop self-organizing network through wireless communication systems, collaborative sense, acquisition and processing the information of the object is perceived in the network coverage region, and sent to the viewer. Sensors, perception of the object and the observer constitute the three elements of the wireless sensor network. Bi-directional communications as a wireless ad hoc network, the sensor network can be done automatically with the greatest flexibility to the networking of the irregular distribution of the various sensors and control node.

Ant colony algorithm is based on this characteristic; imitate the behavior of ants, in order to achieve optimization. This algorithm is different from the traditional programming model, its advantage to avoid the lengthy programming and planning, the program itself is based on certain rules randomly run to find the best configuration. That is, when the program most find the target, the path is almost impossible to be optimal and may even contain numerous errors choice extremely lengthy. However, the program can find food by ants when the principles of the pheromone, continue to go to fix the original route, so that the entire route is getting shorter, that is, the program execution time is longer, and the obtained path is the more likely close to optimal path. The paper presents the optimization algorithm of wireless sensor network node based on improved ant colony.

2. Optimization based on Improved Ant Colony Algorithm

The ant colony algorithm is introduced: the ant colony algorithm to find the shortest path from ant foraging behavior. Ant colony in search of food sent some ants separately wandering around an ant finds food, it returns to the nest notice companions along the way to leave a letter "pheromones (pheromones, pheromone) as an ant colony to the food location mark. Pheromones gradually volatile if the two ants find the same food at the same time, they take a different route back to the nest, then the compare Raowan a road pheromone smell will be relatively light, ant colony will tend to be along another closer route Go to food seat.

Ant colony algorithm, for a single objective optimization problem, the basic ant colony algorithm based on transition probability search mechanism and the pheromone update, helps to guide the ant groups toward the pheromone concentration high regions of the search, the ants gather in space in a certain area, eventually find the global optimal solution. However, for many objective optimization problem, the solution quality is relative, not absolute optimal, therefore, will usually get a Pareto solution set [2]. Because Pareto solution concentration point can be used as the design of the optimal solution, thus requiring the non-inferior solution set to approximate the real Pareto solution set, and obtained Pareto leading to good.

So the pheromone is the only way to communicate between artificial ants. This method of communication in the use of group knowledge plays a vital role in. In addition, ACO also uses the evaporation mechanism, this point corresponding to the real ant evaporation and hormone. The evaporation mechanism makes the ant colony gradually forget the past history, makes the subsequent ants in the search of less affected by the past bad solution, in order to better guide the ant search direction.

The search for the shortest path is with local mobile. Artificial ants and real ants have the same task, i.e. constructed from origin (Termitomyces) to the destination point (food source) shortest path between local mobile manners. Random state is transition strategies. Artificial ants and true ants are in accordance with the probabilistic decision rules from one state to another neighboring state. Probabilistic decision rule is information and local environment information function. State transition process, artificial ants and true ants only use local information, without the use of forward-looking strategies to foresee the future state [3]. Therefore, the strategy used in time and space is local. Different point of artificial is ants and true ants, as is shown by equation 1.

$$\begin{aligned} D(x,y,\sigma) &= L(x,y,k_i\sigma) - L(x,y,\sigma) \\ &= G(x,y,k_i\sigma) * I(x,y) - G(x,y,\sigma) * I(x,y) \\ &= (G(x,y,k_i\sigma) - G(x,y,\sigma)) * I(x,y) \end{aligned} \quad (1)$$

Elitist strategy ant system (Ant System with elitist strategy, ASelite) was among the first to improve the elitist strategy ant system genetic algorithm, traditional genetic algorithms may lead to best adapt to an individual's genetic information elitist strategy idea is lost retain generation in most elite strategy to adapt to the individual ant system after each cycle to give the optimal solution to this additional amount of pheromone solution is called the global optimal

solution (global-best solution) to find out the solution ants called elite ants (elitist ants), with elitist strategy ant system pheromone is updated according to the following formula.

The ant colony algorithm search behavior of ants is concentrated near the optimal solution can improve the quality of solutions and convergence rate, thus improving the performance of the algorithm. This search method cause premature convergence behavior more prone to Max - Min Ant System (Max-Min Ant System, MMAS) This search method can be an effective mechanism to avoid premature convergence together, so that the algorithm optimal performance.

The more ants walk a path, and then choose the path of the greater probability. This will form a positive feedback, through this positive feedback mechanism; the ant can find the optimal path. As shown in Fig. 1, Fig. 1 node B representation of the nest, said the food source node F. At the beginning, from node B released some ants to F, then they will with equal probability selection of B-C-F, B-D-F and B-E-F branch, each ant to release a certain amount in the path through the pheromone, this information is a certain rate of volatile, ant to judge a path to choose the path of pheromone the largest intensity. In the figure, when return to choose B-D-F branch ant to the node A according to the original road, length of the remaining 2, because the longer path, starting from the node A ants have not returned.

After each cycle in order to take full advantage of the optimal solution loop optimal solution and identified so far, only one ant pheromone update. The ant to find out the optimal solution in the current cycle of ants, ants since the beginning of the experiment, the optimal solution is to find out is to avoid stagnation of the search, track the amount of pheromone on the elements of each solution the value range is limited [τ min, τ max] range initialize the pheromone track for the τ max pheromone track update in MMAS only one ant is updated after each cycle information trajectory revised the trajectory update rules are as follows, using the global optimal solution in the ant colony algorithm mainly use iterative optimal solution in MMAS, as is shown by equation 2 [4].

$$W = (I - \frac{Y_1^* Y_1^T}{Y_1^T Y_1^*} - \frac{Y_2^* Y_2^T}{Y_2^T Y_2^*} - \cdots - \frac{Y_M^* Y_M^T}{Y_M^T Y_M^*})\ W_0 \quad (2)$$

Ant according to the concentration of the pheromone to guide their direction of movement tends to move toward the direction of a high concentration of pheromone the same time, the information factors will gradually over time to evaporate, so the length of the path and the path through the ant much impact on the residual strength of the pheromone. In turn, the strength of the pheromone is to guide other ant's action.

In Directed Diffusion, Sink by flooding request to query the data of each node, and in the SPIN of each node broadcasts its data, for the interested nodes to query data. Through the on-demand query mechanism, Directed Diffusion can effectively reduce the unnecessary data transmission, in order to obtain the better energy-saving effect. The main idea of layered routing algorithm for layered routing algorithms by clustering the data cluster correlation cluster heads aggregate and reduce the need to send the Sink data volume, thereby saving energy consumption. This kind of algorithm is a representative algorithm including LEACH, TEEN etc. LEACH (Low-Energy Adaptive Clustering Hierarchy) algorithm is a typical layered.

Ant colony system state transition rules, q is in a random number Q0 [0, 1] interval uniform distribution determines the size of the use and exploration of the new path between the relative importance of the prior knowledge. The state transition rule known as the characteristics of pseudorandom proportional rule: tend to choose short and has a large number of pheromone edge as the direction of movement, ant colony system global updating rule.

Artificial ant has a dual characteristics: on the one hand, they are an abstraction of the real ants, has the characteristics of real ants, on the other hand, they have some characteristics not found in the real ants, these new features make artificial ants in solving actual optimization problem has a better ability to search for better solutions, as is shown by Fig. 1.

Artificial ants and true ants of the same points: 1. Collaborate are a group of individuals. Real ant colony, ACO is composed by a group of artificial ants, synchronous / asynchronous collaboration between artificial ants to find the optimal solution [5]. Single artificial ants can construct a solution of the problem, but only when the number of artificial ants through mutual collaboration in order to find the optimal (suboptimal) solution. Artificial ant's individuals are to collaborate by write / read state variables.2. The use ectohormone traces and the Stemberg mechanisms. Real ants and it is artificial ants by changing access issues digital status information (called pheromones in ACO) to indirect collaboration.

Features: The principle of ant colony algorithm is a positive feedback mechanism or enhanced learning systems; increased by the increase in the number of ants on the optimal path → pheromone strength → later ant's selection probability increases the number of ants' → optimal path greater increase, to achieve the ultimate convergence on the optimal path [6]. It is a general-purpose stochastic optimization method, it absorbs the behavior of ants Special (internal search mechanism), it is the use of artificial ants the simulation ant system (also known as) to solve the problem L artificial ants is not a actual ants simple simulation, it melt into the human intelligence L artificial ants memory; artificial ants are not completely blind; the artificial ants life of the space-time is discrete.

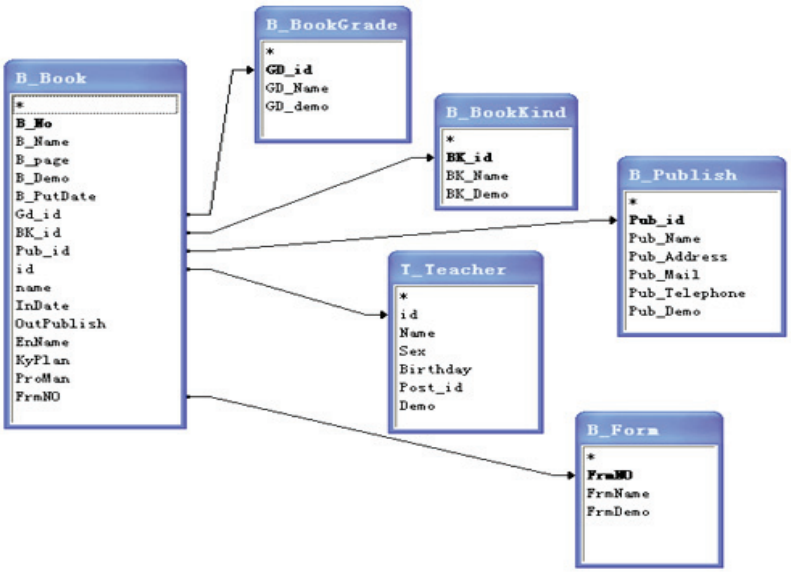

Fig. 1. An ant colony algorithm search behavior of ants.

Based on the above analysis, the basic ant colony algorithm pheromone update makes a new definition. Because it is in the continuous domain, thus distinguished from discrete domains of the situation, the pheromone only retained in the ant position. In order to better guide the ant on the feasible space search, using an external set A (t) for t moment has been found in Pareto solution set. If the ant I to set A (T), the ants in position is now not available, should increase the ant I pheromone, for other ants to ant I position neighborhood search; conversely, if not enter the set A (T), it should be appropriate to reduce the pheromone.

When multiple ant and enter the set A (T), should distinguish each ant pheromone increment to the forefront of Pareto uniformity, the new set A (T) into the ant I and set A (T). Solution of the objective function value of the minimum distance as the ant I in location release pheromones. So, the greater the distance information content is higher, with higher probability guide other ant the ant position neighborhood search as is shown by equation 3.

$$H = \sum_{X \in \Omega_i} \left[\frac{\partial I}{\partial W} \frac{\partial W}{\partial P} \right]^T \left[\frac{\partial I}{\partial W} \frac{\partial W}{\partial P} \right] \tag{3}$$

The artificial ants live in a discrete time, discrete state from a discrete state to another; artificial ants have internal state of the memory capacity of the artificial ants can remember walking along; 3. Artificial ants pheromone are quantity which generates a function of the quality of solutions; 4. The artificial ant's profile pheromone timing depends on the specific problem [7]. For example, most of the artificial ants only after the ants are to find a solution to update the pheromone on the path. Sometimes, in order to improve the efficiency of the ACO, in the algorithm predicted that local optimization and backward strategies.

Ant colony algorithm in the pheromone update, no matter how the solution to search, in these pheromone path will have strengthened to a certain extent, and the solution to information search is not better than has been strengthened, but will with the passage of time is not broken volatile and more difficult to search, so the ant colony algorithm more vulnerable to early found better solutions, to fall into the local optimal solution, in order to overcome this phenomenon, we use the following method to update with the reward and punishment mechanism.

The specific process of the ant colony algorithm:

Start:

1) Initialization: NC = 0, the initial path information elements equal to 0.1, n ant corresponding n cities, all the ants set the initial of Tabu (k);

2) All ants calculated transition probabilities; the roulette wheel to select the next city j and city j joined their respective Tabu (k) table;

3) Update on various lines of pheromone, ant - cycle through route plus a fixed value, Trail route through the ant-quantity plus a fixed value divided by the length.

Where η_{ij} (t) = 1 din is the distance ants i and ants j by the above strategy, the ants will not only have a collection of A (t) is more sparse area search, but also where other ants position search in order to accelerate the convergence speed, and also introduces a random number-based search mechanism for a given [0,1] constant W, when the random number z ($\in$ [0,1]) $\leq$ w, according move the above-mentioned transition probability; while when z> w, optionally selected from the collection of A (t) in the solution as the next step in the moving direction of the ant i, and the neighborhood of the Solutions search summary, solving objective optimization problem ant colony algorithm specific steps are as follows: (1) initialize the ant's position in the search space S as well as a collection of a (t), so that t = 0 (2) t = t +1. (3)

produce random number z, z ≤ w, then the search in accordance with the formula (6); if z> w, in the collection of the solution of A (t) is arbitrarily selected as the next moving direction do neighborhood search set of (4) Update A (t). (5) In accordance with the formula (5) to update the content of the pheromone of the ants. (6) If t is greater than the maximum number of iterations, and exit; otherwise, go to step (2).

Whether you choose to iterative optimal or global best ant pheromone update may lead to the stagnation of the search. Stagnation occurred: a selection in each selected points on the pheromone trajectory significantly higher than the other options [8]. Avoid stagnation state: influence is used to select the next solution elements of probability; it is directly dependent on the pheromone track and enlightening information. By limiting the influence of the trajectory of the pheromone, can be easily avoided by the difference is too large between the tracks of each pheromone trajectory limits.

By virtue of the ability of the path optimization according to the results of the bionics home study, the ants can find the shortest path between the nests with food, the principle is that: ants leave a volatile secretions (pheromone path through the following called pheromones) pheromone disappear over time will gradually volatile. Ants in the foraging process capable of sensing the presence and intensity of this substance, and in order to guide the direction of their own movement, tend toward thistle high strength in the direction of the substances move, i.e. the probability to select the path with the time on this path, is proportional to the intensity of the substance. pheromone intensity higher the path select its ant more, then leave on the path pheromone strength is greater, while the intensity of the pheromone and attract more ants, thereby forming a positive feedback through this positive feedback, the ants can eventually be found that the best path, resulting in the most ants will gothic path, as is shown by Fig. 2.

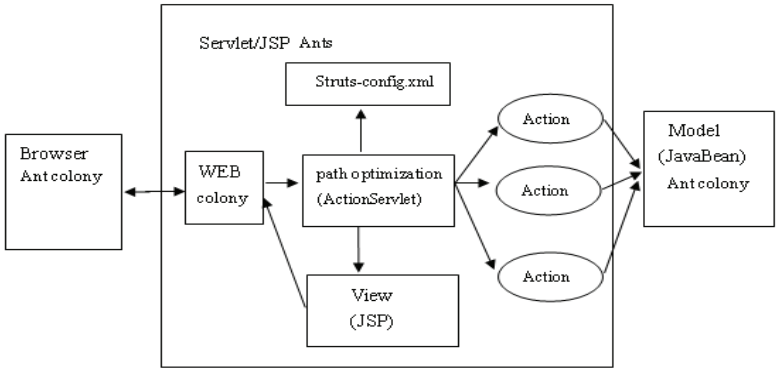

Fig. 2. The principles of rise of the ant colony algorithm research.

With the rise of the ant colony algorithm research, it was found that the ant colony model clustering in some ways closer to actual clustering problem. Ant colony algorithm for cluster analysis, inspired by ants piled their bodies and classification of their larvae. Clustering method based on ant colony algorithm principle can be divided into two types: one is based on the ant heap formation principle to achieve data clustering, another is to use the principle of ants foraging, the use of information to achieve clustering analysis. Data is another important topic of data mining; it is looking for properties in the collection of database objects, and can be divided according to the classification mode.

Gateway (some literature also known as Sink node) is a special node, can communicate with the monitoring center through the Internet, mobile communication network, satellite [9]. Also it can use drones flying over the network, through the gateway acquisition data. The wireless sensor network is still in the research stage, in order to speed up the process of the construction, a lot of foreign demonstration system, relevant theoretical research is also a lot of. In recent years, some domestic research institutes and universities have carried out the research of wireless sensor network theory and application.

The ants have the choice the same line, the line is for the best line; in the actual calculation, given a fixed cycle times but it is difficult to realize in this kind of situation. On the other hand, the actual in him the application, image processing to seek optimal mould plate problems, such as ants and does not require all can find the optimal die plate, only to find that one. The ants are to ask if there is to find the optimal mode plate, and affect the calculation efficiency.

Many applications of wireless sensor networks, and it is typical applications including surveillance, tracking and control. It is generally used in the natural environment monitoring, target tracking, nuclear reactor control, fire monitoring, and traffic monitoring purposes. In a typical business environment, wireless sensor networks are distributed to the region around, to collect data through the

sensor node in it. Area monitoring is a typical application of wireless sensor network. In this application, wireless sensor network is arranged on the need to monitor a phenomenon in the area [10]. For example, the sensor nodes arranged a lot on the battlefield, replace the mine to monitor enemy invasion.

Probability model for estimation of distribution algorithms can be effective pheromone distribution the same time, the distribution of the pheromone can inspire the probability distribution of the model estimation of distribution algorithms, ant colony algorithm pheromone distribution model and estimation of distribution algorithms probabilistic model of phase, as is shown by equation 4.

$$f(t)=f_L(t)+w_L^i(t)+\cdots+w_{M-2}(t)+w_{M-1}(t)=f_L(t)+\sum_{l=L}^{M-1}w_l^i \quad (4)$$

Residual information; the β - inspired importance of information; tabuk - record traversed the city by ant k, called memory list, k = 1,2, ..., m collection tabuk code with the evolutionary process dynamically adjusted after n points in time, all the ants have completed a traverse this case, calculation of the path traversed by each ants Lk of and save the shortest path Lmin = mi The n {of Lk | k = 1,2, ...,m}. ants after the completion of a cycle, the amount of information on each path to be adjusted as follows τ_{ij} (t +1) = (1-ρ) (2) of formula τ_{ij} (t) + $\Delta\tau_{ij}$ ρ ∈ (0,1),represents pheromone τij (t) over time and the degree of attenuation of 1-ρ is the pheromone residual factor start $\Delta\tau_{ij}$ (0) = 0; β is the relative importance of visibility reflect ants choose the path information in the course of the campaign inspired by the emphasis, the greater its value, the state transition probability close to the greedy rules, γ is the parameter indicates the degree of energy is relatively important.

Ant Colony System (Ant Colony System, ACS) ant colony system do the three areas of improvement: state transition rules to provide a better and more rational use of the new path and the use of a priori knowledge about the problem the method global update rules apply only to establish the process of problem solutions, the optimal path of ants in the process of establishing solutions to problems, application of local pheromone update rules.

3. The Optimization Algorithm of Wireless Sensor Network Node

The topology control technology is one of the most important technologies in wireless sensor networks. Generated by the wireless sensor networks, network topology, topological edges between two nodes can communicate directly [11]. If no topology control, all nodes will work with maximum wireless transmission power. In this case, the limited energy of the one hand, the node will be the communication section rapid consumption, reducing the life cycle of the network.

Meanwhile, the radio signal of each node in the network will cover a large number of other nodes, causing wireless signal conflict frequently affect the junction point of the wireless communication quality, reducing the throughput of the network. On the other hand, there are a large number of edges in the network topology will be generated, resulting in a large amount of information of the network topology, routing the computational complexity, waste valuable computing resources. Therefore, we need to study the topology control problem in wireless sensor networks, maintain some of the global nature of the premise of the topology, through adjust nodes transmission power to extend the network life cycle, improve network throughput and reduce network interference, saving the node resources.

Wireless sensor network to monitor the physical world as the main target, is a kind of measurement and control network, node's computing and communication ability is weak, severely limited energy supply. To sum up, the wireless sensor network has the following characteristics: a large scale, high density; no center, self-organization; dynamic network; nodes are resource-constrained; multi-hop routing; based on the data center.

Distributed, self-organizing. The wireless sensor network is a network of peer nodes, there is no central control. Management and networking are very simple and flexible. Not rely on fixed infrastructure, each node has a routing function, by self-coordination, and automatically arranged to form a network, no other auxiliary facilities and artificial means. Using wireless sensor networks collaboration to perception, the perception of the object information acquisition and processing in the network coverage area, and send to the observer [12]. Perception of the object, sensor networks and observer constitute the three elements of the wireless sensor networks, as is shown by Fig. 3.

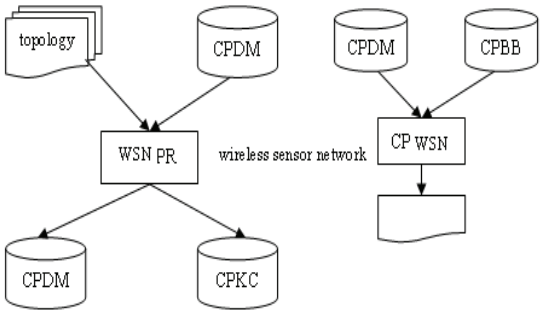

Fig. 3. The three elements of the wireless sensor networks.

A typical wireless sensor network architecture including the sensor nodes, it is aggregation nodes and the management node. A large number of sensor nodes are randomly laid in the monitoring area

network through a self-organizing way. Wireless sensor networks with traditional wireless ad-hoc network, although there are many similarities and common features between the two, there is some essential difference between. Wireless ad-hoc networks through dynamic routing and mobility management technology, to provide users with high-quality data transmission services, is essentially data network node computing and communication ability, and sustained energy supply.

There are many applications of wireless sensor networks in different occasions. In the commercial and industrial fields, some data is difficult to monitor using wired sensors, or the price is too expensive to implement, in this case can be used for wireless sensor networks to monitor. Wireless sensor networks can be deployed in an outdoor environment, they can be placed for years (used to monitor certain environmental factors) without the need to recharge or replace energy equipment. They can form a territorial boundary, and monitor the intruder process of moving (to pass information between the nodes).

Connectivity - in order to achieve mutual communication among the nodes, the topology generated must ensure connectivity; from any one node can send a message to another node. Connectivity is any topology control algorithm must ensure a property. As defined by the UDG graph can know, connectivity of UDG graphs is maximum connectivity network can provide, therefore assume that UDG graph is connected. So, any topology control algorithm is to generate the topology are UDG sub graph.

The characteristics of the wireless sensor network: node miniaturization limited energy supply and harsh environmental conditions node failures, dynamic network topology communications failure problem of a large number of nodes deployed unattended. Topology control algorithm goal is to control the transmission range of the node to satisfy certain properties of the resulting network topology to extend the network life cycle, reduce network interference and improve throughput. Generally assumed junction points are distributed in a two-dimensional plane, all nodes are isomorphic, are the use of non-directional antennas [13]. To graph modeling wireless sensor networks, node i's transmission power Pi is greater than the need to transfer power from node i to node j P_{ij} junction point i to node j between a directed edge. All nodes are generated when the maximum power topology called UDG FIG (Unit Disk Graph). Topology control should be to meet one or several of the following properties of the network topology, as is shown by equation 5.

$$P(k|k) = P(k|k-1) - P(k|k-1)H^T K^T - KHP(k|k-1)^T + K(HP(k|k-1)H^T + R)K^T \quad (5)$$

Wireless measurement system, the support system is very important part of WSN, certainly powered by a battery, and part of data applications, only need a computer to the Internet computer, and the data processing part according to the actual needs of different power supply system. From the figure above, we can see that the biggest advantage of wireless measurement system are three parts to each other there is no hard connection, in other words between them, in addition to the exchange of data, without any physical relationship, so as to effectively avoid the complex inter-interference and the allocation of resources, I think this is the wireless measurement technology vigorously develop key.

Wireless sensor network system is mainly composed of ordinary sensor nodes, aggregation nodes and PC monitoring platform. Ordinary sensor nodes are spread in the observation area, and are responsible for collecting data related to the observed object, and the co-processing of the data after the transfer to the sink node. Aggregation node in the network data is processed and displayed on the PC monitoring platform. Wireless sensor network node design a direct impact on the quality of the entire network, wireless sensor network nodes by the data processing unit, wireless communication unit, sensor unit, the power supply unit group. Under field conditions, the wireless sensor network applications demand real-time monitoring and emergency treatment, the wireless sensor network node design, focusing consider the following aspects: (1) low-power, low-cost, small volume; □ stability; □ scalability and flexibility.

The main challenge is in making low hardware cost and small sensor nodes. Considering this, the sensor node is now mainly at the prototype stage. Miniaturization and low cost solution to hope for a recent and future in microelectromechanical systems (MEMS) and nanoelectromechanical systems (NEMS) field progress [14]. SNM-Sensor Network is Museumtm. Sensor network platform, common general assembly, and it are technology and related topics. And the mainstream computer, there is no official standard for wireless sensor networks, so the interoperability and reusability of hardware and software are low.

WSN node hardware structure, including battery and power management circuits, and it is sensors, signal conditioning circuit, AD conversion devices, memory, microprocessor and RF module. The nodes are battery powered, and once the power is exhausted, the node will lose the ability to work. Power in order to maximize the savings, in terms of hardware design, to maximize the use of low-power devices, RF part of the power cut off in the absence of communication tasks; layers of communication protocol should be energy-efficient in terms of software design, as the center, if necessary can sacrifice some other network performance indicators, in order to obtain higher power efficiency. Sensor battery and power management circuit the memory signal conditioning circuit.

4. The Optimization Algorithm of Wireless Sensor Network Node based on Improved Ant Colony

Ant colony algorithm has its own advantages in the optimization problem to solve the TSP assigned discrete areas of the problem, and shop scheduling problems, and made a lot of progress with the harvest. Optimization for continuous domain, many scholars to study at home and abroad, and it is including real-coded ant colony algorithm and binary encoding of ant colony algorithm. Ant colony algorithm itself is a massive accumulation of positive feedback mechanism optimization algorithm, pheromone pheromone-based and easy to fall into local optimum, resulting in basic ant colony algorithm in continuous domain effect is not obvious.

SOS is an operating system based on event driven operation, the main function of SOS is to support loadable module, a complete system constructed by some small module, construct may occur in the operation of such. In order to support module interface dynamics in, SOS is also committed to support dynamic memory management. BTnut cooperative multi-threading and plain based on C code, a package of development tools and guidelines. At present there is a lot of research into the middleware in the design of middleware for wireless sensor networks [15]. Generally speaking, research direction can be divided into distributed database, mobile agent and an event based model.

Ant colony algorithm is introduced to the discovery of classification rules, the use of the principle of ant colony foraging in the database search, choose randomly generated by a set of rules to optimize coverage until the database can be the set of rules, thus digging out implicit in the database in the rules, to establish the optimal classification model. Ant colony algorithm to search for the initial conditions for the discovery of a set of rules is empty, and the training set contains all the training samples. The ants search time to complete the rule generation, rule pruning pheromone update three tasks. Search to generate a rule and this rule is added to the set of rules while training samples covered by that rule is removed from the training set. If not covered by the number of training samples is greater than the user-defined threshold value, i.e., the maximum is not covered by the number of samples, the above process is repeatedly performed, the final algorithm will be to obtain a set of optimal set of classification rules, as is shown by equation6 [16].

$$\frac{\partial u}{\partial x_1} = u(x_1 + 1, x_2) - u(x_1, x_2) \tag{6}$$

Wireless sensor networks increasingly domestic and international attention as a new information collection technology. This article briefly describes the architecture of the wireless sensor network transport protocol and supporting technologies. Due to the special nature of wireless sensor networks, its development faces many unique issues.

Directed Diffusion is designed for wireless sensor network routing algorithm, which can be viewed as a milepost planar routing protocol of wireless sensor network, then a lot of routing algorithm design is based on the extended to. Its basic idea is a set of nodes generate data (Interest) attribute values are described, the Sink data query request is called "interest" to describe, for example, the region of interest, the name of the event, the acquisition time interval.

Improved ant colony algorithm:

Input: To determine the number of clusters (that is, tracking the number of goals) and predicted by Kaman filter is initialized with the value of each cluster center;

Output: Determine the parameters of the algorithm: the bell-shaped membership function variance, the maximum number of iterations to determine the adjacent iterations membership tends consistent threshold;

1) If Tree contains a single path P;

2) Then for the path P in the junction point of each combination (referred to as β) DO

3) FP[++ length] = item (i);

4) output { item (i) } and its support count [i]/ n;

5) build S T (i) based on Unid_FP-tree;

6) In each layer, the corresponding pixels of the first sub-images obtained from the respective source images respectively corresponding fusion (to the first layer, the first sub-image, for example, the pixels in the pixel until the pixels in the corresponding to fusion, thereby obtaining on the layer corresponding to the pixels at the sub-image details of a band-pass fusion); same processing is performed on the final low-pass filtered image.

7) The β conditions mode structure-based conditional pattern tree Tree and then tectonic β; ∅ ≠ β(8) if Tree then call FP-Max (Tree β β MFI);

8) Else $\{\tau_{ij}(t) = \tau_{ij}(t) + \delta(\tau \max(t) - \tau_{ij}(t))$

9) To generate candidate maximal frequent patterns $C_{max} = \beta$.

In order to verify the performance of the algorithm, we establish a wireless sensor network simulation environment in Matlab. Sensing nodes 40, the communication radius of 60 randomly distributed in a square area of the range of 560 × 850, the sink node coordinates (90, 10). Each experiment is performed 400 times, the experimental results were averaged. This experiment take $\alpha = 2.4$, $\beta = 4$, $\gamma = 5$, $\rho = 2.2$, $Q = 6000$, each node pheromone initialization 7000, each node energy of 300, we assume that each through one node, energy consumption 0.005 % of the original, since the energy consumption is relatively small changes in the experiment of $\gamma = 2$, in the practical application, the value of γ can be based on energy consumption to make appropriate adjustments, or substituting of 70 times, each time released 30 ants, as is shown by Fig. 4.

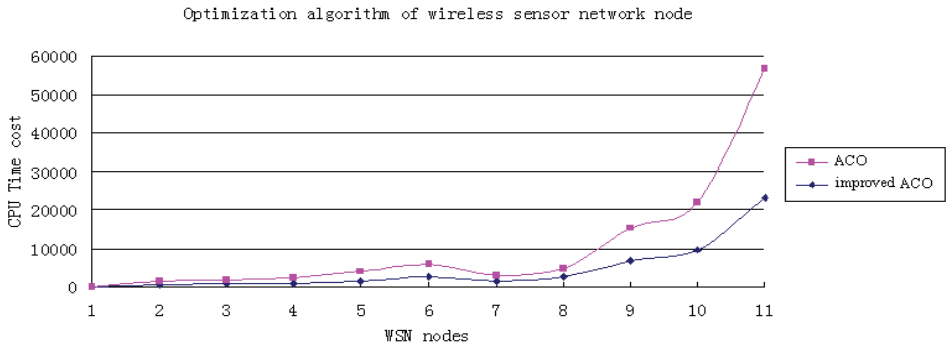

Fig. 4. The optimization algorithm of wireless sensor network node based on improved ant colony.

The paper presents the optimization algorithm of wireless sensor network node based on improved ant colony. To solve this problem, explore other evolutionary algorithm is introduced to solve the pheromone positive feedback accumulated problems, such as the mutation operator of genetic algorithm is introduced into the ant colony algorithm. This hybrid algorithm to some extent alleviated the ant colony algorithm into a local optimum problem, but there is still scope for further research. Draw on the ideas of the hybrid algorithm. This paper introduces the distributed estimation algorithm improvements. Estimation of distribution algorithm is an evolutionary algorithm based on probability model, which itself is based on the statistical characteristics of the ant colony algorithm have great similarity.

5. Conclusions

Wireless sensor network is a large number of tiny sensor nodes with communication and computing power, artificial emplaced, airdrops, artillery projection methods deployed in predetermined monitoring area, constitutes "smart" autonomous monitoring network system. The paper presents the optimization algorithm of wireless sensor network node based on improved ant colony. The ant colony algorithm also known ant algorithm, based on the bionics findings, ants have the ability to find the shortest path between a dangerous thing, and the food source, the principle is that the ants leave a pheromone called path through volatile secretions.

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Using Improved BP Neural Network and Concept Lattice to Construction Smart Sensors System

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Abstract: Smart sensor is information detection, information processing, information memory, logical thinking and judging function of sensor. It not only has the various functions of the traditional sensor, but also has the data processing, fault diagnosis, non linear processing, self correction and man-machine communication. BP (Back Propagation) neural network is a kind of error back propagation training algorithm for the multilayer feed forward network. As the central data structure in formal concept analysis, concept lattice can sign formal description the concept from the denotation and connotation to realize understanding semantic information by computer. This paper uses the implicit structure of three layers of BP neural network to build intelligent sensor network model is improved to complete the prediction of sensor information. The paper presents using improved BP neural network and concept lattice model to construction of smart sensors system. Experimental comparison shows that the method proposed in this paper is very effective to forecast information smart sensing.

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Keywords: Smart sensors, Back Propagation neural network, Concept lattice.

1. Introduction

Integrated smart sensor integrated circuit technology and micro-mechanical technology and sensing element powerful electronic integrated circuits on a chip (or secondary integrated in the same housing), usually with a signal extraction, signal processing, logic, two-way communications and other functions. Sensor and classic, making the smart sensor integrated with small size, low cost, low power consumption, high speed, high reliability, and high precision and powerful advantages.

Smart sensors, intelligent features such as: digital signal output, information storage and memory, logical judgment, decision-making, self-test, self-calibration, self-compensating microprocessor based. Microprocessor-based sensors have been developed from simple digital information processing to the network communication functions, neural networks, fuzzy theory, genetic theory, and the theory of wavelet transform, multi-sensor information fusion and other new technologies of the new theory of gradual improvement of modern smart sensors.

The BP algorithm because of its simplicity, small amount of calculation, the parallel advantages, is now one of the largest and most sophisticated training algorithm using neural network training [1]. The essence of the algorithm is the minimum solution error function, because it uses the steepest descent method in nonlinear programming, modifying the weights according to the direction of the negative gradient of error function.

BP neural network is essentially implements a mapping function from input to output, the

mathematical theory of three layer neural network can approximate any nonlinear continuous function with arbitrary precision. This makes it particularly suitable for solving complex problems of mechanism, namely the BP neural network has strong nonlinear mapping ability.

Concept lattice visualization provides an intuitive method of analysis and observation of the internal relations of the knowledge unit concept lattice layout, however, is quite difficult. In a graphical representation of the conceptual grid, there are two main layout means, one is a manual layout, and another is the use of the automatic layout of the computer. The manual layout of the advantage of being able to modify the position of each node according to the people's aesthetic, resulting in a high-quality concept lattice graphics.

The sensor is a detection device, can feel the measured information, and can be detected by the sense of information, according to the law must transform into the output information required or other signal, in order to satisfy the information transmission, processing, storage, display, record and control requirements. It is the realization of automatic detection and automatic control of the primary link. Compared with the sensor intelligent sensor and the advantage of the traditional, intelligent sensor is popular, mainly because of its multiple functions. Intelligent complex sensor corresponds to the microcomputer and sensor, which is composed of main sensor, auxiliary sensor and microcomputer hardware. The paper presents using improved BP neural network and concept lattice model to construction of smart sensors system.

2. Application of BP Neural Network in development Smart Sensors

Sensor based on semiconductor technique will change this situation, the provision of cheap price, convenient use of the sensor, especially intelligent sensor itself more convenient system and set of yang. And the completion of system development in computer aided design help, realize more complex functions. Compared with the traditional design and it is different application systems without using different sensor. Through the software design to change the sensor function in sensor based on a single, in order to meet the needs of different customers. In this way, intelligent sensor with low cost, high volume can achieve system design, maintenance and function expansion of fast, low cost.

Smart sensors from the compensation and calculation functions for the sensor temperature drift and non-linear compensation has opened up a new road, so that the relaxation of the sensor processing requirements of precision, as long as they can to ensure the repeatability, the use of microprocessor test signal can be calculated by the software to obtain more accurate results. It may also be processed statistically, re-calibration of a sensor, so that it re-effectively.

BP (Back Propagation) neural network is the most in-depth study of artificial neural network, used as a model for a wide range of it, Z is the input, the output vector of the network, each neuron is represented by a node, the network is composed of input layer, hidden layer and output layer nodes, hidden layer can be a layer, can also be a multilayer (the picture is a single hidden layer), anterior layer after layer between the nodes through the weighted coefficient phase connection [2]. BP neural network learning, the input signals from the input layer to the hidden layer to the output layer (forward propagation), if the output layer to obtain the desired output, the learning algorithm: otherwise, go to the end of back propagation.

The key problem of multilayer feed forward neural network is to solve the learning algorithm. Learning algorithm is the main feature of neural network, is also the main topic. Learning concepts from the biological model organism, it is effective in the complex environment and self regulation. Neural network to change its output value, but you can change the transfer function; can only change the input value, and the only way to change the input function to modify the weighting coefficients in the input. So the neural network learning is the process to modify the weight coefficient, the output reaches a desired value, the end of learning. Back propagation neural network structure is as shown in the Fig. 1.

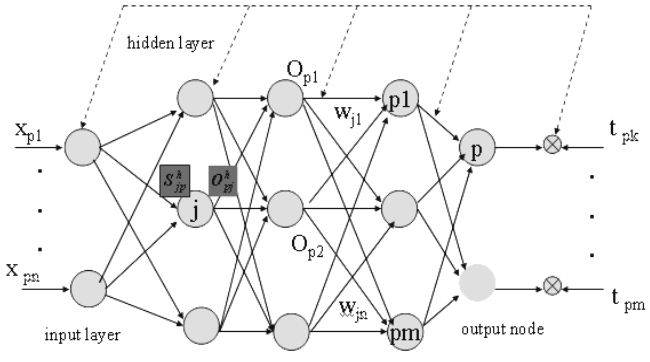

Fig. 1. BP neural network structure diagram.

In addition, the network input output node, a node of hidden layer of a layer or layers; the adjacent layer neuron completely interconnected, nodes of the same layer are independent of each other. The input signal to the input layer, the hidden layer nodes are passed, and finally to the output node: the output nodes in every layer are the input of the next layer.

Implementation of gradient descent algorithm: add mode and batch mode. On the increase mode, network input each submits a, gradient calculation time and update the weights. In the batch mode, when all inputs are submitted after the network was updated. The next two sections will discuss the increase mode and batch mode. Increase model training method (increase model training method (ADAPT) the adapt function is used to increase the training mode of network, it from the training set to accept the network object, network input and the target input, output and error return trained network object, obtained by the weight and bias.

Dubbed Smart Sensors Development and Cooperation in the field bus control system corresponding image processing software, graphic imaging and the promotion of the exhibition situation developed rapidly [3]. Automation instrument constitutes a multi-dimensional image sensor, the smart sensor meter manufacturers original set of production equipment basically reached its most advanced form. Constant plus a piece with a digital bus interface microprocessor, mixed to achieve the flapper assembled and equipped to be able to communicate, control, intelligent sensor system on a chip, self-correcting, self-compensating, self-diagnosis, etc. intelligent software, there are many difficult problems. This is one of the most economical, the system integration aspects (such as sensitive unit, the letter fastest way to build smart sensors, as is shown by equation 1.

$$\begin{cases} \dfrac{dp_0(t)}{dt} = -\lambda p_0(t) + \mu p_1(t), \\ \dfrac{dp_1(t)}{dt} = -\mu p_1(t) + \lambda p_0(t). \end{cases} \tag{1}$$

The rapid development of intelligent sensor network is closely related to the Internet of things, is closely linked with the technical progress of intelligent sensor. Here a concept referred to is the RFID technology. At networking, sensor is the role of information perception system consists of information collection layer and network layer, is the entire network chain total demand of the largest and most basic aspect; at the same time, is also the Internet of things technology support, perceived temperature sensor object information, RFID gives the electronic code.

This paper will be based on the design method of intelligent sensor intelligent sensor characteristics are given, and compared with the general sensor system structure and design method, analyze its advantages. On this basis, it points out that the intelligent sensor network will be the inevitable direction of future sensor development, internal structure models are intelligent sensor network and measurement control system based on field bus network, Internet network model.

Measurement and control system based on distributed intelligent sensor is composed of a certain network will each control node, sensor nodes and a central control unit together constitute. The sensor node is used to realize the parameter measurement and transmit the data to other nodes in the network; control node is according to the need to obtain the required data from the network and the data to formulate the corresponding control method and execution control output according to the. In the system, each sensor node and control node number can be more or less, depending on the requirements of. Network selection can be sensor bus, field bus, can also be the enterprise internal Ethernet, can also be directly is Internet.

Smart Sensor IP core design and SOC build smart sensors related to data acquisition, signal processing (programmable amplification, linearization, signal filtering, signal compensation, artificial neural networks, genetic theory, multi-sensor fusion, fuzzy theory, etc.), data communications, people machine interface and task scheduling [4]. IP core design and SOC build, in order to simplify and reduce the complexity, we use FPGA IP core and based on the ARM7TDMI-SCPU IP core two SOC design approach, which FPGA IP core complete data acquisition and signal processing module, based on the ARM7 IP core to complete the data communication, human-machine interface and task scheduling work.

Implementation of BP neural network is applied in the safety evaluation system of (1) to determine the topology of the network, including the middle hidden layer number, input layer, output layer and hidden layer nodes. (2) the index system is determined by evaluation system includes a feature by using neural network and state parameters for the safety evaluation, we must first determine the internal structure and external evaluation system environment, determined to correctly reflect the main features are evaluation of safety state of object parameters (the number of input nodes, each node actual meaning and expression and so on), and these parameters system (output nodes, each node actual meaning and the way of expression).

$$\frac{dY}{dX} = \frac{a_1 X + b_1 Y}{a_2 X + b_2 Y} = g\left(\frac{Y}{X}\right) \tag{2}$$

Artificial neural network has the unique nonlinear adaptive information processing capability, to overcome the traditional methods of artificial intelligence for the intuition, such as pattern, speech recognition, non structured information processing, so it is successfully applied in the field of neural expert system, pattern recognition, intelligent control,

combinatorial optimization, prediction. Combining artificial neural network with other traditional methods, will promote the development of artificial intelligence and information processing technology development.

Transmission of biological neurons through the synaptic signal is a complex electrochemical process; in the artificial neural network is the simplified simulation into a group of digital signals by learning the rules of the continuous change in the renewal process, this group of numbers stored in the connection weights between neurons. The input layer of the network simulation is the nervous system of sensory neurons; it receives the input signal samples.

Neurons in the network are arranged hierarchically, each neuron only connected with a layer of neurons. Neurons are arranged hierarchically, were composed of input layer, middle layer (also known as the hidden layer can be composed of a plurality of layers and the output layer). Each layer of neurons only accepts from neurons before a layer of input, behind a layer on the front of the layer does not signal feedback [5]. The input mode after sequential communication at all levels; finally get the output in the output layer. Sensor network and BP network are feed forward networks.

BP (Back Propagation) neural network is a neural network learning algorithm. It consists of input layer, the middle layer; the output layer of BP neural network, the intermediate layer can be extended to multilayer. Between adjacent layers of neurons are fully connected, and between each layer of each neuron connectionless network, according to the teaching mode of learning, when learning a provided to the network, each neuron network input response to generate connection weights (Weight). Then according to the output and the actual output error is reduced to the direction from the output layer, the middle layer layer-by-layer correction of each connection weight, back to the input layer. This process is repeated alternately, minimal global error until the network tends to the given value, to complete the learning process, as is shown by equation 3.

$$\frac{dy}{dx}=P(x)y+Q(x)y^n \tag{3}$$

Intelligent sensor design ideas and working principle: the sensing element of physical quantities will be tested into a corresponding electrical signal, if a digital signal directly through the photoelectric coupler to the micro processing parts processing, if the 4-20mA analog signal through the first signal conditioning part, filtering, amplification, and then sent to the micro processing part. The microprocessor is the core of intelligent sensor, which is responsible for sensor measurement data analysis, A/D conversion, storage etc. And then through the communication part of the conversion of electrical signals into 485 signals after treatment, and it is using the RS-485 two buses, exchange of information with the master station.

Man-machine interface mainly design of keyboard and LCD/LED/CRT display interface. The use of ARM7 powerful function of GPIO is not difficult to achieve [6]. Task scheduling of IP include data acquisition, signal processing, data scheduling and communication scheduling and man-machine interface operation etc. We use the open source embedded operating system μ C/OS- □ 2.52 edition as the basis, it can be transplanted into the LPC2106 ARM microprocessor. The research and development is various application software based on μ C/OS-embedded operating system, and it is to complete the work of all kinds of task scheduling and configuration need intelligent sensor.

$$u=\int_0^x xdx+\int_1^y(\frac{x^2}{2}+\frac{1}{y})dy=\frac{x^2}{2}y+\ln y \tag{4}$$

Smart sensor hardware circuit is part of the microprocessor: power supply section; communication section; signal conditioning components. Intelligent sensor can measure parameters of hybrid sensors, so as to further broaden its detection and application field, and the microprocessor intervention makes intelligent sensor capable of real-time processing is more convenient to a variety of signal. In addition, the flexible configuration can not only make the same type of sensor to achieve the best performance, also can make them suitable for different working environment.

The BP neural network has the ability of associative memory and input information to external stimuli [7]. This is because it uses distributed parallel information processing, extraction of information must be associative way, can be related to mobilize all the neurons. BP neural network adaptive training through the previously stored information and learning mechanism, can recover the complete information of the original information and noise interference from incomplete. This ability allows the restoration, language processing, pattern recognition in image has important applications, as is shown by Fig. 2.

The rapid development of the Matlab software provides a convenient means to realize the simulation of neural network theory. The Matlab neural network toolbox broadened the application space of neural network. Neural network toolbox will many of the original manual calculation task to computer, on one hand, improve work efficiency, on the other hand, but also improves the accuracy and precision of calculation, reduce engineering burden.

Neural network toolbox is one of many tools developed in the MATLAB environment. It is based on the theory of artificial neural network; using MATLAB programming language constructs the framework of many typical neural networks and the associated function [8]. The toolbox functions mainly

into two parts. A part of the function for a certain type of neural network, such as the creation of function, BP neural network perception training function etc. While the other part function is general, and it can be used for almost all types of neural network, such as neural network simulation function, initialization function and the training function.

$$\mu(x,y) = e^{\int \phi(y)dy} = e^{\int -\frac{2}{y}dy} = y^{-2} \qquad (5)$$

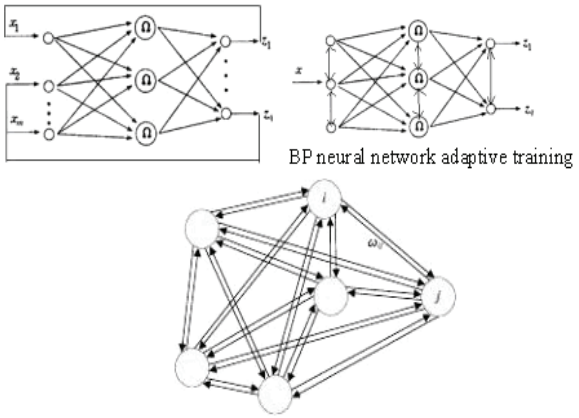

Fig. 2. The improved BP neural network algorithm running structure.

This paper first summarizes the neural network the purpose and significance of the study, introduces the research background and the current status of neural network, analyses the existing problems in the study of neural network, then describes the working principle of the BP neural network algorithm and BP neural network is presented, the limitations of BP network. Based on the BP neural network based, based on the study of BP neural network in function approximation and estimation of sample size of two examples of application. And the analysis conclusion, namely the signal frequency is high, the hidden layer node number, number of hidden units more stronger ability, approximation.

The realization of intelligent sensor system is based on sensor technology, computer technology, signal processing, network control technology, first of all, system. From the perspective of general computer, computer system to process the data is digital signal, and is directly through the external input device, the signal itself will be subject to external equipment limited. But the face for sensor system is analog signal associated with the external environment, some physical quantity related signal with the outside world. This means that signal the presence and the emergence of signals is subject to environmental constraints.

At present, the intelligent sensor system itself, although all are digital, but the communication agreement still needs the help of 40 ~ 60mA standard analog signal to achieve. Is now actively studying the introduction of general field bus digital signal transmission standards related to the structure of some international standardization research machine; however, still mostly uses the remote bus addressing sensor in this transitional phase (HART) protocol. This is a suitable for intelligent sensor communication protocol, is fully compatible with the systems currently used in 4 ~ 20 mA analog signal, analog signal and digital signal can be done at the same time communication, so that different manufacturer's products is universal.

$$y_k = f(s_k) = \frac{1}{1+e^{-(r_k - y_k)}}, \qquad (6)$$

The BP algorithm is to estimate the error directly leading layer output layer by using error output layer, then the error estimation error of a layer. So, we obtain the error estimates for all the other layers. Thus the output error end show transported along with the input signal in the opposite direction step by step to the network to the input of the transfer process. Therefore, people and this algorithm is called back propagation algorithm.

BP model of the I/O problem for a set of samples into a nonlinear optimization problem, it is the most common optimization of gradient descent method. If the neural network as the input to the output of the mapping, this mapping is a highly nonlinear mapping. The design of a neural network expert system focuses on the model of the structure and learning algorithm of choice [9]. Generally speaking, the structure is determined according to the research field and the problems to be solved. Through the analysis of a large amount of historical data and the data of the anterior neural network theory development level, the establishment of a suitable model, and the corresponding learning algorithm based on the model, the network learning process, constantly adjust the network parameters, and output the results meet the requirements.

Multifunctional sensor system consists of a number of sensitive elements of different composition, can be used to measure various parameters with. For example, can be a temperature sensor and a humidity detector configuration together (that is, the thermal sensitive element and a humidity sensor are respectively arranged bearing in the same sensor body) into a sensor, a new way, the new sensor can simultaneously measure temperature and humidity.

3. Concept Lattice model to Building Smart Sensors System

The intelligent degree is proportional to the level of development of the software; I believe that in the near future, computer platform will be widely used by virtual sensors based on software development. Software development tools including design,

management and communication management aspects. At present this kind of tool has begun to appear, Lab view, ActiveX and other software tools can be completed. The function of the software is proportional to the level of development of the software, is used to realize the sensor model, calibration parameters is established, the best calibration model selection.

Data acquisition and data processing of intelligent sensor: it must be pre-processed output signal before intelligent sensors. Because of many kinds of detected signal, a signal output from the analog, digital, the switch quantity, the vast majority of the output of the sensor output signal of the sensor cannot be used directly as input to A/D conversion, must pass various pretreatment circuit of the sensor output signal into a unified voltage signal or periodic signals [10]. Data acquisition: the sensor signal and processing as required by A/D converter bit analog signal, analog to digital the elegant depends on analog converter (A/D) by sampling, quantizing and coding the input signal is transformed into digital signal.

Formal context (formal context) is usually defined as a group of three: K:= (G, M, R), where G and M are set, and R is two Yuan, the relationship between G and M. R G M, where G and M elements are known as the objects and attributes (i.e., gRm (g, m) R) read: object G has an attribute M. If the cardinal is K and it is called an order form. In the premise of not confused, formal context is referred to as the background (context).

$$\begin{aligned} A' &:= \{m \in M \,|\, \forall g \in A \cdot gRm\} \\ B' &:= \{g \in G \,|\, \forall m \in B \cdot gRm\} \end{aligned} \tag{7}$$

In order to describe the model and process model based on concept lattice based knowledge internally consistent, must solve some problems, including the scale control, simplified, background knowledge, knowledge management (metadata) isomorphic judgment, generation of new knowledge and knowledge expression, etc. But in the final analysis, the learning process is the process of generating new knowledge, then, to establish knowledge similarity model of the target is effective to obtain the concept lattice.

In order to batch build concept lattice, mainly to complete two tasks: one is to generate grid nodes all (that is, formal concept set); two is to establish a direct precursor of the lattice nodes / direct successor relationship [11]. So according to these two tasks to complete the order is different, we can use two different ways to complete: one is the first generation of the concept set, and then find out the direct precursor of / between these concepts and the direct successor relationship; another is created each time a small amount of the concept, and the concept of link to node in the collection. The first way is called the task segmentation model.

The basic principle of the design of the intelligent liquid level sensor is to measure the water pressure with a pressure sensor, the output pressure transmitter for analog voltage signal 0-9 V (or analog current signal 4-20 mA), to collect the sensor output voltage and voltage analog signals into digital level signal 0 using ADC0804, then the data will be collected into the SCM minimum system for processing and then converted by RS232 level 1. Sent to the PC machine, machine using VB MS.COM controls write a display interface, PC in the VB program will be converted into digital quantity by the centimeter height and displayed, as is shown by equation 8.

$$\overline{I_k} = \frac{\sum\limits_{x \in \Omega_i} [I_k(X')]}{L \cdot L}, \overline{I_{k-1}} = \frac{\sum\limits_{x \in \Omega_i} [I_{k-1}(X)]}{L \cdot L} \tag{8}$$

Smart Sensor and intelligent actuator and intelligent transmitter-micro sensor (or micro actuator micro transmitter) and it all or part of the processor, it is processing integrated circuit devices on a chip (such as the single chip micro acceleration meter solutions). So the intelligent sensor bionic ability, such as fuzzy logic, proactively identifies environment, and it is automatic adjustment and compensation ability to adapt to the environment, self diagnosis, and self maintenance [12]. Obviously, because of the scale of production and reduce the cost of production requirements, design thought of intelligent sensor, the selection of materials and the production process must as far as possible standard silicon planar process of consistent and integrated circuit chip. In normal process, or processes, and it is process after the completion of some special needs of process, but it should not be too much.

With the development of distributed database technology and computer network technology, distributed data storage and parallel processing has become a reality, how to rationally organize distributed data storage and parallel processing is the urgent task, whether in theory or in technology, there are many problems need to be studied. Because the concept lattice has good mathematical properties and is suitable for batch processing characteristics, therefore, distributed concept lattice is the ideal tool for solving the above problems.

Concept lattice is two L lattice node C1= (O1, D1) and C2= (O2, D2), if D1=D2, we say that C1 is equal to C2; if D1 D2, we say that C1 is greater than C2, or C2 is less than C1, C1= lattice node concept lattice on the L (O1, D1), C2= (O2, D2) and C3= (O3, D3), if O3=O1 O2 D3=D1 D2, ¬, we say that C1+C2 is equal to C3.

Concept lattice is the least upper bound of L (O, f (O)) began. Then generate all of its child nodes and connected to its parent node. Then each node is recursively repeated the process [13]. The key of the algorithm is to generate a child node. The algorithm is, let (X, Y) is the current node, D is the set of attributes. Find all P D-Y, these P can remain

completely on the nature of the X, which is the largest expansion. For any P, P Y is the connotation of the child nodes of the current node. The algorithm is very simple, intuitive, and easy parallelization. The disadvantage is that repeated generates many redundant nodes (Fig. 3).

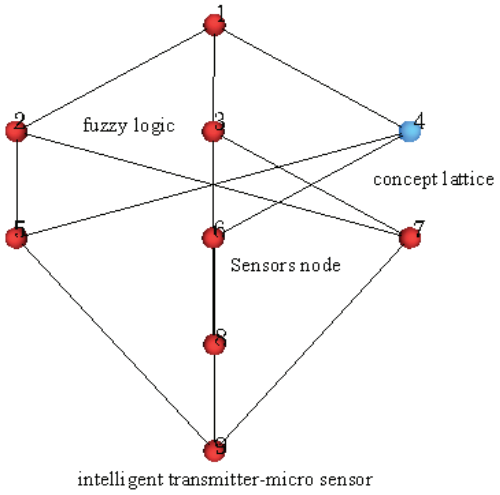

Fig. 3. Building Smart Sensors based on Concept Lattice.

Smart sensor signal processing as required with A/D converter point analog signal, depending on the analog converter (MD) simulation and digital input signal into digital signal, these changes are obtained by sampling, quantizing and coding. Data is processing. A/D data output signal converter to convert the pressure sensor, according to the needs of processing the digital signal, such as scaling, nonlinear compensation, temperature compensation and digital filtering software processing, otherwise it is not directly input supply Yong Cheng sequence using microprocessor, as is shown by equation9 [14].

$$\frac{dp_0(t)}{dt} = -\lambda p_0(t) + u\left(1 - p_0(t)\right) \qquad (9)$$

Smart sensor is information detection, information processing, information memory, logical thinking and judging function of sensor. It not only has the various functions of the traditional sensor, but also with the data processing, fault diagnosis, non linear processing, self correction, self adjustment function and man-machine communication etc. It is a product of microelectronics, micro electronic computer technology and testing technology with the combination of it.

Important characteristics of intelligent sensors, can be used in two ways: the selected function commands issued by the user through the keyboard; the automatic type, data acquisition and processing program has been compiled by the internal function control program coordination, receive or external signal through the IEEE--488 bus, control instruction sent by the intelligent pressure sensor, and then through the self-calibration, tracking, alarm, print output, keyboard, display, A/D conversion circuit and is connected to the 1=I, so as to ensure the orderly operation of intelligent pressure sensor. According to Fig. 2 source program flow diagram design: a calibration of a signal into a digital code a latch output ideal EPROM installed intelligent pressure sensor.

The smart sensor primarily by the information collection circuit, the control unit, keyboard input interface circuit, a display interface circuit and the external communication interface circuit composed of five parts. The high-voltage pulse for driving the ultrasonic probe and it is the probe to convert electrical energy into mechanical energy to generate ultrasonic waves. Ultrasonic pulse signal receiving circuit includes receiving a signal limiter, zoom, compared with single-shot trigger and other sectors, to form a narrow pulse signal.

4. Using Improved BP Neural Network and Concept Lattice Model to Construction of Smart Sensors System

BP network is a multi-layer feed forward neural network, the transfer function of the neurons in the S-type function, so the output of a continuous quantity between 0-1, it can achieve arbitrary nonlinear mapping from input to output. Weights to adjust the back-propagation learning algorithm often referred to as the BP network. Using the input set of output samples to training, and it is in order to enable the network to achieve a given input-output mapping relationship [15]. BP neural network trained for other input can also be given the appropriate output. BP neural network is widely used in function approximation, pattern recognition, data compression, which is a core part of the feed-forward network. BP network is the wider application of the network model, but there are some limitations, such as slow convergence rate, you need to construct a training sample set.

Intelligent temperature sensor based on AT89S52 single chip microcomputer as control core, by a digital temperature sensor DS18B20 measurement of temperature controlled. Peripheral circuit including communication and digital PC display circuit, and it is multi-channel temperature acquisition, output control potential.

Initializes the system's AD converter, and it is the voltage reference system because the input frequency ups to 1.1 MHz, so the system main frequency divider input 8. ADC has no input cache MSP430F2013, cache in cache mode selection in the closed. System for measuring the physical quantities of continuous change, and it is to set up the system for continuous conversion mode [16]. When measuring open interrupt, when measuring results,

generate an interrupt, interrupt flag bit, the measurement results are stored in the conversion result register, when the measured results from the readout register results after conversion, interrupt flag bit is automatically reset. So read the conversion result only when the query interrupts flag, as is shown by equation 10.

$$MI_{FA}(f,a)=\sum_{f,a}P_{FA}(f,a)\ln\frac{P_{FA}(f,a)}{P_F(f)P_A(a)} \quad (10)$$

Concept formation method based on concept lattice, mainly proposed to extract implication rules from concept lattice algorithm, contains problem and use the results of function dependency in relational databases to handle rules, but the implication rules is a deterministic rules, not able to describe probabilistic rules and noise resistance ability, and it is in order to improve the robustness of rule discovery.

Now let us consider L (K2) has a plurality of concept lattice, given L (K1) and L (K2) is the two extention-independent same field lattices, solution of L (K1) L (K2). This algorithm is based on L (K2) was used for constructing concept lattice incremental Godin algorithm to each grid node, insert it into the L (K1) of concept lattice, can get the final L (K1) L (K2).

BP (Back Propagation) neural network is a kind of error back propagation training algorithm for the multilayer feed forward network, is one of the most widely used at present neural network model. BP network can input and output schema mapping relationship between learning and mass storage, without prior to reveal the mathematical equations describing the mapping relation. Its learning rule is to use the steepest descent method, through the back-propagation network to continuously adjust the weights and thresholds of the network, the error square and minimum. Multilayer feed forward BP network is a neural network form the most applied. Multilayer feed forward BP network essentially implements a mapping function from input to output, and the mathematical theory has proved that it has any complex nonlinear mapping function.

BP neural network is a typical global approximation of neural network; the radial basis function (Radial Basis Function, RBF) neural network is a typical artificial neural network. RBF neural network is a neural network proposed by J. Moody and C. Darken in the late nineteen eighties. The basic idea of RBF neural network is used as the "base" of neurons in the hidden layer radial basis function, constitute the hidden layer, hidden layer to the input vector transform, the low dimensional model of the input data is transformed to high dimension space, the low dimensional space of the linear non-separable problem in high dimensional space linear reparability [17]. Basic RBF neural network is three layers feed forward networks with a single hidden layer. RBF hidden layer generally have higher dimension, as input to the hidden layer output between nonlinear transformations; linear transformation, providing a response to activation patterns in the input layer.

Three layer neural networks have strong learning ability. The learning algorithm is three layer BP neural networks. Input layer node number three layers of improved BP neural network for m, the number of nodes in the hidden layer is p, node of output layer is Q, definition of symbols which are as follows.

1) The corresponding number of unit input layer by the sample values; as the activation value, the weight matrix, for the value of the activation of each unit;

2) The input vector of training samples for BP neural network input layer neurons in the input, then the amount of neurons in the hidden layer:

$$b_j=f(a_j)=\frac{1}{1+e^{-(a_j-\theta_j)}}, \quad (11)$$

3) Concept lattice L (K2) grid nodes in ascending order according to the content stored in the queue VEC;

4) Concept lattice is two L lattice node C1= (O1, D1) and C2= (O2, D2), if D1=D2, C1 is equal to C2; if D1 D2, said C1 is greater than C2, less than C1 or C2;

5) According to the output samples of the expected value and the output layer of the poor, to calculate the error signal and is connected on the weights of the output layer neuron threshold related:

$$\delta_k\delta_k=(c_k-y_k)y_k(1-y_k) \quad (12)$$

6) **IF** f*({x*})⊄Intent(inf(L)) **THEN**
IF Extent(inf(L))=Φ **THEN**
Intent(inf(L)):=Intent(inf(L))∪f*({x*})

7) Repeat the calculation (2) - (5) step, until the error is small enough for or zero (at this time).

8) For j←1 to ObjNum－1 Do
For k←j＋1 to ObjNum Do

9) Modified formula weights of neurons in the input layer to the hidden layer neurons of smart sensors.

$$w_{ij}^{(i+1)}=w_{ij}^{(i)}+\alpha\sigma_j I_i \quad (13)$$

10) BP neural network with hidden layer is introduced, and the transfer function for Sigmoid nonlinear function, therefore, its characteristic is it can approximate any continuous function.

Smart sensor parameters may be varied. But from the function module, it mainly includes the data acquisition module, compensation and correction module, data processing module, data communication module, man-machine interface and task management and scheduling module and other

functional units. Thus the intelligent sensor SOC design process based on IP: universal module model firstly to establish intelligent sensor; then the reasonable division of the module function specification, make the interface protocols between modules and standard; then design a series of generic IP kernel; finally the required for IP nuclear build together intelligent sensor system integrity, as is shown by Fig. 3.

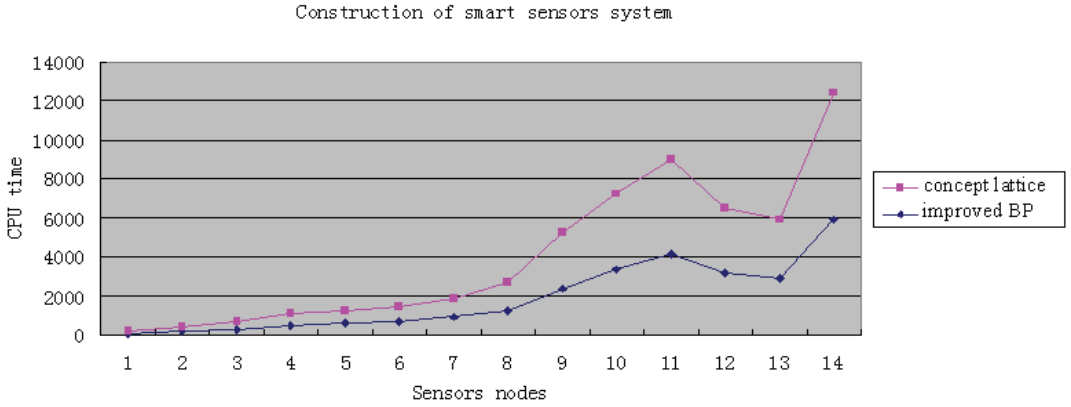

Fig. 4. Comparison results of construction smart sensors system improved BP neural network with concept lattice.

The paper presents using improved BP neural network and concept lattice model to construction of smart sensors system. Smart sensor can not only carry out processing, analysis and control of information, can compensate the measured value and its error, but also can think logically and conclusions, can use to linearism nonlinear signal on the list, with the help of software filter digital signal. In addition, also can realize the nonlinear compensation or compensation in more complex environment by using BP neural network, it is to improve the accuracy of measurement. The improved BP algorithm has the additional momentum gradient descent algorithm, adaptive learning rate gradient descent, conjugate gradient algorithm. Momentum gradient descent with momentum reduces the sensitivity of neural network for details of error curved surface; effectively inhibit the network into a local minimum. The comparison results of construction smart sensors system improved BP neural network with concept lattice are shown in Fig.4.

6. Conclusions

The smart sensor has the diagnosis and self calibration function, and it can be used to detect the working environment. When the work environment nears the limit condition, it will send out an alarm signal, and according to the diagnostic information related to the given input signal analyzer. When the intelligent sensor due to some internal failure and it can use its internal detection link is to find abnormal phenomenon or a faulty component. In this paper, by using BP neural network for forecasting strong sensing. Using the improved additional momentum and adaptive learning rate gradient descent algorithm of BP, determine the neural network model, intelligent sensor prediction. Concept lattice as core data structure in formal concept analysis, symbolic formal description with concepts from the denotation and connotation of the two parties, semantic information to realize the computer can understand.

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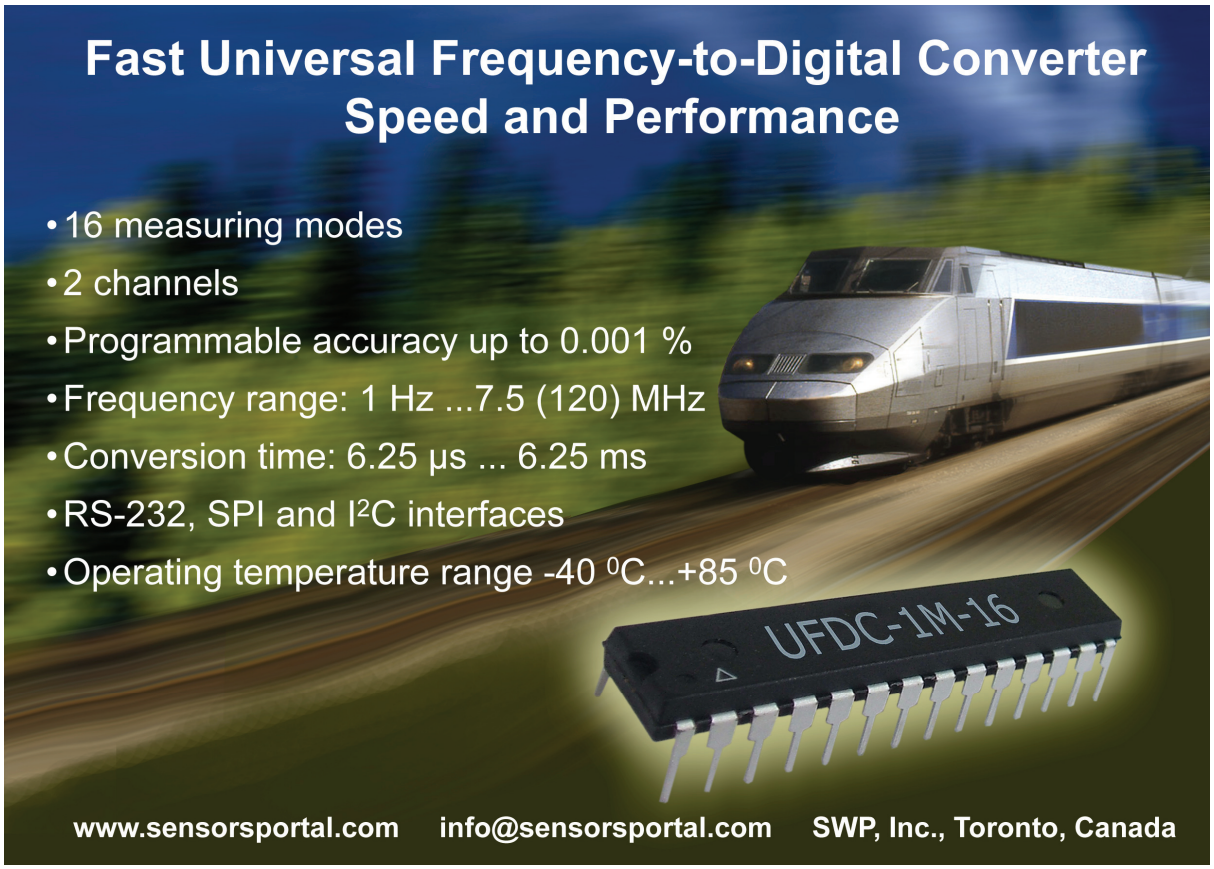

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Connected Target Coverage for Fully Data Gathering in Wireless Sensor Networks

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Abstract: Coverage problem is an important issue in wireless sensor networks, which has a great impact on the performance of wireless sensor networks. Many researches in this area only focus on how to cover all the targets in the network, which ignore considering the problem of how to transmit the sensing data to the sink. In this paper, we propose an energy-effective connected target coverage algorithm named ECTCG to guarantee the sensed data be effectively gathered in mobile wireless sensor networks. We theoretically analyze ECTCG algorithm and conduct extensive simulations to evaluate its performance. Simulation results show that ECTCG outperforms another existing algorithm CWGC by effectively prolonging the network lifetime.

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Keywords: Wireless sensor networks, Target coverage, Data gathering.

1. Introduction

Wireless sensor networks (WSNs) [1-3] have many inexpensive wireless nodes, each capable of collecting, computing, storing, and processing environmental information, and communicating with neighboring nodes [4, 5]. Sensors usually adopt ad hoc networking technology to connect with each other, and transmit the gathered information to one or more than one sink [6, 7].

Target coverage problem is one of the measurements of WSNs QoS (Quality of Service). Given a sensor network, the coverage problem is to determine how well the sensing field is monitored or tracked by sensors, and how to maximize the network lifetime when all the targets are monitored is the most important problem in wireless sensor networks [8-10]. In many practical applications of WSNs, such as forest fires, nuclear leak, etc, we can convert these applications into target coverage problem, so the research of target coverage problem in WSNs has a very important significance.

Recently, the algorithms of target coverage issue in WSNs mainly concern about how to cover all the targets, without consider how to transmit the sensed data to the sink, therefore, in this paper, we will focus on how to effectively transmit the data to the sink based on the connected target coverage problem.

2. Related Work

The target coverage issue is a fundamental problem for wireless sensor networks. There are extensive numbers of studies about the target coverage problem. Reference [11] studied the

connected target coverage problem in wireless sensor networks, and designed a distributed algorithm that did not need the global information of the whole network. Only needed the 3-hops information within neighbors, the algorithm could satisfy the requirement of coverage and connectivity, which was suitable for large-scale intensive sensor network.

Reference [12] proposed a connected target coverage algorithm with multiple sensing ranges of sensors in WSNs. The algorithm firstly constructed a virtual skeleton, and ensured that all nodes either existed in this virtual skeleton or could connect to a node in the virtual skeleton, and then the algorithm considered the goals of how to coverage all the targets in the network. But, it was unnecessary to build a global connected skeleton in many sensor network applications.

The node energy model in [13] took the data receiving energy and data sending energy into consideration, but the algorithm did not pay enough attention to consider the problem of how to transmit the data to the sink node, while only based on other routing mechanism to transmit the sensed data.

On the basis of reference [13], the algorithm in [14] modeled the connected target coverage (CTC) problem as a maximum cover tree (MCT) problem, and also proved that MCT is a NP-complete problem, and then gave an upper bound solution for MCT problem. Then, it developed an approximate algorithm named App_MCT and a greedy algorithm named CWGC. The basic idea of CWGC algorithm is to give each node and edge in the network a weight value, and then greedily choose proper nodes and edges to build a cover tree to execute the work task in each round. Finally, it used the extensive simulations to demonstrate the effectiveness of the App_MCT algorithm and CWGC algorithm.

3. Problem Statement

Before diving into solving the connected target coverage problem, firstly, we will give some fundamental definitions and notations in this section, which will be used throughout this paper, and then model this connected target coverage problem.

3.1. Network Model

Assume that s1, s2, … , sN are N sensors in a WSN and R is the sink. All sensors are randomly deployed to continuously cover M targets p1, p2, … , pM. S and P are two different sets of nodes, where S={ s1, s2,…, sN } (|S|=N) and P={ p1, p2,…, pM } (|P|=M). We use a directed graph G=(V, E) to represent the relationship among S(the set of sensor nodes), P (the set of target nodes) and R (the sink node), where V=S∪P∪R, E consists of transmission edges and coverage edges. If |si—sj|<rc , then (si, sj)∈E; else if |si—pj|<rs, then (si, pj)∈E.

Within the operation time, we use $S_s(\tau)$ to represent the set of nodes participating in the activity of monitoring objectives in the operation time τ , we use $S_r(\tau)$ to represent the set of nodes participating in the activity of transmission data in the operation time τ, and we use $S_a(\tau)$ to represent the set of active nodes in the operation time τ. Obviously, $S_a(\tau) = S_s(\tau) \cup S_r(\tau), S_a(\tau) \subseteq S$.

We have the following three assumptions: (1) each node will not change its Active/Sleep status within the operation time τ; (2) the sampling frequency for each node is the same; (3) the data cannot be aggregated in the network, when a node receives $N*B(\tau)$ data from all of its neighbors (N represents the number of neighbors), the data will be transmitted by the node directly.

3.2. Definitions

Definition 1: Connected Target Coverage (CTC) problem.

Given M targets with known location information and a WSN with N sensors randomly deployed in the area. Schedule the sensors' Active/Sleep activities so that all targets are continuously observed and the network can achieve a maximum network lifetime. The network must satisfy that: (1) each target must be continuously covered by at least one node; (2) the data of the targets can be transmitted to sink through active nodes' one-hop or multi-hops relay.

In order to solve the connected target coverage (CTC) problem, reference [14] modeled this problem as a maximum cover tree (MCT) problem.

Definition 2: Maximum Cover Tree (MCT) problem [14].

Given a directed graph G= (V, E), the initial energy of each node is E_0(s), where V= $S \cup T \cup R$. E consists of the communication edges and monitoring edges. There are a serial trees denoted by $T(\tau_1)$, $T(\tau_2)$,..., $T(\tau_d)$, and the operational time of these trees are denoted by τ_1, τ_2 …, τ_d, the network can achieve a maximum lifetime denoted as L. We can express this problem by the following mathematical representation:

$$\text{Maximize L} = \sum_{i=1}^{d} \tau_i$$

$$\text{Subject to: } \sum_{i=1}^{d} E(s, T(\tau_i)) \le E_0(s), \forall s \in S,$$

where we use $T(\tau) = (S_s(\tau) \cup S_r(\tau), E(\tau))$ to represent the coverage trees constructed in operational time τ . The coverage tree has the following characteristics:

Sink is the root of the tree.

Each leave of the tree is a sampling node.

Each target can connect with at least one sampling node.

Each node in the coverage tree $T(\tau)$ uses the same energy consumption model which is used in [14].

$$E(s,T(\tau))=\begin{cases} e_s B(\tau)+e_{trans}P(\tau), s\in S_s(\tau)\ and\ s\notin S_r(\tau) \\ (e_{trans}+e_r)P(\tau)D(s,T(\tau)), s\notin S_s(\tau)\ and\ s\in S_r(\tau) \\ e_s Bit(\tau)+e_{trans}B(\tau)+ \\ (e_{trans}+e_r)P(\tau)D(s,T(\tau)),\ s\in S_s(\tau)\cap S_r(\tau) \\ 0,\ s\notin S_s(\tau)\ \ and\ \ s\notin S_r(\tau) \end{cases}$$

where $T(\tau)$ represents the trees constructed in operational time τ, $P(\tau)$ is the number of packets collected in operational time τ, e_s and e_r represent the energy dissipated by sensing and transmission of one bit data, respectively. e_{trans} represents the sending energy of a node, $e_{ij}^t = e_t + b \cdot d_{ij}^{\alpha}$ (where s_i is the sender, s_j is the receiver, d_{ij} is the distance between s_i and s_j, α is the attenuation factor of path, e_t and b are constant), so, $D(s,T(\tau))$ represents the number of offspring nodes of s in tree $T(\tau)$.

3.3. Upper Bound for MCT Problem

We will further model MCT problem into a LP problem in this section, and then prove that LP problem is exactly an upper bound for MCT problem. We will find the trees $T(\tau_1)$, $T(\tau_2)$,..., $T(\tau_d)$ in graph G. Then, we will model the MCT problem into a linear programming problem as follows:

L respects the network lifetime, s_i spends the duration time of t_{im} to monitor the target t_m, F_{ij} is the total number of data from s_i flow to s_i flow, F_{iR} is the total number of data flows from s_i to R.

Maximize L

$$\sum_{i=1}^{N} t_{im} - L = 0 \ ; \ \forall p_m \in P \tag{1}$$

$$\sum_{j_i} F_{ij} + F_{iR} - \sum_{j} F_{ji} - f_s \sum_{p_m \in P} t_{im} = 0 \ ; \ \forall s_i \in S \tag{2}$$

$$\sum_{j} F_{ij} \cdot e_{ij}^t + F_{iR} \cdot e_{iR}^t + \sum_{j} F_{ji} \cdot e_r + \sum_{p_m \in P} t_{im} \cdot f_s e_s \le E_0(s_i) \quad \forall s_i \in S \tag{3}$$

Theorem 1: The optimal solution of the above linear programming program is exactly the optimal solution's upper bound of MCT problem.

Proof: If we can prove that one optimal solution of MCT is also one feasible solution of the above LP problem, then, we can get the conclusion that the optimal solution of LP problem must be the optimal solution's upper bound of MCT problem.

Assuming that there are many solutions of MCT problem, such as

$$\{T(\tau_1), T(\tau_2),..., T(\tau_d)\} \text{and} \{\tau_1,..., \tau_d\}.$$

Each target can be covered at least once in the duration of the operating time τ_a, we can get that:

$$\sum_{i=1}^{N} t_{ima} = \tau_a \ , \ \forall p_m \in P \ , \tag{4}$$

where t_{ima} represents the duration of s_i monitoring the target t_m within time τ_a. Equation (4) can also success in other operating time, where $a \in [1,d]$, so, we can get that:

$$\sum_{a=1}^{d}\sum_{i=1}^{N} t_{ima} = \sum_{a=1}^{d} \tau_a \ , \ p_m \in P$$

$$\sum_{i=1}^{N} t_{im} - L = 0$$

That is to say, , thus, Equation (1) is right, and Equation (2) and Equation (3) are also right, so, we can draw the conclusion that the optimal solution of the above linear programming program is exactly the optimal solution's upper bound of MCT problem, and we denote it as L_{LP}.

4. ECTCG Algorithm

Since MCT is a NP-complete problem [14], we present a fast heuristic algorithm named ECTCG (energy-efficient connected target coverage greedy algorithm) to solve the connected target coverage problem.

4.1. Design of ECTCG

The input of ECTGC are S, T, R and τ, the output of ECTGC are a series of cover trees ($T_1,\ldots,T_d$). For the sake of brevity in describing our algorithm in the following, we introduce some symbols that will be used in our algorithm here. Let s_i be the alive nodes set, $w(s_i)$ represents the path weight of s_i in ECTCG. B_i is the contribution of s_i. $R(s_i,T)$ represents the path of s_i to sink R in tree T, where $R(s_i,T)=<s_i,s_j,\ldots,R>$. $\overline{R}(s,T)$ represents the set of node's paths in tree, except the source node s_i and the destination node R.

In a network with N sensors and M targets, the steps of constructing cover trees T_1 ,…, T_d are as follows:

1) At the beginning, the initial energy of each sensor is $E_{0,}$ the number of the rounds is 1, denoted by d=1

2) Uncovered targets in the WSN, denoted by φ^d, sensors in d^{th} cover, denoted C_d, set φ^d= {all targets in the network}, C_d=Ø

Then, follow the below three steps:

Step 1: firstly, algorithm gives each directed edge a weight value, such as the weight value of directed edge $<s_i,s_j>$ can be calculated by:

$$w_{ij} = \frac{e_{ij}^t \times E_0(s_i)}{E_r(s_i)}$$

For each node s_i in the network, find one path with the minimum weight value to the sink (such as using the Dijkstra algorithm)

Step 2: we use a greedy strategy to select nodes to cover all the targets.

(a) When all the targets cannot be completely covered, go to the step (b); otherwise, go to the step 3.

(b) Select a remote target x from all uncovered targets, $\alpha(x)$ is the number of sensors that can cover the target x, and the remote target x has a minimum $\alpha(x)$.

(c) Compute the weight value B_i of each node s_i as follows:

$$B_i = \frac{\theta_i}{w(s_i)},$$

where θ_i is the set of the uncovered targets within s_i' sensing range, and w(s_i) is the path weight value of s_i, which is exactly the sum of weight value of all edges in $R(s_i,T)$.

(d) Select a node s_i with the maximum contribution that can cover the remote target x, then add s_i into C_d.

(e) For node s_j, which satisfy $s_j \in \overline{R}(s_i, T_d)$, update:

$$w(s_j) = w(s_j) + (e_{trans} + e_r)P(\tau) \times w(s_j) / E_r(s_j)$$

and B_j, then go to step (a).

Step 3: Add the selected sampling node and $\overline{R}(s,T)$ into tree T, then update the residual energy of sensors according to their energy model.

In every round, the algorithm will be finished in the following time:

$$\tau_d = \min(\tau, \min_{s \in T_d}(\frac{E_r(s)}{E_s(\tau, T_d)}\tau))$$

Steps (2)-(3) are iteratively executed until all the targets can no longer be continuously covered. A specific description of ECTCG is shown in Fig. 1.

Algorithm ECTCG:
1. set sense range as r and energy of each sensor as e;
2. set the initial energy of each sensor s_i as $E_{0,}$ $E_r(s_i)=E_0$
3. $d = 1$
4. while all the targets can be covered by active nodes, do
5. **step 1:** for each edge (s_i,s_j) in the network, compute $w_{ij} = e_{ij}^t \times E_0(s_i) / E_r(s_i)$
6. for each node s_i, find out one path with the smallest weight value to sink.
7. **step 2:**
8. $\varphi^d = \cup P_j$, （j=1...M）, $C_d = \varnothing$, $T_d = \varnothing$, $\tau_d = \tau$
9. for each node s_i, compute $B_i = \frac{\theta_i}{w(s_i)}$
10. while $\varphi^d \neq \varnothing$ do
11. select a remote target x from all the uncovered targets, which satisfy that:
$\alpha(x) < \alpha(y)$, where $y \in \varphi^d - \{x\}$ (if there exists more than one target x, select the nodes x that has the minimum ID number.)
12. construct a set S', any node s_i in set S' can cover target x, then select s_i from S' such that s_i can cover x and satisfy max B_i among all the sensors
13. $C_d = C_d \cup s_i$
14. $\varphi^d = \varphi^d - \rho_i$
15. for each node s_j that satisfy $s_j \in \overline{R}(s_i, T_x)$ do:
16. update $w(s_i) = w(s_i) + (e_{trans} + e_r)B(\tau) \times w(s_i) / E_r(s_i)$ and B_j
17. end while
18. **step 3:**
19. for each node s_i in C_d do: $T_d = T_d \cup R(s_i, T_d)$
20. for each node $s_i \in T_d, \tau_d = \min(\tau_d, \frac{E_r(s_i)}{E_s(\tau_d, T_d)}\tau_d)$
21. for each node $s_i \in T_d$, $E_r(s_i) = E_r(s_i) - E(s_i, T_d(\tau))$
22. $d = d + 1$
23. end while

Fig. 1. ECTCG algorithm.

4.2. Complexity of ECTCG

It is easy to know that the duration time for each round is τ or when a node is dead, so we can define the number of cover trees within the range of $N + L_{LP} / \tau$. The maximum complexity of each round is MN, therefore, the complexity of ECTCG is $O((N + L_{LP})MN / \tau)$, the same as that of CWGC.

5. Performance Evaluation

In this section, we will compare ECTCG with another typical algorithm CWGC, ECTCG and CWGC adopts the same methods to generate the

skeleton of the tree to transmit the data, but they use different greedy strategies to select the source nodes. In the simulations, we will evaluate the metrics of network lifetime among above two algorithms.

We simulate a network with sensors and targets randomly deployed in a 100 m × 100 m area, the sink is located in the center of the area, and its coordinate is (50,50). We assume that each node has the same initial energy of 20 Joule (J), the other parameters are set as: $e_t = 50nJ/bit$, $b = 100pJ/m^4$, $\alpha = 4$, $e_r = 150nJ/bit$, $e_s = 150nJ/bit$, the data sampling frequency is 10 kbps, all the nodes have a same sensing radius of r_s and a some transmission radius of r_c, where r_s=20 m and r_c=40 m.

5.1. The Relationship between ECTCG and Parameter τ

At first, we study the performance of ECTCG by varying the value of parameter τ . There are 60 sensors and 20 targets that are randomly scattered in the area, we set τ as τ =0.1 % L_{LP} , 0.2 % L_{LP} , 0.4 % L_{LP} , 0.6 % L_{LP} , 0.8 % L_{LP} , 1 % L_{LP} , 2 % L_{LP} , 4 % L_{LP} , 6 % L_{LP} , 8 % L_{LP} and 10% L_{LP} respectively. The simulation results are shown in Table 1.

Table 1. The relationship between lifetime of ECTCG and τ .

τ/ L_{LP}	ECTCG	τ/ L_{LP}	ECTCG
0.1 %	2968.8	1 %	2892.4
0.2 %	2956.1	2 %	2501.1
0.4 %	2940.9	4 %	2404.4
0.6 %	2923.7	6 %	2304.7
0.8 %	2902.4	8 %	2301.5

From Table 1, we can see that the network lifetime becomes larger when the parameter τ becomes smaller. The smaller the τ , the more trees can be generated by ECTCG algorithm; the cost of the algorithm will become larger. So, we will set τ =1 % L_{LP} in the following simulations to balance the network lifetime and the cost of the algorithm.

5.2. The Performance Comparison between ECTCG and CWGC

We assume there are 20 targets in the area, and we vary the number of sensors from 50 to 130 with an increment of 10, respectively. The network lifetime of these two algorithms under different number of sensors are shown in Fig. 2.

From Fig. 2, we can see that, when compared with CWGC algorithm, ECTCG can enlarge the network lifetime to a certain extent as well as ensure all the targets be covered and the network be connected.

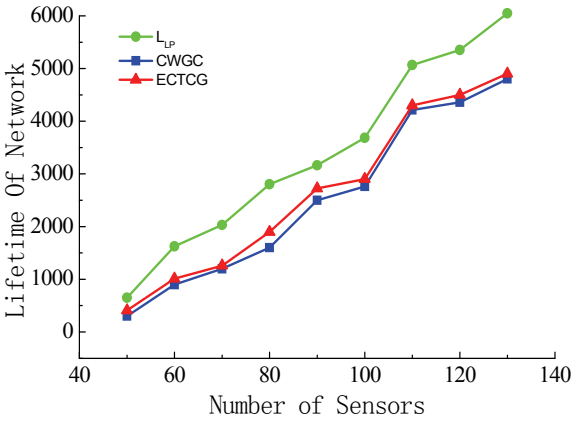

Fig. 2. The comparison of network lifetime between ECTCG and CWGC.

5.3. The Performance Comparison with Different Size of the Network

We compare the performance of ECTCG and CWGC under different size of network, and here we set two scenes, scene 1 (network of small size): there are 80 sensors and 10 targets in this scene; scene 2 (network of big size): there are 200 sensors and 25 targets in this scene. We set network delay as 20 and 10, respectively. The simulation results of network lifetime of different algorithms are shown in Table 2.

Table 2. The relationship between network lifetime and the size of the network.

	Lifetime (scene 1)	Lifetime (scene 2)
ECTCG	2460.3	6312.2
CWGC	2190.8	5324.5

In Table 2, ECTCG achieves a larger network lifetime than CWGC in the same complexity of algorithm, no mater how does the network size vary. Obviously, ECTCG gains a good performance than that of CWGC.

6. Conclusions

In this paper, we study the connected target coverage problem in wireless sensor network. We model the connected target coverage problem as a maximum cover tree (MCT) problem, and then we give an upper bound value for MCT problem. Since MCT is a NP-complete problem, we then develop a fast heuristic algorithm, named ECTCG to solve the MCT problem. Simulation results show that ECTCG can achieve a better performance than other existing algorithms.

Acknowledgements

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MRAS Based Speed Sensor-less Vector Controlled Induction Motor Using Modified Adaptive Mechanism

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Abstract: To improve operational stability of control system induction motor, a speed sensorless vector control system for induction motor based on DSP (TMS320F2808) was presented, fluxes estimation and speed observer based on PI adaptive method; model reference adaptive system (MRAS), a speed sensorless field-oriented control (FOC) system for induction motor was adopted. The speed estimation was based on the difference between two fluxes estimators; reference and adaptive model. The adaptive mechanism of the conventional estimator was replaced with a booster module driven by error signal from the two models. The proposed control algorithm, simulations, implementation data, diagrams and test results were given. It shows good performance for the sensorless vector control system in torque, speed and robustness. *Copyright © 2013 IFSA.*

Key words: Field oriented control (FOC), Sensor-less, Induction motor, Operational stability, Estimators.

1. Introduction

High-performance motor control is characterized by smooth rotation over the entire speed range of the motor, full torque control at low speed, fast accelerations and decelerations. To achieve these operational stability qualities, speed sensorless vector control techniques are used for induction motor. Several algorithms have been applied to solve stability problem associated with speed sensorless vector control induction machine at low speed. Low-Pass Filters (LPF) with very low cut-off frequency have been used to replace the pure integrator, but they introduce phase and gain errors due to their natural delay which causes problems in the frequency range below the filter cut-off frequency [1, 2], this work introduced field oriented control (FOC) with a booster module as the adaptation mechanisms in placement of "gain constants and low-pass filter" of the back-EMF-based MRAS control algorithm for sensorless control of induction motor. This cuts down on tuning time while providing a good and fast response. The simulation and implementation result were shown in this paper that proved the cost effective and dynamic response of the algorithm.

2. Field Oriented Control Theory

With the advent of FOC by Balaschke has made control system more flexible. The basic idea of the vector control (FOC) is to provide an algorithm capable of decomposing a stator current into a

magnetic field-generating part (i_d) and a torque-generating part (i_q), both components can be controlled separately after decomposition. The structure of the motor controller become simple as that for a separately excited DC motor allowing accurate transient and steady state manipulation [3,4]. The proposed sensorless vector control scheme is illustarted by Fig. 1.

The machine equations in the stator reference frame, written in terms of space vectors, are:

$$\bar{V}_s = R_s \bar{I}_s + \frac{d\bar{\varphi}_s}{dt} \tag{1}$$

$$0 = R_r \bar{I}_r + \frac{d\bar{\varphi}_r}{dt} - j\omega_m \varphi_r \tag{2}$$

$$\varphi_s = L_s \bar{I}_s + M\bar{I}_r \tag{3}$$

$$\varphi_r = L_r \bar{I}_r + M\bar{I}_r \tag{4}$$

$$\frac{d\Omega}{dt} = \frac{T - T_L}{J} \tag{5}$$

$$T = P\frac{M}{JL_r}(\varphi_{rd} i_{sq} - \varphi_{rq} i_{sd}) \tag{6}$$

These expressions signify the control strategy based on the orientation of the flux vector along d-axis which can be expressed by considering:

$$\varphi_r^{d,q} = (\varphi_{rd}^2, 0)^T$$

Equations (5) and (6) can be written in the rotor reference frame with respect to the axis components, leads to:

$$\frac{d\varphi_{rd}}{dt} + \frac{1}{\tau_r}\varphi_{rd} = \frac{M}{\tau_r} i_{sd} \tag{7}$$

$$T = P\frac{M}{JL_r}\varphi_{rd} i_{sq} \tag{8}$$

These equations represent the basic fundamental principle of FOC and are derived based on the decoupling control concept where the fluxes values and orthogonal spatial angles between them and MMF are achieved, in the current mode, the total rotor flux-linkage is aligned with the d-axis component [2, 3]. This explicitly illustrated in the Fig. 1.

The flux estimation is given as:

$$\frac{d\bar{\varphi}_s}{dt} = u_s - R_s \bar{i}_s + u_{off}\,, \tag{9}$$

where u_{off} is the unavoidable errors.

$$\bar{\varphi}_r = \frac{L_r}{M}(\bar{\varphi}_s - \sigma L_s \bar{i}_s) \tag{10}$$

The flux estimator was considered to be ideal, being the effects due to parameter variations at low speed.

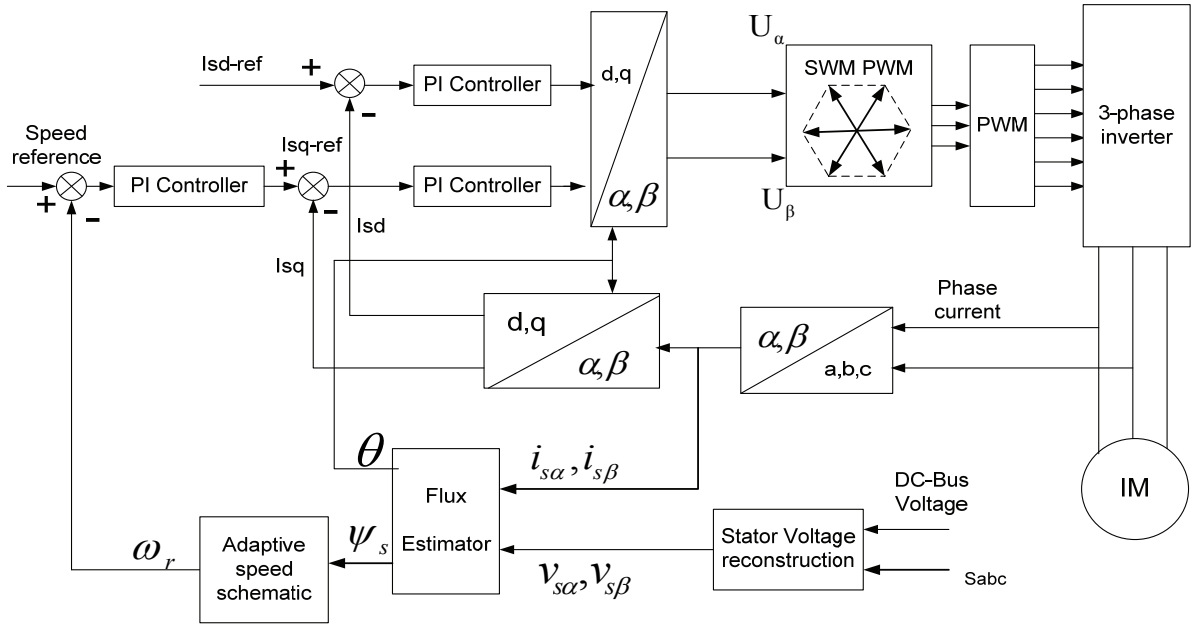

Fig. 1. Block Diagram of sensorless vector control IM drive.

3. Model Reference Adaptive System

Model Reference Adaptive Systems (MRAS) are used to estimate quantities using a reference model and an adaptive model. The difference between the outputs of the two models drives an adaptive mechanism (booster module) that provides the quantity that is to be estimated. Different approaches have been developed using MRAS, such as rotor flux-linkage estimation-based and back-EMF-based MRAS (reactive-power-based) [5, 6]. In this thesis we used the improved back-EMF-based MRAS and we proposed the module to replace the adaptation gain constants and low-pass filter in the adaptive mechanism to produced high dynamic response.

3.1. Rotor Flux-linkage Estimation-based

It uses state observer model with current error feedback and rotor current models for flux estimation The reference model is given by:

$$\dot{\hat{\psi}}_{rd} = \frac{L_r}{L_m}\left(\int(\hat{u}_{sd} - R_s\hat{i}_{sd})dt - \sigma L_s\hat{i}_{sd}\right) \quad (11)$$

$$\dot{\hat{\psi}}_{rq} = \frac{L_r}{L_m}\left(\int(\hat{u}_{sq} - R_s\hat{i}_{sq})dt - \sigma L_s\hat{i}_{sq}\right) \quad (12)$$

The adjustable model is given by:

$$\dot{\bar{\psi}}_{rd} = \frac{1}{T_r}(L_m i_{sd} - \bar{\psi}_{rd} - \omega_r T_r\bar{\psi}_{rq}) \quad (13)$$

$$\dot{\bar{\psi}}_{rq} = \frac{1}{T_r}(L_m i_{sq} - \bar{\psi}_{rq} - \omega_r T_r\bar{\psi}_{rd}) \quad (14)$$

The rotor speed is obtained from the adaptation mechanism as follow:

$$\hat{\omega}_{re} = \left(k_P + \frac{k_i}{p}\right)(\psi_{rq}\hat{\psi}_{rd} - \psi_{rd}\hat{\psi}_{rq}) \quad (15)$$

3.2. Back-EMF-based MRAS (Reactive-Power-based)

The model expression is given as follows.
The reference model is given by:

$$e_d = u_{sd} - R_s i_{sd} - L_s' i_{sd} \quad (16)$$

$$e_d = u_{sq} - R_s i_{sq} - L_s' i_{sq} \quad (17)$$

The adjustable model is given by:

$$e_d = \frac{L_m}{L_r}\frac{L_m i_{sd} - \hat{\psi}_{rd} - \omega_r T_r\hat{\psi}_{rq}}{T_r} \quad (18)$$

$$e_d = \frac{L_m}{L_r}\frac{L_m i_{sd} - \hat{\psi}_{rd} - \omega_r T_r\hat{\psi}_{rq}}{T_r} \quad (19)$$

The rotor speed is obtained from the estimator as follow:

$$\hat{\omega}_{re} = K_p\left(\hat{e}_{\beta s}\tilde{e}_{\alpha s} - \hat{e}_{\alpha s}\tilde{e}_{\beta s}\right) + K_i\int\left(\hat{e}_{\beta s}\tilde{e}_{\alpha s} - \hat{e}_{\alpha s}\tilde{e}_{\beta s}\right)dt, \quad (20)$$

where K_p and K_i are the proportional and integral gain constants respectively [3].

The use of booster module Fig. 3 eliminated the effect of tuning in the adaptive mechanism. The error between the reference and adaptive outputs, along with the reference speed (N_{ref}) is passed to drive the booster block. The initial condition of both signals was kept at zero. The rate limiter restricts the change of the signal passed to it by limiting the slope. The output of the rate limiter is calculated as follows:

$$O_{i/p}(i) = \nabla t.\gamma + N(t-1) \ , \quad (21)$$

$$O_{i/p}(i) = \nabla t.\sigma + N(t-1) \quad (22)$$

$$O_{i/p}(i) = N(i) \quad (23)$$

The upper limit is called the rising slew parameter (δ) and the lower limit is the falling slew parameter (γ). N refers to the input to the rate limiter. The output is passed to a Zero Order Hold (ZOH) to generate continuous time input by holding each sample value constant over one sample period. The ZOH also acts as a hypothetical filter that gives a piece-wise signal as is demonstrated by equation (24)

$$O_{ZOHo/p}(t) = \sum_{n=-\infty}^{\infty} N_{in}(n).rect\left(\frac{t-nT}{T} - \frac{1}{2}\right) \quad (24)$$

Finally, the estimated speed is calculated as:
Estimated Speed (ω_{est})

$$= \omega_{ref}(i) - \omega_{booster}(i) \quad (25)$$

$\omega_{booster}$ is the speed error detected in the booster module.

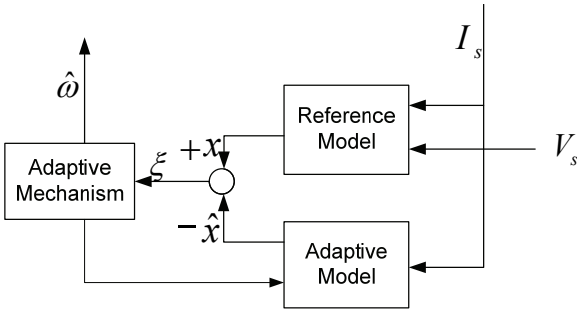

Fig. 2. Configuration of the conventional MRAS.

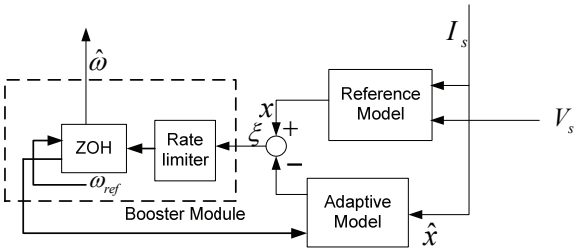

Fig. 3. Configuration of the Booster module MRAS.

4. Simulation Result and Analysis

Simulation was performed to verify the dynamic behavior and operational stability of the drives in responds to back-EMF based MRAS with substituted booster module for adaptive mechanism, using the MATLAB/Simulink. The induction motor used for the implementation of the proposed algorithm has the rated values P_N = 2.2 kW, f_N = 50 Hz, V_N = 220 V, p=2 pole pairs, and parameters Rs=0.877 Ω, Rr=1.4752 Ω, Ls = Lr = 1.4752 H, Lm = 0.8428 H, rated speed =1750 rpm. The measured results are shown in Fig (4-7), the adaptation gain constants (K_p and K_i) tuning difficulty were able to be avoided. This was achieved by passing output of the reference and adaptive model blocks to drive the booster module that minimized the estimated speed error to nearly zero, it acts as filter. Referring to equation (19), it can be seen that at every change in the reference speed, the values of proportional and integral gains (Kp/Ki) has to be continually changing and do not yield in convergence in the speed and fluxes. The adoption of the booster module in place of the gains yields very good performances as shown by the simulation result.

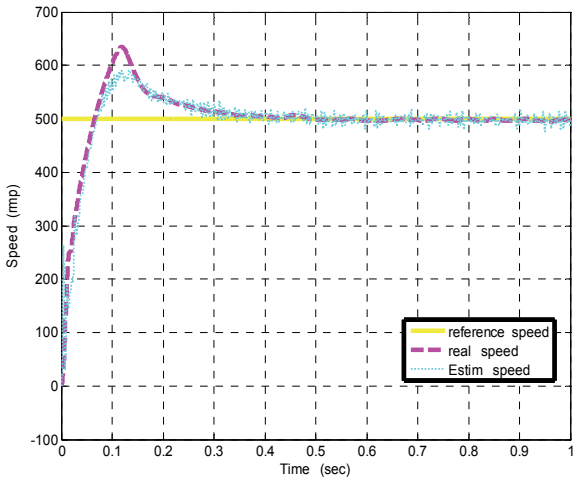

Fig. 4. Real, Estimated speed at reference speed of 500 rpm.

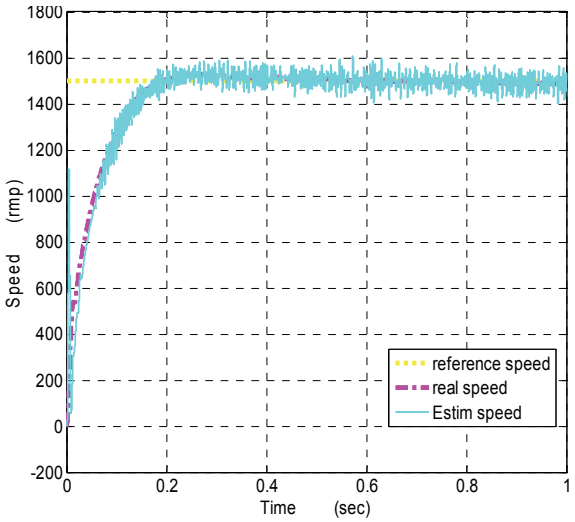

Fig. 5. Real, Estimated speed at reference speed of 1500 rpm.

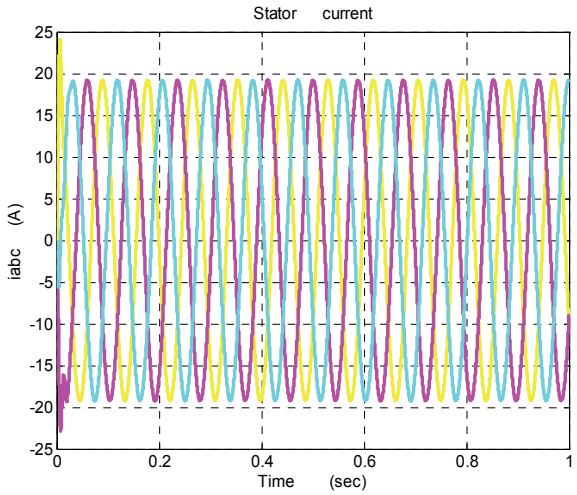

Fig. 6. Instantaneous three phase Stator currents.

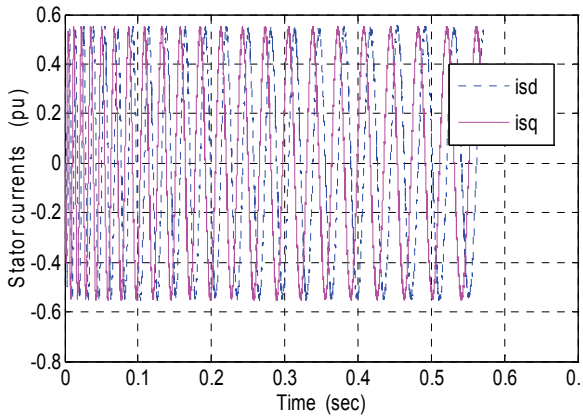

Fig. 7. Stator current in two coordinate axis (dq).

5. Experimentation Results

The proposed control schemes were implemented on a low cost Texas instruments' DSP (TMS320F2808) based PWM inverter system, to regulate the motor speed and current when running on load. The control intervals of the adaptive observation system are set at 100 μs and the control intervals of the speed control loop are set at 5 ms. In this paper' the integration of the observer was transferred to the "booster module" with the initial value setting to zero. The DSP performs the calculations required for the speed estimator and the flux observer, the sampling period is 10 μs, and the test results are showed in the Fig. 8 and 9.

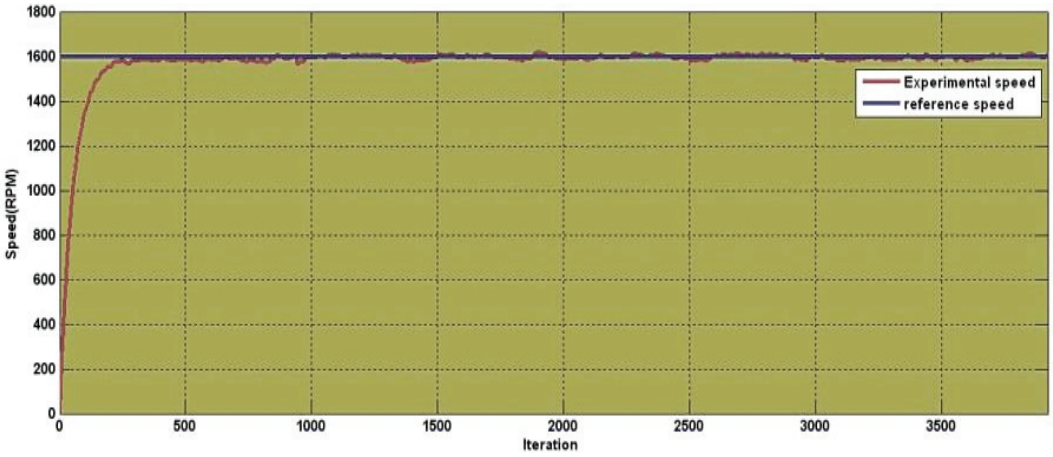

Fig. 8. Experimental and reference speed at 1600 rpm.

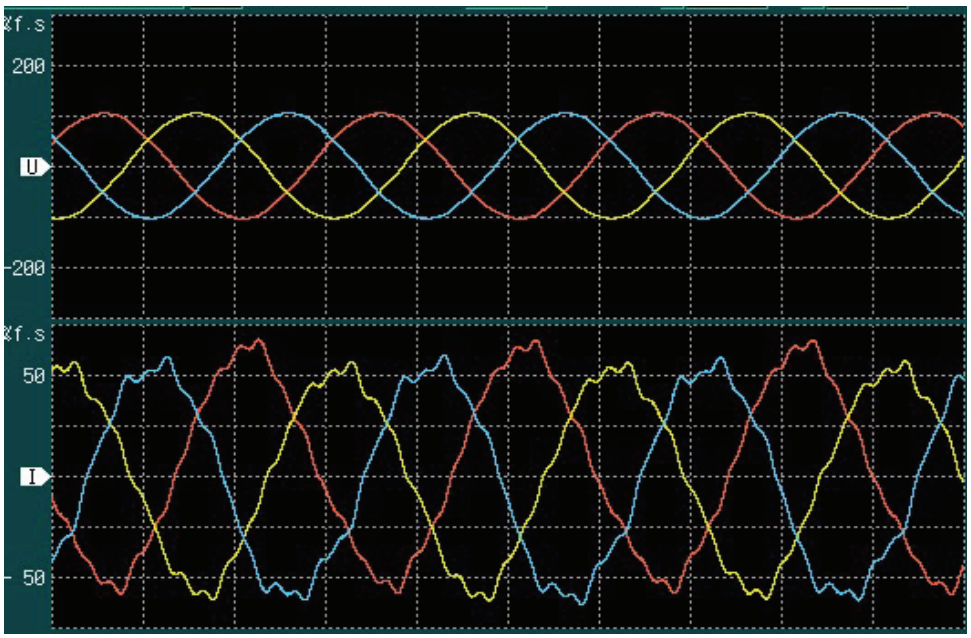

Fig. 9. Instantaneous three phase Stator Voltages (Volt) & currents (Ampere).

6. Conclusion

The concept of speed estimation based on Booster module MRAS, which uses the back-EMF-based MRAS (reactive-power-based), the estimator has performed very well in the wide range of speed references in the sensor-less vector control drives system. The estimator works in a stable way even for relatively big parameter variation and the stability points were detected. The drive system works well for low speed reference with load torque, with limited error of about ±45rpm. The analysis confirmed very good dynamic responds and robustness of the Booster module for adaptive mechanism.

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Modeling and Simulation of Intra-Chip Wireless Propagation Channels for Hybrid WiNoC

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Abstract: The performance of traditional NoC architecture is limited in terms of throughput, delay and power consumption due to multi-hop wired links. The Wireless Network-on-Chip (WiNoC) has demonstrated their advantages as an emerging interconnect and communication architecture for the new-generation multi-core systems. In this work, we systematically analyze the propagation path of hybrid WiNoC under two on-chip antennas. Based on the deduction of reflection coefficient using 3-D Descartes coordinate system and the ray tracing, we model the intra-chip wireless propagation channel for hybrid Wyncote ensure the reliable communication, we evaluate the Bit-Error-Rate (BER) performance, delay spread and coherence bandwidth. Especially, the simulation based on the model shows that the Signal-Noise-Ratio budget at the end of receiver must be greater than 15 dB to ensure the BER performance for the on-chip wireless communication. It also demonstrates that the path loss for optical on-chip antenna could be improved about 6dB by elevating the height of antenna pairs. The channel coding scheme used for improving the system performance has been analyzed in detail. Besides, we propose an adaptive power control mechanism to maintain high performance by making a balance between BER, operating frequency and transmitted power.

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Keywords: Wireless network-on-chip, Multi-path propagation, On-chip antenna, Path loss, Transmission coefficient.

1. Introduction

As the feature size of transistors is scaling down, chip designs are moving towards integrating a large number of IP cores onto a single die or System-on-Chips (SoCs) to satisfy the individual and multifunctional demands. The performance of SoCs will be limited by the ability to efficiently interconnect IP cores to accommodate their communication requirements. Due to the stringent power consumption limitation, the shared-bus has been shown to be unable to supply SoCs with both sufficient bandwidth and low latency. Network-on-Chip (NoC) has emerged as a promising interconnection architecture and communication platform for complex SoCs [1-2]. Although various topologies for NoC have been proposed to advance the design of the inter-core communication infrastructure [2], the data exchange between two distant blocks causes high latency and power consumption arising from multihop communication. According to the International Technology Roadmap

for Semiconductors (ITRS) in 2011, for the long term, metal material innovation with traditional scaling will no longer satisfy performance requirements of future designs and new interconnect innovations are needed [3].

Recently, several different emerging interconnect technologies have been considered to solve the performance limitation due to long range multi-hop wire line links. To provide ultralow latency and low power, the express virtual channels between selected nodes by inserting long range links in the a regular mesh networks is introduced in [4, 6]. It is shown that by using express virtual channels to connect distant cores in the network, it is possible to improve the NoC performance in terms of throughput, delay and power consumption. The performance of NoC is further improved by using global Lines for flow-control signals [5]. However, such solutions are difficult to alleviate the problems of skew, jitter and delay caused by long wired communication channel. Although these seem to be good methodologies that significantly improve performance when compared with a traditional wired NoC, it is not enough if the number of nodes increases to thousand or more. In addition, 3D integration using through silicon vias (TSVs) is adopted as a popular design methodology to reduce the latency and hops [7]. Despite their advantages, 3D NoC bring the new process challenges, such as finite thickness of IC, inter-layer contact patterning and low yield caused by vertical via failures. The heat dissipation caused by increased power density in 3D structure is also a serious problem. During to traditional wired interconnects based only on TDMA for data transmission, a new interconnect scheme named Radio Frequency Network-on-Chip (RFNoC) by using modern multiple access algorithms including CDMA, FDMA and microwave transmission line (MTL) to boost the bandwidth has been proposed [8, 9]. Although the channel has many benefits by inserting long on-chip multi-band MTL between distant blocks and also uses existing CMOS technology, it still require the long transmission lines to work as the wave guides. Optical interconnect NoC as a revolutionary approaches to improve performance in terms of bandwidth latency and power consumption [10, 11]. Despite optical interconnect has some advantages as a viable interconnect option, the optical modulators and detectors need much deeper investigation.

Wireless interconnect has been introduced to on-chip communication first in [12] for distributing clock signals. Then wireless interconnect has many advantages by comparing with other interconnects in [13]. Soon after a new on-chip communication system, dubbed Wireless Network-on-Chip (WiNoC) has been proposed in [14]. Since the achieved transmission range of UWB-based antenna is about 1 mm [15-17], which means in a chip whose area is 20 mm × 20 mm, this technology principally gives rise to multi-hop communication when transmitting data. Inspired by this wireless communication scheme, the chip multi-processors (CMPs) have more performance gains by using a scalable micro WiNoC interconnect structure [18]. In many cases, the additional latency of on-chip interconnects can have a significant impact on system performance. To further improve the performance of WiNoC, multi-hop wireless channels are replaced by inserting single-hop long-range wireless links [20-26]. Mass experiments show that WiNoC by using on chip antenna outperforms its more conventional wired counterpart [20, 21, 23], such as wired NoC, RFNoC, ONoC and so on. Thus it can be seen that WiNoC as a emerging interconnect solution has huge potential compared to traditional NoC to become the mainstream interconnect architecture and bridge the widening gap between computation requirements and communication efficiency for complex SoC system. The critical difference is the model of propagation channel between wired NoC and WiNoC.

However, aside from previous work in [25, 28], the model of wireless multi-path propagation channel for WiNoC have not been carefully investigated, despite its importance of intra-chip wireless interconnect. In order to more accurately simulate and evaluate the actual performance of system, the multi-path propagation channel is necessary to be explored and evaluated carefully for a given WiNoC. In this paper, we analyze on-chip antenna since it is the critical component for intra-chip wireless communication. A system-level model of wireless multi-path propagation channels for WiNoC is presented.

The remainder of the article is organized as follows: Section 2 presents two different on-chip antennas and their cross-sectional view. Section 3 presents the model of multi-path propagation channel for intra-chip wireless communication based on the reflection coefficient using 3-D Descartes coordinate system and ray tracing. The evaluation and simulation is reported in Section 4. Finally, Section 5 offers a brief set of conclusions and the future work.

2. On-Chip Antenna for WiNoC

The performance of intra-chip wireless interconnect is mainly dependent on antenna as the critical component for intra-chip wireless channel, thus we will ultimately need to incorporate antenna characteristics. Over the past few years, there considerable efforts have been done in terms of feasibility, measurement and signal propagation mechanisms of on-chip antenna for realizing the intra-chip wireless interconnect [27-34]. The on-chip antennas could be classified into two categories: millimeter-wave on-chip antenna operating at GHz frequency [30-32] and carbon nanotubes (CNTs) as optical on-chip antenna operating at THz frequency [33-34]. For the former, the transmission of radio signal is mainly realized with surface-wave rather than free-space wave through carrying out a large number of experiments and measurements [30, 31]. The simulated results show that the dominant

medium supporting the electromagnetic wave propagation is the silicon substrate [31]. For the latter, the carbon nanotubes (CNTs) as optical on-chip antenna operating at THz frequency transmit the optical signal at the true velocity of light through free-space [23, 33-34].

To make integration of on-chip antennas viably and obtain low-cost, it must be compatible with the substrate using mainstream silicon technologies. The cross-sectional views for integrated two different antenna pairs on silicon wafer with a dielectric layer inserted between the substrate and heat sink are shown in Fig. 1(a) [32] and Fig. 1(b).

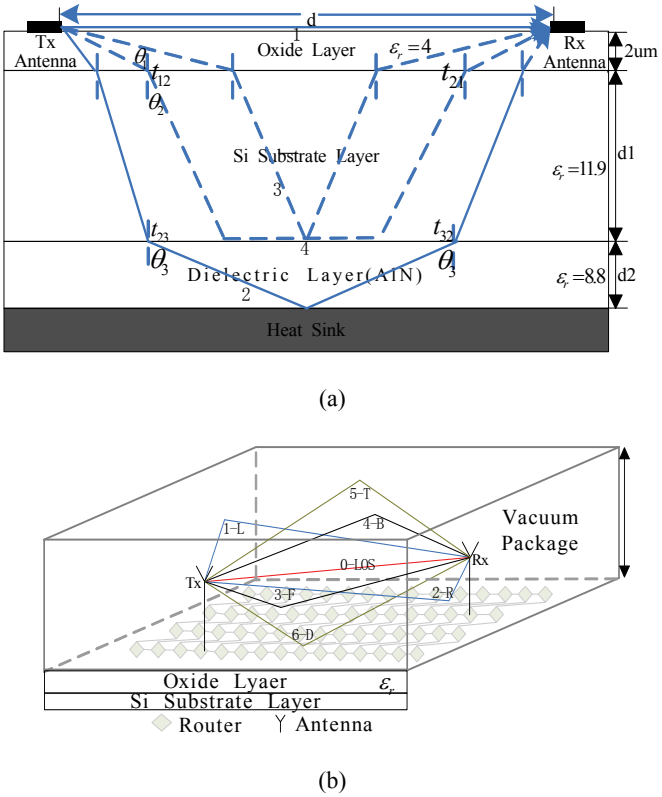


Fig. 1. Cross-Sectional View and Multi-Path Channel of WiNoC for Integrated Antennas on Silicon Wafer (a) Integrated Millimeter-Wave Antenna Pairs Operating at GHz, (b) Integrated Optical On-Chip Antenna Pairs Operating at THz.

3. Modeling of Multi-Path Propagation Channel

The wireless channel will be largely time invariant since there will be no motion for intra-chip transceivers. According to the classical theory of wireless communication system, the received signal is the sum of all the possible multi-path signal via a series of attenuation, time delay and phase shift. The typical mathematical model for multi-path channels is discrete multi-path time domain model. The multi-path channels could be seen as a linear band-pass filter. The shock response of multi-path channels is given by:

$$h(t) = \sum_{i=0}^{N-1} a_i e^{-j\varphi_i} \delta(t - \tau_i) \quad (1)$$

where i and N represent the i[th] path and the total number of multi-path, respectively; i=0 represents the first arrival path; τ_{RMS} is relative time delay to the first arrival path; a_i and φ_i represent the attenuation coefficient for amplitude and phase for the i[th] path at the time delay τ_{RMS}.

The frequency domain transfer function for multi-path channels is given by:

$$H(f) = \int_{-\infty}^{+\infty} h(t) e^{-j2\pi ft} dt = \sum_{i=0}^{N-1} a_i e^{-j\varphi_i} e^{-j2\pi f \tau_i} \quad (2)$$

The received multi-path signals will produce time delay. The Root-Mean-Square Delay Spread (RMS-RDS) weighted proportional to the energy in the reflected paths is given by:

$$\tau_{RMS} = \sqrt{\sum_{i=0}^{N-1} p_i \tau_i^2 - (\sum_{i=0}^{N-1} p_i \tau_i)^2}, \quad (3)$$

where

$$p_i = \left| a_i \right|^2 \Big/ \sum_{i=0}^{N-1} \left| a_i \right|^2$$

The maximum data transfer rate depends on root-mean-squared delay spread τ_{RMS}, which is the root-mean-square value of the delay of reflections, weighted proportional to the energy in the reflected waves. It will induce the frequency-selective fading and inter-symbol interference (ISI) if the data transfer rate is greater than the $1/\tau_{RMS}$.

3.1. Path Analysis of Multi-Path Propagation

To evaluate the impact on the performance of antenna integrated on a silicon wafer, here, we will carefully analyze the propagation path for two different on-chip antennas. To the millimeter-wave antenna pairs integrated on the oxide layer, the four possible paths for signal propagation as shown in Fig. 1(a). The path 1 represents the direct path between two antennas. It is called surface wave which may suffer attenuation due to the silicon substrate. The attenuation mainly depends on the distance between antennas, frequency and conductivity. Path 2 is the wave going through the dielectric layer and reflected back by the metal chuck. Among all these four paths, path 3 and path 4 are negligible because of the lossy silicon substrate. In this case, only path 1 and path 2 dominate [35]. The path which travel through the top oxide layer is neglected because the oxide thickness is negligibly small compared to a wavelength.

In Fig. 1(a), the electric field of surface wave in direct path 1 can be listed as [32]:

$$E_{sur} = E\frac{1}{d}e^{-\gamma_{sur}\cdot d} \tag{4}$$

where $\gamma_{sur} = \alpha + j\beta$ is the complex propagation constant of surface wave. α and β represent the attenuation coefficient for amplitude and the propagation constant for phase shift. d denotes antenna separation distance.

In path 2, the wave first goes into the silicon substrate, then into the insulating dielectric layer (e.g., AlN), and finally is reflected back by the heat sink to reach the receiving antenna. Similarly, the electric field formula of the wave in path 2 can be deduced as:

$$E_2 = \frac{E}{d}T_{12}T_{23}T_{32}T_{21}e^{[-\gamma_{si}\cdot 2d_1/\cos\theta_2]}e^{[-j\beta_{idl}\cdot 2d_2/\cos\theta_3 - j\pi]}, \tag{5}$$

where *d1* and *d2* represent the thickness of silicon substrate and thickness of insulating dielectric layer. γ_{si} and β_{idl} is complex propagation constant in silicon substrate and the propagation constant in insulating dielectric layer.T_{12}, T_{23}, T_{32}, and T_{21} represent the transmission coefficients. The attenuation coefficient α and propagation constant β are defined as follows:

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right]^{\frac{1}{2}} \tag{6}$$

$$\beta = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}+1\right]^{\frac{1}{2}} \tag{7}$$

According to above analysis, the total electric field of the wave for millimeter-wave antenna pairs operating at GHz frequency is given by

$$E_{G\text{-}total} = E_{sur} + E_2 \tag{8}$$

However, as shown in Fig. 1(b), there are seven different paths to CNTs as the optical on-chip antenna in free-space. It has become a viable candidate to create a vacuum for line of sight (LOS) communication operating at THz/optical frequency by elevating the chip packaging material from the substrate. From the electric field, the total electric field intensity as a function of transmission distance could be expressed by:

$$E_{T\text{-}total} = \frac{E_d}{d}e^{-\gamma d} + \sum_{i=1}^{6}\frac{E_i}{L_i}R_{0i}e^{-rL_i} \tag{9}$$

where E_d is the electric field intensity of LOS communication. *i* represents the i^{th} surface of all six surfaces(i.e., Left, Right, Front, Back, Top or Down surface corresponding to the path 1-L, 2-R, 3-F, 4-B, 5-T, 6-D in Fig. 1(b), respectively) of the chip packaging. The *d* is the LOS distance between a transmitting antenna and a receiving antenna. L_i is the reflection path length traveled by the wave through the i^{th} surface.R_{0i} is the reflection coefficient while ray travel from free-space to the i^{th} surface of packaging material. In free-space, where the attenuation constant α =0, so the complex propagation constant $\gamma = \alpha + j\beta = j\beta$.

3.2. Reflection and Transmission Coefficient based on 3-D Descartes Coordinate System

Though there is no moving for the intra-chip communication, the multipath propagation will be induced by the reflections of dielectric surfaces. According to wireless communication system theory, the received signal could be deduced by reflection and transmission coefficient based on ray tracing. The simplified two ray reflection model as shown in Fig. 2.

Since the angle of incidence θ_1 is difficult to obtain in reality, 3-Dimension descartes coordinate system has been adopted to express the reflection

coefficient. We suppose that the value of descartes coordinate for T_x antenna and R_x antenna are (x_t, y_t, z_t) and (x_r, y_r, z_r), respectively. Here, z_t and z_r could be seen as the heights of transmitting and receiving antenna. H is the height of vacuum packaging above the oxide layer.

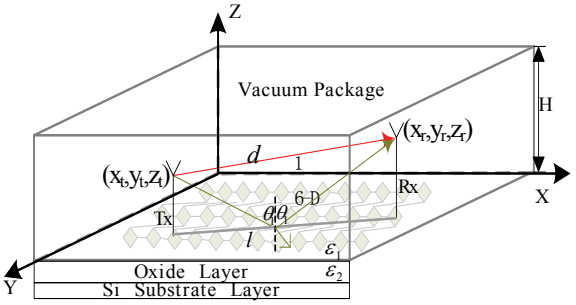


Fig. 2. Two Ray Reflection Model for WiNoC.

The l is the projection of the direct LOS length d on the X-Y plane. The direct LOS length d can be expressed by

$$d = \sqrt{l^2 + (z_r - z_t)^2}\,, \tag{10}$$

where $l = \sqrt{(x_r - x_t)^2 + (y_r - y_t)^2}$, the relationship between distance l and angle of incidence θ could be expressed by

$$l = (z_r + z_t)tg\theta \tag{11}$$

The reflection coefficient R_{06} (i.e., the path 6-D traveled from free-space to down-surface) and transmission coefficient T_{12} usually depend on the permittivity ε of the surface materials, the angle of incidence θ_1 and polarization mode, are defined as [36]:

$$R_{06//} = \frac{\cos\theta_1 - \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2\theta_1}}{\cos\theta_1 + \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2\theta_1}} \tag{12}$$

$$R_{06\perp} = \frac{\frac{\varepsilon_2}{\varepsilon_1}\cos\theta_1 - \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2\theta_1}}{\frac{\varepsilon_2}{\varepsilon_1}\cos\theta_1 + \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2\theta_1}} \tag{13}$$

where $R_{06//}$ or $R_{06\perp}$ is reflection coefficient at down surface for the horizontal or vertical polarization, respectively. The transmission coefficient for the horizontal or vertical polarization could be computed according to the eqs. (12) and (13).

$$1 + R_{//(\perp)} = T_{//(\perp)} \tag{14}$$

According to the above deduce, so the length of path 6-D could be expressed as:

$$L_6 = \sqrt{l^2 + (z_r + z_t)^2} \tag{15}$$

Combining eqs. (11)-(13), the reflection coefficient formula using 3-Dimension Descartes coordinate system can be deduced as follows:

$$R_{06//} = \frac{(z_r + z_t) - \sqrt{(\frac{\varepsilon_2}{\varepsilon_1} - 1)l^2 + \frac{\varepsilon_2}{\varepsilon_1}(z_r + z_t)^2}}{(z_r + z_t) + \sqrt{(\frac{\varepsilon_2}{\varepsilon_1} - 1)l^2 + \frac{\varepsilon_2}{\varepsilon_1}(z_r + z_t)^2}} \tag{16}$$

$$R_{06\perp} = \frac{\frac{\varepsilon_2}{\varepsilon_1}(z_r + z_t) - \sqrt{(\frac{\varepsilon_2}{\varepsilon_1} - 1)l^2 + \frac{\varepsilon_2}{\varepsilon_1}(z_r + z_t)^2}}{\frac{\varepsilon_2}{\varepsilon_1}(z_r + z_t) + \sqrt{(\frac{\varepsilon_2}{\varepsilon_1} - 1)l^2 + \frac{\varepsilon_2}{\varepsilon_1}(z_r + z_t)^2}} \tag{17}$$

The other reflection coefficients (e.g., R_{01}~R_{05}) and path length (e.g., L1~L5) could be computed as above.

3.3. Path Loss Model for Wireless Propagation Channel

In classical antenna theory, the power is directly proportional to the square of the electric field intensity. The figure of merit for an on-chip antenna pair is the power transmission gain between a receiving (P_R) and a transmitting antenna (P_T).It is represented using Friis transmission formula [33] and shown in (18):

$$\frac{P_R}{P_T} = |S_{21}|^2 = (1 - |\Gamma_t|^2)(1 - |\Gamma_r|^2)G_t G_r (\frac{\lambda}{4\pi d})^2 e^{-2\alpha R} \tag{18}$$

where Γ_t or Γ_r is the reflection coefficient for the transmitting or receiving antenna, respectively. G_t and G_r are the transmitting and receiving antenna gain. λ and d represent the wavelength based on the effective dielectric constant ε_r and the antenna separation distance. α is the attenuation constant. In free-space, where the attenuation constant α=0. S_{21} is the transmission coefficient.

The path loss for wireless multi-path propagation is calculated from ray tracing model. To GHz millimeter-wave antenna pairs, the two-ray path loss PL_G combining eqs. (4), (5) and (8) is given by

$$PL_G = (4\pi)^2 \Bigg/ \left| \frac{\lambda_{sur} \cdot G_d \cdot e^{-\gamma_{sur} d}}{d} + T \frac{\lambda_{si} G_2 \cdot e^{[-\gamma_{si} \cdot 2d_1/\cos\theta_2]} e^{[-j\beta_{dll} \cdot (2d_2/\cos\theta_3) - j\pi]}}{d'} \right|^2, \tag{19}$$

where λ_{sur} and λ_{si} are the wavelength of surface wave and silicon substrate, respectively. T is the products of transmission coefficient for each layer.

For the same theory , the path loss PL_T combining the eqs. (6), (7) and (9) to the THz optical on-chip antenna pairs in free-space is given by:

$$PL_T = \left(\frac{4\pi}{\lambda}\right)^2 \Bigg/ \left| \frac{G_{t0}G_{r0}e^{-j2\pi d/\lambda}}{d} + \sum_{i=1}^{6} R_{0i} \frac{G_{ti}G_{ri}e^{-j2\pi L_i/\lambda}}{L_i} \right|^2, \quad (20)$$

where λ is the wavelength in free-space. According to eqs. (15)-(17), R_{0i} and L_i could be computed based on 3-D descartes coordinate system and ray tracing. G_{ti} and G_{ri} is the transmitting antenna and receiving antenna field radiation pattern functions for the i^{th} path.

3.4. Delay Spread and Coherence Bandwidth

The Doppler spread effects arise from the multi-path propagation are negligible since the fixed position of transmitting and receiving antenna for intra-chip wireless communication, however the delay spread is a very important parameter must taken into consideration for multi-path channel. No serious ISI is likely to occur if the symbol duration is longer than the RMS delay spread. The coherence bandwidth B_c as another important channel parameter is given by

$$B_c \approx \frac{1}{\tau_{RMS}} \quad (21)$$

The receiver will not identify the symbol correctly as propagation channel producing serious signal distortion and high BER if the symbol rate is bigger than the coherence bandwidth B_c. Channel coding mechanism and equalization technology are needed to solve the ISI if the maximum symbol rate R_s is bigger than the coherence bandwidth B_c. However, these measures are beyond the scope of this paper. The B_c is easy to obtain according to eqs. (3) and (21) for two different on-chip antenna pairs. The maximum symbol rate R_s is less than B_c.

3.5. SNR and BER Performance of WiNoC

The performance for hybrid WiNoC system depends on the signal-to-noise ratio (SNR) and BER. It is found that the signal is distorted by the thermal noise and switching noise within the packaging [28]. The SNR for the receiver is defined as

$$SNR = \frac{E_{rb}}{N_0 + S_0}, \quad (22)$$

where N_0 and S_0 represent the power spectral density of thermal noise and switching noise. However, the wireless channels are in very high frequency band and the switching noise coupled to the antenna is common-mode, thus the intra-chip switching noise power spectral density to the WiNoC is negligible.

In equation (22), the average energy per bit at receiving end, E_{rg} , is calculated as [28]:

$$E_{rb} = E_{tb} + G_{tr} + G_r, \quad (23)$$

where E_{tb} is the transmitted energy per bit, G_{tr} is the average value of the transmission gain between the transmitting and receiving antennas, and G_r is the gain of the receiver.

The thermal noise power spectral density, N_0 , is calculated as

$$N_0 = K T_0 F = K T_0 \left(\frac{T_{antenna}}{T_0} + F_r\right), \quad (24)$$

where K is the Boltzmann constant, T_0 is the room temperature taken as 290 K typically, $T_{antenna}$ is the temperature of antenna assumed to be 330 K, and F_r is the receiver noise figure (taken as 4 dB generally). This assumption is supported by a recent study of CMOS integrated circuit.

The wireless channels for WiNoC have different bit-error rate function at different modulation mode (e.g., OOK, BFSK, BPSK). The on-off keying (OOK) and binary phase-shift keying (BPSK) are usually adopted with respect to wireless channels for WiNoC [21, 25-26].

To simulate a WiNoC system, some necessary changes should be made since there are differences between the WiNoC and NoC architectures, such as the channel's character, number of nodes, combination of protocols (intra- and inter- subnet) used in the system and routing algorithms. Still, there is no general simulator for WiNoC for public use.

Establishment of accurate wireless channel model has important guidance values for dynamic power management of wireless link, obtaining low bit-error rate, overcoming inter-channel interference (ICI) and ISI.

4. Simulation Results and Discussion of Wireless Multi-Path Channel Model

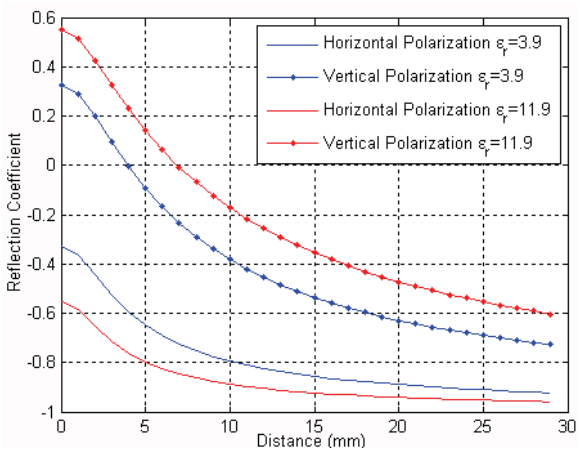
In this section, we simulate and analyze the characteristics of the proposed on-chip wireless channel model, including reflection coefficient versus distance, path loss as a function of operating frequency or distance, and the comparison of BER under different coding. The BER performance has been greatly improved by making a trade-off between operation frequency and transmitted power.

4.1. Reflection Coefficient Versus Antenna Separation Distance

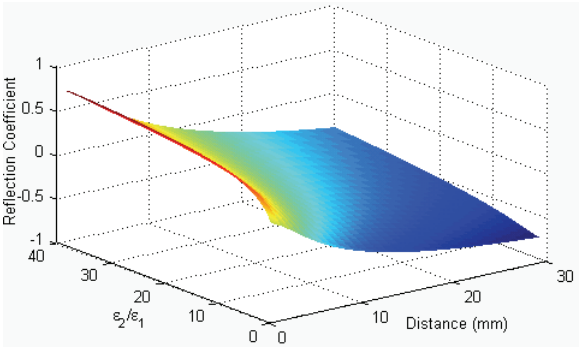
According to eqs. (10)-(17), it can be seen that the reflection coefficients depend upon the dielectric

constant ε_r and the optical antenna separation distance d. The simulation results are illustrated in Fig. 3(a) and 3(b).From the Fig. 3(a), we can observe that the value of $R_{//}$ is negative and decrease significantly with distance, and also the $R_{\perp}$ pass through the zero and decrease with the distance at different value of ε_r. The $R_{\perp}$ tends to be negative only when separation distance is large enough, e.g., 4-7 mm.

Since the current in vertical polarization way is less than that in horizontal polarization way, the vertical polarization is adopted to reduce propagation attenuation induced by the dielectric surfaces in this experiment. Fig. 3(b) shows that the variation of the reflection coefficient over the 20 mm × 20 mm die area using vertical polarization. It's evident that the value of $R_{\perp}$ increases with the $\varepsilon_r=\varepsilon_2/\varepsilon_1$ and decreases with the optical antenna separation distance.



(a)

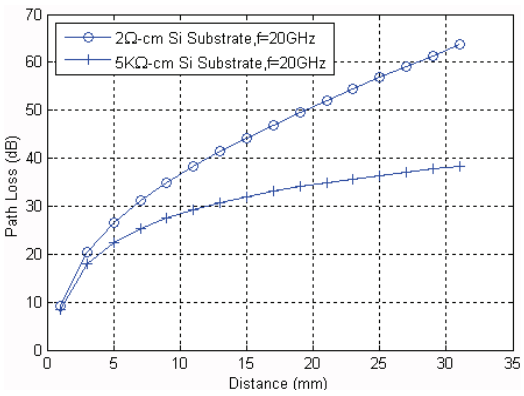


(b)

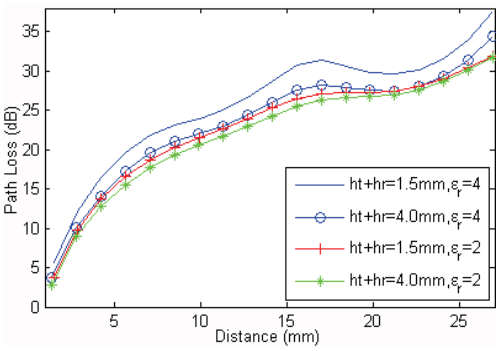
Fig. 3. Simulated (a)Horizontal polarization and Vertical polarization versus distance d, and (b)Vertical polarization versus distance and $\varepsilon_r=\varepsilon_2/\varepsilon_1$.

4.2. Path Loss Versus Distance, Antenna Height and Frequency

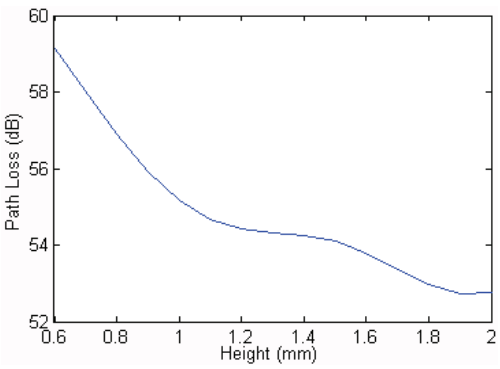
Fig. 1(a) shows that the oxide layer of thickness 2 μm was grown on the silicon substrate and a dielectric layer was used at GHz frequency to improve isolation and attenuation. The path loss for wireless propagation channel as a function of separation distance between antennas has already been shown in Fig. 4 according to eqs. (19) and (20). Fig. 4(a) and 4(b) show that the path loss increase obviously with separation for both millimeter-wave on-chip antenna and CNTs as optical on-chip antenna. It's found that the path loss could be reduced about 12 dB on average by increasing the resistivity of silicon substrate from 2 Ω-cm to 5 kΩ-cm.



(a)



(b)



(c)

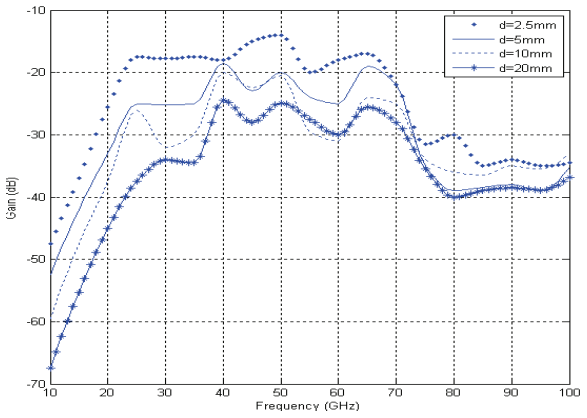
Fig. 4. (a) Path loss versus distance for different silicon substrate resistance at GHz frequency; (b) Path Loss versus distance and height of optical on-chip antenna operating at THz frequency; (c) Path loss versus sum of height of transmitting and receiving antenna pairs.

As illustrate in Fig. 4 (b), the path loss of on-chip optical antenna varies with dielectric constant and the sum of the height of transmitting and receiving antenna pairs. It's feasible to improve the path loss about 6-10 dB on average by increasing the height of antenna pairs and using low-loss dielectric layer .

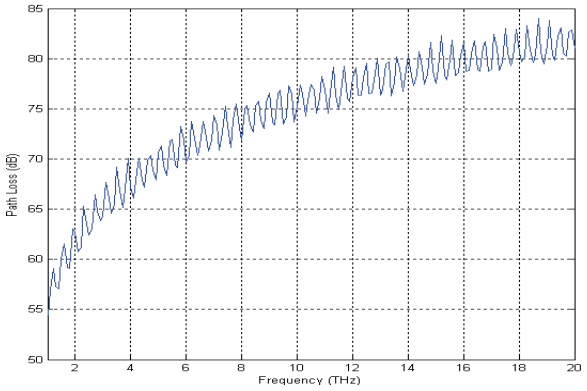
Moreover, from Fig. 4 (c) it can be observed that the path loss in THz frequency range could be improved about 6 dB by elevating the height of antenna pairs from 0.6 mm to 2 mm. But this improvement is achieved at the cost of area overhead.

Besides those, the path loss as a function of frequency for two types of on-chip antennas, biased on full electromagnetic-wave simulation software, Ansoft HFSS, are illustrated in Fig. 5. High resistivity silicon substrate (ρ = 5 kΩ-cm) is used for the simulation of wireless channel. As can be seen, the path loss doesn't always increase with the frequency for the millimeter-wave antenna. From the simulation results in Fig. 5 (a) it is observed that the gain is high in the window's region. It suggests that the operating frequency of on-chip wireless communication should be allocated within the high gain window for achieving better performance. Furthermore, it is found that the transmission gain decreases with distance especially in window's region and increases obviously with the frequency range of 10-28 GHz.

As can be seen from Fig. 5 (b), the path loss in THz frequency range versus distance goes through ups and downs and has many minimax values as a result of the interaction between in-phase and anti-phase of reflection paths. It also demonstrates that the path loss doesn't present window's region because the THz operating frequency is far beyond the resonant frequency of antenna.



(a)



(b)

Fig. 5. (a) Gain versus frequency for different separation distance at GHz frequency (b) Path Loss versus frequency operating at THz.

4.3. RMS-RDS and Coherence Bandwidth

Apart from path loss, delay spread is another important factor for an on-chip wireless propagation channel. In an on-chip wireless communication system with high bit rate, the reliability and efficiency will be reduced as frequency-selective fading and ISI. As can be seen from Fig. 6, the τ_{RMS} and coherence bandwidth B_c for optical antenna using vertical polarization as a function of distance don't always increase or decrease with symmetric distance.

It's apparent that while distance is approximately equal to 10.8 mm, the maximum of the τ_{RMS} is about 3.03 ps, and also the B_c shows the minimum under the maximum of τ_{RMS}. Moreover, the τ_{RMS} for WiNoC using optical antenna is less than that for WiNoC using millimeter-wave antenna [29]. Furthermore, it proved that the signal propagation through free-space for optical on-chip antenna is fast than that through silicon-substrate for millimeter-wave antenna.

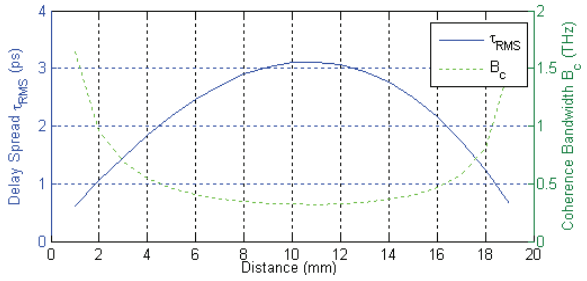


Fig. 6. Delay spread and B_c versus separation distance for optical antenna using vertical polarization.

4.4. Simulation and Discussion of SNR and BER

To further estimate the characteristics of the proposed on-chip wireless channel model, the BER performance is must be considered. The BER of the wireless communication for WiNoC is totally determined by the SNR of the receiver. Especially, the coding scheme and SNR must be analyzed to ensure the quality of service QoS for supporting low BER communication.

The simulation based on Simulink software was carried out to estimate the BER performance. Fig. 7 shows the block diagram of on-chip wireless communication system. The modulated data go through both AWGN and Wireless Multi-Path Channel. In our simulation, we assumed that the output power Pt is set to be -15 dBm for the transmitter, and the thermal noise of substrate is set to be 174 dBm/Hz. Thus, the SNR at the end of receiver can be calculated as:

$$SNR_t = P_t\text{-PathLoss-}(-174)\text{-NF-}10\log B \quad (25)$$

where NF is the noise figure of the receiver including LNA and Mixer. B is the bandwidth.

The required SNR was adopted for the targeted BER of 10^{-14}, which is the BER of traditional RC wires [18]. Fig. 8 (a) illustrates the BER performance of on-chip wireless modulation/demodulation system with different coding methods and without coding, while silicon substrate resistance is equal to 5 kΩ-cm. As shown in Fig. 8, it's evident that the RS or Hamming coding scheme improve the BER performance, especially when the SNR is greater than 5 dB or the separation distance is less than 20 mm. From the point of communication distance, it suggests that it's very suitable for the hybrid WiNoC by inserting long-range wireless links in place of multi-hop wire line links. So, it becomes essential to apply channel coding scheme as simple as possible for improving the BER performance of WiNoC.

As illustrated in Fig. 9, we can observe that the BER performance will be improved significantly by increasing the operating frequency from 10 to 20 GHz. Again, simulation in Fig. 9 indicates that channel coding scheme offers a great advantage in term of BER performance improvement especially when the SNR value at the receiver is higher than 5 dB.

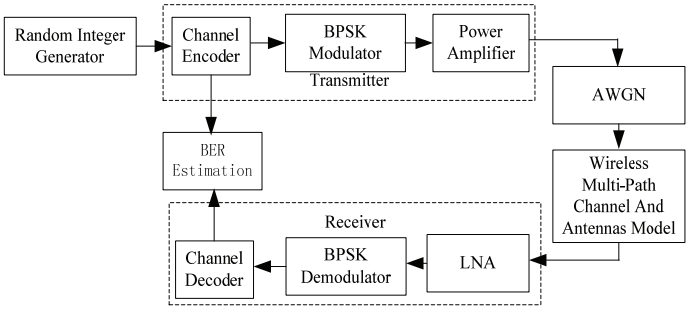


Fig. 7. Simulation block diagram for wireless modulation/demodulation system.

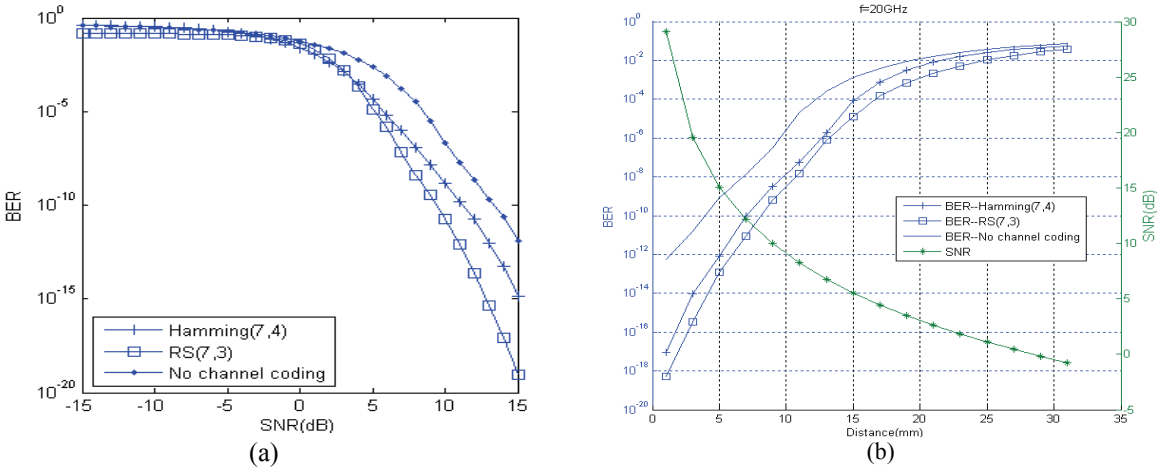


Fig. 8. In GHz frequency range (a) BER performance versus SNR, (b) BER performance and SNR versus separation distance.

4.5. A Trade-off between BER, Operating Frequency and Transmitted Power

As can be seen from Fig. 8(b), although the maximum communication distance of hybrid WiNoC is less than 28 mm, the BER of the WiNoC is less than 10^{-3} at P_t=-15 dBm, which is not enough for hybrid WiNoC. This value of BER is far higher than that of traditional wired links (about 10^{-14}). To maintain the same low BER as wired NoC, from Fig. 8(a) we can observe that the SNR_t budget at the end of receiver must be greater than the minimum SNR_{min} (about 15 dB).

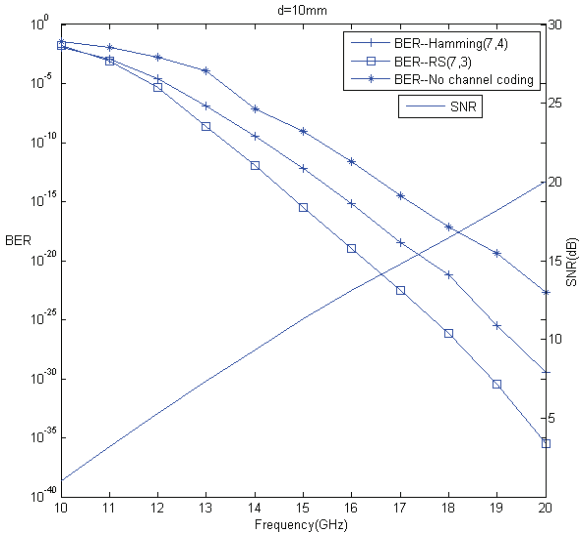


Fig. 9. BER performance as a function of GHz operating frequency at a distance of 10 mm.

Traditionally, in order to ensure enough power and BER performance for all the transmissions, the worst-case power loss in the WiNoC is considered, and the P_t is set to a fixed value offering the worst-case power for the all communication. Thus, it will causes the changes of SNR within a large dynamic rang for receiver. This will produce unnecessary power consumption for some data traffic. As shown in Fig. 8 and Fig. 9, we could make a direct tradeoff between BER, operating frequency and transmitted power to realize adaptive power control. Depending on the separation distance, it is possible to consume more or less power and bandwidth to compensate for the BER. In other words, for a short distance communication ranging from 0 mm to 5 mm with high SNR, the transmitter will maintain a low P_t and operating frequency. Likewise, for a a medium-distance ranging from 5 mm to 15 mm, only the operating frequency will be increased. For a long distance that is greater than 15 mm, both operating frequency and P_t will be enhanced to ensure the lower power consumption and BER performance.

5. Conclusions and Future Work

In this paper we systematically analyzed the propagation path of hybrid WiNoC under two on-chip antennas, including millimeter-wave on-chip antenna operating at GHz frequency and optical on-chip antenna operating at THz frequency. Base on those analyses, we modeled the wireless multi-path propagation channel for hybrid WiNoC. Especially, the path loss model has been present based on reflection coefficient using 3-D Descartes coordinate system for optical on-chip antenna. To ensure the reliable communication, the BER performance, delay spread and coherence bandwidth have been evaluated in detail. The simulation results demonstrate that the path loss could be reduced about 12 dB on average by increasing the resistivity of silicon substrate. It also shows that the path loss in THz frequency range could be improved about 6 dB by elevating the height of antenna pairs from 0.6 mm to 2 mm. Especially to improve the BER performance, we proposed a channel coding scheme to realize error control. Furthermore, through analyzing the influence factors of BER, the simulation based on the model shows that the SNR budget at the end of receiver must be greater than 15 dB. In order to ensure the overall performance for the on-chip wireless communication. We proposed an adaptive power control mechanism by making a balance between BER, operating frequency and transmitted power.

In ongoing and future investigations, we intend to evaluate the area and power consumption overhead under different coding scheme for improving the BER performance. Furthermore, we will improve the coding with no handshaking communication to reduce the communication delay and power.

Acknowledgements

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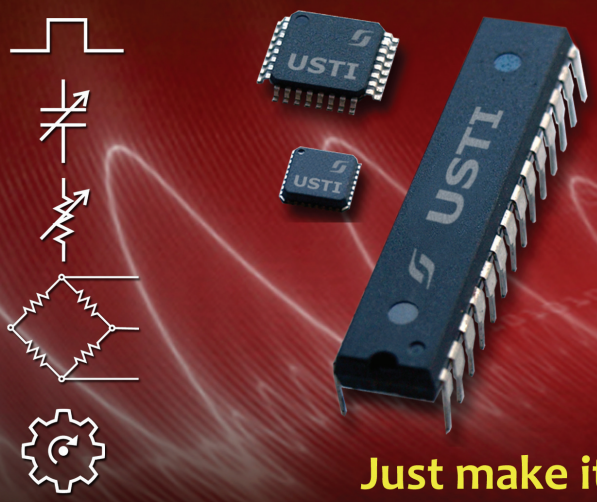

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Construction of Smart Sensor Networks Data System Based on Integration Formalized BCCS Model

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Abstract: Smart sensor means having information detection, information processing, information storage, logical thinking and judgment capabilities of the sensor. In this paper, CCS and B pass-by-value method and language, by giving a lightweight description language to achieve the BCCS model can be used to describe concurrent or distributed systems. The paper presents construction smart sensor networks data system based on integration Formalized BCCS model. The experiment shows that the B method combined with CCS on the powerful formal description of actual smart sensor networks data system capacity. This smart sensor can calculate the value of a physical quantity by using compensation sensor network data in order to improve intelligence. *Copyright © 2013 IFSA.*

Keywords: Smart sensors, B method, Value-passing CCS, BCCS, Sensors network.

1. Introduction

Smart sensors not only visual, touch, hearing, smell, taste function, and should have a memory, learning, thinking, reasoning and judgment "brain" capability. The former is accomplished by conventional sensors. Here, the functional configuration of conventional sensors include sensitive components, and analog conditioning circuit (ADC), an objective that sensitive components will be described with the physical state of the environment or the characteristic parameters of circuit elements into or state parameter, conditioning circuit parameters of the circuit converted into a voltage signal and normalized to meet the dynamic range of the ADC. ADC output intelligent processor should be intelligent digital signal processing.

Smart sensor means having information detection, information processing, information storage, and logical thinking and judgment capabilities of the sensor. It not only has a variety of conventional sensors function, but also with data processing, fault diagnosis, nonlinear processing, self-tuning, self-tuning and human-machine communication and other features. It is microelectronics, micro-computer technology and detection technology combined with the product. Early intelligent sensor is the sensor output signal is processed and converted by the interface to the microprocessor after the arithmetic processing part.

Our research combines methods and language of value-passing CCS and B given a lightweight describe language, achieve the BCCS model can be used to describe the concurrent or distributed systems. On the basis of the return value CCS operational semantics, combined with B on the definition of the abstract data structure, and it is the system constraints and the functional processing given BCCS operational semantics [1].

The BCCS model make up the traditional single method separates the various facets of the information systems inadequate. The BCCS model,

two important facets of information systems has been unified, with B to describe the function, these functions are defined by the operation and status of modeling, with value-passing CCS to describe the behavior of the system characteristics, such as concurrency, real-time, the distribution of B described the function to define the value of the value-passing CCS. Expect to take the BCCS model, can reduce the difficulty of formal methods to describe and verify large concurrent software systems, while improving the reliability of concurrent or distributed software development.

Smart sensor further includes means to push. The design of this type of sensor has begun to change methods, not as much as a simple hardware configuration, the need for a simple software processing channel Movement, the corresponding output signal is a digital signal; from the "live part line" concept made after the field bus-based measurement and control system has been widely used, corresponding to the sensor design also proposed new requirements. From a development perspective, the future of independent single sensor will be less and less use of the occasion, more of multi-sensor system applications in order to achieve multi-parameter measurement and control of multiple objects. Measurement and control of the exchange of information is at the bottom line of the main unit through the field to complete. Data exchange is mainly through the Intranet and other networks to achieve. The paper presents construction smart sensor networks data system based on integration Formalized BCCS model.

2. BCCS Syntax and Operational Semantics Based on CCS and B Method

2.1. Research of CCS and B Method

Concurrent increase in the semantics of basic B, the original B method of operation (Operations) clause to event (EVENTS) clause to increase the dynamic clause (DYNAMICS), seen purely as a B abstract machine (AMN) is an interactive system. So Event B is similar to the Petri net description of dynamic, complicated process tool, but also inherited the function of B in the data description, and its basic structure, but also to some extent affected the simplicity of the method B, ease of use, more importantly, is such that the B in the description of the sequential model, the method becomes complicated and incomprehensible.

BCCS model uses the idea of integrated value-passing CCS and B language, with the B language to extend the ability of the CCS data and function definitions, and B method extended through CCS clear that the ability of the operational semantics and describe concurrent systems, forming amore expressive formal development method to describe the problem from a more facets [2]. BCCS model describes the syntax of the language, but it lacks the support of the semantic, combined with B on the definition of the abstract data structure on the basis of the return value CCS operational semantics constraints and the functional processing of the system, given the operation of the BCCS semantic, further portrayed BCCS model.

Smart sensors according to the given environmental conditions conventional sensors and a priori knowledge, the processor using the digital calculation, the hardware automatically compensate for conventional sensors linear, nonlinear and drift caused by environmental factors and signal distortion, to best recovery test signal [3]. Calculation method implemented in software, hardware defect compensation to achieve the purpose of the software.

The B method originally proposed by J.-R. Abrial, it is a theory and method of the formal development of computer systems used in the real-time domain has been applied. B, mentioned in this article is actually short for B technology in software engineering methods and tools set, it contains the B method tool B, B Tool Box is a sound practice-oriented software process based on the mathematical theory of technology B used symbols and supports most of the software process: requirements analysis, specification, software design, implementation and maintenance of the step-by-step structure of layered software along with step-by-step validation and verification of the guiding principles of the B method [4]. The B tool box includes a lot of tools, all the tools integrated in a window-based software development environment, these tools can be integrated to run automatically, you can support the entire process of the use of the B method development software tool B is gradually constructed to support software the confirmation process available static analysis, dynamic analysis using simulation technology, integrated theorem proving correctness proof.

B method using a simple pseudo programming language to describe the demand model, and software design and implementation, this language is an abstract machine symbols (AMN, Abstract Machine Notes), AMN type detection support the specification, dynamic verification of mathematical proof ensure that the design process indeed ended.

The BCCS system, in order to achieve lightweight, only need a pass value of a subset of the CCS, and therefore value-passing CCS to do some cutting (Let's Take C_CCS to represent the cutting CCS). Let C_CCS detailed description given several constructors. C_CCS and gives the operational semantics, operational semantics and basic CCS operational semantics similar. CCS introduction to the standard we can also see that it is based on two grammar classes, a set of processes, and a set of actions. Us and said two grammar C_CCS set. Here is syntax defined C_CCS, as is shown by Fig. 1.

$CCS_prog_cut = Agent_Decl_cut \ldots Agent_Decl_cut$

$Agent_Decl_cut = C_Agent_cut \triangleq Agent_cut$

$Agent_cut = C_Agent_cut$

$Action_cut.Agent_cut$

$\overline{Action_cut}.Agent_cut$

$\tau.Agent_cut$

$Agent_cut + Agent_cut$

$Agent_cut \mid Agent_cut$

$Agent_cut \setminus resSet_cut$

o

$resSet_cut = \{Action_cut, \ldots, Action_cut\}$

Fig. 1. The defined C_CCS the syntax.

Finally, take a look at the dynamic semantics of CCS, CCS is a function of language, so the concept of the state not. People interested, if an action is performed before the process of expression is, then what will become of expression after the implementation of the action process. Actions usually mark, which, possibly with the parameter value refers to the action [5]. Mark indicates that the implementation of the action. This is essentially a reduction action; these rules can be a reduction rule, as is shown by equations.

$$a(x_1,\ldots,x_n).e \xrightarrow{\partial(v_1,\ldots,v_n)} e[v_1x_1,\ldots,v_n/x_n] \quad (1)$$

$$\overline{a}(f_1,\ldots,f_n).e \xrightarrow{\overline{\partial}(f_1,\ldots,f_n)} e \quad (2)$$

$$\tau.e \xrightarrow{\tau} e \quad (3)$$

$$\frac{e_1 \xrightarrow{\mu v} e_1'}{e_1 + e \xrightarrow{\mu v} e_1'} \quad (4)$$

$$\frac{e_2 \xrightarrow{\mu v} e_2'}{e_1 + e_2 \xrightarrow{\mu v} e_2'} \quad (5)$$

$$\frac{e_1 \xrightarrow{\mu v} e_1'}{e_1 \mid e_2 \xrightarrow{\mu v} e_1' e_2} \quad (6)$$

$$\frac{e_2 \xrightarrow{\mu v} e_2'}{e_1 \mid e_2 \xrightarrow{\mu v} e_1 e_2'} \quad (7)$$

$$\frac{e_1 \xrightarrow{\mu v} e_1', e_2 \xrightarrow{\mu v} e_2'}{e_1 \mid e_2 \xrightarrow{\mu v} e_1' e_2'} \quad (8)$$

Through the use of a hybrid model-based language is VDM formal methods in the process algebra CCS. It constructs a mixed-language based on CCS and VDM to CCS-defined value. All operations on the data in this language VDM to define and call by value-passing CCS, the semantic object is Milner defined CCS semantics instead of VDM.

CCS is a functional language, its basic ingredient is called process (or stroke), process expressions containing free variables, the combination of the process is the process, the combination of depth is also arbitrary [6]. CCS has a pair of communication primitive groups label set, which is a set of names, and the anti-set of names $\overline{\Delta} = \{\overline{\alpha} \in \Delta\}$, which are complementary, and the set is called the label set, its elements unified to represent $\Diamond = \Delta \bigcup \overline{\Delta}$, on a self-remapping, indicated by $\overline{\overline{\lambda}} = \lambda$.

B tight management control over the entire software process description stage early in software development specifications, developers will control the user needs to verify the consistency of the test system demand model, at each stage of development, each design description and specification requirements document control its consistency, the lowest level of the design components from the pre-defined and reliable assembly of parts library, and translated into executable code, reliable predefined parts (many parts for the B tool-sets generated by testing or reuse application), the new lower layer member can be further added to these added components may be provided by the existence of reliable components and development, and to test their properties.

The constant part of the environment is with the definition of the B environment. With the handling of the environment and the process in the of Bruns early formal operational semantics "parameter residual problem, it is because the process to introduce the values of local variables are allowed memory caused by the environment. BCCS model environment to describe this constant environment, one is described by B the global variables, and the other is a local variable of a process, which ensures that the two does not occur previously mentioned, due to the "Parameter residue" the cause of the conflict.

ZCCS this article is a hybrid integrated formal methods on the basis of value-passing CCS, to borrow a Z language model (schema) and schema text to extend the value-passing CCS value. The advantage of this approach and our article, and it is the idea of a clear, easy-to-use structure. But due to lack of Z Language modular capacity, it is difficult to identify the impact of a state model for all modes of operation, can not support the reuse of specifications, the Z language difficult to be handled directly by the computer and other defects, and limits the practical application of ZCCS in large systems. In order to overcome these deficiencies of the Z language, we use the B method and language instead of Z and CCS do hybrid BCCS model, so ensure that the next premise of ZCCS advantages, B and language is also

very good to solve due to a defect of the Z language itself insufficient of the ZCCS model.

Smart sensors the sensor during use all the problems involved includes. Shown sensitivity, zero drift, calibration, etc., even if a single physical measurement sensors must also have a standard interface to obtain the required information (such as temperature, humidity, etc.) in order to achieve and calibration of measured physical quantity. On many occasions, plus the output display unit, this single Smart sensor system has already covered the concept of traditional instruments.

2.2. BCCS Syntax and Operational Semantics

The BCCS model is more similar to the syntax and semantics of value-passing CCS. In the BCCS, the value of the value-passing CCS expression to describe B, which include: process and input action parameter declaration; parametric process and output action arguments (assignment); condition statement predicate expression. In order to better describe the syntax of the BCCS, its operational semantics, we need to define the BCCS syntax met language.

The input parameters for the action in the Bruns value-passing CCS is not global, but also for any action with the Senate, not to do a symbolic statement, nor the value of these symbols can be obtained for a correct description. In BCCS model, the parameters can be defined in any output action.

Deduction is assuming that you want to prove a conjecture, for example, it is rarely isolated to you, in most cases, the need to prove something (also known as the premise or assertion) relies on certain assumptions, the proof can assume that they are true, some when the suspect is under proof, also said that by the assumptions to deduce conclusions recorded as the following formula.

$$HYP \vdash P \tag{9}$$

$$\frac{}{P \vdash P} \tag{10}$$

$$\frac{HYP \vdash PHY \; P \subseteq HYP'}{HYP' \vdash P} \tag{11}$$

$$\frac{P \subseteq HYP}{HYP \vdash P} \tag{12}$$

$$\frac{HYP \vdash P \;\; PHY, P \subseteq Q}{HYP \vdash Q} \tag{13}$$

The process can be similar to the definition of Bruns in CCS, with B to describe the grammar class B described. In a description of the list, you can introduce a collection of axioms description to define constants for the conventional definition and context. In BCCS, before the introduction into the process with B to define the structures previously mentioned.

A very important value-passing is CCS operator. Action drive, the state of the process can be constantly changing, we discuss the action parameter expression and next, we look at the CCS action itself what should in BCCS model expression.

Another problem is that in an environment with the handling of the environment and the process must be recorded multiple concurrent state of the process variables. This means that if the variable is shared by multiple parallel processes, so it is necessary to perform the conversion different process to ensure that the only variable. In the BCCS, based on the above-mentioned environment constructed into a binary tree, in order to better simulate with operator | or \ process environment, and the environment of different processes in different branches, on the front of the mentioned problem, which is reflected in the operational semantics of inference rules.

Traditional value-passing CCS, the representation of the action is a very abstract concept, and do not have a detailed description of the list. In the within the BCCS model, we put the CCS action through an implicit function converted to a B abstract machine defined operation (OPERATION) [7]. The implicit meanings mentioned here BCCS model description of a concurrent system, you can direct the operations defined in the B abstract machine as CCS action. Here there is a problem in the abstract machine B categories, but not on the input and output events CCS in the operation of the input and output strict classification. There are two ways to solve this problem, a semantic representation of extended abstract machine B, B abstract machine classification of input and output events, the second method is to add a descriptor events using B abstract machine specify the operation as an input or output action.

B inside a few very important binary relation, we study involving several were to be introduced, for each operator, first introduced its syntax, then a non-formal explanation to each operator. We first introduce a binary relation from one set to another set (), and then introduce a relationship of inverse ($\leftrightarrow$), also known as anti-relationship, the next is the relationship between the domain (dom) and range (ran), then the relationship between composite (;) and constant relationship (id) and finally, the relationship between the various restrictions ($\lhd, \rhd$). The table is not the same each other, and the situation p, q, s and t are as follows (Table 1).

The BCCS The operational semantics based Bruns research in the semantics of value-passing CCS and B semantic foundation up to construct. Defined value in BCCS general description from B, rather than by value CCS values defined, so that the expression that defines the content richer, of course, if CCS can define, we do not limit it. A clear description of the BCCS operational semantics given

26 operating rules, used to describe the behavior may occur in all BCCS description. Operational semantics allows us to better understand the BCCS, and ensure that the BCCS syntax correctness, completeness and consistency.

Table 1. Binary relation defined table.

Grammar	Definition
$u \leftrightarrow v$	$\mathrm{P}(u \times v)$
p^{-1}	$\{b,a \mid (b,a) \in v \times u \wedge (a,b) \in p\}$
$dom(p)$	$\{a \mid a \in u \wedge \exists b \cdot (b \in v \wedge (a,b) \in p)\}$
$ran(p)$	$dom(p^{-1})$
$p\,;q$	$\{a,c \mid (a,c) \in u \times w \wedge \exists b \cdot (b \in v \wedge (a,b) \in p \wedge (b,c) \in q)\}$
$q \circ p$	$p\,;q$
$id(u)$	$\{a,b \mid (a,b) \in u \times u \wedge a = b\}$
$s \lhd p$	$id(s);p\ \{a,b \mid (a,b) \in p \wedge a \in s\}$
$p \rhd t$	$p;id(t)\ \{a,b \mid (a,b) \in p \wedge b \in t\}$

The environment is similar to the constant part B on the definition of environment [8]. Bruns early formal operational semantics with the handling of the environment and the process, as well as on the expression of the reference rules appear the same problem is the parameter residual "This is because the process of the introduction of the value of the local variable is allowed the memory caused in the environment. If the process follows the formula is the equation 14.

$$in(x : \{0,1,2\})A \tag{14}$$

Three free appears likely to have different values in the elements of the set of constraints may be derived from the constant introduction of a process parameter, before migrating to this point in the process, the other two parameters are the individual input action binding. Through this example, you can clearly see that the component of both processes in their own environment, the constraint set also has its own environment. Let us assume that a process form, we must have an environment to describe the status of its variables, with its constant environment, to provide a value for the free variable, as is shown in Fig. 2.

How to define the object described in the description of the list just beginning moments. If you do not have this information as the basis behind the objects described would be meaningless. Here, we assume that all statements are closed, that is, it should not contain any free variables. We can consider that the initial environment is empty that is, starting from a particular environment; you can use a non-formal methods.

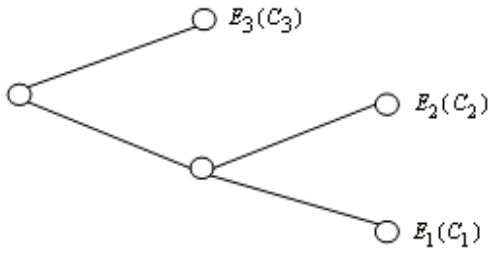

Fig. 2. The defined C_CCS the syntax.

We give the axiomatic structure is based on the simplified classical set theory, set theory and mathematical fields, we prefer the formal application, so are more concerned with the notation. Suppose for all BCCS grammar class can collection description, and each grammar fragment corresponds to a unique element of this collection, based on set theory and operational semantics is put the grammar variables described as a value expression [9]. In addition, it is assumed that a collection of the range of each variable, so the premise of the rule of each reference is $v \in S$.

BCCS described in Part B may be constructed to the global variable environment, CCS portion may be configured to a process environment, the descriptor list can be generated environment, where "local variable environment corresponding to an empty (this is because the initial process is not the environment branch so is empty).

Parameters of the process of constant, if there is an environment in which the local environment is described by the B abstract machine simple environment, global variable environment overloaded extension from the context of the environment, the value of the expression and characteristics of tulles the same value, which belongs to a local environment can migrate through action to reload, constants and parameters in the process environment bindings to the process, you can migrate to the after action.

SFSB instance pool, the EJB container provides a method similar to the operating system in a "context switch" approach for SFSB. In order to limit the SFSB number of instances running in memory, the EJB container needs to be SFSB session state saved to hard disk or other storage source, a process known as aestivation, SFSB instance customers call it up again , passivity session state back to the SFSB instance, a process known as activation (activation).Note that activate SFSB the instance is not necessarily SFSB instance is originally used, however, the new instance of the SFSB session state is still SFSB instance is activated session state.

This system has two components, sampling (samples) components periodically sampled radius and angle of the object, and accept conversion trigger (convert) to the conversion component (converter). Converter component to accept and it is converting them to a Cartesian coordinate output. The sampling component sampling radius and angle can be in any

order. Conversion component in addition to the polar coordinates to rectangular coordinates output, but also remember the last time the converted value for the next conversion before receiving a repeat (repeat) trigger output was recently Cartesian coordinate values.

Ordinary input action prefix, for any without operator | and \ local environment is a simple environment (do not call the function), this environment is the environment of a B abstract machine described here is not (environment tree).This rule means that the process environment can be migrated to the action (input).

$$\frac{\{\alpha\} = ran(Inv \circ id(\{\alpha\}))}{<<BE, BE', AE>, \bar{a}.F> \xrightarrow{\alpha} <<BE, BE', AE>, F>} \quad (15)$$

There are three types of distinct reference to the rules defined in the value-passing CCS description of the meaning of the first class describe how inherited value-passing CCS extended environment, where the environment contains a set of processes and a set of constants, symbol sets, action sets [10]. The second category describes how by converting a (environment, process) migrate to another (environment, process).The third category describes how to represent the value of the expression in the CCS.

The semantics of the CCS semantics and BCCS biggest difference is the third type of reference rules. Its expression is in the BCCS B given directly. Another difference is in BCCS introduced on the environment, in the discussion with parametric input operation, automatically includes any need to determine the type information of the range of parameters, since the BCCS B abstract machine the description portion and is defined on its basis's already made a judge.

A smart sensor node is composed of three parts: the traditional sense of the sensor, the network interface and a processing unit. According to different requirements, these three components can be used together to form different composite chips, it can be a single type. First, the sensor measuring a physical quantity to be converted to electrical signals by A/D converted digital signal through the microprocessor data processing (filtering, calibration) will send the results to the network, and the network data exchange is not the network interface module.

3. Construction Smart Sensor Networks Data System Based on Integration Formalized BCCS Model

Smart sensors to have a certain judgment, analysis and information processing capabilities, according to the work of the whole system regulating the various parts of the data transfer with the host computer, the system operates in a low power state and the optimal transmission efficiency optimized state. Currently there are several ways to achieve intelligent sensors, one is the common sensor and a microprocessor with digital bus interface coupled with signal conditioning circuitry combined together as a whole constitutes an intelligent sensor system, this non-integrated smart sensor is field bus control system, driven by the rapid development of the situation to develop. Another is the integration of smart sensors; this smart sensor system is sensitive components, signal conditioning circuitry and microprocessor unit integrated in a chip composition. In addition it was also taken semi-integrated manner, these several components into a few chips to form to facilitate their use of smart sensors.

Smart sensor system is the use of micro-machining technology and LSI technology, the use of silicon as a basic material to produce sensitive components, signal conditioning circuits and micro processing units and integrate them in a chip composition. This makes the intelligent sensor reaches the miniaturization, structural integration, thereby improving the accuracy and stability. Sensitive components constitute arrays, with the corresponding image processing software; you can achieve a number of graphic imaging and form image sensors.

The basic idea of the operating semantic description language abstract way the effect of the implementation of each component, avoid the semantics described depends on the language used in the specific computer. With BCCS operational semantics can better understand the BCCS model, to ensure the correctness of the syntax, completeness and consistency.

EJB container defines a method B abstract machine, portray some operations (actions), these operations are basically restore some of the ways or APIs EJB3, reflects Beans may have experienced in their life cycle event (action). With these events in the life-cycle process, the use of CCS can easily describe the EJB3.0 SFSB distribution of its concurrency features. By a formal description of the SFSB lifecycle, in the EJB3 specification of the SFSB lifecycle process more clear and consistent, showing BCCS powerful formal description of the actual system capacity.

This design of smart sensor system has the following characteristics: (1) the use of commonly used industrial site Mudbugs communication protocol, the programmable controller and communicate with other devices, using the RS-485 bus structure, get rid of the previous multi-line multi-point type sensor connection structure, simplifying the number of signal cable industrial site, standardized site layout, ease of installation, commissioning; (2) real-time, accurate and stable digital signal acquisition and 4-20 mA analog signal; (3) in order to avoid interference from external electrical signals, and does not affect the use of high communication baud communication effects, using high-speed opt coupler 6N137 the MAX485 with microcontroller PIC16F687 optical isolation. Select

both isolated power module for more stable, lower power consumption.

From smart sensor network applications, its general use in the automation field of measurement and control level, between the Ministries of cable needed to connect through live together. For different applications, now there are many different bus standard protocols. To ensure that the design of the sensor is difficult to fully meet these protocols, interface issues which must be considered. This is a smart sensor network sensor biggest difference with the ordinary.

The use of formal methods for software development for the design, should be used in the discussion of "ideal" process algebra to describe concurrent structure and communication, but must rely on the description of the model-oriented method to declare the operation behavior. This paper presents a hybrid B language and process algebra CCS to build a BCCS model. In this model, we are given a language, basically mentioned in the first question can be resolved, and that value-passing CCS can provide rich description of data, while a partial solution to the second problem, that is the same model within demand for describing the behavior of the model-based demand, as the BCCS implemented software environment in the current study did not give, so the second question is just on the level described [11].

Let's further describe the dynamic part of the SFSB life cycle. Instance SFSB has not yet been instantiated that is no longer the system memory which does not exist. Useful resources through the introduction of a new instance are injected into the data members of the Bean, Bean class defines the @PostConstruct callback method, and the container will call the method.

$$ini_SFSB \equiv newIns\tan ce().rdy1_SFSB(nb) + 0$$

$$rdy1_SFSB(nb) \equiv {}_{\text{if}}\ opt_di\ {}_{\text{then}}$$
$$dependencyInjection.rdy2_SFSB(dib)$$
else
$$rename.rdy3_SFSB(nb)$$
$$rdy2_SFSB(nb) \equiv {}_{\text{If}}\ opt_pc\ {}_{\text{then}}$$
$$postconstruct.rdy_SFSB(beanIns)\ {}_{\text{else}}$$
$$rename.rdy_SFSB(beanIns)$$
$$rdy3_SFSB(nb) \equiv {}_{\text{if}}\ opt_pc\ {}_{\text{then}}$$
$$postconstruct.rdy_SFSB(beanIns)\ {}_{\text{else}}$$
$$rename.rdy_SFSB(beanIns)$$

Instance in the ready state, he can be free to accept a call request from the client; can be transactional or non-transactional nature. Timeout is removed or call @PrePassivate method the corresponding entering not instantiate or passivity.

Smart sensor network data can be easily detected by real-time processing of large amounts of data may also need to store them. The main purpose of storing large amounts of information is to prepare for later queries, this type of information include the history and information about the device probing analysis of the results of indexing, intelligent sensor with a digital communication interface, this interface can be directly through the computer to which it belongs communications and liaison and exchange of information. In addition, intelligent sensor information management program is very simple and convenient, for example, the detection system can be remote control, or the lock mode, work, and the measured data can be sent to the remote users.

$$\frac{BCCS_\text{Prog} \vdash BCCS_Env,\quad B \vdash BE\ \ CCS \vdash AE}{BCCS \vdash < BE, \{\}, AE >} \tag{16}$$

With the sensor technology and computer network technology, including micro-mechanical systems and the associated connector, the signal processing technology, the development of network interface smart sensors, and put them on the network and for distributed control has become possible measure and gradually heating up [12]. The Smart sensor network can be composed of multiple high-precision, powerful control network. And low cost, installation and configuration, upgrades and maintenance. Smart sensor networks to make it more complete monitoring capabilities are widely used in automotive, industrial process control, health care, environmental monitoring and so on.

Assuming that the shape, such as a process or just a constant environment (i.e. the environment), but is in a constant environment, and the below-mentioned process brought. This can be achieved by a rule, with the non-formalized in terms of the method is: if the operator with a | or \ and the local environment of the process to migrate to a new process with the new local environment are possible, with the local environmental process can be done, which is behind the operational semantics rules Rule21.

Because of a timeout when the instance is in the passive state, the same will be removed, of course, can also be accepted the method @PostActivate re-return to the ready state.

The final with a SFSB's life cycle can be defined with the following equation. Because a SFSB is equivalent to a client on a server extension, then the following equation can also be regarded as the state in which the client.

Smart sensor system can be measured under feelings and follow certain rules to convert the signal device or devices, usually by sensitive components and conversion components, seen from the system, the sensor is a detection device being measured can feel information, and can feel detecting information according to certain rules converted into electrical or other forms of information output required to meet the information transmission, processing, storage,

display, recording and control requirements. The importance of the sensor is embodied in its automation, scientific testing, measurement accounting, monitoring and diagnosis systems such as the basic link, as is shown by Fig. 3.

The paper presents construction smart sensor networks data system based on integration Formalized BCCS model. As computer hardware and software technology penetration, the microprocessor and sensors can be combined to produce with a certain data processing capabilities, and can self-test, self-calibration, self-compensation of a new generation sensors, smart sensors. Smart sensor network data processing occurs on the sensor development had a profound impact. Smart sensor network communication technology gradually matured and penetrated into all walks of life, all kinds of high reliability, low power, low cost, micro-volume network interface chip is developed.

The BCCS model is given the operational semantics to build a relatively complete the BCCS theoretical model. In this model, it is co-organized by the B and CCS a clear framework for the description of the data and behavior of complex systems. However, the relative B language and CCS speaking of BCCS the syntax is slightly more complex. BCCS in more detail characterizes the operation, but relatively CCS speaking, these operations will produce a more complex process. If the result of the operation is to be determined, will be relatively better, however, if the operation result is not determined, this problem will become more prominent.

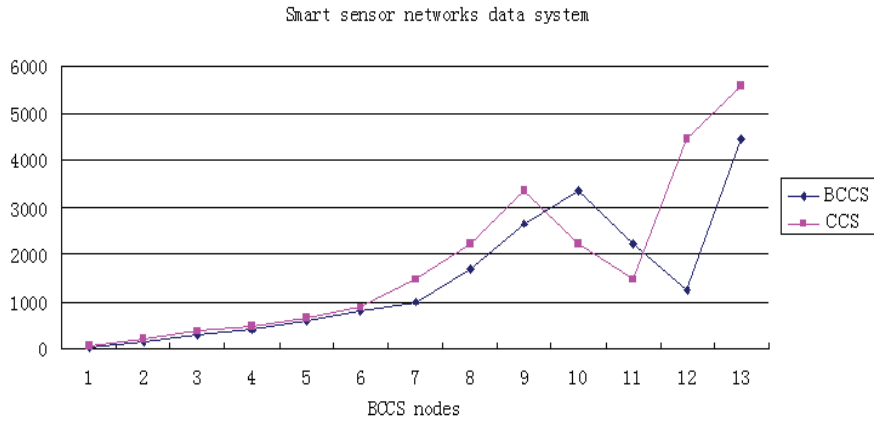

Fig. 3. Comparison results of smart sensor networks data system based on Formalized BCCS with CCS.

4. Conclusions

The main features of Smart sensor with high accuracy; has a good dynamic response speed; high reliability and stability; high SNR and high resolution; adaptive capacity; prices low, cost. Smart sensor to achieve the main way to achieve non-integrated: the traditional sensors, signal conditioning circuits, microprocessors and other combinations which constitute the system as a whole. Is an original technology based on the most economical and efficient way. The final realization of the basic norms is formal description on the EJB3 SFSB life cycle. EJB container defines a B abstract machine, and at the same time to portray some operations (actions), these operations are basically restoring the EJB3 some way (or APIs) reflects Beans may have experienced in the course of its life cycle event (Action). With these events in the life-cycle process, the use of CCS can easily describe the characteristics of the EJB3 SFSB the distribution of its concurrency. By the formal description of the SFSB life cycle so that the EJB3 specification SFSB the lifecycle process more clear and consistent, and also shows the B method combines CCS powerful formal description of the actual system capacity.

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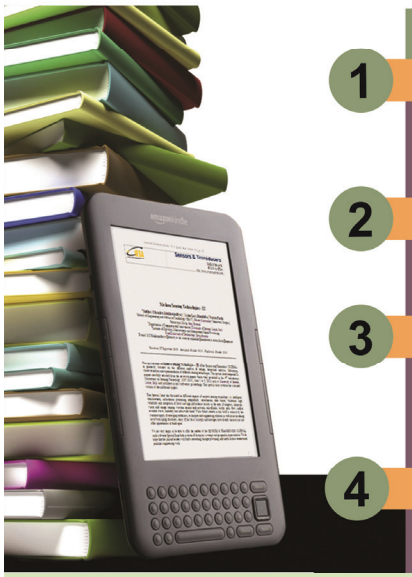

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Development Novel Organization Structure of Wireless Sensor Network Protocol Based on ZigBee Technology

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Abstract: ZigBee technology is a kind of two-way wireless communication technology in short distance, low complexity, low power consumption, low data rate, low cost, which is suitable for the automatic control and remote control. The wireless sensor network may be arranged in a hostile environment, in order to prevent the supply into the fake information to the network, need to realize the security of multicast source authentication based on wireless sensor network. Therefore, this paper presents implementation scheme of wireless sensor network technology 2.4 GHz ZigBee and 433 MHz combined. Each sensor node can be collected, simple computing environment data and to communicate with other nodes. The paper put forward development novel organization structure of wireless sensor network protocol based on ZigBee technology. The experimental results show that the terminal node has extremely low power consumption, long time work continuously, and this method has good self-organization, self-healing function. *Copyright © 2013 IFSA.*

Keywords: Wireless sensor network, ZigBee, Sensor nodes.

1. Introduction

As a result of computation, communication and sensor three technologies combined with wireless sensor network, is a novel technology about acquiring and processing information. Due to the improvement of recent micro manufacturing technology, communication technology and battery technology, ability makes tiny sensor has sensing, wireless communication and information processing. This kind of sensor not only can sense and detect the target object and environment change, and processing the data collected, and the processed data in wireless transmission mode to the data collection center or base station [1]. This tiny sensor is usually composed of sensor unit, data processing unit and communication unit, sensor, data processing unit and communication modules of tiny nodes through the self-organizing way constitute a network integrated random distribution.

As a newly emerging technology, wireless sensor network to establish a good, good robustness is still faced with many challenges. But due to some special characteristics, are very different design method of wireless sensor network and the existing wireless network. For example, since the sensor nodes in sensor network distribution is dense, so the data management and processing technology of large range. Secondly, the wireless sensor network node deployment in general human to reach and contact area is facing great challenges to maintain the sensor network node. In addition, the power consumption is a very important issue, wireless sensor node is a tiny device, with limited power, in some applications, replace the power supply is almost impossible.

ZigBee technology is a kind of two-way wireless communication technology in short distance, low complexity, low power consumption, low data rate, low cost, which is suitable for the automatic control and remote control, can be embedded in various devices, and supports for the geographical location. With respect to various wireless existing wireless communications technology, ZigBee technology will be the lowest power consumption and cost of technology. The ZigBee protocol is suite by the application of the application of high standards, convergence layer, network layer, data link layer and physical layer.

The data link layer and medium access control layer provides reliable communication channel for the neighbor nodes, in the MAC protocol, node to determine its ability to access communication channel by monitoring the neighbor node sends data. The carrier sense method is particularly vulnerable to denial of service attack is DOS. Methods using carrier sense in certain MAC layer protocol to neighboring nodes use the channel. When the channel conflict, nodes use a binary value index regression algorithm to determine the time to send data, conflict the attacker only needs to produce a byte can destroy the entire packet [2]. Because as long as part of the data conflict will cause the receiver to check data packets and do not match. In response to control information ACK receiver will send the data conflict sending node according to the binary exponential regression algorithm to choose the transmission time.

In wireless sensor networks, data processing by the node itself, the purpose is to reduce the amount of data transmitted in wireless links, only the relevant information with other nodes in the link transmission. Characteristics of data-centric is another feature of wireless sensor networks, because the nodes not planned in advance, but the node location is not determined in advance, so that there are some nodes due to the occurrence of many wrong or cannot carry out the specified task and stopped running. In order to monitor the target object in the network, configuration of redundant nodes is necessary, communication and collaboration, sharing of data between nodes that can, can guarantee to obtain the monitored object more comprehensive data. Wireless sensor network is a wireless network composed of a set of ZigBee nodes with Ad Hoc, its purpose is to perceive objects regional cooperative sensing, collecting and processing information in the network coverage, and released to the observer sensor, object perception and observation. The paper put forward development novel organization structure of wireless sensor network protocol based on ZigBee technology.

2. Structure of Wireless Sensor Network Protocol Analysis

Sensor network deployment of large-scale networks in complex environment, real-time data acquisition and processing brings hope. But at the same time, WSN is usually deployed in unattended maintenance, can not control the environment, besides the general wireless network facing the information disclosure, information tampering, replay attack, denial of service and other threats, WSN faces the sensor node easy for an attacker to physical manipulation, and get all the information stored in the sensor node, so as to control the part of the network threat. The user can not be accepted and deployed without a sensor network to solve the security and privacy issues, so in the WSN protocol and software design, should fully consider the security problem of WSN may face, and the safety mechanism is integrated into the system design. Only in this way, can promote the extensive application of sensor networks, otherwise, can only be deployed in Sensor Network Limited, controlled environments, the ultimate aim and sensor networks -- to achieve universal computing and become an important way of people's life is contrary.

The data link layer in the protocol stack usually provides two main services: media access control (MAC) and error control [3]. In a variety of MAC, carrier senses multiple accesses (CSMA) in ad-hoc sensor networks are the most commonly used. This is mainly because it is easy to implement, but more important is that it can improve the rate of channel multiplexing large network.

Wireless sensor nodes by multi-hop forwarding, through the access network gateway node, management, classification, processing of sensory information in the task manager node network, then the sensing information to the application user. Sensor nodes first acquisition such as sound, light and distance environment related data, and these data are sent to the gateway node simple processing. Wireless sensor networks usually have two kinds of application modes: active polling mode, passive mode. Active mode requires the gateway node active polling for each sensor nodes to get the news, while passive mode requires that in the event of a sensor node occurs, the gateway node can be timely response. Each sensor node data can be combined, which greatly improves the efficiency of sensor network. Of course, this also requires sensor nodes to computing ability, as is shown by equation 1.

$$\begin{aligned}\mu_{s|a} &= E\{s(k) \mid a(k)\} \\ &= \mathrm{M}^{-1}\{\beta(k)^{\mathrm{T}}\Sigma_{\varepsilon(k)}^{-1}\left(a(k)-\alpha(k)\right)+\frac{s_0(k)}{\sigma_{s(k)}^2}\}\end{aligned} \quad (1)$$

In view of the environment and structure state monitoring, we design a universal wireless sensor network hardware platform; the hardware platform is composed of a number of sensor nodes, Sink nodes with wireless receiving function and a computer. Wireless sensor nodes distributed in the monitoring area, performing data acquisition, processing and wireless communication, Sink node receives the sensor data to the wired way and sends the data to the computer.

The design goal of traditional MAC protocol is to maximize throughput, minimize the delay and fairness. About MAC layer protocol for WSNs design is to minimize the energy consumption, which determines that it should be appropriate to reduce the throughput and delay. Because the WSNs node always cooperates to complete an application task, so the usual fairness is not the main problem. In addition, some typical WSNs application also proposed is different from traditional wireless network design requirements for the MAC layer protocol.

Integrated circuits, micro electromechanical systems and the development of communication theory led to the emergence of wireless sensor networks [4]. This wireless sensor network is composed of many sensor nodes self-powered. Each sensor node can be collected, simple computing environment data and to communicate with other nodes and the outside world. The sensor node characteristics of sensor networks make many sensors can be made of high quality through collaborative work, and consists of a collection of good fault tolerance of the system. It is because of these advantages, in recent years there have been many wireless sensor network applications based on distributed detection and rescue, emergency rescue and disaster relief, intelligent home furnishing such as chemical and biological weapons attacks, as is shown by equation 2.

$$\hat{f}_n = f(u_n) = f[\sum_{k=1}^{K} w_k \sum_{m=1}^{M} x_m \psi(\frac{x_m - b_k}{a_k})] \quad (2)$$

The use of radio frequency module, there are two problems to be paid attention to: the first is when the transmission is completed, must send the enable pin and transmit data pin is set to low level. Otherwise one will consume energy, on the other hand, the RF module will always send a single carrier frequency signal, the disturbance to the surrounding nodes work; second is the wireless transmission module to transmit state from the standby state, there are about 20 s between the delay. In the meantime, input module data cannot be sent correctly, so ready to send before, should be sent to set high in advance.

Wireless ad hoc networks (mobile Ad-Hoc network) are a kind of tens to hundreds of nodes by multi-hop, mobile wireless communication network, dynamic peer-to-peer network. Its purpose is to through dynamic routing and mobility management technology, multimedia information transmission with quality of service requirements flow. Usually the node with energy supply continued. Wireless sensor networks with wireless ad hoc networks have similarities, but there is great difference. Sensor network is integration of the monitoring, control and wireless communication network system, the number of nodes is large, a network with thousands or even tens of thousands of node; node distribution is more intensive; due to the depletion of environmental impact and energy, nodes are more prone to failure; environmental interference and node failure caused by the change of the network topology; in general, most of the network node is fixed.

The majority of distributed MAC protocol uses carrier sense or collision avoidance mechanism and the use of additional signaling information to solve the hidden and exposed node problem. Competition is a random access MAC protocol based on nodes need to transmit data, using the wireless channel by means of competition. IEEE802.11 MAC protocol based on Carrier Sense Multiple Access with collision avoidance (Carrier Sensor Multiple Access with Collision Avoidance, CSMA/CA) is based on MAC protocol is a typical competition. Based on IEEES02.11 MAC protocol, the researchers put forward many sensors for sensor network contention based MAC protocol, S-MAC protocol, T-MAC protocol: for example, ARC-MAC protocol, Sift-MAC protocol, Wise-MAC protocol [5].

$$J_M R^* J_M (w_b^B)^* = J_M r_b^* = r_f = R w_f \quad (3)$$

Wireless sensor network architecture consists of three main parts: the sensor node, terminal node (Sink) and object of observation. Sensor nodes scattered collection and observation of the object of related data observed in the region, and after coordination to deal with data transfer to Sink. Sink can realize the communication of sensor network node and task management through Internet or a communications satellite.

The network layer is responsible for routing lookup and packet transmission. Since a large number of nodes in a sensor network deployment group are random, so look at the mesh network of multi-hop routing is very difficult, when the node failure or redeployment route maintenance and repair (self-healing) will be equally difficult. In the past few years there has been a large number of can support distributed routing algorithm for ad hoc mullion network. In general, these routing algorithms can be divided into two categories: active (proactive) and passive (reactive). In a proactive routing protocol, all nodes in the network are often maintained a route between the source address and destination address list, regardless of the need of these routing.

$$\sigma_\Omega^2 = \frac{1}{A_\Omega} \sum_{(x,y)\in\Omega} [(I(x,y) - \bar{I}_\Omega]^2 \quad (4)$$

Wireless sensor network consists of a large number of sensor nodes and a base station (BS), the base station node communication with other network access, environment monitoring sensor node and will collect the data to the base station. However, its limited energy, directly transmit the data to the base station will consume a lot of energy (Fig. 1). The routing method for multi-hop is not ideal, because the nodes closest to the base station will soon die for routing a received data, resulting in the later arrival

data cannot be passed to the base station. Routing method of other nodes in the PEGASIS, only with communication neighbor nodes, nodes take turns sending the fusion data to BS, ant colony algorithm considering the energy of each node in the route to choose the shortest path at the same time, based on consumption, so as to select the path more appropriate.

The hardware part of the gateway mainly consists of a central processing unit, a storage unit, frequency transceiver module and GPRS communication module, as shown in Fig. 1. The central processing unit of gateway is mainly used to treat from the data collected by sensors and finish some control function. The central processing unit of function realization is still using FPGA in Alter Cyclone series and NIOS soft core embedded processor [6]. This design makes it can complete some functions of rich applications in low cost, low power conditions. In addition, it can be integrated many peripheral interface, USB2.0 interface and Ethernet interface.

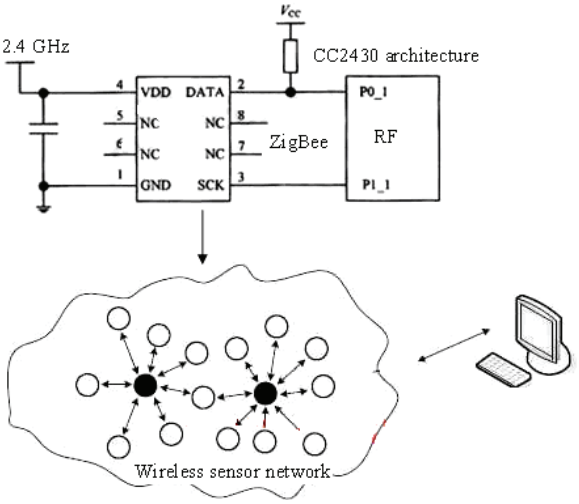

Fig. 1. The structure of Wireless sensor network based on CC2430 of ZigBee.

Wireless sensor network information with the external network or terminal connection between the need to achieve through the Sink node, Sink node is the wireless sensor network and cable connection, is responsible for sending the command (such as query, ID address allocation etc.), lower nodes receive requests and data, with data fusion, request the arbitration and the routing function, is one of the most important parts in wireless sensor networks. We design a Sink node with USB data and RS232 data, two kinds of data can be switched by a switch, for easy connection to the Sink and the external network or between terminals.

The basic features of WSN are energy limited. The MAC protocol as much as possible to save energy, such as reducing conflict and crosstalk, reduce the duty cycle and avoid long distance communication. The agreement shall also include the compromise mechanism, the user can improve the throughput, lower delay in energy saving and make a choice between. In addition, protocol designers should pay attention to energy is not readily available. Node is in a sleep state or the cause of death unknown.

The specific page programming operation is as follows: Command stage, to I/O port to send the page programming operation of the first command word (Ox80), said is page programming operation. The address phase, continuous send 4 address, K9Fl208 address register receives the address value, waiting to receive the data; when the data bus to send data, receive data until receiving continuous K9F1208, page second programming command word (0x10), that is the end of waiting to receive data state; R/B signal will remain "busy" for a period of time, then R/B is ready. The last bus issues a read status command word (Ox70), the K9F1208 command register to receive and respond to the command, said state data operation successfully to I/O port to send (Ox00) or said state data operation failure (OxO1), as is shown by equation 5.

$$f(x) = \frac{1}{nh^d} \sum_{i=1}^{n} K(\frac{x - x_i}{h}) \tag{5}$$

Nodes are battery-powered, the battery energy Co., and nodes may work in the inaccessible region, inconvenience frequent replacement battery. So in the design, energy conservation is the need to give priority to a problem [7]. First of all, SCM should perform tasks with the fastest speed, once it is possible to enter power saving mode. In the energy-saving mode, through the management circuit, will cut off the power supply device in addition to outside of the mcu. Enter power saving mode, if the monitoring center needs to access the node, through the RF transceiver module to awaken the node MCU.

Sensor nodes first acquisition such as sound, light and distance environment related data, and these data are sent to the gateway node simple processing. Wireless sensor networks usually have two kinds of application modes: active polling mode, passive mode. Active mode requires the gateway node active polling for each sensor nodes to get the news, while passive mode requires that in the event of a sensor node occurs; the gateway node can make timely response. Each sensor node data can be combined, which greatly improves the efficiency of sensor network. Of course, this also requires sensor nodes to computing ability.

Since the communication module to transmit power of wireless sensor nodes, nodes and the duty ratio is very small, a lot of data cannot be transmitted out at the same time, so it is necessary to have a manageable memory for storing the data collected, temporary or need other nodes forwarding the data collection. The 512 KB serial FLASHAT45DB041

data stored in this design. Compared with the common data memory, the chip has the characteristics of low power consumption, small volume, serial interface; the external circuit is simple, suitable for sensor nodes.

Two kinds of special security protocol in Wireless Sensor Networks: encryption protocol SNEP security network (Sensor Network Encryption Protocol) and stream authentication protocol time efficient tolerate packet loss based on TESLA. The function of SNEP is to provide node to receiver data authentication, encryption, refresh, TESLA is the function of broadcasting data authentication. Because wireless sensor networks may be arranged in a hostile environment, in order to prevent the supply into the fake information to the network, need to realize the security of multicast source authentication based on wireless sensor network [8]. But because of the wireless sensor networks, can not use public key cryptosystem, so multicast source authentication is not easy to realize. Security protocols for sensor network SP INK gives uTESL A multicast source authentication based on, the scheme is the improvement of the TESLA protocol, which is suitable for sensor networks.

Usually, in wireless sensor networks, a large number of sensor nodes are densely distributed in an area, the message may be required after a certain number of nodes to reach the destination, but also due to the dynamic nature of the sensor network, so there is no fixed infrastructure, so each node needs to have routing function. Since each node is a potential routing node, and therefore more vulnerable to attack. The main attack types of wireless sensor network.

$$\mu_{s_0} = E_{\hat{k}}\{s_0(\hat{k})\} = \sum_{\hat{k}\in\mathcal{R}_{\hat{k}}} p(\hat{k}) s_0(\hat{k}) \qquad (6)$$

Distributed routing algorithm for mobile ad hoc networks most are using and the development of mesh network, network structure based on the plane, and whether it is active or passive routing. Since the network is not stratified, each node acts as a relay of other nodes, which bear the responsibility of the same. In this flat network using full distributed routing algorithm, all nodes are not transmitted must actively monitor channel, in order to realize the relay. Therefore, power distributed routing algorithm in mesh networks have higher. Use of a star - mesh mixed structure can develop an intelligent routing, achieve high efficiency, reduce delay and enhanced connectivity.

Due to the limited routing list storage space each sensor, the passive routing can provide more compact solutions for sensor network applications. The transmission of information is to serve as a small number of nodes data collection station, which can effectively solve the time delay problem of passive routing. Each central station for communication information is collecting adjacent regions.

3. Research of Wireless Sensor Network Protocol Based on ZigBee Technology

Wireless sensor network is a computer network arises independently, its basic unit is a node, and the node integrates sensor, microprocessor, wireless interface and power module four. Computer network technology in the traditional industry has a mature solution could be used in wireless sensor networks. But the uses and advantages of wireless sensor network based on the development of special, communication protocol and routing algorithm has become a subject of urgent research in the field of the wireless sensor network.

ZigBee supports the star network; peer-to-peer network and hybrid network 3 network topology structures. Fig. 2 is a hybrid ZigBee network. Each network has its own advantages. A star network with a powerful master device is as the network center, responsible for the coordination of the whole net work and it is the master device other or from the devices in their coverage range [9]. The network control and synchronization are relatively simple, suitable for relatively few occasions' equipment quantity. Peer-to-peer network is divided into point to point and cluster tree 2, is composed of a main device connected to the. This network can provide higher reliability. Star network and peer-to-peer network is formed by the combination of the hybrid network, subnet to star connection, the main devices in a peer-to-peer manner. The network application is complex on the network request. General in the real application environment, mixed type has more practicability.

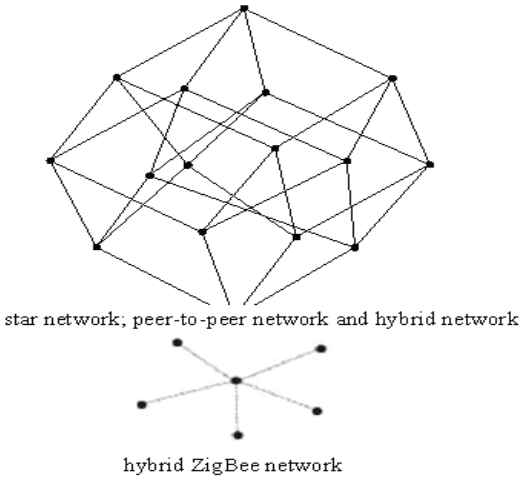

Fig. 2. The hybrid ZigBee network figure.

We design node implementation mechanism is serial communication module based on ZigBee transmission module to replace the traditional, the information collected data wirelessly sends out. The node that contains the ZigBee wireless transmission module, microcontroller module, and it is the sensor module and interface circuit, DC power supply

module and an external memory. In order to reduce the cost of sensor node, to reduce the volume of sensor nodes, data transmission and processing of our company introduced the use of highly integrated SoC chip CC2430 to realize the sensor node.

Mainly for K9F1208 to operate the page read and page programming operation. Fig. 2 is a standard page NAND Flash read timing diagram. The page read operation is as follows: Command stage, in the chip select signal CE effective, first command allows signal CLE effectively, the write signal WE, chip ready signal R/B is set high, says ready; and sending a read operation command to the I/O port (0xOO or OxO1), that is the read operation. The address phase and, at this time chip select effective, effective address enable signal ALE, the write signal WE remain effective, continuous send 4 address; K9F1208 address register receives the address value, the R/B signal will remain "busy" for a period of time, then R/B is ready [10]. The last is the data output stage, each read effective signal is asserted low effective, will output a set of data. And so on until the entire data output end, as is shown by equation 7.

$$P_{LP}(\theta)=\frac{1}{\left|G_{LP}\right|^2}=\frac{1}{\left|a^H(\theta)W\right|^2} \tag{7}$$

ZigBee technology adopts IEEE 802.15.4-2003 standards of physical layer and media access control layer as ZigBee physical layer and media access control layer, ZigBee alliance in the provisions of the framework of network layer and application layer; ZigBee technology has powerful function of the equipment of the Internet, he supported the star structure (Star), reticular formation (Mesh) and cluster (Tree) three kinds of self-organizing wireless network type, especially the network, he has the reliability of network robustness, strong.

From the point of view of maintaining routing security angle, find the safest possible route to ensure the security of network. If the routing protocol is damaged, transmitted message tampering, so no security at all for any application layer packet. This paper introduces a method called "have senses of security routing" (SAR), the idea is to find out the relationship between the real values and node, and then use the true value to generate a secure routing. The method to solve two problems and it is namely how to ensure the information security of data transmission in the safety path and routing protocol in WSN.

CC2430 is the first to comply with the ZigBee standard 2.4 GHz single chip (System On Chip, SOC), a wireless network node is suitable for various ZigBee or similar to the ZigBee, including the coordinator, router and terminal node, chip using the previous CC2430 architecture, in a single chip integrated ZigBee radio frequency (RF) transceiver, memory and the micro controller, in sleep mode, the whole chip flow consumption is less than 0.5 μ A, integrated timer and a lot of resources on chip.

The MAC layer: follow the IEEE 802.15.4 protocol for wireless, wireless data link between the equipment establishment, maintenance and end of pattern recognition, data transmission and reception, optional slot, low transmission delay, support a variety of network topology, network of each device for a 16 bit address addressing. The network layer: the establishment of a new network, the processing nodes enter and leave the network, according to the arrangement of nodes of the network type of protocol stack, the network coordinator node distribution of address, to ensure synchronization between nodes, network routing, ensure the integrity of the data, use the optional AES-128 for communication encryption, as is shown by equation 8.

$$I_\sigma(i,j)=\sigma_{\Omega_{ij}},\ \Omega_{ij}=\{(i-1)l+1\le x<il,\ (j-1)l+1\le y<jl\} \tag{8}$$

In most sensor network deployment, the network topology is unpredictable, and after deployment, role in the network topology of the whole network, sensor nodes are constantly changing, so unlike the wired network, wireless network that most of the network equipment configuration fully, pre configuration of sensor node is limited in scope, a lot of network parameters, key are formed of sensor nodes in the deployment after consultations.

In the system of larger and more complex (such as a manufacturing site), the central point of control is likely to exceed the coverage of ZigBee network, and may even be placed in another building. So, the PAN coordinator may be required for communication with a central control point through the cable connection [11]. Because the application of Ethernet in the industrial market is more and more popular, so in most cases, Ethernet is the most likely choice. Application of Ethernet has two potential implications for network design: one is to consider the required processing Ethernet interface processor bandwidth; two is to drive the Ethernet interface, the network will need the corresponding driver and protocol stack, which increase the system PAN controller needs of program memory.

4. Development Novel Organization Structure of Wireless Sensor Network Protocol by ZigBee

Sensor network is used to collect the information as the main objective; the attacker can obtain this sensitive information by eavesdropping, added illegal node forged mode, if the attacker knows how to get the related algorithm limited information from multiples information, then the attacker can be derived through the massive effective information acquisition information. The issues of private general

sensors, and not through the sensor network to obtain is unlikely to be collected information, but the attacker through remote monitoring WSN, resulting in a large number of information, and according to the specific algorithm analysis problems of the private. So the attacker does not require physical contact sensor nodes, is a low risk, anonymous access private information. Transmission of remote monitor can also make a single attacker access to multiple nodes information.

The central control center is connected with multiple sink nodes through the network, between the sink node and sensor nodes exchange information through the wireless ZigBee technology, wireless sensor nodes with a radio transceiver is responsible for sensing and processing and transmission of data to the sink node; information control center through the network to obtain the collected, realize the effective control and management on the scene.

The sink node distribution in sensor network is mainly used for reporting receiving sensor data, and the fusion processing, to the transmission module of wireless communication data, transmitted to the central control center through the network information [12]. A connection between ZigBee module and MCU through asynchronous serial port to realize the communication between them, the speed is 95.4 kB / s, MCU completes the communication the sink node and central control center control module, the sensor network in the distribution of multiple sink nodes, so a 32 bit MCU to use software interrupt for ID the sink node to upload data to the sink node polling scanning, data can be orderly, complete by MCU treatment after. As the sink nodes in sensor networks is the gateway between the sensor nodes and the network.

$$\rho(y) \equiv \rho[\hat{p}(y), \hat{q}(y_0)] = \sum_{b=1}^{m} \sqrt{\hat{p}_b(y)\hat{q}_b(y_0)} \quad (9)$$

Although the ZigBee technology is the ideal solution for wireless sensor networks, but in the actual engineering application also has the side of his lack of it. The use of ZigBee in the global range of frequency is 2.5 GHz, which belongs to the category of the characteristics of microwave, high frequency, wavelength is short, straight line, in the direction of propagation is almost not open around the obstacle, the transmit power and frequency on the ZigBee node is very low, which causes, ZigBee wireless signals PENETRATE obstacles capability is very limited. Although can be increased through the arrangement of ZigBee routing nodes to steer clear of obstacles, but this will increase the capacity of the network and network costs, and some places are not allowed to assign a network node. Therefore, this paper presents implementation scheme of wireless sensor network technology 2.4 GHz ZigBee and 433 MHz combined.

Developed a ZigBee to Ethernet module, this module is mainly through the TCP / IP protocol and uploaded to the Internet using ZigBee wireless sensor network to collect information, whether you live in that corner of the world, are available through ZigBee to Ethernet module for remote real-time monitoring. Can also through the GSM network, the system adopts TC35 as data transmission terminal, can transmit data in sensor networks quickly, reliably. Use MSP430MCU to control the TC35 module to complete the communication of the sink node and the central control center.

Wireless sensor network based on ZigBee has the advantages of low power consumption, low cost, small size, can realize the acquisition and processing the monitoring region signal in special environment. With the proposed self-organizing wireless network technology mature and new energy solutions, the application of wireless sensor networks in various areas of the life.

Modular design of various nodes in the system hardware, the ZigBee module object such as shown in Fig. 3 on the left, the CC1200 module as shown in Fig. 3 as shown in the right. The compact and reasonable is structure and small volume. In the sunny weather conditions in the open area, effective communication distance is measured between the 120 ZigBee node m, the effective transmission distance of up to 433 MHz RF module 400 m; working under laboratory conditions, ZigBee wireless signals can penetrate 1 concrete wall, 433 MHz RF signal can penetrate 4 concrete wall, effectively to overcome the 2.5 GHz RF signal penetrable weakness. In the power supply circuit of ZigBee terminal nodes connected in series 2 resistor with an oscilloscope connected, measured in the working process of the terminal node.

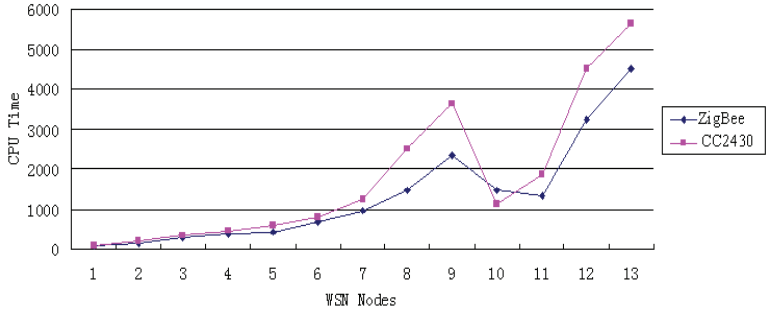

Fig. 3. Comparison organization structure of wireless sensor network protocol based on ZigBee with CC2430.

The Linux open-source is operating system. Linux is a network operating system environment, especially suitable for network application. Linux is a complete TCP/IP protocol stack, while supporting various network protocols, such as PPP protocol stack, making it easy to implement the GPRS dial function. Because the Linux open source, users can easily on the basis of the development of their applications.

The coordinator through 433 MHz RF, send the data to the concentrator, and eventually to management database. The ZigBee terminal nodes move far away, beyond the ZigBee network coverage, returned to the network coverage, can continue to work; close the router R1, again after the opening, the terminal node E2, E3 still can send data to the coordinator through R1; close the router R1, as long as the E2 the coordinator, E3 C visual range, E2, E3 can directly send data to the coordinator. The above results validated the ZigBee wireless network self-organization and self-healing characteristics of good.

6. Conclusions

Future sensor networks generally have hundreds of thousands of sensor nodes, it is difficult to monitor and protection on each node, and each node is a potential point of attack, can be an attacker to physical and logical attack. In addition, the sensors are usually deployed in unattended maintenance environment, it is more convenient for attackers to capture sensor node. The paper put forward development novel organization structure of wireless sensor network protocol based on ZigBee technology. In this paper, based on the ZigBee wireless network communication technology and 433 MHz radio communication technology is to realize the wireless sensor network. The experimental results show that, the terminal node has extremely low power consumption; long time the whole network can work continuously and stably, and has good self-organization, self-healing function.

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Intelligent Meter Based on ZigBee Wireless Transmission

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Abstract: Application of ZigBee technology combined with the power acquisition module and S3C2440 chip design a household electricity information acquisition and processing terminal system is described in the article. The system using the TI company Z-Stack protocol, and combines CC2530 RF chip to realize the remote meter reading, real-time data interaction, meter parameter setting and control commands sent, etc. *Copyright © 2013 IFSA.*

Keywords: ModBus, ZigBee, CC2530, Intelligent, Meter.

1. Introduction

With the rapid development of computer technology and communication technology, various kinds of intelligent equipment with embedded systems incorporated in the people's life [1], has promoted the convergence of computing, communications, consumer electronics industry, and thus promotes the people's life and work dramatically changes.

In recent years, there is a shortage of electricity, power cuts and other phenomena in our country. In order to make reasonable use of electricity, our country began to implement ladder electricity price for residents, so that we can improve energy efficiency.

2. Composition of System

The composition of the system structure as shown in Fig. 1, the system consists of power acquisition module, ZigBee wireless transmission module and the terminal receiving device.

2.1. Power Acquisition Module

The acquisition module uses RN8209G which is produced by the RENERGY company as shown in Fig. 2, this module current measurement range is 5 mA to 30 A, voltage measuring range is AC76 ~ 260 V, output interface have two different types, named RS485 and TTL (5 V) respectively,

communication protocols using the ModBus protocol, baud rate from 1200 bps to 9600 bps, default to 4800 bps, real-time collection voltage, current, power factor, power and other parameters, the error is 0.1 %, operating temperature from -30 ° C to 70 ° C.

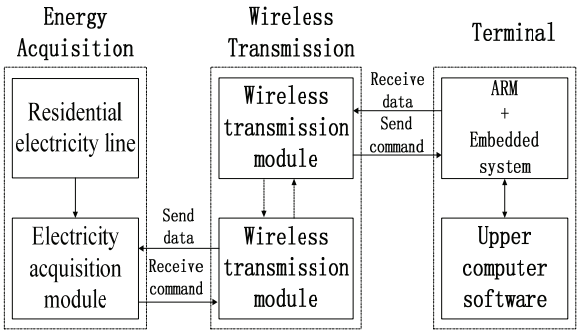


Fig. 1. The composition of the system structure.

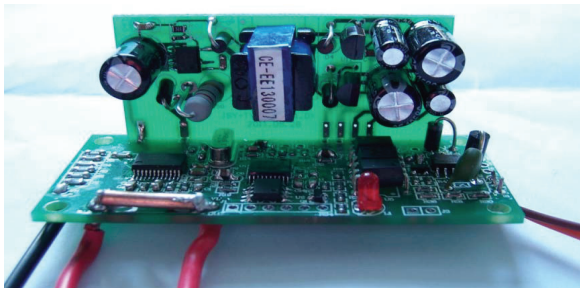


Fig. 2. RN8209G acquisition module.

2.2. Wireless Communication Module

Wireless communications can use ZigBee, Bluetooth, WI-FI and infrared technology, etc. ZigBee technology is a short distance, low complexity, low power, low data rate, low-cost wireless network technology; it is a kind of technical scheme between wireless tag technology and Bluetooth technology, mainly used for the wireless connection from close range [2]. The outstanding characteristic of this technology is applied simple, long standby time, network capability, high reliability and low cost, our system uses the ZigBee CC2530 communication module as shown in Fig. 3, through the RS232 interface connected to the power acquisition module and ARM respectively, based on TI's official protocol stack to achieve power acquisition module with terminal wireless communication.

The software design of ZigBee module consists of the initialization and system management master program, clock module, power management, communication module [3]. The transmission of data and command is according to the ZigBee protocol to communicate. ZigBee sensor on the program design process as shown in Fig. 4, Based on the ZigBee protocol Stack (Z -Stack), the control terminal of the ZigBee module coordinator as a wireless network device automatically complete the establishment of the network, ZigBee module with power acquisition module link as the terminal equipment (end device) by binding way to join the network, and initialization, data transmission and processing tasks news, etc.

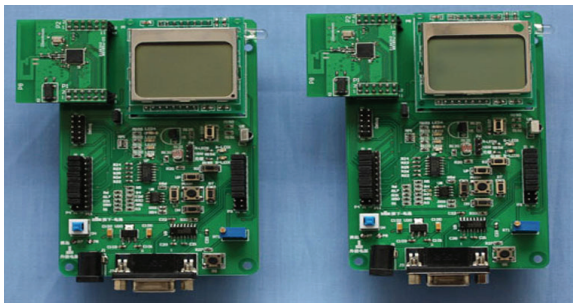


Fig. 3. CC2530 communication module.

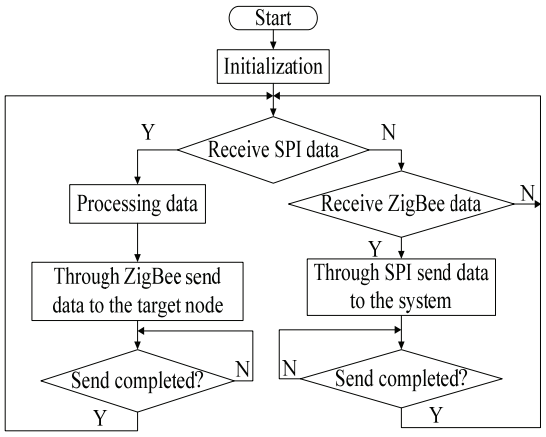


Fig. 4. ZigBee sensor on the program design process.

The establishment of a network:

MeterDataApp_NwkState = DEV_INIT;

ZDOInitDevice()

Network automatically created by the coordinator, and the initialize of the device.

Add a system task:

const pTaskEventHandlerFntasksArr[] = {macEventLoop,nwk_enent_loop,Hal_procesdsEvent ,APS_event_loop}

The task initialization:

void osalInit Tasks(void);

Binding request:

ZDApp_SendEndDeviceBindReq();

typedef struct{ZDEndDeviceBind_t ED1;uint8 state;.....}

Function body contains the binding nodes to join the network relevant information, including the NWK address, IEEE address, EndPoint ID, etc [4].

If joining the network successfully, it returns: Assoc Cnf Success.

Data sent through call AF_DataRequest () function to achieve.

void MeterDataApp_SendTheMessage(void)
{if(AF_DataRequest(&MeterDataApp_DstAddr, (byte)osal_strlen(theMessageData)+1,(byte*)&theMessageData,&MeterDataApp_TransID.....}

DstAddr: The address of the target node. Len: The length of data sent. TransID: Message ID, Play the role of the counter.

If the data request is sent successfully, the function returns: afStatus_SUCCESS.

Ubinding:

Terminal equipment issued Ubinding request: Unbind_req((uint16)0x0022).

Exit the network:

End_Device_Bind_rsp(End_Device_Bind_req|ZDO_RESPONSE_BIT)

The coordinator response to the terminal equipment unbind request[5].

2.3. The Terminal Equipment

The terminal receiving device uses the 2440 ARM9 development board which is produced by the Friendly ARM company as shown in Fig. 5, it uses Samsung S3C2440 as the microprocessor, having an external memory (64 M SDRAM, 256 M NandFlash), LCD interface, 100M Ethernet RJ-45 interface, serial port, USB port, SD memory card interface, clock source, DE interface, audio output and other rich resources, the operating system can run WINDOWS CE, Linux, UCOSII embedded systems.

Fig. 5. S3C2440 ARM9 development board.

Mainly realized function is real-time receive and storage data which is collected by power module , the power acquisition module parameters Settings, and automatic alarm to remind the residents with electricity close to the ladder to adjust the price.

3. Communication Protocol of System

3.1. Introduction of the ModBus Protocol

ModBus communication protocol is divided into RTU protocol and ASCII protocol. ASCII protocol compared with RTU protocol have the start and end tags, so when making process can be more convenient, and because the transmission of ASCII characters are visible, so when debugging is more intuitive. In addition, the LRC check is easy relatively. But every byte of RTU transmission ASCII data is to use two bytes to transfer, so that its transmission efficiency is lower. So, in general, if the need of transfer data is small ,we can consider to using the ASCII protocol, if they desired transfer large amount of data, it is best to use the RTU protocol, our system uses ModBus-RTU Transfer Protocol.

3.2. Communication Protocol

When communication command sent to the power acquisition module, conform to the corresponding address code (the system power acquisition module address is 01) device receives the command, and remove the address code, read the information, If there is no error, then execute the corresponding task; Then the result of execution is returned to the sender, The returned information includes address, function code, data, and error check code. If the error does not send any information, data frame format as shown in Table 1.

Table 1. Data frame format.

Address Code	Function Code	Data Area	Check Code
8 bits	8 bits	N×8 bits	16 bits

Address Code: The address code is the first byte (8 bits) of the message frame from 0 to 255. This byte indicates that the slave address set by the user will receive the information sent by the host. Each slave must have a unique address code, only the corresponding address code from machine to respond to the loopback. When send information back from the machine, the address code indicates that the information from where.

Function Code: The host sends the function code tells the machine what tasks to execute. Table 2 lists the function code has a specific meaning and operation.

Table 2. Function code.

Code	Meaning	Operation
03	Read data	Read the current register one or more binary values
06	Reset a single register	Write the binary value of set to a single register
06	Reset a single register	Write the binary value of set to a single register

Data Area: The data area contains the machine what action to perform, or send back information collected from the machine. These information may

be a numerical value or the reference address, etc. For example, the function code tells the machine to read the value of the register, the data area must contain the start address of the register to read and read length. For different machines, has a different address and data information.

Check Code: The sender or the receiver can use check code to judge whether receive information error. Sometimes, due to electronic noise or other information, minor changes will occur in the process of transmission, check code ensure that error messages do not work in the process of sending or receiving. So that increases the safety and efficiency of the system, error check adopts CRC-16 check code. RN8209G collected data are stored in the corresponding register inside real-time, the module effective register as shown in Table 3.

Table 3. RN8209G effective register.

Register address	Function Description
0000H (read only)	Model 1 value is 9000
0001H (read only)	Model 2 value is B131H
0002H (read only)	Voltage range
0003H (read only)	Current range
0004H (readable and writable)	The default value is 0105 H. The default address is 01 H. The default format is 8, N, 1, 4800bps. Description: The 8 bits of high byte is the address, 1~127; The high 2 bits of low byte is the data; The low 4 bits of low byte is the baud rate: 3-1200 bps, 4-2400 bps, 5-4800 bps, 6-9600 bps.
0048H (read only)	Voltage
0049H (read only)	Current
004AH (read only)	Active power
004BH (read only)	High bits of electrical energy
004CH (read only)	Low bits of electrical energy
004DH (read only)	Power factor

3.3. Command Parsing

Send the command: 01 03 00 48 00 06 45 DE (Read 6 registers starting from 0048).

Receive Data: 01 03 0C 57 E4 09 D9 02 2D 00 00 03 40 03 D6 3F 17.

57 E4 corresponding 0048 register, that is the voltage: 0x57E4=22500, and then divided by 100 is equal to 225 V, each register data is 2 bytes, similarly according to the above formula can be derived the other data.

4. Design of Terminal Software

The software development of intelligent terminal communication module is based on Win CE 6.0 embedded operating system. Its architecture adopted independent programming language and Windows-compatible API way, ensure that Windows CE.net components and ROM to adapt to the requirements of the limited storage space and a variety of chip [6]. Development of intelligent home appliance control terminal system, first of all, we need according to the hardware platform and applications to customize the Windows CE.net operating system kernel, development of intelligent handheld terminal application on the basis of complete custom operating system kernel, combined with underlying drivers of the chip, and design software of the intelligent meter. The intelligent control terminal of the system design as shown in Fig. 6 mainly includes three different parts: current parameter, ladder electricity price and historical curve.

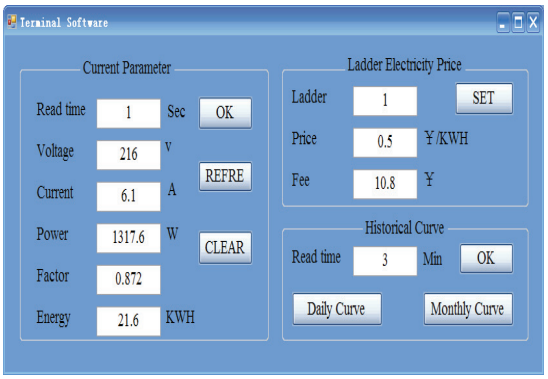


Fig.6. The intelligent control terminal design.

The function of current parameter section is real-time to read the voltage, current, power factor, power, and data collected by the power module.

The function of ladder electricity section is to display the current ladder, electricity prices and the monthly electricity fee, when the month electrical energy close to the ladder adjust price then alarm to remind automatically, In addition, according to local electricity fee standard is not the same, could set the range and price of each ladder as shown in Fig. 7.

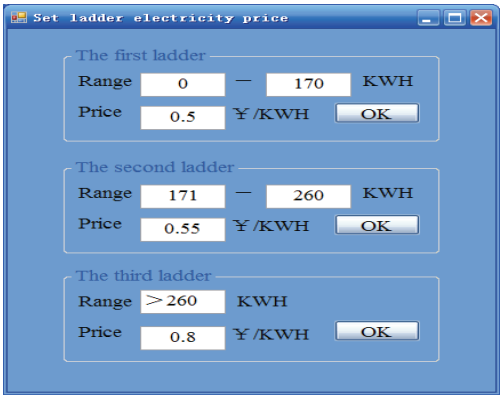
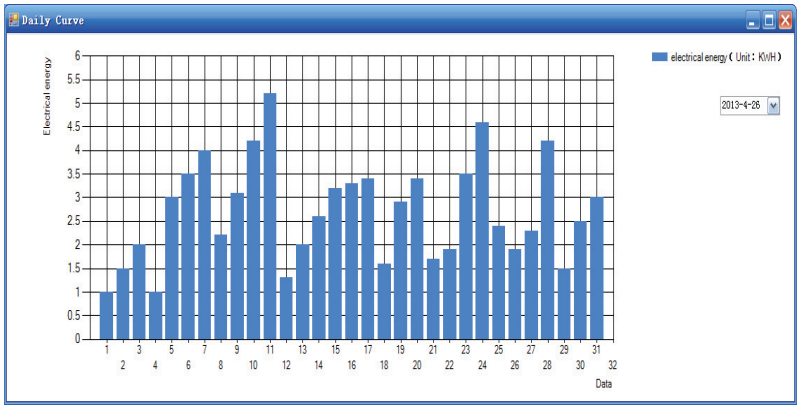
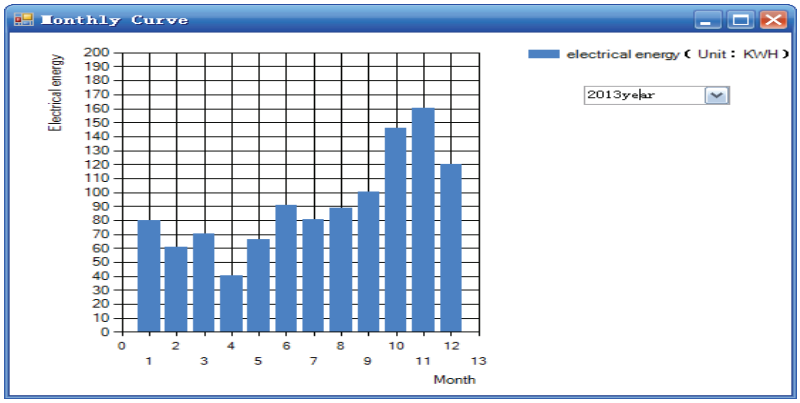


Fig. 7. Set the range and price.

The function of historical curve section is to analyze the user's historical data in the form of a histogram, including daily curve and monthly curve as shown in Fig. 8 (a, b).



(a). Daily curve.



(b). Monthly Curve.

Fig. 8. Historical curve.

5. Conclusions

In this paper, a intelligent meter system is designed with S3C2440, RN8209G power acquisition module as the core, through the ZigBee wireless communication mode, intelligent control terminal can realize remote meter reading, meter control, information query, ladder electricity price charging and early warning, historical data curve display, etc. Great convenience to the lives of the residents, while adapting to the needs of the highly market-oriented electricity trading, prompting residents to a more rational use of electrical appliances, energy conservation, reducing electricity bills.

Acknowledgements

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Development of Precession Instrumentation System for Differentiate Walking from Running in Race Walking by Using Piezoelectric Sensor

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Abstract: In this paper, a precession instrumentation system for measurement and characterization of human foot stress has been developed. The system is very useful to differentiate walking from running in race walking sport, by identification of the foot loss of contact with the ground. The system was developed in two main units: sensing unit (RTU) and signal processing unit (MTU). The RTU consist of piezoelectric stress sensor module and data acquisition module. The data acquisition is developed based on microcontroller to achieve the system as small as possible, but powerful. The MTU consist of computer system and several program procedures to identify the foot stress signals and to differentiate walking from running. Data communication between RTU and MTU can be conducted through wires by using RS485 protocol, or wirelessly by adding a pair of radio transceiver device. The system can be operated by online and real time. Result of the experiments show that the developed system has good performance. *Copyright © 2013 IFSA.*

Keywords: Race walking, Judging, Instrumentation, Piezoelectric sensor, Signal processing.

1. Introduction

To get better achievements in the race walking competitions, understanding of the rules and certainty in judging is very important. Judging of the race walking is based on the IAAF rule, no.230; which states: "Race Walking is a progression of steps so taken that the walker makes contact with the ground, so that no visible (to the human eye) loss of contact occurs. The advancing leg must be straightened (i.e. not bent at the knee) from the moment of first contact with the ground until the vertical upright position" [1]. Hence, there are only two basic rules: first is the knee cannot be bent from the moment until the vertical upright position of the body, and second is the walker must always have at least one foot on the ground, and this is the feature that differentiates race walking from running [2, 3].

Fig. 1 shows how to differentiate walking from running. This figure is taken from Race Walk Officiating Handbook, USA Track & Field 2008 [4]. It can be seen, that in walking event, the walkers must always have at least one foot on the ground, and in running event, the walkers has "flying" condition.

Furthermore, although the judging of the competition currently is based only on view of the human eye, but it is necessary the existence of an instrument that can be used to for a more rigorous identification. It is very useful not only to give certainty justification in competition, but also can be

used by athlete during practices in order to avoid unwanted disqualification when competition. To provide the intended instrument, some researchers have proposed several systems [5-7], of course each of them with own merits and limitations.

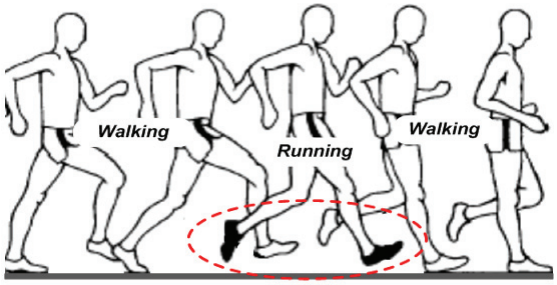

Fig. 1. Differentiates walking from running.

In this paper, a precession instrumentation system for measurement and characterization of human foot stress is proposed. The system has capabilities to differentiates walking from running by identify loss of contact with the ground.

2. Design of the Instrumentation System

Fig. 2 shows a block diagram of the proposed instrumentation system. It is consists of several modules that can be grouped into two units: remote terminal unit (RTU) and master terminal unit (MTU). The RTU is sensing unit; it is built of sensor module and data acquisition module. The MTU is control unit; it is a personal computer device that equipped with signal processing program procedures. Data communication between RTU and MTU can be conducted through wires by using RS-485 protocol, or wirelessly by adding a pair of RF radio transceiver device.

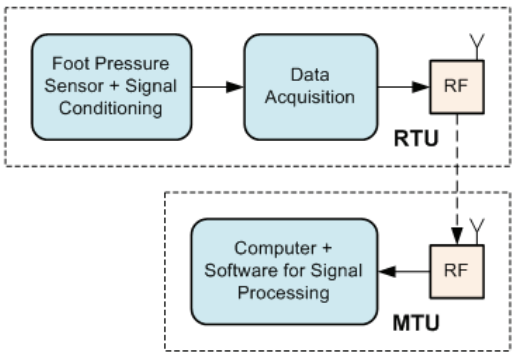

Fig. 2. Block diagram of the instrumentation system.

In the RTU, sensor module senses the presence of foot stress that indicates foot makes contact with the ground. There are four points to be measured, whose function will be explained later. Output signals from each sensor then conditioned by signal conditioning circuits (SC). The data acquisition module converts analog signals from SC into digital form, and then transmits the data to the MTU. In the signal processing module, the data will be identified by certain rules and be decided of whether a person was being walking or running conditions, included the time of events.

2.1. Sensor Module

The sensor module is feet pressure sensors, which deals with foot stress signal sensing and conditioning. To sense existence of the foot stress, we put four sensor elements at four selected points. The first sensor is placed in the front section of right foot (call as RF), the second sensor at the back section of right foot (RB), the third sensor at the front section of left foot (LF), and the fourth sensors in the back section of left foot (LB), as shown in Fig. 3.

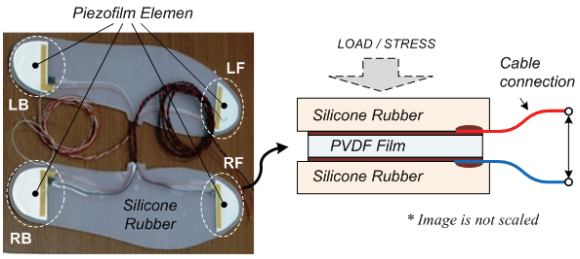

Fig. 3. Sensor elements, positions and mechanism.

Sensing element is made of PVDF film, which is a kind of piezoelectric material. In the both sides of PVDF film connected to metal cable to make electric connection. The film clamped by silicone rubber in both sides to make elastic when stepped on. In here, PVDF film will generate electric charge on both surfaces when loaded or subjected by mechanical stress [8]. If the stress occurs in the thickness or *z*-direction only, then the equation for the voltage mode can be expressed by:

$$V = g_{33}\sigma_3 t \tag{1}$$

V : open loop voltage (V);
σ_3 : mechanical stress for z-axis (N/m^2);
g_{33} : piezofilm stress constant in z-axis (V.m/N);
t : PVDF film thickness (m);

The piezoelectric sensor can be modeled as a voltage source (V_p) coupled in series with a capacitor (C_p) [9]. And by the principles of parallel plate capacitor, value of C_p has a linear relationship with the area of piezoelectric elements. Therefore, for the same stress, the sensor with a larger area will generate a larger voltage.

Furthermore, piezoelectric sensors require high input impedance signal conditioning circuit in order to suppress loading effects caused by measuring device. In this case, we use a charge amplifier circuit as shown in Fig. 4. In addition, because the charge amplifier is operated on inverting mode, so it need an inverting amplifier circuit to return the signal to its normal phase, as well as for signal amplifications.

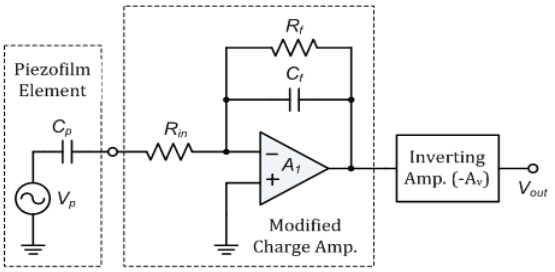

Fig. 4. Signal conditioning circuit.

The output voltage (V_{out}) of the circuit in Fig. 4 can be expressed as:

$$V_{out} = A_v \frac{Z_{(R_f,C_f)}}{Z_{(C_p,R_{in})}} V_p \tag{2}$$

Completion Eqs. 2 is given in [10]:

$$V_{out} = A_v \frac{C_p}{C_f} V_p \text{ ; for } (\omega_1 << \omega << \omega_2), \tag{3}$$

where

$$\omega_1 = \frac{1}{C_f R_f} \text{ and } \omega_2 = \frac{1}{C_p R_{in}} \tag{4}$$

Eqs. 3 can be interpreted that the output voltage of the sensor depends on the area of the piezoelectric element (equivalent C_p) and magnitude of the foot stress (equivalent V_p).

2.2. Data Acquisition Module

The data acquisition (DAQ) module was designed with the following assignments:

- To sampling the analog signals (RF, RB, LF, and LB) from sensor module with a certain sampling time;
- To convert the analog signals into digital data;
- To send the digital data to the MTU via RS485 network (wire) or radio transceiver (wireless).

The main component of DAQ module is a microcontroller PIC16F876. It has some features inside such as 5-channels ADC with 10 bits resolution, TIMER, USART, 8K × 14 words of flash program memory, 368 × 8 bytes of RAM [11]. Fig. 5 shows the DAQ module that used in this project.

It should be explained here that the internal ADC of microcontroller only capable recognize positive signals (typically in the range of 0 to 5 Volts). Hence, before converted by the internal ADC, the analog signal from the SC should be increased to a certain dc level. This is accomplished by dc-offset circuit. We use 2.5 volts for offset, with the intention that the signal has same swings in up and down.

In view of the microcontroller based DAQ system, firmware is required to manage hardware system driving, signals capturing, and data transmitting (to MTU). The firmware is developed by the use of MPLAB assembly codes.

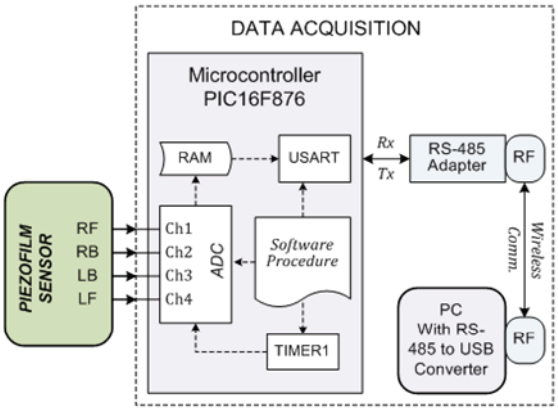

Fig. 5. Data acquisition module.

3. Signal Characterization

In order to understanding the differences between walking and running, firstly we take the voltage signals from foot sensors by using multi channels digital oscilloscope. For this purpose, the sensor module is inserted into the shoes, and then used for walking and running activities. Signals are recorded in the four data points that have been determined (RF, RB, LF, and LB).

Results of experiments are given in Fig. 6 and Fig. 7 for walking and running activities respectively. In those figures, when the sensor pressed by the foot (at the ground), it will generate electric voltages, and when the foot release the stress, the voltage will down rapidly. In here, one cycle is a complete one-step event, such as for RB-RF-LB-LF, and back to RB.

To simplify the captured signals that do not look complicated, we add the signals of each foot; so henceforth we have two signals only, e.g. the right foot (R) and the left foot (L) signals, as shown in Fig. 8 for walking and Fig. 9 for running. According to these figures, we have four important events:

(1). RB hit the ground (RBU)
(2). RF release the ground (RFD)
(3). LB hit the ground (LBU)
(4). LF release the ground (LFD)

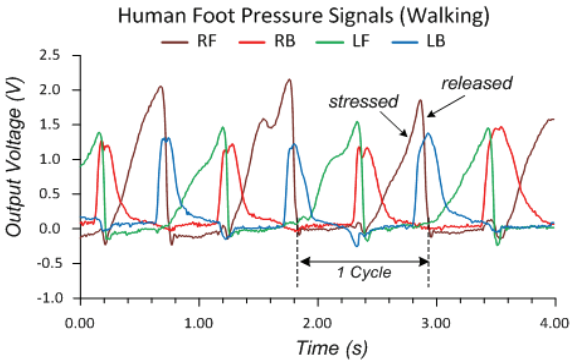

Fig. 6. Human walking signals.

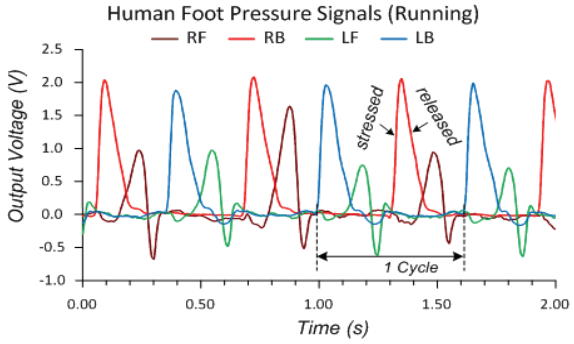

Fig. 7. Human running signals.

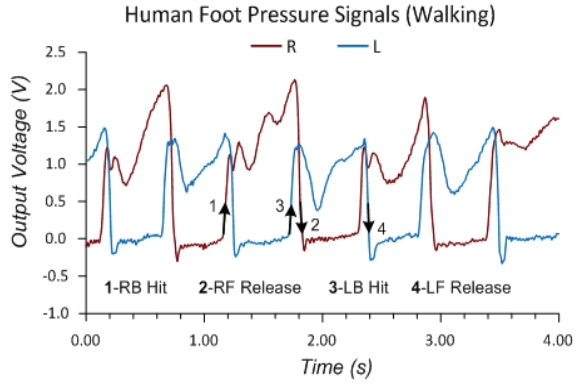

Fig. 8. R-L signals when walking.

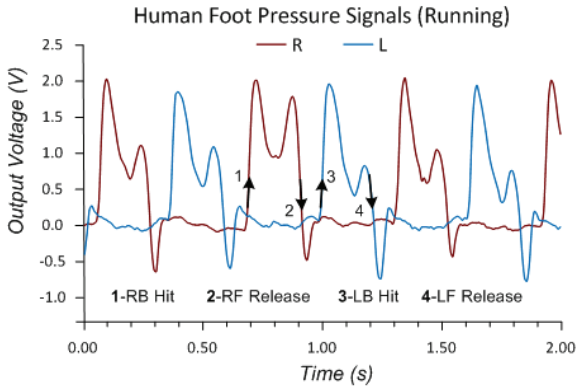

Fig. 9. R-L signals when running.

Here, we can differentiate walking and running activities clearly. We can see that time series of the walking activity in one-cycle is (1-3-2-4) or (RBU – LBU – RFD – LFD), and time series of the running activity in one-cycle is (1-2-3-4) or (RBU – RFD – LBU – LFD). Time within interval (1-2) is the right foot makes contact with the ground, and time within interval (3-4) is the left foot makes contact with the ground. Transition step from right foot to left foot is given by (2) to (3) and transition step from left foot to light foot is given by (4) to (1).

Moreover, transition from right foot to left foot and vice versa is an important thing to make differences between walking and running activities. In walking events, transition from the right foot to the left foot, and conversely, has overlapping time ((3) precedes (2) or (1) precedes (4)). It means, in walking event, at least one foot is always has contact with the ground. But, in running events, transition from the right foot to the left foot, and conversely, has no overlapping contact time. Then, we can say that the running event has "flying" condition.

4. Signal Processing

Based on the information that has been presented above, the following description will be given several procedures for signal processing.

4.1. Threshold Voltage

Threshold voltage (V_T) is needed in order to make the signal only has two values. It is very important to be used for time identification of the events of RBU, RFD, LBU, and LFD.

Rules of the program procedure is if voltage of the signal sampling is higher than V_T, then value of the signal sampling becomes "high" (5 Volts), if lower than V_T, then it is become "low" (0 Volt). Fig. 10 may useful to help clarify this statement. In here XY is the original signal, and XYN is new corresponding signal after threshold procedure applied.

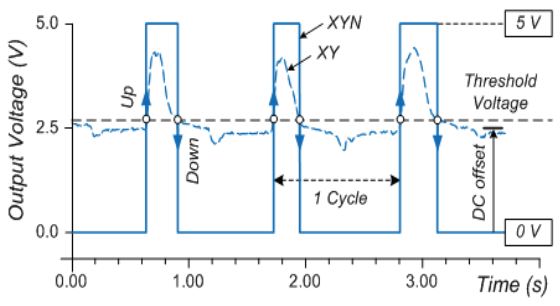

Fig. 10. Rule of the threshold voltage.

Hence by the rule, four original signals (RF, RB, LF, and LB) and **n** number of sampling, the program procedure is given below:

for n:= 1 to max do
if RF[n] > V_T then RFN[n]:=5; else RFN[n]:=0;
if RB[n] > V_T then RBN[n]:=5; else RBN[n]:=0;
if LF[n] > V_T then RFN[n]:=5; else LFN[n]:=0;
if LB[n] > V_T then RFN[n]:=5; else LBN[n]:=0;

4.2. Determination of Rise & Fall Time

As discussed earlier, to identify and make differences between walking and running activities, we can use foot transitions while the stepped. Transition from right foot to left foot is given by RFD (2) to LBU (3), and transition from left foot to right foot is given by LFD (4) to RBU (1). Please see again Fig. 8 and Fig. 9.

Time identification is made by detection of two pairs of the signals: RFD-LBU and LFD-RBU, what is come first. Then, the time when RFD and LFD going down, as well as RBU and LBU going up, are must be determined. So we have four important time events as shown in Fig. 11 and Fig. 12, there are:

- TRU, time for RBN going up
- TRD, time for RFN going down
- TLU, time for LBN going up
- TLD, time for LFN going down

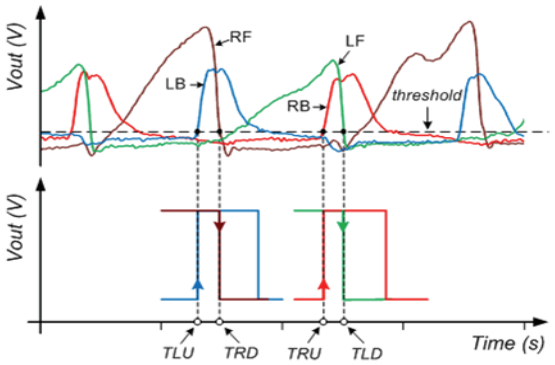

Fig. 11. Time series of walking.

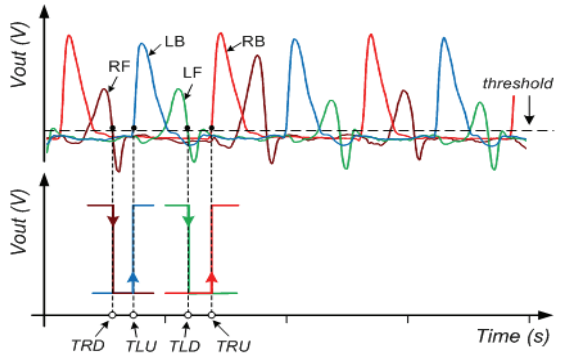

Fig. 12. Time series for running.

Rise time ("going up") is pointed by changes in the voltage value of the signal from 0 Volt become 5 Volts, and the fall time ("going down") is changes the voltage from 5 volts become 0 volt. Then, the procedure to detect the rise time and the fall time is given as below:

{Transition from Right to Left}
RBU:=0; LFD:=0; //how many transitions?
for n:=1 to max do
if (RBN[n-1]=0) and (RBN[n]=5) then
RBU:=RBU+1; TRU[RBU]:=Time[n];
if (LFN[n-1]=5) and (LFN[n]=0) then
LFD:=LFD+1; TLD[LFD]:=Time[n];

{Transition from Left to Right}
LBU:=0; RFD:=0; // how many transitions?
for n:=1 to max do
if (LBN[n-1]=0) and (LBN[n]=5) then
LBU:=LBU+1; TLU[LBU]:=Time[n];
if (RFN[n-1]=5) and (RFN[n]=0) then
RFD:=RFD+1; TRD[RFD]:=Time[n];

4.3. Decision of Running from Walking

Decision of running from walking is made by using time determination that has been described in Fig. 11 (walking) and Fig. 12 (running). Based on those figures, time series in the walking activity is TLU-TRD-TRU-TLD- and so on, whereas time series in the running activity is TRD-TLU-TLD-TRU- and so on.

However, the signal must be paired; wherein TLU must be paired with the TRD, and the TRU must be paired with TLD. So the "pulse" that does not have a pair should be discarded. Fig. 13 shows the unpaired signal both in walking and running events.

Rearrange time of the signals for discard unpaired pulse is given below:

if (RBU<LFD) then N1:= RBU else N1:= LFD
if (LBU<RFD) then N2:= LBU else N2:= RFD

for i:=1 to N1 do
if (ABS(TRU[i]-TLD[i])) > (ABS(TRU[i+1]-TLD[i]))
then TRU[i]:=TRU[i+1];
if (ABS(TLD[i]-TRU[i])) > (ABS(TLD[i+1]-TRU[i]))
then TLD[i]:=TLD[i+1];
for i:=1 to N2 do
if (ABS(TLU[i]-TRD[i])) > (ABS(TLU[i+1]-TRD[i]))
then TLU[i]:=TLU[i+1];
if (ABS(TRD[i]-TLU[i])) > (ABS(TRD[i+1]-TLU[i]))
then TRD[i]:=TRD[i+1];

Finally, the procedure to identify the existence of running events is:

for n:= 1 to N1 do
if (TRU[n]>TLD[n]) then Alert:="RUNNING"
for n:= 1 to N2 do
if (TLU[n]>TRD[n]) then Alert:="RUNNING"

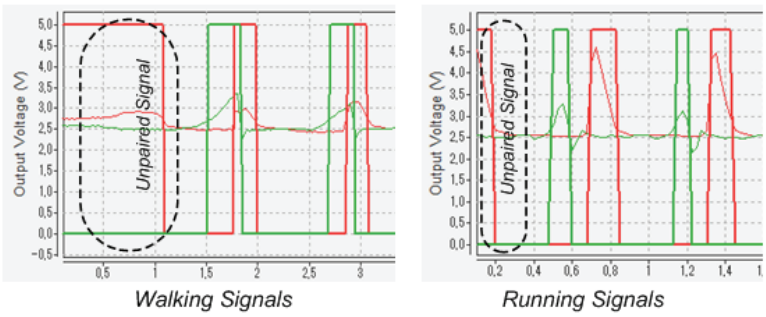

Fig. 13. Unpaired pulse in walking and running events.

4. Implementation of the System

Implementation diagram of the instrumentation system to identify of walking or running events by online and real time is given in Fig. 14. The sensor system is used to detect the foot stresses and convert to the voltages within range of 0 to 5 volts, depend on stress quantity. Data acquisition (DAQ) converts analog signals from sensor to the 10 bits digital data, then transmitted to the MTU by wire or wirelessly by using radio transceiver (RF). On the MTU, data is received by computer system via RF transceiver, and then processed by using developed signal processing software. The program procedures as described above are used to identify running from walking events.

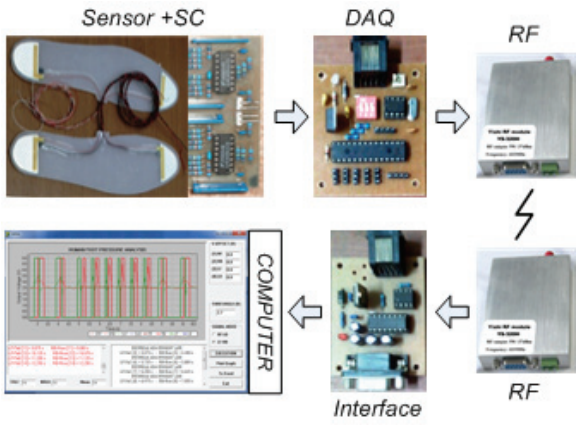

Fig. 14. Implementation of the system.

In the developed system above, DAQ has five-channel analog inputs, which are used to capture the output signals from the sensor module. The uses of the five channels are as follows:

- Ch-1 for Right Foot - Front Section (RF)
- Ch-2 for Right Foot - Back Section (RB)
- Ch-3 for Left Foot - Front Section (LF)
- Ch-4 for Left Foot - Back Section (LB)
- Ch-5 for Reference (REF)

Ch-5 (REF) is used to check the value of the dc-offset voltage, to ensure that the Ch-1 up to Ch-4 has given the appropriate voltage offset.

Implementation of instrumentation system to measure and record of the walking and running activities, is carried out by inserting the piezoelectric sensor into the shoes, and topped with a pedestal to keep the sensor in safe condition. Then, the shoe is used for walking or running. Result of experiments is given in Fig. 15. In this figure, brown color signal is RF, red color is RB, green color is LF, and blue color is LB. By providing dc-offset of 2.5 Volts, then the signal will fluctuate on that value.

By provide threshold voltage at 2.7 Volts, signals in Fig. 15 then process become two level signals (pulses) as shown in Fig. 16. However, in here only two signals that displayed, while to displays the other two signals, can be done by clicking on the options provided by the program. By looking at the figure, we can immediately differentiate between walking and running. In walking signals, appears the overlapping “pulses” at the time of the shift legs. But in running signals, the overlapping pulses is does not appears, and it has delay time between LF and RB (see black circles).

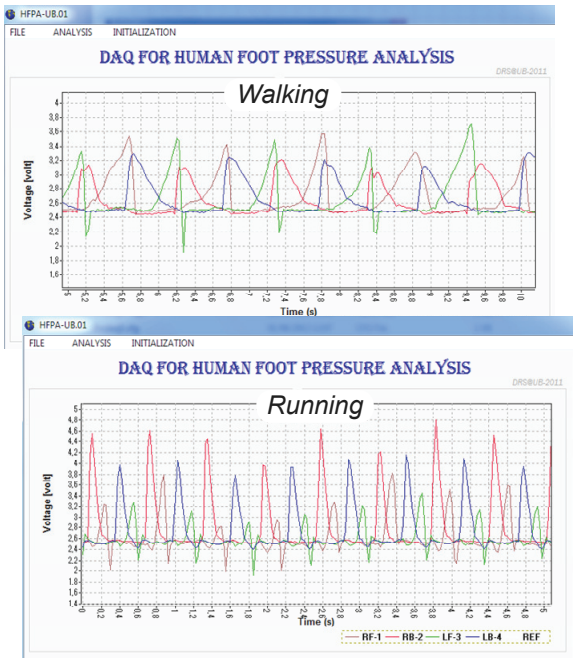

Fig. 15. Signal capturing by developed system.

Fig. 16. Signal identification being walking or running.

Fig. 17 shows when both activities are recorded. First part of the recorded signals is walking events, middle part is running events, and last part is walking events again. The time when running event occurs is given in black circle. Here, it appears clearly that the developed system has ability to detect the presence of running events, including when it happened.

Moreover, some researchers also explained that the identification of running and walking activities can be carried out by using the frequency response, in which the frequency of the walking is in the range of (0.5 - 1.5) Hz, while the running is in the range of (1.0 - 3.0) Hz [12]. In here, by using the developed signal processing software procedure, it can be given clearly as shown in Fig. 18.

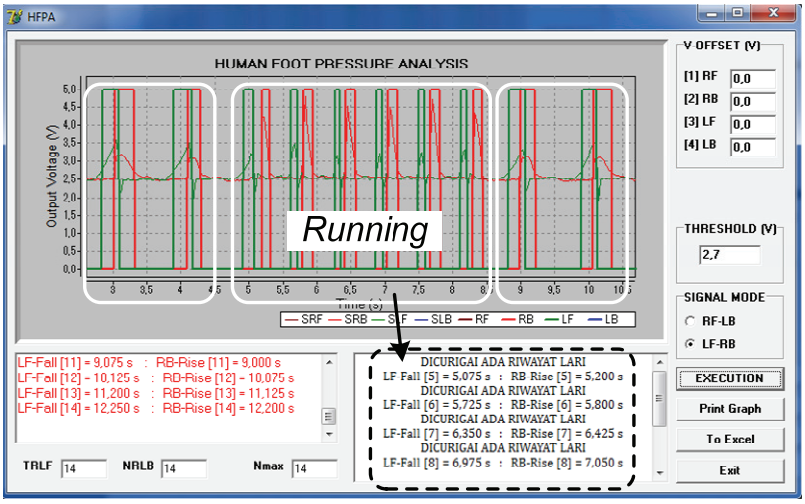

Fig. 17. Signal identification being walking or running.

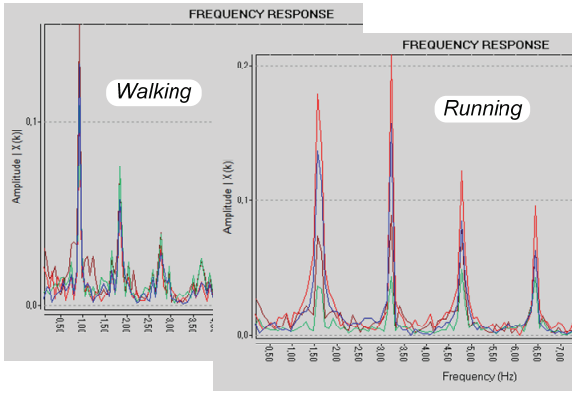

Fig. 18. Frequency response of walking and running.

In addition, the developed instrumentation system may also be used in medical fields, particularly for gait analysis and learn about biomechanics of the foot. Some publications of the research in this field can be found in [13-16].

5. Conclusions

To improve performance in race walking sport, the use of instrumented judging system becomes important. It is very useful not only for athletes, but also for coaches and judges to avoid unnecessary disqualification. In this project, a precession instrumentation system to identify walking or running activities has been developed. The principle of the instrumentation system is to measure foot stress at four places simultaneously, and by the use of measuring signals then detects the existence of loss of foot contact with the ground.

The system was developed in two main units: sensing unit (RTU) and signal processing unit (MTU). The RTU consist of piezoelectric stress sensor module and data acquisition module. The data acquisition is developed based on microcontroller to achieve the system as small as possible, but powerful. The MTU consist of computer system and several program procedures to identify the foot stress signals

and to differentiate walking from running. Data communication between RTU and MTU can be conducted through wires by using RS-485 protocol or wirelessly by using radio telemetric system. The system can be operated by online and real time.

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Water Quality Data Measurement and Analysis System Equipped in Underwater Navigation Robot

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Abstract: This paper introduces a water quality data measurement and analysis system equipped in underwater robot. The system consists of sonar navigation module, inertial navigation module equipped in underwater robot, data processing module and sensor detection module. Sonar navigation module and inertial navigation module equipped in underwater robot are separately used to detect current speed, azimuth and depth in different latitudes and longitudes. And then we strike differences of the measured data from two modules above, the difference will be feedback to the inertial navigation module after the filtering process to obtain a more precise location information. Sensor detection module uses the water quality sensors equipped in underwater robot to measure water quality parameters of the current location. Underwater robot data processing module matches the water quality data of the current position to its tracking latitude and longitude, forming water quality of the latitude and longitude distribution data. Underwater robot float to water surface, transmit water quality data to the shore-based facility micro-processor before moving on to the next testing spot. Shore-based facility has its own computer, our self-developed Density Map software is embedded in it. Water quality data from different testing point can be visually displayed. The feasibility of the system design and the rationality of the water quality parameter measurement results were verified by experiments and on-site tests.
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Keywords: Water quality monitoring, Underwater robot, Underwater navigation and positioning, Water quality in Longitude and Latitude.

1. Introduction

In today's society, environmental issues have become one of the most important issues in the sustainable development of human society. The pollution of water resources is a part of the environmental pollution. Water is the basic element of human survival, water quality must be tested timely and effectively so that we human beings can take remedial measures in appropriate time.

The traditional method of detecting water quality data is: workers choose several testing spots in a river, recording water temperature, turbidity, and blue-green algae, chlorophyll, dissolved oxygen, PH, ORP data with sensors, and then make data sheets to analysis water quality of various spots [1-2]. The testing spots are quite limited using the traditional method, which also lacks the macro analysis of specific water environment. It is difficult to grasp the latitude and longitude distribution of various water quality parameters, let along predicting its future distribution trends.

Underwater navigation robot can work under complex water environment below a certain depth for

a long time, and conduct a wide range of water quality test, which are easily operated and labor-saving. They have been widely used today.

Underwater navigation is a key issue to be solved of the underwater robot water quality testing system. It includes how to get navigation trajectory and the latitude and longitude of each testing spots, how to match a variety of water quality parameters from all kinds of sensors with each specific latitude and longitude. The single navigation technology has low position accuracy, poor reliability and fault tolerance phenomenon. In addition, the unpredictable underwater environment, navigation error will be easily accumulate with time and traveling distance, eventually leading to the low positioning accuracy [3].

Aiming at the aforementioned drawbacks above, this paper introduces a measurement and analysis system for water quality data of different latitudes and longitudes. A serious of experiments have verified the feasibility and accuracy of the system design, and resolved the issue to a large extent at this stage.

2. System Design

In view of the disadvantages above, the system has adapted multiple navigation facilities to coordinate their work together and track its speed and position by using adaptive navigation algorithm to get the location (the current latitudes and longitudes) of the testing spots accurately. When Global Navigation Satellite System （GNSS） signal exists, it can help to optimize robot position. Then water quality parameters are obtained and match to the latitudes and longitudes. Density Map shows the testing results intuitively, from which we can conduct comprehensive analysis in any areas we have tested. The schematic diagram of the overall system design is shown in Fig. 1, it includes sonar navigation module, inertial navigation module equipped in underwater robot, data processing module and sensor module. Sonar navigation module and inertial navigation module combine with the filter information fusion system to get underwater robot's position accurately. Data processing module consists of underwater robot built-in microprocessor and shore-based facility computer, the former deprives water quality parameters and matched them to corresponding latitudes and longitudes. While the latter helps to display data on the screen, so that workers can analyze conveniently. Sensor module includes all kinds of sensors and analog-to-digital conversion module.

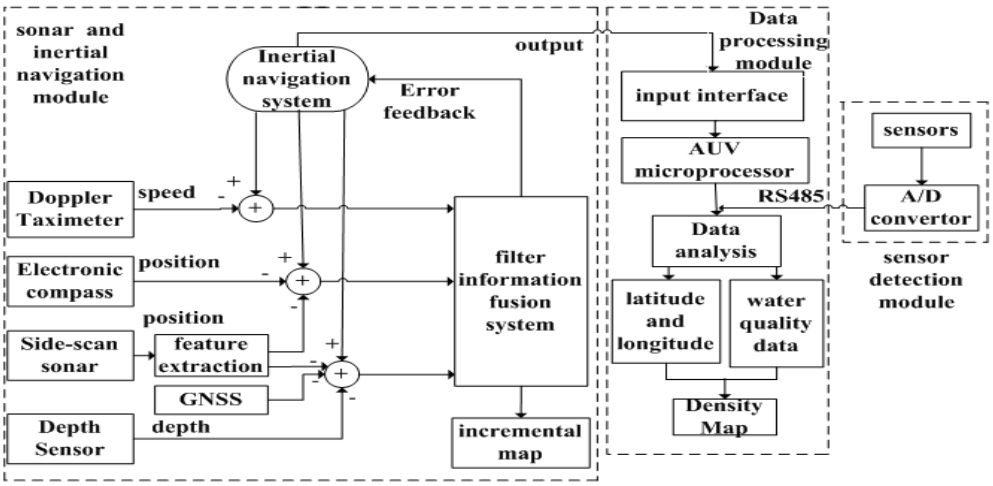

Fig. 1. Schematic diagram of the overall system design.

3. Underwater Robot Navigation System

Sonar navigation module and inertial navigation module detect current speed, position and depth respectively, and then the system compute speed, position and depth differences respectively from two modules. These differences will be processed by filter information fusion system [4]. Filter information fusion system output its processing results back to inertial navigation module. The patched and superimposed speed orientation, depth information is sent to the underwater robot built in microprocessor. Considering the obstacles will appear when the robot navigates autonomously, the obstacle diagnostic method is added to SLAM algorithm in order to ensure safer navigation of the robot.

3.1. Underwater Robot Simultaneous Localization Algorithm

The positioning of underwater robot refers to determine the location of the moving robot relative to the global coordinates in the two-dimensional (three-dimensional) working environment, and its own posture. It is the basic steps when the robot is moving.

Underwater robot works in an unknown underwater environment, without any human intervention. We need to get the real-time speed, orientation, depth information, and incrementally create environment maps, which at the same time can help to position itself more accurately. In order to obtain a more optimized estimate, we adapt fusion algorithm (Kalman filter) [5-6] to fuse the same parameter deprived from different modules. The algorithm use simultaneous localization and mapping (SLAM) algorithm Kalman filtering theory and as the core basis to get a more accurate position.

SLAM algorithm includes prediction and updating two stages. Control signal from underwater robot input to the system state equation to predict position and map features [7]. Updating stage includes adding new features, deleting disappeared features and updating repeated observed features.

Coordinates of underwater robot position and environment features are expressed in a state vector [8]. During underwater robot's motion, system state vector is represented as:

$$\hat{X}^{B} = \begin{bmatrix} \hat{X}_{R}^{B} \\ \hat{X}_{F_1}^{B} \\ \cdots \\ \hat{X}_{F_n}^{B} \end{bmatrix}, \tag{1}$$

where $\hat{X}_R^B$ is the pose vector of underwater robot, $\hat{X}_{F_i}^B$ is the vector of all environmental characteristics [8] of F_1~ F_n, covariance matrix of the system is:

$$P^{B} = \begin{bmatrix} P_R^B & \cdots & P_{RF_n}^B \\ \cdots & \cdots & \cdots \\ P_{F_nR}^B & \cdots & P_{F_n}^B \end{bmatrix}, \tag{2}$$

where P_R^B, $P_{RF_n}^B$, $P_{F_n}^B$ subarrays denote underwater robot, environment features, difference matrix between environment features. They all use ground absolute coordinate system B as reference coordinate system, an extended Kalman filter to estimate the state variables and covariance matrix P [9-10]. Between prediction and updating, we need to finish feature matching or data relating. Only matching features can be used to update the position and feature map [11]. Underwater robot will be interfered during its motion, so the motion model is:

$$X_{R_k}^{R_{k-1}} = \hat{X}_{R_k}^{R_{k-1}} + V_k, \tag{3}$$

where $X_{R_k}^{R_{k-1}}$ is the displacement value from time k-1 to time k, DVL provided the UNDERWATER ROBOT speed to estimated displacement value of the underwater robot in time k:

$$\hat{X}_{R_k}^{R_{k-1}} = [V_x \Delta T \;\; V_y \Delta T \;\; 0], \tag{4}$$

where ΔT is the time interval [12] from time k-1 to time k, V_x, V_y is the underwater robot speed value measured from the DVL sensor with noise. Speed value integrated in time interval ΔT can get underwater robot displacement value, assuming that the gain of the angle is very small and appears in the form of system noise. Covariance matrix of displacement estimation σ_{V_X}, σ_{V_y} is the standard deviation of the noise-containing speed. σ_{Φ_r} is angle noise variance to simulate the change amount of the underwater robot azimuth. Q_k is zero mean additive white Gaussian noise.

$$Q_k = \begin{bmatrix} (\sigma_{V_X}\Delta T)^2 & 0 & 0 \\ 0 & (\sigma_{V_y}\Delta T)^2 & 0 \\ 0 & 0 & (\sigma_{\Phi_r}\Delta T)^2 \end{bmatrix} \tag{5}$$

In order to obtain the ultimate underwater robot position vector in the absolute coordinate system B, we need a synthetic transformation of the coordinate space, signed with the signal $\oplus$, convert formula is defined as:

$$\hat{X}_{R_{k|k-1}}^{B} = \hat{X}_{R_{k-1}}^{B} \oplus \hat{X}_{R_k}^{R_{k-1}} = \begin{bmatrix} \hat{X}_{R_{k-1}}^{B} + V_x \Delta T \cos\hat{\phi}_{R_{k-1}} - V_y \Delta T \sin\hat{\phi}_{R_{k-1}} \\ \hat{Y}_{R_{k-1}}^{B} + V_x \Delta T \sin\hat{\phi}_{R_{k-1}} - V_y \Delta T \cos\hat{\phi}_{R_{k-1}} \\ \hat{\phi}_{R_{k-1}} \end{bmatrix} \tag{6}$$

$$P_{R_{k|k-1}}^{B} = J_{1\oplus} P_{R_{k-1}}^{B} J_{1\oplus}^{T} + J_{2\oplus} Q_k J_{2\oplus}^{T}, \tag{7}$$

where $J_{1\oplus}$, $J_{2\oplus}$ are the Jacobian determinants of spatial coordinates composite transformation.

$$J_{1\oplus}\left\{\hat{X}_{R_{k-1}}^{B}, \hat{X}_{R_k}^{R_{k-1}}\right\} = \begin{bmatrix} 1 & 0 & V_x\Delta T\sin\hat{\phi}_{R_{k-1}} - V_y\Delta T\cos\hat{\phi}_{R_{k-1}} \\ 0 & 1 & V_x\Delta T\cos\hat{\phi}_{R_{k-1}} - V_y\Delta T\sin\hat{\phi}_{R_{k-1}} \\ 0 & 0 & 1 \end{bmatrix} \tag{8}$$

$$J_{2\oplus}\left\{\hat{X}^{B}_{R_{k-1}},\hat{X}^{R_{k-1}}_{R_k}\right\}=\begin{bmatrix}\cos\hat{\phi}_{R_{k-1}} & -\sin\hat{\phi}_{R_{k-1}} & 0\\ \sin\hat{\phi}_{R_{k-1}} & \cos\hat{\phi}_{R_{k-1}} & 0\\ 0 & 0 & 1\end{bmatrix} \quad (9)$$

The complete system state vector expression is:

$$\hat{X}^{B}_{R_{k|k-1}}=\begin{bmatrix}\hat{X}^{B}_{R_{k-1}}\oplus\hat{X}^{R_{k-1}}_{R_k}\\ \hat{X}^{B}_{F_1}\\ \cdots\\ \hat{X}^{B}_{F_n}\end{bmatrix} \quad (10)$$

In order to constantly update and create map, the new environment features need to be added to the state vector, feature points corresponding to the underwater robot's position is represented by vector $\hat{X}^{B}_{F_i}$, covariance matrix is $P^{B}_{F_i}$, the augmented matrix vector is:

$$\hat{X}^{B}_{aug}=\left[\hat{X}^{B}_{R}\quad\hat{X}^{B}_{F_1}\quad\cdots\quad\hat{X}^{B}_{F_n}\quad\hat{X}^{B}_{new}\right]^{T} \quad (11)$$

where

$$\hat{X}^{B}_{new}=\hat{X}^{B}_{R}\oplus\hat{X}^{R}_{new} \quad (12)$$

the updating covariance matrix is:

$$P^{B}_{aug}=F_{aug}P^{B}F^{T}_{aug}+G^{B}_{aug}P^{R}_{new}G^{T}_{aug}, \quad (13)$$

where

$$F_{aug}=\begin{bmatrix}I & 0 & \cdots & 0\\ \cdots & \cdots & \cdots & \cdots\\ 0 & 0 & \cdots & I\\ J_{1\oplus} & 0 & \cdots & 0\end{bmatrix};G_{aug}=\begin{bmatrix}0 & \cdots & 0 & J_{2\oplus}\end{bmatrix}$$

P^{R}_{new} is the new covariance matrix of the feature vector.

In Fig. 2, it shows that the robot in the two-dimensional space is simulation effect picture of SLAM algorithm. The movement location of the robot is updated by the measured value in every state, then according to the location of the robot the coordinates of feature point are measured. In the figure, the blue point is the route sign testing point for robot, and the red point is the obstacle testing point. The red curve is a work route set for robot in advance and the orange thick line is a navigation route obtained based on SLAM algorithm. Through MATLAB simulation, it shows that the route is better consistent with the one set beforehand. But when there are too many environmental characteristics, the calculation spends a lot of time, and it is easy to cause offset. The algorithm is worth optimizing.

3.2. Underwater Obstacles Identification

It is necessary for underwater robot to have the ability of obstacles-identification during a travel underwater. The obstacle data obtained from the sensors must match the obstacle in the pre-processing map, if the successful matching number reaches a certain threshold, it proves that the obstacles do exist. And then do SLAM calibration process, in order to improve the diagnostic accuracy of obstacle information. When the same obstacle within the detection range of the sensor, it should be collected many times continuously in theory, however the obstacle added to the pre-processing map has not been collected many times continuously, then the match fails, which can diagnose that the obstacle information is not accurate, we should remove it from the pre-processing map. Now we need to set two thresholds of obstacle judgment $m\,\&\,n$, if the matching number is less than m, we remove the obstacle from the pre-processing map, if the matching number is greater than n, it judges that the obstacle does exist.

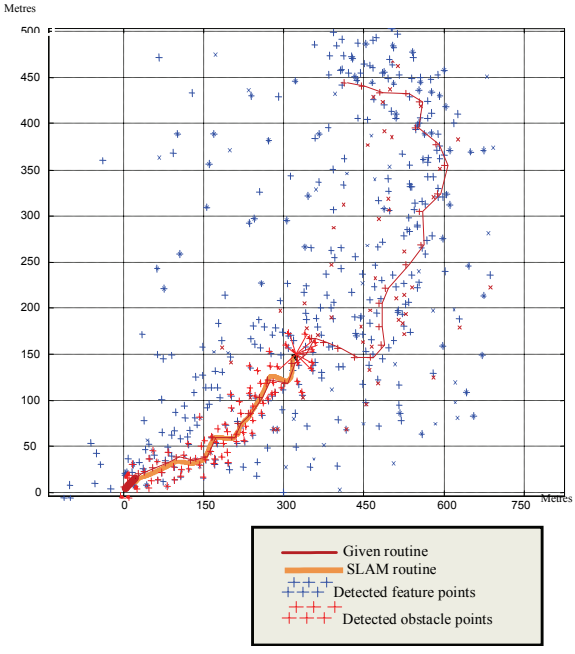

Fig. 2. Underwater Robot SLAM Incremental map simulation diagram.

4. Water Quality Measurement and Latitude and Longitude Match

4.1. Water Quality Measurement

The sensor probe integrates 7 kinds of sensors, they are: temperature, turbidity, blue-green algae, chlorophyll a, dissolved oxygen, PH, ORP. The test data from sensors is converted by A/Dmodule [13], and then the data transfer through RS485 communication interface to the underwater robot built-in microprocessor. RS485 level conversion chip choose MAX485. It has the advantages of low power consumption, automatic hibernation. In order to avoid mutual interference of the signals [14], and also to avoid failure of one signal affect other road signals, and the brightest chips are mutually independent in power supply, and the junction of the signal is opto-isolated [15,16]. This method can improve the reliability and stability of the system.

4.2. Water Quality Match Latitude and Longitude

Underwater robot built-in microprocessor contains a sensor database, the database mainly receives the test point data of each latitude and longitude, and at the same time open up enough memory to store the water quality parameters, the memory space is named with the current latitude and longitude. At the end of the test, the database now has stored all water quality parameters in different latitudes and longitudes. The research and development of Density Map software is based on this database created above. Firstly, the software automatically draws a contour map of the testing area, the then calls database data. The software shows different shades of color in the corresponding region according to the concentration of various water quality parameters, which can visually analyze the current water quality conditions .It also provide a research basis to the relevant departments. The data processing flow is shown in Fig. 3.

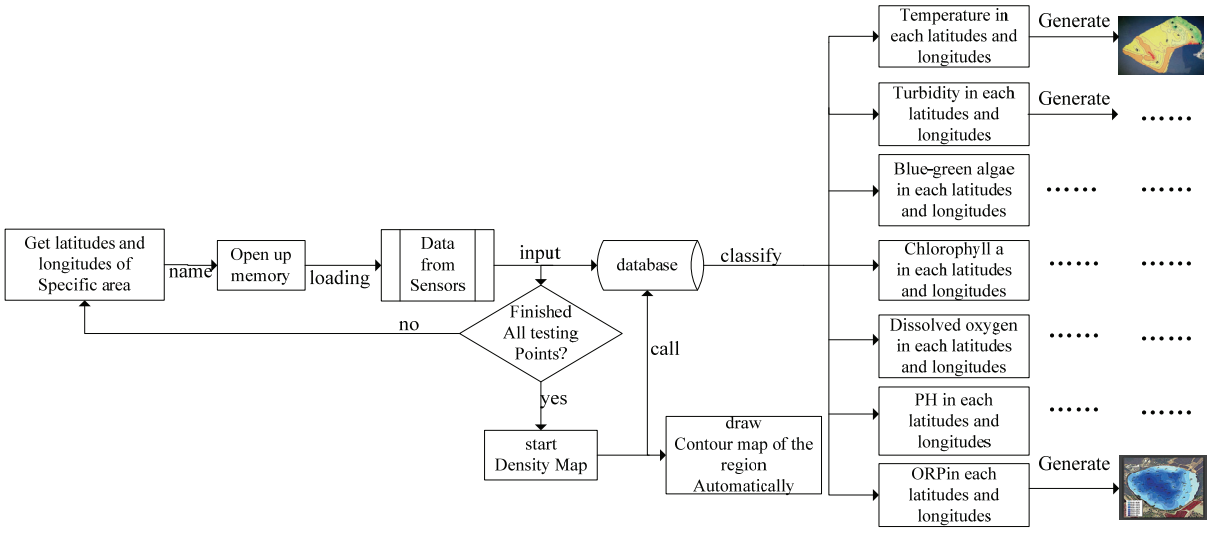

Fig. 3. The data processing flow of the system.

5. Experiments and Analysis

In November 2012, our groups have conducted several experiments in Nanjing External Qinhuai River of China. And at the same time we carry two kinds of the market water quality detector for simultaneous detection, which has initially verify the reasonableness of the design and the feasibility of the testing results.

5.1. Implementation Methods

The experiment was divided into two parts. The first part: the research group uses our own underwater robot to conduct water testing. The robot's hardware configuration mainly includes: 10 ADCP Doppler velocity taximeter instruments; 4 speed beam and a vertical (down) beam sounding and altimetry; electronic compass, model: Honeywell, HMR3000 electronic compass; side-scan imaging sonar, model: Delta T multibeam sonar; the depth sensor, model: Canada NetMind. The overall hardware device is shown in Fig. 4 The second part: we choose two kinds of water quality testing instrument sold in markets to do a comparative analysis. The two water quality testers (Fig. 5) are: XZ-0111 type 11 parameter water quality analyzer; 5B-3B (V8) multi-parameter water quality analyzer. They can not navigate autonomously, so we use fixed-point detection method.

Fig. 4. Prepare debugging.

5.2. Analysis

At the end of the underwater robot navigation, water quality parameters have been stored in the shore-based facility computer. Fig. 6 shows the Density Map software operation interface, the software is developed into to match the test parameters with the latitude and longitude match. As can be seen from Fig. 6, according to the needs of the environment of the area we test, we can selectively obtain the required measurement parameters. In the experiments, we choose temperature, dissolved oxygen, and turbidity three kinds of water quality parameters. Fig. 6 displays the concentration of each area, color difference represents different concentration, the darker the area, indicates that the concentration of the substance of the region is the deeper. Density Map not only show the latitude and longitude of each parameter diagram, we can also choose the drop-down box inside the data sheet directly to obtain the other parameters in different latitude and longitude.

In the experiment, we select the XZ-0111 type 11 parameter water quality analyzer and-5B-3B (V8) multi-parameter water quality analyzer for reference. Test results are as follows.

Tables 1-4. We treat the average of XZ-0111 and 5 B – 3 B measuring data as a standard reference value, then the error calculated as: Error=|underwater robot testing data - standard| /standard.

From the Tables 1 - 4, we can see, robot water quality parameters measurement error is always maintained within ± 5 %, to some extent shows the feasibility of our design. However, the biggest advantages of this system are that it can be combined with shore-based facility, and get the latitude and longitude matching diagram of water quality parameters.

Fig. 5 (a). XZ-0111.

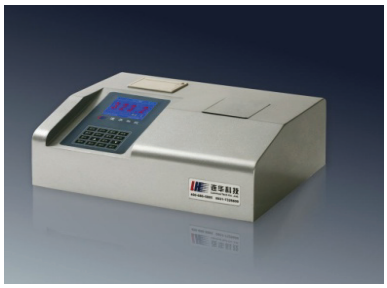

Fig. 5 (b). 5B-3B.

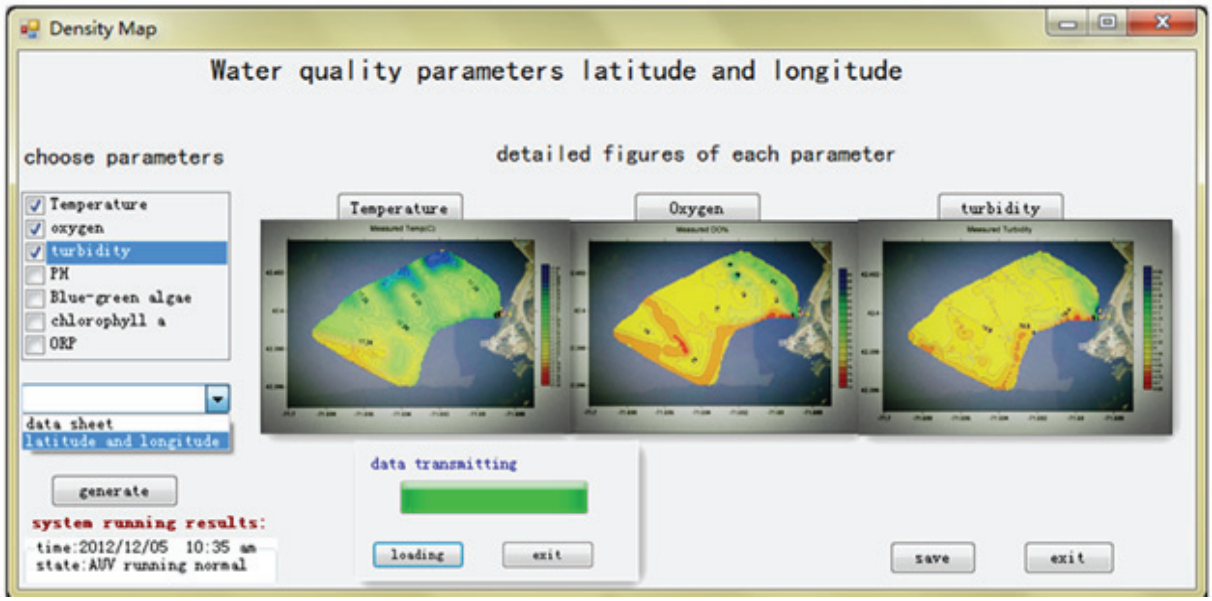

Fig. 6. Measuring results displayed on Density Map.

Table 1. 118.31°E, 32.04°N, 10 am.

Instrument	Tempera-ture, °C	Dissolved oxygen, mg/L	Turbidity NTU
XZ-011	13.7	9.2	20.5
5B-3B	13.9	9.5	21
Robot	13.6	9.8	20.2
Error	1.44 %	4.81 %	2.65 %

Table 2. 118.28°E, 32.27°N, 11 am.

Instrument	Tempera-ture, °C	Dissolved oxygen, mg/L	Turbidity NTU
XZ-011	14.1	8.8	22.5
5B-3B	14.5	8.9	21
Robot	14.4	8.8	21.2
Error	0.7 %	0.56 %	2.6 %

Table 3. 119.05°E, 31.52°N, 1 pm.

Instrument	Tempera-ture, °C	Dissolved oxygen, mg/L	Turbidity NTU
XZ-011	12.7	9.5	18.6
5B-3B	12.4	9.9	18.8
Robot	12.4	9.4	18.9
Error	1.2 %	3.1 %	1.1 %

Table 4. 119.14°E , 32.15°N, 2 pm.

Instrument	Tempera-ture,°C	Dissolved oxygen, mg/L	Turbidity NTU
XZ-011	12.9	9.2	21.5
5B-3B	12.6	8.9	21.8
Robot	13.0	9.3	21.3
Error	2.0 %	0.56 %	2.8 %

6. Conclusion

This paper innovatively put forward a water testing system equipped in underwater robot, getting rid of the traditional fixed-point measuring mode. The measuring method and accuracy have been greatly improved. However, there are still large room for improvement in data acquisition and processing.

In future research work, the following can be drawn with respect to the reach examined:

- The accurate and high efficient wireless communication between underwater robot and shore-based facility is very important to improve system performance, which can reduce memory spending.
- The present system now can choose to measure specific parameters, if the measurement conditions change, the other parameter measurements maybe required to measure, sensor needs to increase, the more kinds of data fusing together is a major difficulty.
- When there are too many environmental characteristics, the calculation spends a lot of time, and it is easy to cause offset. The algorithm is still worth optimizing.
- Underwater robot positioning is still a problem, and how to further precise positioning, coordinate the various navigation facilities working together is an important task for future research.

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Recent Advance of Biosensor Instruments in Pesticide Residues Rapid Detection

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Abstract: In recent years, efforts have been reported to develop portable biosensor instrument in the detection felid such like pesticide residues. There is a great potential in the applications of biosensor technologies for rapid detection of pesticide residues in food and environment. In other words, meantime, biosensor instrument has wide application in the instruments market.

This paper presents an overview of various transduction systems of instruments, such as based on electrochemical, optical, piezoelectric, and nanomechanics methods' biosensor instruments, which have been reported in the literature in the design and presentation of biosensor instruments for pesticide detection. Various immobilization protocols used for formation of a biorecognition interface are also discussed. Moreover, in the design of biosensor instruments, regeneration technology, include signal amplification, miniaturization, and biosensor devices are evaluated for the development and applications of these biosensors. It can be concluded that despite some limitations of the pesticide residues detection technologies, these biosensors for pesticide monitoring are becoming more and more relevant in environmental and food analysis, that is to say, with the development of means of biosensors for detection monitoring , corresponding biosensor instruments are also becoming one essential tools in society. *Copyright © 2013 IFSA.*

Keywords: Electrochenmical biosensor, Optical biosensor, Biosensor instruments, Pesticide residues.

1. Instruction

Pesticides (herbicides, fungicides, insecticides) are widely used in agriculture and industry due to their high insecticidal activity [1-3].

Pesticides derived from synthetic chemicals are essential inputs in increasing agricultural production by preventing control pest and crop losses before and after harvesting. One-third reduction in crop yield would be happened if pesticides are not used against pest [4, 5].

The pesticides is an important management tool to boost agricultural productivity-increase crop yield and reduce post-harvest losses, especially in a world facing a hunger crisis and famine. Due to high efficiency for insect elimination, easy synthesis, and low cost, different kinds of pesticides are the most widely used in agriculture [6, 7].

Numerous chemical compounds, routinely used in agriculture and chemical industry, can form persistent toxic residues in air, soil and water, thus polluting large geographical areas [8]. However, overuse of these pesticides results in pesticide residues in food, water and environment, and leads to a severe threat to human health due to their high toxicity to acetylcholinesterase (AChE), which is essential for the functioning of the central nervous system in humans [9, 10]. Numerous analysis methods such as

gas chromatography [11], high-performance liquid chromatography [12], capillary electrophoresis [13], flow injection immunoanalysis [14-16] and fluorimetry [17] have been developed for detection of pesticides residues. These analytical techniques have been described and reviewed extensively in the literature [18]. These methods are very sensitive and reliable, but they are time-consuming and expensive. Moreover, they can only be performed by highly trained technicians and are not convenient for on-site or in-field Detection. However, false positive may easily appear and this method also need some improvements (e.g. for continuous detection). These methods have some drawbacks such as poor selectivity, high cost, slow response, poor stability and time-consuming [19]. Moreover, they can only be performed by highly trained technicians and are not convenient for on-site or in-field detection, which limit their application for real-time detection.

Owing to a great mass of pesticides generally being used, there is an increased interest for developing rapid screening systems for their detection [20]. In recent years, biosensors are potentially useful as suitable complementary tools for the real-time detection of pesticides residues and have been an active research area for some years [21]. Biosensors are designed to detect and/or quantify target molecules such as those for medical use and environmental monitoring [22, 23]. Various biological recognition elements, including cofactors, enzymes, antibodies, microorganisms, organelles, tissues, and cells from higher organisms, have been used in the fabrication of biosensors [24, 25].

For that reason, they need some improvements (e.g. for continuous detection). Biosensors are potentially useful as methods to quickly detect pesticides and have been an active research area for some years [26]. Biosensors have been defined as analytical devices which comprise of two components: biomaterial and transducer. That is to say, biosensors used as a device which tightly combine biorecognition elements with physical transducers for detection of the target compounds [27, 28]. Lately, many biosensors are used for pesticide detection which are based on the inhibition reaction or catalytic activity of several enzymes in the presence of pesticides [29-33].

According to the classification of detecting signals, mainly divided as enzyme-based biosensors, immunosensors ,microbial sensors and cell-based biosensors. Enzyme-based biosensors for pesticide determination are reviewed in the literature [34, 35]. Enzyme-based biosensors (e.g. acetylcholinesterase biosensor) for pesticide determination have been widely reported in the literature [36-39]. Since a number of pesticides have a similar mode of action affecting the activity of the same enzyme, most of enzyme-based biosensors are used for screening purposes and are unspecific for individual pesticides. They can only detect total pesticide content and do not provide specific information about a particular pesticide [20]. Enzyme-based biosensors for pesticide determination have caused public interest due to their reliability, fast response, high sensitivity and selectivity. In recent years, enzyme-linked immunosorbent assays (ELISA) have grown rapidly as tools for pesticide measurement.

Immunosensors have been used to detect or quantify the specific pesticide based on the binding interactions between immobilized biomolecules (Ab or hapten) on the transducer surface with the analyte of interest (hapten or Ab), resulting in a detectable signal [41]. The sensor system takes advantage of the high selectivity provided by the molecular recognition characteristics of an Ab, which binds reversibly with a specific hapten. They consist of two process, a molecular recognition process, for sensing the specific Ag-Ab binding reaction at the surface of receptor, and a signal-transfer process, for responding to changes in an electrochemical, optical, spectroscopic, or electrical parameter of the receptor caused by the specific binding [42]. They appear to be appropriate for identification of a single pesticide or, in some cases, small groups of similar pesticides in environmental monitoring as they are rapid, specific, sensitive and cost-effective analytical devices [43].

Furthermore, with the development of biosensor detecting technologies, generating biosensor devices in biosensor instruments against pesticide molecules has been relatively mature, which has provided further impetus in this area. Excellent reviews that focused on biosensor instruments for pesticide monitoring were described in the Literatures. However, there is a time gap between current status in the field and the most recent reviews. In this review, several types of biosensor instruments developed for the different biosensors in pesticide analysis are highlighted. Various biosensor instruments used for pesticide residues in foods and vegetables are also discussed. In addition, the development of rapid determination and techniques of regeneration, signal amplification, miniaturization are evaluated for the development and applications of these biosensor instruments.

2. Classification of Biosensor

Biosensors have been defined as a device that tightly combine biorecognition elements with physical transducer by catching process reaction between sensitive element and target objects, then convert them to continuous or discrete electrical or optical signals which can become useful information. In general, immunosensors can be distinguished from immnunoassays where the transducer is not an integral part of the analytical system. The biorecognition element determines the degree of selectivity or specificity of the biosensor, whereas the sensitivity of the biosensor is greatly influenced by the transducer. Typically, biosensors classified in some ways. Depending on different kinds of biorecognition element, biosensor can be classified into enzyme sensor, microbial sensor, organal sensor, tissue sensor, immunosensor. According to the transduction mechanism, biosensors can be further classified into

electrochemical, optical, piezoelectric and nanomechanic immunosensors. Electrochemical transducers, classified as amperometric, potentiometric, conductometric, capacitive and impedimetric measure changes in current, potential (voltage), conductance, capacitance and impedance respectively [44-46].

With the development of biosensors, biosensor instruments have been appeared emerge a historic moment. Based on different biorecognition elements, biosensor instruments will be described in the following sections.

3. Instrument of Biosensor for Detection

Over the last decade, biosensors instruments based on biosensor theory have emerged as a promising technique for pesticide residues analysis, environmental monitoring, food quality control and military investigations. The sensitivity of the biosensor is greatly influenced by the transducer, and the biorecognition element determines the degree of selectivity or specificity of the biosensor [47]. These analytical instruments are designed to complement or replace the existing reference analytical methods (chromatographic and coupled chromatographic-spectrometric procedures) by simplifying or eliminating sample preparation, thus decreasing the analysis time and cost. Most biosensor instruments reported to date are utilizing different detection principle. Based on the transduction mechanism, biosensors can be further classified into electrochemical, optical, piezoelectric biosensor instruments as described in the following sections.

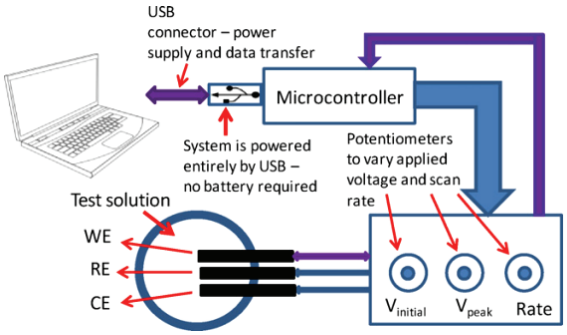
3.1. Instruments of Electrochemical Biosensors

Electrochemical biosensors have emerged the past few years as the most promising alternative to detect pesticide due to the high inherent to the electrochemical detection and the possibility of portability and miniaturization. Also the oldest and most common methods used in biosensors. They can determine the level of pesticides by measuring the change of potential, current, conductance, or impedance caused by the electrochemical reaction or immunoreaction. Thus, instruments of electrochemical biosensors also have an up and coming value of research. The instruments combine the high specificity of traditional electrochemical methods with the data acquisition, and thus exhibit great advantages. They have more advantages such like not affected by sample turbidity, quenching, or interference from absorbing and fluorescing compounds commonly found in biological samples as well as electrochemical biosensor instruments. But there still remains the big challenge in high-performance, cost-effective field analysis.

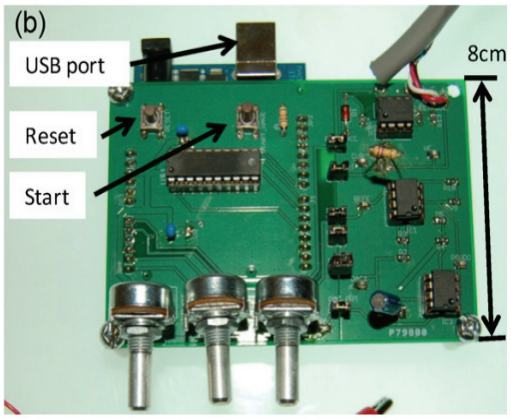
3.1.1. Instrument Based on Potentiometric Biosensors

Potentiometric biosensors are based on measuring the changes in potential induced by the label used, which occur after the specific binding of the Ab-Ag. They measure the potential across an electrochemical cell containing the Ab or Ag, usually by measuring the activity of either a product or a reactant in the recognition reaction monitored. Examples of these kinds of biosensors instrument system for the determination of the pesticides residues have been reported [48, 49]. A flow-injection system with dual amperometric and potentiometric OPH biosensors for the simultaneous and rapid measurements of OP compounds was described. The independence of the two analytical signals obtained with the dual-transducer system was illustrated [50]. The potentiometric biosensor responds favorably to all OP compounds, reflecting the pH changes associated with the OPH activity, the amperometric device displays well-defined signals only towards OP substrates (pesticides) liberating the oxidizable p-nitrophenol product. The potentiometric detection has been accomplished with a silicon-based pH-sensitive electrolyte-insulator-semiconductor (EIS) transducer, operated in the constant-capacitance (ConCap) mode. Electrolyte-insulator-semiconductor measurements were performed using the Con Cap mode, with a capacitance of 22.0 nF (in connection to total impedance of 15 kΩ). The sample injections were initiated after a 30 min stabilization of the bias voltage drift in the presence of the following carrier solution. Details of such EIS operation were described elsewhere [51]. Demonstrated a portable USB-based electrochemical biosensor prototype for point-of care testing (POCT) was designed, fabricated and tested with a model analyte. The device implements cyclic voltammetry (CV) measurements by means of a portable potentiostat in conjunction with a miniaturized electrochemical cell. The microelectronic circuit consists of a single-supply potentiostat and I/V converter, an analog-to-digital converter (ADC), and a microcontroller unit (MCU); power and controlled entirely by USB the device is suitable for field testing alongside any portable personal computer (PC). Using a simple operation protocol we can adjust the voltage range and scan rate of cyclic voltammetric measurement. The data acquisition and display can be captured using Microsoft Excel, a commonly available program on most of today's PCs. They designed a portable potentiostat that would execute cyclic voltammetric measurements. In order to create such a power Portable system, we opted for a USB powered version so there would be no need for an external supply, doubling also as the communication bus to the PC for data logging and analysis. Their system is compatible with any computer that has a USB ported is equipped with the Microsoft Excel program. However, in order to process the data and calculate the under the anodic peak, additional software is needed; we used Sigma

Plot (Systat Software, Inc.) to accomplish this task [52].



(a)

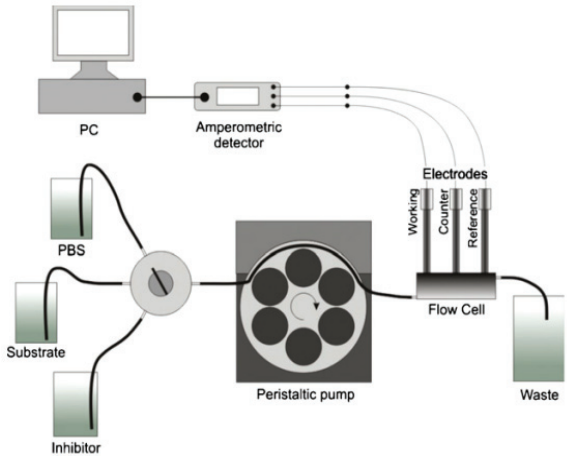


(b)

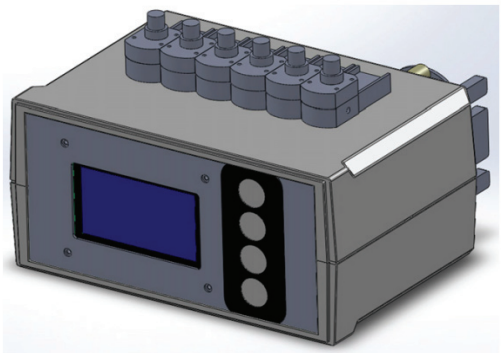
Fig. 1. (a) Overall design of the USB-powered portable electrochemical biosensor prototype; (b) Photograph of the potentiostat prototype showing the function keys/ports.

3.1.2. Instrument Based on Amperometric Biosensors

Among electrochemical biosensors, amperometric AChE biosensors as a combination of enzymatic reactions with the electrochemical methods have shown satisfactory results for pesticides analysis, where the enzyme activity was employed as indicator of quantitative measurement of insecticides [53]. When AChE is immobilized on the working electrode surface, its interactions with the substrate of acetylthiocholine (ATCl) produce the electro-active product of thiocholine [54]. The reaction equation is shown as follows [55-57]. Amperometric sensors are a subclass of electrochemical sensors. For the simultaneous analysis of several samples using only one device to developed a multichannel biosensor [58]. The detection limit for free 2,4-D in water was 0.1 ng/mL. The advantages of the presented electrochemical detector were high stability and sample through put, low detection limit, the ability to be repeatedly used, and no need for regeneration. One example is that Ivaylo Marinov et al. describe a flow-injection system with integrated amperometric biosensor featuring a replaceable AChE-immobilzed membrane with incorporated gold nanoparticles [59]. The flow-injection system comprised the following elements: three electrodes-a platinum working electrode, a standard calomel electrode (SCE) and an auxiliary platinum wire electrode; a flow-cell with a working volume of 0.275 mL; a peristaltic pump and an amperometric detector (Palm Instruments BV, The Netherlands). The flow-injection system configuration is presented in Fig. 2 (a). The biosensing device results in an integrated, automatic and portable system for environmental and Agri-food application (Fig. 2).



(a)



(b)

Fig. 2. (a) Flow-injection system; (b) 3D computer aided design (CAD) view of the biosensor device.

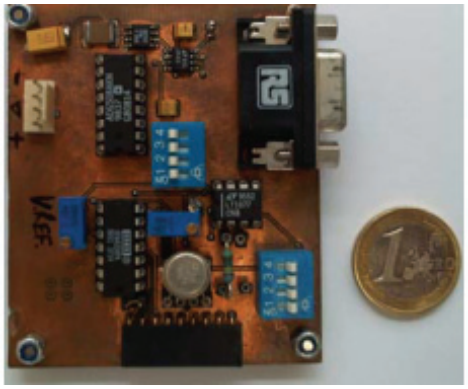
The working conditions were optimized so that the throughput of the analytical system was improved and the incubation, measurement and reactivation time of one analytical cycle was reduced twice in comparison with our previous work [60]. Where analytical measurements were conducted under static conditions. Anti-cholinesterase activities (expressed by the detection limits and the bimolecular inhibition constants) of three pesticides-paraoxon ethyl, monocrotophos and dichlorvos as well as their mixtures were assessed by the use of a flow-injection

system. This competitive behavior is worth being investigated in detail since it may present an opportunity for the discrimination between two or more pesticides present in an analyzed sample by employing chemometrics and only one type of acetylcholinesterase. The differentiation could be based on the reduction of the anti-cholinesterase activity of a supplementary inhibitor (pesticide) with known and high inhibition potency. Another example is that Viviana et al. has designed and developed one miniaturized biosensing array system was designed to be flexible, modular and small, with six independent sensing cells equipped with optical excitation and detection, current measurement and flow control systems [61]. In this study, multitask biosensor for the detection of endocrine disrupting chemicals is proposed. The sensing system employs an array of biological recognition elements. Amperometric and optical transduction methods are provided in an integrated biosensor together with flow control systems Fig. 2 (b), the biosensor system was able to detect most of the chemicals analyzed with very high sensitivity. Atrazine and diuron were detected with a limit of detection of 0.5 nM, with an RSD% less than 5 %; paraoxon and chlorpyrifos were revealed with a detection of 5 μM and 4.5 μM, respectively, with an RSD% less than 6 %; catechol and bisphenol were identified with a limit of detection of 1 μM and 35 μM respectively, with an RSD % less than 5 %.

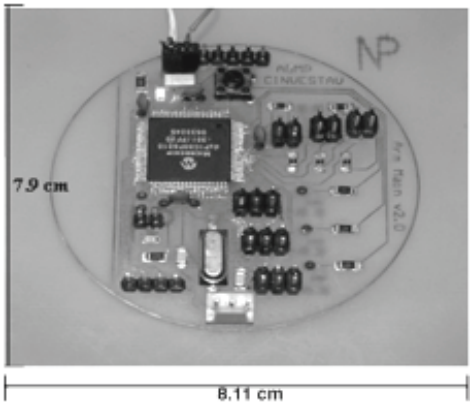
AChE-based biosensors have a major drawback: they give a sum parameter of AChE-inhibition without any qualitative or quantitative information about the individual analytes. One approach to solve this problem involves the application of multi-sensor arrays that are combined with the data processing of artificial neural networks. Using micro or nanosensor arrays will likely become a new trend. Thus, the ability to construct arrays of enzymes will likely allow current multianalyte detection of several compounds to be expanded to accommodate the analysis of perhaps hundreds or thousands of separate compounds [62]. One of the challenges that must be met for this type of system would be the development of parallel computational methods to convert electronic responses for each analyte into meaningful concentration data. In this respect progress has been reported for artificial neural network implementation in single low-cost chip for the detection of insecticides by modeling of enzymatic sensors response [63, 64]. The design of single chip to selectively quantify mixtures of the pesticides chlorpyrifosoxon and chlorfenvinfos by an artificial neural network implementation was shown in Fig. 3B, and the dedicated system was shown in Fig. 3C [65].

3.1.3. Instrument Based on Impedance Analysis

Recent development in the area of electrochemical biosensor instruments with impedance methods of is very encouraging and offers potential advantages.



(a)



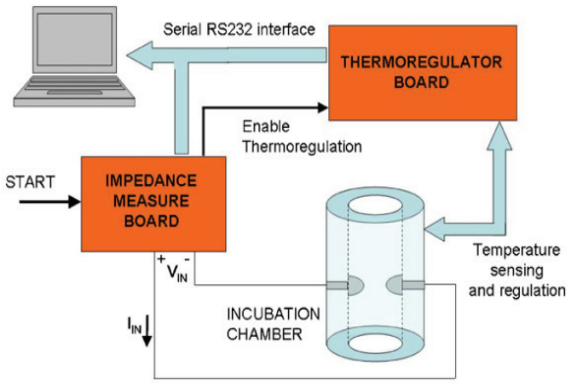
(b)

(c)

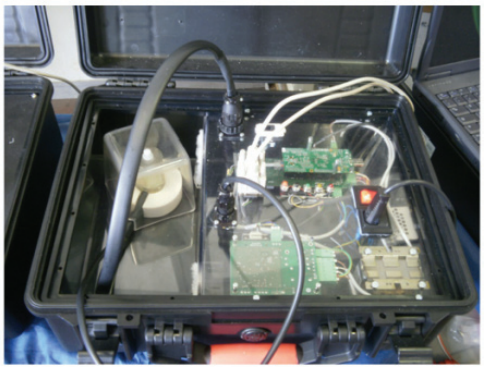
Fig. 3. Picture of the miniaturized electronic plate that functions as a potentiostat.

Usually, the concepts of impedance, conductance, capacitance, and resistance are different ways of monitoring the test system and are all interrelated [66]. Impedance biosensors instruments measure different changes of an electrical field. Electric impedance spectroscopy (EIS) is a sensitive technique, which detects the electrical response of the system studied after application of a periodic small amplitude AC signal. M. Grossi et al. developed an embedded portable biosensor system for bacterial concentration. The portable biosensor system described in this work, represented in Fig. 4, features two circuit boards: one, dedicated at maintaining the SUT at the target temperature by means of an adhoc algorithm running on the microcontroller ATmega168 by Atmel (California, USA); the other, based on the microcontroller ARM STR912 by ST Microelectronics (Agrate Brianza, Italy), is used for

signal analysis and the calculation of the impedance and its components. The SUT incubation chamber features a couple of stainless steel electrodes for the electrical measures, a temperature sensor and a heating system for thermal regulation.



(a)



(b)

Fig. 4. Photograph (a), and schematic (b) of the biosensor system presented in the work.

The system is composed of an impedance measurement board, a thermoregulation board and an incubation chamber containing the sample under test. The system is suitable for applications in the industrial field (in particular the dairy one is of interest here), as well as for environmental monitoring (for instance for the case of water microbial screening). The greatest advantage of the presented system is that it is portable and low cost. On the contrary, most of the biosensors for microbial concentration determination on the market are essentially bench top instruments to be used within microbiology laboratories [67].

3.2. Optical Biosensor Instruments

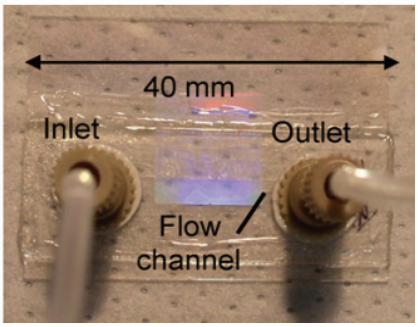
Compared with electrochemical approaches, optical devices are the second most commonly used transducers. A general optical sensor system consists of a light source, a number of optical components to generate a light beam with specific characteristics and to direct this light to a modulating agent, modified sensing head, and a photodetector [20]. Different techniques can be used for creating an optical change e.g., optical waveguide light mode spectroscopy, total internal reflection fluorescence, and surface plasmon resonance. More advantages like compactness, flexibility, resistance to electrical noise and a small probe size to be offered by optical biosensors. It is important to wash the surface before reading of the signal.

3.2.1. Optical Waveguide Lightmode Spectroscopy (OWLS)

The OWLS technique is a new sensing technique using evanescent field for the in situ and label-free study of surface processes at molecular levels. It is based on the precise measurement of the resonance angle of linearly polarized laser light, diffracted by a grating and in coupled into a thin waveguide layer. Actually, in coupling is a resonance phenomenon that depends on the refractive index of the medium covering the surface of the waveguide occurs at a defined angle of incidence**.** In the waveguide layer, light is guided by total internal reflection to the edges where it is detected by photodiodes. In recent years, efforts have been reported to develop portable sensors using fully packaged approaches based on surface plasmons [68-71] or to reduce size by integration of sample handling on-chip (e.g. in ring resonators [72]) yet these systems do not achieve a truly hand-held size ordo so at expense of lack of multiplexing or non-scalable device manufacturing [73, 74]. Sonia Grego et al. previously fabricated grating-integrated waveguide devices and characterized their Sensing performance using benchtop components including a wavelength swept laser source for telecommunication testing [75]. That work demonstrated a large wavelength scan at kHz speed driven by an external voltage source and was realized as a benchtop sensor apparatus. In 2012, another paper from Sonia Grego et al. describes a miniaturized system serving as optical reader of the waveguide devices including a new optical source and photodetector as well as multichannel capability. The configuration is designed and the components are selected with the Ultimate objective of integration into a hand-held device interrogating disposable sensor chips. The miniaturized system performance is improved compared to our previously reported values to a detection limit of 1×10^{-5} RIU and an on-chip reference configuration is demonstrated [76].

3.2.2. Surface Plasmon Resonance (SPR) Instruments

SPR has an inherent advantage over other types of biosensors in its versatility and capability of monitoring binding interactions without the need for labeling of the biomolecules.



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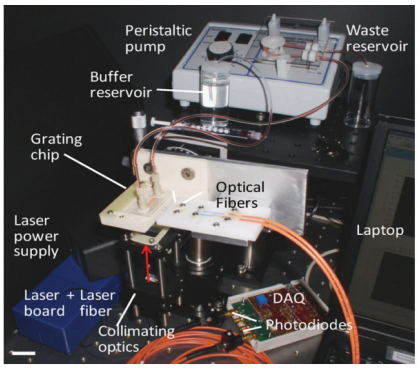


(b)

Fig. 5. (a) Photograph of the assembled cartridge, (b) Photograph of a benchtop prototype including all the components for the TWIST instrument with two sensor outputs. Scale bar is 2.5 cm. Food coloring is used to highlight the flow path.

It is versatile owing to its outstanding attributes of miniaturization, reliable portable instrumentation, and automation. Monitoring of the pesticide chlorpyrifos in water samples was performed using SPR immunosensors [77]. The chlorpyrifos derivative was immobilized onto the gold-coated sensing surface and competed with free chlorpyrifos for binding to the Ab, and as a result, increasing concentrations of chlorpyrifos will reduce the SPR signal. Other examples of single and multianalyte assays for simultaneous detection of different pesticides by SPR were reported by the same research group [78]. This portable biosensor based on SPR technology could provide a highly sensitive detection of pesticide analytes at nanogramper liter levels. The stability of the biosensor was proved by performing 15 series of daily measurements. Another sensitive and reusable SPR based immunosensor was developed for the determination of 2,4-D [79]. The SPR sensor was capable of detecting part per billion levels of 2,4-D in 20 min and the regeneration ability enabled the achievement of as many as 20 measurement cycles. SPR biosensors may suffer from disturbances caused by changes in the refractive index or temperature. The use of a reference surface makes it possible to separate signals related to binding events from signals caused by differences in refractive index between a sample and running buffer. A laboratory prototype of the SPRCD sensor has been developed. Initial version of the sensor consisted of a microfluidic cartridge incorporating the SPRCD structure and six microfluidic channels and a compact SPRCD reader [80].

Recently, capabilities of the SPRCD sensor (Fig. 6) have been further expanded. A sophisticated temperature control employing two peltier elements was incorporated to suppress the effects of the ambient temperature on the sensor response. In addition, to ensure reproducible placement of the SPRCD cartridge in the reader, a motorized system for the loading of the cartridge in the SPRCD reader was developed. Upon loading, the cartridge is automatically connected to the fluidic system of the SPRCD reader [81].

Fig. 6. A laboratory prototype of the compact SPRCD sensor.

3.3. Other Optical Biosensor Instruments

The fiber-optic biosensor consisting of AChE-immo-bilized LB film was constructed for simple and direct detection of orgarnophosphorus compounds in contaminated water. The configuration of sensor system is schematically shown in Fig. 7. The three kinds of solutions (distilled water, o-nitrophenyl acetate, and sample solution) were prepared. The absorbance change of product to be induced directly by the inhibition of organophosphorus compounds on immobilized enzyme without a pH indicator was successfully detected by the sensor system. And the proposed biosensor could successfully detect the organophosphorus compounds up to 2 ppm and the response time to steady signal of the sensor was about 10 min [82].

Fluorescence biosensor research was to develop, from first principles, a low cost, potentially portable, capable of detecting low levels of contaminants and compounds of interest in food analysis. From first principles, about L32 cm × W24 cm × H16 cm prototype biosensor detection unit was constructed that was composed of an illumination component, a sensor unit, a flow cartridge and a peristaltic pump, as illustrated in Fig. 8. The prototype fluorescent biosensor detection unit has been successfully developed that exhibits highly acceptable baseline stability, and in its current form, i.e., a manual system,

shows excellent signal repeatability. This unit has been tested using two models and has proven to be capable of detecting binding and inhibition with both compounds with a good degree of repeatability [83].

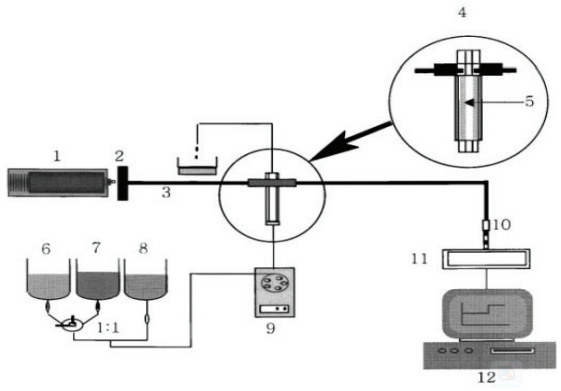
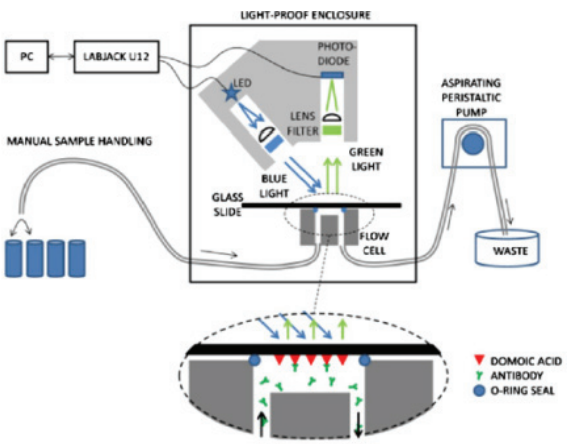
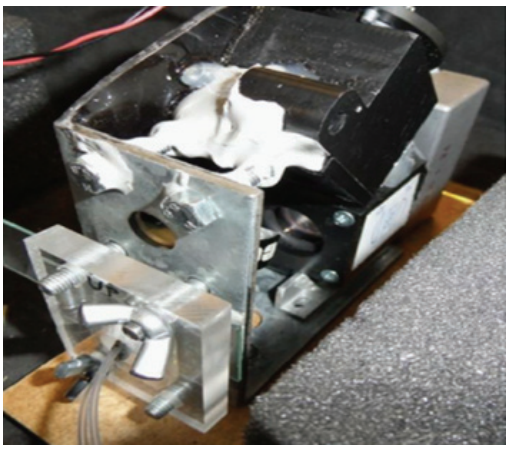


Fig. 7. The experimental set-up for biosensor: (1) xe lamp; (2) 400 nm filter; (3) optical fiber; (4) reactor containing AChE LB film; (5) immobilized enzyme layer; (6) sample; (7) distilled water; (8) sub-strate; (9) peristaltic pump; (10) photodiode; (11) A/D converter; (12) computer.



(a)



(b)

Fig. 8. (a) Diagram of the biosensor detection unit showing light source, sensor unit, flow cartridge and glass slide integrated within a light-proof enclosure (L 32 cm ×W 24 cm × H 16 cm) and connected to a PC via a labjack U12 data acquisition device. (b) Photograph showing clamping of glass slide within the biosensor.

C. Ercole et al. describes the properties of a biosensor for the determination of E. coli cell number in water samples. The potentiometric alternating biosensing (PAB) system utilized is based on a transducing element (light addressable potentiometric sensor (LAPS)) detecting pH variations due to NH_3 production by an urease–E. coli antibody-conjugate [84, 85]. In this work, C. Ercole et al. demonstrate how an immunoassay-based PAB sensing system was able to specifically detect E. coli strains in water. This system is quite flexible, manageable and easy to construct utilising commercial electronic components and it can be further improved [86].

On top a detailed drawing of the measuring chamber is reported with the sensing electrode made of silicon, silicon dioxide and silicon nitride. In the bottom, the signal measurements performed by a synchronous demodulation techniques and the conditioning electronics blocks are shown (Fig. 9).

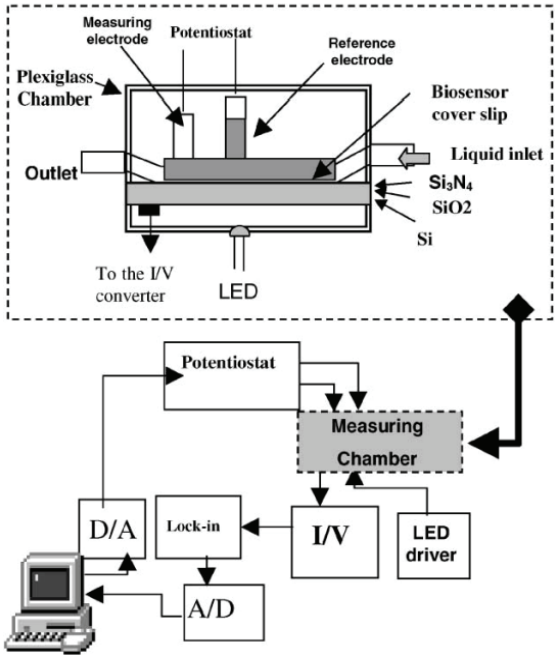


Fig. 9. Block diagram of the transducer system.

4. Immobilization Protocols

In terms of the development of electrochemical biosensors, the Ab/hapten immobilization onto a transducer or a support matrice is a key step in optimizing the analytical performance, such as response, reproducibility, stability, selectivity and regeneration. Some requirements should be met for one good immobilization method: (1) be simple and fast; (2) produce immobilized reagents that are stable and do not leach from the substrate; and (3) maintains its biological integrity flexibility, and proper active

site orientation toward the bulk solution. Thereby, Ab/hapten immobilization has been a critical issue in immunosensor technology [87-91]. Ab/hapten immobilization consists of physical adsorption and chemical binding, which depends on the driving force [92]. In general, they mostly fall into following methodologies.

1) Physical adsorption: Generally physical adsorption is based on interactions ,such as van der Wals forces, electrostatic interactions and hydrophobic interactions ,between the Ab/hapten and the transducer. Physical adsorption is simple and easy, but nonspecific attractive forces easily causes Ab/hapten desorption [93]. It is especially common on hydrophobic polymer surfaces. The main advantages of this mode of immobilization are its rapidity and simplicity, while its main drawbacks are random orientation and weak attachment. Gobi et al., created a functional sensing surface of the immunosensor by immobilizing an ovalbumin conjugate of 2,4-D (2,4-D-OVA) by simple physical adsorption on a thin-film gold chip. It has been established that the Au surface of the sensor chip was completely covered by 2,4-D-OVA up to a monomolecular layer and that the 2,4-D-OVA immobilized sensor chip was highly resistive to non-specific binding of proteins [94].

2) Covalent coupling: Ab/Ag can be covalently linked to the surfaces of a transducer through formation of a stable covalent bond between functional groups of Ab and the transducer. On the one hand, the procedure provides increased stability of the Ab ,but on the other hand covalent coupling is decreases the activity of Ab/Ag and is generally poorly reproducible. Blocking steps are usually necessary to limit the non-specific binding. An example of where this approach has been exploited is illustrated by immobilization of 2,4-D on the surface of amino-silanized optical fibers [95].

3) Self-assembled monolayer (SAM): Self-assembled monolayers (SAMs) have aroused much interest due to their potential applications in biosensors, biomolecular electronics and nanotechnology. This has been largely attributed to their inherent ordered arrangement and controllable properties. SAMs can be formed by chemisorption of organic molecules containing groups like thiols, disulphides, amines, acids or silanes, on desired surfaces to fabricate immunosensors [96]. This technique was used to immobilize analyte derivatives onto the surface of gold-coated sensors. The immunosurface formed can be used for over 120 cycles In addition to these conventional methods, new materials, such as nanoparticles have been employed in immobilizing Ab when constructing immunosensors Nanoparticles are used as solid phase due to their large surface area in a small fluid volume and good biocompatibility. Biological interactions, such as biotin/streptavid in interactions can be used to easily immobilize the Ab on the surface of nanoparticales. One of the most advantageous features of this system is that although the affinity constant between avidin and biotin is rather high, the bonding is of non-covalent nature, which allows for multiple washing and reuse of the same sensing device. Recently, describes the development of an electrochemical immunosensor for the analysis of atrazine associated to biotinylated-Fab fragment K47 antibody. The sensors are based on mixed self-assembled monolayer consisting of 1,2 dipalmitoyl-sn-glycero-3-phosphoethanolamine-N-(biotinyl) (biotinyl-PE) and 16-mercapto -hexadecanoic acid (MHDA) [97].

In addition to these conventional methods, new materials, such as nanoparticles have been employed in immobilizing Ab when constructing immunosensors Nanoparticles are used as solid phase due to their large surface area in a small fluid volume and good biocompatibility. Biological interactions, such as biotin/streptavid in interactions can be used to easily immobilize the Ab on the surface of nanoparticales. One of the most advantageous features of this system is that although the affinity constant between avidin and biotin is rather high, the bonding is of non-covalent nature, which allows for multiple washing and reuse of the same sensing device.

5. New Trends

5.1. Miniaturization

Miniaturization is expected to have a marked impact in the development and applications of every kind of biosensor instruments. Miniaturization of a biosensor instruments not only reduces the size of detection device and integrates all steps of the analytical process into a single-sensor device, but also reduce the difficulty of experiment technology and the requirement of experimenter. Thus, it results in reduction of both the time and cost of analysis. Moreover, it is expected to lead to a further portability for in vivo sensing and in-field applications. The miniaturization trend of biosensor instruments is adapted to the miniaturization of microfabrication and nanofabrication techniques, such as micro-electromechanical systems (MEMS). MEMS combined with microelectronic circuitry are sometimes referred to as “smart sensor systems for sensor instruments”, which in turn can be configured into highly portable and inexpensive handheld instrumentation [98].

There still have some challenges remain to be overcome before miniaturized sensors instrument can be commercially applied. For one thing, miniaturization coupled with higher detection sensitivity places serious demands on both the design of instruments and methodology inherent in Ab/Ag immobilization. Furthermore, the practical application of miniaturization requires the use of complete analysis systems for sample handling, such as pumping, filtering, valving, and sample conditioning [99].

To the best of our knowledge, due to the high miniaturization and portability, immunosensors based

on SPR are now the most well known in biosensors, and then SPR instruments could faster become a popular technology in domain of detection. The combination of SPR immunosensor with flow-injection analysis (FIA) techniques offers great potential in future devices for pesticide monitoring.

5.2. High Throughput of Detection Samples

AChE-based biosensors instruments have a major drawback: they give a sum parameter of AChE-inhibition without any qualitative or quantitative information about the individual analytes. That leads to the same flaw to enzyme-based biosensor instruments. One approach to solve this problem involves the application of multi-sensor array that are combined with the data processing of artificial neural networks. Using micro or nano-sensor arrays will likely become a new trend.

5.3. Integration of Detection System

One of the challenges that some new type biosensor instruments must be met for this type of system would be of the development of parallel computational methods to convert electronic responses for each analyte into meaningful concentration data. Recently, silica based monoliths, coupled with micro-fluidic devices, have been used as an attractive alternative to packed columns for the analysis of proteins, peptides and nucleic acids with special features of low diffusion resistance during mass transfer, controllable porosity and low back pressure compared to packed columns.

5.4. Signal Amplification

Signal amplification is crucial for obtaining low detection limits in biosensors. Most amplification schemes for biosensor instruments rely on target labeling. Thus, amplification techniques lay outside the domain of label-free immunosensors. To amplify the immunoreaction signal, many methods have been proposed. Generally, methods for amplifying the immunoreaction signal are based on the detection of electro-active species, which are generated in the presence of tracer coupled to the Ag or Ab.

5.5. Real Samples Detection

The majority of biosensor instruments reported to date have been designed for detection of pesticides in environment. Application to other matrices such as food samples (fruits and vegetates) has been restricted due to the problems related to the use of these devices in the presence of organic solvents extracts. Several works reported that enzyme biosensors can function in a mixed aqueous-organic phase in low amounts of organic solvents [100, 101].

6. Conclusions

Biosensor instruments based on electrochemical, optical, and piezoelectric biosensors are strong candidates for screening pesticide residues and they become more and more relevant in environmental and food analysis, especially electrochemical biosensors. Compared to chromatography and other methods, the strengths of biosensor instruments can be described as follows:

-They are same selective and sensitive as traditional detection analytical techniques instrument, such as gas chromatography, etc.

-They can be carried out for use in the field.

-They can work with complete automation and give the results after a short period of time.

In particular, biosensor instruments based with electrochemical, optical, and piezoelectric transducers have the potential to achieve the low limits of detection imposed by legislation. Despite the promise of immunosensors, they do have certain limitations. For example, few biosensor instruments are commercially available at the present time and are yet to be established as research or routine tools, due to a lack of validated protocols for a wide range of sample matrices.

However, these biosensor instruments still face many challenges hindering their real applications.

-The data are analyzed on the electrochemical analysis instrument, and they are not enough miniaturization and portable.

-They have not specific and suitable real samples pretreatment.

-They only give a simple parameter of AChE-inhibition without quantitative information and high throughout detection.

Acknowledgements

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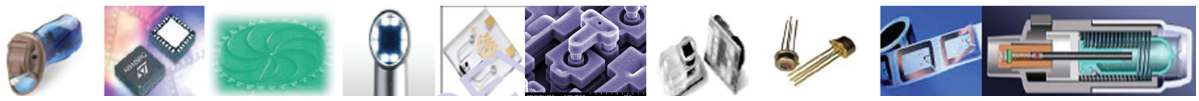

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Electrochemical Sensors of Cyclic Voltammetry to Detect Cd(II) in Blood Medium

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Abstract: An electrochemical investigation of cadmium species has been carried out by using cyclic voltammetry (CV) at different modified glassy carbon electrodes (GCE). The modified CNT/GCE, C_{60}/GCE and modified activated carbon (AC) on GCE as AC/GCE were used as working electrodes in CV. One application of these electrodes is detection of Cd(II) ion in blood medium using CV technique. Electrodes response was obtained for the oxidation and reduction peaks of Cd(II) ion in blood medium at modified CNT/GCE, C_{60}/GCE, and AC/GCE. A well defined oxidation and reduction current peaks appeared at -0.54 and -0.67 V versus Ag/AgCl with a current enhancement and peak potential shift toward higher potential due to nano materials (CNT and AC) comparison with C_{60}/GCE and bar GCE. Besides that, the presence of CNT or AC on the GCE in blood media caused an increase of the oxidation and reduction current peaks of Cd(II) ion (current enhancement) by about five times compared to the use of other modified electrodes. A linear relationship (R^2=0.9567, Y=2779.9X+5.5598) was observed for the plot of current (μA) versus concentration range of 8×10^{-5}-1×10^{-4} M of cadmium ion in blood medium using CNT and AC modified GCE. Based on the background noise of 50 data points, adjacent to the oxidation peak of Cd(II), and 3 σ/slope, a detection limit of 8×10^{-5} M was determined. So it can be said that these modified electrodes could be used as good sensors in CV for the detection of traces of cadmium ion in blood medium. *Copyright © 2013 IFSA.*

Keywords: Sensors, Blood medium, Cd(II), CNT/GCE, C_{60}/GCE, AC/GCE.

1. Introduction

A highly toxic material of heavy metals in environment is cadmium (Cd). Due to their extreme toxicity, this metal must be detected at very low levels in biological fluids such as blood. Nanoanalytical based sensors that work with complex biomatrices such as blood, urine, or saliva are being developed and validated and will improve our ability to make definitive associations between chemical exposures and disease [1-4].

Nanomaterials have become an extremely popular theme in recent electrochemical sensing research, due to their electrical conductivity, unique structural and catalytic properties, high loading of biocatalysts, good stability and excellent penetrability. Carbon nanotubes (CNTs) can be used as electrode materials with useful properties for various potential

applications including miniature biological devices. These sensors achieved higher response current, low work potential and low interference. Carbon nanotube was used to modify glassy carbon electrode sensor, which performed the electroreduction at a low operating potential. In general, voltammetric sensors examine the concentration effect of the detecting species on the current-potential characteristics of the reduction or oxidation reaction involved [5-7].

Voltammetric behaviors of Cd(II) ion in the presence of a ligand with glutathione were studied using cyclic voltammetry (CV). The coordination chemistry of reduced glutathione is of great importance as it acts as an excellent model system for the binding of metal ions. It was observed that an addition of glutathione as a ligand to solution containing Cd(II) with sulphate as a supporting electrolyte caused an increase in the reduction current of Cd(II) by several factors and also with a slight cathodic shift in the reduction peak potential of Cd(II). Further assessment of the chemical and physical conditions that may favor optimum current enhancement was done by studying the effect of varying pH, supporting electrolyte concentration of ligand and metal ion, interfering ions and scan rate [8].

It is well known that lead and cadmium can be determined with good sensitivity using anodic stripping voltammetry (ASV). The optimization of the analytical parameters pH, electrolyte composition, and deposition time and potential was studied. The method was successfully applied to the determination of cadmium and lead in lake waters and seawater after UV digestion [9].

The aim of this work was to compare three electro-chemical instruments [a standard potentiostat (Autolab), a commercially available miniaturized potentiostat (PalmSens) and a homemade micropotentiostat] for easy to use and sensitive determination of cadmium(II) and lead(II) ions. The lowest detection limits (hundreds of pM) for both metals was achieved by using of the standard potentiostat, followed by the miniaturized potentiostat (tens of nM) and the homemade instrument (hundreds of nM). Nevertheless, all potentiostats were sensitive enough to evaluate contamination of the environment, because the environmental limits for both metals are higher than detection limits of the instruments [10].

The simultaneous determination of copper, lead and cadmium by anodic stripping square wave voltammetry in HAC-NaAC buffer solution (pH=4.5) was studied. The method was rapid, simple and accurate, and could be used for the determination of copper, lead and cadmium in vinegar [11].

The preparation of Hg(II)-modified multi walled carbon nanotube (MWCNT) by reaction of oxidized MWCNT with aqueous Hg(II) was carried out. The Hg(II)-modified multi walled carbon nanotube (Hg(II)/MWCNT) dispersed in Nafion solution was used to coat the polished graphite electrode surface. Differential pulse anodic stripping voltammetry of ppb levels of cadmium and lead using the modified electrode yielded well-defined peaks with low background current under a short deposition time. The determination of Pb(II) and Cd(II) in tap water and Pb(II) in human hair samples was carried out [12].

The estimation of Pb, Cd, Cu, Zn, Fe, Se concentration in the tap water of Jeddah city in Kingdom Of Saudi Arabia was accomplished using electrochemical methods. The obtained results were lower than the average range of these elements in the maximum concentration as they were allowed to be by The World Health Organization (WHO) [13].

An electrochemical sensor for the detection of cadmium ions is described using a gold electrode. A cadmium ion forms a complex with glutathione via the free sulfhydryl group and also to the carboxyl groups. This complex ion is reduced by linear and Osteryoung square wave voltammetry with a detection limit of 5 nM [14, 15].

Heavy metal contaminations create high risk of health hazards across the world. Toxic metals deactivate valuable enzymes of the human body, causing a health hazard. In this paper we have reported a monitoring system for detection of a few heavy metal ions (e.g., mercury, cadmium, and arsenic). The process utilizes deactivation of the enzyme urease by the metal present in the sample. The three electrode screen-printed assembly with working electrode made of rhodinized (10 %) carbon was used to oxidize ammonia produced by deactivated enzyme and urea. The system was thus suitable for amperometric measurement. It was also durable and could be used without any sample preparation. Calibration curves were constructed for pure samples of various concentrations (0–20 ppb) of the three heavy metal ions [16].

In this work, CNT, C_{60} and AC were modified GCE by mechanical and solution evaporation methods to be used as good sensors for detecting traces of Cd^{2+} in blood medium by cyclic voltammetric technique [17, 18].

2. Materials and Methods

2.1. Materials

CNT (Fluka, 98 %), C_{60} (Mark, 99 %) and AC (Fluka, 98 %). Other chemicals and solvents were used of annular grade and as received from the manufacturer. Distilled water was used for the preparation of aqueous solutions. All solutions were deairated with oxygen free nitrogen gas for 15 minutes prior to making the measurement.

2.2. Instruments

Electrochemical workstations of Bioanalytical System Inc. USA: Models BAS CV 50 W with potetiostate driven by electroanalytical measuring

software was connected to PC computer to perform cyclic voltammetry (CV), an Ag/AgCl (3M NaCl) and Platinum wire were used as a reference and counter electrode respectively (Instruction manual, 1996). The working electrodes used in this study were GC electrode and modified GCE with CNT by mechanical attachment method (CNT/GCE) (Scholz and Lange 1992; Tan *et al,* 2000). Another modified electrode with C60 has evaporated on the GCE (C60/GCE) (Tan *et al,* 2003). C60/Li+/GCE and CNT/Li+/GCE were prepared by the doping of Li+ ion on to C60/GCE and CNT/GCE via 10 potential cycling between +600 to -600 mV in presence of 0.1M LiOH during cyclic voltammetry. A platinum wire (1 mm diameter) counter electrode and an Ag/AgCl (3 M NaCl) reference electrode were used in CV analysis.

2.3. Electrodes

There are two methods for modification of working solid electrodes:

1. A Mechanical Attachment technique (MA): was used which involved the pressing of a clean GCE surface onto a few mg of CNT powder placed on a filter paper.

2. Solution evaporation technique: This method includes application of a 2 µL of saturated C_{60} in acetonitrile and subsequently dried by hot air blower before placing in voltammetric cell.

3. Results and Discussion

3.1. Effect of Different Modified Electrodes

Fig. 1 shows that the redox peaks of Cd^{2+} were considerably enhanced by 4-5 times, when the modified CNT/GCE and AC/GCE were used in comparison with the C_{60}/GCE and GCE. The result confirms the electro-catalytic activity of CNT and AC were also exerted on the redox of Cd (II) under the conditions of cyclic voltammetry. The degree of sensitivity/electro-catalytic response for the different electrodes increases in the oxidation-reduction current peak in order of:

$$AC/GCE > CNT/GCE > C_{60}/GCE > GCE$$

Also, it seems from the redox current peaks of the cyclic voltammogram, the two oxidation and two reduction peaks for the redox of cadmium as in the following equations:

$$Cd^{2+} + e^- \rightarrow Cd^+ \quad \text{reduction} \quad (+0.6\ V) \qquad (1)$$

$$Cd^+ + e^- \rightarrow Cd^o \quad \text{reduction} \quad (-0.8\ V) \qquad (2)$$

$$Cd^o \rightarrow Cd^{2+} + 2e^- \quad \text{oxidation} \quad (-0.6\ V) \qquad (3)$$

$$Cd^+ \rightarrow Cd^{2+} + e^- \quad \text{oxidation} \quad (+0.4\ V) \qquad (4)$$

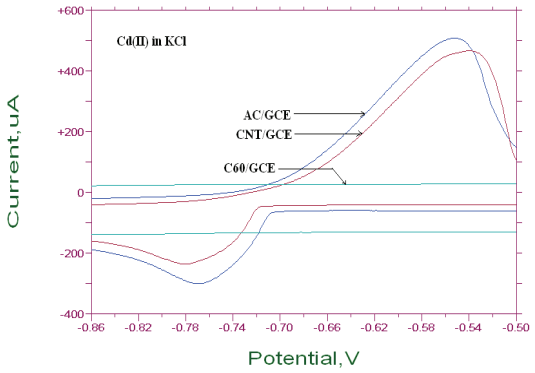

Fig. 1. Voltammogram for the redox current at different modified GCE in 1mM Cd(II) with 0.1 M KCl as supporting electrolyte versus Ag/AgCl as reference electrode and scan rate 100mVsec-1.

3.2. Applications of Electrochemical Sensors in Blood Analysis

The glassy carbon electrode modified with electrodeposited CNT and AC nanoparticles showed excellent electrocatalytic activity towards oxidation and reduction high peaks of Cd(II) in blood sample and a wide range of concentration. This electrode can be used for determination of micromolar or nanomolar concentration ranges of Cd(II) using cyclic voltammetry technique. The electrochemical system will be applicable for analysis of Cd(II) in blood samples containing different reducing compounds and other interferences exists in sample matrix. Furthermore, the modification procedure offers considerable simplicity and economy of electrode preparation as compared to other electrochemical methods for Cd(II) detection in blood medium.

Fig. 2 shows the affect of the modified GCE with nonmaterial work as a good sensor for the detection of the cadmium ion in blood medium. The sensitivity of this modified electrode is 0.9567 as shown in Fig. 3. The activated carbon and CNT used as electro catalyst for the detection of the contamination of blood with heavy metal ions such as cadmium.

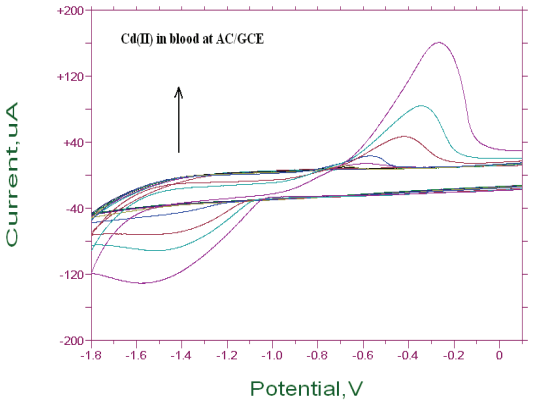

Fig. 2. Cyclic voltammogram for the redox peaks of Cd^{2+} (0.1 -5 mM) in Blood sample at scanning rate 100 mv s-1 using AC/GCE.

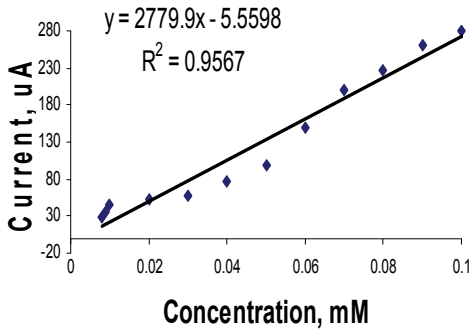

Fig. 3. Plot of oxidation current versus different concentration 0.008-0.1 mM $CdCl_2$ with 4mM AA in 0.1 M KCl using AC/GCE versus Ag/AgCl.

3.3. Scanning Electron Microscopy (SEM)

Scanning Electron Microscopy (SEM) the fractured surfaces of the nanocomposites were studied using a JEOL attached with Oxford Inca Energy 300 EDXFEL scanning electron microscope operated at 20 to 30 kV. The scanning electron photographs were recorded at a magnification of 1000X to 6000X depending on the nature of the sample. SEM analysis was carried out to investigate microcrystals. Samples were dehydrated for 45 minutes before being coated with gold particles using SEM coating unit baltic SC030 sputter Coater. SEM was used to examine the morphology of CNT microcrystals by mechanical attached on a graphite electrode surface before and after electrolysis by cyclic voltammetry. Fig. 4, 5 and 6 are SEM of CNT, C60 and AC respectively, attached and evaporated before electroanalysis with ions on to a 6 mm diameter basal plane graphite electrode which exhibits an array of microcrystals with 0.1-2 μm diameter.

4. Conclusions

CNT and AC modified GCE has been successfully fabricated by the mechanical method. Other modified electrode is C_{60} on GCE as C_{60}/GCE fabricated by solution evaporation method which is shown to be able mediated effectively in redox of Cd^{2+} with significant current enhancement in blood medium. The redox peaks of Cd^{2+} was dependent on the different concentrations (traces) in blood sample. This method for measurement of the effect redox peaks of Cd^{2+} ions formed on modified electrode surfaces as sensor electrodes for the detection of the cadmium ions in blood media. In this method, depending on the redox current we evaluated the determined traces of Cd^{2+} in blood samples at the sensitive modified electrodes. So we can say that these modified electrodes can be used as sensors for the detection of the traces of heavy metals such as cadmium in blood media due to environmental contamination.

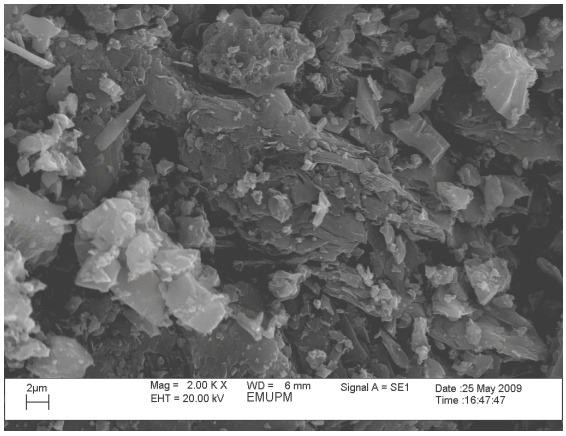

Fig. 4. SEM of the CNT microcrystal attached to a graphite electrode surface via solvent cast on to 5mm diameter basal plane graphite electrode.

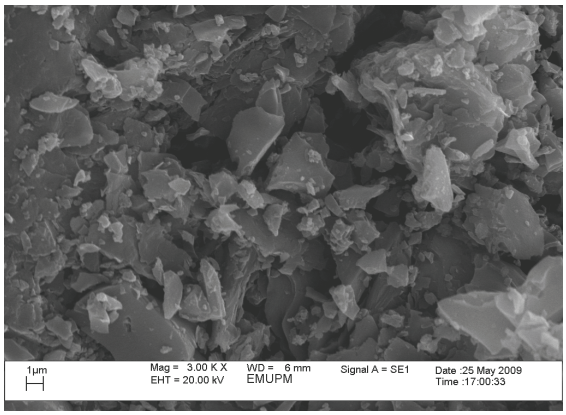

Fig. 5. SEM of the C_{60} microcrystal attached to a graphite electrode surface via solvent cast on to 5mm diameter basal plane graphite electrode.

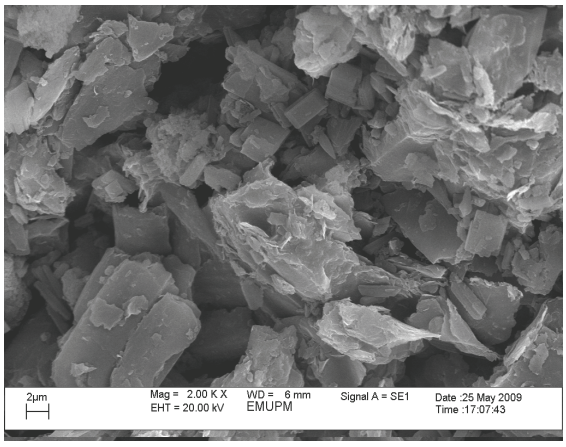

Fig. 6. SEM of the AC microcrystal attached to a graphite electrode surface via solvent cast on to 5mm diameter basal plane graphite electrode.

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Research on Digital Robot Aided Surgery System Based on Biomechanical Information

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Abstract: Aimed to develop a virtual robot aided surgery system which can represent the reality surgery process and realize force feedback by digital human tissue biomechanical information and outer force feedback instrument, human tissue geometrical and biomechanical modeling, biomechanical model calculation, force feedback instrument design, control algorithm are described. Finally, experiment was conducted after local area network is connected between haptic device control system and virtual environment. The results show this virtual robot can be controlled by master slave systems and realize visual feedback and force feedback. *Copyright © 2013 IFSA.*

Keywords: Biomechanical information, Force feedback, Robot aided surgery, Virtual surgery system, Control system.

1. Introduction

The development of modern science and technology more and more reflects the interdisciplinary characteristics. Virtual surgery, as a developing research direction, is a new field which integrates medical science, biomechanics, mechanics, materials science, computer graphics, computer vision, mathematical analysis, robotics and many other disciplines. Its purpose is to use computer technology to simulate and guide various processes involved in medical surgeries, which can be classified as preoperative, intraoperative, postoperative processes according to time, or be classified as surgical planning, surgical rehearsal, surgical teaching, surgical skills training, intraoperative guidance, postoperative rehabilitation according to surgical purpose.

In 1996, Satave [1] proposed the concept about the three generations of medical simulation system framework at the Fourth International Conference on Medical Virtual Reality. In the three generations of medical simulation system framework, the first generation medical simulation system focused on the performance of geometric characteristics of the human body, which applied the concepts of roaming and immersion to human anatomy data sets and provided a limited user interaction, has been applied to the education and training of health care workers. The physical characteristics of the different organization were taken into account in the second generation of medical simulation system during the organization modeling. The functional essence of human organs, for example, cutting off the blood vessels (physical phenomenon) may impact on blood pressure (physiological phenomenon) and then affect the normal function of other organs (physiological function), was taken into account in the third generation of medical simulation system. Overall, the third generation of medical simulation system is

closest to physiological functions and biological functions of human body, and it is the ultimate research goals of medical simulation system. Currently, most the researches on virtual surgery focus on the second generation of development framework, that is physics simulation of the biomechanical properties of organ and tissue.

Orthopedic surgery is the most kinds of operation in hospitals. But precise reposition of broken bones, which is the major procedure before fixing in fracture surgery, is always challenges to most surgeons, sometimes even the experienced experts. The dragging is the popular antagonistic action on the beginning of reposition, which is used to deal with retraction and imbedding. The strength of traction should be gradually, continuously increased to avoid steep rising. The demand for pulling manner, pulling strength and distance is varied with breaking position. So it will make much sense to develop an anatomically realistic biomechanical model for surgical training, surgical rehearse and telesurgery. Especially, it is more meaningful in robot aided surgery system, as it can guide the control strategy for medical robots.

Based on the above background, this paper completed the design of digital robot-assisted surgery system based on biomechanics signal for the robot-assisted fracture traction reduction surgery platform.

2. Biomechanical Model

The force analysis of leg in tibia and fibula fracture traction reduction surgery is shown in Fig 1. Parallel robot applied traction to the human leg calcaneus by Kirschner nail and the talus was connected with the calcaneus, then was the fractured lower half fibula. Because the fractured lower half fibula is separated from the fractured upper half fibula, so the stress can't be transmitted directly but only by muscle adhered thereon. The fixing force of fractured upper half fibula was mainly derived from bracket fixed on the upper half curs as well as the strap fixed on lower half thigh. Overall, either muscle or bone was substantially exposed to a force along the direction of the lines of force, as shown by the blue line in the figure below.

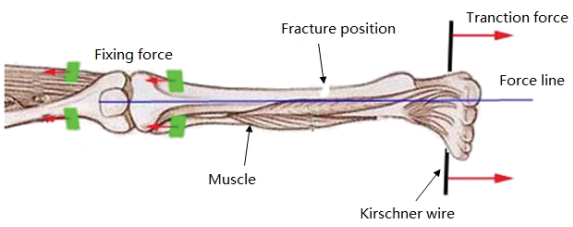

Fig. 1. The force analysis graph of leg during detracting.

Although there are many physiological tissues in human bogy leg, such as bones, muscles, nerves, blood vessels, lymph, fat, skin and so on, neither their anatomical structures nor their biomechanical properties presented in surgery differ from one another, this point is significant in the force analysis on legs through the traction reset operation, their functions are different as well. For example, the nerves, blood vessels and fact, almost have no contribution to the generation of traction during surgery, so they can be ignored in modeling process. While the bones and muscles play relatively significant roles during surgery.

2.1. Bone Biomechanical Properties Equation

The bones are generally composed of cortical bone and cancellous bone and its major components are hydroxyapatite and collagen fibers. The hydroxyapatite is very hard, its elastic modulus along the axial direction is 165 GPa; the collagen fibers do not consistent with the Hooke's law, but its tangent modulus of elasticity is about 1.24 GPa. The overall elastic modulus of the bone falls in between of the above two, the overall mechanical property is better than any of the two, because the soft collagen fibers can prevent the hard materials from brittle fracture and the hard hydroxyapatite can prevent the soft materials from yielding.

In actual surgery, X-ray was used to detect the leg bones, and we found that the bone tissue has no significant deformation. Therefore, the isotropic linear elastic model was used to define the materials to reduce the calculation amount. The following table1 is the basic material properties parameters of bone determined through research results of Jason, Zhang Ming et al, Hong Kong University of Science and Technology [2].

Table 1. Material properties and element types of bones in the finite element model.

Name	Unit type	Young's modulus	Poisson's ration
Bone	Solid185	7300	0.3

For the purposes of simplify the model and reduce the calculation amount, the isotropic linear elastic model has been widely used in simulation of biological finite element, especially in the cases of small deformation and small displacement [3]. The elastic potential energy of the isotropic material W (X) is a quadratic function of the infinitesimal strain tensor:

$$W(X)=\frac{\lambda}{2}(trE_L)^2+\mu trE_L^2, \quad (1)$$

where λ,μ are the Lame constants used to characterize the elastic constants of the elastomer, which are functions of the elastic modulus E and Poisson's ratio.

2.2. Muscle Biomechanical Properties Equation

The muscle contains many muscle fibers which are small subunits, and this is the reason the muscle surface has striped appearance. The muscle fiber is about a few centimeters of cylindrical structure, it has many long strips of subunits inside, namely, myofibril. According to the currently accepted mechanism of muscle contraction – myofilament sliding theory, these myofibrils are the real elements of contraction mechanism. When the muscle is in a resting state (without muscle activities dominated by nerve awareness), the elasticity of muscle is mainly generated by muscle myofibrils of the muscle. The muscle myofibrils mechanically arranged in parallel to form the parallel elastomeric element. Therefore, during the construction of the finite element model of the muscle fibers, the anatomical structure of the finite element units should be consistent with that of the muscle fibers on both shape and size as much as possible.

The linear elastic materials model was used for the representative tendon of beam element to show the biomechanical properties. The Mooney-Rivlin incompressible hyperelastic model was used for the muscle fiber model, which could be well described the mechanical properties with deformation less than 150 % and can fully meet the property computing needs of the practical application. The Mooney-Rivlin two-parameter model was used in this model in order to guarantee the computational efficiency. The formula is as follows:

$$W = \mathrm{C}_{10}(\overline{I_1} - 3) + \mathrm{C}_{01}(\overline{I_2} - 3) + \frac{1}{d}(J-1)^2 \quad (2)$$

where W is the strain energy density function. C_{10}, C_{01} are the mechanical properties constants. $\overline{I_1}$ and $\overline{I_2}$ are the first and second deformation tensor invariants respectively.

According to the research results of the computer science professor J. Teran of Stanford University [4], the material parameters of the muscle fiber and the tendon in the muscle model were determined in Table 2.

Table 2. Material properties and element types of muscles in the finite element model.

Component	Element type	Young's modulus E (MPa)	Cross-sectional area (mm^2)
Muscle	Solid185	Mooney-Rivlin Hyper elastic model	
Tendon	Link10	350	290.7

2.3. The Biomechanical Properties Equation of the other Soft Tissues

The other soft tissues including fat, blood vessels, nerve and so on. Due to their CT images are more blurred than that of the muscles, we can't get the three-dimensional reconstruction. They were all regarded as an isotropic material in this model, besides the bones, muscles and tendons, all other space surrounded by skin are filled with such soft tissue.

Generally, the stress-strain relations of these soft tissues do not meet the Hooke's law and can obtained through experiment only. As shown in Fig. 2, the experiment data were obtained by the Lemmon et al, Health Research Center of Pennsylvania State University [5]. Wherein, the F2, F3 and F5 respectively refer to the stress-curves at 2-fold, 3- fold and 5-fold normal tissue stiffness.

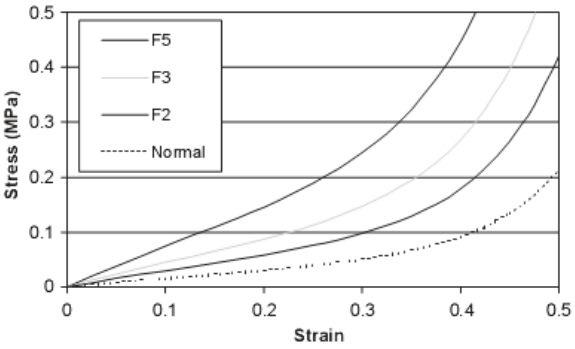

Fig. 2. Nonlinear stress–strain response of soft tissue.

The second-order Polynomial Form hyperelastic model was used to simulate the material properties of the soft tissue.

$$W = \sum_{i+j=1}^{N} C_{ij}(\overline{I_1} - 3)^i(\overline{I_2} - 3)^j + \sum_{k=1}^{N} \frac{1}{d_k}(J-1)^{2k} \quad (3)$$

Where W is the strain energy density function. C_{ij} is the mechanical properties constant. $\overline{I_1}$ and $\overline{I_2}$ are the first and second deformation tensor invariants respectively. Available parameters are shown in Table 3.

Table 3. The coefficients of the hyperelastic material model used for the encapsulated soft tissue.

Coefficients	normal	F2	F3	F5
C_{10}	0.08556	0.17113	0.25669	0.42782
C_{01}	-0.05841	-0.11683	-0.17524	-0.29207
C_{20}	0.03900	0.07800	0.11700	0.19499
C_{11}	-0.02319	-0.04638	-0.06957	-0.11594
C_{02}	0.00851	0.01702	0.02553	0.04256
D_1	3.65273	1.82636	1.21758	0.73055
D_2	0.00000	0.00000	0.00000	0.00000

3. Finite Element Model

The main task in reset operation of bone is to separate the two broken leg bones and then align them. As shown in Fig. 3, the upper half bone of the lower leg was fixed to the bracket with a bandage. The lower half bone was fixed on the parallel robot using the Kirschner nail.

Fig. 3. Load conditions in surgery.

Fig. 4 shows the finite element model determined by actual state in surgery. On the fixed point with bandages on upper half of the lower leg, the degree of freedom was defined as 0. Two forces were applied to both ends of the Kirschner nail with the same direction and half-intensity of traction force of parallel robot. Fig. 5 shows obtaining the information between position and the force according to the finite element biomechanics calculation.

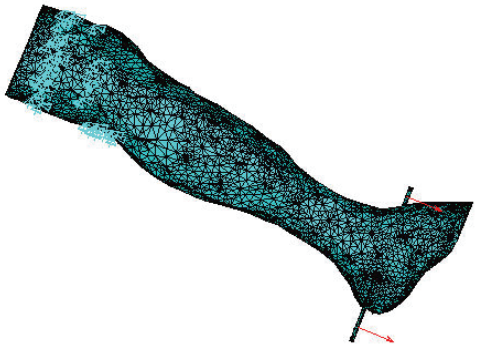

Fig. 4. Finite element model.

4. Design of Force Feedback System

4.1. Design of Haptic Device Mechanism

The haptic device was shown in Fig. 6. This parallel haptic device was consisted of moving platform and the base connected by the kinematic chains composed of three groups of the compression apparatus and connecting rod, the stiffness of this parallel mechanism was good, and its structure was compact, ensuring its sufficient degrees of freedom.

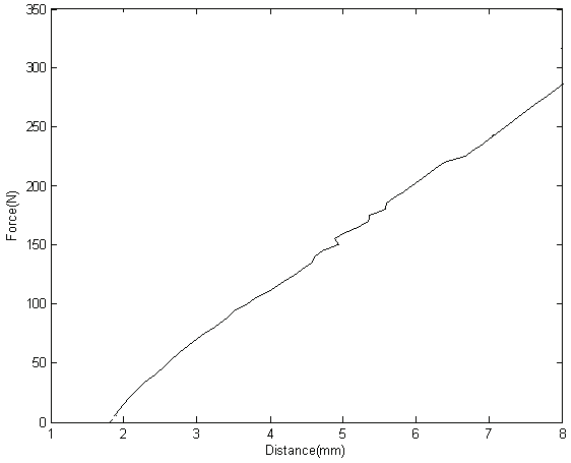

Fig. 5. Force-distance curve.

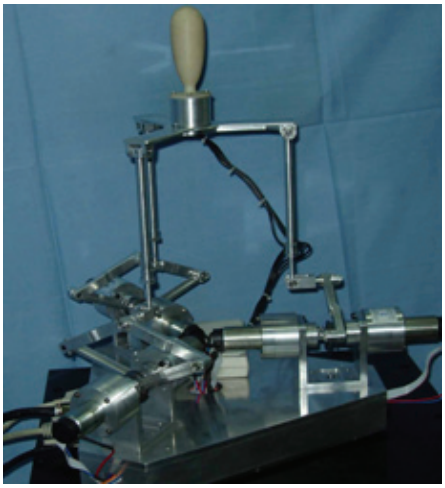

Fig. 6. Parallel haptic device.

Due to the motor fixed on base, the inertia of this mechanism was greatly reduced, and the force feedback property was enhanced. Through the application of pantograph structure, and the connection between RRR type kinematic pair and connecting rod as well as the connection between the other end of the connecting rod and upper mounting plate by a revolute pair, thus the parallelogram compression apparatus link mechanism was limited to rotate within three planes parallel to base, thus the manipulation on the rotation angle of connecting rod of base motor, the position and shape can be adjusted. Six-axis driving motor used DC torque motor, through the connection between harmonic reducer and various link mechanism, and the drive of motor by DC and the actuation of harmonic reducer, the systematic backlash can be avoided, and the motion accuracy of haptic device can be improved.

Further, the structure of this mechanism is simple, and its manufacturing and operation is relatively easy. Its accuracy can also be guaranteed. The parameters of the workspace of the haptic device are shown in Table 4.

Table 4. Workspace parameters of the haptic device.

Rotation angle vertical to X-axis	Rotation angle vertical to Y-axis	Rotation angle vertical to Z-axis	X-axis	Y-axis	Z-axis
$\leq 35^0$	$\leq 35^0$	$\leq 35^0$	±150 mm	±150 mm	180-350 mm

4.2. Haptic Device Control Structure

In order to achieve master-slave mode of force feedback and meet the requirements such as haptic device position accuracy, and rapidity, based on the ideology of the position loop force control, the control system used computer cascaded structure, which was divided into four levels: current loop, velocity closed loop, position closed loop and force closed loop. Force closed loop contains position loop, the realization of force control is based on the realization of position control, as can be seen from Fig. 7, the structural relationships between four closed loops were shown.

4.3. Design and Implementation of Systemic Control Arithmetic

The control structure based on position loop was applied to the parallel haptic device control system. The feedback force produced by driven robot and the human tissue effect was taken as the desired force of the system F_d, the interaction force of the operator and the haptic device detected by the force sensor on the parallel haptic device was F. The difference between the F_d and F was taken as the system control inputs, so as to form a closed loop servo of system force. The structure is shown in Fig. 8. The characteristics of its control structure include [6-10]:

1) The control inputs transform the force control into the robot's speed control similar to the form of damping control, rather than explicitly act on each joint in form of torque.

2) The force feedback directly compares with the desired force and takes the difference as the input of the force controller, rather than become the input item of the speed command, thereby forming a force control servo loop.

3) The movement control of main hand in workspace under the dynamics decoupling conditions, the multidimensional force of workspace can be decomposed into force control of dimensional independent force, and a variety of control law of force control can be achieved.

5. Achievement of Digitization Robot-Assisted Surgery System Based on Biomechanics Signal

The virtual surgery system is an integration of virtual reality technology and the modern medicine.

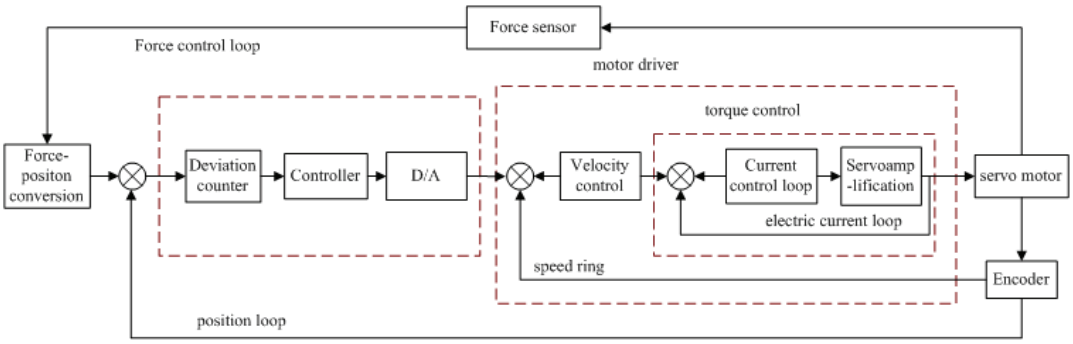

Fig. 7. Layers of control system.

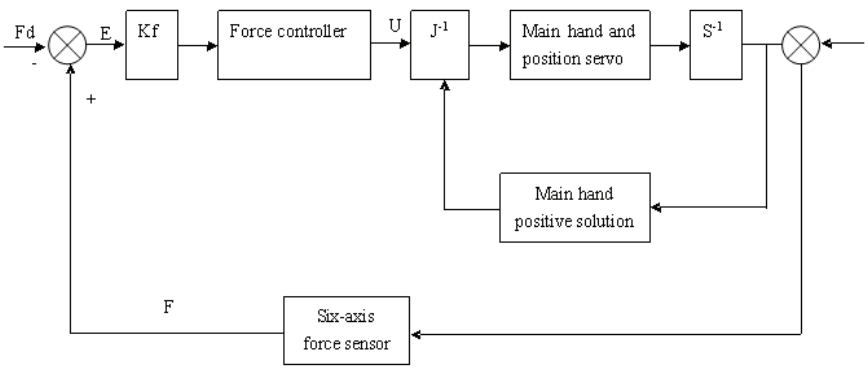

Fig. 8. Constructor of control arithmetic.

The users can feel immersion (such as sense of vision, force and tactile), realism and real-time interaction in a virtual environment, and the object's status in virtual environment can be changed through virtual tools [11-16]. The framework of virtual surgery system is showed in Fig. 9 and master slave control operation system is showed in Fig. 10. In the experiment, when the haptic device control system and virtual environment were successfully connected through the LAN, the haptic device control system will collect the movement positions and postures of the parallel haptic device every 15 ms and send the data to the virtual environment end through network, at the same time, the virtual environment end will create a separate thread to receive signals, the 6 positions and postures data was loaded into the paralleled surgery robot platform after the signals were successfully analyzed in accordance with the established way, thus paralleled surgery robot platform produced movement consistent with the haptic device. The displacement between the broken bones can be calculated based on the movement of the parallel surgery robot platform, and the force required in the current circumstance can be obtained after input the displacement into the neural network model based on the learning of finite element model, and the force required can be fed back to the operator through the parallel haptic device, thus completed a full closed loop.

The operator will also get visual feedback from the virtual environment when got the force feedback. The parallel robot in virtual environment will update the position and posture based on the signal sent by outer parallel haptic device. In addition, the force feedback can be displayed on the operator interface in real-time, it can be also reflected intuitively through the JFreeChart control and bone color changes, as shown in Fig. 11.

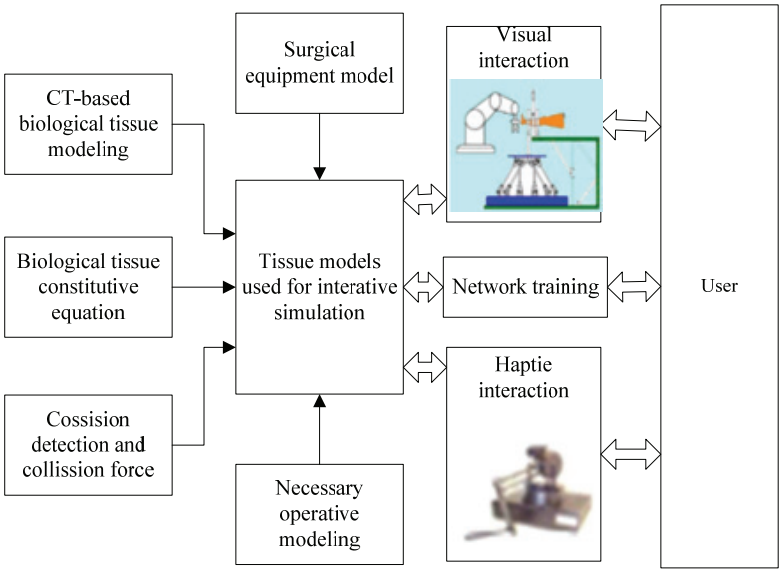

Fig. 9. Virtual surgery system.

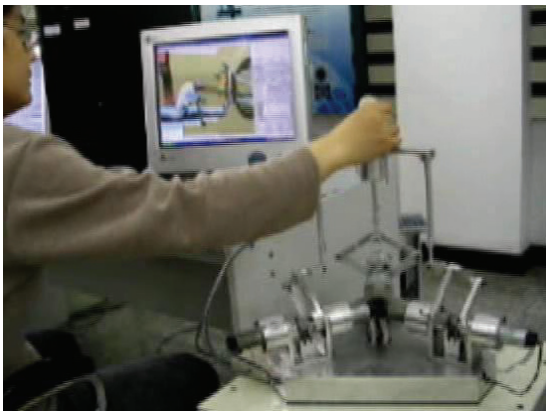

Fig. 10. Master slave control operation.

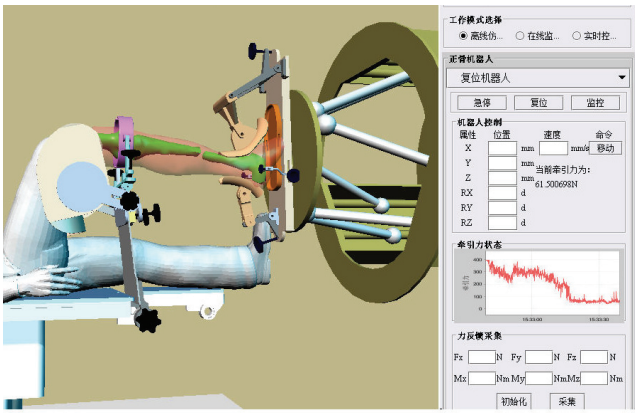

Fig. 11. Interface of virtual surgery system.

6. Conclusions

Through the establishment of biomechanics characteristic equation, finite element calculation of biomechanical equations and the design of force feedback device, digitization robot-assisted surgery system based on the biomechanical signal was achieved, the system is able to obtain force position changes information and visual feedback in surgery, and the robot control parameters can be adjusted at any moment based on the feedback information. The program also can be applied to non-robot surgery system, the users can feel the force status of surgical equipment during surgical procedures through the feedback device. The major conclusions include:

1) Multi-level human tissue biomechanical models of bone, tendon, muscle, skin and other soft tissues were established through the analysis of force status of corresponding human tissues in the bonesetting experiments.

2) The mechanical body design, control system design of the haptic feedback device were complete and design of systemic control arithmetic were achieved, the virtual master-slave operation tests on digitization robot-assisted surgery system based on biomechanics signal were established, and the feasibility of parallel haptic device control arithmetic was verified through completing the relevant tasks using parallel haptic device control the virtual robot.

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Analysis on AUV Heading Control System Based on RS-Chaos-LSSVM

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Abstract: Aimed at the submarine uncertainties and model's impreciseness, an integrated RS-Chaos-LSSVM model is put forward because of nonlinearity and time-variability of AUV heading control system. The LSSVM is a typical nonlinear regression modeling based on small sample. Its parameters are optimized by Chaos algorithm. It is to gain the optimal model and get the higher accurate. By using Rough theory the monitor data attribute of AUV is reduced to eliminate the redundant information to improve efficiency. Taken a certain typed AUV as an example, the two controller compare results .The results show the RS-Chaos-LSSVM controller markedly reduce the time. And the heading error and control rudder are simulated in the ideal and ocean current environment. In the ideal environment the two controllers all have good performances. But in the ocean current environment the simulation results show the integrated controller improve the model's stability and accuracy. The overshoot, the rise and adjust time reduced greatly. In a word the integrated RS-Chaos-LSSVM heading predictive controller has better performances especially strong anti-interfere capability.

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Keywords: AUV, Heading Control, RS, LSSVM, Chaos optimization.

1. Introduction

Since last decades, AUV (Autonomous Underwater Vehicle) has gradually replaced part of people's labor in the special works, such as dirty, tired, bitterness, dangerous or repetitive etc. It has developed more widely, more intelligence. Thus it is necessary to design an autonomous and precise heading control system.

In reference [1] AUV was dynamically controlled by using a PID controller. The results are valid. But it cannot compensate for un-modeled hydrodynamic forces or unknown disturbances dynamically. And the parameters need rich experience to be determined. Reference [2] described the NN was used in the heading control. The simulation accuracy results are satisfied. But the poor real time is a really problem, the system stability and convergence have the ascension of the larger space. Reference [3] involved the Fuzzy PID controller. It was relatively simple and easy to determine the unknown parameters. However, to the parameter fluctuations the real-time robustness was not strong and the subjective factors were introduced. The sliding mode controller was introduced to control the AUV yaw control in reference [4]. The simulation results

showed that the method had good robustness on the un-modeled dynamics and uncertainty environmental parameters. But there existed dithering phenomenon and it belonged to off-line control. And as the AUV is a long development cycle and high cost. The heading control is a typical of small sample condition problem of pattern recognition. Combined with the characteristics of various methods the AUV heading control based on RS-Chaos-LSSVM is forward.

The LSSVM (Least Square Support Vector Machine, LSSVM) has good generalization ability, small sample of control, fast calculation speed, nonlinear system modeling. It is a powerful tool. But it has a direct influence on the parameters of the control effect. The model's parameters are optimized by using COA (Chaos optimization algorithm), which is to control the heading model accurately. The redundant information is reduced by using RS (Rough Sets). The extracted decision rules are used to train LSSVM model. So the AUV heading control model can be built by RS-Chaos-LSSVM. It is to explore a new effective control method.

The rest of the paper is organized as follows. The next section is AUV heading model. The RS-Chaos-LSSVM Model is in the Section 3. The RS theory and the LSSVM algorithm model are brief introduced. And the Chaos-LSSVM and the RS-LSSVM Predictive Model are given, respectively. The AUV heading control is in the Section 4. It is the key part. It contains an identifier LSSVM identifier and a controller LSSVM controller. In Section 5 the analysis simulation results are given to demonstrate the compound RS-Chaos-LSSVM heading controller. In Section 5, conclusions are drawn.

2. The AUV Heading Model

It is vital to the AUV heading control especially far sailing. The integrative controller to a certain control law must be driving course control actuators-vertical rudders. It makes it run in the prior planning path according to a certain performance index.

2.1. AUV Heading Control System Model

A pair of rudder located on the AUV rear as the main actuator to the heading control. The Fig. 1 is the AUV heading control system organization chart. The whole control system consists of AUV uncertain nonlinear dynamics model, measure parts, heading controller, vertical rudders and some disturbing factors such as wave and current etc.

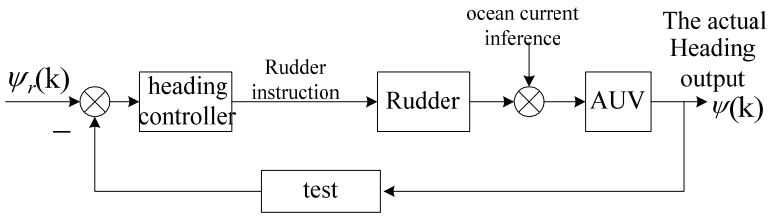

Fig. 1. The organization chart of the AUV heading control system.

2.2. AUV Nonlinear Mathematic Model

An AUV complete six degree-of-freedom mathematical model can be found in reference [1] and [12]. They are surge, sway, roll, pitch heave, and yaw motion equation. Suppose the couple between horizontal and vertical plane is omitted. It is very convenient that research two motions, respectively. The horizontal plane motion equations can be simplified formula are as follows.

$$(m+\lambda_{11})\dot{u} = T - C_{xs}\frac{1}{2}\rho v^2 S \tag{1}$$

$$\begin{aligned}&(m+\lambda_{33})\dot{v}_z - (mx_c+\lambda_{35})\dot{\omega}_y \\ &= mv_x\omega_y + \frac{1}{2}\rho v_T^2 S(C_Z^{\beta}\beta + C_z^{\delta_r}\delta_r + C_z^{\bar{\omega}_y}\bar{\omega}_y)\end{aligned} \tag{2}$$

$$\begin{aligned}&(J_{yy}+\lambda_{55})\dot{\omega}_y - (mx_c+\lambda_{35})\dot{v}_z + mx_c v_x\omega_y \\ &= \frac{1}{2}\rho v_T^2 SL(m_y^{\beta}\beta + m_y^{\delta_r}\delta_r + m_y^{\bar{\omega}_y}\bar{\omega}_y)\end{aligned} \tag{3}$$

$$\dot{\varphi} = \omega_y \tag{4}$$

$$\dot{X}_0 = v_x\cos\varphi - v_z\sin\varphi \tag{5}$$

$$\dot{Z}_0 = -v_x\sin\varphi + v_z\cos\varphi \tag{6}$$

$$v = \sqrt{{v_x}^2 + {v_z}^2} \tag{7}$$

$$\beta = \arctan(\frac{v_z}{v_x}) \tag{8}$$

where v_x (m/s) is the horizontal speed, v_z (m/s) is the vertical velocity, ω_y (rad) is the rotation angular velocity of Y axis, β (deg) is the sideslip angle, θ (deg) is the pitch, ψ is the yaw φ (deg) and the roll φ (suppose $\varphi = \omega_x \neq 0$), X_0 (m) is the x component, Y_0 (m) is the y component, δ is the rudder angle, ρ (kg/m^3) is the density of water, λ

is the additional mass, m is the mass of AUV, L is the long of vehicle, S is the surface area of AUV, J is the rotational inertia [12].

3. The RS-Chaos-LSSVM Model

The LSSVM is the typical nonlinear system, fast calculation modeling based on the AUV small sample. The LSSVM parameters can be optimized by using the Chaos theory. The control stability accuracy has improved. But the identification control speed is greatly reduced. Rough set can reduce the attribute value. The system real time can be improved.

3.1. The RS Theory

The RS theory can extract the decision rule set by reducing knowledge, eliminating redundant information and seek the questions of the classification rule set in the premises of keeping the classification without any prior knowledge [5]. So it is widely applied to the pattern recognition of various pattern identifications.

The Fig. 2 shows the establishment of RS-LSSVM predictive model. The acquired data of AUV are reduced by using RS theory to obtain an extracted rule set. It can be used to train LSSVM model. The trained model can be adopted to control AUV's heading motion.

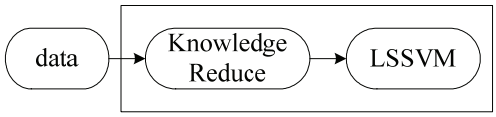

Fig. 2. The RS-LSSVM predictive control model.

The acquired data can be mapped as a knowledge retrieval system $S =< U, A, V, f >$, where U is an universe, $A = C \bigcup D$ is the attribute set, subset C is called as the conditional attribute, D is called as the decision attribute, $C \bigcap D = \varnothing$, $V = \bigcup_{r \in R} V_r$ is the attribute value set, V_r represents the value of the attribute r, $f : U \times R \to V$ is an information function. The steps of knowledge reduction can be described as follows [5].

Input the knowledge system S with the conditional attribute set $C = \{a_1, a_2, \cdots a_n\}$ and the decision attribute set $D = \{Dec\}$.

Step1: calculate C positive field $pos_C(D)$ in D;

Step2: eliminate the attribute a_i, $C_t = C - \{a_i\}$ from C;

Step3: calculate C_t positive field $pos_{C_t}(D)$ in D;

Step4: if $pos_{C_t}(D) = pos_C(D)$, then wipe out a_i, else remain a_i;

Step5: repeat traverse all attributes according to the procedures above.

Output one of the relative reductions of the conditional attribute set C to the decision attribute D.

3.2. The LSSVM Algorithm Model

Least Squares linear System is leaded in SVM. It replaces the traditional quadratic programming, aims to solve the question of the classification and function estimation. The system has the advantages of high procession-precision, stabilized performance, high speed. The structure drawing as followings:

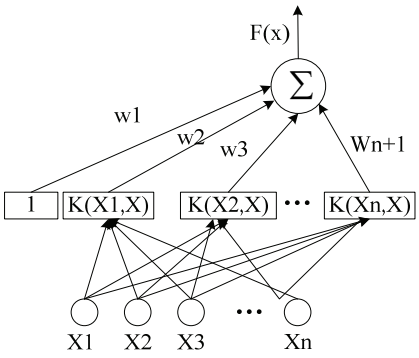

Fig. 3. The structure drawing of the LSSVM algorithm.

LSSVM algorithm has the advantages of high procession precision and stable performance [7]. The rule set extracted by using RS is taken as the input of LSSVM to train it. Suppose the training sample is a n-dimensional vector, N samples form the sample set D, ($D = \{(x_i, y_i) | i = 1, 2, \cdots N\}$), $x_i \in R^n$ $y_i \in R$, where x_i is an input data, and y_i the corresponding output value. LSSVM can map the data into a high dimensional feature space by using a nonlinear mapping ϕ. Then, the equation of linear regression is

$$y(x) = w^T \phi(x) + b, \tag{9}$$

where $\omega \in R^{n_h}$ is the weight vector, b is the constant bias.

The learning process can be translated into an optimization problem according to the principle of structural risk minimization.

$$\min J(w, e) = \frac{1}{2} w^T w + \frac{1}{2} C \sum_{i=1}^{N} e_i^2$$

$$\text{s.t. } y_i = w^T \varphi(x_i) + b + e_i, \; i = 1, 2, \cdots N$$

$\varphi(x)$: $R^n \to R^{n_h}$ is a mapping function in the kernel space, $e_i \in R$ is the error variance, J is the loss function, C is the adjustable regularized parameter.

A Lagrangian function can be introduced to solve the above optimization problem, which can transform the optimization with constraint to the optimization without constraint.

$$L(w,b,e;a) = J(w,e) - \sum_{i=1}^{N} \alpha_i (w^T \varphi(x_i) + b + e_i - y_i) , \tag{10}$$

where $\alpha_i \in R$ is the Lagrangian multiplier. The partial derivatives of L with respect to ω, b, e_i and α_i are found out according to Karush-Kuhn-Tucker (KKT) condition. The middle variables ω and e_i are eliminated, and the solution of α_i and b can be obtained

$$\begin{bmatrix} 0 & \eta \\ s & \Omega + \gamma^{-1} I \end{bmatrix} \begin{bmatrix} \alpha \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ Y \end{bmatrix},$$

where
$\Omega \in R^{N \times N}$, $\Omega_{km} = \phi^T(x_k)\phi(x_m)$, $Y = [y_1, \cdots, y_N]^T$, η is the N-dimensional column vector, namely, η =[1…N]), $\alpha = [\alpha_1, \cdots a_N]$, I is the confirmed matrix, s is the N-dimensional row vector, namely, $s = [1 \cdots N]^T$. So the function estimation can be expressed as

$$\hat{y}(x) = \sum_{i=1}^{N} \alpha_i K(x_i, x_j) + b , \tag{11}$$

where $K(x_i, x_j)$ is the kernel function that meets Mercer requirements. Gaussian radial basis function (RBF) is selected as the kernel function in this paper.

$$K(x_i, x_j) = \exp(-\left\| x_i - x_j \right\|^2 / 2\delta^2) , \tag{12}$$

where δ is the kernel factor, i.e. the width of the kernel function.

3.3. The LSSVM Algorithm Model

The LSSVM model's parameters are optimized by COA (Chaos optimization algorithm). The regularized parameter C and the kernel factor δ are very important. The chaos theory can optimize the parameters of LSSVM model. The LSSVM parameters are acted as the combinatorial optimization problem. And then the objective function is built on it. The chaotic state brings to the optimization variables by similar carrier wave. The ergodic scope of chaotic motion extends to the upper and lower limits of the optimizing variable. A more detailed explanation follows below. Define the optimization problem as

$$\min J = \min f(z_i) = \frac{1}{L} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{13}$$

$$s.t. \;\; C \in [C_{\min}, C_{\max}], \sigma \in [\sigma_{\min}, \sigma_{\max}], \tag{14}$$

where y_i is the $i-th$ known sample output value of the samples $\{x_i, y_i\}_{i=1}^N$, $\hat{y}_i$ is the output prediction of the former, which can be calculated from equation (11), $f(z_i)$ is the regression of LSSVM, z_i is the optimized variable.

Logistic mapping is chosen to produce the chaos variables.

$$x_{n+1} = \mu x_n (1 - x_n) \tag{15}$$

where the initial value $x_0 \in (0,1)$, and x_0 is not equal to 0.25, 0.5 or 0.75.

Logistic mapping is a full mapping in interval $(0,1)$ when $\mu = 4$, a system goes into chaos state. x_n is the ergodic in the interval $(0,1)$.

It can be seen from (11) and (12) that the objective function J of the optimization problem (13) is the function of C and δ. J will decrease and converge to its minimum J^* if searching ergodicly in C and δ definition domains by using the chaos variable. The corresponding minimum C^* and δ^* will be the optimal parameters of the LSSVM. Specific Chaos-LSSVM algorithm procedure is as follows.

Step1: Initialization. Input data: sample set $\{x_i, y_i\}_{i=1}^N$, optimization number A1 (large number), $n1 = 0$, the optimal index J^* (appropriate positive), two random number in the $(0,1)$ interval x1, x2;

Step 2: Generating chaotic variables. The n chaotic variables $x_{n,i}$ are selected by carrier-wave way. The n optimization variable generating chaotic variables $z_{n,i}$ are introduced, respectively. Then

$$x_i^* = z_i^*$$

$$z_{n,i} = c_i^r + d_i^r x_{n,i} \tag{16}$$

Step 3: Iterative search chaotic variables.
If $f(z_{n,i}) < f^*$, then $f^* = f(z_{n,i})$, $z_i^* = z_{n,i}$
else abandon $z_{n,i}$;
Step 4: In turn iteration $n = n + 1$;

$$x_{n,i} = \mu x_{n-1,i} (1 - x_{n-1,i}) ;$$

Step 5: Repeat step2-4 till it remains unchanged so far in a certain step 2;
Step 6: If $n < A1$, then turn to step 2;
Step 7: return to the optimal δ^* and C^*, α_i and b.

3.4. The RS-LSSVM Predictive Model

The AUV depth predictive control flow chart shows in the Fig. 4.

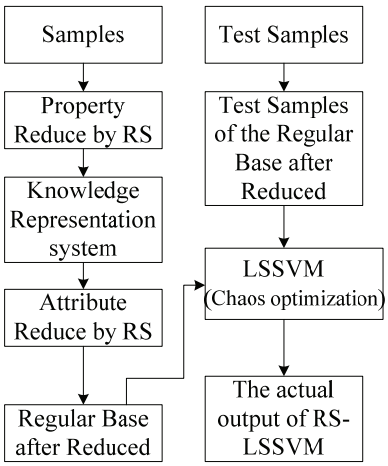

Fig. 4. The AUV predictive algorithm flow chart of RS-LSSVM.

1) Analysis the collect data, select the sample set;
2) Attribute value reduction form the knowledge representation system by the RS;
3) Attribute reduction to the knowledge representation system form the extract regulated set by the RS;
4) The LSSVM model is build. The appropriate kernel function and the parameters are determined by the chaos theory. The reduced rules set train the LSSVM model till the satisfied result.
5) The RS-LSSVM predictive model is tested and verified by the test samples.
6) The whole predictive results analysis.

4. The RS-Chaos-LSSVM Model

The LSSVM model is the kernel of the heading control system. It contains an identifier LSSVMI and a controller LSSVMC.

4.1. The AUV LSSVMI

The vehicle heading motion model is approximated by using LSSVMI. Its inputs are the actual heading signals $\psi(k)$, rudder signal $\delta(k)$ and roll angle $\varphi(k)$.

$$X=[\psi(k),\psi(k-1),\cdots\psi(k-n+1),\delta(k-1),\cdots,\\ \delta(k-m+1),\varphi(k),\varphi(k-1),\cdots,\varphi(k-p+1)], \quad (17)$$

where n, m and p are the orders of yaw, rudder and roll signals, respectively. The output of LSSVM model is the system output yaw signal, namely heading signal. Given the train samples

$$\{(X_1,\psi_1),(X_2,\psi_2),\cdots,(X_l,\psi_l),\}\subset R^N\times R,$$

$$X_1,X_2,\cdots,X_l$$

as the input vector at $k+l-1$, they are denoted as

$$X_1=[\psi(k),\psi(k-1),\cdots\psi(k-n+1),\delta(k-1),\cdots\\ \delta(k-m+1)],\varphi(k),\varphi(k-1),\cdots,\varphi(k-p+1)] \quad (18)$$

$$X_2=[\psi(k),\psi(k+1),\cdots\psi(k-n+2),\delta(k),\cdots\\ \delta(k-m+2)],\varphi(k+1),\varphi(k),\cdots,\varphi(k-p+2)] \quad (19)$$

……

$$X_l=[\psi(k+l-1),\psi(k+l-2),\cdots\psi(k-n+l)\\ \delta(k+l-2),\cdots,\delta(k-m+l)],\varphi(k+l-1),\\ \varphi(k+l-2),\cdots,\varphi(k-p+l)] \quad (20)$$

where $\psi_1,\cdots,\psi_l$ is the output from $k-1$ to $k+l-1$, namely, $\psi_1=\psi(k-1),\cdots,\psi(k+l-1)$.

The AUV heading predictive control model is obtained according to equation (11).

$$\psi_m(k)=f(X(k))=\sum_{k=1}^{n}\alpha_k K(X(i),X(k))+b \quad (21)$$

4.2. The AUV Integrated LSSVMC System

The AUV Chaos-LSSVM close-loop control scheme mainly governs the heading control of AUV, as shown in Fig. 5.

In Fig. 5 the $\psi_r(k)$ is the preset heading value. The rudder $\delta(k)$ is the controlled quantity. The $\psi(k)$ is the actual yaw angle. The LSSVM model parameters are optimized by Chaos theory. The AUV input is the weight sum between LSSVMC and LSSVMI.

$$\delta(k)=\beta\delta_L(k)+(1-\beta)\delta_C(k), \quad (22)$$

where $\delta_L(k)$ is the LSSVM output model. $\delta_C(k)$ is the LSSVMC output of the Chaos optimizing. β is the identification accuracy factor of the LSSVM.

The online modified model parameters are predictive by using the chaos LSSVM. The accurate heading control is obtained. The LSSVM model is build according to the historical data before starting up. The model parameters C, σ, α_i and b are saved in the CPU. After the running, the present moment controlled and the output variable covers the previous data.

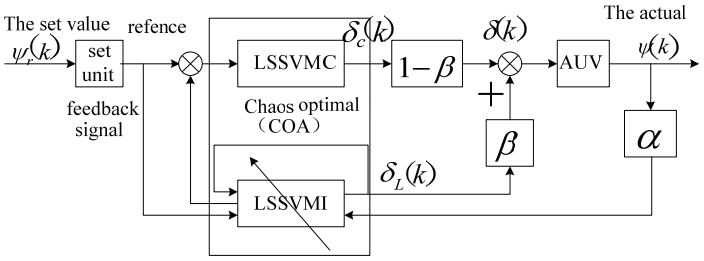

Fig. 5. The vehicle heading control based on Chaos-LSSVM.

Every certain of the time the Chaos-LSSVM identifier updated sample set and build the new model. At the next sample time the new model is substitute for the older one. The sufficient stability and accurate are ensured. So the AUV dynamical model is online learning by the RS-LSSVMI. It is constantly approximation the nonlinear mapping characteristics. The RS-LSSVMC gives the control signal. It makes the AUV heading control satisfying precision according to the random instructions.

4.3. The RS-Chaos-LSSVM Heading Model

The predictive model based on RS-Chaos-LSSVM is shown in Fig.. 6. The bigger left, big left, small left, less left, keeping, less right, small right, big right and bigger right form the control decision attribute set D . The yaw, rudder and ocean current etc. form the conditional attribute set C . The classification can be conducted in two steps. First, the redundant attributes are removed in the sample pretreatment. The invalid data are eliminated. The attribute of data set are reduced by using rough sets theory. Second, the LSSVM training model is established. It will further classify the reduced data; the results and the classification accuracy are outputted. They are as shown in Fig. 6.

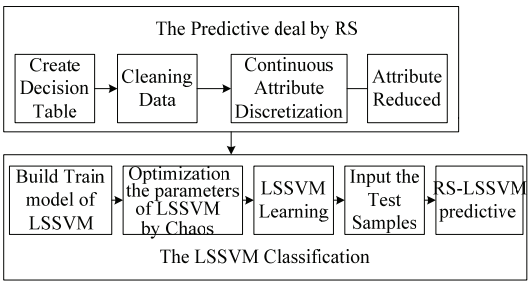

Fig. 6. The heading prediction method based on RS-Chaos-LSSVM

5. The Analysis of Simulation Results

To verify the RS-Chaos-LSSVM heading controller's effectiveness, taking a certain vehicle as objective, the yaw control simulation is conducted. Set the horizontal rudder is [-35^o ~ +35^o]. The average speed is about 2 m/s. the given heading angle is 20^o. Select 300 data as the training samples, the other 200 data as the test samples. The simulation results are described as follows.

5.1. The Compare between Chaos-LSSVM and RS-Chaos-LSSVM Controller

To improve the real timing the AUV integrated heading controller is put forward between the RS theory and Chaos-LSSVM. The Table 1 shows the heading simulation time results between the Chaos-LSSVM and RS-Chaos-LSSVM controller in the foregoing environment.

Table l. The compare results between Chaos-LSSVM and RS-Chaos-LSSVM.

	Train error	**Test error**	**Test time**	**Train time**
Chaos-LSSVM	0.1087	0.1682	89.91s	64.65s
RS-Chaos-LSSVM	0.1102	0.1697	62.13s	36.88s

The Chaos-LSSVM controller has the good performance for the special model. Such as the dynamic un-model, model inaccurate, interference or other uncertain factors. But the learning convergence speed is slow. It is difficult to meet the real-time control requirements. The RS theory is fused. Namely it is the integrated RS-Chaos-LSSVM controller. It can save about the half train and test time. The scheme markedly improves the control performance of the AUV heading control.

5.2. The Heading Control Simulation in Ideal Environment

Suppose there is no current interference. The certain AUV heading control simulation results are as shown in below Figs. 7, 8 and 9 in the foregoing environment, respectively.

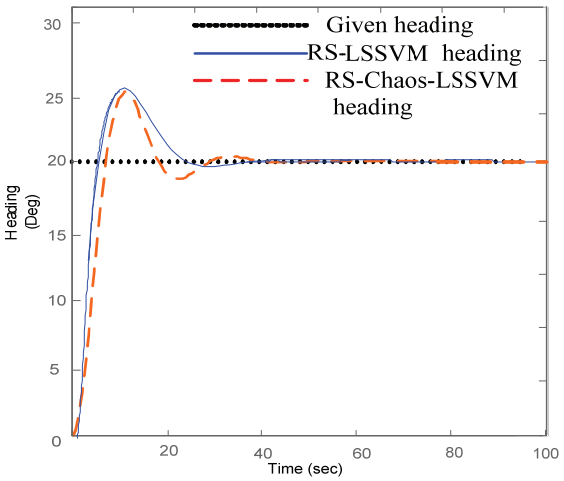

Fig. 7. The heading controller based on RS-LSSVM and RS-Chaos-LSSVM without interference.

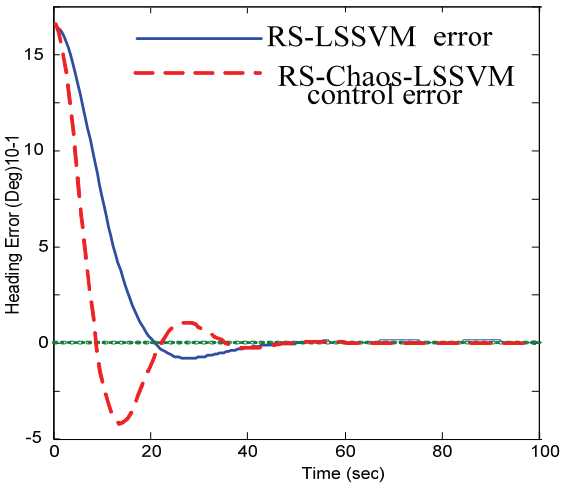

Fig. 8. The heading error controller based on RS-LSSVM and RS-Chaos-LSSVM without interference.

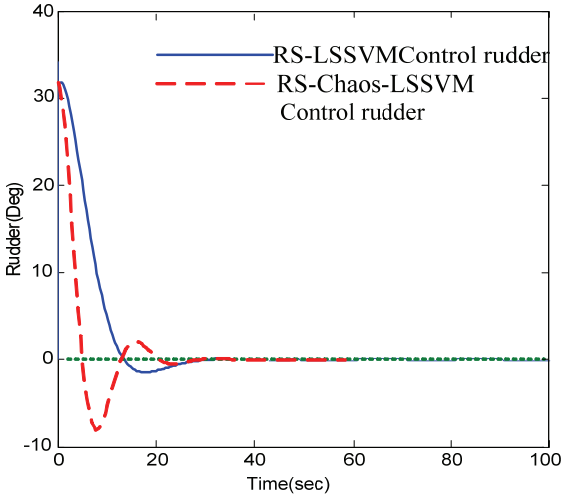

Fig. 9. The control rudder based on RS-LSSVM and RS-Chaos-LSSVM without interference.

The two max yaw angles are all approximately 25^0. The both overshoot are about 25 %. And that the RS-LSSVM controller has a bit faster tracking speed. The RS-Chaos-LSSVM and RS-LSSVM controller stabilization time is about 45 s and 38 s, respectively (see Fig. 7, the heading error in the RS-LSSVM controller is significantly less than the RS-Chaos-LSSVM one. The error is approximately 5 and 2 degree, respectively (see Fig. 8), the RS-LSSVM controller steering varying instruction is not much easier than the RS-Chaos-LSSVM (see Fig. 9).

In short conclusion the RS-LSSVM heading controller has little better performance than the comprehensive one in ideal environment.

5.3. The Heading Control Simulation in Ocean Current Environment

There are some disturbance ocean currents. Such as suppose ocean current is 60^o, the 2 sea wave power. The given yaw angle is the 20^o. The both simulation curves are as following Figs. 10, 11 and 12.

There are external environment interferences. The hydrodynamic coefficient affects the model, which is not precise. So the RS-LSSVM heading controller has the bigger errors than the RS-Chaos-LSSVM. In the Fig. 10 the yaw angle accurate of the RS-Chaos-LSSVM heading controller has the better than the other one. The overshoot of the RS-LSSVM heading controller is about 50 %. The other one is only 15 % yet; the two adjust time is 25 s and 75 s, respectively (see Fig. 10). The heading error of RS-LSSVM controller is about 9 degree, the other one is approximately 3 degree in the Fig. 11; The RS-LSSVM control rudder is bigger than the other one in the Fig. 12. From the three figures there are more the control yaw angle accurate and the robust in the RS-Chaos-LSSVM heading controller.

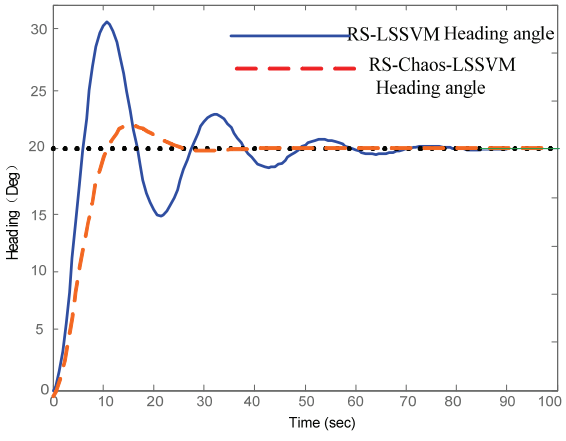

Fig. 10. The heading controller based on RS-LSSVM and RS-Chaos-LSSVM in ocean current environment.

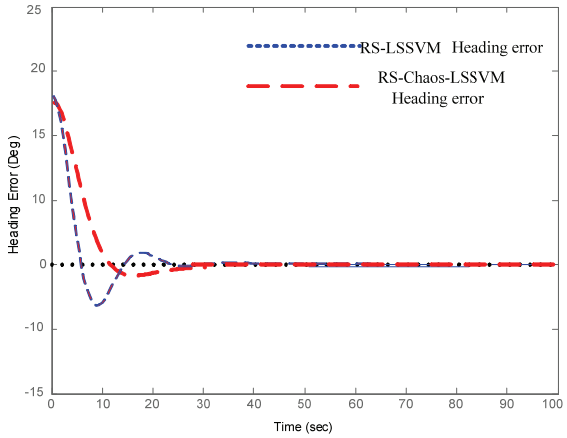

Fig. 11. The heading error controller based on RS-LSSVM and RS-Chaos-LSSVM in ocean current environment.

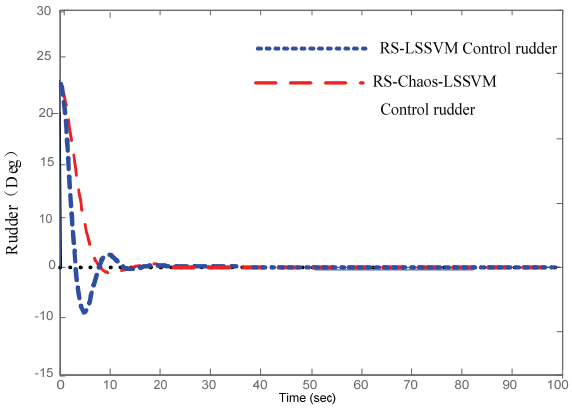

Fig. 12. The control rudder based on RS-LSSVM and RS-Chaos-LSSVM in ocean current environment.

In conclusion, suppose the AUV hydrodynamic coefficient is constancy, not vary. The outside current disturbing influences are ignored. The RS-LSSVM heading controller has good performances. But it is valid within a certain range of work; however the actual ocean current environment is very complicated. The control performances turn worse when the system model changed or dynamical imprecise. The RS-Chaos-LSSVM controller can realize the real-time online optimization control. The control performances are obviously superior to the RS-LSSVM one especially in special circumstances. Such as: dynamic un-modeled, uncertain inference etc.

6. Conclusion

The AUV heading control is the typical nonlinear system. The control research has very important theoretical and practical significance. Fused the RS, Chaos theory and LSSVM model, the integrated RS-Chaos-LSSVM controller has been forward. The optimal model has been ensured by using the Chaos theory in the ocean interference environment. The better stability and accuracy have been gained. The overshoot of the integrated RS-Chaos-LSSVM heading control is only 15 % in the ocean interference environment The RS theory can be attributing and attribute value reduction. The redundant information has been eliminated. The real time has been improved. The adjust time has been reduced about 50 second. The simulation results have been showed the integrated RS-Chaos-LSSVM heading control has the better control stability and accuracy, at the same time the good real time has been gained in the uncertain interference environment.

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Analysis and Numerical Simulation of the Stable Helical Vortex in TCP Flow in Vertical Screw Conveyor

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Abstract: The air in vertical screw conveyor creates flows such as Taylor-Couette-Poiseuille flow and turbulent flow with the increase of rotational speed of screw shaft. The stable helical vortex in TCP flow is beneficial to improve the transporting efficiency of granular materials. This paper described the N-S equations of incompressible steady flow (air) in vertical screw conveyor and deduced the distribution solution of velocity of stable helical vortex in TCP. And this paper also analyzed the relationship between the Reynolds number Re and parameters such as rotational speed of screw shaft, shaft diameter of screw shaft, screw pitch. Then this paper simulated different flows using FLUENT software and obtains the velocity vector chart and streamline chart with different rotational speed of screw shaft in vertical screw conveyor. The effect of simulation verified the correctness of theoretical analysis and recommended the critical Reynolds number Re_c suitable for appearance of stable helical vortex in TCP flow in the conveyor. *Copyright © 2013 IFSA.*

Keywords: Vertical screw conveyor, Taylor-Couette-Poiseuille flow, Stable helical vortex, N-S equations.

1 Introduction

The gap in two relative rotating concentric cylinders produce a secondary flow (the famous Taylor vortex), which is known as Taylor-Couette flow (TC) [1-3]. When the outer cylinder is stationary, if the inner cylinder rotates at a low rotating speed, the flow moves circularly in horizontal plane around the cylindrical axis, which is called the basic couette flow. When the angular velocity of the inner cylinder reaches a critical value, couette flow begins to lose stability. Then the new axisymmetric steady flow emerges, which is distributed vortex along the axial direction and adjacent vortex are reverse. The flow above is called as Taylor-Couette flow, which only moves along circumferential direction without axial direction. If an axial force is exerted on the movement, Taylor-Couette flow achieves simultaneously circumferential speed and axial speed, which is called as Taylor-Couette-Poiseuille flow (TCP) [4-7]. TCP flow can appears some vortexes such as laminar Taylor vortex, stable helical vortex and fluctuant helical vortex with the increase of Reynolds number [8-10]. When the Reynolds number continues to increase, the flow will appears wide range of disorder, which is called as the turbulent flow.

If we change modeled structure of the classic couette flow, translating the pressure forced from external into the pressure formed by own movement, which makes the flow moves upwards spirally. So adding vane to the inner cylinder in the modeled structure of the classic couette flow, which also is the structure of vertical screw conveyor. The theory of vertical screw conveyor will be described by the classic couette flow model modified. The high-speed rotation of screw shaft enables the flow of air around it to move, generating external pressure effecting on the classic couette flow. The external pressure is easy to be translated into the pressure formed by own

movement. The vertical screw conveyor is a conveyor without flexible traction components. It drives materials to move relaying on the rotation of the screw shaft in a closed duct. The traditional conveying theory is described as follows: the centrifugal force generated by rotational screw vane effecting on granular materials and the force of friction between granular materials and the duct overcome gravity of granular materials, which make the granular materials are not dropped, but are lifted [11-14]. When the granular material is small, the interaction between granular materials and air becomes not be ignored. And the movement becomes more complex.

The theoretical and experimental research on the screw conveyor was started before appearance of screw ship unloader. Western scholars Vierling and Sinha put forward elemental point theory of screw conveying mechanism, which simplified the granular material on the surface of the screw as a single particle to analyze its movement and force. They introduced the concept of critical speed, and put forward the analytical method of solving rising speed of granular material. But they ignored the diversity of composition of particles, and the pressure distribution of granular material on the pipe wall of vertical screw conveyor and the surface of the screw. Swedish scholar Nilsson and Finland scholar Rademacher took granular material as ideal fluid by use of fluid mechanics theory. They analyzed the regularity of flow of granular material, and got analytical solution of free surface and pressure distribution. But they ignored the difference between granular material and ideal fluid, which inevitably lead to error between theoretical analysis and reality. In view of the limitations of these theories, the western academics Hans Gabier and Bttcher took particles as the object, and put forward the stack state of granular material on the surface of screw will change with the rotational speed of screw. The theoretical result is closer to reality, but still is not conform to the actual situation of the screw ship unloader [15-16].

The existing theory of vertical screw conveyor ignores that the stable helical vortex in TCP of air can be formed under the action of the high-speed rotating screw shaft in the vertical screw conveyor, but whose flow is beneficial to improve the transporting efficiency of granular materials. This paper studies formation conditions of the stable helical vortex in TCP of air, which lay the foundation for study of gas-solid two-phase stable helical vortex in TCP.

2. Mathematical Equations and Characteristic Parameter

2.1. Mathematical Equations

Navier-Stokes equations for incompressible viscous fluid in the form of cylindrical coordinates are as follows:

$$\begin{cases} \dfrac{\partial u_r}{\partial t}+u_r\dfrac{\partial u_r}{\partial r}+\dfrac{u_\theta}{r}\dfrac{\partial u_r}{\partial \theta}-\dfrac{u_\theta^2}{r}+u_z\dfrac{\partial u_r}{\partial z} \\ =f_r-\dfrac{1}{\rho}\dfrac{\partial P}{\partial r}+\mu\left(\nabla^2 u_r-\dfrac{2}{r}\dfrac{\partial u_\theta}{\partial \theta}-\dfrac{u_r}{r^2}\right) \\ \dfrac{\partial u_\theta}{\partial t}+u_r\dfrac{\partial u_\theta}{\partial r}+\dfrac{u_\theta}{r}\dfrac{\partial u_\theta}{\partial \theta}-\dfrac{u_r u_\theta}{r}+u_z\dfrac{\partial u_\theta}{\partial z} \\ =f_\theta-\dfrac{1}{r\rho}\dfrac{\partial P}{\partial \theta}+\mu\left(\nabla^2 u_\theta-\dfrac{2}{r}\dfrac{\partial u_r}{\partial \theta}-\dfrac{u_\theta}{r^2}\right) \\ \dfrac{\partial u_z}{\partial t}+u_r\dfrac{\partial u_z}{\partial r}+\dfrac{u_\theta}{r}\dfrac{\partial u_z}{\partial \theta}+u_z\dfrac{\partial u_z}{\partial z} \\ =f_z-\dfrac{1}{\rho}\dfrac{\partial P}{\partial z}+\mu\nabla^2 u_z \end{cases}, \quad (1)$$

where $\nabla^2=\dfrac{\partial^2}{\partial r^2}+\dfrac{1}{r}\dfrac{\partial}{\partial r}+\dfrac{1}{r^2}\dfrac{\partial^2}{\partial \theta^2}+\dfrac{\partial^2}{\partial z^2}$, r, θ, z are the cylindrical coordinate of the spatial point, u_r, u_θ, u_z are three coordinate components for speed, ρ is the fluid density, μ is the kinetic viscosity coefficient, P is the dynamic pressure for viscous fluid.

As shown in Fig. 1, we set the radius of screw shaft and the inner walls of duct are R_1 and R_2 respectively and the screw shaft rotates at an angular velocity ω. And we set origin on the screw shaft and Z-axis on the direction of screw shaft. (r,θ,z) is set as the column coordinate vector. The physical component of the fluid velocity is $\mathbf{u}=(u_r,u_\theta,u_z)$.

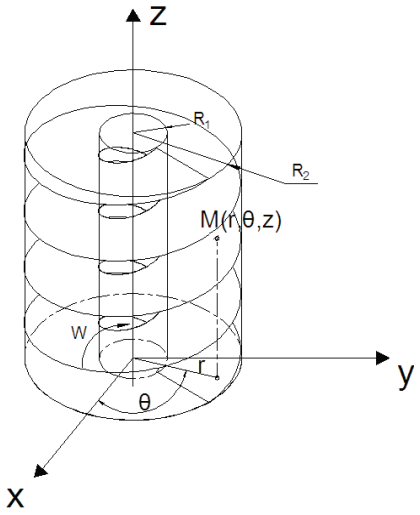

Fig. 1. Chart for the vertical screw conveyor in the form of cylindrical coordinates.

If mass force of air in vertical screw conveyor is ignored and the angular velocity of the screw shaft is constant, then $f_r=f_\theta=f_z=0$, $\dfrac{\partial}{\partial t}=0$.

Applying the initial condition above to Eq.1, the corresponding N-S equations can be expressed as follows:

$$\begin{cases} u_r\dfrac{\partial u_r}{\partial r}+\dfrac{u_\theta}{r}\dfrac{\partial u_r}{\partial \theta}-\dfrac{u_\theta^2}{r}+u_z\dfrac{\partial u_r}{\partial z} \\ =-\dfrac{1}{\rho}\dfrac{\partial P}{\partial r}+\mu\left(\nabla^2 u_r-\dfrac{2}{r}\dfrac{\partial u_\theta}{\partial \theta}-\dfrac{u_r}{r^2}\right) \\ u_r\dfrac{\partial u_\theta}{\partial r}+\dfrac{u_\theta}{r}\dfrac{\partial u_\theta}{\partial \theta}-\dfrac{u_r u_\theta}{r}+u_z\dfrac{\partial u_\theta}{\partial z} \\ =-\dfrac{1}{r\rho}\dfrac{\partial P}{\partial \theta}+\mu\left(\nabla^2 u_\theta-\dfrac{2}{r}\dfrac{\partial u_r}{\partial \theta}-\dfrac{u_\theta}{r^2}\right) \\ u_r\dfrac{\partial u_z}{\partial r}+\dfrac{u_\theta}{r}\dfrac{\partial u_z}{\partial \theta}+u_z\dfrac{\partial u_z}{\partial z} \\ =-\dfrac{1}{\rho}\dfrac{\partial P}{\partial z}+\mu\nabla^2 u_z \end{cases} \quad (2)$$

If the flow in vertical screw conveyor is stable helical vortex in TCP, the initial conditions are described as: $\frac{\partial}{\partial\theta}=\frac{\partial}{\partial z}=\frac{\partial}{\partial t}=0$, $u_r=0$, $u_\theta=u_\theta(r,z)$, $u_z=u_z(r,z)$, $P=P(r,z)$.

Applying these conditions to Eq. 2, Eq. 2 can be simplified as follows:

$$\begin{cases} -\dfrac{u_\theta^2}{r}=-\dfrac{1}{\rho}\dfrac{\partial P}{\partial r} & (3\text{-}1) \\ 0=\mu\left(\dfrac{\partial^2 u_\theta}{\partial r^2}+\dfrac{1}{r}\dfrac{\partial u_\theta}{\partial r}-\dfrac{u_\theta}{r^2}\right) & (3\text{-}2) \\ 0=-\dfrac{1}{\rho}\dfrac{\partial P}{\partial z}+\mu\left(\dfrac{\partial^2 u_z}{\partial r^2}+\dfrac{1}{r}\dfrac{\partial u_z}{\partial r}\right) & (3\text{-}3) \end{cases}$$

Eq.3-2 can be changed to:

$$\mu\left(\frac{\partial^2 u_\theta}{\partial r^2}+\frac{1}{r}\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r^2}\right)=\frac{\partial}{\partial r}\left(\frac{\partial u_\theta}{\partial r}+\frac{u_\theta}{r}\right)=\frac{\partial}{\partial r}\left(\frac{\partial(ru_\theta)}{rdr}\right)=0$$

$$\frac{\partial(ru_\theta)}{r\partial r}=C_1(z)$$

$$\therefore u_\theta+r\frac{\partial u_\theta}{\partial r}=rC_1(z)$$

$$\therefore u_\theta(r,z)=\frac{r}{2}C_1(z)+\frac{C_2}{r} \quad (4)$$

If Mass force of air in vertical screw conveyor is ignored when screw shaft rotates, the force of friction and the centrifugal force effecting on air can be ignored. So the air suffers mainly the thrust perpendicular to screw surface, obtaining upward velocity and rotating around the screw shaft. The distribution of velocity is as shown in Fig. 2.

Where Φ is the screw angle, u_c is the velocity perpendicular to the screw surface generated by thrust. It can be decomposed into two velocity component: velocity component u_θ along the direction of θ and velocity component u_z along the direction of z.

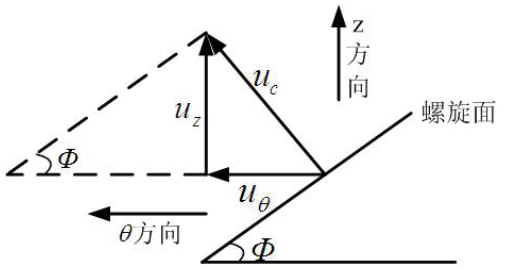

Fig. 2. The distributing diagram of velocity of air in vertical screw conveyor.

Thus the boundary conditions for stable helical vortex in TCP in vertical screw conveyor is

$$\begin{cases} u_\theta\big|_{r=R_1}=\omega R_1 \\ u\big|_{r=R_2}=0 \\ tan\Phi=\dfrac{u_\theta}{u_z}=\dfrac{S}{2\pi r} \end{cases}$$

where ω is the angular velocity of screw shaft, Φ is the screw angle, S is the screw pitch.

Applying the boundary conditions to Eq.4:

$$\begin{cases} \dfrac{R_1}{2}C_1(z)+\dfrac{C_2}{R_1}=\omega R_1 \\ \dfrac{R_2}{2}C_1(z)+\dfrac{C_2}{R_2}=0 \end{cases}$$

$$\therefore C_1(z)=-\frac{-2\omega R_1^{\,2}}{R_2^{\,2}-R_1^{\,2}}$$

$$C_2=\frac{\omega R_1^{\,2}R_2^{\,2}}{R_2^{\,2}-R_1^{\,2}}$$

$$\therefore u_\theta=\frac{r}{2}C_1(z)+\frac{C_2}{r}=\frac{\omega R_1^{\,2}(R_2^{\,2}-r^2)}{r(R_2^{\,2}-R_1^{\,2})} \quad (5)$$

Thus the distribution solution of velocity for stable helical vortex in TCP in vertical screw conveyor is

$$\begin{aligned} u&=\sqrt{u_\theta^{\,2}+u_z^{\,2}}=u_\theta\sqrt{1+\frac{4\pi r^2}{S^2}} \\ &=\frac{\omega R_1^{\,2}(R_2^{\,2}-r^2)}{r(R_2^{\,2}-R_1^{\,2})}\sqrt{1+\frac{4\pi r^2}{S^2}} \end{aligned} \quad (6)$$

2.2. The Characteristic Parameter of Hydrodynamics

The characteristics of hydrodynamics of TCP flow is described by the Reynolds number *Re*.

Reynolds number can distinguish flow characteristics and whether the flow is laminar or turbulent. So the critical parameters of different flow in vertical screw conveyor can be described by the Reynolds number.

Calculation formula of Reynolds number of abnormal pipe is as follows:

$$Re = \frac{\bar{u}d}{v} \tag{7}$$

where $\bar{u}$ is the average velocity perpendicular to the flowing section, v is the kinetic viscosity coefficient, d is the hydraulic diameter of abnormal pipe, which is defined as follows:

$$d = \frac{4A_s}{x} \tag{8}$$

where A_S is the area of flowing section, x is the wetted perimeter.

The formula of flow passing flowing section in unit time is as follows:

$$Q = \iint_{A_S} \bar{u} dA_S = \iint_A \bar{u}_n dA \tag{9}$$

where A is the section of flow in any direction, that is $dA_s = \cos(\bar{u}, n)dA$. $\bar{u}_n$ is average normal velocity perpendicular to A, that is $\bar{u}_n = \bar{u}\cos(\bar{u}, n)$. So Eq.7 and Eq.8 can be changed to:

$$Re = \frac{\bar{u}_n d}{v} \tag{10}$$

$$d = \frac{4A}{x} \tag{11}$$

The formula of Reynolds number in vertical screw conveyor is deduced according to Eq. 10 and Eq. 11. The axial section of vertical screw conveyor as is shown in Fig. 3. The section between a pitch as shown as the shade region in Fig. 3 is analyzed. The flow is split longitudinally along the screw axis.

Hydraulic diameter in vertical screw conveyor is

$$d = \frac{4A}{x} = \frac{4(R_2 - R_1)S}{2(R_2 - R_1 + S)} = \frac{2S(R_2 - R_1)}{R_2 - R_1 + S} \tag{12}$$

Reynolds number in vertical screw conveyor is

$$Re = \frac{\bar{u}_n d}{v} = \frac{2\bar{u}_n S(R_2 - R_1)}{v(R_2 - R_1 + S)} \tag{13}$$

As shown in Fig. 3, where R_1 and R_2 are the radius of screw shaft and the inner wall of duct respectively, S is the screw pitch, v is he kinetic viscosity coefficient, $\bar{u}_n$ is average normal velocity perpendicular to the section, that is average velocity component $\bar{u}_\theta$ along the direction of θ.

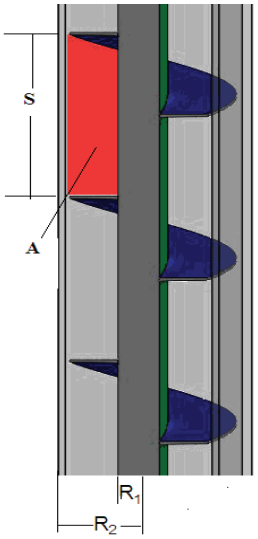

Fig. 3. The section between a pitch in vertical screw conveyor.

Let $\bar{u}_n = \bar{u}_\theta = \frac{1}{R_2 - R_1}\int_{R_1}^{R_2} u_\theta dr$, applying Eq.5 to it, then

$$\bar{u}_n = \frac{1}{R_2 - R_1}\int_{R_1}^{R_2} u_\theta dr = \frac{1}{R_2 - R_1}\int_{R_1}^{R_2} u_\theta dr =$$
$$= \frac{\omega {R_1}^2}{({R_2}^2 - {R_1}^2)(R_2 - R_1)}\int_{R_1}^{R_2} \frac{{R_2}^2 - r^2}{r} dr =$$
$$= \frac{\omega {R_1}^2}{({R_2}^2 - {R_1}^2)(R_2 - R_1)}({R_2}^2 \ln r - \frac{1}{2}r^2)\Big|_{R_1}^{R_2} =$$
$$= \frac{\omega {R_1}^2[{R_2}^2 \ln\frac{R_2}{R_1} - \frac{1}{2}({R_2}^2 - {R_1}^2)]}{({R_2}^2 - {R_1}^2)(R_2 - R_1)} \tag{14}$$

3. Fluent Numerical Simulation and Analysis of Result

3.1. Calculation Model

Building model as shown in Fig. 4, the gap G between screw vane and inner wall of duct is 2 mm, the radius of screw shaft R_1 is 50 mm, the radius of inner wall of duct R_2 is 155 mm, the screw pitch S is 110 mm and the model height is 330 mm. Simulating the model above using FLUENT, whose material is the air, that is the condition of vertical screw conveyor is idling. The kinetic viscosity coefficient v of air between screw shaft and inner wall of duct is 1.4850e-05 m^2/s. The duct is motionless while the screw shaft rotating. We can obtain different velocity vector chart and streamline chart under different rotating speed by calculation of k-e turbulence model.

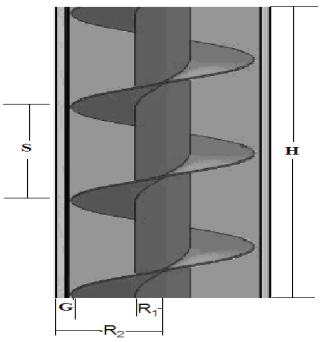

Fig. 4. The calculation model of vertical screw conveyor.

3.2. Mesh Generation

After modeling of main body of screw conveyer is set up by SOLIDWORK, it is imported to the former processor of GAMBIT to complete modeling of other parts, mesh generation and set up of boundary conditions.

When the screw vane rotates with the screw shaft, shape of the flow is changed along with time. Thus dynamic mesh is needed here. Compiling the file as the followings:

((dong 3 point
(time 0 1 60)
(omega_v $-\omega$, $-\omega$, $-\omega$)),

where ω is the angular velocity of screw shaft. It need read different file when the angular velocity is different.

3.3. Fluent Numerical Simulation

When simulating the flow of air in vertical screw conveyor by FLUENT software, the output wanted is the velocity vector chart and the streamline chat. And the critical Reynolds number is discussed between the stable helical vortex in TCP flow and other flow in vertical screw conveyor.

When the rotational speed of screw shaft is 0.5 r/s, the average velocity $\overline{u}_n$ along the direction of θ is 0.057 m/s by Eq. 14. The Reynolds number *Re* worked out is 412 by Eq. 13, whose velocity vector chart and the streamline chat is as shown in Fig. 5.

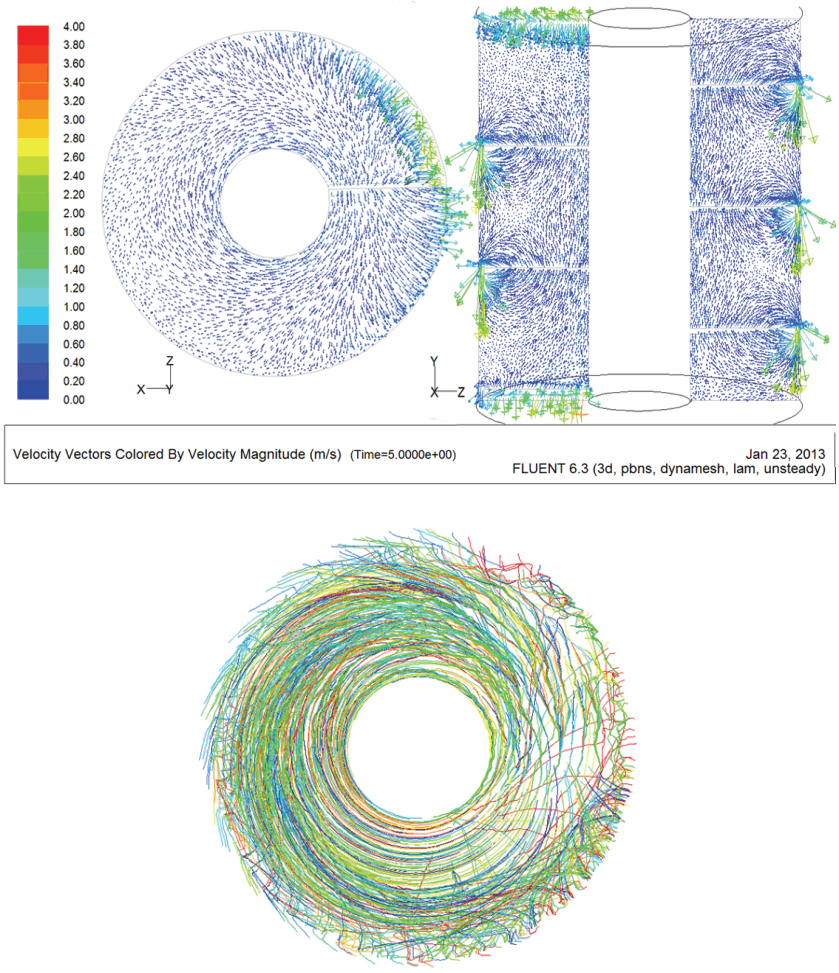

Fig. 5. The velocity vector chart and the streamline chat when the rotational speed of screw shaft is 0.5 r/s.

The velocity of air is as small as in the range of 0 m/s ~ 0.3 m/s because of small rotational speed. The flow in the streamline chat is stable. The flow shows laminar flow model but confused streamline between the screw vane and the inner wall of duct. The air obtains upward velocity and rotates around the screw shaft, which belongs to the stable helical vortex in TCP flow. The flow generates some vortex between the screw vane and the inner wall of duct, whose velocity is bigger.

When the rotational speed of screw shaft is 3r/s, the average velocity $\overline{u}_n$ worked out is 0.342 m/s and the Reynolds number *Re* worked out is 2470. The velocity vector chart and the streamline chat are as shown in Fig. 6. The velocity of air is in the range of 0 m/s ~ 1.8 m/s, which is bigger near the screw shaft. The flow in the streamline chat is stable, which shows laminar flow model. The air obtains upward velocity and rotates around the screw shaft, which belongs to the stable helical vortex in TCP flow.

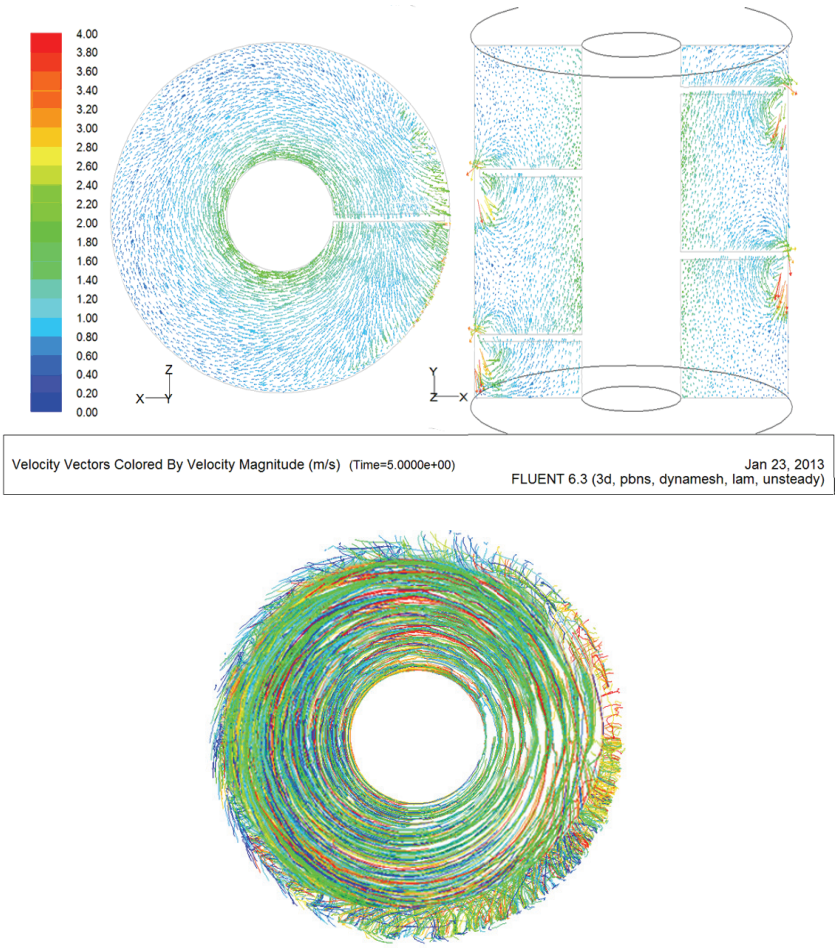

Fig. 6. The velocity vector chart and the streamline chat when the rotational speed of screw shaft is 3 r/s.

When the rotational speed of screw shaft is 6 r/s, the average velocity $\overline{u}_n$ worked out is 0.684 m/s and the Reynolds number *Re* worked out is 4950. The velocity vector chart and the streamline chat are as shown in Fig. 7. The velocity of air is in the range of 0 m/s ~ 3.5 m/s, which is bigger near the screw shaft. The flow in the streamline chat is stable, which shows laminar flow model. The air obtains upward velocity and rotates along the screw shaft, which belongs to the stable helical vortex in TCP flow. But compared with the streamline in Fig. 6, the flow generated some bigger random motion near inner wall of duct, which indicates the appearance of turbulence.

When the rotational speed of screw shaft is 9 r/s, the average velocity $\overline{u}_n$ worked out is 1.025 m/s and the Reynolds number *Re* worked out is 7410. The velocity vector chart and the streamline chat are as shown in Fig. 8. The velocity of air is in the range of 0 m/s ~ 5.2 m/s, which is bigger near the screw shaft. The streamline is disordered except for the streamline near the screw shaft. The flow layers adjacent produce slide and mix, which lead to damage of the laminar flow. The flow makes random motion and belongs to turbulence.

When the rotational speed of screw shaft is 1r/s, 2 r/s, 4 r/s, 5 r/s, the flow is similar to the flow in Fig. 6, which belongs to the stable helical vortex in TCP flow.

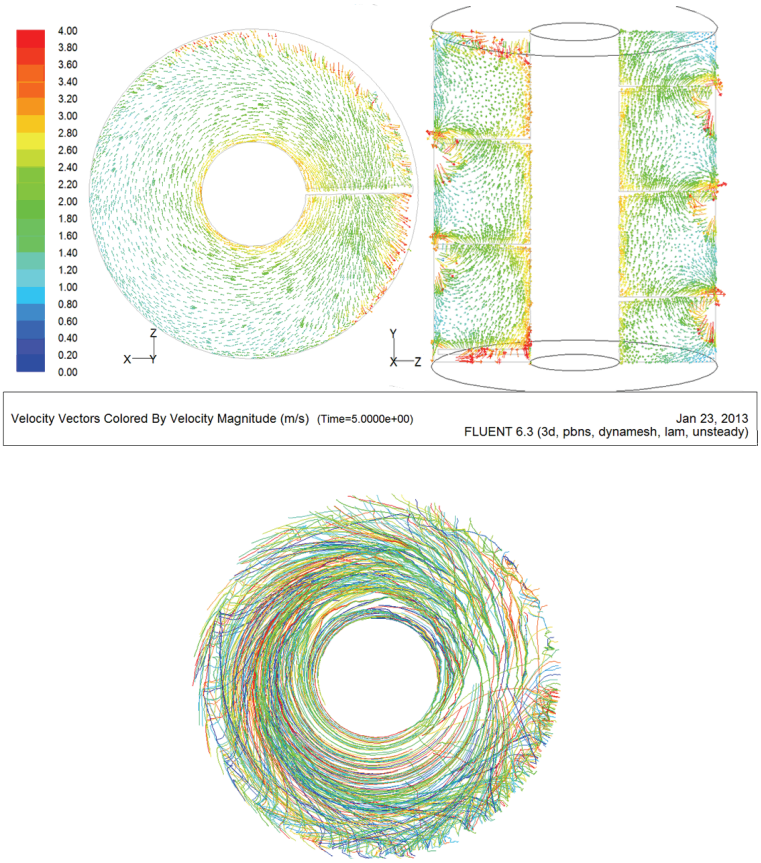

Fig. 7. The velocity vector chart and the streamline chat when the rotational speed of screw shaft is 6 r/s.

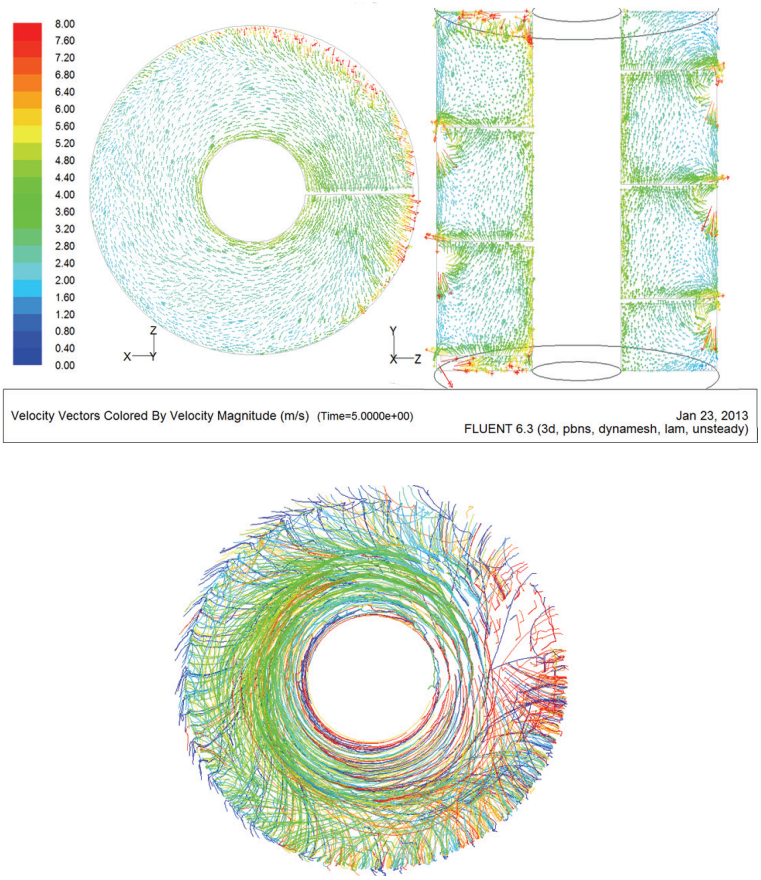

Fig. 8. The velocity vector chart and the streamline chat when the rotational speed of screw shaft is 9 r/s.

3.4. Analysis of Result

The flow in vertical screw conveyor belongs to the stable helical vortex in TCP flow when the Reynolds number *Re* within the range of 5000, which can improve the transporting efficiency of granular materials. So the critical Reynolds number Re_c suitable for appearance of stable helical vortex in TCP flow in the conveyor is 5000. When the Reynolds number *Re* is bigger than 5000, the flow may is adverse to transportation of granular materials.

The flow belongs to the stable helical vortex in TCP flow when the rotational speed of screw shaft is 3 r/s by result of numerical simulation. Taking five points from the flow above, the velocity of the numerical simulation and theoretical analysis by Eq. 14 are compared. The result is as shown in Table 1. The velocity error is small. And the velocity is bigger near the screw shaft, which proves the validity of theoretical analysis.

Table 1. Velocity comparison between numerical simulation and theoretical analysis.

The distance from the center line of the screw shaft (m)	0.055	0.079	0.103	0.127	0.150
The velocity of numerical simulation (m/s)	1.72	1.33	1.05	0.64	0.18
The velocity of theoretical analysis (m/s)	1.702	1.349	0.989	0.574	0.11

4. Conclusions

This paper describes the N-S equations of incompressible steady flow (air) in vertical screw conveyor and deduces the distribution solution of velocity of stable helical vortex in TCP. And this paper also analysis the relationship between the Reynolds number *Re* and parameters such as rotational speed of screw shaft, shaft diameter of screw shaft, screw pitch. Making numerical simulation and analysis to the flow of air in vertical screw conveyor, the effect of rotational speed on the flow is focused. Finally this paper recommends the critical Reynolds number Re_c suitable for appearance of stable helical vortex in TCP flow in the conveyor and providers the basis for architectonics of vertical screw conveyor.

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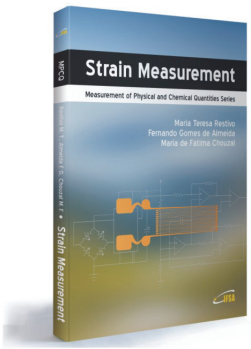

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Capacitors Optimization Placement in Distribution Systems Based on Improved Seeker Optimization Algorithm

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Abstract: An improved seeker optimization algorithm (SOA) was proposed for capacitors optimization configuration in distribution network in this paper, which was a classical non-linear discrete problem. To aim at the flaws of traditional SOA that it was easy to fall into the local extremum in later stage of optimization process, Simulated Annealing (SA) algorithm was introduced by setting the Boltzmann simulated annealing operator to select a new solution to enhance the global search ability of the algorithm. In addition, search step-length was adjusted using automatic adjustment factor so that non-linear discrete problem can be solved better. Considering the economy as the desired aim function, and power flow, voltage level, and as well as capacitors capacities as the constraints, thus a corresponding mathematical model was established for distribution grid capacitors optimization. Finally, the simulation results for IEEE 33-node distribution system show that the improved SOA is very effective and feasible, and is a quite ideal swarm intelligence algorithm.

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Keywords: Distribution grid, Capacitors, Optimization placement, Seeker optimization algorithm (SOA), Simulated annealing (SA).

1. Introduction

In distribution grid, the optimal placement of the capacitors is to determine the best installation location, number, capacity and type of the capacitor so as to minimize the sum of the operating costs and the investment costs of the system [1]. For distribution grids, it is quite essential to perform capacitor optimization configuration to reduce the power loss and to improve the voltage quality. However, the most direct and effective role is to strengthen the safe economic operation of distribution grids. With the complexity of power system loads increasing, and power grids interconnection, and as well as the access of new energy resources, the traditional optimal algorithms show their limitations and shortcomings increasingly. Presently, the traditional methods to allocate the capacitors are relatively backward, which rely mainly on relevant procedures and previous configuration experience. With the development of computer and artificial intelligence technology, artificial intelligence optimization algorithm is getting the attention of more and more scholars, and they have made good application effects in many areas [2, 3]. Based on it, many novel methods to solve the problem have proposed in recent years. In [4], combined with genetic algorithm (GA), fuzzy method for capacitors

optimization configuration is expressed as a multi-objective determination problem, where three different targets are considered respectively: the cost of grid loss, the capacitor cost, and the voltage drop, then measuring the satisfaction level of the feasible solutions using these targets, finally, the most satisfactory solution is accepted. In [5], ant colony algorithm is presented to solve the problem, where to take the grid loss as the aim, and to consider the voltage of each node and system power flow as the constraints, and still to consider the distribution network reconfiguration, and thus it better solves the capacitors optimization configuration problem. In [6] an artificial immune algorithm is proposed to solve this problem. Rather than the other methods, the important advantage of immune algorithm lies in its multifunction, i.e., it can investigate many parameters in the target function. In this study, six distinct objectives are considered in all such as the cost of grid loss, voltage profile, power factor, utilities development, and the cost of capacitors including the purchase and installation, and so better optimizing results are acquired. In recent years, some new optimal algorithms are proposed to solve this problem, for instance, plant growth simulation algorithm [7], and shuffled leaping frog algorithm [8]. All of these methods mentioned above have achieved a certain effect, but some deficiencies still are exposed out in process of seeking optimization, for example, it is easy to get into local extremum, and especially in later stage of optimization process, and as well as slow in convergence speed. To change the situation and achieve more satisfactory results, in this paper, the seeker optimization algorithm (SOA) is proposed to solve the puzzle. In process of optimization, to aim at the flaws of SOA, the paper implements the improvements, i.e., the idea of simulated annealing algorithm is introduced into at the late stage of the iteration calculation, and Boltzmann simulated annealing operator is added to strengthen the local search ability of the algorithm so that the seeker will not cluster and fall into stagnation or local extremum. In this algorithm the step-length update formula is improved so as to strengthen its nonlinear optimization ability when the SOA search step-length is very close to zero, but no globally optimal solution is searched at the end of the search under some circumstances. Finally, through applying the improved SOA to optimize the capacitor configuration in IEEE 33-node distribution system, the related simulation results show the method can resolve the puzzle better.

2. Mathematical Model on Capacitors Optimization Configuration

2.1. Aim Function

The aim function to be selected is based on economic considerations in the paper, such as the active grid loss, and the purchasing, installation and maintenance cost of the capacitor. In addition, some constraints are also taken into account, for instance, voltage and power factor constraints. The aim function may be written as

$$\min v = k_{\mathrm{e}} \sum_{i=1}^{n_{\mathrm{t}}} t_i P_{\mathrm{Loss}}(x_i, q_i) + \sum_{k=1}^{n_{\mathrm{c}}} f(q_{k,0}), \quad (1)$$

where k_{e} expresses the cost of unit electric energy consumption, and n_{t} denotes the number of the load level, and t_i means the duration time of the load level i, and $P_{\mathrm{loss}}(x_i,q_i)$ is the active power loss of the system under the load level i, x_i=[P_i,Q_i,V_i], P_i expresses the active power injection vector of each node under the load level i, Q_i expresses the reactive power injection vector of each node under the load level i, V_i expresses the voltage under the load level i, q_i expresses the capacitor configuration state vector under the load level i, and n_{c} denotes the sites number of the installed capacitors, and $f(q_{k,0})$ is the total cost of the installed capacitor at the site k.

Seen from the above equation, the aim function consists of two aspects, one is the active power loss cost of distribution grid, and the other one is the capacitor-related cost.

2.2. Aim Function

There are three constraints to require to be considered, they are respectively defined below.

1) Power flow constraint:

$$P_{\mathrm{fiow}}(x_i, q_i) = 0, \quad (2)$$

where i denotes the load level, and x_i denotes the location information of the seeker, and q_i means the capacitors capacity under the load level i.

2) Voltage constraint:

$$U_{i\min} \le U_i \le U_{i\max}, \quad (3)$$

where $U_{i\max}$ represents the upper limit of the voltage on the bus i, and $U_{i\min}$ presents the allowed lower limit of the voltage.

3) Capacitor capacity constraint:

$$0 \le q_{k,i} \le q_{k,0}, \quad (4)$$

where $q_{k,0}$ denotes the total capacity of the installed capacitors in terms of the site k, and $q_{k,i}$ denotes the installed capacity of the installed capacitors under the load level i for the site k.

3. SOA and its Improvement

SOA was proposed based on intelligent random searching behavior of human beings in recent years

[9]. SOA is an intelligence optimization algorithm with the colony based, and simulates the random search behavior which is described using human natural language, and applies fuzzy reasoning to establish model as its guiding ideology. Different form the existing optimal algorithms, SOA is based on the fuzzy intelligent search behavior of the human beings. Each individual is treated as an independent individual during searching optimum, but the individual has the communication, collaboration, learning, and reasoning abilities. SOA evolution is considered as the locomotion of all the individuals who participate in the searching event. It obtains search experience through cognitive learning between the individuals, and the social learning between the groups when randomly searching, and selects the forward direction, determines the changes of the seeker sites and completes the search of the targets in accordance with human intelligent thought.

3.1. Basic SOA

In SOA, each seeker is seen as an independent individual in the colony, and the teams seeking optimization is enhanced in the algorithm, and the optimal solution for the target problem actually is the best position searched by the seekers. The current location of the seeker i is defined by $x_i=(x_{i1},x_{i2},\ldots,x_{iD})$, where $i\in[1,s]$, s is the size of the populations, and D is the dimensions of the search space. During the seeking optimum, the best position of the individual is expressed using $P_{i,\text{best}}$, and the best position of the adjacent populations is expressed by $G_{i,\text{best}}$, The neighborhood can be defined depending on specific optimization problem, and the search team is divided into three sub-population on average in this paper and each neighborhood comprises of all individuals belonging to the one sub-population. The location update formula of SOA can be simply expressed as:

$$x_{ij}(t+1)=x_{ij}(t)+\alpha_{ij}(t)d_{ij}(t), \tag{5}$$

where $\alpha_{ij}(t)$ express the seeking step-length, $d_{ij}(t)$ express the seeking direction, $x_{ij}(t)$ express the position at time t, i=1,2,…,s, j=1,2,…,D.

3.2. Determination of Seeking Step-Length

As determining the search step-length the reasoning and approximation capability of fuzzy system is adopted [10]. The search rules are described using human natural language instead of computer language to enable SOA to be able to simulate human fuzzy intelligent search behavior. A simple fuzzy rule can be expressed as the forward step being shorter if the fitness value of the function is smaller. In the paper, Gaussian function is selected as the fuzzy membership function of the forward step being shorter as follows.

$$\mu(\alpha)=\mathrm{e}^{-\alpha^2/2\delta^2}, \tag{6}$$

where α express the step-length, and δ is the undetermined parameter of the Gaussian function. According to (6), obviously, the probability outside the interval $[-3\delta, 3\delta]$ is less than 0.0111. Hence, the minimum subjection degree $\mu_{\min}$ is defined as 0.0111. Under normal circumstances, when $\mu_{\max}$=1, the step of the best individual is 0. However, in order to speed up the convergence rate and make the best individual also has a certain step, in this paper $\mu_{\max}$=0.9.

The scope of the aim function value requires setting alone for a variety of optimization problems. In this paper, the values of the aim problems are arranged in descending order, and are converted into a series of natural numbers in sequential order from real numbers, and are severed as the inputs of fuzzy reasoning to establish a general system suitable for commonly-used optimization problems. The following functions are selected as the membership functions of fuzzy variable of the function fitness value being smaller.

$$\mu_i=\mu_{\max}-\frac{s-I_i}{s-1}(\mu_{\max}-\mu_{\min}), \tag{7}$$

$$\mu_{ij}=\mathrm{R}(\mu_i,1), \tag{8}$$

where I_i is the series number of the aim function value of the seeker at the current site $x_i(t)$ in descending order, and the function $\mathrm{R}(\mu_i,1)$ is a value randomly generated belonging to $[\mu_i, 1]$. Seen from (7), it can simulate the randomness of human seeking behavior. The method to obtain the search step-length is described as follows. Firstly, the μ_{ij} is determined in accordance with (7) and (8). Then the j^{th} dimensional step-length is determined according to the following formula.

$$\alpha_{ij}=\delta_i\sqrt{-\ln(\mu_{ij})}, \tag{9}$$

where δ_i expresses the Gaussian function parameter which is determined by

$$\vec{\delta}_i=\varphi\,\mathrm{abs}(\vec{x}_{\min}-\vec{x}_{\max}), \tag{10}$$

where $\vec{x}_{\min}$ and $\vec{x}_{\max}$ separately represent the two locations with the minimum and the maximum function value that the seeker i searched in its population, and φ is applied to adjust the search precision, with the deepening of the search it linearly decreases from 0.9 to 0.1 to balance local and global exploration and development capabilities, and as well as to achieve the change from rough to fine searching stage, and the signal abs means taking absolute function value.

3.3. Determination on Seeking Direction

The forward direction of the seeker is determined by the so-called experience gradient which is obtained through the evaluation on the locations changes in its own and other seekers searching history [11]. It mainly adopts the three directions, i.e., egocentric direction $d_{i,\text{ego}}$, altruistic direction $d_{i,\text{alt}}$, and pre-action direction $d_{i,\text{pro}}$ [12, 13], which are represented as follows.

$$d_{i,\text{ego}}(t) = p_i - x_i(t), \tag{11}$$

$$d_{i,\text{alt}}(t) = g_i - x_i(t), \tag{12}$$

$$d_{i,\text{pro}}(t) = x_i(t_1) - x_i(t_2), \tag{13}$$

where t_1, $t_2 \in \{t, t\text{-}1, t\text{-}2\}$, $x_i(t_1)$ and $x_i(t_2)$ respectively express the best position and the worst position in terms of $\{x_i(t\text{-}2), x_i(t\text{-}1), x_i(t)\}$, and $p_{i,\text{best}}$ and $g_{i,\text{best}}$ respectively express the best position of the individual and the population so far today. The final search direction of every seeker is determined by the random weighted geometry sum of the above three directions, which can expressed by

$$d_i(t) = \text{sign}(\omega d_{i,\text{pro}} + \phi_1 d_{i,\text{ego}} + \phi_2 d_{i,\text{alt}}), \tag{14}$$

where φ_1 and φ_2 are two unrelated random numbers within the interval [0, 1], and ω is the inertia weight, which decreases linearly from 0.9 to 0.1 with the deepening of the search process.

3.4. Determination on Seeking Direction

Although SOA can solve the discrete nonlinear optimization problem more effectively, but it still explore some deficiencies during applications, such as premature and slow convergence speed at iteration late period. To solve the problem, Boltzmann simulated annealing operator is here applied to select a new solution after the update of the seeker location. To some extent it plays a significant role that we first accept a deterioration solution to get a better solution, similarly, and the requirement of population diversity is also met. In this algorithm, the determination on the initial temperature and the annealing operation are most critical. Too high or too low initial temperature will affect the initial performance of the original algorithm, which will directly lead to ineffective search or take a longer time to search. The initial temperature can be determined based on the relative performance of the initial population. Here the sum of fitness functions is selected as the initial temperature below [14].

$$t_0 = \frac{f_{i\min} - f_{j\max}}{\ln(f_{i\min} / f_{j\max})}, \tag{15}$$

where $f_{i\min}$ and $f_{i\max}$ are the minimum fitness function value and maximum fitness function value in the initial population, respectively. It can minimize the flaws existing in the initial performance of the algorithm if selecting the initial temperature according to (15). The annealing temperature is determined by

$$t_{k+1} = \alpha t_k, k \geq 0, 0 \leq \alpha \leq 1, \tag{16}$$

where γ denotes the selected factor ratio, and k is the iteration times.

The probability for each seeker to be selected is judged using Boltzmann annealing operator after the temperature changes.

$$p_i = \frac{\exp\left(-\frac{f(i) - \overline{f}}{t_k}\right)}{\sum_{j=1}^{s} \exp\left(-\frac{f(i) - \overline{f}}{t_k}\right)}, \tag{17}$$

where $\overline{f}$ express the average fitness of the population. Seen from (17), that the simulated annealing-based individual searching method enables the populations choose the deterioration solution during the optimization process to prevent the seeker to fall into local extremum.

3.5. Improvement on Search Strategy

A step fitness factor $u(d_i)$ is added to strengthen the nonlinear optimization ability of the algorithm and to improve the step-length, the improved step-length formula is as follows.

$$\alpha_{ij} = \delta_{ij} u(d_i) \sqrt{-\ln(\mu_{ij})}, \tag{18}$$

$$u(d_i) = \begin{cases} 3.55 + \alpha, & d_i \leq R_1 s \\ 3.55, & R_1 s < d_i < R_2 s \\ 3.55 - \beta, & d_i \geq R_2 s \end{cases}, \tag{19}$$

where the α and β are the adaptive adjustment parameters, and defined as α=β=1.05, R_1 and R_2 are the adaptive control parameters, and define as R_1=1/7, and R_2=3/7, and d_i is the corresponding number arranged by each seeker in the group in descending order, and s is the population size, and define as 36.

The search direction $d_{ij}(t)$ and the search step-length $\alpha_{ij}(t)$ of the jth dimension of each step of the seeker i at time t are then calculated based on the above equations, separately, and $\alpha_{ij}(t) \geq 0$, $d_{ij}(t) \in \{-1, 0, 1\}$. From $d_{ij}(t)$, we can know the seeker should keep stationary, or moving along the coordinate j in the positive direction and moving along the coordinate j in the negative direction.

Location updating of the seeker is performed after both the search direction and the step-length are determined, the updating formula is as follows.

$$\Delta x_{ij}(t+1) = \alpha_{ij}(t) d_{ij}(t), \quad (20)$$

$$x_{ij}(t+1) = x_{ij}(t) + \Delta x_{ij}(t+1), \quad (21)$$

We replace the search locations of the two worst seekers in the sub-population with the search locations of the two best seekers in the other two sub-populations after evaluation process each time to share information of the seekers between the diverse populations, which not only avoids premature to some extent, but also achieves the sharing of the social information between the populations.

4. Capacitors Optimization Placement for Distribution Grid

The Fig. 1 is the improved SOA flow chart diagram.

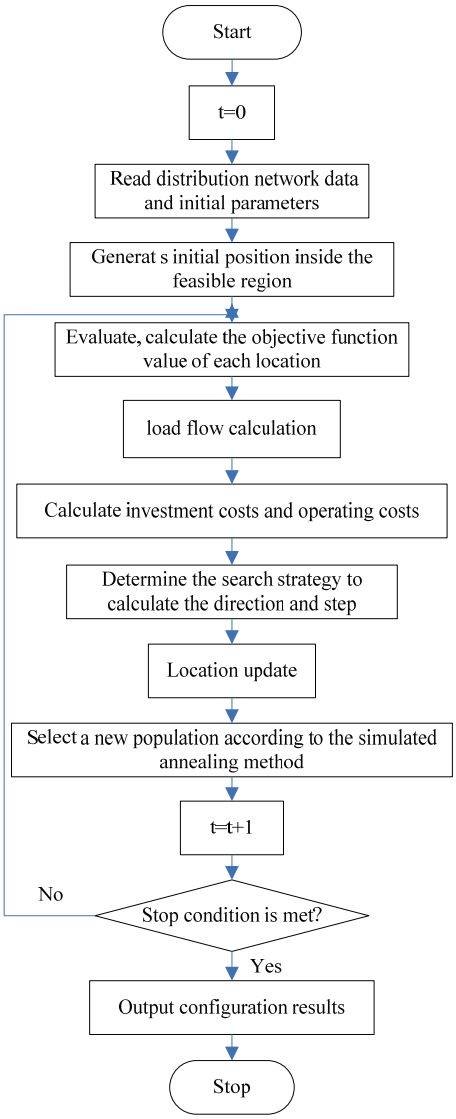

Fig. 1. Improved SOA flow chart diagram.

In this paper an improved power flow calculation method is applied to meet the power flow constraint equations [15], which ensure the speed and accuracy of the power flow calculation. The program adopts integer codes to simplify the calculation process, and the fixed-capacity capacitors are grouped to meet diverse capacity constraints. The range of 0.95 to 1.05 reference voltage of distribution grid is given for the node voltage constraints of distribution grid. The capacitors installed under light loads are used as fixed capacitors, and the number of capacitors installed is not bigger than the number of switching capacitors which are necessary under medium loads and heavy loads. The number of capacitors required under light loads should be firstly calculated in the first configuration, and the required capacitors under light loads are treated as reactive power supply, and on the basis of it the switching capacitors under medium loads and heavy loads are reasonably allocated. Improved SOA flow chart is shown in Fig. 1.

5. Example Analysis

IEEE 33-node standard distribution system is applied to verify the effectiveness of Improved SOA, and whose topology is shown in Fig. 2.

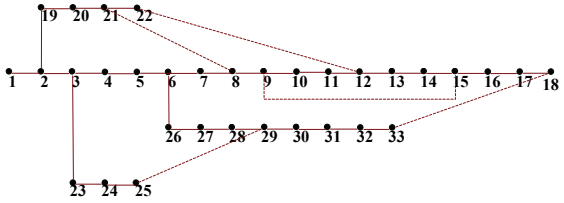

Fig. 2. IEEE33-node distribution system.

In Fig. 2, except that node 1 is the balanced node that cannot be configured with capacitors, the other 32 nodes can be configured with capacitors. The power reference value of the system is defined as 1.0 MVA, and the major parameters are defined as: capacitor capacity with 30 kvar is considered as a unit, the installation cost of the capacitor is expressed by RMB 500 yuan/unit, and a fixed capacitor cost is RMB 650 yuan/unit, and a switching capacitor cost is RMB 1250 yuan/unit, and annual maintenance cost of a capacitor is RMB 150 yuan/unit, and the price of electricity is RMB 0.6 yuan/kW·h. Each node can be installed with 8 capacitors at most which includes the switching and fixed capacitors. The duration time and the proportions of the different load grades are shown in Table 1.

Simulated annealing, SOA and improved SOA are respectively used to calculate the capacitors optimization allocation in IEEE 33-node system. Table 2 shows the details of the optimal configuration of the capacitors trough improved SOA, and

Table 3 is the results comparison of grid loss reduction among them, and Table 4 is the results comparison of voltage stabilization, and Table 5 is the results comparison of investment operating costs after optimal configuration.

Table 1. Duration of different loads.

Load level	Duration/h	Proportion
Light load	5260	0.75
Medium load	24520	1
Heavy load	5260	1.25
Four years statistics outcome		

Table 2. Details of the capacitors configuration of each node.

Node number	The number of installed units			Installed capacity (kvar)
	Light load	Medium load	Heavy load	
2	0	0	3	90
3	0	1	7	240
4	0	5	3	240
5	0	0	8	240
6	0	2	4	180
7	0	6	1	210
8	0	6	1	210
9	7	1	0	240
10	0	1	3	120
11	0	4	4	240
12	0	4	2	180
13	0	0	2	60
14	1	3	4	240
15	1	0	2	90
16	0	2	1	90
17	3	0	5	240
18	0	4	4	240
19	0	0	3	90
20	0	1	1	60
21	1	1	2	120
22	0	1	1	60
23	0	1	7	240
24	0	0	3	90
25	0	6	0	180
26	0	1	7	240
27	0	1	7	240
28	0	8	0	240
29	2	4	1	210
30	1	6	1	240
31	3	3	0	180
32	1	5	2	240
33	4	4	0	240
The total number of units	24	81	89	
Total compensation capacity				5820

Table 3. Active power loss comparisons after the configuration.

Load level	Active power loss/kW			
	Before optimization	SA	SOA	Improved SOA
Light load	89.13	65.13	62.88	60.46
Medium load	254.48	188.64	193.75	176.33
Heavy load	433.27	424.89	426.54	421.31

Table 4. Minimum voltage comparison after configuration.

Load level	Minimum voltage/%			
	Before optimization	SA	SOA	Improved SOA
Light load	92.18	95.13	95.11	95.26
Medium load	87.97	95.02	95.07	95.32
Heavy load	83.75	95	95	95.03

Table 5. Total investment and operation cost comparisons after the configuration.

Total costs/Ten thousand yuan in RMB			
Before optimization	SA	SOA	Improved SOA
449.38	429.39	418.21	387.03

Fig. 3 shows the optimization process with three algorithms. We can see from Table 3 that the reduction on the active grid loss of the distribution grid is very significant after the configuration, and it can save RMB 623,500 yuan within 4 years, which accounts for 13.87 % of the total cost before optimal configuration. It also can be seen from the Table 4, the node voltage quality of the distribution network after compensation has been improved, dramatically, and all the node voltages are qualified, but simulated annealing algorithm has a good performance in voltage optimization at the expense of the economy. At the same time, we also can be seen from Fig. 3 that the optimization curve of SOA declines fast in iteration 50 times, this shows the optimization mechanism of SOA is much better than the other two. After iteration 100 times, when other algorithm exhibits cease in converge performance, there are still changes in terms of improved SOA, which shows that the improved SOA is not easy to fall into local optimum compared with the other two algorithms. On the whole, the Improved SOA can get the best solution after 200 times iteration. The comprehensive comparisons indicate that the improved SOA has shown feasibility and superiority in solving this problem.

Taking into account the randomness of the three algorithms, I conducted 20 operations using the three different algorithms, and the result is can be seen from the Table 6. Then I sorted out and calculated the

result, and the mean value，standard deviation and variance of three algorithms are got, which can be seen from the Table 7.

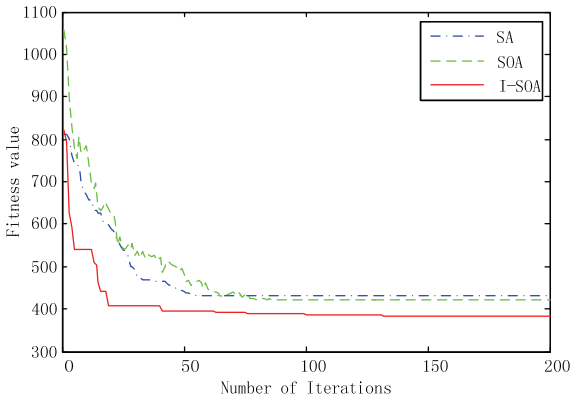

Fig. 3. Optimization procedure with three algorithms.

Table 6. Three algorithms optimal function value of 20 samples.

Number of calculations	SA	SOA	Improved SOA
1	458.5	416.34	397.71
2	426.85	416.75	392.47
3	411.2	401.93	397.71
4	455.29	415.72	384.32
5	441.46	408.47	389.28
6	444.49	434.53	396.99
7	454.75	409.31	397.22
8	421.86	413.6	383.22
9	402.7	408.48	385.04
10	406.4	428.19	392.09
11	447.7	440.23	400.21
12	416.1	409.45	399.4
13	437.46	424.11	392.19
14	437.78	431.05	383.39
15	429.79	427.23	389.32
16	419.53	399.59	396.23
17	405.04	419.49	383.36
18	456.19	432.21	397.26
19	456.82	433.15	386.13
20	424.89	438.38	392.49

Table 7. Mean value, standard deviation and variance of three algorithms 20 samples.

Algorithm	Mean value	Standard deviation	Variance
SA	432.74	18.88	356.53
SOA	420.41	12.25	150.15
Improved SOA	391.80	5.92	35.10

We can see from Table 3 that the mean value of improved SOA is smaller than SOA and SA, which can proof that the improve SOA is suitable for solving such problems, and the standard deviation and variance of it is much lower than the other two algorithm, this result shows that the Stability of improve SOA is better than SOA and SA, the improvement is effective.

6. Conclusions

In this paper, an optimized mathematical model based on economy for the capacitors of the distribution network has been established first, and the seeker optimization algorithm is selected as a solution for the characteristics of discreteness and larger space of the solution of the configuration. Simulated annealing operator is added when improving the algorithm to improve the overall search ability and to avoid the premature in the optimization process and the search strategy has been improved so that it is more suitable for solving nonlinear discreteness. The distribution network is optimized with the improved algorithm, which has an ideal effect both on loss reduction and on voltage optimization and shows the feasibility of the improved algorithm on advancement and technology.

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SAR Target Recognition Using Improved Fuzzy Neural Network

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Abstract: Target recognition in high-resolution synthetic aperture radar (SAR) images is a challenging task, because SAR images have higher ambiguity for different target, which will reduce the correct recognition rate. This paper presents an improved SAR recognition algorithm based on fuzzy neural network (FNN), which deals with the ambiguity SAR target recognition very well. This improved FNN system improves learning algorithm and structure which has fuzzy multi-input and fuzzy multi-output. This paper takes the MSTAR data as the test data in simulation to show that this improved fuzzy neural network obtains a higher correct recognition rate. *Copyright © 2013 IFSA.*

Keywords: Fuzzy neural network, Higher ambiguity, Correct recognition rate, SAR image.

1. Introduction

Radar systems are important sensors because they can detect the remote target day and night, and not affected by fog, clouds and rain. Along with the development of radar technologies, as well as with increasing demands for target identification in radar applications, automatic target recognition (ATR) using synthetic aperture radar (SAR) has become an active research area [1]. The latest theoretical developments in classification methodologies quickly have been applied to the SAR ATR design [2].

However, SAR target recognition is still a tough task due to the high variation of image signatures caused by different target orientations, incidence angles, and resolutions [3]. Generally, target recognition consists of three processes: preprocessing, feature extraction and classification. Feature extraction and classification are two principal steps, which mainly determine the recognition performance of a target recognition algorithm [4].

SAR images are highly sensitive to the target aspect because of shadowing effects, interaction of the signature with the environment, projection of a three-dimensional scene onto a slant plane and other reasons related to the aspect dependence of radar cross-sections. The ability to discriminate between targets in SAR imagery also varies greatly with target aspect [5]. Therefore, using multiple views of a target in different aspects may provide more robust classification performance than only using a single view. Different aspects of same target would have huge dissimilarity, and different targets seem semblable in various aspects. A lot of classifiers have been applied in SAR target recognition and each of them has their own strengths and weaknesses. The correct recognition rate greatly suffers from the illegibility [6].

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Fuzzy logic and neural network are complementary technologies, and they can provide us with rich functionality when the two technologies are integrated together. Fuzzy logic was proposed by Zadeh, it has the ability of human thinking for objective world. Fuzzy logic can offers a high level framework for approximate reasoning that can commendably perform the task with uncertainty and imprecise information, and offers requisite control principles [7-8]. Neural network provide nonlinear mapping functionality, self-learning ability and changeable architectures that can easily adapt to contextual input and feedback [9-10]. By integrating the two technologies, the fuzzy neural network (FNN) system is endowed with a twofold advantage. FNN model can provide a fuzzy logic system with the powerful computation and learning capabilities of neural networks, and provide neural networks with "human-like" reasoning capabilities of fuzzy logic systems. Nowadays, a lot of new concepts, such as fuzzy model and innovative architecture about FNN have been developed, and FNN has been widely used in many fields, such as pattern recognition [11-14], modeling, and diagnostics. Takagi-Sugeno fuzzy reasoning model [15] is the most commonly used fuzzy model in fuzzy neural network, it can extract the fuzzy rules and identify the fuzzy parameter directly from the input-output data.

This paper is organized as follows. Section 2 includes a short description of fuzzy neural network. Section 3 illustrates the improved fuzzy neural network proposed in this paper. Section 4 shows the recognition performance of improved FNN for SAR images. The conclusions are given in Section 5.

2. Description of Fuzzy Neural Network Model

2.1. Structure of Fuzzy Neural Network

A fuzzy neural network has four layers including input layer, fuzzy layer, fuzzy inference layer and de-fuzzy layer, Fig. 1 shows the structure of fuzzy neural network.

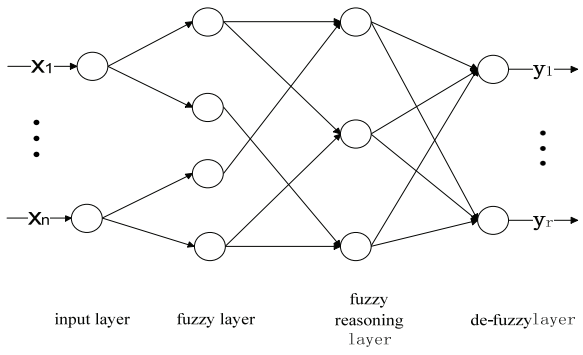


Fig.1. Structure of fuzzy neural network.

Input layer: The input layer nodes which transfer input value to the next layer directly connect with the input vector $X=[x_1\ x_2 \cdots x_k]$, and the number of nodes in this layer is the dimension of X.

Fuzzy layer: Each node represents a linguistic variable in this layer and its function is calculating membership $\mu_{A_j^i}(x)$ of each linguistic variable. In this paper, Gauss function is used as the membership function. The gauss membership function is expressed as:

$$\mu_{A_j^i}(x_j) = \exp(-(\frac{x_j - c_j^i}{b_j^i})^2), \quad (1)$$

where A_j^i, which is the i[th] linguistic variable of x_j - the fuzzy set defined in the universe of discourse, c_j^i is the center value of membership function and b_j^i is the width of membership function.

Fuzzy reasoning layer: Each node in this layer represents a fuzzy rule. It is used to match the fuzzy rules, and calculate the fitness of each rule by product operation or minimal operation, the rules is shown as follows:

$$\omega^i = \mu_{A_1^i}(x_1) * \mu_{A_2^i}(x_2) * \cdots * \mu_{A_k^i}(x_k) \quad (2)$$

De-fuzzy layer: It calculates the total output as:

$$y_i = \sum_{i=1}^{n} \omega^i (p_0^i + p_1^i x_1 + \cdots + p_k^i x_k) / \sum_{i=1}^{n} \omega^i \quad (3)$$

2.2. Learning of Fuzzy Neural Network

Fuzzy neural network design the control system by input/output data and this process is defined as learning process. Gradient descent algorithm is widely used in parameter adjusting process of ordinary neural network for its applicability, as well as in fuzzy neural network. The objective of Gradient descent is using the gradient of weights to update the weights. With the repetitious training of sample, fuzzy rules can be adjusted by the updating weights. The learning algorithm is expressed as follows:

1) Expectation Error

$$e = \frac{1}{2}(y_d - y_c)^2, \quad (4)$$

where y_d is the expected output, y_c is the actual output, e is the error between both.

2) Coefficient Updating

$$p_j^i(k) = p_j^i(k-1) - \alpha \frac{\partial e}{\partial p_j^i}, \quad (5)$$

$$\frac{\partial e}{\partial p_j^i} = (y_d - y_c)\omega^i / \sum_{i=1}^{m} \omega^i \square x_j \,, \tag{6}$$

where P_j^i is the coefficient of neural network, α is the learning rate.

3) Parameter Updating

$$c_j^i(k) = c_j^i(k-1) - \alpha \frac{\partial e}{\partial c_j^i} \tag{7}$$

$$b_j^i(k) = b_j^i(k-1) - \alpha \frac{\partial e}{\partial b_j^i} \tag{8}$$

where c_j^i is the center of membership function; b_j^i is the width of membership function.

Parameter training process will finish one time after parameter is updated one time by expressions (5)-(8), then the neural network will finish the training process again and again until the expectation error satisfies the request of precision. The learning algorithm is based on the thinking of BP neural network, and it updates parameter by Gradient descent.

3. The Improved Fuzzy Neural Network

As described in section II, The learning rate and step will not change in the training process of conventional fuzzy neural network, since the majority of training data have high difference from others, and in order to increase learning efficiency, the learning rate and step will mainly satisfy these data. Thus, the conventional FNN cannot accurately recognize some higher ambiguity data. For these data, there have little difference between the distance from it to its own class center and the distance from it to error recognition class center. When these higher ambiguity data are trained by conventional network, they would not be trained adequately and even learned mistakenly for the fast speed of gradient descent which also can induce a concussion in training process. What's more, if the training data is too many as well as the ambiguity data, the conventional FNN will gain a low correct recognition rate. Aiming at these problems, and in order to increase correct recognition rate, this paper proposed an improved fuzzy neural network with improved structure and learning algorithm.

3.1. Improved Network Structure

Compared with the conventional network structure, each last output of the improved structure is an ambiguity result on each type, not an accurate judge, so it is a fuzzy multi-input and fuzzy multi-output structure. It is shown as Fig. 2.

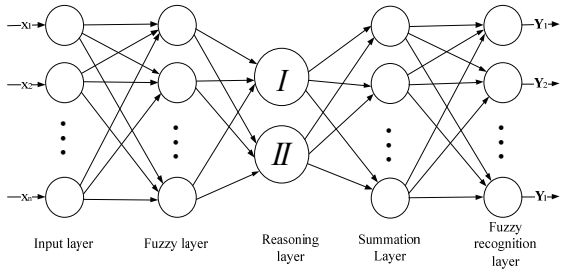


Fig. 2. Structure of improved fuzzy neural network.

Input layer: the number of input variable is *n,* and the corresponding nodes number is *n*.

$$O_a^1 = x_a \qquad a = 1, 2, \cdots, n \tag{9}$$

Fuzzy layer: as described in section II, the improved network will also adopt gauss function as the membership function in this layer.

$$f_{aj} = \exp(-\frac{(x_a - c_{aj})^2}{\delta_{aj}^2}), j = 1, \cdots, u \,, \tag{10}$$

where f_{aj} is the membership from x_a to j^{th} neural node, c_{aj} is the center value of membership function, and δ_{aj}^2 is the width of membership function, u is the number of neural nodes in this layer, and the output of j^{th} neural node is expressed as follows:

$$O_j^2 = \exp(-\sum_{a=1}^{n} \frac{(x_a - c_{aj})^2}{\delta_{aj}^2}), j = 1, \cdots, u \tag{11}$$

Reasoning layer: to gain a better training result on higher ambiguity data, the improved network choose summation ($\sum$) and product ($\prod$) operation as reasoning rule since they have more parameters in training process. The rules are expressed as follows:

$$\begin{aligned} O_{\Sigma}^3 &= w_{11} O_1^2 + w_{12} O_2^2 + \cdots + w_{1u} O_u^2 \\ &= \sum_{j=1}^{u} w_{1j} O_j^2 \end{aligned} \,, \tag{12}$$

$$\begin{aligned} O_{\Pi}^3 &= w_{21} O_1^2 \square w_{22} O_2^2 \square \cdots \square w_{2u} O_u^2 \\ &= \prod_{j=1}^{u} w_{2j} O_j^2 \end{aligned} \,, \tag{13}$$

where $w_{kj}, k = 1, 2$ is the weight parameter.

Summation layer: it is used to calculate the summation between output of product node and output of summation node in former layer, this layer is expressed as follows:

$$O_m^4 = \alpha_m O_\Sigma^3 + (1-\alpha_m) O_\Pi^3, m = 1, \cdots, u \quad (14)$$

Fuzzy recognition layer: the output is an ambiguity or probability information which relative to each recognition category. Based on different type data, the last recognition judge can be finished from the ambiguity information by different method. The output is shown as:

$$\begin{aligned} y_i &= \sum_{m=1}^{u} O_m^4 w_{mi} + \theta_i \\ &= \sum_{m=1}^{u} (\alpha_m \prod_{j=1}^{u} w_{1j} O_j^2 + (1-\alpha_m) \prod_{j=1}^{u} w_{2j} O_j^2) w_{mi} \\ &\quad + \theta_i \end{aligned} \quad (15)$$

3.2. Improved Learning Algorithm

As described in the improved network structure, four kinds of parameters should be trained, that are the mean c_{aj} and variance δ_{aj}^2 of membership function, the weight w_{kj} in reasoning layer, the weight w_{mi} in fuzzy recognition layer. The improved learning algorithm is shown as follows:

$$\begin{aligned} c_{aj}(n+1) - c_{aj}(n) &= -\eta(n) \Delta c_{aj}(n) \\ \eta(n) &= (4/3)^{\lambda} \eta(n-1) \\ \lambda &= (sign[\Delta c_{aj}(n) \cdot \Delta c_{aj}(n-1)] - 1)/2 \end{aligned} \quad (16)$$

$$\begin{aligned} \delta_{aj}(n+1) - \delta_{aj}(n) &= -\eta(n) \Delta \delta_{aj}(n) \\ \eta(n) &= (4/3)^{\lambda} \eta(n-1) \\ \lambda &= (sign[\Delta \delta_{aj}(n) \cdot \Delta \delta_{aj}(n-1)] - 1)/2 \end{aligned} \quad (17)$$

$$\begin{aligned} w_{kj}(n+1) - w_{kj}(n) &= -\eta(n) \Delta w_{kj}(n) \\ \eta(n) &= (4/3)^{\lambda} \eta(n-1) \\ \lambda &= (sign[\Delta w_{kj}(n) \cdot \Delta w_{kj}(n-1)] - 1)/2 \end{aligned} \quad (18)$$

$$\begin{aligned} w_{mi}(n+1) - w_{mi}(n) &= -\eta(n) \Delta w_{mi}(n) \\ \eta(n) &= (4/3)^{\lambda} \eta(n-1) \\ \lambda &= (sign[\Delta w_{im}(n) \cdot \Delta w_{im}(n-1)] - 1)/2 \end{aligned} \quad (19)$$

It is shown that the learning rate will change by the change of positive and negative of dynamic variable from expressions (16)-(19). With an opposite change of gradient, the learning rate will decrease by 25 %, this can effectively control the concussion produced by training process.

The concrete updating process of parameter δ_{aj}^2 is shown as follows:

$$\delta_{aj}(n+1) - \delta_{aj}(n) = -\eta(n) \cdot \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial \delta_{aj}(n)} \quad (20)$$

It is shown that the updating process needs to solve the partial derivative of variable in denominator from expressions (20). Owing to this process, the computational complexity would be increased and the learning efficiency would be decreased. Aiming at the problem, this paper regards the reciprocal of δ_{aj} as an independent variable and uses the reciprocal to finish training process. The reciprocal is denoted by b_{aj}, and the membership function becomes as follows:

$$f_{aj} = \exp(-(x_a - c_{aj})^2 b_{aj}^2) \quad (21)$$

Then the updating process of parameter c_{aj} and b_{aj} become as follows:

$$c_{aj}(n+1) - c_{aj}(n) = -\eta(n) \cdot \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial c_{aj}(n)} \quad (22)$$

$$b_{aj}(n+1) - b_{aj}(n) = -\eta(n) \cdot \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial b_{aj}(n)} \quad (23)$$

As the above analysis, in expression (20), δ_{aj} has a third power and a concussion will be induced when δ_{aj} become smaller in training process. However, in expression (23), parameter b_{aj} does not appear in the reciprocal and just has a third power. Thus, the computational complexity will be decreased and a concussion will be controlled effectively when δ_{aj} become smaller.

4. Simulation Results

The MSTAR (three types of ground vehicles in the moving and stationary target acquisition and recognition) data is a standard dataset in the SAR ATR community, allowing researchers to fairly test and compare their ATR algorithms. A variety of feature extraction algorithms have been used in SAR target recognition. This paper takes the Hu moment invariants as the SAR target feature [16]. The geometry moment is written as

$$m_{pq} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^p y^q f(x, y) dx dy, \quad (24)$$

where $f(x,y)$ are two dimensionality random functions. Its center moment is

$$\mu_{pq} = \sum_x \sum_y (x-\bar{x})^p (y-\bar{y})^q f(x,y), \quad (25)$$

where $\bar{x} = \frac{m_{10}}{m_{00}}, \bar{y} = \frac{m_{01}}{m_{00}}$.

Then, it is feasible to get the first four moment, they are shown as follows:

$$\begin{aligned}
&\mu_{00} = m_{00} = \mu \\
&\mu_{01} = \mu_{10} = 0 \\
&\mu_{20} = m_{20} - \mu\bar{x}^2 \\
&\mu_{11} = m_{11} - \mu\bar{x}\bar{y} \\
&\mu_{02} = m_{02} - \mu\bar{y}^2 \\
&\mu_{12} = m_{12} - m_{02}\bar{x} - 2m_{11}\bar{y} + 2\mu\bar{x}\bar{y}^2 \\
&\mu_{03} = m_{12} - 3m_{02}\bar{y} + 2\mu\bar{y}^3 \\
&\mu_{21} = m_{21} - m_{02}\bar{y} + 2m_{11}\bar{x} + 2\mu\bar{x}^2\bar{y} \\
&\mu_{30} = m_{30} - 3m_{20}\bar{x} + 2\mu\bar{x}^3
\end{aligned} \quad (26)$$

The normalized central moment is

$$\eta_{pq} = \mu_{pq} / \mu_{00}^r \quad (27)$$

where $r = (p+q+2)/2$, $p+q = 2,3,\cdots\cdots$.

The Hu moment invariants can be expressed as follows:

$$\phi_1 = \eta_{20} + \eta_{02} \quad (28)$$

$$\phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2 \quad (29)$$

$$\phi_3 = (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2 \quad (30)$$

$$\phi_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2 \quad (31)$$

$$\begin{aligned}
\phi_5 = {} & (\eta_{30} + \eta_{12})(\eta_{30} - 3\eta_{12}) \,|\, (\eta_{30} + \eta_{12})^2 \\
& -3(\eta_{21} + \eta_{03})^2 \,| + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03}) \\
& |\, 3(\eta_{30} + \eta_{12})^2 - (\eta_{21} - \eta_{03})^2 \,|
\end{aligned} \quad (32)$$

$$\begin{aligned}
\phi_6 = {} & (\eta_{20} + \eta_{02}) \,|\, (\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2 \,| \\
& + 4\eta_{11} \,|\, (\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03}) \,|
\end{aligned} \quad (33)$$

$$\begin{aligned}
\phi_7 = {} & (\eta_{30} + \eta_{12})(3\eta_{12} - \eta_{30}) \,|\, (\eta_{30} + \eta_{12})^2 \\
& -3(\eta_{21} + \eta_{03})^2 \,| + (3\eta_{12} - \eta_{30})(\eta_{21} + \eta_{03}) \\
& |\, 3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2 \,|
\end{aligned} \quad (34)$$

The four moment invariants which are obtained from the affine transformation are expressed as [17]

$$I_1 = \frac{\mu_{20}\mu_{02} - \mu_{11}^2}{\mu_{00}^4} \quad (35)$$

$$\begin{aligned}
I_2 = {} & (\mu_{03}^2\mu_{30}^2 - 6\mu_{30}\mu_{21}\mu_{12}\mu_{03} + 4\mu_{30}\mu_{12}^3 \\
& +4\mu_{21}^3\mu_{03} - 3\mu_{21}^2\mu_{12}^2) / \mu_{00}^{10}
\end{aligned} \quad (36)$$

$$\begin{aligned}
I_3 = {} & (\mu_{20}(\mu_{21}\mu_{03} - \mu_{12}^2) - \mu_{11}(\mu_{30}\mu_{03} - \mu_{21}\mu_{12}) \\
& +\mu_{02}(\mu_{30}\mu_{12} - \mu_{21}^2)) / \mu_{00}^7
\end{aligned} \quad (37)$$

$$\begin{aligned}
I_4 = {} & (\mu_{20}^3\mu_{03}^2 - 6\mu_{20}^2\mu_{11}\mu_{12}\mu_{03} - 6\mu_{20}^2\mu_{02}\mu_{21}\mu_{03} \\
& +9\mu_{20}^2\mu_{02}\mu_{12}^2 + 12\mu_{20}\mu_{11}^2\mu_{21}\mu_{03} + 6\mu_{20}\mu_{11}\mu_{30}\mu_{03} \\
& -18\mu_{20}\mu_{11}\mu_{02}\mu_{21}\mu_{12} - 8\mu_{11}^3\mu_{03}\mu_{30} - 6\mu_{20}\mu_{02}^2\mu_{30}\mu_{12} \\
& +9\mu_{20}\mu_{02}^2\mu_{21}^2 + 12\mu_{11}^2\mu_{02}\mu_{30}\mu_{12} - 6\mu_{11}\mu_{02}^2\mu_{21}\mu_{03} \\
& +\mu_{02}^2\mu_{03}^2) / \mu_{00}^{11}
\end{aligned} \quad (38)$$

This paper takes the moment invariants as the feature to SAR target recognition, which is

$$X = [\phi_1 \ \phi_2 \ \phi_3 \ \phi_4 \ \phi_5 \ \phi_6 \ \phi_7 \ I_1 \ I_2 \ I_3 \ I_4 \ \mu \ Var]',$$

where μ, Var is the mean and variance of the two dimensionality random functions.

This paper builds a simulation recognition system for MSTAR image, its structure is shown as Fig. 3.

This paper uses 1286 images for simulation, these images are from 15° and 17° aspects of three targets (T72, BMPZ, BTR7O). Simulation system random chooses 1000 images as the training data and other 286 images as test data. The simulation has taken iteration training in 100 times by using the learning algorithm which is detailed in part 3. The iteration learning curve is shown as Fig. 4, which shows the iterative result of normal learning and improved learning algorithm.

It can be found that the improved learning algorithm can contribute less error and more stable in iterative procedure. And the recognition result is shown as Fig. 5.

Table 1 shows the difference recognition results of some illegibility SAR images between conventional and improved FNN.

For the table, it is shown that the correct recognition rate has been increased in the improved FNN. The FNN recognition can contribute 84.7 % for the MSTAR and the improved FNN recognition can get 93.4 %.

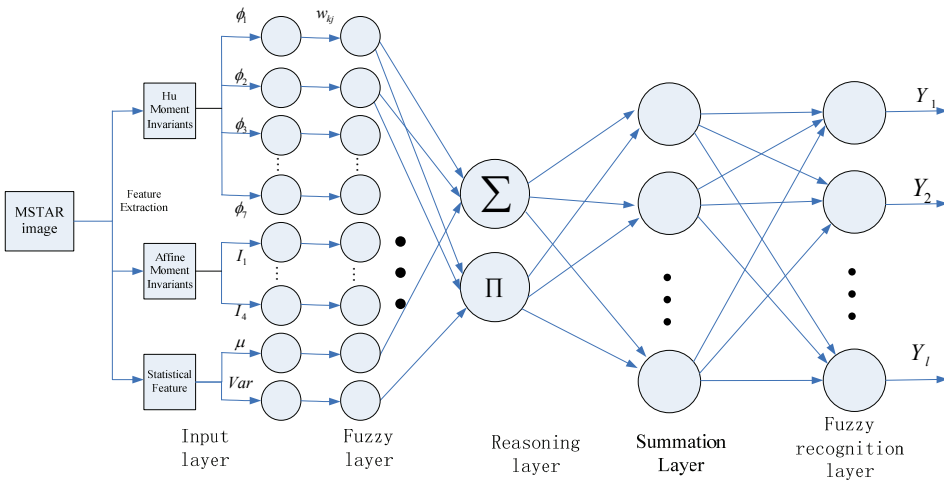
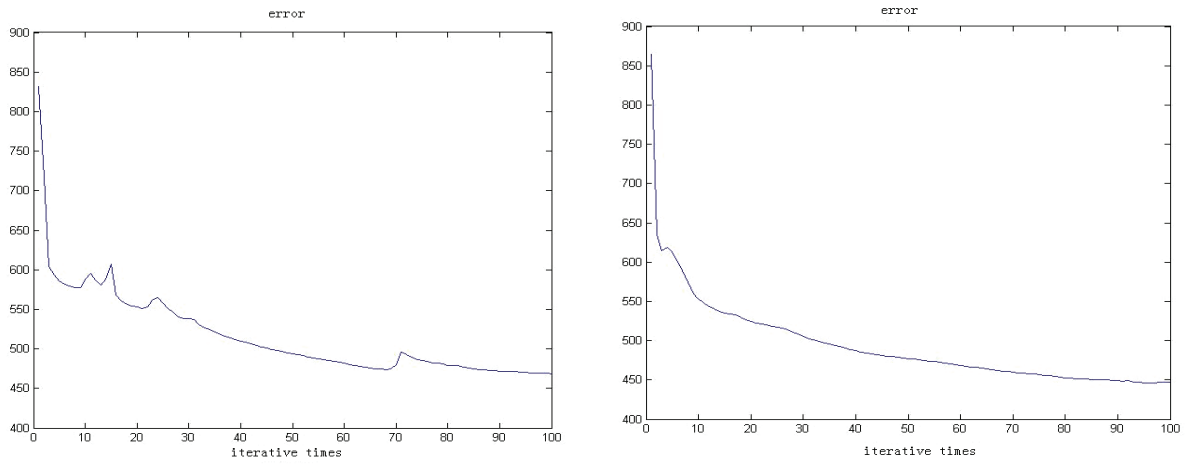


Fig. 3. MSTAR image recognition system structure.



(a) (b)

Fig. 4. Error curve, (a) error curve of conventional learning algorithm, (b) error curve of improved learning algorithm.

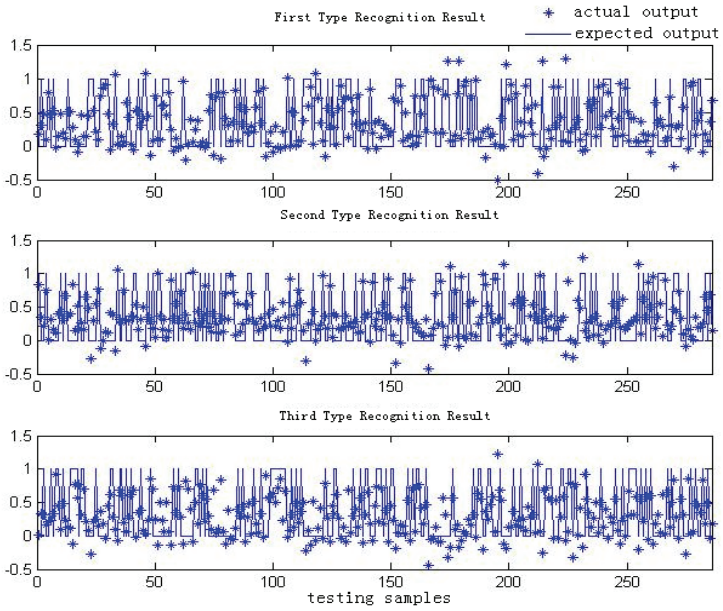


Fig. 5. Actual output and expected output of testing sample.

Table 1. Recognition results of some illegibility SAR images.

	Conventional FNN output	Improved FNN			Actual output
		Probability of 1st type	Probability of 2nd type	Probability of 3rd type	
1	[0 0 1]	0.528038	-0.01213	0.476288	[1 0 0]
2	[0 1 0]	-0.10639	0.545774	0.547673	[0 0 1]
3	[1 0 0]	0.511021	0.53006	-0.04534	[0 1 0]
4	[0 0 1]	0.47564	0.005308	0.525628	[1 0 0]
5	[0 1 0]	0.035892	0.426995	0.535446	[0 0 1]
6	[0 0 1]	0.430008	0.182494	0.385733	[1 0 0]
7	[1 0 0]	0.401752	0.151182	0.460472	[0 0 1]
8	[0 1 0]	0.365494	0.339978	0.292665	[1 0 0]

5. Conclusions

This paper proposes an improved fuzzy neural network for SAR image recognition, and the proposed network includes an improved structure and learning algorithm. At the last layer of improved structure, the output is fuzzy information which contains the probability on each type. The improved learning algorithm modifies the learning rate and parameters. This paper uses the MSTAR SAR images data for simulation, and the results verify that the improved FNN recognition can contribute higher recognition rate than conventional FNN, especially in illegibility images data.

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The Research of PPM Modulation in Space Laser Communication Transmitting System

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Abstract: This paper uses the method of pulse MOPA fiber laser with the PPM modulation to achieve. This paper first describes the characteristics of space laser communication requirements and technical issues that need to be addressed, And then, according to the requirement analysis the MOPA transmitter program, and prove its accord the high power and the high rate what the space optical communication requirements. Through the analysis for PPM modulation mode, the MOPA type fiber laser based on the PPM modulation system is designed. Through the simulation verify its can reach 1 Mbps transmission rate and 5W average power. *Copyright © 2013 IFSA.*

Keywords: Laser communications, PPM, Fiber laser, Modulation, MOPA.

1. Introduction

In space Laser communication, we generally hope that the light source we use can meet the characteristics of high operating frequency, good output beam quality and narrow linewidth [1]. High-power laser light source is a key factor for space laser communication. The high-power fiber laser is a new light source, the fiber laser have a great advantage in beam quality, optical transmission characteristics, reliability and the size and also other areas in the same optical output power. High rate modulation technique makes very important impact on the performance of the laser communication system, it is one of the key technologies of optical communication [2]. Selecting the appropriate modulation and demodulation method depends on the channel characteristics of the system, so that the signal and the channel characteristic can match each other to make the way of the space laser link achieve communication more effectively and more reliably. Therefore, researchers on the modulation technique of fiber laser which can adapt to space optical communication transmission characteristics will be of great significance for the space laser communication's pre-research to application [3].

The fiber laser is characterized by large power to comply with the conditions of space optical communication, but it is difficult to achieve a high rate because of low repetition frequency [4]. PPM can achieve a high data transfer rate with a small light power in a given laser pulse repetition frequency, so it compensates for the disadvantage of the fiber laser. And the advantages of higher power utilization, high transmission efficiency and anti-interference ability are widely used in optical communication systems. So technology combined with a fiber laser and PPM modulation will become the hot in space optical communication.

2. Selection and Analysis of a Fiber Laser

For space laser communication tasks, the generation of power is very difficult and expensive, it requires the laser emission source with a higher transmit power and emission efficiency. Laser emission needs to consider peak-average power ratio, up to the peak power, the electro-optical conversion efficiency and total power consumption. Compromise considerations between various existing laser techniques to make a reasonable design choice [5], average power is sufficient, or the high-performance pulse lasers which are from the low power oscillator amplified to the required power level can satisfy space application requirements.

Due to the modulation option in this paper is the PPM modulation, first of all the type of fiber laser should be selected pulsed fiber lasers .In addition, the PPM modulation techniques must increase the laser pulse repetition frequency in order to obtain high code rate. After calculation, we require the laser peak power of 1000 W, repetition frequency of 200 kHz and the pulse width of 15 ns in order to achieve the transmission rate of 1 Mbps. For this, we propose the master oscillator power amplifier (MOPA) [6]. Seed source can guarantee that the output characteristics of the pulse of the laser beam for pulse fiber laser-based on MOPA technology. The optical fiber power amplifier can ensure that the output laser power and energy characteristics thereby obtaining excellent laser pulse characteristics and can greatly improve the output energy of the laser (power).

2.1. Selection of Seed Source

Seed source determines output laser pulse characteristics of pulse fiber laser [7]. This paper analyzes the characteristics of the fiber-optic laser modulation, then we must first start from the seed source of fiber lasers to analyze the output characteristics of the fiber laser, so the selection of seed source was particularly important in this paper.

Fiber acousto-optic Q-switched laser has output wavelength of 1070 nm nearby, just easy to zoom in doped Yb quartz fiber gain bandwidth [8]. And because of the direct fiber output we can achieve all-fiber structure by directly weld with the amplification system. However, due to the characteristics of the fiber itself, its cavity length is longer, so it is difficult to obtain a narrow pulse width output. At the same time due to the self- clamping effect, it is difficult to obtain a smooth pulse waveform output. Based on the above considerations, a small laser diode (LD) pumping shipped acousto-optic Q-switched *Nd:YAG* laser [9] (Fig. 1) as the source of the seed of the MOPA system.

As shown in Fig. 1, the laser source consists of the pump source, the coupled system, the laser crystal, the resonant cavity and the drive source. The laser source uses semiconductor pump technology, end-pump technology and acousto-optic Q-switched technology, it also specially designs laser crystal resonator and the driving source in order to obtain high efficiency, high power and high beam quality. where the pump source is selected semiconductor laser (LD) of 808 nm, so that the life of the laser source is up to 10,000 hours or more; Since the LD of 808 nm emission spectra fall exactly on the absorption peak vicinity of *Nd:YAG* crystal, the laser crystal selected *Nd:YAG*, by adjusting the working temperature of the LD it can make their emission wavelength be accurately adjusted to the absorption peak of the *Nd:YAG* crystal, thus greatly increasing the utilization of the pumping light; Seed source uses acousto-optic Q-switched intracavity modulation and take measures to compression chamber, so that the pulse width of signal light is maintained at less than 20 ns; Due to the high conversion efficiency of the laser source, the thermal effects of *Nd:YAG* crystal are reduced, combined with end-pump technology, and laser collimator technology makes laser source to achieve proximal film output, beam divergence angle compression to less than 1.5 mrad and the light emitting surface less than 2 mm.

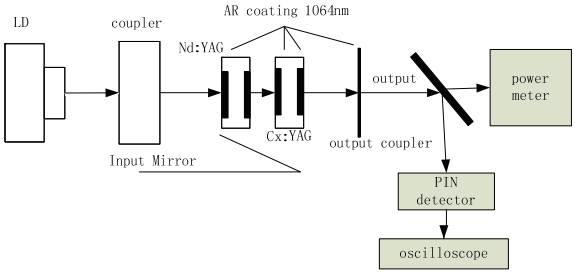

Fig. 1. Schematic diagram of experimental system of passively Q-switched laser.

In the operational state of the Q-switched, the stable Q-switched pulse output can be obtained when the modulation frequency is adjusted between 200 kHz - 600 kHz. As fixed modulation frequency is 1 kHz, the relations of the pulse width of output laser and the average power changing with the pump power and the repetition rate are shown as Fig. 2.

As can be seen from Fig. 2, pulse peak power gradually increases with the increase of the pump power, eventually stabilized. This is due to the gradual increase in the pumping rate, the level accumulated inversion particle of Yb3+ gradually increased until saturation, resulting in the above-mentioned changes of the pulse peak power. Changes in the pulse width is gradually reduced and stabilized.

2.2. Design of MOPA System

MOPA optical fiber laser is a laser emitting system on the basis of the seed source of a pulse adding one or several optical fiber laser amplifier systems, so the seed source can be obtained the laser output with higher energy of the single pulse, high

average output power and high beam quality through several stages of amplification. Laser amplifier requires working material with sufficient population inversion to ensure the gain is sufficient to overcome the loss of the amplifier inside when the signal passes. According to the requirements of the system parameters, the fiber lasers based on MOPA architecture is shown in Fig. 3.

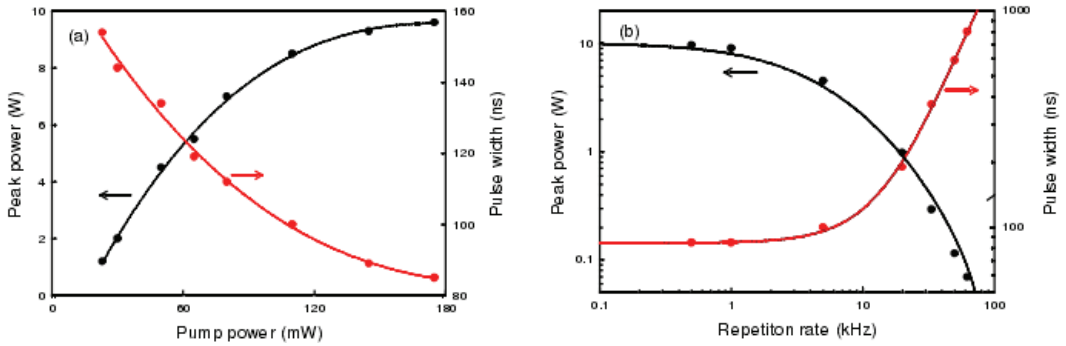

Fig. 2. Peak power and pulse width as a function of pump power and repetition rate.

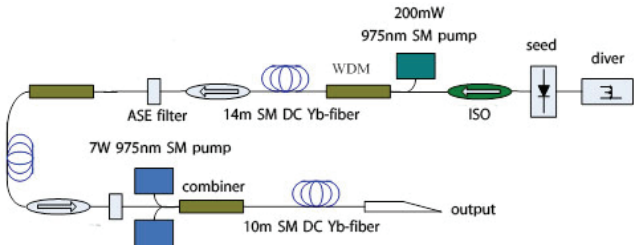

Fig. 3. Schematic of pulse fiber laser.

As shown in Fig. 3, the seed laser uses the way of acousto-optic Q-switched so that there is a large enough output power. So this paper uses two-stage amplification structure as in literature. The output signal light is coupled into the gain fiber by the optical coupler system after the optical isolator. The first level is the preamplifier system, the amplifier consists of a LD laser whose center wavelength is 975 nm as its pumping source, its output power is 30 W. Second level uses two LD lasers whose output power is 7 W, and the center wavelength is 975 nm as its pumping source. We obtained the second stage amplification of the pulse output using a Single mode doped double-clad chest fiber of length 16 m, and weld directly with the seed laser and gain fiber splicing through fiber coupler. Amplification stages used in the system adopt the structure of the backward-pumped, namely the signal light and the pumping light is coupled into the fiber respectively from both ends of the double clad fiber, this structure is conducive to obtain higher magnification and output power for the optical fiber amplifier.

The output power changing with the pump power of fiber laser primary amplification based on the MOPA architecture when it realizes the stable clamping is shown in Fig. 4:

The output power changing with the pump power of fiber laser secondary amplification based on the MOPA architecture when it realizes the stable clamping is shown in Fig. 5.

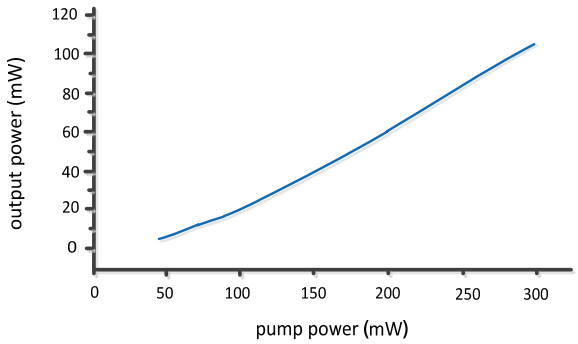

Fig. 4. The output power changing with the pump power in fiber laser primary amplification.

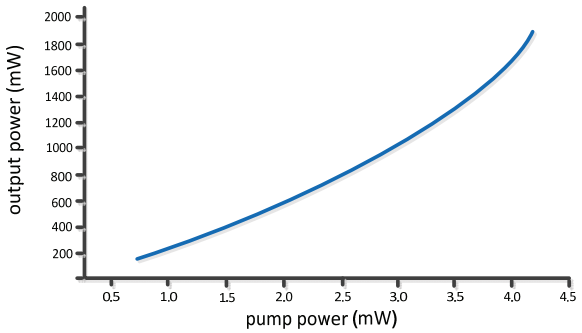

Fig. 5. The output power changing with the pump power in fiber laser secondary amplification.

The comparison can be seen by the previous figure, the pulse width of the output pulse is broaden and the smoothness of the curve is also declined after a primary and secondary amplification.

3. The PPM Modulation Analysis

Normally, we call the process of loading transfer information to the laser radiation as laser modulation, the device to complete the process is called a laser modulator, the laser is called carrier for playing the role of the "carry" information. Information passed is usually referred to as a modulated signal. Modulated laser is called a modulated wave or the modulated light. The modulation characteristic is defined as the relationship of the output signal of the laser with the changes in the modulated signal. This paper uses the pulse signal modulate laser. Laser output signal is observed by adjusting the various pulse repetition frequencies so that the laser works stably in the working point of the highest modulation efficiency.

3.1. Basic Principles of PPM

PPM [10] is a form of signal modulation in which M message bits are encoded by transmitting a single pulse in one of 2^M possible time-shifts. This is repeated every T seconds, so the information bits have been transmitted by an L-PPM modulation signal is $\log_2 M$. If the n-bits data could be written as $M=(m_1, m_2, ...m_n)$, then the mapping coded relation of the PPM modulation could be written as: $\phi: l = m_1 + 2m_2 + 2^{n-1}$, $m_n \in \{0,1\cdots n-1\}$, For a 4-PPM modulation, if M=(0,0), then L=0; if M =(1,0), then L=1; if M =(0,1), then L=2; if M =(1,1), then L=3; Then the slot positions of 0, 1, 2, 3 correspond as shown in Fig. 6.

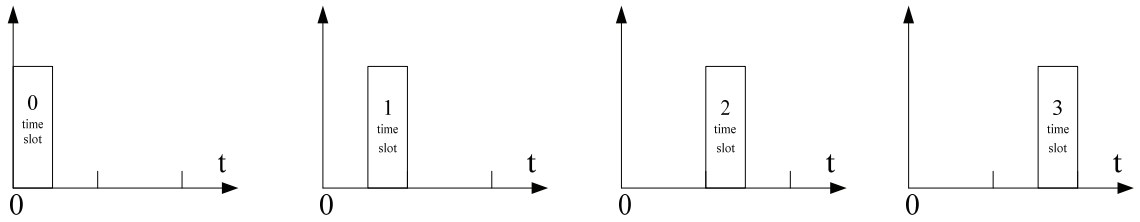

Fig. 6. L-PPM pulse position modulation diagram.

If a symbol rate of the digital baseband signal is $R_b=1/T_b$, the ideal low communication channel bandwidth it requires is $B=R_b/2$. The symbol rate of OOK signal and L-PPM signal respectively is $R_b=1/T_b$ and $R'_b=1/T'_b$. If messenger rate is same, $T'_b.L=T_b \log_2 L$, $B/B'=\log_2 L/L$ ($L=2^n$, n is integer.) L-PPM modulation has high requirements for the bandwidth.

Units pass the letter rate is an important parameter, namely the transmission bits number of per second per hertz γ.

$$\gamma = R/B\left(bit \cdot s^{-1} \cdot Hz^{-1}\right) \tag{1}$$

where R is the transmission rate, B is the signal bandwidth, and the units respectively are $bit \cdot s^{-1}$ and Hz. If the optical communication systems use a pulsed laser, the pulse duration is τ, the signal bandwidth is approximated as:

$$B(Hz) = \frac{1}{\tau}(s) \tag{2}$$

If the duty cycle of the OOK modulation symbols is τ_p, its unit messenger rate is:

$$\gamma_{OOK} = \left(1/T\right)/\left(1/\tau\right) = \tau_p \tag{3}$$

Signaling the number of slots in a single pulse position modulation is $L=2^n$,

Duty cycle is the same with OOK modulation, the width of the 2^n slots is $T_{PPM} = 2^n \cdot \tau / \tau_p$ .So units passing letter rate is:

$$\gamma_{PPM} = n \cdot \tau_p / 2^n \tag{4}$$

From the above analysis, in the same passing letter rate, the bandwidth occupied by the L-PPM modulation is increased and the required transmission rate is also increased compared with OOK modulation [11].

3.2. PPM Modulation Error Analysis

People can know that we usually describe the anti-interference ability of channel coding by using Hamming distance from error-correction coding theory. But when the signal has been modulated, BER of the systems depends on the Euclidean distance of signal sequence, the role of coding is to increase the Euclidean distance in signal sequence, thus to improve bit-error performance of the systems. If choosing the PPM modulation for the optical communication systems, no matter what the

magnitude of the Hamming distance between the coded sequences is, in a PPM symbol, the Euclidean distance is a constant, and they cannot correspondent to each other, the equivalence between Euclidean distance and Hamming distance makes the impairment of the coding gain.

The essential of a PPM is a kind of partition modulation, the receiver needs to judge the positions of the high level in 2^M slots, so there may be the following as: When the PPM symbol errors occur, the error numbers of corresponding to the coding information will appear in different situations, it may err in an unitary bit, or continuous with *M*-bits errors, which causes partial "error diffusion" in the *M*-bits messages. This is one of the reasons why the coding gain would be impaired.

3.3. The Options Program of PPM with Fiber Laser

The paper uses high-speed FPGA design the PPM modulation system to driven fiber Q-switched laser, the fiber laser power is amplified using MOPA structure and PPM modulated signal is loaded on top of the fiber laser by external modulation to achieve PPM modulation technology of high-speed and high-bit-rate. The program structure is shown in Fig. 7.

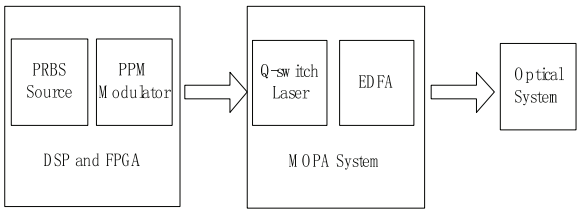

Fig. 7. Overall program of modulation system.

The modulation system is composed of three parts, including FPGA modulation driver, MOPA and an optical system. FPGA generates the PRBS sequence signal and the PRBS is mapped into the PPM code sequence. The PPM code sequence is used to drive the Q-switched lasers. Q-switched laser has drawbacks of low repetition frequency and small output power, so we choose the PPM modulation technology and an optical amplifier of MOPA architecture to solve the above drawbacks.

Through modulating a transmitting signal in the laser communication system, an easier transmitting signal form is converted to improve signal noise immunity and transmission efficiency, thereby increasing the SNR and eventually achieving the increase in the communication distance. To achieve this modulation system, the key lies in how the system convert binary signal into a PPM pulse signal [12]. Fig. 8 is a block diagram of the modulation system.

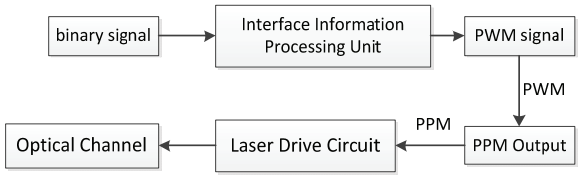

Fig. 8. The block diagram of PPM modulation system.

If binary information received directly is converted into the PPM pulse, first of all we must make a serial-to-parallel conversion for binary data, then a series designed of counter and comparator is made to achieve the PPM signal. Circuit logic is very complex. Firstly we do a transformation for binary information, the binary information first coded as PWM [13] (Pulse Width Modulation). Then using the correspondence relationship between the waveform of PPM and PWM, the PWM wave is converted into PPM wave through hardware trigger circuit, so that the PPM modulation can be relatively easy to achieve. The diagram of PWM to PPM wave transition is shown in Fig. 9.

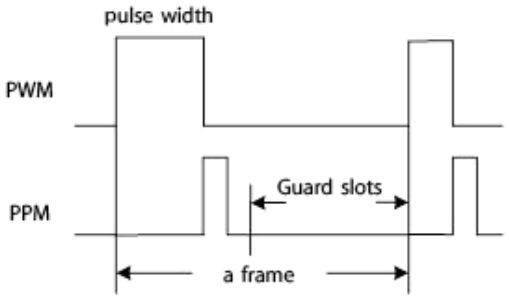

Fig. 9. Diagram of PWM to PPM wave transition.

The pulse width modulated signal (PWM) is a series of pulse whose width is changing. There is only one pulse at each frame period. The frame period is fixed. In time period composed by a series of time slots, i.e. in a fixed frame period consists only one slot position on the signal and only one of each frame period. L-PPM signal transmits data by pulses at different position. Both have in common is that the pulses are in the cycle of fixed length, i.e. within one frame period has one and only one pulse. The different points between the two are that the PPM transmits different data by the position of the pulse signal in a frame period, and the PWM wave transmits different data by pulse width changing. So that the system designs of the PPM modulation is converted to the design of the PWM modulation.

Two key problems exists when using FPGA generates the PWM wave, they are the cycle and pulse width of the PWM wave. The data sent by the host computer and hexadecimal numbers PPM modulated will affect the cycle and pulse width of the PWM wave .Disposed PPM modulation hexadecimal number is L, and the n-bit data to be transmitted is

M=(m_1, m_2, ...m_n). A PPM frame period is divided into information segment and protective segment, the information segment consists of L time slots, a slot width is denoted by τ, the guard period is denoted by T_D, the frame cycle of the system can be obtained as follows:

$$T_{PPM} = L * \tau + T_D \tag{5}$$

The frame rate of the system is:

$$f_R = \left(L * \tau + T_D\right)^{-1} \tag{6}$$

The messaging rate of the system is:

$$R = \left(\log_2 L\right) \Big/ \left(L * \tau + T_D\right) \tag{7}$$

The slot position corresponding to each group of sending data is denoted by L, the data and the position L mapping coding relationship is:

$$L = m_1 + 2 \cdot m_2 + ... + 2^{n-1} \cdot m_n \tag{8}$$

The frame cycle of the PWM and PPM are the same, as shown in Fig. 9 that is T_{PWM}=T_{PPM}, T_{PPM} can be obtained by formula (9).

$$T_{PPM} = T_{PWM} = 2 \cdot TPER \cdot T_{OSC} \tag{9}$$

TPER is the value of the timer period register. T_{OSC} is crystal cycle, PWM pulse width can be obtained by formula (10):

$$T_{ON} = 2 \cdot \left(TPER - CMPR\right) \tag{10}$$

CMPR is the value of register which needs to be calculated by the encoded data, and then be set by the software. Once T_{PPM} is known, *TPER* can be launched by the formula 9 then it can also be set by software. T_{ON} can be launched by the formula 10 with CMPR and TPER. Cycle and pulse width of the PWM wave are resolved, then PWM wave can be produced by designation.

3.4. PPM System Model

This paper designs 32-ary PPM modulation system, i.e. L=32. There is one pulse at 32 signaling time slots, where the laser repetition frequency is 200 kHz, one frame period is set as 10, then the protection time slot is T_D=5 μs, the clock frequency is set as 80 MHz, T_{OSC} =12.5 ns. As shown in Fig. 10, one frame period is divided into two parts, they are respectively 32 protection time slots and 32 communication slots, each time slot is T_s=5 μs /32=15.6 ns. So the design parameters of PPM modulation system are [14]:

1) Time slot frequency is f_s=64 MHz, frame frequency is 100 kHz;
2) Frame period is $T_f = \left(32 + 32\right) \times T_S = 10\mu s$;
3) A light pulses sends 5 bit data information in a cycle, the data are encoded in correspondence with a certain position of 32 communication slots one-to-one;
4) From the above analysis, a frame transmits 5 bits data, PPM modulation rate of the system can be drawn according to the formula (7):

$$R = \log_2 L \Big/ T_f = 5bit \Big/ 5.096\mu s = 1\,Mbit \Big/ s$$

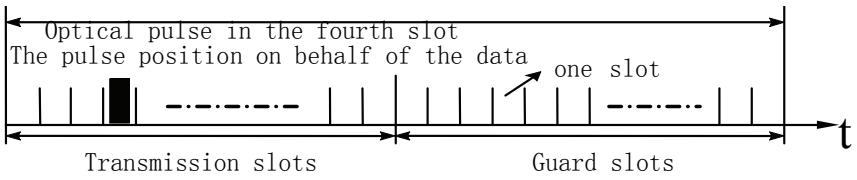

Fig. 10. The frame structure of the 32 - PPM modulation.

PPM modulation scheme was first proposed by the JRP tierce applied to space communications. Using this modulation requires a minimum average optical power in a given laser repetition frequency, and the maximum data transfer rate. In theory, the data transfer rate can reach thousands of bits per second. PPM modulation uses the intermittent cyclical light pulses as information carrier, the binary information of source control pulse position. The principle of PPM is that divided a period of time into M equal portions, each part is called a time slot, at one time slot sends a pulse within a frame time. The frame time is a PPM signal, it contains L time slots and a guard time.

With the development of high-speed digital signal processor (DSP) and field programmable gate array (FPGA) [15], it is easy to build modem system based on the DSP processor and FPGA at this stage as to achieve the PPM modulation with high performance of transmitting and receiving by software programming. In the digital transmitter and receiver of the PPM modulation, the received signal is sampled using a fixed clock frequency, then the sampled signal is processed using a DSP and FPGA to achieve signals decision decoding and signal synchronization. It is easier to achieve the PPM optical transmitter and receiver than using pure hardware.

This paper proposes the PPM all-digital modem system solutions according to the characteristics of the PPM modulation system with high-speed processing power of the DSP and flexible FPGA [16]. As shown in Fig. 11, the PPM modulation and demodulation system is divided into three parts, they are DSP control unit, FPGA processing unit and peripheral circuits and optical system. DSP realizes the function of modulation, it encodes the received binary data for PWM to achieve the control of the system, the communication with the host, the signal processing and analog-to-digital conversion; FPGA mainly realizes the function of the demodulation to extract slot synchronization signal and frame synchronizing signal and to decode the PPM signal. The circuit comprises the PWM trigger producing PPM narrow pulse circuit, the laser driver circuit, an optical system and the transmission channel, APD photo detector circuit.

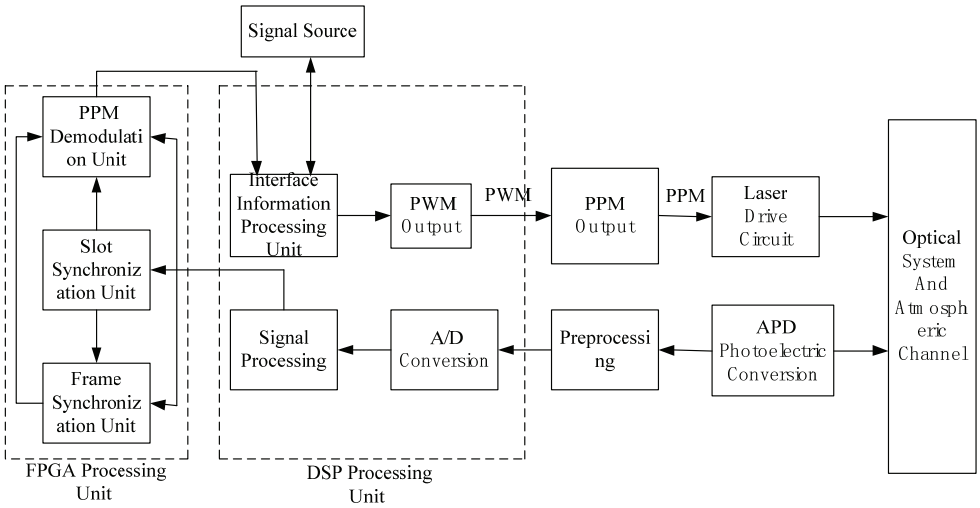

Fig. 11. PPM modulation and demodulation system model.

The focus of this thesis is to achieve the PPM modulation system using DSP and FPGA through detailed analysis of the principle of PPM modulation. According to PPM modulation system in this paper, the transmitted signal is modulated into PPM signals, PPM signal is transmitted into current or voltage by the laser driving circuit, the information is loaded into the light pulses which laser emits by controlling the laser directly. The present PPM modulation system is an internal modulation system, its essence is to convert the received binary signal into a PPM pulse to achieve the modulation of the laser, PPM modulation system principles will be introduced in the back [17].

4. PPM Modulation Simulation

4.1. PPM Modulation Experiments

The experimental equipment includes: a computer, a DSP system board, 5 V DC switching power supply, a DSP emulator and an oscilloscope. Oscilloscope selects Tektronix DPO7104, it has a 1GHz with three bandwidth model, it achieves real-time sampling rate of 10 GS/s in 4 channels to be able to observe transient phenomena. PWM waveform and PPM waveform can be observed in oscilloscope.

Computer transmits binary data to the DSP system board through the serial port. DSP system board convert binary data into the PWM pulse, then the PPM pulses is achieved by the trigger shaping, the three waveforms are displayed by oscilloscope. At last we verify the feasibility of the system and if it can achieve the correct modulation through comparative analysis.

As shown in Fig. 12, computer transmits binary data through the serial port. The waveform of the input random binary information can be displayed by the oscilloscope.

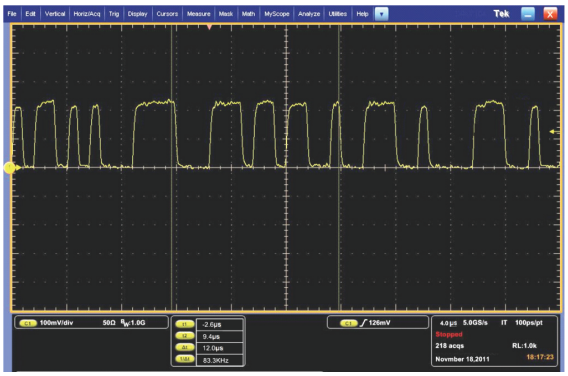

Fig. 12. Input data waveform chart.

PPM modulation is pulse modulation to a corresponding time slot position of some frame on the process which is actually a process of counting the output pulse. PPM modulation software modules achieved by the PWM modulation. PWM modulation module receiver the information and modulate it to the PWM wave to transmission. After a preliminary design the completion of algorithm, simulation of the system software. PWM modulation of the binary data output waveform through DSP processing board is displayed on the oscilloscope, as shown in Fig. 13, the oscilloscope interface is divided into 10-frame cycles, a fixed time period composed by a series of time slots is 4 μs. Pulse is in the cycles of this fixed length. There is only one pulse in each cycle, there is a total of 10 PWM output pulse waveform, and the pulse width is different. PWM wave transmits data by different pulse width, the pulse width corresponds with the binary information one-to-one, and so the PWM modulation in this verification is successful.

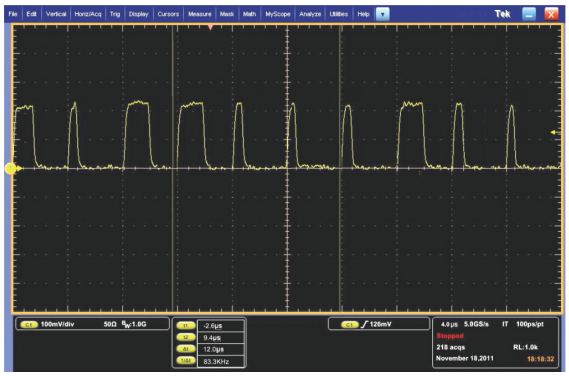

Fig. 13. PWM output waveform chart.

PPM is generated when PWM pulse through trigger shaping circuit at the falling edge of the PWM wave. The oscilloscope interface is divided into five-frame cycles in order to clear display PPM pulse on the oscilloscope and a frame is 4 μs, as shown in Fig. 14. This paper designs the 32-PPM modulation, one frame period is divided into signaling time slots and protection time slots, where the signaling time slots and protection time slots are both 32, one frame period is divided into 64 time slots, the first 32 slots is signaling time slot, the latter 32 slots is the protection time's lot. It can be observed by an oscilloscope, there is only one slot position having the signal in a fixed frame period, and the pulse position is different, the pulse position corresponds with transmitted information, PPM modulation transmits data through pulse in different positions, Fig. 14 shows that the modulation of the binary information is correct.

Binary data is converted into the PPM pulse signal after the PPM modulation, and the pulse position is determined by the transmitted data. Because of 32-PPM modulation system, 5 bit data is transmitted once a time, frame period is 4 μs .Thus modulation rate of 1Mbit/s can be achieved, modulation rate can meet the needs of the space optical communication. The PPM pulse level is TTL level, it becomes PECL level after level conversion which matches with driver chip ADN2841. So single-mode semiconductor laser can be driven, the information is loaded into the pulse of light to be emitted. The average power can reach 5 W after amplification, so that the modulation of the optical communication can be realized.

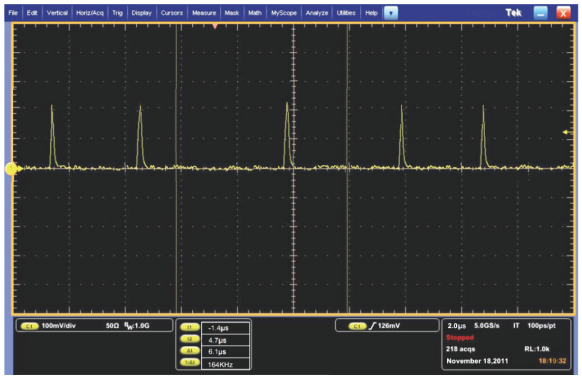

Fig. 14. The PPM modulation output waveform.

4.2. Analysis of Fiber Lasers with PPM Simulation

Simulation of the overall system design is shown in Fig. 15. Since there is not optical fiber laser whose repetition frequency is 200 kHz in this software, so the light source is a combination of the bit sequence generator and the optical pulse generator. The wavelength of the seed light source is 1064 nm, the input current is 25 mA, the peak current is set as 1.5 A, and the peak power is 800 mW, and then from the relationship between the average power and peak power: $P=P_w\tau/T$. The pulse width is 30 ns, T = 4 μs, then P is 6 mW. The light pulse emission wavelength is set as 1064 nm, and the transmit pulse power is 6 mW.

The signal and pump source are coupled into the Yb-doped optical fiber amplifier, the wavelength of pump light is 975 nm, the pump power is 200 mW, the fiber length is 14 m. After the primary amplification, the output power reaches 28 mW, as shown in Fig. 16.

The second stage amplifier uses two-way pump ways, the length of Ytterbium fiber is 10 m, the wavelength of forward pump light is 975 nm, pump power is 200 mW, the wavelength of backward pump light is 915 nm, pump power is 6 W, output power reaches 5.271 W after two zoom, as shown in Fig. 17.

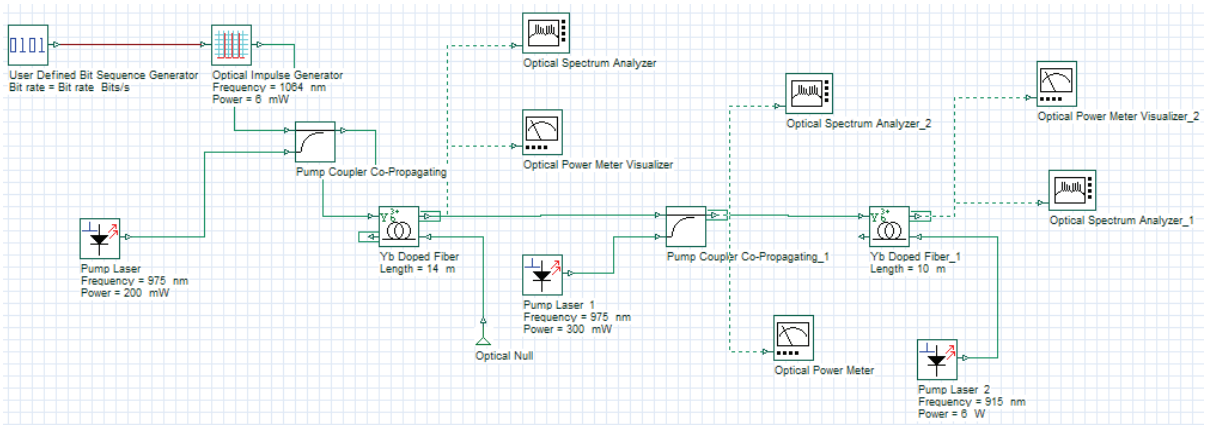

Fig. 15. The overall design of the simulation system.

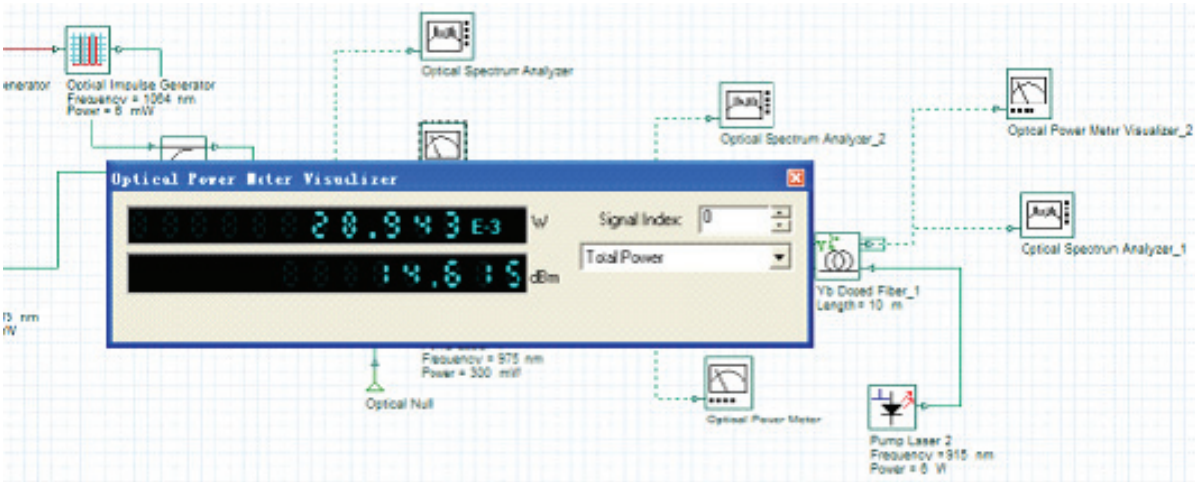

Fig. 16. The output power after the primary amplification.

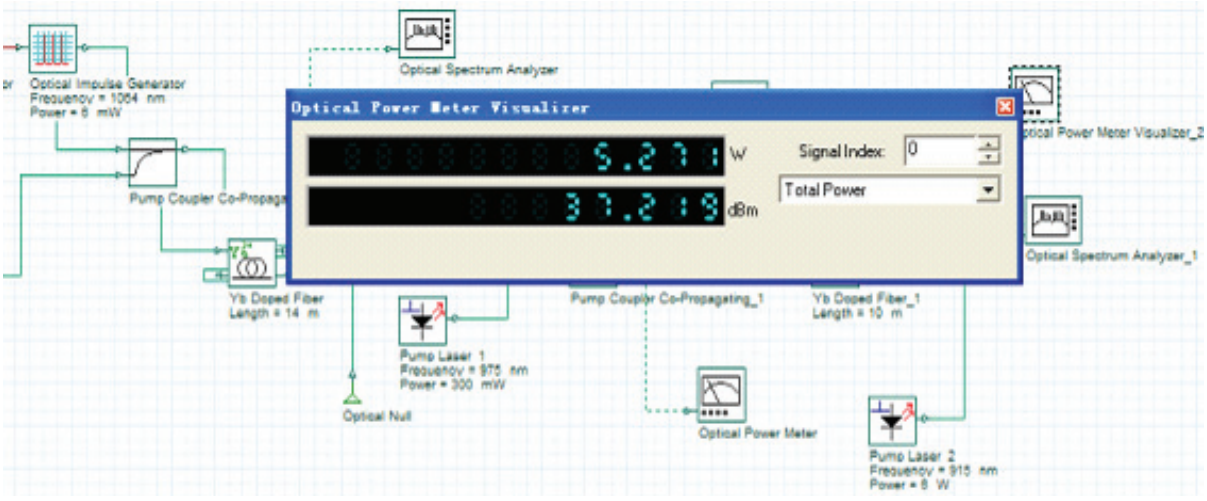

Fig. 17. The output power after the secondary amplification.

5. Conclusions

From this analysis process and experiment, we endeavor to seek some superiority of fiber laser with PPM modulation. And in the course of this exploration, we are able to see some ways in space laser communication.

This paper first describes the characteristics of space laser communication requirements and

technical issues that need to be addressed. And then, according to the requirement analysis the MOPA transmitter program, and prove its accord the high power and the high rate what the space optical communication requirements. Through the analysis for PPM modulation mode, the MOPA type fiber laser based on the PPM modulation system is designed. Through the simulation verify its can reach 1 Mbps transmission rate and 5 W average power.

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Analysis on Temperature Control of Tubular Furnace Based on Fuzzy Control

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Abstract: By analyzing of the tube-type heating furnace features and process, the temperature heating of the furnace was unidirectional, large inertia, heavy lag and changeable, it is hard to build an exact mathematical model confirmed parameters. Applying traditional control theory and method is difficult to achieve perfect control effect. We studied the application of the fuzzy control in tubular furnace temperature control. Then according to the dynamic response process of the second order system, fuzzy control rule table is established. The input and output relationship of fuzzy control rule table is introduced applying of synthesis reasoning rules, so as to calculate the fuzzy output vector corresponding to the deviation and the deviation change rate, and then fuzzy judged by using the center of gravity method, finally the adjustment amount of time interval for push rod is obtained. The feeding time of push-off arm was adjusted to control the heat-up time of each blank in the tubular heater dynamically, and ensures it reach the set temperature when coming out of the tubular furnace. This paper, finally, analyzes the factors that influence the system error and processes them adopting error analysis method, so as to improve the control precision of system. *Copyright © 2013 IFSA.*

Keywords: Fuzzy control, Feeding time, Heat-up time, Error analysis, Control precision.

1. Introductions

A great many metal work pieces was finished by investment casting, machining and post processing, yet investment casting had many shortcomings: blank ingredients and organization structure were not uniform, cutting workability was bad, utilization rate of raw materials was low and so on, now the workability and quality of the products were improved by adopting the bar heating and then precision forging. A key factor influencing the forging bar's quality is the temperature control when heating the bar. At present the heating way we adopted is intermediate frequency induction heating way. Induction heating output electricity power via the tubular furnace made by induction circle and heat up metal bar by electromagnetism induction. Because the temperature heating of the furnace was unidirectional, large inertia, heavy lag and changeable, it is hard to build an exact mathematical model confirmed parameters. Applying traditional control theory and method is difficult to achieve perfect control effect.

This text analyses tubular finance's working character and adopts the fuzzy control algorithm to solve the heating temperature control problem, enhancing system's controllability and reliability.

2. The Analysis of Tubular Intermediate Frequency Induction Furnace

Furnace mainly consists of material pushing system, intermediate frequency heating power supply, induction circle, stove and temperature control system.

Medium frequency is 1500 Hz, the length of work piece base material determined by the standard of each group product. The number of the work pieces in the stove cannot beyond 10, the diameter of the material among 80~90 mm. When out of stove, the surface temperature of material must among the deviation limits set point allowed, it means that system ask for the base material controlled by constant temperature. This system have two different control way. The first is pushing material pushing use constant beat, on the basis of temperature when work pieces out stove to adjust the heat source power. The second is heat source power setting a fixed power output, according work pieces temperature to adjust pushing beat to change work pieces' heating time. In most cases, furnace source power is heavy, adopting the first method source power would adjust frequently, when power is weak, because the conduction angles of the silicon controlled in rectifier bridge is small, harmonic wave is big and easy to affect the steady of electrified wire netting[1-3]. Thus this text chooses the second control way.

The structure of tubular furnace is show in Fig. 1.

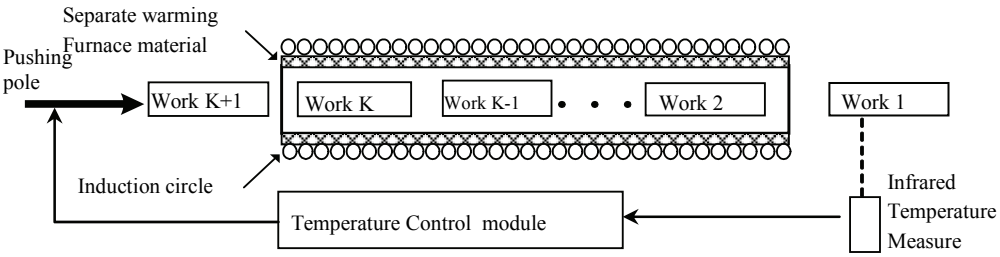


Fig. 1. Tubular furnace structure.

The part of main power source consists of three-phase power source, three-phase controlled rectifier, DC filter, and intermediate frequency inverter and induction furnace and so on. The part of control consists of automatic send\out system, intermediate frequency power source control system and temperature control system. Intermediate frequency power source control system manages the part of rectification and contra variant, and ensures the induction furnace output power keep constant in heating process. Intermediate frequency control system protects the whole induction power source overheating, overflow and phase lack at the same time. Temperature control system adopt infrared measuring temperature sensor to measure each material's outing temperature, then rely on the setting temperature to adjust pushing beat.

Through the above analysis, we get systemic electrical frame diagram as show in Fig. 2.

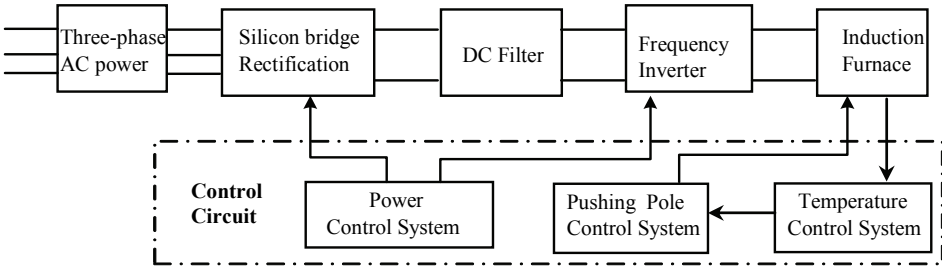


Fig. 2. System hardware frame.

3. The Study Work Pieces Temperature Control Algorithm

3.1. The Analysis of System Fuzzy Control

In consideration of rapid control and easy achievement, this text builds a two dimension fuzzy controller to control the temperature. In fuzzy control, input temperature deviation et and temperature alteration etc, adopt incremental control by changing Δt to adjust pushing beat time jp_t, system's setting temperature and sampling actual temperature are *x*, *y*, *k* means the K^{th} calculated data [4-5].

There be:

$$et(k) = x(k) - y(k) \tag{1}$$

$$etc(k) = et(k) - et(k-1) \tag{2}$$

Based on the fuzzy subset setting rule, set the fuzzy language variable of *et, etc, Δt* respectively be

ET, ETC, ΔT. Theirs change limits respectively define 14, 13, and 13 grade fuzzy domain:

$$X_{et}=\{-6, -5, -4, -3, -2, -1, -0, +0, 1, 2, 3, 4, 5, 6\};$$
$$Y_{etc}=\{-6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\};$$
$$Z_{\Delta T}=\{-6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$$

The language described fuzzy subset of *ET, ETC* and *ΔT* respectively be:

$$ET=\{NB, NM, NS, NO, PO, PS, PM, PB\};$$
$$ETC=\{NB, NM, NS, ZO, PS, PM, PB\};$$
$$\Delta T=\{NB, NM, NS, ZO, PS, PM, PB\}$$

This tubular furnace temperature control error demand is ±5, so temperature deviation basic domain setting is [-5, +5], quantize in fuzzy domain [-6, 6], temperature deviation quantification factor Ket=6/5=1.2. Because warming speed is not fast, setting temperature deviation change rate basic domain is [-2, +2], difference in temperature change rate fuzzy domain is [-6, 6], deviation change quantification factor K_{etc}=6/2=3. The adjustment basic domain output quantity beat time based on systemic fixed power and material's volume, thus here on setting [0, 100].

According to above setting, the transformation of fuzzy control physical quantity as show in Fig. 3.

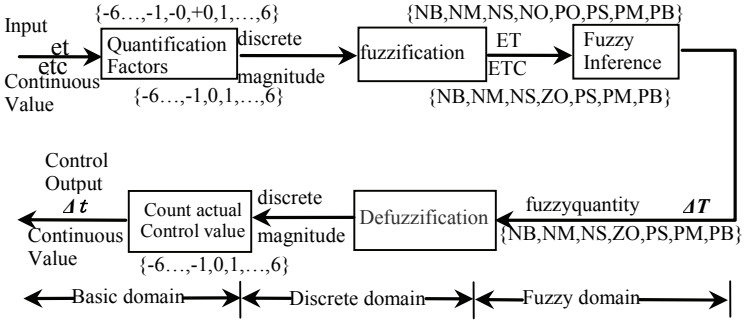


Fig. 3. Transformation figure of vague control physical quantity.

Because fuzzy control character is not sensitive about fuzzy language membership function form, mainly depend on the number of membership function and the fuzzy domain of each function [20]. According to the confirmed principle of membership function of fuzzy function: membership function must meet the demand of convex fuzzy set; in large deviation area we adopt low resolution fuzzy set, in small deviation area we adopt high resolution fuzzy set; the two spacing fuzzy membership set should not try to intersect, making sure one point not be covered more than two in domain, avoiding fuzzy membership set concept contradictory [6]. Thus fuzzy variable *ET, ETC, ΔT* adopt triangular MF, fuzzy variable membership curves as show in Fig. 4.

According to MF curve we can get the membership degree of each fuzzy variable, on the basis of fuzzy set's Zadeh express method, there be:

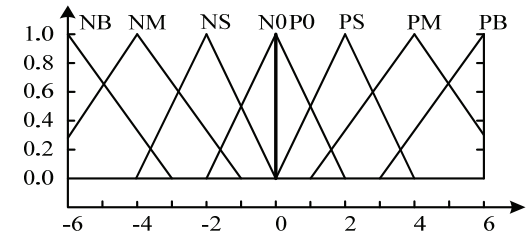
$$NB_{ET} = \frac{1.0}{-6} + \frac{0.7}{-5} + \frac{0.3}{-4}$$

$$NM_{ET} = \frac{0.3}{-6} + \frac{0.7}{-5} + \frac{1.0}{-4} + \frac{0.7}{-3} + \frac{0.3}{-2}$$

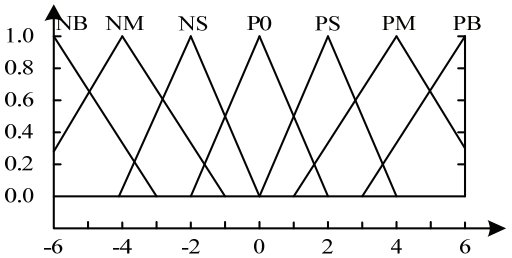
$$NS_{ET} = \frac{0.5}{-3} + \frac{1.0}{-2} + \frac{0.5}{-1}$$

$$N0_{ET} = \frac{0.5}{-1} + \frac{1.0}{0}$$

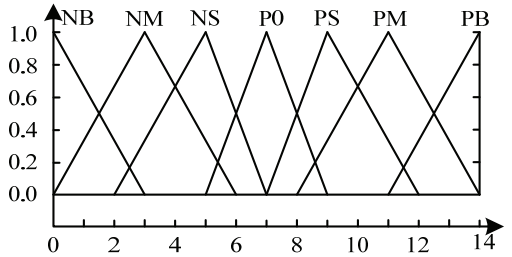
…



a. Subjection function curve of *ET*



b. Subjection function curve of *ETC*



c. Subjection function curve of *ΔT*

Fig. 4. Subjection function curve.

On the basis of temperature deviation fuzzy set expression we can clear up and get MF assignment table as show in Table 1. Successively write the expression of fuzzy set *ET, ETC, ΔT,* the same management we can get their MF assignment table.

For example second order system's step leap response curve, systemic trends character curve as show in Fig. 5, mainly consists of 4 cases.

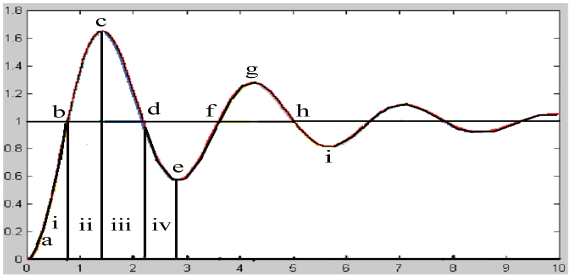


Fig. 5. Second order systems step response curve.

A. *et > 0 etc < 0 (i)*;
B. *et < 0 etc < 0 (ii)*;
C. *et < 0 etc > 0 (iii)*;
D. *et > 0 etc > 0 (iv)*;
The crossing, highest and lowest
Crossing:1.*e>0→e<0,Δe<o(b, f)*
2.*e<0→e>0,Δe>0(d, h)*
Highest: *Δe=0,e<0(c, g)*
lowest: *Δe=0,e>0(e, i)*

According to the direction and size of *ET, ETC,* determine the control rule as follows:

in i area when et is large, to shorten rise time *Δt<0*;when e is small, to prevent overshoot *Δt=0*;

in ii area to reduce overshoot, *Δt<0*;

in iii area when et is large, to shorten fall time *Δt<0*; when et is small reaching set point *Δt=0*;

in iv area to reduce lowest deviation *Δt>0*;

For crossing, highest and lowest, if et and etc both be 0 Δt=0;

Crossing: 1.*et >0→et<0, etc<0 (b, f), Δt <0*
2. *et <0→et>0, etc<0(d, h), Δt>0*;

Highest: *Δe=0,e<0(c, g), Δt<0*

Lowest: *Δe=0, e>0(e, i), Δt>0*

According above rules we can design fuzzy control principle table as show in Table 1.

Table 1. Vague control principle list.

ET / ETC	NB	NM	NS	ZO	PS	PM	PB
NB	PB (R1)	PB	PB	PB	PM	ZO	ZO (49)
NM	PB (R2)	PB	PB	PM	PM	ZO	ZO
NS	PM	PM	PM	PS	ZO	NS	NS
NO	PB	PM	PS	ZO	NS	NM	NB
PO	PB	PM	PS	ZO	NS	NM	NB
PS	PS	PS	ZO	NS	NM	NM	NM
PM	ZO	ZO	NM	NM	NB	NB	NB
PB	ZO	ZO	NM	NB	NB	NB	NB (R56)

According to the sampling get et and figure out etc, then quantize them corresponding get an element *ET* and *ETC* in vague domain [7-10]. Through the Table 1, it can be found that the condition part of fuzzy relation R was comprised by the two conditions connected by using the "AND" operator. The satisfaction of general conditions can be calculated with the minimum of the subcondition membership function. For example, the subordinate degree of the two conditions for R_m (m=1,2…56) can be calculated by the following formula.

$$\mu_{ETm\cap ETCm}(X_{et},Y_{etc})=min\{\mu_{ETm}(X_{et}),\mu_{ETCm}(Y_{etc})\} \tag{3}$$

The general conditions of R_m can be expressed as two-dimensional matrices:

$$\begin{aligned}&\mu_m(X_{et},Y_{etc})=min\{\mu_{ETm}(X_{et}),\mu_{ETCm}(Y_{etc})\}\\&=min\{[\mu_{ETm}(X_{et1}),\cdots\mu_{ETm}(X_{et14})]^T\times[\mu_{ETCm}(Y_{etc1}),\cdots\mu_{ETm}(Y_{etc13})]\}\\&=\begin{bmatrix}min\{\mu_{ETm}(X_{et1}),\mu_{ETC1}(Y_{etc1})\},\cdots min\{\mu_{ETm}(X_{et1}),\mu_{ETC1}(Y_{etc13})\}\\min\{\mu_{ETm}(X_{et2}),\mu_{ETC1}(Y_{etc1})\},\cdots min\{\mu_{ETm}(X_{et2}),\mu_{ETC1}(Y_{etc13})\}\\\vdots\\min\{\mu_{ETm}(X_{et14}),\mu_{ETC1}(Y_{etc1})\},\cdots min\{\mu_{ETm}(X_{et8}),\mu_{ETC1}(Y_{etc13})\}\end{bmatrix}\end{aligned} \tag{4}$$

It will be receiving "(7×8)×1" matrices (formula 5) that write the row element of the "7×8" two-dimensional matrices from the first row in turn in rank.

$$\mu_m(X_{et},Y_{etc})=\begin{bmatrix}min\{\mu_{ETm}(X_{et1}),\mu_{ETC1}(Y_{etc1})\}\\\vdots\\min\{\mu_{ETm}(X_{et1}),\mu_{ETC1}(Y_{etc13})\}\\min\{\mu_{ETm}(X_{et2}),\mu_{ETC1}(Y_{etc1})\}\\\vdots\\min\{\mu_{ETm}(X_{et2}),\mu_{ETC1}(Y_{etc13})\}\\\vdots\\min\{\mu_{ETm}(X_{et14}),\mu_{ETC1}(Y_{etc13})\}\\\vdots\\min\{\mu_{ETm}(X_{et14}),\mu_{ETC1}(Y_{etc13})\}\end{bmatrix} \tag{5}$$

The "if-then" relation may be represented by the cross product of conditions and conclusion in fuzzy logic, the membership function of cross product is the minimum of the membership function of conditions and conclusion. So, the fuzzy relations "R_m" can be expressed as follow.

$$\begin{cases}R_m=(ET_m^T\times ETC_m)^T\times\Delta T_m\\\mu_{Rm}(X_{et},Y_{etc},Z_{\Delta T})=min\{min\{\mu_{ETm}(X_{et}),\\\qquad\qquad\mu_{ETCm}(Y_{etc})\},\mu_{\Delta Tm}(Z_{\Delta T})\}\end{cases} \tag{6}$$

The formula 6 can be written as the matrix format as the formula 7. According to the formula 7, every rule can be computed.

$$\mu_{Rm}(X_{et},Y_{etc},Z_{\Delta T})=\begin{bmatrix} min\{min\{\mu_{ETm}(X_{et1}),\mu_{ETCm}(Y_{etc1})\},\mu_{\Delta Tm}(Z_{\Delta T1})\},...,min\{min\{\mu_{ETm}(X_{et1}),\mu_{ETCm}(Y_{etc1})\},\mu_{\Delta Tm}(Z_{\Delta T15})\} \\ ... \\ min\{min\{\mu_{ETm}(X_{et1}),\mu_{ETCm}(Y_{etc2})\},\mu_{\Delta Tm}(Z_{\Delta T1})\},...,min\{min\{\mu_{ETm}(X_{et1}),\mu_{ETCm}(Y_{etc2})\},\mu_{\Delta Tm}(Z_{\Delta T15})\} \\ ... \\ min\{min\{\mu_{ETm}(X_{et2}),\mu_{ETCm}(Y_{etc2})\},\mu_{\Delta Tm}(Z_{\Delta T1})\},...,min\{min\{\mu_{ETm}(X_{et2}),\mu_{ETCm}(Y_{etc2})\},\mu_{\Delta Tm}(Z_{\Delta T15}) \\ ... \\ min\{min\{\mu_{ETm}(X_{et14}),\mu_{ETCm}(Y_{etc13})\},\mu_{\Delta Tm}(Z_{\Delta T1})\},...,min\{min\{\mu_{ETm}(X_{et14}),\mu_{ETCm}(Y_{etc13})\},\mu_{\Delta Tm}(Z_{\Delta T15})\} \end{bmatrix} \quad (7)$$

Fuzzy relation matrix of System is the result of all control rules connected by the "OR" operator.

$$\begin{cases} R=R_1\vee R_2\vee...\vee R_{56} \\ \mu_R(X_{et},Y_{etc},Z_{\Delta T})=max\{\mu_{R1}(X_{et},Y_{etc},Z_{\Delta T}),\mu_{R2}(X_{et},Y_{etc},Z_{\Delta T}), \\ \qquad ...,\mu_{R56}(X_{et},Y_{etc},Z_{\Delta T})\} \end{cases} \quad (8)$$

For a given input (X_{et}, Y_{etc}), the fuzzy output for the R_m is as follow.

$$\Delta T_m=(X_{et}\times Y_{etc})\circ R_m \quad (9)$$

The output of controller can be written as the formula (10).

$$\begin{cases} \Delta T=\bigcup_{m=1}^{56}[(X_{et}\times Y_{etc})\circ R_m] \\ \mu_{\Delta T}(\Delta T)=max\{min\{min\{\mu_{ETi}(X_{et}),\mu_{ETCi}(Y_{etc})\}, \\ \qquad \mu_R(X_{et},Y_{etc},Z_{\Delta T})\}\} \end{cases} \quad (10)$$

If the output of controller for the given (X_{et},Y_{etc}) input is

$$\Delta T=\{r_0,r_1,r_2,\cdots r_{14}\} \quad r_n\in(0,1)(n=0,1\cdots 14),$$

the fuzzy vector can be written as the

$$\Delta T=\frac{r0}{-6}+\frac{r1}{-5}+\frac{r2}{-4}\cdots+\frac{r14}{6} \quad (11)$$

Output from the defuzzification process of controller output by using the weighted average method can be calculated as the follow.

$$\Delta T=\frac{\sum_{n=1}^{14} r_n\bullet Z_{\Delta T\,n}}{\sum_{n=1}^{14} r_n} \quad (12)$$

Then last result of controller is the follow formula.

$$\Delta t=\Delta T K_{\Delta t} \quad (13)$$

Then put the MCU control data into a table form and write on SCM's program storage, realizing locating method in SCM [4].

3.2. The Self-tuning Algorithm of Control Rule

i) Property test.

Owing to the fact that initial control rules are rough, it can not achieve satisfactory performance requirements, and to achieve better control effect, you must modify the control rules, which is necessary to test the performance of the controller [11]. Typically, the control effect and performance requirements deviation of error *et(nT)* and error rate of change of *ce (nT)* to test, set up et and *etc(nT)* belongs to fuzzy set *ET (nT)* and *ETC (nT)*. Through the test of the deviation *ET (nT)* and *ETC (nT)*, on the controller output is corrected, the correction of the amount of *P (nT)* is expected to improve the control effect of the system, i.e.

$$ET(nT)\rightarrow ETC(nT)\rightarrow P(nT)$$

$$(ET(nT)=NB)\rightarrow(ETC(nT)=NS)\rightarrow(P(nT)=PB)$$

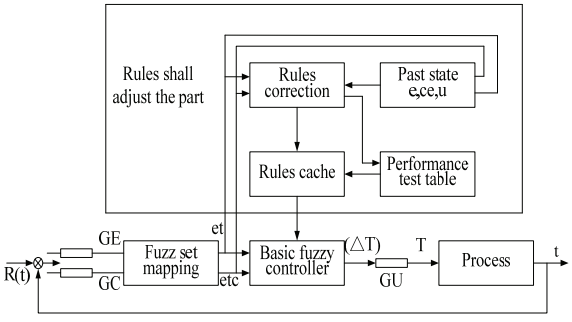


Fig. 6. Control Rules Fuzzy Logic Controller.

ii) Correction of control input.

According to each of the performance measurement of output *P (nT)*, we can calculate for the process control of the amount of correction of the input *r (nT)*, which the purpose is that the output characteristic can be changed later. We can discuss in following situations:

a) SISO system.

$$r(nT)=k_p(nT)$$

If input and output which frequently change in the process can be normalized to the maximum, then *k = 1*.

b) MIMO system.

For a MIMO system, we have the input-output relationship

$$X = F(X^T, U^T), \tag{14}$$

where X is the output vector of the system, and U is the input vector, then through a sampling period, the incremental relationship is:

$$\Delta X \approx TJ\Delta U = M\Delta U \tag{15}$$

J represents the Jacobian matrix in the formula. The formula which determines the incremental model ranging from the input change ΔU and the output change ΔX ,which can be called the process of incremental model. Thus (Process) correction of the amount of input *r (nT)* can be obtained by the above incremental model

$$r(nT) = M^{-1}P(nT) \tag{16}$$

In actual operation, because they do not know the exact model of the system, and therefore do not know the coupling coefficient between the input and output. Thus typically we can be based on experience for the understanding of the system, assuming a coupling coefficient, and then through a self-learning process to overcome the inaccuracy of the assumed [12].

c) Delay system.

As to the system which is larger delay, we need to determine control action of which moment should be responsible for the poor performance, which is defined by time delay period mT of the offline measurement-delay.

iii) Amendments to the rules of control.

We use the example of SISO system as an introduction to the amendment process of the control rules. The delay is assumed that the system of m sampling period, the control input of the sampling time *(nT-mT)* have a greatest impact on the performance of process output at time nT. If the initial rule of the control rule table as follows:

$$ET(nT-mT) \to ETC(nT-mT) \to \Delta T(nT-mT)$$

Through nT time characteristics test should be amended as:

$$ET(nT-mT) \to ETC(nT-mT) \to V(nT-mT)$$

and

$$\begin{aligned} ET(nT-mT) &= F\{et(nT-mT)\} \\ ETC(nT-mT) &= F\{etc(nT-mT)\} \\ \Delta T(nT-mT) &= F\{\Delta t(nT-mT)\} \\ V(nT-mT) &= F\{\Delta T(nT-mT)+r(nT)\} \end{aligned}$$

Write the sentence form, is shown as:

$$\begin{aligned} &IF \quad e = ET(nT-mT) \quad and \quad etc = ETC(nT-mT) \quad THEN \\ &\qquad \Delta t = \Delta T(nT-mT) \end{aligned} \tag{17}$$

The revised results as:

$$\begin{aligned} &IF \quad e = ET(nT-mT) \quad and \quad etc = ETC(nT-mT) \quad THEN \\ &\qquad \Delta t = V(nT-mT) \end{aligned} \tag{18}$$

Using the relation matrix can be expressed as:

$$R'(nT) = ET(nT-mT)\times ETC(nT-mT)\times U(nT-mT) \tag{19}$$

and

$$R''(nT) = ET(nT-mT)\times ETC(nT-mT)\times V(nT-mT) \tag{20}$$

Let *R(nT)* is the current moment controller relation matrix, *R(nT+T)* is a modified matrix, realizing the conversion equations can be expressed the following statement:

$$R(nT+T) = \{R(nT) \ \ is \ \ not \ \ R'(nT)\} \ \ else \ \ R''(nT)$$

Also available for set representation:

$$R(nT+T) = \{R(nT) \wedge \overline{R'(nT)}\} \vee R''(nT)$$

General method for correcting controller type is proposed by E. H. Mamdan, this time the new relationship matrix is obtained, then according to the synthesis of *et (nT)*, *etc (nT)* and *R (nT+T)*, obtained fuzzy grade control quantity, After decisions to draw firm control, added it to the system, to complete a control action.

As nearly above, this method is time-consuming and accounted for a large amount of storage space. Here are both intuitive and simple and practical algorithms.

iv) A new control rule modification algorithm.

Here need to supplement equation (16) is the actual process input control signal, and if it is converted to fuzzy control. The output of the system, *r (nT)* should be divided by *GU*, as

$$r'(nT) = \frac{1}{GU}M^{-1}P(nT) = (GU \bullet M)^{-1}P(nT) = M_1^{-1}P(nT), \tag{21}$$

where *M1=GU·M*.

We want to revisit the relationship between the original rules and the new rules.

$$r(nT) = r'(nT)$$

The original rules:

$$T'\{et(nT-mT), etc(nT-mT)\} = \Delta T(nT-mT)$$

The new rules:

$$T'\{et(nT-mT), etc(nT-mT)\} = V(nT-mT)$$

But we found that, If we simply modified according to the above formula ,there will be such a problem, If two consecutive side was even in the delay period (*mT*), the two times are the same, namely *et(nT-mT)*, *etc(nT-mT+T)* and *ET (nT-mT+T)*, *ETC (nT-mT+T)* are equal, then because of the (*nT*) time will change

$$T'\{et(nT-mT),etc(nT-mT)\}=\Delta T(nT-mT)$$

into

$$T'\{et(nT-mT),etc(nT-mT)\}=V(nT-mT)$$

As in (*nT+T*), the original rules no longer exists, in accordance with the revised rules, the new rules

$$T'\{et(nT-mT+T),etc(nT-mT+T)\}$$
$$=T'\{et(nT-mT),etc(nT-mT)\}=V(nT-mT)$$

and then amended as follows:

$$T'\{et(nT-mT+T),etc(nT-mT+T)\}$$
$$=T'\{et(nT-mT),etc(nT-mT)\}=V'(nT-mT)$$

The equation

$$V'(nT-mT)=F\{\Delta t(nT-mT+T)+r(nT+T)\}$$
$$=F\{\Delta t(nT-mT+T)+r(nT)+r(nT+T)\} \quad (22)$$

Taking into account the above, we change the rule modification method.

The original rules:

$$T'\{et(nT-mT),etc(nT-mT)\}=U(nT-mT)$$

The new rules:

If in the past *(nT-mT)* of mT delay period, there is a *et (nT-mT-iT)*, *etc (nT-mT-iT)* and *et(nT-mT)* and *etc(nT-mT-iT)* are the same, then in the second amendment, the rules are:

$$T'\{et(nT-mT),etc(nT-mT)\}$$
$$=\{\alpha T'\{\{et(nT-mT),etc(nT-mT)\}+(1-\alpha)V(nT-mT)\}$$

Otherwise:

$$T'\{et(nT-mT+T),etc(nT-mT)\}=V(nT-mT)$$

Among them, $0<\alpha<1$, $i=1,\cdots,m-1$.

Now our new algorithm can be summarized as follows

Step 0: To establish buffer rushed 3 * 2m units of data storage.

$$\begin{cases} et(nT),et(nT-T),\cdots,et(nT-2mT+T) \\ etc(nT),etc(nT-T),\cdots,etc(nT-2mT+T) \\ \Delta T(nT),\Delta T(nT-T),\cdots,\Delta T(nT-2mT+T) \end{cases} \quad (23)$$

Step 1: Calculating *et(nT)*, *etc(nT)*;

Step 2: Look-up table $T'(et,etc)$, to get $\Delta t\ (nt-mT)$,

Look-up table $T''(et,etc)$, to get $p(nt-mT)$,

Calculating $V(nT-mT)=\Delta t\ (nT-mT)+r(nT)$;

Step 3: Modifying the control decision tableT'(e,c):

For i=1 to i=m-1

{Compare *et(nT-mT)*,*etc(nT-mT)* and *et(nT-mT-iT)*,*etc(nT-mT-iT)*;if the same, then Flag=1;else Flag=0};

If flag=1, then

$$T'_{nT+T}\{et(nT-mT),etc(nT-mT)\}$$
$$=\alpha T'\{et(nT-mT),etc(nT-mT)\}+(1+\alpha)V(nT-mT)$$

Otherwise,

$$T'_{nT+T}\{et(nT-mT),etc(nT-mT)\}=V(nT-mT)$$

Step 4: To determine the control amount of $U^*(nT+T)$ at the time of *(nT+T)*

$$\Delta T(nT+T)=\Delta T(nT+T)\times G\Delta T=T'_{(nT+T)}\{et(nT),etc(nT)\}\times G\Delta T$$

There $T'(et,etc)$ is the control decision table, $T''(et,etc)$ is the performance test correction table, These stored in the computer in advance.

4. The Design of Fuzzy Controller

This system use MCU to design a temperature controller which mainly consist of micro control device, infrared temperature survey and signal disposal circuit, beat drive module, keyboard and display module and power supply module [13-14]. System temperature set and initial beat time input by keyboard, the temperature of furnace exit work pieces are measured by infrared temperature measure instrument and go through signal disposal circuit transformed voltage span SCM AD needed. According to measured temperature went through fuzzy control to get the beat time to ensure heat work pieces reaching set temperature. Software uses modules structure programming, mainly consists of keyboard input /output, show, temperature measure and filtration, fuzzy control arithmetic subroutine. To decrease the SCM's program difficulty, control quantity output was calculated off-line of fuzzy control's 56 rules and put them into a table [15]. The controller output in MCU is a hexadecimal data, so the Control variable should be converted into hexadecimal. The MCU domain is [00H, 01H, …, FFH], so the Conversion relationship is formula 24.

$$D_{MCU_Table}=(D_{control_var}+6)\times 256/12$$
$$D_{MCU_Table}: Data\ of\ MCU\ table \quad (24)$$
$$D_{control_var}: Data\ of\ \Delta t\ table's\ control\ variable$$

5. Analysis of the Temperature Error of Workpiece Experimental Test Result

5.1. Analysis and Treatment of Temperature Error of the Workpiece

According to control theory the precision of the system depends on the accuracy of given and feedback, so the heating furnace's control temperature precision is mainly decided by the given value and measured value of the temperature. In order to reduce the factors that may influence system error, the set point of temperature is given by keyboard to eliminate the given error. So the factors that mainly influence system error are in the link of temperature measurement [16-17].

Thermometry link consists of temperature transmitting (temperature signal is converted to 4 ~ 20 mA) and AD collecting process, so the errors of temperature measurement are mainly stochastic nonlinear error, gross error and the nonlinearity of data transformation [18]. The response time of thermometer is 50 ms, and each time when the work pieces being push out from heating furnace, the passing time in thermometer is 1 ~ 2.5 seconds (the length of each work piece is different), so each work piece's temperature measurement time is 20 ~ 50 times. In order to gain effective measurement data, the measure standard is based on the shortest work piece measured 15 times. According to the characteristics of random error, the random error of temperature measurement generally obey normal distribution, when the confidence interval is [-3σ, 3σ], the confidence probability is 99.7 % as shown in Fig. 7.

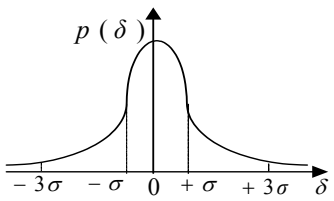


Fig. 7. Probability density curve of random error.

The measured temperature of each work pieces are *y1,y2,...yi* (*i=1,2...15*), the average value of each work piece's temperature is follow.

$$\overline{y} = \frac{1}{15}\sum_{i=1}^{15} y_i$$

The measured temperature root-mean-square error is got using Palin jess formula

$$\sigma = \sqrt{\frac{\pi}{2}}\frac{\sum_{i=1}^{n}|y_i - \overline{y}|}{\sqrt{n(n-1)}} \approx \frac{5}{4}\frac{\sum_{i=1}^{n}|y_i - \overline{y}|}{n - \frac{1}{2}} \qquad (25)$$

When $|y_i - \overline{y}| > 3\sigma$, then y_i is out of confidence interval, it will be deleted from measured data. Then recount $\overline{y}$ *and* σ until all left measuring point dropping in confidence interval. At present temperature measurement result is $\overline{y} \pm \sigma$.

During temperature measurement gross error adopts Wright's standards, firstly calculating Bessel data of measured value,

Among measured data, if y_i fits $|y_i - \overline{y}| > 3\sigma$ condition, this measured value will be deleted. Rest data will be processed by random error.

$$\sigma = \sqrt{\frac{\sum_{i=1}^{15}(y_i - \overline{y})^2}{15-1}} \qquad (26)$$

Measuring temperature data's nonlinearity dispose adopts segmented correction method [19]. According to the accuracy requirement, $Y_i=f(y)$ is divided into n segment. And the dots in $Y_i=f(y)$ *1, 2, 3, ...n* are calculated correspondingly getting dots 1', 2', 3', …..n' in $Y_{2i} = K_{2i}y$. When n is large enough, each segment on $f(y)$ can be seen as a straight line, with slope K_{1i}. as shows in the Fig. 8.

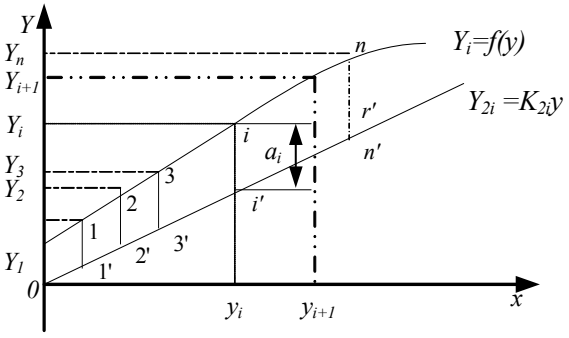


Fig. 8. Temperature acquisition nonlinear correction method.

Y_i—the ith segment linear equation is got as follows:

$$Y_i = Y_{i0} + K_{1i}(y - y_i)$$

Y_{i0} is the initial value for this segment. K_{1i} is straight line slope for ith segment. Corresponding to y_{2i} the straight line equation of the ith segment is as follows:

$$Y_{2i} = (Y_i - a_i) + K_{2i}(y - y_i),$$

where a_i is the difference between *i* and *i'* segment, K_i is the difference of slope the between i^{th} and i'^{th} segment, i.e.

$$K_i = K_{1i} - K_{2i}$$

By the three formulas above:

$$Y_{2i} = (Y_i - a_i) + (K_{1i} - K_{2i})(y - y_i)$$
$$= [Y_i + K_{1i}(y - y_i)] - [a_i + K_i(y - y_i)]$$
$$= Y_{1i} - [a_i + K_i(y - y_i)]$$

5.2. The Analysis of Experimental Test Result

Using this controller in tubular furnace, setting work piece warming time 1100 s and initial beat 45 seconds, containing three work pieces in this furnace, gathering 20 work pieces' temperature during warming process, we get experimental graph show in Fig. 9.

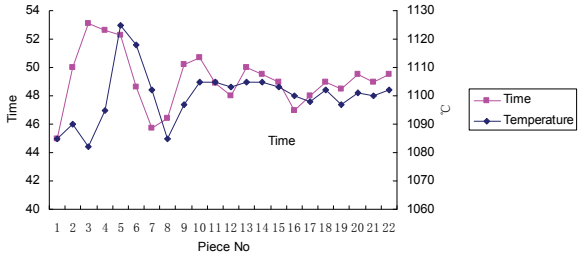


Fig. 9 The curve of push-time and temperature of workpieces.

6. Conclusions

This paper presents a tubular furnace temperature control system based on fuzzy control theory. On the basis of the features of tubular furnace, a fuzzy controller was designed applying two dimensions fuzzy control theory. The control rules were modified by the self-tuning algorithm of control rule during the controller running. Experiment indicated that induction furnace based on fuzzy control has the characters of high control accuracy, strong anti-interference ability and so on.

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An Investigation on Test Signal Transmission at Temperature Changing Environment

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Abstract: In vibration test, the signal transmission is affected significantly by high ambient temperature. The basis of on traditional transmission theory, the telegraph operator equation is satisfied for analyzing signal feature. When ambient temperature increases from low to high, transmission parameters RLCG change correspondingly. Considering circuit feature of signal transmission, Kirchhoff's Law is applied to establish a discrete temperature-time signal transmission model. And the FDTD method is applied to give a new iterative algorithm for achieving the solution of voltage u\and current. The signal transmissions with different temperature changing process are realized using above simulation. The results show that the amplitude of output signal is gradually attenuate with the rise of conductor temperature. Compared with amplitude at 30 °C, the attenuate degree comes to 10 % at 300 °C. The amplitude attenuation is more evident due to the coexistence of temperature rise and drop. The corresponding experiment has been conducted and the results can meet that of proposed model. It provides a new method for improving the accuracy of vibration test signal.
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Keywords: Vibration test, Signal transmission, Temperature changing environment, FDTD, Temperature gradient, Signal amplitude.

1. Introduction

The test signal processing and analysis are the key component of the power generating machine test system. Many effective methods have been proposed to improve the accuracy of signal analysis [1]. In fact, it is not enough for analyzing signal feature only using above processing technique. The change of ambient temperature also takes important effect on signal transmission path [2], which would ultimately change the original features of test signal. Thus, to obtain exact feature of signal, it is significant to study on changing characteristics of test signal transmission in temperature-changing environment.

In terms of signal transmission, past studies have shown that the response characteristic of test signal transmission is mainly related to signal frequency. And researches are concentrated on high speed and microwave circuits [3]. Signal transmission models can be established on the base of the telegraph operator equation [3]. Taking into account the solving complexity of the equation, a variety of numerical methods are widely developed from time domain or frequency domain. For example, finite different time domain method (FDTD) [4, 5], transmission line method (TLM) [6, 7], Inverse Fast Fourier Transform [8] (IFFT), numerical Laplace inverse transform method [9]

(NLIT) and modal analysis method [10], were proposed to realize signal transmission analysis in specific conditions. Actually, signal transmission equation is the solution of circuit response. And then applying Spice-like equivalent circuit model, transient response analysis can be achieved by converting transmission parameters into pure circuit components [10]. These methods are valid for transmission network with high frequency distribution parameters, and can be also applied to analyze transient response of origin and terminal signal. However, the test signal of power system is different to microwave and high speed signal. The frequency of test signal is often mid or low frequency which is lower than that of high speed signal. And the transient response due to change of signal frequency is not obvious. So applying the previous methods to establish signal transmission model in high temperature environment has certain limitations.

In the paper, we cast the problem in change of signal response resulting from outer temperature field. Considering temperature-changing characteristic of transmission parameters, a new transmission model using FDTD is proposed to get test signal response changing with temperature. Corresponding experiment is carried out to prove the feasibility of simulation analysis.

2. Theory Model of Signal Transmission

The actual test signal is affected by many interfering factors. In the article, we only take change of ambient temperature as main influencing factor. First, the test signal network is given, shown as Fig. 1.

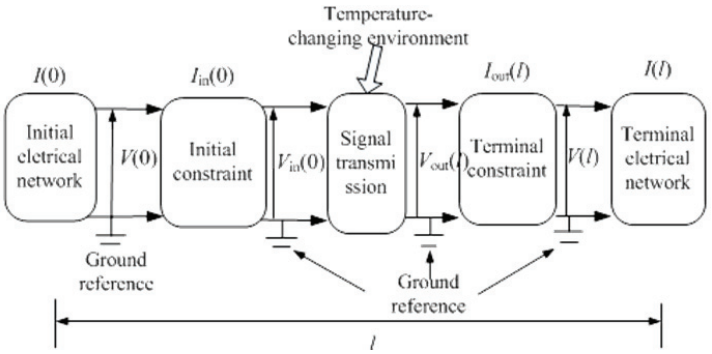

Fig. 1. Chart of test signal network.

In Fig. 1, $V(0)$ and $I(0)$ are input signals collected by the sensor, $V(l)$ and $I(l)$ are terminal signals. The transmission cable would change due to increase of ambient temperature.

When signal is transmitting in cable, the telegraph operator equation is satisfied for analyzing signal feature. The defined equation is followed

$$\begin{cases} \dfrac{\partial V}{\partial z}+\left(R+L\dfrac{\partial}{\partial t}\right)I=0 \\ \dfrac{\partial I}{\partial z}+\left(G+C\dfrac{\partial}{\partial t}\right)V=0 \end{cases} \tag{1}$$

Denote voltage V and current I in cable along the direction of z. Electric resistance R, electric capacity C and inductance L, electric conductance G are unit length transmission parameters of cable. From Eq. (1), it can be seen that no external input source is applied in signal transmission process. So, according to Fig. 1, it needs to put the signals collected by the sensor as input source, and new equation is obtained

$$\frac{\partial V(z,t)}{\partial t}+L\frac{\partial I(z,t)}{\partial t}+RI(z,t)=V_{\mathrm{F}}(z,t) \tag{2a}$$

$$\frac{\partial I(z,t)}{\partial t}+C\frac{\partial V(z,t)}{\partial t}+GV(z,t)=I_{\mathrm{F}}(z,t) \tag{2b}$$

Known from Eq. 2, $V(z,t), I(z,t)$ are related to time t. The $V_{\mathrm{F}}(z,t)$ and $I_{\mathrm{F}}(z,t)$ are the outside incentive source voltage and source current when time is t, corresponding to $V(0), I(0)$ in Fig. 1. In thermal environment, R, L, C, G are the function of temperature T. Substitute them into equations, and the Eq.2 becomes

$$\frac{\partial V(z,t)}{\partial t}+L(T)\frac{\partial I(z,t)}{\partial t}+R(T)I(z,t)=V_{\mathrm{F}}(z,t) \tag{3a}$$

$$\frac{\partial I(z,t)}{\partial t}+C(T)\frac{\partial V(z,t)}{\partial t}+G(T)V(z,t)=I_{\mathrm{F}}(z,t) \tag{3b}$$

,

Here, there are three parameters: t, z and T in transmission equation. So, it is more difficult to solve signal transmission equation than that with two parameters. According to the change of ambient temperature, a reasonable signal transmission model is given, and corresponding conversion method is presented using FDTD to realize iterative solution of signal transmission equation.

3. Model of Signal Transmission

Actually, the test environment is complex and diversified. So the temperature of selected environment is satisfied as Fig. 2.

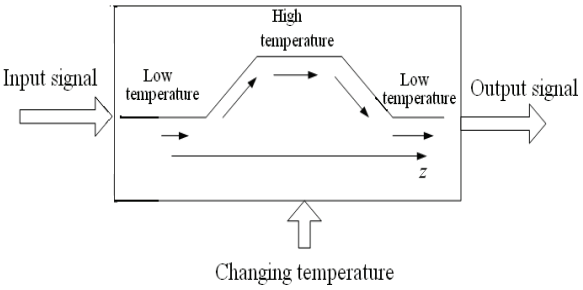

Fig. 2. Temperature gradient model.

Know from Fig. 2, to ensure the instrument work normally, the temperature in both ends of cable is low, setting as 20 °C. In the middle part, ambient temperature is high, more than 200 °C. Heat exchange results in change of cable temperature. The cable temperature is gradually changeable from low to high or from high to low.

3.1. Theoretical Model

For the signal transmission model shown as Fig. 2, transmission parameters have different values in different temperature. So, traditional integral method is hard to use to solve equation Eq. 3. On the basis of time and space discretization, temperature T is also discretized by the small load step to get a more precise result. Set $\triangle T$ as discrete temperature step. Now, parameters of R, L, C, G are expressed as $\triangle R(T)$, $\triangle L(T)$, $\triangle C(T)$, $\triangle G(T)$ in $\triangle T$ Then any part of is selected to analyze, and a transmission model with temperature gradient is shown in Fig. 3.

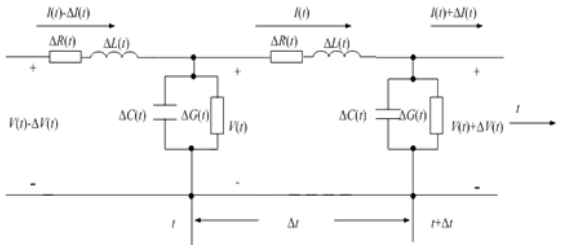

Fig. 3. Equivalent circuit diagram of temperature gradient model.

In Fig. 3, $V(T)$ and $I(T)$ are the initial voltage and current respectively, and $V(T)+\Delta V(T)$, $I(T)+\Delta I(T)$ are that at output ends. Applying Kirchhoff' s law in whole circuit, transmission equation Eq. (3) are converted as

$$\begin{aligned}&\frac{\partial V(T,t)}{\partial T}+R(T)\cdot I(T)+L(T)\frac{\partial I(T,t)}{\partial T}=V_F(T,t)\\&\frac{dI(T,t)}{dT}+G(T)\cdot V(T)+C(T)\frac{\partial V(T,t)}{\partial T}=I_F(T,t),\end{aligned}\tag{4}$$

where, $V(T,t),I(T,t)$ express the distribution of voltage and current when wire temperature is T in time t; $V_F(T,t),I_F(T,t)$ are the outside incentive voltage and current source of unit length in time t. it is clear that the transmission equation of Eq.(4) contains two variables of time t and temperature T. So, it can be defined as temperature-time discrete model.

3.2. Iteration Solution

Applying partial ΔT section to the entire temperature field, the distribution of voltage and current are obtained by defined nodes, shown as Fig. 4. In charm of Fig. 4, only temperature and time are discrete representing as $\Delta T/2$ and $\Delta t/2$. And the voltage nodes and current nodes are cross-arranged, which the current node is in the middle of two voltage nodes.

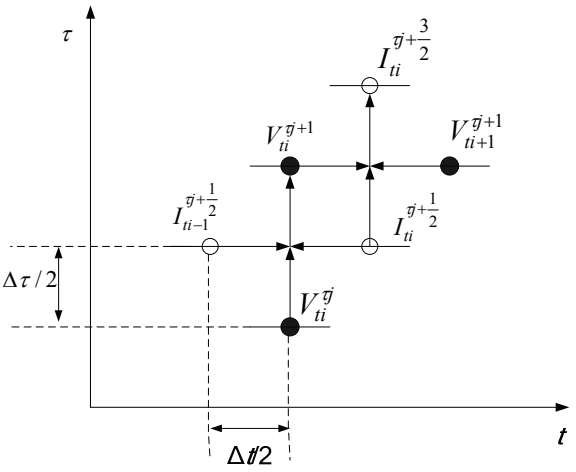

Fig. 4. Discrete chart of voltage and current on temperature and time.

Here, V_{Ti}^{tj} and I_{Ti}^{tj} are the voltage and current in different time and temperature respectively. Set V_{Ti}^{tj} and I_{Ti}^{tj} meet

$$\begin{aligned}V_{ti}^{\tau j}&=V((ti)\Delta t,j\Delta\tau)\\I_{Ti}^{tj}&=I((Ti+1/2)\Delta t,j\Delta\tau)\end{aligned}\tag{5}$$

Then Eq. (4) can be obtained as follows

$$\frac{V_{ti+1}^{\tau j+1}-V_{ti}^{\tau j+1}}{\Delta t}+L_{ti}\cdot\frac{I_{ti}^{\tau j+\frac{3}{2}}-I_{ti}^{\tau j+\frac{1}{2}}}{\Delta\tau}+R_{ti}\cdot\frac{I_{ti}^{\tau j+\frac{3}{2}}+I_{ti}^{\tau j+\frac{1}{2}}}{2}=\frac{V_{\mathrm{F}}^{\tau j+\frac{3}{2}}+V_{\mathrm{F}}^{\tau j+\frac{1}{2}}}{2}$$
$$\frac{I_{ti}^{\tau j+\frac{1}{2}}-I_{ti-1}^{\tau j+\frac{1}{2}}}{\Delta t}+G_{ti}\cdot\frac{V_{ti}^{\tau j+1}-V_{ti-1}^{\tau j}}{\Delta\tau}+C_{ti}\cdot\frac{V_{ti}^{\tau j+1}+V_{ti}^{\tau j}}{2}=\frac{I_{\mathrm{F}}^{\tau j+1}+I_{\mathrm{F}}^{\tau j}}{2} \tag{6}$$

The further organization is carried out for formula, and we have

$$\left(L_{ti}\frac{\Delta t}{\Delta\tau}+\frac{R_{ti}}{2}\Delta t\right)I_{ti}^{\tau j+\frac{3}{2}}=\left(L_{ti}\frac{\Delta t}{\Delta\tau}-\frac{R_{ti}}{2}\Delta t\right)I_{ti}^{\tau j+\frac{1}{2}}-\left(V_{ti+1}^{\tau j+1}-V_{ti}^{\tau j+1}\right)+\frac{\Delta t}{2}\left(V_{\mathrm{F}}^{\tau j+\frac{3}{2}}+V_{\mathrm{F}}^{\tau j+\frac{1}{2}}\right) \tag{7a}$$

$$\left(C_{ti}\frac{\Delta t}{\Delta\tau}+\frac{G_{ti}}{2}\Delta t\right)V_{ti}^{\tau j+1}=\left(C_{ti}\frac{\Delta t}{\Delta\tau}-\frac{G_{ti}}{2}\Delta t\right)V_{ti}^{\tau j}-\left(I_{ti}^{\tau j+\frac{1}{2}}-I_{ti-1}^{\tau j+\frac{1}{2}}\right)+\frac{\Delta t}{2}\left(I_{F}^{\tau j+1}+I_{F}^{\tau j}\right) \tag{7b}$$

3.3. The Initial and Boundary Conditions

Known from Fig. 3, the Initial conditions of signal transmission meet T_1=T_0, and there is a source matched resistance R_{S}, so conditions meet

$$G_{t0}=\frac{1}{R_{\mathrm{S}}} \tag{8a}$$

$$\frac{C_{t0}}{2}\Delta t=C_{ti}\Delta t \tag{8b}$$

$$I_{\mathrm{F}}=\frac{V_{\mathrm{S}}}{R_{\mathrm{S}}\Delta t} \tag{8c}$$

Taking the boundary condition Eq. (8) into Eq. (7a), Voltage and current can be expressed at T_1 °C

$$\left(C_{t0}R_{\mathrm{S}}\frac{\Delta t}{\Delta\tau}+1\right)V_{t1}^{\tau j+1}=\left(C_{t0}R_{\mathrm{S}}\frac{\Delta t}{\Delta\tau}-1\right)V_{t1}^{\tau j}-2R_{\mathrm{S}}I_{t1}^{\tau j+\frac{1}{2}}+\left(V_{\mathrm{S}}^{\tau j+1}+V_{\mathrm{S}}^{\tau j}\right) \tag{9}$$

When the matched resistance R_{L} is at T_{N+1}°C, the boundary conditions becomes

$$I_{\mathrm{tN+1}}=0 \tag{10a}$$

$$G_{\mathrm{tN+1}}\Delta t=\frac{1}{R_{\mathrm{L}}} \tag{10b}$$

$$I_{\mathrm{F}}\Delta t=\frac{V_{\mathrm{L}}}{R_{\mathrm{L}}} \tag{10c}$$

$$\frac{C_{\mathrm{L}}}{2}\Delta t=C_{\mathrm{T}i}\Delta t \tag{10d}$$

By Eq. (7b) and Eq. (10), the equation at T_{N+1}°C can be solved as followed formula.

$$\left(C_{tN}R_{\mathrm{L}}\frac{\Delta t}{\Delta\tau}+1\right)V_{tN+1}^{\tau j+1}=\left(C_{tN}R_{\mathrm{L}}\frac{\Delta t}{\Delta\tau}-1\right)V_{tN+1}^{\tau j}+2R_{\mathrm{L}}I_{tN}^{\tau j+\frac{1}{2}}+\left(V_{\mathrm{L}}^{\tau j+1}+V_{\mathrm{L}}^{\tau j}\right) \tag{11}$$

3.4. Iterative Algorithm Based on Temperature-time

Form T_2 °C to T_N °C, there is no external excitation acting on wire, with V_{F} =0 and I_{F}=0. Applying initial condition of Eq. (8) and boundary conditions of Eq. (9), the discrete iterative equations of signal transmission are given as follows

$$V_{t1}^{\tau j+1}=\left(C_{t0}R_S\frac{\Delta t}{\Delta\tau}+1\right)^{-1}\left\{\left(C_{t0}R_S\frac{\Delta t}{\Delta\tau}-1\right)V_{t1}^{\tau j}-2R_SI_{t1}^{\tau j+\frac{1}{2}}+\left(V_S^{\tau j+1}+V_S^{\tau j}\right)\right\} \tag{12a}$$

$$V_{tN+1}^{\tau j+1}=\left(C_{tN}R_{\mathrm{L}}\frac{\Delta t}{\Delta\tau}+1\right)^{-1}\left\{\left(C_{tN}R_{\mathrm{L}}\frac{\Delta t}{\Delta\tau}-1\right)V_{tN+1}^{\tau j}+2R_{\mathrm{L}}I\tau_{tN}^{\tau j+\frac{1}{2}}+\left(V_{\mathrm{L}}^{\tau j+1}+V_{\mathrm{L}}^{\tau j}\right)\right\} \tag{12b}$$

$$V_{ti}^{\tau j+1}=\left(C_{ti-1}\frac{\Delta t}{\Delta\tau}+\frac{G_{ti-1}}{2}\Delta t\right)^{-1}\left\{\left(C_{ti-1}\frac{\Delta t}{\Delta\tau}-\frac{G_{ti-1}}{2}\Delta t\right)V_{ti}^{\tau j}-\left(I_{ti}^{\tau j+\frac{1}{2}}-I_{ti-1}^{\tau j+\frac{1}{2}}\right)\right\} \tag{12c}$$

$$I_{ti}^{tj+1/2}=\left(L_{ti}\frac{\Delta t}{\Delta\tau}+\frac{R_{ti}}{2}\Delta t\right)^{-1}\left\{\left(L_{ti}\frac{\Delta t}{\Delta\tau}-\frac{R_{ti-1}}{2}\Delta t\right)I_{ti}^{\tau j-\frac{1}{2}}-\left(V_{ti+1}^{\tau j}-V_{ti}^{\tau j}\right)\right\} \tag{12d}$$

To ensure the astringency of iterative equations, the ΔT and Δt are small enough to get a good

precision of solution. In Eq. (12), not only R, L, C and G are related to temperature, but also the voltage and current change with increase of temperature. Thus, compared to equation with constant parameters, it is harder for the solution of Eq. (12).

4. Simulation Analysis

According to Fig. 2, a signal transmission model is established, shown in Fig. 5. In the circuit, the temperature change of cable is obvious from T_0 to T_{max} and from T_{max} to T_0.

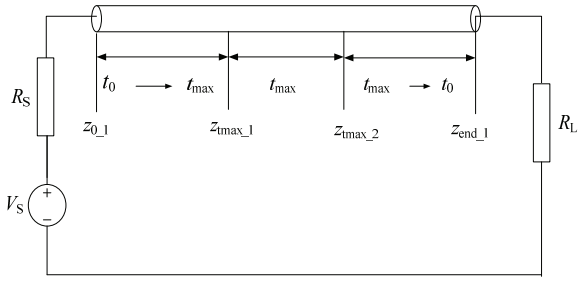

Fig. 5. Signal transmission model changing with temperature.

Here, Vs is the signal input source; Rs and R_L are initial and terminal resistance respectively; z_{0_1} is start point of cable with temperature of T_0 °C; z_{Tmax_1} and z_{Tmax_2} are the position corresponding to temperature of T_{max} °C ; z_{end_1} is terminal point of cable with temperature reduced to T_0°C. In whole test process, high temperature coaxial-cable is applied for signal transmission. The corresponding parameters characteristics of RLCG are referred in reference [11]. It can be seen from Fig. 5, the transmission network of test signal is composed of three parts: temperature rise of cable along direction of z, maintained at high temperature and temperature drop of cable along direction of z. Here the above three process can be solved respectively, and the total signal response is also able to analyzed.

4.1. Change of Signal Amplitude

To avoid signal delay caused by too long cable, the length of transmission wire is set as 10 m. The input signal at z_{0_1} is sinusoidal signal with amplitude of 1 V and frequency of 1000 Hz. The signal is collected at z_{end_1}. Initial resistance R_S is equal to terminal and R_L, which are 50 Ω. The circuit satisfies impedance matching, and magnitude of output signal is half of that of input signal.

In the paper, initial temperature of the wire is 20 °C, and the maximum of temperature is 300 °C. Applying all the conditions, iterative equation of Eq. (12) is solved by Matlab programming. First, feature of signal transmission at single temperature environment is discussed, and Fig. 6 shows the comparison of signal response at 30 °C and 300 °C.

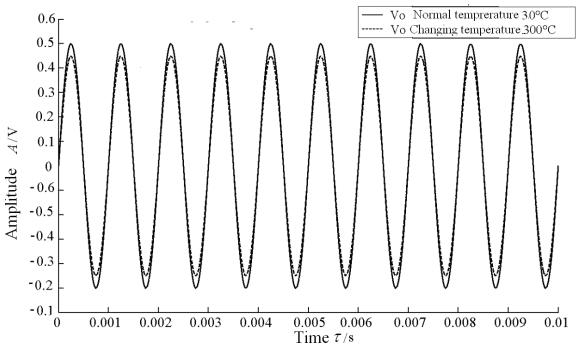

Fig. 6. Signal response of 1000 Hz at 20 °C and 100 °C.

As shown by the Fig. 6, the change of temperatures puts an important influence on signal output response Vo. Here, the solid and dashed lines show the output response at 30 °C and 300 °C separately. It can be observed that amplitude of signal response at 300 °C drops significantly, reached to 95.036 % of that at 30 °C. From the results, it can be demonstrated that temperature effect for output signal can be not ignored. And the reason of amplitude reducing is mainly from temperature-changing characteristic of RLCG. Compared to amplitude, phase deviation of signal is smaller effected by temperature variation. Therefore, we will put main attention to analyze change of signal amplitude caused by temperature rise or drop.

4.2. Amplitude Change at Different Temperature

In test environment of Fig. 5, change of cable temperature is composed of three processes successively. The obtained signal response is consistent with the above processes. Detailed analysis is given below.

1) Temperature rise of transmission path: As can be seen that the temperature of cable keeps constant of 30 °C from 0 to $z_{0\text{-}1}$, and temperature increases gradually from 30°C to 300 °C in part of $z_{0\text{-}1}$~$z_{Tmax\text{-}1}$.
2) Temperature decrease: In part of $z_{Tmax\text{-}2}$~$z_{end\text{-}1}$, temperature of cable decreases from high to constant, that is 300 °C at $z_{Tmax\text{-}2}$ and 30 °C at $z_{end\text{-}1}$. It is worth noting that input signal at $z_{Tmax\text{-}2}$ is considered as same to that of temperature rise process.
3) Temperature rise-constant-decrease: The exact model meeting to original model of Fig. 5 is composed of temperature rise, decrease and the middle part with high temperature. Firstly, the attenuation of signal transmission is taken place due to rise of temperature. Because transmission

parameters remain constant at a certain temperature, it can be considered the signal transmission without attenuation from $z_{\text{Tmax-1}}$ to $z_{\text{Tmax-2}}$. And then, the initial input signal in temperature decrease process is not same to that of temperature rise process, but is the signal of temperature rise after attenuation.

To discuss signal response at different $T_{\max}$, we take $T_{\max}$ as a set of temperature vector arranged from low to high. And then, V_{o_T} is set as amplitude of signal response at $T_{\max}$, and V_{o_30} is output signal at 30 °C. To obtain more comparison of response δ_T is defined as the percentage of signal amplitude, following as

$$\delta_T = \left(1 - \frac{V_{o_T}}{V_{o_30}}\right) \times 100\% \quad (13)$$

Therefore, the attenuation of signal response can be represented using δ_T , and Table 1 provides the attenuate values in three processes at different temperature.

Table 1. Comparison of signal response attenuation at different process.

$T_{\max}$ (°C)	30	50	100	150	200	250	300
δ_T at temperature rise /%	0.13	0.59	1.45	2.32	3.16	4.00	4.96
δ_T at temperature decrease /%	0.09	0.56	1.41	2.28	3.12	3.96	4.92
δ_T at total process /%	0.26	1.17	2.84	4.55	6.18	7.8	9.65

It can be seen from Table 1, the attenuate degree of δ_T is approximately same in processes of temperature rise and decrease. When ambient temperature is below 50 °C, the signal attenuation is small and can be ignored. However, the attenuate amplitude becomes accretive gradually over 100 °C, with the attenuate of 5 % at 300 °C comparing to room temperature. Taking into account transmission parameters remain unchanged in thermostat stage, test signal is considered as no attenuate transmission. Here, output signal of temperature rise process becomes the input signal of temperature decrease process, which would result in the greater attenuation of signal. In Table 1, the attenuation of signal is proportional to temperature rise, and the higher the temperature is, the greater the attenuation of signal. Especially at 300 °C, signal amplitude is only 90 % of that at 30 °C, reduced by 10 %. So, when ambient temperature is over 100 °C, the attenuation of signal amplitude should be taken seriously enough. It also shows that it is undoubtedly indeterminate to judge the structural feature using such signal.

5. Experiment in Temperature Changing Environment

In fact, the influence of environmental temperature on test signal is achieved through the effect of temperature on the wire. The experiment is carried out on the basis of test bench of vibration test. The impact of temperature on signal is analyzed by measuring the structural vibration parameters and heating wire locally.

5.1. Experiment System

The experiment system is composed of three components: vibration test system, environmental temperature changing system and data acquisition and processing system. In vibration testing system, choose the steel plate as experiment component, and acceleration transducer is applied to test and output two signals. In temperature changing part, the wires are placed in iron drum, and adjustable electric furnace is used to heat wires to make temperature rise, shown as Fig. 7.

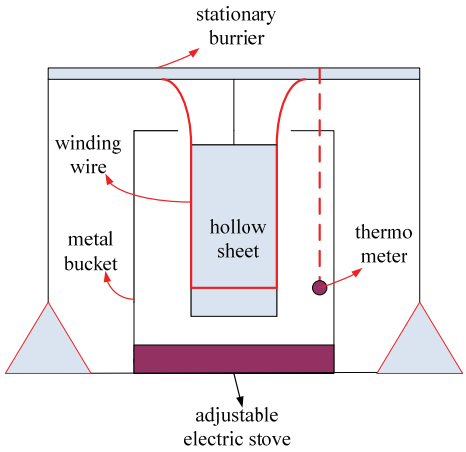

Fig. 7. Test of temperature changing environment.

It is shown from Fig. 7 that temperature of red horizontal wires is at the highest temperature. Moreover, temperature of surrounding wire reduces gradually due to barrel heat dissipation from low to high. The temperature-changing process is good meet to the analytical model. The specific experimental procedure is as follows.

First, open the adjustable electric furnace and keep it at certain power. Test the ambient temperature constantly, until the thermometer Index is constant. Start the vibration test system. Keeping the signal power amplification constant, two signals are obtained: the original signal 1 of no temperature impact and test signal 2 influenced by ambient temperature. All the data is collected and recorded by computer.

5.2. Analysis of Experiment Data

It is worth noting that adjusting the electric power can cause temperature change of the barrel, which would also generate different highest temperature of the wires. Considering the effect from noise signal, the collected signal is filtered by proper band-pass and band-stop filter. The output signals of 100 Hz at different temperature are shown in Fig. 8.

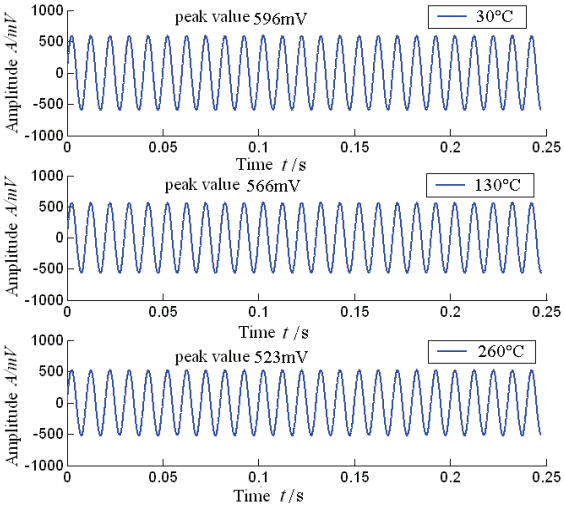

Fig. 8. Signal response after filter at different temperature.

Here, ambient temperature is set to 30 °C. The results from Fig. 8 show that average amplitude of output signal is reduced from 596 mV at 30 °C to 523 mV at 260 °C, with the attenuation degree of 12.25 %. It also indicates that average amplitude of signal is gradually reduced with temperature rise.

The simulation analysis results and the test results were compared using the above-mentioned principle, and then the change results while it is 1000 Hz were shown in Fig. 9.

In experiment, different signals at several temperatures are compared to deeply discuss temperature influence on signal amplitude. Same as above, ambient temperature is 30 °C as a reference, and the amplitude percentage of δ_t is used to represent signal change. When the signal frequency is 1000 Hz, all the experiment and simulation results can be shown using Fig. 9.

As can be seen from the Fig. 9, the amplitude variation obtained by simulation is approximately linear distribution, and is inversely proportional to temperature change. Relatively speaking, the experimental results are fluctuated within a certain range. The reason is believed that only a single signal transmits in the wire ignoring other interference signal in simulation analysis, which is an ideal case analysis. In actual test process, certain error is evitable in experiment. And it is more difficult to completely eliminate effect of interference signal by filtering and other technology. Although, the relationship between signal and temperature is very clear and is nearly consistent with simulation results. At 260 °C, the average amplitude of simulation and experimental results are respectively 91.99 % and 87.75 % of that at 30 °C. Relatively, experimental results changes significantly and signal amplitude has a greater attenuation than simulation results.

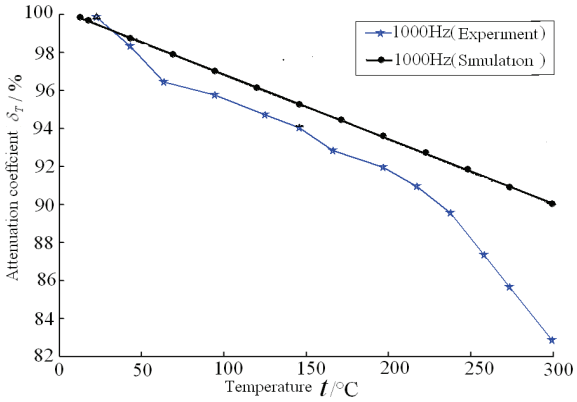

Fig. 9. Experiment and simulation results.

6. Conclusions

For the general vibration testing system, the influence, such as ambient temperature, is often ignored. And if considering temperature, it is also hard to get the specific influence of temperature variation on signal response using conventional signal processing method. In the paper, the FDTD method is successfully applied to achieve the simulation analysis, and corresponding experimental test is carried in temperature changing environment. Firstly, a circuit network is established to express signal transmission part of vibration testing system visually. Applying the temperature-changing characteristics of RLCG parameters, a new approach based on FDTD is proposed to create iteration equations. A transmission model with temperature rise and decrease is given using simulation analysis. From simulation results, we can found that signal amplitude doesn't keep unchanged in whole process. Signal amplitude has a greater attenuation in above temperature change process. The higher the temperature is, the more attenuate the signal amplitude has. When reach to 300 °C, the signal amplitude is only about 90 % of that at 30 °C. This is fully shown that characteristic of original signal has changed from energy point. In the paper, corresponding experiment is given and it has same temperature characteristic to simulation. By using conduction of heat, temperature of wire increases and decreases gradually in test experiment. The signal feature changing with temperature is consistent both in simulation and experimental results. Therefore, it is better to improve the precision of signal analysis when considering the influence of ambient temperature to signal.

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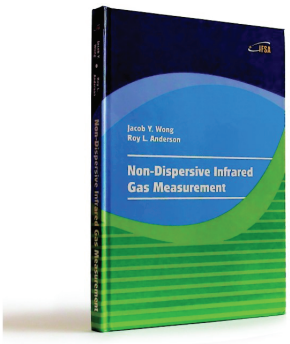

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Study of Orthogonal Face Gear's Contact Spots and Transmission Errors

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Abstract: By single-gear pair contact analysis, this paper, based on the Hertz theory, gets the mesh density of finite element solver in convergence case and then studies contact spots with multi-gear pairs. Through simulation analysis, this paper studies the face gear transmission error in different conditions, and this paper studies the face gear transmission error in different conditions. Then come to a conclusion that: In a set of three different load cases, the greater the torque load is applied to the face gear, the greater the transmission error there is, but the speed of face gear almost has no effect on the gear transmission error. When the face gear is installed with shaft angle of 0.02°, its transmission error magnitude will be 2.0~2.3 times bigger compared with installed without any error. The study can provide guidance for the design, production and practical application of the face gear. *Copyright © 2013 IFSA.*

Keywords: Face gear, Simulation analysis, Contact spot, Transmission error.

1. Introduction

The tooth contact and transmission errors of face gear are related to the vibration, noise and transmission precision of face gear's transmission process. Because face gear is becoming more and more important in the use of aviation industry and precision mechanical transmission in recent years, face gear with quality of low noise, small vibration, stable transmission and high transmission precision is widely required. So the research of the tooth contact and transmission error has very important significance.

Many scholars have done a lot of study about face gear. Claudio Zanzi [1] modified the profile of the pinion of face gear pair and did the analysis of tooth contact and tooth root bending stress. Litvin [2] studied the face gear contact stress and bending stress by finite element method. The experts of face gear are mainly focused on face gear processing, modification, surface contact stress and face gear strength. But they haven't studied tooth contact and transmission errors of face gear deeply. This paper studies contact sport's coordinates of face gear and its realization of visualization through comparing the theory and finite element simulation, gear transmission error under standard installation in different working conditions, but also studies the transmission error of face gear in different coaxial angle installation error. This paper provides guidance for face gear design, noise reduction etc. So the study of this paper has practical significance

2. Mathematical Modeling and Working Load of Orthogonal Face Gear

The tooth working area of face gear is generated by involute tooth profile of the cylindrical gear slotting tool. Transition curved surface is

generated by the top edge of Slotting tool. Tooth fillet of face gear is decided by the tip structure of slotting knife. Tooth surface equation of face gear [5] is as (1) shows:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r_{bs}\cos\phi_2\cos\gamma(\cos\phi_\theta+\theta_s\sin\phi_\theta)-r_{bs}\sin\phi_2(\sin\phi_\theta-\theta_s\cos\phi_\theta) \\ -\frac{r_{bs}(i_{s2}+\cos\gamma)\cos\phi_2}{\cos\phi_\theta} \\ r_{bs}\sin\phi_2\cos\gamma(\cos\phi_\theta+\theta_s\sin\phi_\theta)+r_{bs}\cos\phi_2(\sin\phi_\theta-\theta_s\cos\phi_\theta) \\ -\frac{r_{bs}(i_{s2}+\cos\gamma)\sin\phi_2}{\cos\phi_\theta} \\ r_{bs}\sin\gamma(\cos\phi_\theta+\theta_s\sin\phi_\theta)+\frac{r_{bs}(i_{s2}+\cos\gamma)\cos\gamma}{\sin\gamma\cos\phi_\theta} \end{bmatrix} \quad (1)$$

Because two tooth surface of a face gear tooth are symmetrical about surface yoz, so when change the sign of θ_s, the mathematical model of the other side of the face gear tooth can be got. The equation is as (2) shows:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r_{bs}\cos\phi_2\cos\gamma(\cos\phi_\theta-\theta_s\sin\phi_\theta)-r_{bs}\sin\phi_2(\sin\phi_\theta+\theta_s\cos\phi_\theta) \\ -\frac{r_{bs}(i_{s2}+\cos\gamma)\cos\phi_2}{\cos\phi_\theta} \\ r_{bs}\sin\phi_2\cos\gamma(\cos\phi_\theta-\theta_s\sin\phi_\theta)+r_{bs}\cos\phi_2(\sin\phi_\theta+\theta_s\cos\phi_\theta) \\ -\frac{r_{bs}(i_{s2}+\cos\gamma)\sin\phi_2}{\cos\phi_\theta} \\ r_{bs}\sin\gamma(\cos\phi_\theta-\theta_s\sin\phi_\theta)+\frac{r_{bs}(i_{s2}+\cos\gamma)\cos\gamma}{\sin\gamma\cos\phi_\theta} \end{bmatrix} \quad (2)$$

In the above formulas (1), (2), x, y, z are coordinate points of the tooth surface, r_{bs} is the tools' radius of base circle, φ_2 is turned angle of the face gear, r = 90°, i_{s2} is the transmission ratio of the face gear and the tool, φ_s is turned angle of the tool, φ_{s0} is the angle between tool tank symmetrical line and involute starting point, θ_s is the angle between involute starting point and generating line, and $\varphi_\theta = \varphi_s + \theta_s + \theta_{s0}$

Using the software of MATLAB calculates tooth surface discrete coordinate data of face gear and put the data into Pro/E to get the figure of tooth surface distribution points. It is shown in the following Fig. 1.

Because the data of points is obtained from the tooth surface equation, so the graphics can well reflect the tooth surface profile of face gear. By using Pro/E, connect points into lines, process lines into surfaces, and finally get single tooth model of face gear through substantiated operation. Use Pro/E software to model the involute spur pinion, and assemble face gear with Involute cylindrical pinion through Pro/E software, Specific assembly model are as shown in Fig. 2.

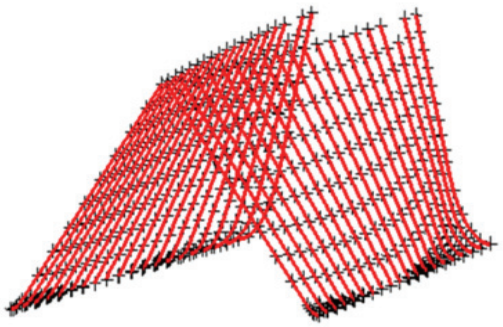


Fig. 1. Tooth surface discrete points of face gear.

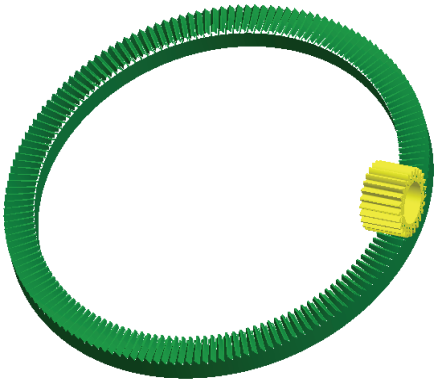


Fig. 2. Assembly drawing of face gear.

Using simulation technology to study face gear transmission error, we apply the speed of: 100 r/min, 200 r/min, 400 r/min to the pinion, and apply the resistance moment of: light load: 5E5 $n \cdot mm$

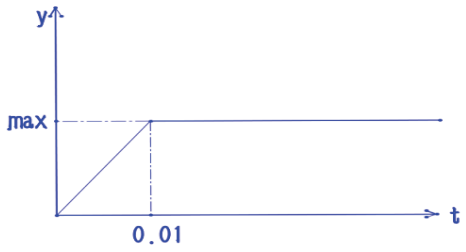


Fig. 3. Loading curve.

In Fig. 3, abscissa refers to the loading time which unit is seconds, ordinate refers to the speed of active pinion or the torque applied to large gear. As the graph shows that, from 0 s to 0.01 s [6], loadings linearly increase, after 0.01 s loadings reach maximum and remain unchanged. Considering the solving characteristics of the software ABAQUS, we choose such a method to apply loads to avoid the non-convergence problems caused by applying loads too large at first. Through changing the speed of pinion and the torque applied to large gear, this paper studies the transmission error when the speed of the pinion is 200 r/min, the resistance moment of large

gear is different, and when pinion is in the above three different speeds, the resistance moment of large gear is the same. The basic parameters of orthogonal face gear are shown in Table 1. Because the contact spots can reflect the face gear's loading capacity and transmission accuracy, it is necessary to study the contact spots of face gear before the study of its transmission error.

Table 1. Transmission parameters of orthogonal face gear.

Name	Number of teeth	Width (mm)	Pressure Angle (°)	Modulus (mm)	Inner diameter (mm)	External diameter (mm)	Shaft Angle (°)	Gearblank thickness (mm)
Face gear	160	89	25	6.35	471	560	90	40
Pinion	25	90	25	6.35	——	——	90	——

3. Study of Face Gear Contact Spots

According to the principle of tooth contact analysis (TCA), two gears are continuous tangential contact in gear meshing transmission. So in the fixed coordinates system of two gear tooth surfaces, the two surfaces have public contact points in random moment. Because they have common normal line at contacting point, solving simultaneous equations of two gear tooth surface can get one of the basic equations of the tooth contact analysis:

$$\begin{aligned} \vec{r}_h^{(1)}\left(\theta_{a1},\phi_{b1},\varphi_1\right) &= \vec{r}_h^{(2)}\left(\theta_{a2},\phi_{b2},\varphi_2\right) \\ \vec{n}_h^{(1)}\left(\theta_{a1},\phi_{b1},\varphi_1\right) &= \vec{n}_h^{(2)}\left(\theta_{a2},\phi_{b2},\varphi_2\right) \end{aligned} \tag{3}$$

In the equations, Superscript "1" and "2" separately refer to cylindrical pinion and face gear, the subscript means that each vector is in the same coordinate. S_h, $(\theta_{a1},\varphi_{b1})$ and $(\theta_{a2},\varphi_{b2})$ refer to the tooth surface coordinates equation of cylindrical pinion and face gear, φ_1 and φ_2 mean the angle turned by the drive gear (pinion) and driven gear (face gear) which both meet the requirements of the transmission ratio. Because of $\left|\vec{n}_1^{(d)}\right| = \left|\vec{n}_2^{(d)}\right| = 1$, Equations (2) can be divided into five scalar equations. A coordinates of contact point of two gear tooth surfaces can be got with fixed value of φ_1. Then use a certain step length to change the value of φ_1. Use these values to do an iterative solution until the coordinates of contact point exceed the efficient frontier of face gear's tooth height. These instantaneous contact points of tooth surface form the face gear's contact path. In the actual bearing case, tooth surface of face gear has a certain degree of elastic deformation. At each contact point, we can get the value of instantaneous contact ellipse's short semi-shaft and long semi-shaft. The series of contact ellipse imprints form the tooth surface contact area of face gear. By coordinate transformation of the tooth surface contact equation, we can get three coordinate component formulas of face gear's tooth surface contact trajectory. These formulas are listed below

$$\begin{aligned} \vec{n}_{xf}^{(1)}\left(\phi_1^{'},\theta\right) &= \vec{n}_{xf}^{2}\left(\phi_2^{'},\theta_s,\phi_s\right) \\ \vec{n}_{yf}^{(1)}\left(\phi_1^{'},\theta\right) &= \vec{n}_{yf}^{2}\left(\phi_2^{'},\theta_s,\phi_s\right) \\ \vec{n}_{zf}^{(1)}\left(\phi_1^{'},\theta\right) &= \vec{n}_{zf}^{2}\left(\phi_2^{'},\theta_s,\phi_s\right) \end{aligned} \tag{4}$$

Assume that angle $\varphi_2^{'}$ is a known parameter. So that, giving a fixed angle $\varphi_2^{'}$, a group of coordinate data can be got. In this way, we can get a contact point. Use a certain step length to change the value of $\varphi_2^{'}$. Then we can work out a series of tooth contact points of face gear. According to the formulas and using software MATLAB, We can get the coordinate values of contact trajectory [4]. As is show in Table 2.

Table2. Contact trajectory coordinates component (mm).

x	*y*	*z*
9.7622	-497.8976	-85.649
8.8386	-497.7834	-83.6832
7.9181	-497.7032	-81.7174
7.3056	-497.6686	-80.4069
6.0825	-497.6447	-77.7858
4.8604	-497.6813	-75.1647
3.6374	-497.7784	-72.5437
2.4116	-497.9359	-69.9226
1.1811	-498.1538	-67.3015
0.254	-498.3568	-65.3357
-0.3665	-498.5109	-64.0252
-0.9893	-498.68	-62.7147
-1.6146	-498.8641	-61.4041
-2.2425	-499.0632	-60.0936

According to Table 2, use software MATLAB to display the coordinate values of tooth surface contact points in the tooth surface. Specific is shown in the following Fig. 4.

According to the L. L ITVIN's book [7], which puts forward the contact ellipse, as is shown in the following Fig. 5.

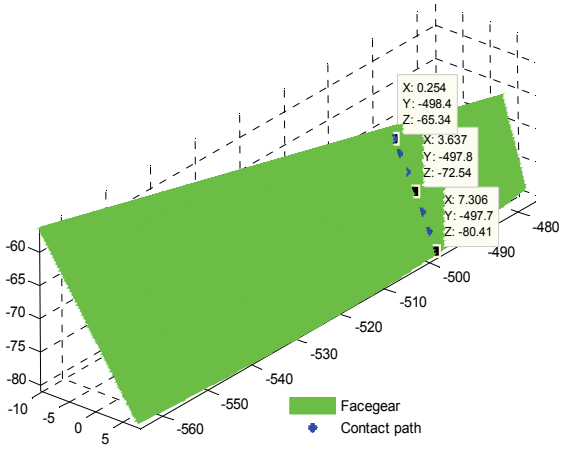


Fig. 4. Contact trajectory.

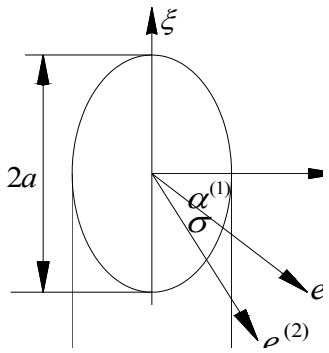


Fig. 5. Contact ellipse.

We can get the length of two half shafts a and b, which are in face gear meshing area.

$$a = \left|\frac{2\delta}{k_a}\right|^{1/2}, \quad b = \left|\frac{2\delta}{k_b}\right|^{1/2}, \tag{5}$$

where δ is the total indentation of the gear contact. Because it is single tooth contact, the load distribution factor is approximated as 1[8] in this paper. To get contact spots, the theoretical calculation of torque is calculated by the following formulas (6) and (7) in the condition that the speed of pinion is 300 r/min and the power of the pinion is 1100 kW.

$$T = \frac{P}{\omega} \tag{6}$$

$$\omega = \frac{2\pi n}{60} \tag{7}$$

In the above formulas, n is speed of the pinion, ω is the angular velocity, P is power. After solving the formulas, the value of torque is 350.14 $n \cdot m$.

k_a, k_b are induced normal curvature of long and short half shaft direction in the meshing area. Get the direction of principal curvatures in the tooth surface contact area and the length of short semi-shaft and long semi-shaft. Then the shape of contact ellipse can be got. Use software MATLAB to apply a torque of 350.14 $n \cdot m$ to the pinion. According to the coordinate trajectory of tooth surface's contact spots in face gear, we can get the coordinate points of contact ellipse's short semi-shaft and long semi-shaft specific is shown in Table 3. Using software MATLAB can draw the face gear's contact trajectory, which shown in Fig. 6.

Table 3. Coordinates of contact points and the length of long and short shaft.

x	y	z	A	b
7.3056	-497.6686	-80.4069	6.6208	0.5917
6.0825	-497.6447	-77.7858	6.6140	0.4881
4.8604	-497.6813	-75.1647	6.6059	0.4573
3.6374	-497.7784	-72.5437	6.5923	0.4150
2.4116	-497.9359	-69.9226	6.5892	0.4031
1.1811	-498.1538	-67.3015	6.5567	0.3717
0.2540	-498.3568	-65.3357	6.5446	0.3452

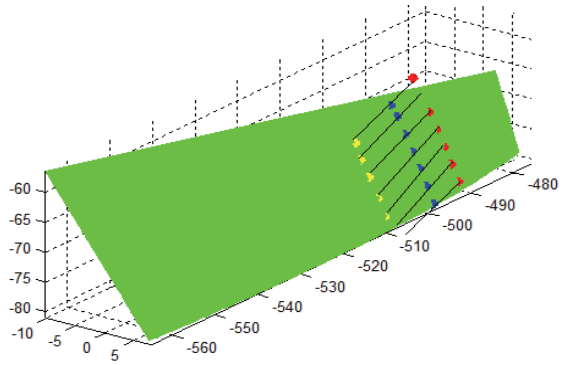


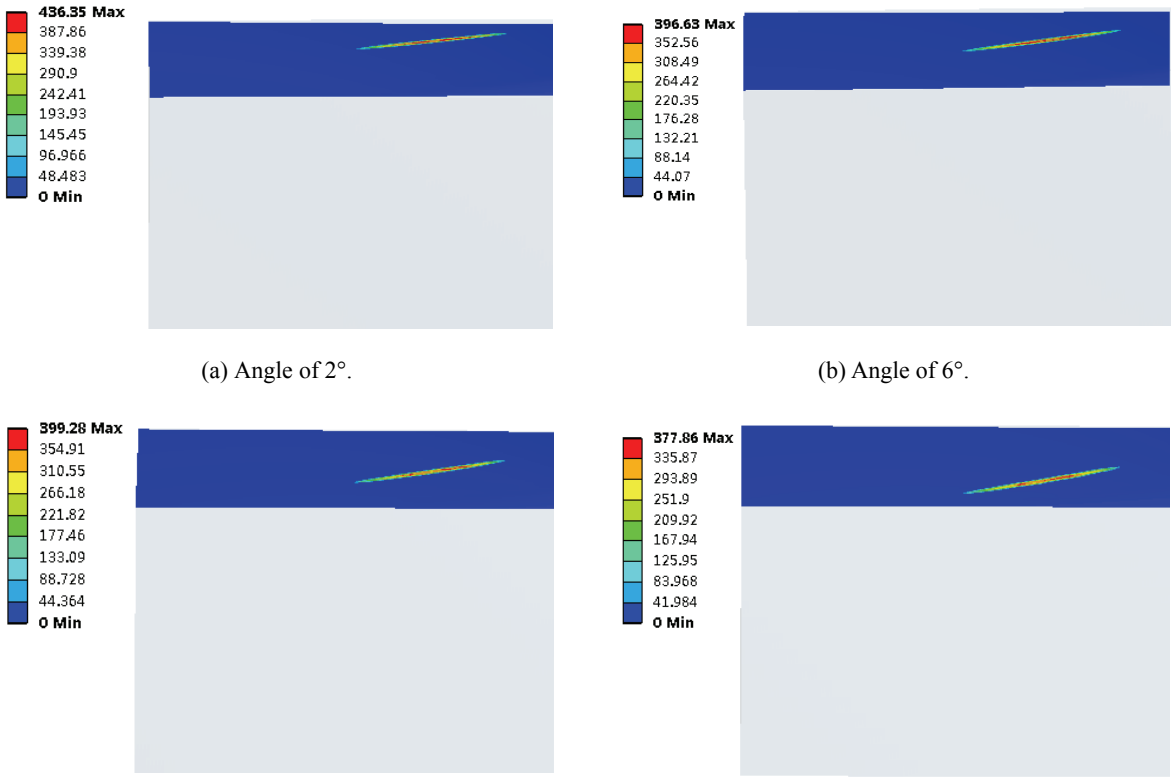
Fig. 6. Contact area of face gear.

The above conclusion is only considering the contact of a pair of tooth. Theoretically get the contact spots and the contact trajectory of face gear. Comparing the theoretical results, use finite element analysis software ANSYS WORKBENCH to analysis single tooth's static contact of face gear pair. Simulation materials for big wheel and pinion are structural steel. Simulation boundary condition is to constraint gearwheel's all degrees of freedom, and give the pinion a certain torque.

Based on the Hertz theory [3], the simulation results converge, when contact surface grid reach 0.2 mm. Under the action of torque, the pinion's each tooth turns an angle of 360/z in a circle, which means the turned angle of each tooth is 14°. Change the pinion 2° each time, so that we can simulate the whole process of dynamic meshing.

In the whole meshing process of the pinion, we study one of its teeth by the simulation. And for the face gear, the contact stress of tooth surface on one sides shown in Fig. 7.

Approximately, put the four contact trajectories shown above on one tooth surface. Then the distribution of contact lines of one tooth surface can be got, as is shown in Fig. 8.



(a) Angle of 2°.

(b) Angle of 6°.

(c) Angle of 10°.

(d) Angle of 14°.

Fig. 7. Contact stress of tooth surface under different angles.

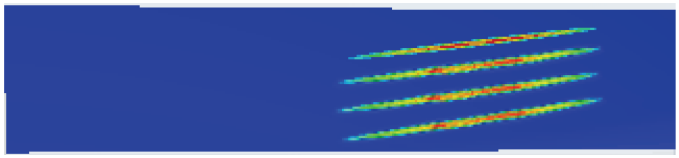
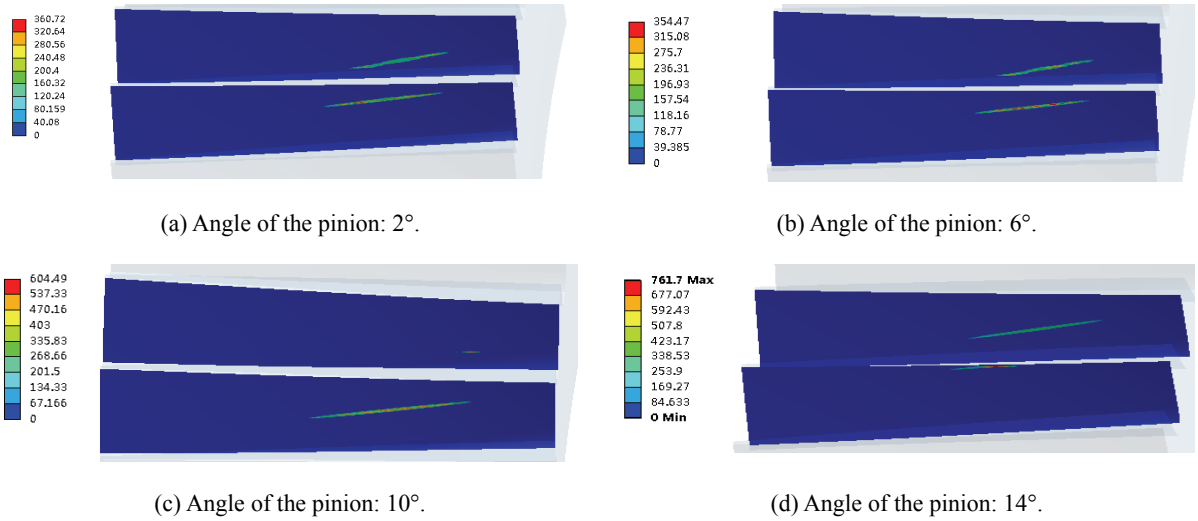


Fig. 8. Contact trajectory distribution of the same tooth surface.

Compare Fig. 6 and Fig. 8. Without considering the influence of gear contact ratio, the face gear's contact spots are close to the small end-face and the tilt direction of contact line is from the small end-face to the large end-face when study a pair of tooth.

Under actual working conditions, contact ratio should be considered. So it is necessary to study the tooth surface contact state of multi-tooth working condition when we study the face gear's contact spots. Using the way of getting contact spots of single-tooth condition, we study the contact spots and contact path of multi-tooth condition in different angles. Specific is shown in following Fig. 9.

Approximately, put the contact trajectories which are in different angles on one tooth surface. Specific is shown in the follow Fig. 10.



(a) Angle of the pinion: 2°.

(b) Angle of the pinion: 6°.

(c) Angle of the pinion: 10°.

(d) Angle of the pinion: 14°.

Fig. 9. Contact trajectory of tooth surface under different angles.

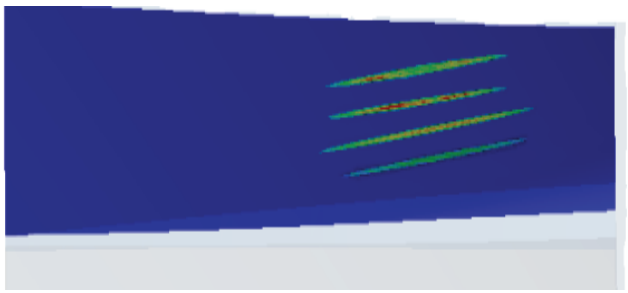


Fig. 10. Contact trajectory of the same tooth surface.

Compare Fig. 8 and Fig. 10. The contact ratio is considered in Fig. 10. So the trend of contact line trajectory becomes long gradually then becomes shorter, which is more accord with the actual meshing process of face gear.

Study on visualization of face gear's theoretical contact spots and face gear's contact areas after its elastic deformation by using software MATLAB. And use software to get the face gear's contact areas of simulation. So the theory and simulation can get verified with each other. Through the study of the contact spots of face gear, may provide certain theory reference for face gear vibration, noise and transmission error.

4. The Study of Face Gear Transmission Error

The ideal gear transmission is conjugate. When involute pinion turns an angle, the face gear (driven gear) also turns a corresponding angle. These two angles have an inverse relationship with two gear tooth number. But actually because of the elastic deformation produced on face gear, the transmission ratio is changed in the loading analysis. When the pinion turns an angle, the deviation of large gear's actual angle and theory angle can be described by the following formula (8):

$$\delta = (\varphi_2 - \varphi_2^0) - z_1(\varphi_1 - \varphi_1^0)/z_2 \tag{8}$$

In the above formula: φ_1, φ_2 are actual angle of the pinion and the large gear. φ_1^0, φ_2^0 are the initial position of two gears. $\frac{z_1}{z_2}$ is the theoretical transmission ratio of the gear pair.

This paper uses dynamic implicit algorithm to study the angular displacement error of face gear with software ABAQUS. To achieve higher precision, all simulations are analyzed by using hexahedron element. Enlarged view of local grid is shown in following Fig. 11.

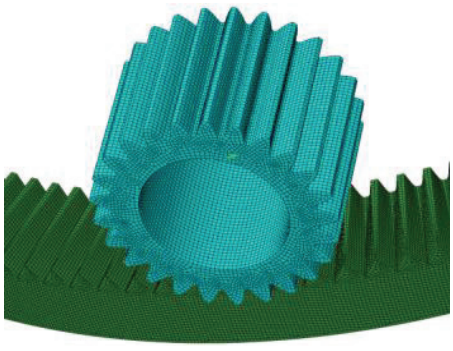


Fig. 11. Model of face gear with mesh.

Based on Hertz theory, we study the tooth surface contact stress through refining mesh size of tooth surface from 1mm to 0.2 mm. In the process, the tooth surface contact stress gradually reaches the theoretical value, but the amount of calculation is increased accordingly. Considering the calculation precision and calculation time, we choose 0.2 mm to be the mesh size. Then we set the boundary conditions of simulation analysis. Set reference point in the pinion's circle center and the large gear's circle center respectively. Use the function of coupling to couple the reference point with the gear's inner surfaces respectively. Apply the pinion's speed to its reference point and apply the torque to the large gear's reference point. Constrain the reference points' degrees of freedom, release the pinion's tangential degrees of freedom and release the large gear's tangential degrees of freedom. Apply a speed of 200r/min to the pinion and apply the resistance moment of: light load: 5E5 $n \cdot mm$, moderate load: 3E6 $n \cdot mm$ and heavy load: 6E6 $n \cdot mm$ to the large gear. The setting is shown in the following Fig. 12.

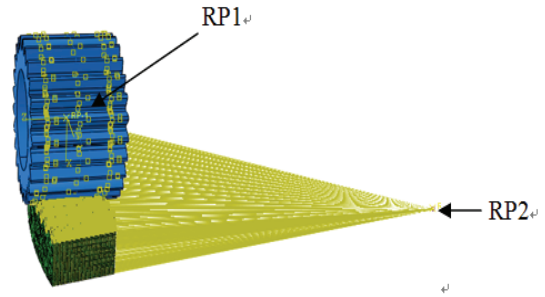


Fig. 12. Boundary conditions of face gear.

Apply the angular velocity on the reference point (RP1) of the pinion's circle center and apply the torque on the reference point (RP2) of large gear's circle center. Study the transmission error while keep the speed of the pinion unchangeable and change the resistance moment of the large gear. We can get the transmission error curves under different conditions. The curves are shown in the following Fig. 13.

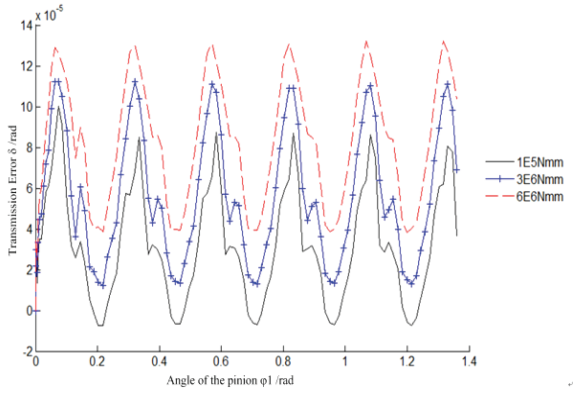


Fig. 13. Transmission error curves under different loads.

From the Fig. 13 we can see that: the transmission error of face gear is small, the shape of error curves approximates to parabola, transmission error curves under three kinds of loads tend to be uniform, and the fluctuation range of the error curves becomes larger as the load is high. In the analysis process, we use the way, which is shown in Fig. 3 to load. So when the pinion's angle turns from 0 to 0.2 radian, the load applied to the face gear increases linearly. Because the load is not stable, there producibility of the curves before 0.2 radian is worse than the reproducibility after 0.2 radian.

This paper studies the characteristic of face gear's transmission error when we apply low speed: 100 r/min, medium speed: 200 r/min, high speed: 400 r/min to the pinion and apply the constant torque of 5E5 $n \cdot m$ to the large gear. The curves are shown in following Fig. 14.

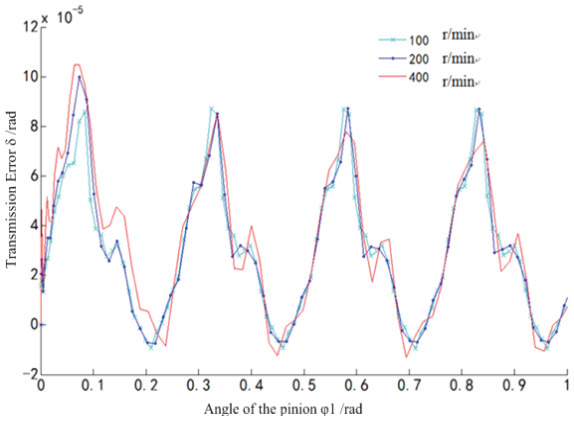


Fig. 14. Transmission error curves under different speeds.

From the Fig. 14 we can see that: when the pinion is in different speeds, the shape of error curves approximates to parabola. The difference between error curves is very small. The absolute differences of adjacent peaks in the curves are all between 9.6E-5 rad and 9.9E-5 rad. In transmission process, we can know that the change of drive gear's speed almost has no influence on face gear's transmission error. In the analysis process, we use the way which is shown in Fig. 3 to load. So when the pinion's angle turns from 0 to 0.2 radian, the load applied to the face gear increases linearly. At this time, the fluctuation range of the error curves increases as the speed becomes higher and the characteristics of gear vibration are more obvious. When the pinion's angle is greater than 0.2 radian, the load is constant and the transmission is stable. This study also explains that face gear pair is suitable for high speed transmission, its noise and vibration are small, and transmission is stable.

Through simulation technology we study the tooth surface transmission error of the face gear with staggered shaft installation error [9]. The angles are $\gamma = 0.02°$, $\gamma = 0°$ and $\gamma = +0.02°$. Use software ABAQUS to respectively extract angle curve of the

large gear and pinion. Then according to formula (6), we get transmission error curves of the face gear under three different shaft angles. The curves are shown in the following Fig. 15.

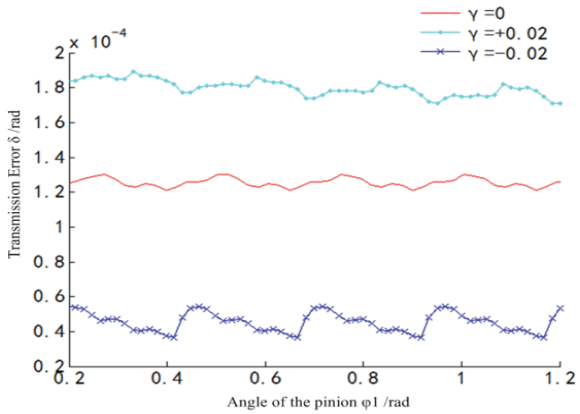


Fig. 15. Transmission errors curves under different shaft angles.

Put the three transmission error curves together. Compare the fluctuation under the three different angles. The curve is shown in the following Fig. 16.

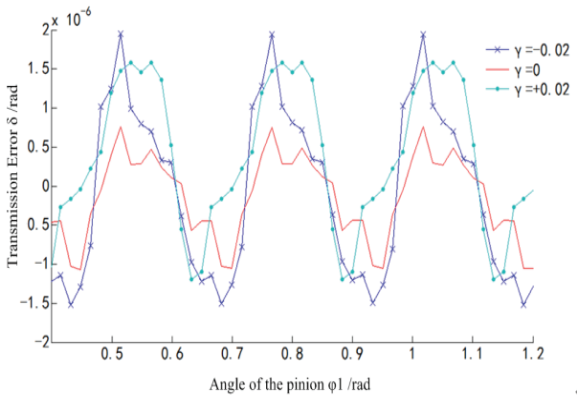
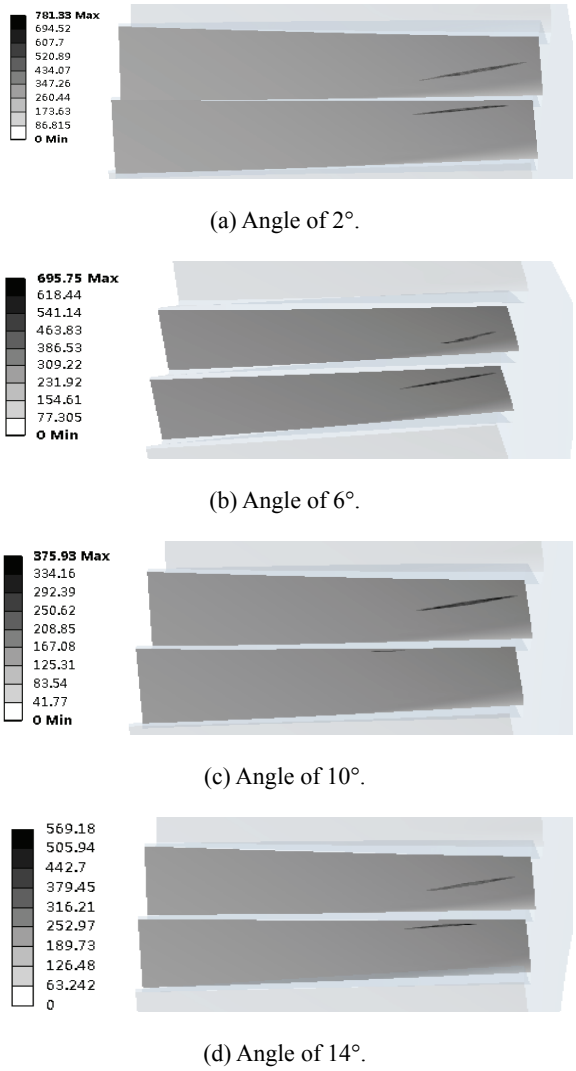


Fig. 16. Comparing of transmission errors curves under different shaft angles.

From the Fig. 16 we can see that when the face gear pair has installation error, the amplitude of transmission error increases obviously. The amplitude under the condition of having the installation error is 2.0 ~ 2.3 times bigger than the amplitude under the condition of standard installation. This accounts for that the vibration and noise will aggravate and the gear transmission precision will decline when the face gear works with installation error.

When the gear pair has a positive installation error: γ = + 0.02, the instantaneous contact stress of face gear under different angles is shown in the following Fig. 17.



(a) Angle of 2°.

(b) Angle of 6°.

(c) Angle of 10°.

(d) Angle of 14°.

Fig. 17. Instantaneous contact stress of face gear with positive installation error.

Put the contact trajectories which are in different angles on one tooth surface. Specific is shown in the following Fig. 18.

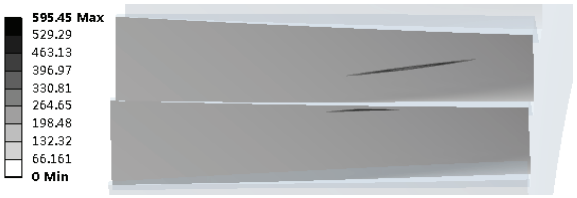
Fig. 18. Contact trajectory of the same tooth surface with positive installation error.

Compare Fig. 10 and Fig. 18, we can see that when the pair has a positive installation error, the tooth contact position is moving to face gear's inner surface and the edge contact is easily happened to the tooth. So the gear vibration and transmission error are increased.

When the gear pair has a negative installation error: $\gamma = -0.02$, the instantaneous contact stress of face gear under different angles is shown in the following Fig. 19.

(a) Angle of 2°.

(b) Angle of 6°.



(c) Angle of 10°.

(d) Angle of 14°.

Fig. 19. Instantaneous contact stress of face gear with negative installation error.

When the pair has a negative installation error, we study the contact spots in the same tooth surface through simulation analysis. The trajectories are show in following Fig. 20.

Fig. 20. Contact trajectory of the same tooth surface with negative installation error.

The face gear contact trajectory moves from its inner diameter to outer diameter when the face gear has negative installation error. The tooth thickness of face gear is not uniform, so when the gear has negative installation error, the gear clearance near the outside part increases in the ocess of gear transmission, tooth collision becomes more severe and vibration and noise are more serious.

5. Conclusion

This paper uses MATLAB to calculate the theory discrete points of the tooth surface of face gear. Through software Pro/E, we get the 3D model of face gear. Work out the coordinates of gear contact points and realize the visualization of face gear contact trajectories by the calculation of contact ellipse. Using software ABAQUS, we work out the face gear's transmission error curves and angular velocity error curves under different working conditions. And through simulation analysis, we get the transmission error curves of face gear with different shaft angle errors. This paper can provide a favorable reference for face gear design, processing and noise reduction.

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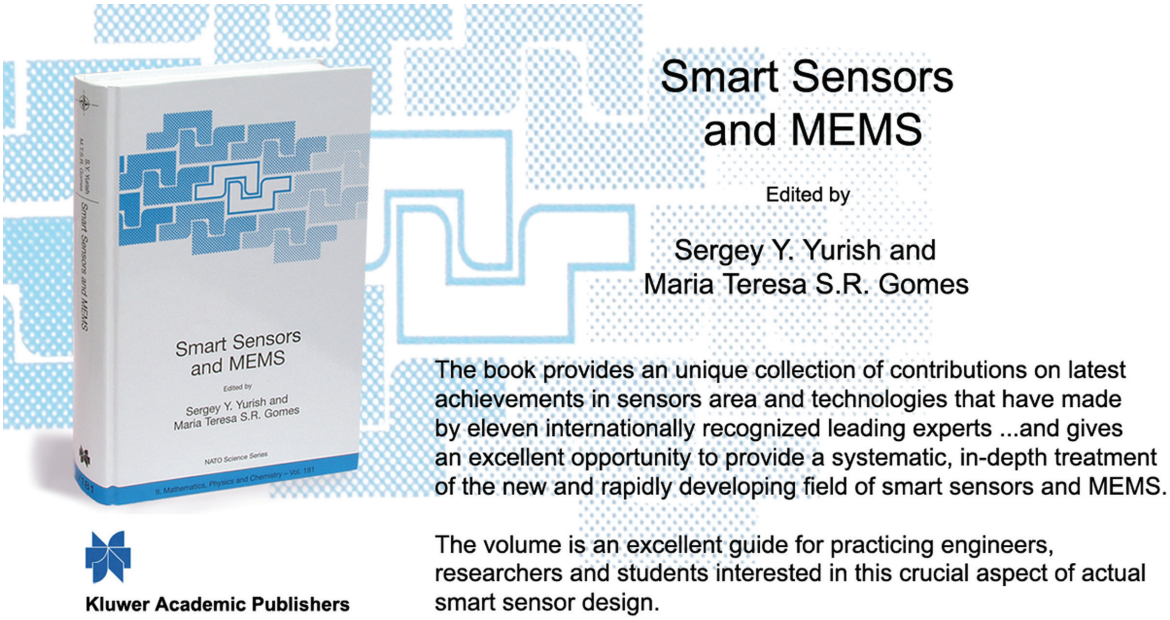

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Genetic Algorithm-based Hybrid Methods for a Flexible Single-operation Serial-batch Scheduling Problem with Mold Constraints

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Abstract: Scheduling in a manufacturing system is currently a research hotspot. This paper proposes a flexible, single-operation, serial-batch scheduling problem with mold constraints that arise from injection molding production. In such a scheduling problem, three decision problems exist: product batch splitting, sequencing, and resource selection. A divide-and-conquer framework is introduced to solve the three decision problems. A hybrid algorithm combining the genetic algorithm for sequencing with the heuristic algorithm for resource selection is developed. We design an experimental case to which the corresponding simulation scheduling process is applied. We present a detailed analysis of the simulation results, which indicate that the solution framework and the hybrid algorithm are effective. *Copyright © 2013 IFSA.*

Keywords: Serial-batch scheduling, Mold constraints, Injection molding production, Genetic algorithm, Dispatching rule.

1. Introduction

The scheduling problem in a manufacturing system has been extensively studied over the past decades. Based on the processing method, scheduling problems can be classified into two types: uniprocessing and batch processing scheduling problems. In batch processing scheduling, jobs are processed in batches. This process indicates that either two or more jobs are processed simultaneously in a machine or jobs in the same family are processed continuously and serially in a machine. The first case is generally referred to as parallel batch scheduling, whereas the latter is called serial-batch scheduling (SBS). Batch processing scheduling has attracted considerable research interest for its practical engineering applications.

In 1984, Santos studied single-machine scheduling problems with batching and sequencing decisions with lead-time considerations. Thereafter, over 100 studies on batch scheduling have been conducted. A great deal of research has been performed by Kovalyov, Brucker, and Potts, who studied molds, algorithms, as well as the complexity of SBS for single [1, 2] and for parallel machines [3, 4]. The models presented were characterized by the availability of all jobs at time zero, set-up time consideration, and weighted delay-associated objective functions. The algorithms that were commonly adopted in previous studies include the dynamic programming, branch and bound, simulated annealing, taboo search, and genetic algorithms [2-6], with dynamic programming being the most widely applied. Moreover, for single-machine and multi-

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machine SBS, Ghosh developed a backward dynamic programming model [7], which is an improvement over dynamic programming [8], as well as a pseudo-polynomial dynamic programming method [9]. Webster *et al.* studied the complexity of SBS for parallel machines [10]. A single-machine SBS for flow time minimization was explored by Mason *et al.* [11]. Cheng also conducted a great deal of research on batch scheduling, particularly on single-machine SBS problems [6, 12, 13]. Cheng proposed several kinds of dynamic programming methods that considered pseudo-polynomial-time.

In this paper, we address a new, flexible, single-operation SBS problem with mold constraints. In this problem, multiple products each have a batch with a predetermined size, a release time, a due date, and an importance measurement (i.e., weight). The jobs for all products have only one operation (e.g., injection processing) that has to be processed on a machine by using a mold based on the product family. The number of available molds for a family job may be more than one. A job can be processed by a machine selected from a specified machine set. However, the matching degree in terms of technology, processing economics, and processing velocity for a machine within the same job-specified machine set may differ. Moreover, differences exist in terms of the colors of jobs belonging to different product families. When a machine processes a deep-color job before a light-color job, this machine has to be cleaned. For the scheduling problem, the objective is to minimize the total weighted tardiness (TWT) of all jobs.

The remainder of this paper is organized as follows: Section 2 states the proposed scheduling problem. Section 3 presents a detailed solution to such problem. A case study on the scheduling problem is described in detail in Section 4. Section 5 presents the scheduling results and corresponding analysis. Finally, the conclusion is presented in Section 6.

2. Problem Description

The problem addressed in this paper can be defined as follows: n batches of products $P = \{P_1, P_2, \cdots, P_i, \cdots, P_n\}$ are to be processed. The families and due dates of different batches are not the same. In a shop, *ma* machines $Ma = \{Ma_1, Ma_2, \cdots, Ma_{ma}\}$ and *mo* sets of molds $Mo = \{Mo_1, Mo_2, \cdots, Mo_{mo}\}$ can be used. For product P_i, we denote n_i as its batch size and w_i, co_i, p_i, r_i, and d_i as its weight, color, job processing time, release time, and due date, respectively. When a job is to be processed, a corresponding mold must be equipped to the corresponding machine. For a job of P_i, the mold can be selected from

$$Mo_i = \{Mo_{i1}, Mo_{i2}, \cdots, Mo_{ij}, \cdots, Mo_{iq_i}\},$$

where q_i denotes the number of available molds for P_i. The processing machine will be one among those in a machine set expressed as the following matrix:

$$Ma_i = \begin{bmatrix} Ma_{i1} \\ Ma_{i2} \\ \vdots \\ Ma_{ik} \\ \vdots \\ Ma_{im} \end{bmatrix} = \begin{bmatrix} Ma_{i11} & Ma_{i12} & \cdots & Ma_{i1l} & \cdots \\ Ma_{i21} & Ma_{i22} & \cdots & Ma_{i2l} & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ Ma_{ik1} & Ma_{ik2} & \cdots & Ma_{ikl} & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ Ma_{im1} & Ma_{im2} & \cdots & Ma_{iml} & \cdots \end{bmatrix} \quad (1)$$

In the machine matrix, we divide the machines into *m* grades according to their technological and economic matching degree with P_i. All machines in $Ma_{i1} = \{Ma_{i11}, Ma_{i12}, \cdots, Ma_{i1l}, \cdots\}$ have the best matching degree, whereas all machines in $Ma_{im} = \{Ma_{im1}, Ma_{im2}, \cdots, Ma_{iml}, \cdots\}$ have the worst matching degree.

To solve this scheduling problem, we initially split all product batches into several lots, after which we schedule all the lots. The scheduling process involves the sequencing of lots and the selection of the optimal machine and the optimal mold, such that the TWT of all products is minimized [14]. Let s_i denote the processing starting time of P_i; and let c_i and $\vec{t}_i$ be the processing finishing time and the weighted tardiness, respectively. The TWT $\vec{T}$ of all products can thus be derived as follows:

$$\vec{T} = \sum_{i=1,2,\cdots,n} \vec{t}_i = \sum_{i=1,2,\cdots,n} w_i \cdot \max(0, c_i - d_i) \quad (2)$$

3. Problem Solving Methods

3.1. Solving Framework

To solve the given scheduling problem, three decisions have to be made: (1) splitting of all product batches into lots; (2) optimally sequencing all lots; and (3) selecting the optimal processing machine and mold. A divide-and-conquer framework is introduced to address the aforementioned decisions (Fig. 1).

First, all batches of products were split based on a splitting rule. We then sequenced all product lots by using the genetic algorithm, where a chromosome corresponds to a lot sequence. Finally, a chromosome is decoded according to the decision rule and resource selection, while simultaneously calculating the processing time intervals calculating. The splitting result, the lot sequence, and the schedule comprise the solution to the scheduling problem.

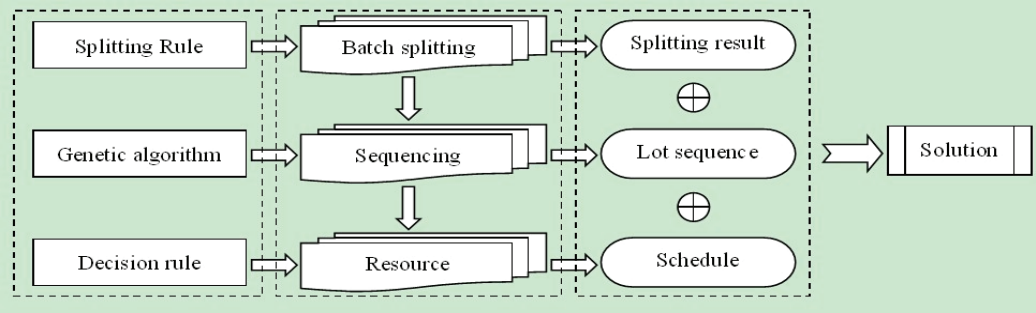

Fig. 1. Framework for solving the scheduling problem.

3.2. Product Batch Splitting

As previously mentioned, when a job is to be processed on a machine, a mold must be equipped beforehand. Thus, this procedure requires a relatively long setup time. When a machine processes a job belonging to one product family that is then immediately preceded by a job from another product family, the disassembly of the previous mold and the assembly of the subsequent one would cost a great deal of time. To reduce setup times while improving processing efficiency by fully utilizing mold resources, we split the product batches based on the following rules:

For P_i, if only one mold is available, i.e., $q_i=1$, then the batch of product P_i need not be split, and the products should consecutively be processed on a machine with the same mold.

For P_i, if more than one mold is available, i.e., $q_i > 1$, the batch of product P_i can possibly be split into several lots for concurrent processing on different machines with different molds. Thus, we split the batch according to the equipartition principle into q_i lots that are denoted as $P_i^{x_i}$, $x_i = 1,2,\cdots,q_i$. Let $b_i^{x_i}$ denote the size of $P_i^{x_i}$. Then,

$$b_i^{x_i} = \begin{cases} \lfloor n_i/q_i \rfloor & x_i = 1,2,\cdots;q_i-1 \\ n_i-(q_i-1)\cdot\lfloor n_i/q_i \rfloor & x_i = q_i \end{cases}, \quad (3)$$

where $\lfloor x \rfloor$ is the floor function about x.

Notably, given that splitting precedes resource selection, we split a product batch into small lots, such that for product P_i, the number of lots equals the static available quantity of mold q_i. Thus, the decision space for subsequent scheduling will be enlarged. However, considering dynamic factors such as mold maintenance and breakdown when scheduling, the dynamic available quantity of a mold may be only one. Thus, all lots may be assigned to the same mold and the same machine for consecutive processing.

3.3 Genetic Algorithm for Lot Sequencing

To sequence all product lots, we adopt the genetic algorithm based on symbol encoding, where every chromosome represents a sequencing scheme. By scheduling the sequencing scheme, we can make the processing resource decision as well as calculate the processing times and TWT.

3.3.1. Chromosome Encoding

By splitting all product batches, we can derive the number of product lots $P_i^{x_i}$, where $i = 1,2,\cdots,n$, and $x_i = 1,2,\cdots,q_i$. Let N_{sub} denote the number of lots of all product batches. We thus obtain

$$N_{sub} = \sum_{i=1,2,\cdots,n} q_i \quad (4)$$

Every legal chromosome is an orderly symbol string comprising $P_i^{x_i}$ ($i = 1,2,\cdots,n; x_i = 1,2,\cdots q_i$) with length N_{sub}. For example, three product batches exist: P_1, P_2, and P_3, each of which are split into three lots: P_1^1, P_1^2, and P_1^3; P_2^1, P_2^2, and P_2^3; and P_3^1, P_3^2, and P_3^3, respectively. Fig. 2 illustrates a case of a legal chromosome.

P_1^2	P_3^3	P_2^1	P_1^3	P_2^2	P_3^2	P_2^3	P_3^1	P_1^1

Fig. 2. Case of a legal chromosome.

3.3.2. Selection Operator

In this paper, we use tournament selection as the selection operator. Commonly used methods include binary and ternary tournament selection. Evidently, a greater number of chromosomes participating in a tournament results in the higher homoplasy and higher average fitness of such chromosomes. Thus,

the evolutionary procedure will facilitate rapid convergence. However, this procedure may cause population diversity to worsen and catastrophe to occur. To achieve a tradeoff between convergence rate and population diversity, we found that an appropriate number of individuals to be considered in tournament selection may be derived according to the population size as follows:

$$\Pi = \begin{cases} 2 & if: popu \le 20 \\ 4 & f: popu \ge 40 \\ 3 & else \end{cases}, \quad (5)$$

where $popu$ denotes the population size. For example, let $popu = 20$, and let binary tournament selection be adopted. For each instance that two chromosomes are drawn through to simple random sampling with replacement, the fitness of each chromosome is evaluated, and the individual with the higher fitness is retained for the subsequent genetic operation. This process is repeated until 20 chromosomes are selected.

3.3.3. Crossover Operator

In the proposed genetic algorithm, a chromosome is a symbol string comprising the representation of all product lots. Thus, a legal chromosome must satisfy the completeness and exclusiveness criteria. Part mapping crossover is adopted to ensure that the offspring are legal. We take a sample that includes three product batches: P_1, P_2, and P_3, each of which are split into three lots. Fig. 3 (a) represents two parent individuals. After applying part mapping crossover, we derive the offspring individuals as shown in Fig. 3(b).

parent 1	P_1^2	P_3^3	P_2^1	P_1^3	P_2^2	P_2^3	P_3^2	P_3^1	P_1^1
parent 2	P_2^3	P_1^1	P_1^3	P_3^2	P_3^1	P_2^2	P_1^2	P_3^3	P_2^1

(a) parent chromosomes.

offspring 1	P_2^1	P_3^3	P_1^3	P_3^2	P_3^1	P_2^2	P_1^2	P_2^3	P_1^1
offspring 2	P_3^1	P_1^1	P_2^1	P_1^3	P_2^2	P_2^3	P_3^2	P_3^3	P_1^2

(b) offspring chromosomes.

Fig. 3. Example of part mapping crossover.

3.3.4. Mutation Operator

Mutation is implemented in the manner similar to a basic swap operation, i.e., a chromosome is randomly selected while two loci are randomly generated. Genes located in the two loci are then mutually exchanged. Taking the offspring chromosome 1 in Fig. 3 as an example, let the loci be 3 and 6. The resulting mutation is then shown in Fig. 4.

P_2^1	P_3^3	P_2^2	P_3^2	P_3^1	P_1^3	P_1^2	P_2^3	P_1^1

Fig. 4. An example of mutation.

3.4. Resource Decision

In Subsection 3.3, we discussed that sequencing only provides the dispatching order for all product lots. Hence, we should schedule all lots so that the most suitable resources, i.e., processing machine and mold, could be assigned and processing time intervals could be calculated. Considering the technological and economic matching degree of machines with respect to the product family and TWT objective, we introduce a two-step combination rule: Earliest Tardiness Time (ETT) + Best Matching Degree (BMD), which operates as follows: when a product lot is to be scheduled, a processing resource combination (a machine and a mold) with the least weighted tardiness will be assigned for the lot. If more than one processing resource combination with the same least weighted tardiness exists, then the one with the highest matching degree will be selected. On the other hand, if more than one processing resource combination with the same least weighted tardiness and the same highest matching degree exists, any one can be selected at random. When deciding on resources, we must calculate the weighted tardiness of the lot. Thus, the setup, starting, and finishing times are introduced in the following subsections.

3.4.1. Setup Time

Molds are indispensable technological equipment for injection processing. Before a job is processed, a mold must be set up on a machine. Setup time, which is usually exceeds the processing time of a single job, is this required. Given that a mold can be assembled before a job arrives and that setup times are independent of the number of jobs processed on the mold, setup time should not be integrated with processing time. The composition of setup times can be classified into four cases: a machine processes a product lot without any forerunner, because of which assembling time occurs; a machine successively processes two lots of products belonging to the same family, because of which no setup time occurs; a machine successively processes two lots of products belonging to different families and the color of the former product is the same as or lighter than that of the latter, because of which the disassembling time of the mold for the former product lot and the assembling time of the mold for the subsequent product occur; and finally, the color of the former product is darker than that of the latter, because of which the setup time will include the disassembling,

assembling, and washing times. Let the product lot to be scheduled now be P_i^*. The selected machine is Ma_i^* ($Ma_i^* \in Ma_i$); the selected mold is Mo_i^* ($Mo_i^* \in Mo_i$); and the previous product lot and its mold are P_{i-1}^* and Mo_{i-1}^*, respectively. We then provide the following conventions:

If two molds are the same, $Mo_i^* = Mo_{i-1}^*$;

If the color of product P_i is lighter than that of product P_{i-1}, $co_i < co_{i-1}$; and if the color of product P_i is the same as or darker than that of product P_{i-1}, $co_i \geq co_{i-1}$.

For product P_i, we denote the assembling, disassembling, and washing times as t_{i-asm}, t_{i-dasm}, and t_{i-wash}, respectively. The setup time, denoted as $t_{i-setup}$, can then be presented as

$$t_{i_aux}=\begin{cases} t_{i-asm} & if: P_{i-1}^* = empty \\ 0 & if: Mo_i^* = Mo_{i-1}^* \quad and \quad co_i \geq co_{i-1} \\ t_{i-wash} & if: Mo_i^* = Mo_{i-1}^* \quad and \quad co_i < co_{i-1} \\ t_{i-asm}+t_{i-dasm} & if: Mo_i^* \neq Mo_{i-1}^* \quad and \quad co_i \geq co_{i-1} \\ t_{i-asm}+t_{i-dasm}+t_{i-wash} & if: Mo_i^* \neq Mo_{i-1}^* \quad and \quad co_i < co_{i-1} \end{cases} \tag{6}$$

3.4.2. Starting Time

In the proposed scheduling problem, for each product lot to be processed, the following conditions must be satisfied: (1) the product lot reaches the corresponding machine; (2) at least one corresponding mold is available at each moment; (3) at least one corresponding machine is available at each moment; and (4) the mold is assembled on the machine. With respect to product lot P_i^*, the available time of the mold is denoted as $t_{Mo_i^*-avail}$. Thus, the starting time, denoted as s_i^*, can be computed by

$$s_i^* = \max\{r_i, t_{i-asm} + \max(t_{Mo_i^*-avail}, c_{i-1} + t_{i-wash} + t_{i-dasm})\} \tag{7}$$

which is illustrated in Fig. 5.

3.4.3. Processing Times of a Product Lot

In injection molding production, a mold usually has a multi-cavity feature, such that it can simultaneously process more than one job according the number of cavities. Let ω_i^* denote the number of cavities of Mo_i^*, such that ω_i^* identical jobs can be processed in an injection technological cycle. The processing time for these ω_i^* jobs is equal to that for processing a single job, i.e., p_i. As previously mentioned, the processing velocity for a machine within the same job-specified machine set may be different. Thus, we adopt the velocity factor v_i^* for Ma_i^* to measure processing velocity. For the product lot P_i^*, the size of which is b_i^*, the processing time bp^{i*} is calculated by

$$bp_i^* = \left\lceil \frac{b_i^*}{\omega_i^*} \right\rceil \cdot p_i / v_i^*, \tag{8}$$

where $\lceil x \rceil$ is the ceiling function about x.

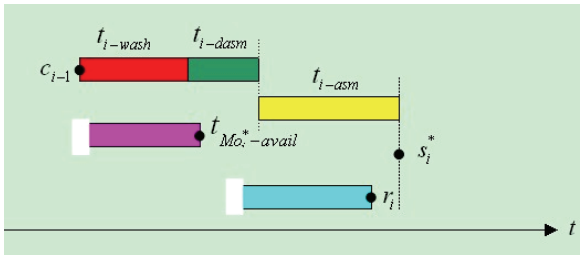

Fig. 5. Illustration of starting time computation.

3.4.4. Finishing Time and Weighted Tardiness

For a product lot P_i^*, the finishing time c_i^* can be calculated based on the starting and processing times by using the following formula:

$$\begin{aligned} c_i^* &= s_i^* + bp_i^* \\ &= \max\{r_i, t_{i-asm} + \left\lceil \frac{b_i^*}{\omega_i^*} \right\rceil \cdot p_i / v_i^* \\ &+ \max(t_{Mo_i^*-avail}, c_{i-1} + t_{i-wash} + t_{i-dasm})\} \end{aligned} \tag{9}$$

and the weighted tardiness $\vec{t}_i^*$ can be computed by

$$\vec{t}_i^* = w_i \cdot \max(0, c_i^* - d_i) \tag{10}$$

3.4.5. Resource Selection

After generating a processing sequence for the product lots, the most suitable resources should be assigned for each lot. According to the resource

decision combination rule EET+BMD, we can derive the resource selection algorithm as procedure 1.

Procedure 1 [Procedure for resource selection]

$\vec{t}_{\text{benchmark}}$: benchmark of weighted tardiness

$Ma_i^{selected}$: selected machine

$Mo_i^{selected}$: selected mold

Begin

$\vec{t}_{\text{benchmark}} = +\infty$

For k=1 to k=m

For l=1 to l= y_k

$Ma_i^* := Ma_{ikl}$

For j=1 to q_i

$Mo_i^* := Mo_{ij}$

$\vec{t}_i^* = w_i \cdot \max(0, c_i^* - d_i)$

If ($\vec{t}_i^* < \vec{t}_{\text{benchmark}}$)

$\vec{t}_{\text{benchmark}} := \vec{t}_i^*$

$Ma_i^{selected} := Ma_i^*$

$Mo_i^{selected} := Mo_i^*$

End if

End for

End for

End for

End

Referring to the definition of the technological and economic matching degree of machines, we can easily determine that matching degree can be measured by using the loop variable k. A smaller value of k denotes a higher matching degree. In the resource selection procedure, loop variable k progressively increases as the loop runs, such that resources with a higher matching degree will be selected automatically.

4. Case Study

4.1. Case Information

In this section, we discuss a case to illustrate the scheduling problem and the proposed solving methods. In the proposed case, 15 batches of products are to be processed in a shop with eight sets of machines and 13 sets of molds. The information on product batches is presented in Table 1. The details on the machines and molds are presented in Table 2 and Table 3, respectively. The washing times necessary for a machine to process two different colors of products successively are given in Table 4. Sometimes, a color taboo exists for when two products are processed successively in the same machine. For example, if the color of the previously processed product is black, the immediate successor should not be a white product. To implement such a taboo, we set an appropriate duration for the washing times, e.g., 1000 units of time, as shown in Table 4.

Table1. Information on product batches.

Basic information								Resource information	
ID	**Color**	**Batch size**	**Weight**	**Processing time**	**Release time**	**Due date**		**Machine set**	**Mold set**
P1	white	500	1	2	0	600		$\begin{bmatrix} Ma_1 & Ma_2 \\ Ma_3 & Ma_4 \end{bmatrix}$	$\{Mo_1, Mo_2\}$
P2		500			0	800			
P3	yellow	1000	2	3	10	600		$\begin{bmatrix} Ma_1 & Ma_2 \\ Ma_5 & Ma_6 \end{bmatrix}$	$\{Mo_3\}$
P4	blue	1000	1		5	1300		$[Ma_3 \quad Ma_4]$	$\{Mo_4\}$
P5		1000			5	1800			
P6	red	800	3	4	20	1000		$\begin{bmatrix} Ma_5 & Ma_6 \\ Ma_7 & Ma_8 \end{bmatrix}$	$\{Mo_5\}$
P7	black	500	2	3	10	800		$[Ma_1 \quad Ma_2]$	$\{Mo_6\}$
P8		500			10	900			
P9		500			10	1000			
P10	white	2000	2	6	50	2600		$\begin{bmatrix} Ma_1 & Ma_2 \\ Ma_5 & Ma_6 \end{bmatrix}$	$\{Mo_7, Mo_8\}$
P11	white	1200	1	4	0	1200		$\begin{bmatrix} Ma_7 & Ma_8 \\ Ma_3 & Ma_4 \end{bmatrix}$	$\{Mo_9\}$
P12	red	800	1	3	0	1500		$[Ma_7 \quad Ma_8]$	$\{Mo_{10}\}$
P13		800			0	2000			
P14	blue	1500	1	2	30	1000		$\begin{bmatrix} Ma_5 & Ma_6 \\ Ma_1 & Ma_2 \end{bmatrix}$	$\{Mo_{11}, Mo_{12}\}$
P15	yellow	600	2	6	20	1200		$[Ma_3 \quad Ma_4]$	$\{Mo_{13}\}$

Table 2. Information on machines.

ID	Initial available time	Velocity factor
Ma_1	10	0.8
Ma_2	0	1.0
Ma_3	0	1.2
Ma_4	50	1.0
Ma_5	20	1.0
Ma_6	10	1.5
Ma_7	100	0.8
Ma_8	50	1.0

Table 3. Information on molds.

ID	Number of cavities	Initial available time	Assembling times	Disassembling times
Mo_1	4	0	50	20
Mo_2	4	10	50	20
Mo_3	4	15	40	10
Mo_4	4	8	60	20
Mo_5	4	0	80	20
Mo_6	4	20	50	20
Mo_7	4	12	70	30
Mo_8	4	18	70	30
Mo_9	4	0	30	10
Mo_{10}	4	30	35	10
Mo_{11}	4	0	50	20
Mo_{12}	4	0	50	20
Mo_{13}	4	20	45	20

Table 4. Washing time for color switching.

Color of previous product	Color of subsequent product				
	white	yellow	red	blue	black
white	0	0	0	0	0
yellow	10	0	0	0	0
red	20	10	0	0	0
blue	30	20	10	0	0
black	*1000*	30	20	10	0

4.2. Product Batch Splitting and Chromosome Encoding

According to the proposed rules for the splitting of product batches, P1, P2, P10 and P14 can each be split into two product lots, such that all batches would initially have two sets of available molds. However, other product batches cannot be split, given that each of them has only one set of mold is available. Thus, we have 19 product lots to be scheduled. A legal chromosome with 19 genes is illustrated in Fig. 6.

P_1^2	P_3^3	P_2^1	P_1^3	P_2^2	P_3^2	P_2^3	P_3^1	P_1^1

Fig. 6. Example of a legal chromosome.

5. Simulation Scheduling and the Analysis of the Results

5.1. Simulation Scheduling and the Results

To analyze and evaluate the proposed scheduling methods, we develop a simulation scheduling system using Java programming language. We conduct 10 runs of simulation scheduling for the above case using the system. In our simulation scheduling runs, the operation parameters of the genetic algorithm are set as follows: population size is 20, crossover probability is 0.8, mutation probability is 0.1, and the maximum iterations are 200. The elitist selection strategy is adopted in the genetic algorithm. Results of the 10 runs of simulation scheduling are shown in Table 5. In these results, the optimal fitness, i.e., the minimal TWT, is 937.6, and the corresponding solution is illustrated in Fig. 7.

Table 5. Results of 10 runs of simulation scheduling based on the genetic algorithm.

Number of runs	Results		
	Best fitness	Generationth for the best fitness	Run times(s)
1	937.6	29	0.422
2	937.6	29	0.438
3	937.6	26	0.375
4	937.6	43	0.438
5	937.6	18	0.344
6	937.6	37	0.438
7	1356.3	69	0.594
8	937.6	14	0.344
9	1356.3	59	0.516
10	937.6	18	0.422

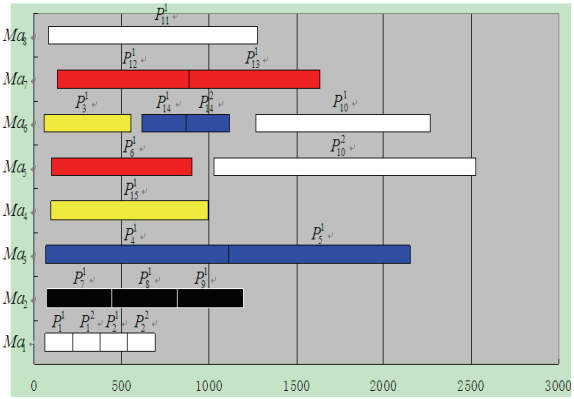

Fig. 7. Gantt chart of the best scheduling solution based on the genetic algorithm.

In our simulation scheduling, dispatching rules, with which we aim at evaluating the proposed genetic algorithm, are also adopted in product lot sequencing. Based on the features of the proposed scheduling problem, we propose two dispatching rules: Shortest

Estimate of Batch Processing Time (SEBPT) and Least Estimate of Batch Slack (LEBSL). For a product lot $P_i^{x_i}$, the estimates of batch processing time and of batch slack, respectively denoted by $b\hat{p}_i^{x_i}$ and $b\hat{s}_i^{x_i}$, are defined as follows.

Estimate of batch processing time: For any of a product lot, multiple processing machines are available, such that the processing times are machine-dependent. We can only derive an estimate of the batch processing times before the resource decision is made. By ignoring the velocity factor, $b\hat{p}_i^{x_i}$ can be derived as

$$b\hat{p}_i^{*} = \left\lceil \frac{b_i^{*}}{\omega_i^{*}} \right\rceil \cdot p_i \tag{11}$$

Estimate of batch slack: Given that the starting and processing times of a product lot are dependent on the selected machine and the selected mold, slack cannot be calculated accurately before the resource decision is made. We thus provide the estimate of batch slack as follows:

$$b\hat{s}_i^{x_i} = d_i - (r_i + b\hat{p}_i^{x_i}) \tag{12}$$

With respect to SEBPT, the priority of a product lot is given as $Z_i^{x_i} = b\hat{p}_i^{x_i}$, and with respect to LEBSL, the priority is given as $Z_i^{x_i} = b\hat{s}_i^{x_i}$. The product lot with the least $Z_i^{x_i}$ is chosen for loading.

Table 6 presents the detailed results of the rule-based simulation scheduling. The Table 6 shows that the TWTs for SEBPT and LEBSL are 2285 and 4343.8, respectively.

Table 6. Results of rule-based simulation scheduling.

Product lots	Results based on SBPT (TWT=2285.0)				Results based on LEBSL (TWT=4343.8)			
	Machine	Mold	Starting time	Finishing time	Machine	Mold	Starting time	Finishing time
P_1^1	Ma_1	Mo_1	60.0	217.5	Ma_2	Mo_1	50.0	176.0
P_1^2	Ma_1	Mo_1	217.5	375.0	Ma_2	Mo_1	176.0	302.0
P_2^1	Ma_1	Mo_1	375.0	532.5	Ma_1	Mo_1	1218.8	1376.3
P_2^2	Ma_1	Mo_1	532.5	690.0	Ma_1	Mo_1	1376.3	1533.8
P_3^1	Mo_6	Mo_3	55.0	555.0	Ma_6	Mo_3	55.0	555.0
P_4^1	Ma_3	Mo_4	895.0	1936.7	Ma_4	Mo_4	110.0	1360.0
P_5^1	Ma_3	Mo_4	1936.7	2978.3	Ma_4	Mo_4	1360.0	2610.0
P_6^1	Ma_8	Mo_5	130.0	930.0	Ma_5	Mo_5	100.0	900.0
P_7^1	Ma_2	Mo_6	70.0	445.0	Ma_1	Mo_6	70.0	538.8
P_8^1	Ma_2	Mo_6	445.0	820.0	Ma_2	Mo_6	588.8	963.8
P_9^1	Ma_2	Mo_6	820.0	1195.0	Ma_2	Mo_6	963.8	1338.8
P_{10}^1	Ma_5	Mo_7	972.0	2472.0	Ma_6	Mo_8	1015.7	2015.7
P_{10}^2	Ma_6	Mo_8	645.0	1645.0	Ma_5	Mo_7	1030.0	2530.0
P_{11}^1	Ma_4	Mo_9	80.0	1280.0	Ma_3	Mo_9	30.0	1030.0
P_{12}^1	Ma_7	Mo_{10}	135.0	885.0	Ma_7	Mo_{10}	135.0	885.0
P_{13}^1	Ma_7	Mo_{10}	885.0	1635.0	Ma_7	Mo_{10}	885.0	1635.0
P_{14}^1	Ma_5	Mo_{11}	70.0	446.0	Ma_1	Mo_{12}	618.8	1088.8
P_{14}^2	Ma_5	Mo_{11}	446.0	822.0	Ma_6	Mo_{11}	615.0	865.7
P_{15}^1	Ma_3	Mo_{13}	65.0	815.0	Ma_3	Mo_{13}	1085.0	1835.0

5.2. Analysis of Results

5.2.1. Comparative Analysis of the Results of Genetic Algorithm-based and the Rule-based Scheduling

The minimal TWT for genetic algorithm-based scheduling is 937.6, and those for SEBPT and LEBSL are 2285 and 4343.8, respectively. Genetic algorithm-based scheduling evidently outperforms rule-based scheduling. An effective dispatching rule should generally reflect the effects of the main factors on scheduling. However, in the proposed scheduling problem, several factors affect the scheduling results including product batch information, multiple machines with different technological and economic matching degree, multiple molds, and sequence-dependent washing times. Thus, developing a rule with single-valued priority including all influencing factors is difficult. Thus, achieving a satisfactory scheduling result by using rule-based sequencing for the scheduling problem proposed in this paper is difficult.

5.2.2. Analysis of the Results of Genetic Algorithm-based Scheduling

Table 5 shows that when population size is 20 and the maximum iterations are 200, eight out of 10 runs of simulation scheduling achieve the best fitness, i.e., 937.6 and two runs achieve a fitness of 1356.3. We conduct another scheduling test with population size 50 and maximum iterations of 200, in which all 20 runs achieved the best fitness at 937.6. The results indicate that the genetic algorithm-based solving methods proposed in this paper have good stability.

Fig. 7 illustrates the best scheduling solution through a Gantt chart. Based on Fig. 7, the detailed analyses are given as follows:

In the case with mold assembling times, the starting time for the first product lot on a machine lags behind the initial available time of the machine. For example, the initial available time of Ma_1 is 10. However, the assembling time of Mo_1, which is assigned to the first product lot P_1^1, is 50. Thus, the starting time of P_1^1 is 60 (10 plus 50).

On Ma_5, the red product lot P_6^1 is processed immediately before the white product lot P_{10}^2. This order is unreasonable because red is darker than white, thus requiring washing times. However, we note that no tardiness occurred with the two product lots in this processing sequence. However, if we switch the sequence, P_6^1 will be delayed. Furthermore, given that the weight of P_6^1 is relatively large (w_i =3), a large weighted tardiness will be incurred for P_6^1. The same condition also appears on Ma_6 (P_{14}^2 precedes P_{10}^1).

As mentioned in Subsection 3.2, splitting a product batch into small lots aims at enlarging the decision space for scheduling. If an idle mold is available for a product lot at its scheduling moment, parallel processing of the same family of product lots may occur. Otherwise, different lots belonging to the same product batch will be scheduled to be processed consecutively on the same machine to minimize setup times. For example, product batch P_1 is split into two lots, P_1^1 and P_1^2. However, these lots are assigned to the same machine (Ma_1) and processed consecutively. The same condition also occurs for P_2 and P_{14}. With respect to P_1 and P_2, such results not only reduce setup times but also ensure their due date (i.e., tardiness is zero). For P_{14}, tardiness occurs for product lot P_{14}^2. However, to prevent the delay of other product lots, the tardiness of P_{14}^2 will be larger if we assign it to another machine. Fig. 7 shows that parallel processing occurs for product batch P_{10}, which is split into two lots and assigned to different machines (Ma_5 and Ma_6) to processed using different equipped molds (Mo_7 and Mo_8). Such an arrangement ensures P_{10}'s due date. Otherwise, if P_{10} is not split, tardiness will be inevitable.

P_3^1 is arranged for processing on machine Ma_6. This arrangement seems unreasonable because the matching degree of Ma_6 is lower than those of Ma_1 and Ma_2, which also belong to the processing machine set of P_3^1. However, Ma_6 has the largest velocity factor (i.e., 1.5), such that the due date of P_3^1 can be ensured. If P_3^1 is assigned to Ma_1 or Ma_2, the respective velocity factors of which are 0.8 and 1.0, tardiness will inevitably occur for P_3^1.

6. Conclusions

This paper addressed a flexible, single-operation SBS problem with mold constraints arising from injection molding production. To solve this problem, we propose solving methods including genetic algorithm-based sequencing and rule-based resource decision. A case with 15 product batches is studied by conducting simulation scheduling. In the case study, we also propose two rules, SEBPT and

LEBSL, which are adopted during product lot sequencing. The results and the analysis show that the genetic algorithm-based sequencing method has good stability and evidently outperforms the rule-based sequencing method.

Acknowledgements

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Application of New Matching Technique in Doherty Amplifier

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Abstract: According to the shortage of the traditional offset line in Doherty power amplifier, a new offset line technique is proposed to match carrier amplifier with the load and to improve the performance of the Doherty amplifier. By simulation of the computer software, a higher efficiency is obtained using the new offset line comparing the two kinds of offset lines. The new offset line technique could be applied in the system with high linearity and low power operation. *Copyright © 2013 IFSA.*

Keywords: Doherty power amplifier, Offset line, Analogous offset line.

1. Introduction

With the steady and effective process of social informatization, transmission network, content diversity, terminal mobile and commercialization operation, in recent years people have seen the fast replacement and development of high speed mobile communication system and broadband multimedia broadcast system. Along with the signal bandwidth of wireless transmission is continuously increasing, the inherent frequency selective fading of wireless channel and narrow band interference effects become the main factors of wireless transmission effect.

In the modern wireless communication system, Doherty technology [1] is one kind of amplifier design technology with best commercial prospect, because of its simple structure, low cost, is easy to implement, the influence to system linearity is relatively minor and is suitable for high efficiency transmission of modern wireless signal with Peak to Average Power Ratio.

This paper firstly introduce the fundamental principles and offset line technology of Doherty power amplifier, secondly put forward the concept and calculation method of analogous offset line according to shortage of low power line carrier offset line, and compare the system efficiency joined the traditional offset line with the system efficiency joined class offset line by the simulation design for Doherty amplifier, the result show that the performance of the low power system can be improved after joining analogous offset line.

2. Fundamental Principles of Doherty Power Amplifier

The classical structure of Doherty power amplification circuit [2-4] is shown in Fig. 1. It's consists of two amplifier, respectively called carrier amplifier and peak amplifier, and the carrier amplifier works in class AB or class B, peak amplifier works in C. The two amplifiers connect in parallel, the cascading quarter wavelength transmission line after carrier amplifier play a role in impedance transformation and the quarter wavelength transmission line before the peak amplifier contribute in the phase compensation. The

peak amplifier begin to work when the carrier amplifier is close to saturation, so that by reducing the carrier amplifier's apparent impedance makes the carrier amplifier output more current when reaching saturation. Using this method can make the power amplifier gain high efficiency in saturated output point back within 6 dB.

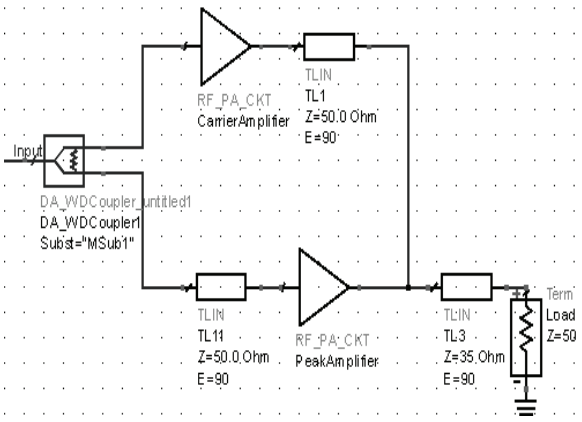


Fig. 1. Structure diagram of Doherty power amplifier.

Doherty power amplifier has three working status: low power state, which only carrier amplifier works, peak amplifier output end opens, carrier amplifier output impedance value is twice than the best impedance, and the high impedance makes the output current of carrier amplifier is only half of the maximum when they close to saturation, at this time, the efficiency achieves to the first peak point; middle power state, which the peak amplifier began to work when the carrier amplifier is working in saturated state, the load traction technology make the carrier amplifier output more current with invariant output voltage constant, and the system efficiency always stay at higher level; high power state, which the two amplifier both are working in saturated state, output current reaches the maximum, and the system achieves the second efficiency peak point.

3. Analogous Offset Line Technology

The following problems of Doherty amplifier in the actual application exist [5]: peak power amplifier doesn't work in low power state, certain open circuit resistance and the parasitic capacitance exists between leakage and source, through the matching network, the load as a carrier amplifier will show a impedance value, which leads the power of the carrier amplifier leak to the peak power amplifier; in high power state, carrier amplifier's output impedance Z_{opt} matches to 50 Ω, and in low power state, carrier amplifier's load impedance change into 100 Ω, Z_{opt} and load do not match. In view of the above questions, the literature [6-8] mentioned the offset line method to solve the problems, namely add a transmission line with 50 Ω characteristic impedance behind the carrier power amplifier and peak power amplifier, respectively called carrier offset line and peak offset line in this paper.

For peak offset line, due to the peak amplifier's output impedance has very low resistance and strong capacity in low power status, as point A shown in Fig. 2, the output impedance can be adjusted to high resistance area by adjusting the 50 Ω transmission line, and the peak amplifier is approximately open, set output impedance of peak amplifier output end point A in Fig. 3 for ZA, reflection coefficient.

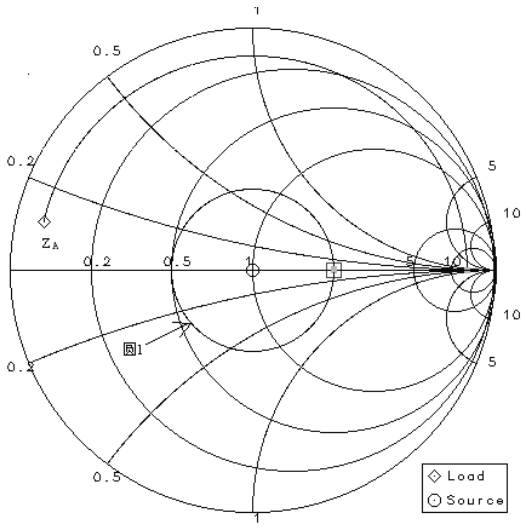


Fig. 2. Schematic diagram of offset line.

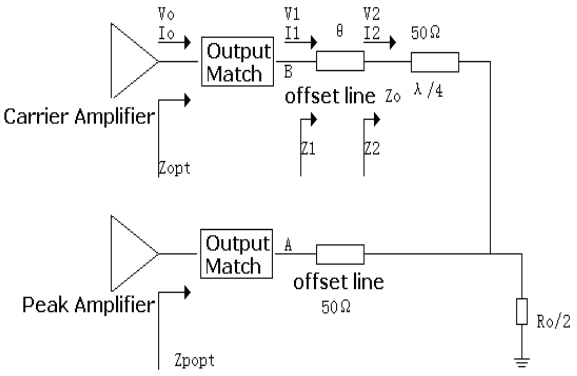


Fig. 3. Schematic diagram of analogous offset line calculation.

$$\Gamma_A = \frac{Z_A - 1}{Z_A + 1} = |\Gamma_A| \angle \phi_A \tag{1}$$

The length of peak amplifier's line offset:

$$L_{opt} = \frac{\phi_A}{4\pi} \lambda \tag{2}$$

Among them, β is the phase constant, λ is the wavelength and φ is phase.

For carrier offset line, suppose Z_{popt} is the best output power impedance for carrier power amplifier when load in traction, Z_{eopt} is the best efficiency impedance for carrier power amplifier when its output power is half of the saturation power, Z_{popt} and Z_{eopt} are impedance seen in the carrier power amplifier drain. The point Z_{popt} is matched to 50 Ω through the matching network in high power status, the point Z_{eopt} must be matched to 100 Ω if early saturation is going to be made to obtain greater efficiency in low power. When Z_{eopt} is matched to round 1 which is shown in Fig. 2 [9] through the same matching network, Z_{eopt} can be matched to 100 Ω by adjusting carrier offset line length.

In actual cases, the output impedance of Z_{eopt} through the matching network are not round 1 shown in Fig. 2, and adjusting the carrier offset line length cannot make Z_{eopt} matching to 100 Ω. In order to solve this problem, this paper puts forward the concept of analogous offset line, and changes the characteristic impedance and length of offset line at the same time, which making the carrier power amplifier is matched with 100 Ω loads in low power.

Suppose ABCD matrix of matching network designed by Z_{eopt} is:

$$M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \tag{3}$$

To the matching network, there is:

$$\begin{pmatrix} V_0 \\ I_0 \end{pmatrix} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \begin{pmatrix} V_1 \\ I_1 \end{pmatrix} \tag{4}$$

When is in low power, choose Z_{opt}=Z_{eopt}, according to $Z_{opt} = \frac{V_0}{I_0}$, $Z_1 = \frac{V_1}{I_1}$, there is

$$Z_1 = \frac{Z_{eopt} m_{22} - m_{12}}{m_{11} - Z_{eopt} m_{21}} \tag{5}$$

$Z_1 = R_1 + jX_1$, to analogous offset line, there is

$$\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} \cos\theta & jZ_0 \sin\theta \\ j\frac{1}{Z_0}\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} V_2 \\ I_2 \end{pmatrix} \tag{6}$$

substitute $Z_1 = \frac{V_1}{I_1}$, $Z_2 = \frac{V_2}{I_2} = 2R_0$ (R_0=50 Ω), the following result is got:

$$\begin{cases} Z_0 = \sqrt{\frac{2R_0 X_1^2}{R_1 - 2R_0} + 2R_0 R_1} \\ \theta = \arctan \frac{(R_1 - 2R_0) Z_0}{2R_0 X_1} \end{cases} \tag{7}$$

where Z_0 and θ are respectively the characteristic impedance and angle of analogous offset line. By joining analogous offset line, the carrier power amplifier's best efficiency impedance Z_{eopt} and 100 Ω loads are matched in low power, and prodigious efficiency can be obtained. In high power status, because Z_{eopt} and Z_{popt} differ rarely while $real(Z_{eopt}) < real(Z_{popt})$, the calculated Z_0 will close to 50 Ω, and the influence on the system's performance is not big [13].

4. Computing Software Simulation

ADS software is chosen to do the simulation, and LDMOS tube MRF6VP3450H of Freescale, which is selected. It is a pipe, with 470-860 MHz working frequency band, 22 dB gain, 450 W maximum output power. It is easy to realize Doherty amplifier by using it [10, 11].

First of all, quiescent point is determined by dc simulation, and carrier power amplifier is working in class AB, with 2.8 V grid voltage and 48 V drain voltage. Peak power amplifier is working in class C, with 1.4 V grid voltages and 48 V drain voltage [12].

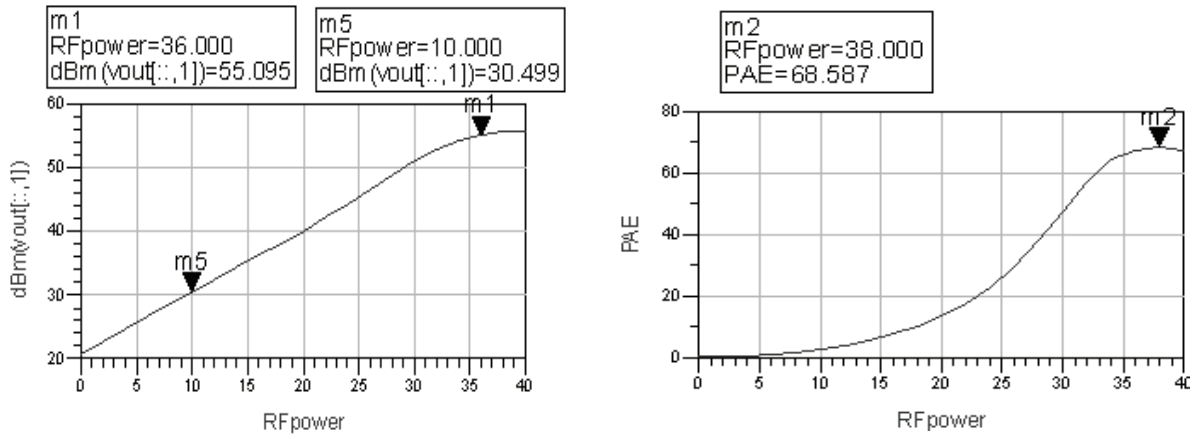
Then draw power amplifier's input impedance Z_{source} = 2.247-j1.5456, the best output impedance Z_{popt} = 2.34+j1.981, and the optimum efficiency impedance Z_{eopt} = 1.517+j2.643 in half power output through the source traction and load traction. Using Z_{source} and Z_{popt} to design input/output matching network, and the single tube simulation performance curve is shown in Fig. 4.

The designed Doherty amplifier is shown in Fig. 5. The carrier power amplifier and the peak power amplifier use the same input/output matching network, and the input power is allocated to the carrier power amplifier and peak power amplifier through equal power allocation such as the Wilkinson power splitters [14].

According to the offset line calculation method, the peak offset line is 51.58 mm, and carrier offset line is 52.43 mm. analogous offset line Z_0 = 40.36 Ω and θ = 56.41 ° are calculated from the formula (7), namely the width and length of transmission line is respectively 2.42 mm and 46.00 mm.

The Power Added Efficiency curve of Doherty amplifier power concluded by simulation is shown in Fig. 6. Among them, the diagram Fig. 6 (a) is simulation results where the carrier power amplifier uses 50 Ω offset line. Fig. 6 (b) is simulation results where the carrier amplifiers use analogous offset line [15].

Comparing Fig. 6 (a) and Fig. 6 (b), it can be seen that the characteristic impedance of class Z_0 offset line has better performance than 50 Ω offset line in low power. There is 3.956 % improvement in efficiency at the first peak point, and a 9.592 % improvement in efficiency when the output power is 50 dBm, and a 4.447 % improvement in efficiency when the output power is 45 dBm. In the medium power and high power the efficiency declines slightly, but the efficiency is still very high.



(a) Output Power. (b) Power Added Efficiency.

Fig. 4. Simulation performance of single tube.

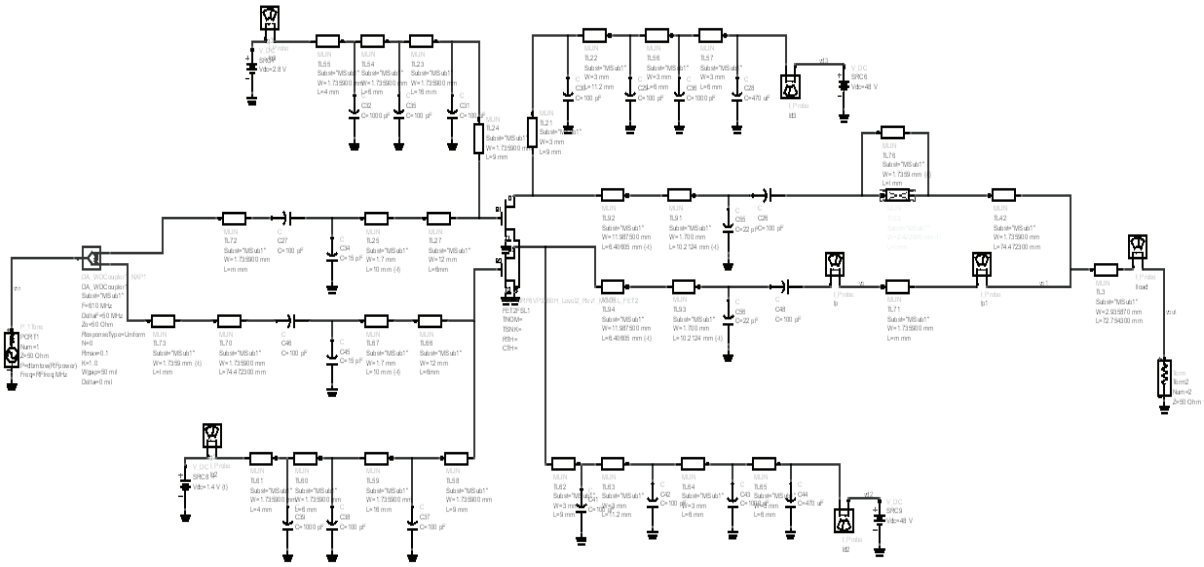
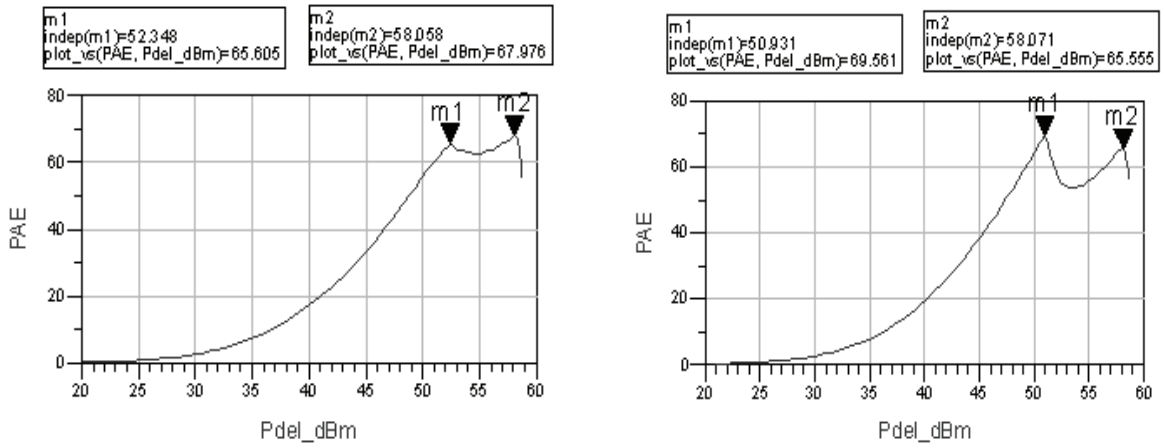


Fig. 5. Simulation schematic diagram of Doherty power Amplifier.



(a) 50 Ω offset line. (b) Class Z_0 analogous offset line.

Fig. 6. Power Added Efficiency of Doherty power amplifier.

5. Conclusion

By using analogous offset line technology, the efficiency performance of Doherty amplifier in low power is considerably improved. In order to obtain higher output power, combined with balanced amplifier technology, 150 W balanced Doherty power amplifier is designed in UHF frequency band. All the indexes of this system meet the requirements for power amplifier of the digital TV system, and this system is with practical application value.

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The Low Power Design Model Based on RNS-FIR

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Abstract: High power problem become one of the most important bottleneck of the IC industry. FIR filter is a most basic element in a digital signal processing system. In this paper, we managed to design a new FIR filter that can have low power. We use a residue number system to design a FIR filter in order to reduce power. In our model, we designed a serial modulus adder and a new way to compute modulus bit by bit. We use a way of looking up table in place of multiplication. By experiment verification, these methods in each module can effectively cut down the power and area. *Copyright © 2013 IFSA.*

Keywords: Low power, Filter, RNS, FIR, Modulus, Parallel, Prefix, Adder.

1. Introduction

1.1. Background

With the rapid development of IC technology, high power problem become one of the most important bottleneck of the IC industry. It is imperative to carry out a low power design technology [1].

The main factors drive low power researches are the wide application of mobile communication equipment and its sluggish progress on battery [2]. Besides, low power can not only reduce the costs on packaging and cooling but also the cost of high performance cluster system. Through dropping the chip temperature, a lower power can make the system have a less fault rate so that improve the reliability. In terms of environmental protection, a lower power decreases noise pollution and cuts down the radiation on human body [3].

For the moment, there are several primary strategies for low power. Firstly, is to reduce the capacitance of the chip and encapsulation [2]. This method is very effectively, just expensive. Secondly is to decrease the supply voltage. This can work well, too. But as a consequence, it causes a decline in performance. Then is strategy on power management. The technology has been used widely [4]. Its efficiency depends on the application of statistical features and optimization on strategies. Finally, to use various novel circuit structures. It has a low cost and extensive application. There is a large space on optimization [5]. Above all, it is one of the most efficient ways of depressing power consumption. In this paper, I adopt the method of using various novel circuit structures. In this way, the power consumption of the FIR filter can be dropped significantly.

1.2. FIR Filter

FIR filter is a most basic element in a digital signal processing system. It has any amplitude frequency and strict linear phase frequency. At the same time, its unit sample response is limited, thus filter is a stable system [6]. Therefore, FIR filter has a

range of applications in communication, image processing, pattern recognition and other fields. The model in this paper takes a 24 shaping filter as an example, do some research on how to reduce the power consumption of FIR filter using a residue number system design. First we designed a serial modulus adder and a parallel prefix modulus adder. Then I chose the serial modulus adder as a better one after compared. Second, we have a new way to compute modulus bit by bit. Third, we use a way of looking up table in place of multiplication. All these methods have effective functions so that they can cut down the filter's power and area a lot.

2. Model and Method

2.1. Theory of Algorithms

The principle of a decimal general filter is:

$$y(n)=\sum_{i=0}^{23} c_i x_i(n)\,, \tag{1}$$

where $x_i(n)$ is the filter inputs with 16 bits wide, $c_0, c_1 \ldots c_{23}$ are filter coefficients with 12 bits wide. $y(n)$ is filter output with 16 bits wide.

Below is the calculation formula of FIR filter based on a cardinal M. Y is the filter output, c_i is the filter coefficients, x_i is filter input, N is the number of filter and M is a cardinal of RNS [7].

$$Y=\left|\sum_{i=0}^{N-1} C_i X_i\right|_M \tag{2}$$

Define a discriminant:

$$\varsigma_i(g)=\begin{cases}1 & if\ \left|C_i\right|_M=g, g=0,1,2,\ldots..M-1\\ 0 & otherwise\end{cases} \tag{3}$$

Only when $\left|C_i\right|_M=g$, $\varsigma_i(g)$ =1, otherwise $\varsigma_i(g)$ =0.

Using the discriminant, we continue to simplify the expression of Y

$$Y=\left|\sum_{g=0}^{M-1} g\sum_{i=0}^{N-1}\varsigma_i(g)X_i\right|_M \tag{4}$$

$$Y=\left|\sum_{g=1}^{M-1} g\sum_{i=0}^{N-1}\varsigma_i(g)X_i\right|_M \tag{5}$$

Divide x_i into single bits

$$Y=\left|\sum_{g=1}^{M-1} g\sum_{i=0}^{N-1}\sum_{j=0}^{L-1}2^j\varsigma_i(g)x_{i,j}\right|_M \tag{6}$$

Transform the order of sum

$$Y=\left|\sum_{j=0}^{L-1}2^j\sum_{g=1}^{M-1}\sum_{i=0}^{N-1} g\varsigma_i(g)x_{i,j}\right|_M \tag{7}$$

Divide (g=1 to M-1) into g=1 to (M-1)/2 and g=(M+1)/2 to M-1

$$Y=\left|\sum_{j=0}^{L-1}2^j\sum_{i=0}^{N-1}\left[\sum_{g=1}^{(M-1)/2} g\varsigma_i(g)x_{i,j}+\sum_{g=(M+1)/2}^{M-1} g\varsigma_i(g)x_{i,j}\right]\right|_M \tag{8}$$

Using the following formula, we go on to split from g=(M+1)/2 to M-1

$$\left|\beta X\right|_M=\left|\beta+(M-\beta)\bar{X}\right|_M,\ X\in(0,1) \tag{9}$$

Here $\bar{X}$ is a reverse of X so that $\left\langle \bar{X}*X\right\rangle_M=1$

$$\begin{aligned}&\left|\sum_{g=(M+1)/2}^{M-1} g\varsigma_i(g)x_{i,j}\right|_M=\\ &=\left|\sum_{g=(M+1)/2}^{M-1}\left[g+(M-g)\overline{x_{i,j}}\right]\varsigma_i(g)\right|_M\\ &=\left|\sum_{g=(M+1)/2}^{M-1} g\varsigma_i(g)+\sum_{g=(M+1)/2}^{M-1}(M-g)\overline{x_{i,j}}\varsigma_i(g)\right|_M\end{aligned} \tag{10}$$

Thus we abstracted a constant coefficient which can be searched in a table when implemented.

$$\alpha=\left|\sum_{g=(M+1)/2}^{M-1} g\varsigma_i(g)\right|_M \tag{11}$$

By DA algorithm, we can transport Y into following formula, and then make coefficients coalesced.

$$Y=\left|\sum_{j=0}^{L-1}2^j\sum_{i=0}^{N-1}\left[\alpha_i+\sum_{g=1}^{(M-1)/2} g\varsigma_i(g)x_{i,j}+\sum_{g=1}^{(M-1)/2} g\overline{x_{i,j}}\varsigma_i(M-g)\right]\right|_M \tag{12}$$

$$=\left|\sum_{j=0}^{L-1}2^j\sum_{i=0}^{N-1}\left[\alpha_i+\sum_{g=1}^{(M-1)/2} g\left(\varsigma_i(g)x_{i,j}+\overline{x_{i,j}}\varsigma_i(M-g)\right)\right]\right|_M \tag{13}$$

S_j is actually a convolution sum of single bits.

$$S_j=\left|\sum_{i=0}^{N-1}\left[\alpha_i+\sum_{g=1}^{(M-1)/2} g\left(\varsigma_i(g)x_{i,j}+\overline{x_{i,j}}\varsigma_i(M-g)\right)\right]\right|_M \tag{14}$$

Below is a bit combination, then we finally found out the results.

$$Y = \left| \sum_{j=0}^{L-1} 2^j S_j \right|_M \tag{15}$$

Next step is scaling.

$Y = X / K$, K is the scaling constants, Y is what we need.

In the end, transport the data of residue number system to decimal ones by CRT [6] (Chinese Remainder Theorem).

$$\langle X \rangle_M = \left\langle \hat{m}_1 \left\langle \hat{m}_1^{-1} \right\rangle_{m1} r_1 + \hat{m}_2 \left\langle \hat{m}_2^{-1} \right\rangle_{m2} r_2 \right\rangle_M \tag{16}$$

In this formula, $\langle X \rangle_M$ represents the results of X mod M, $M=m_1m_2$, $\hat{m}_i = M/m_i$, $\hat{m}_i^{-1}$ is reverse of $\hat{m}_i$ so that $\left\langle \hat{m}_i^{-1} * \hat{m}_i \right\rangle_{m_i} = 1$. Besides, $r_1=<X>m_1$, $r_2=<X>m_2$, $m_1=M/m_2$, $m_2=M/m_1$.

2.2. Architecture of Model

First of all, convert input data to binary residue number. In the process of the filter will be a single bit split step. Then do a convolution operation of a single bit. After that, merge single bit together and scaling them. Finally, switch the data from residue number system to decimal system using CRT (Chinese Remainder Theorem). The details of RNS-FIR frame can be seen in Fig. 1.

In Fig. 1, the subscript stands for the i^{th} remainder branch; the superscript stands for the i^{th} bit. For the 1^{st} and 2^{nd} branches i=0, 1, ... 4. For the rest branches i=0, 1, ... 5.

The model in this paper picks six cardinals (25-3, 25-1, 25+1, 25+3, 26-3, 26-1). The range of computation covers all bit wide of residue addition and residue multiplication, so the calculation will not overflow.

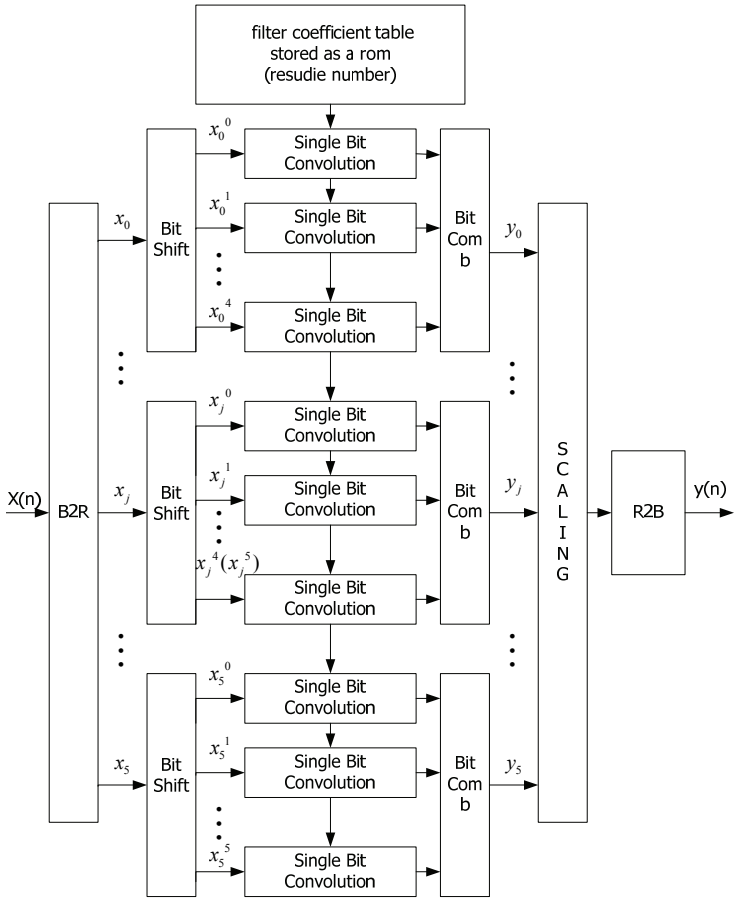

Fig. 1. RNS-FIR frame.

3. Low Power Method in Each Module

3.1. Low Power in Residue Adder

A RNS needs a number of addition operations. Traditional adder has to extend bits wide while residue adder can keep the bits scale [8] stably by modulus operation. Thus, the complexity of computing can be cut down. As a residue adder finishes a summation operator and a modulus operation at the same time, the complexity can be reduced further.

In this paper, I design two modules of residue adder. One is a serial modulus adder and the other is a parallel prefix one.

3.1.1. Serial Modulus Adder

The first scheme combined modulus principle and the ordinary adder in QUARTUS platform.

Residue adder theory is:

Respectively set up the two numbers A, M, M is cardinal, $T = 2^n - M$, S is result [9].

$$when \quad A + B + T > 2^n \tag{17}$$

$$S = A + B + T \tag{18}$$

$$otherwise \quad S = A + B \tag{19}$$

That is if output carries of $(A + B + T)$ is "1", the result will pick low-order *n* bits of $(A + B + T)$.

If output is not "1" the result will pick high-order *n* bits of A+B. The details on the RTL circuit can be seen as Fig. 2.

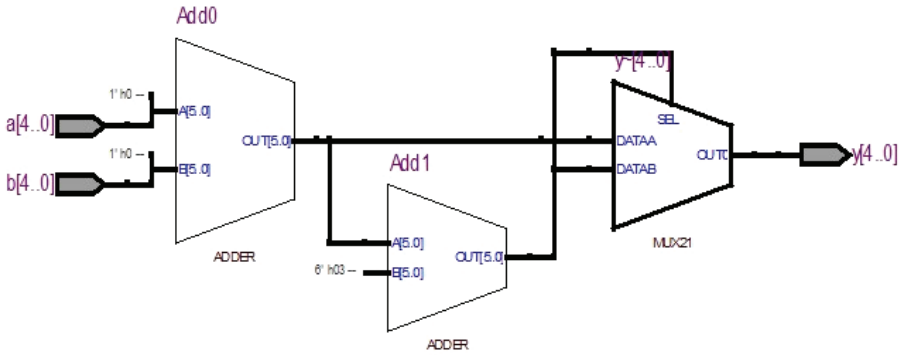

Fig. 2. Serial modulus adder's RTL.

3.1.2. Parallel Prefix Modulus Adder

The second scheme uses all basic logic units. It can achieve a faster computing speed. But it needs more logic resources.

Its principles are:

First of all, it is based on the principle of parallel prefix addition.

Generate prefix pairs used for prefix operation of $(A + B)$

$$(g_i, p_i) = (a_i b_i, a_i \oplus b_i), \tag{20}$$

g_i stands for the i[th] carry generation.

p_i stands for the i[th] carry transmission [10].

Then we have carry information of each bit.

$$\begin{cases} c_1 = g_0 \\ c_i = g_{i-1} + \sum_{j=0}^{i-2} \left(\prod_{l=j+1}^{i-1} p_l g_j \right) \quad i \geq 2 \end{cases} \tag{21}$$

This is each bit prefix operation in its group. After that, with every bit's carry information, we obtain summation result of each bit.

$$s_i = c_i \oplus p_i \tag{22}$$

On the basis of front information and the modulus theory, we revise the results according to the top bit carry value. Then real carry value of each bit is generated. In this way, we avoid to compute respectively carry values of $(A + B + T)$ and $(A + B)$. So the delay is decreased.

Firstly, get pairs of carry generation and carry transmission.

$$\begin{cases} (g_0, p_0) = (a_0 b_0, \overline{a_0 \oplus b_0}) & i = 0 \\ (g_i, p_i) = (a_i b_i, a_i \oplus b_i) & i = 1, 2, ..., k-1 \end{cases} \tag{23}$$

By observation, you can found that $M = 2^k + 1$, $T = 2^n - 2^k - 1$ so its binary representation is $\underbrace{000...001}_{n-k\,bits}\underbrace{00...001}_{k\,bits}$, therefore, we regard them as an add operation with a default carry both at the 0th bit and the k[th] bit.

When i=k, k+1, …. n-1, calculate reserve carry (g', p')

$$(g_i^{'}, p_i^{'}) = (a_i b_i, a_i \oplus b_i) \quad i = k, k+1, ..., n-1 \tag{24}$$

Add the pair $(g_i^{'}, p_i^{'})$, get pairs of carry generation and carry transmission eventually.

$$\begin{cases} (g_k, p_k) = (a_k + b_k, \overline{p_k^{'}}) & i = k \\ (g_{k+1}, p_{k+1}) = (p_{k+1}^{'} g_k^{'}, (\overline{g_k^{'}} + g_0)(p_{k+1}^{'} \oplus g_k^{'}) + g_k^{'} \overline{g_0} p_{k+1}^{'}) & i = k+1 \\ (g_i, p_i) = (p_i^{'} g_{i-1}^{'}, p_i^{'} \oplus g_{i-1}^{'}) & i = k+2, ..., n-2 \\ (g_{n-1}, p_{n-1}) = (a_{n-1} b_{n-1} + p_{n-1}^{'} g_{n-2}^{'}, p_{n-1}^{'} \oplus g_{n-2}^{'}) & i = n-1 \end{cases} \tag{25}$$

The output of highest bit is the bottom carry of A+B+T

$$c_{out} = g_{n-1} + \sum_{i=0}^{n-2}\left(\prod_{j=i+1}^{n-1} p_j g_i\right) \tag{26}$$

Do a revise once.

$$\begin{cases} c_1^{'} = g_0 \\ c_i^{'} = g_{i-1} + \sum_{j=0}^{i-2}(\prod_{l=j+1}^{i-1} p_l g_j) \quad i = k+1,...,n-1 \end{cases} \tag{27}$$

Do a second revise
For i=1, 2, …, k-1, the carry keeps the same
For i=k, k+1, …, n-1, set another variable Z

$$\begin{cases} z_k = p_{k-1}(p_k \oplus c_k^{'}) \\ z_i = \overline{\prod_{j=k+1}^{i} p_j} + p_0(p_k \oplus c_k) \quad i = k+1,...,n-1 \end{cases} \tag{28}$$

$$\begin{cases} z_k = p_{k-1}(p_k \oplus c_k^{'}) \\ z_i = \overline{\prod_{j=k+1}^{i} p_j} + p_0(p_k \oplus c_k) \quad i = k+1,...,n-1 \end{cases} \tag{29}$$

The corrected results are as follows via the variable Z.

$$\begin{cases} c_i = c_i^{'} \quad i = 1,...,k-1 \\ c_k = c_k^{'}(c_{out} + p_{k-1}) \quad i = k \\ c_i = c_i^{'}(c_{out} + z_i) \quad i = k+1,...,n-1 \end{cases} \tag{30}$$

Set the output sum is S, finally is a sum operation.

$$\begin{cases} s_0 = c_{out} + \overline{p_0} \\ s_k = c_k \oplus c_{out} \oplus \overline{p_k} \\ s_i = c_i \oplus p_i \quad i \neq 0,k \end{cases} \tag{31}$$

The comparing resources of our parallel prefix modulus adder and serial modulus adder can be seen in Table 1.

Table 1. Logic cost of two modulus adders.

Total logic elements	Parallel prefix modulus adder	Serial modulus adder
Total combinational functions	22	17
Dedicated logic registers	0	0

The simulation results of parallel prefix modulus adder and serial modulus adder can be seen in Fig. 3.

In the Fig. 3, x1, x2 is input data, y1 stands for parallel prefix modulus adder outputs, y2 represents serial modulus adder output. It is obvious in the diagram that there is no big advantage for parallel prefix modulus adder on time of computing. But it cost more logic units. Consequently, application of a serial modulus adder can effectively reduce the power consumption of the whole system.

3.2. Low Power Strategy in B2R

3.2.1 Modulus Operation Bit by Bit

This part is to transmit binary data to residue data. The input is computed by modulus operation under each cardinal. Its remainder is divided into six roads to be passed to the next module. Our cardinals are (25-3, 25-1, 25+1, 25+3, 26-3, 26-1). After calculation, this group of cardinals can cover whole computing wide, so that we can solve the problem of overflow.

A modulus operation involves division And the division isn't belong to basic computing operation. For this reason, if we directly use modulus operator, it will occupy more logic and slow down the speed. In this paper, we take a look-up table method instead of modulus operation which has a division. The details of the process can be seen in Fig. 4.

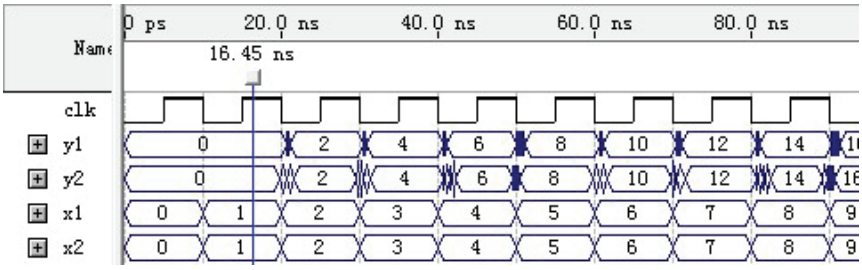

Fig. 3. Simulation results of two modulus adders.

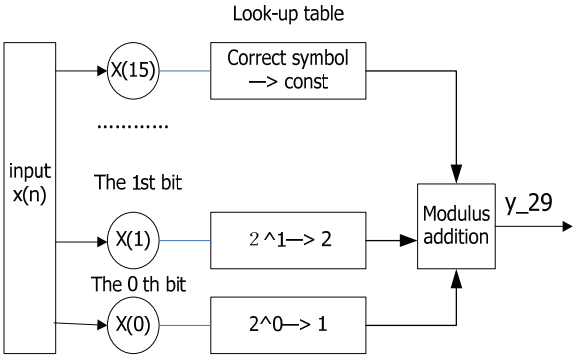

Fig. 4. Modulus process.

$x(n)$ is filter input with 16 bits wide and it's a signed data. The table lists all binary remainder of 16 bits $2^0, 2^1, 2^2 \cdots 2^{15}$ under a cardinal. There are six tables in total. For each bit of $x(n)$, if that is "1", look up the remainder in the table, if that is "0", the default remainder is "0". Add all the 16 remainders using our modulus adder. In this way, we get a remainder under a cardinal. In order to enhance speed, computing of six look-up tables is parallel.

3.2.2. Correct Symbol

As the input is a signed number, a negative input's modulus operation is quite different from positive inputs [11]. So we find a way of correcting symbol. This method can share the same logic circuit with negative modulus operation and positive modulus operation. There is no need to do another modulus operation for negative inputs as usual.

Principle is as follows:

Set the input data is x, when x is negative, there is a relationship between the true code of x's absolute value and complement code of x. In the following formulas, the subscript t stands for true code and the subscript c stands for complement code.

$$\left[|x|\right]_t + \left[x\right]_c = 2^{16} \tag{32}$$

Do a transposition:

$$\left[x\right]_c - 2^{16} = -\left[|x|\right]_t \tag{33}$$

The right of the equation $-\left[|x|\right]_t$ is exactly our input x, set out cardinal is r, do a modulus operation on both sides of equation.

$$\left\langle \left[x\right]_c - 2^{16} \right\rangle_r = \left\langle -\left[|x|\right]_t \right\rangle_r \tag{34}$$

The right side of the equation $\left\langle -\left[|x|\right]_t \right\rangle_r$ is just our output remainder.

In order to simplify the logic, we add an r in left side of equation according to the remainder principle so that we can get rid of subtraction.

$$\left\langle \left[x\right]_c + \left(r - 2^{16}\right) \right\rangle_r = \left\langle -\left[|x|\right]_t \right\rangle_r \tag{35}$$

Decompose the modulus in the left side.

$$\left\langle \left[x\right]_c \right\rangle_r + \left\langle r - \left\langle 2^{16} \right\rangle_r \right\rangle_r = \left\langle -\left[|x|\right]_t \right\rangle_r \tag{36}$$

Since r is known, $\left\langle 2^{16} \right\rangle_r$ is also a definite number, so you can get value of $\left\langle r - \left\langle 2^{16} \right\rangle_r \right\rangle_r$ then store in the table.

And $\left\langle \left[x\right]_c \right\rangle_r$ can be used as a positive number to do modulus operation.

For any input data, what need is only to detect the top bit. If it's "0", then it's positive, the remainder is "0". If it's "1", then it's negative, we look up to the table to fetch its remainder. Thus, this symbol correct operation is the same as other bits' modulus operation. So they can be done in one table.

3.3. Look-up Table Replace of Multiplication

3.3.1. Single Bit Convolution Section

Our cardinals are {27, 29, 31, 35, 59, 61}. Four of them are 5 bits while two of them are 6 bits. In the process of convolution, there will be a large number of multiplications. In order to reduce complexity of computing, we use a look-up table instead of multiplier.

In the session of single bit convolution, each bit of x will be combined to original data after it is convoluted. These operations are

$$y = \left\langle \sum_{i=0}^{5} x_i \times 2^i \right\rangle_r \tag{37}$$

where x_i is the convolution result of each bit, it has 6 bits wide. The details can be seen in Fig. 5.

This left shift is equivalent of a multiplier of 5×5, the final data wide will be extended to 10 bits. This is a huge cost. Here we have a improve method. As the result of single bit addition is less then 2 bits. So we can make an exhaustion. Every case of sum which multiplied 2^i is listed in a table. So just look at the sum of single bit and then look up the table. The details can be seen in Fig. 6.

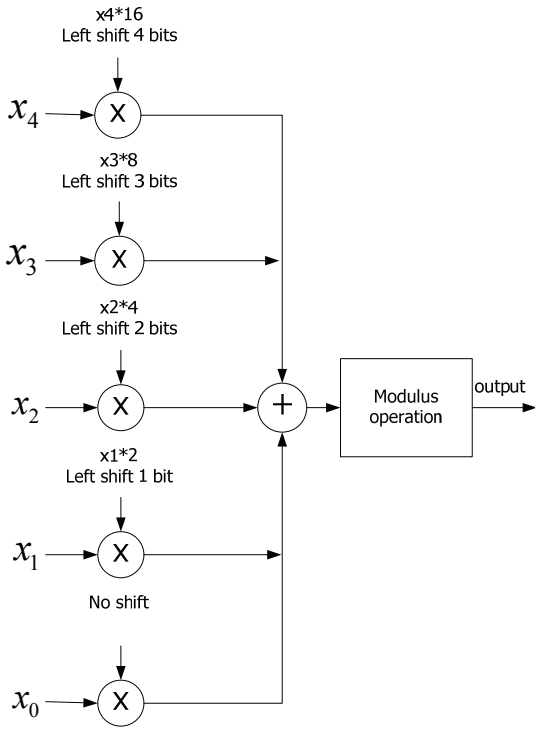

Fig. 5. Convolution process.

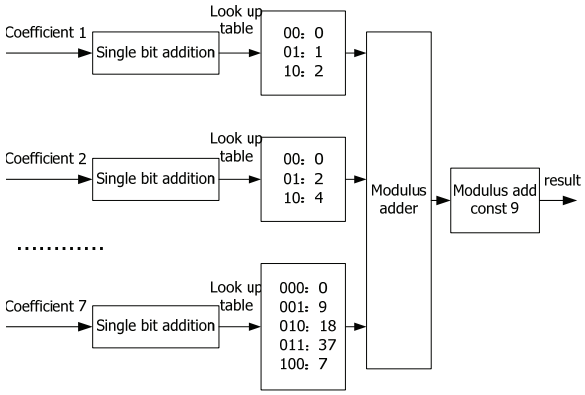

Fig. 6. Look-up table convolution.

3.3.2. The Session of Scaling

$Y=X / K$, K is the scaling constant, Y is what we need [7]. Set X is a number before scaling, Y is X's result after scaling. K is an inverse element in cardinal M_i. Set X is a number before scaling, Y is X's result after scaling. K is an inverse element in cardinal M_i. The details on the RTL scaling circuit is shown in Fig. 7.

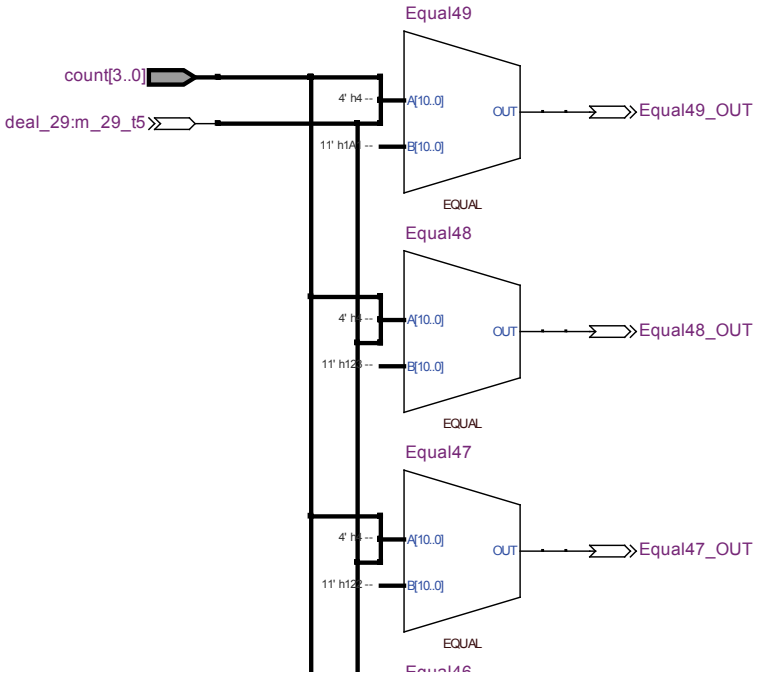

Fig. 7. Scaling RTL.

4. Experimental Verification

Based on the Design Compiler, we take respectively the SMIC 0.13 um and 0.18 um standard library under 100 MHz clock. We do a synthesis both on RNS FIR and decimal FIR. In order to compare more accurately, the decimal FIR has an identical function as RNS-FIR. It is also a 24 shaping filter and its 24 coefficients are the same as our RNS-FIR's. Under the same 100 MHz clock, they have totally the same input data and output data. Their area and power consumption are shown in Fig. 8 and Fig. 9.

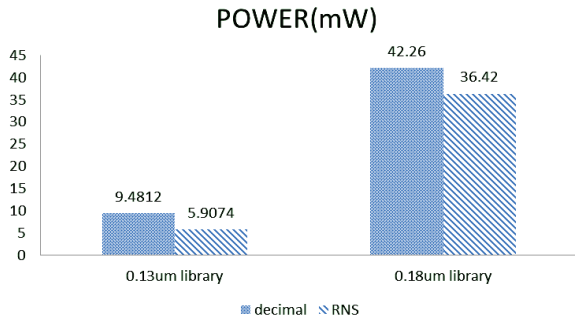

Fig. 8. Power comparisons of RNS and decimal FIR filters.

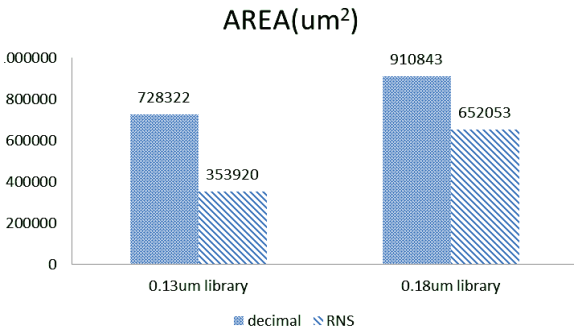

Fig. 9. Area comparisons of RNS and decimal FIR filters.

From Fig. 8 and Fig. 9, the methods in this paper can reduce effectively the power cost by 37.69 % under 0.13 um library and 13.82 % under 0.18 um library Compared to decimal FIR's power. On power area, we decrease by 51.4 % under 0.13 um library and 28.41 % under 0.18 um library compared to decimal FIR's power area.

After computing, on every area (*um*), they can cut down the power by 22 % less than decimal FIR filter under 0.13 um library and 16.94 % under 0.18 um library.

5. Conclusions

A residue number system is used to design a FIR filter to reduce power by verification from experience. The serial modulus adder and a look-up table are useful to cut down the power and area.

However, there is still some disadvantage in our design. As we have large amount of single bit computing and convolutions, so the delay in our system is longer than the decimal filter. Nonetheless, we have great throughput. So this model will be more suitable for large data computing system.

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Research in the Operational Modal Analysis of Rotating Machinery Reducer

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Abstract: In operational modal analysis process of the rotating machinery, it is always difficult to adopt usual modal analysis methods in parameter correction because of the modal parameter identification problems caused by harmonic modes. A technique to distinguish harmonic and structural modes is proposed here, that is, through frequency domain decomposition (FDD) of the vibration signals power spectral density matrix and probability density function, the two components (structural modes and harmonic modes) are successfully separated from the overall responses. The experiment indicates that structural modal parameters of the reducer can be obtained in real environment, which provides accurate references to the continuous mining machine reducer parameter correction. The method provides an efficient path for the modal analysis of rotating machinery. *Copyright © 2013 IFSA.*

Keywords: The continuous mining machine reducer, Operational mode, Frequency domain decomposition, Modal parameter identification, Probability density function.

1. Introduction

After the physical prototype of continuous coal mining machine gear reducer is manufactured, there are always problems exist in its operation because of design defects, machining errors, and so on, as might cause even more serious faults, hence, design parameters correction is needed. Inherent characteristics of the equipment can be better accessed to by modal analysis method. Modal analysis method developed in recent years can simulate real environment to analyze the modal parameter of the equipments, which provides accurate references for parameter correction and design optimization of the prototype machinery.

Operational Modal Analysis (Operational Modal Analysis short for OMA) is a technology that extracts the model parameters based on the response signal [1]. Compared with traditional method of Experimental Modal Analysis [2], OMA does not need special incentive devices, and it is especially suitable for large equipments in operation because it does not affect the normal work; as can not only satisfy the actual boundary conditions, but also obtain complete modal parameters [3]. For rotating machineries like internal combustion engines, gas turbines, reducers, and so forth, operational modal parameters obtained is more practical than those under artificial incentive work conditions [4].

The premise of Operational Modal Analysis is that the equipments are excited to white noise

excitation (random excitation). However, inherent and harmonic modals (pseudo modal) [5] exist in the analysis results because of the simultaneous existing of random and cycle incentives in the operation of rotating machineries, which causes difficulties in obtaining modal parameters; hence, this method is seldom adopted in rotating machinery.

However people are still attempting to seek proper methods to apply work modal because of its inherent advantages. Rune Brincker (Denmark) extracted operational modal through stochastic subspace method on motor engines and diesel engines at work, in which, optimized parameters were obtained. [6-9]. Zheng Min (China) did operational modal tests on a model plane by using cross-correlated complex exponential methods [10], and the results obtained are in accordance with those through the optimal tuning vibration method. Dong Shu Wei (China) filtered harmonic signals and identified harmonic components in response signals by using improved algorithm of least squares complex frequency domain, some achievements were realized, yet, this method is not applicable in practical engineering because of its calculation complexity [11].

In view of the above, this paper proposes a new method to identify operational modals, that is, combining frequency domain decomposition algorithm and probability density statistics together to decomposition and extract singular value in the power spectral density matrix of vibration signal obtained by operational modal tests. Through the statistical features of the probability density curves of the signals, the inherent and harmonic modals can be identified. This can be applied in the work modal analysis of continuous mining machine reducers to extract the two different modals successfully; as can establish a better path and references for the obtaining of real inherent modal parameters of rotating machineries.

2. Frequency Domain Decomposition Theory and Solution of the Modal Parameters

Frequency Domain Decomposition (short for FDD) is one method that decomposition singular value (short for SVD) [12-14] in the response power spectral density matrix of discrete frequency.

In a modal analysis, the relationship between output response signals and input excitation signals could be expressed as equation (1):

$$G_{yy}(j\omega) = \overline{H}(j\omega)G_{xx}(j\omega)H(j\omega)^T, \tag{1}$$

where $G_xx(j\omega)$ and $G_{yy}(j\omega)$ are the power spectral density matrix of input and output respectively in the form; $H(j\omega)$ is the frequency response matrix, symbol "—" and "T" stand for complex conjugation and transposition.

When the input is white noise excitation, $G_{xx}(j\omega) = C$ is the constant, set the output response signals as equation (2):

$$\{y(t)\} = \{y_1(t), y_2(t), \cdots, y_n(t)\}^T \tag{2}$$

Then the output response power spectral density matrix could be expressed as equation (3):

$$G_{yy}(j\omega) = \begin{bmatrix} G_{11}(j\omega) & G_{12}(j\omega) & \vdots & G_{1n}(j\omega) \\ G_{21}(j\omega) & G_{22}(j\omega) & \vdots & G_{2n}(j\omega) \\ \cdots & \cdots & \vdots & \cdots \\ G_{n1}(j\omega) & G_{n2}(j\omega) & \vdots & G_{nn}(j\omega) \end{bmatrix} \tag{3}$$

Diagonal element in the matrix of $G_{yy}(j\omega)$ is auto spectrum of single measuring point signal, other elements is cross spectrum of different measuring points.

Modal parameters are obtained based on frequency response function. Modal parameters of the frequency response function are usually in the form of a residue and poles, as equation (4) below:

$$H(j\omega) = \sum_{k=1}^{n} \frac{R_k}{j\omega - \lambda_k} + \frac{\overline{R}_k}{j\omega - \overline{\lambda}_k} \tag{4}$$

$\lambda_k = -\sigma_k + j\omega_{dk}$ is the pole of the frequency response function in form above; $R_k = \psi_k \gamma_k^T$ is the residue of frequency response function matrix; n is the number of system modes; σ_k is the first order modal attenuation coefficient k; ω_{dk} is the damped natural frequency; ψ_k and γ_k are the modal vibration shape vectors and modal participation factor respectively.

Bring equation (4) into equation (1), then equation (1) could be written in a minimum fraction form as equation (5) below:

$$G_{yy}(j\omega) = \sum_{k=1}^{m} \left(\frac{A_k}{j\omega - \lambda_k} + \frac{\overline{A}_k}{j\omega - \overline{\lambda}_k} + \frac{B_k}{-j\omega - \lambda_k} + \frac{\overline{B}_k}{j\omega - \overline{\lambda}_k} \right) \tag{5}$$

$A_k = \dfrac{R_k C R_k^H}{2\sigma_k}$ is the k order residue matrix response function in the form, B_k is the conjugate of transpose of A_k. When the damping is small, the residue matrix of response function is proportional to the system modal vector, which could be expressed as equation (6):

$$\lim_{damping \to light} A_k = \psi_k \gamma_k^T C \gamma_k \psi_k^T = d_k \psi_k \psi_k^T \tag{6}$$

Among which d_k is proportional constant.

In a certain affirmative frequency, only one order or two orders modal poses an important influence on the response of system usually. We write these modal as sub(ω). So the response spectrum matrix can be written as the final form, shown by equation (7):

$$G_{yy}(j\omega) = \sum_{k \in sub(\omega)} \left(\frac{d_k \psi_k \psi_k^T}{j\omega - \lambda_k} + \frac{\overline{d}_k \overline{\psi}_k \psi_k^T}{j\omega - \overline{\lambda}_k} \right) \quad (7)$$

To extract the modal parameters, singular value is decomposition (SVD) in response spectrum matrix of equation (7). Then equation (8) is gotten as below:

$$G_{yy}(j\omega) = \Phi \Sigma \Phi^H \quad (8)$$

In the form above $\Phi\Phi^H = I$, Φ is shown as equation (9), I is unit matrix.

$$\Phi = \left[\{\phi_1\} \{\phi_2\} \{\phi_3\} \cdots \{\phi_n\} \right] \quad (9)$$

The decomposed singular value Σ stands for a series of single freedom system auto power spectrum density function. Singular value curve peak corresponds to the first order modal frequency of the K. The corresponding singular vector ϕ_1 decomposed is the modal vibration mode of estimation. The time domain signal is gotten by the function of single freedom degree of spectral density being transformed of inverse Fourier transform. The rest modal parameters of response could be obtained through its autocorrelation function.

3. The Statistic Characteristics of Random Response and Harmonic Response

For rotating machinery, under circumstances of coexistence of random excitation and harmonic excitation, the probability density function distribution form would not be changed after the signal is processed for filtering, sampling and etc, of which when the structure is inspired by Enough independent random load, the response of the structure could be regarded as a linear superposition of multiple excitations of random. Its probability density function obeys Gaussian distribution approximately, meaning a normal distribution. The normal distribution is displayed as the unimodal on the graphics. The curve of Theory Probability Density is shown as Fig. 1 (a).

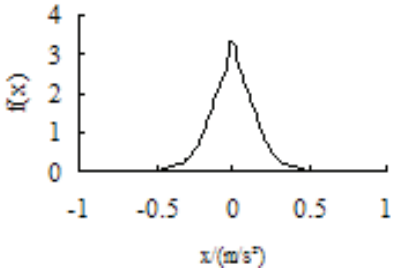
As for probability distribution of harmonic response (production after cycle motivation), assuming that two functions of probability density of random variables X and Y are $f_X(x)$, $f_Y(y)$ respectively, even $x = h(y)$, then form (10) was established.

$$f_Y(y) = \begin{cases} f_x[h(y)]h'(y) & \alpha < y < \beta \\ 0 \end{cases} \quad (10)$$

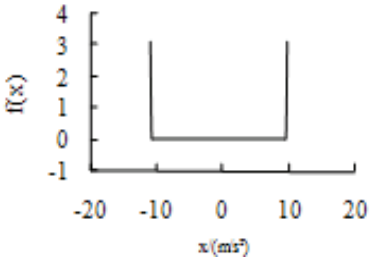
In the equation above, $h'(y)$ is the differential coefficient of $h(y)$; α, β are value ranges of y. Assuming that the simplest harmonic response is being considered firstly, $y = g(x) = a\sin(\omega x)$, take the probability density of x is constant value $f(x)$. The probability density function of harmonic response is expressed as equation (11).

$$f_Y(y) = f_X = g^{-1}\left(\frac{1}{\omega}\arcsin(y/a)\right) = f_X \frac{1}{\omega a} \frac{1}{\sqrt{1-\left(\frac{y}{a}\right)^2}} \quad (11)$$

When $y \to \pm a$ in equation (11), $f_Y(y)$ is going to tend to infinity, there will be two different peaks in the probability density function curve of harmonic response, of which curve of the theory probability density is shown as Fig. 1 (b) [15].



(a) Random Response



(b) Harmonic Response

Fig. 1. Theoretical probability density curves.

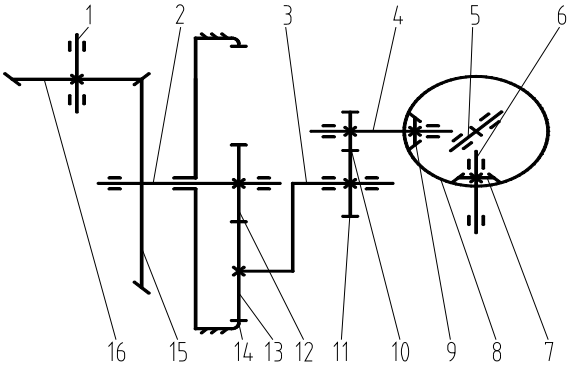
Based on the theoretical analysis above, this article is going to apply frequency decomposition technique combining with statistical property method into parameters identification of the modal analysis of some even mining machine reducer, which would distinguish each order modal of authenticity (inherent modal and harmonic modal) effectively.

4. Operational Modal Test and Parameter Identification

In the operating process of the gear reducer, Meshing point position and the number of teeth on the mesh change periodically in the process of gear transmission, cause that stress of gear change with stiffness cyclically, which makes the vibration of the reducer, contain both random vibration and vibration cycles. So at this time the modal parameters obtained include inherent modal and harmonic modal.

Some even mining machine speed reducer is a five class transmission reducer studied by this paper, of which structure is complicated, of which transmission diagram is shown as Fig. 2. After the completion of the reducer physical prototyping, there are a lot of problems, in order to get the real inherent modal parameters of the reducer entities, the operational modal test of the reducer has been done under different operational conditions in this paper.

The reducer is installed under the condition of simulation in the experiment to maximize boundary of simulation of the operational condition. A total of 72 sensors are arranged as the measuring points of response on the reducer in the experiment to meet the requirements of operation modal test. Distribution of measuring points set covers range of equipment transmission ratio, and is able to simulate the shape of equipment, shown as Fig. 3.



1. I Axis , 2. II Axis , 3. III Axis, 4. IV Axis, 5. V Axis, 6. VI Axis, 7. Gear 10, 8. Gear9, 9. Gear8, 10. Gear7, 11. Gear6, 12. Gear5, 13. Gear4, 14. Gear3, 15. Gear2, 16. Gear1.

Fig. 2. Transmission diagram of the reducer.

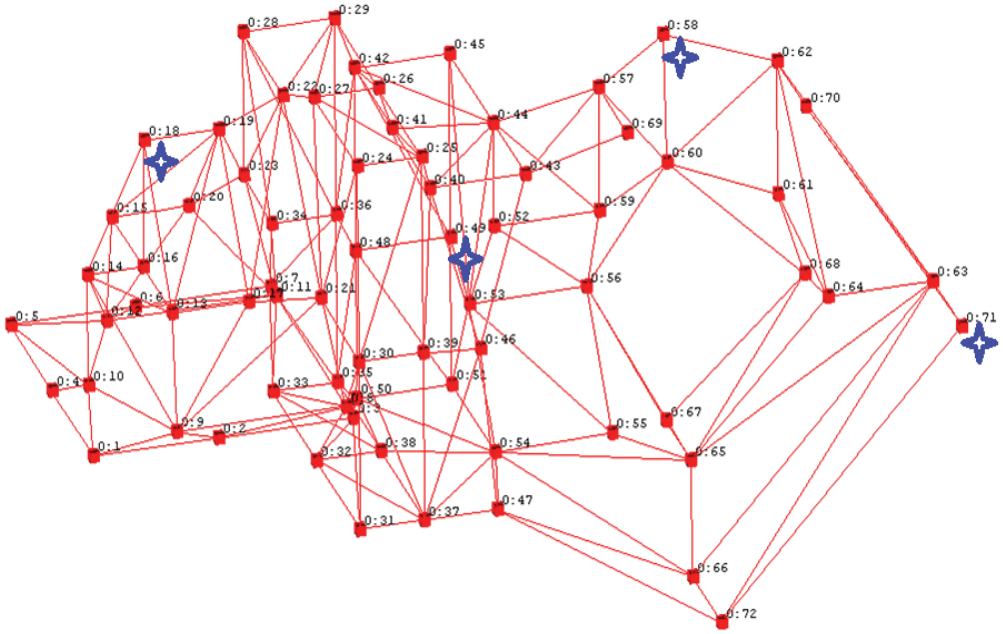


Fig. 3. The distribution of sensors.

Response signals are picked up by using American KITSLER three-way acceleration transducer. Vibration direction of the sensor must agree with the vibration direction of the system to be measured at this point when installing. Data collection is finished by using data collection analyzer. The test time of the test is set to 2 seconds. Sampling frequency is 8192 Hz. Frequency resolution is 4 Hz. The main purpose that the parameters above are set is to ensure that all the information would be picked up within the scope of the operation frequency of the equipment.

Take the vibration data of the measuring points of response into equation (3) to get the power spectral density matrix of the vibration data and then take which into equation (8) again to decomposition singular value for power spectral density matrix of measured data to get the singular value decomposition (SVD) curve. In order to make the singular value curves seen clearly, this article only

chooses four curves to show (They are measuring points of 18, 49, 58, 71 of star location shown in Fig. 3, the vibration amplitude of the four points is large, and which is far away from the station, the spatial resolution of vibration mode is high.) shown as Fig. 4.

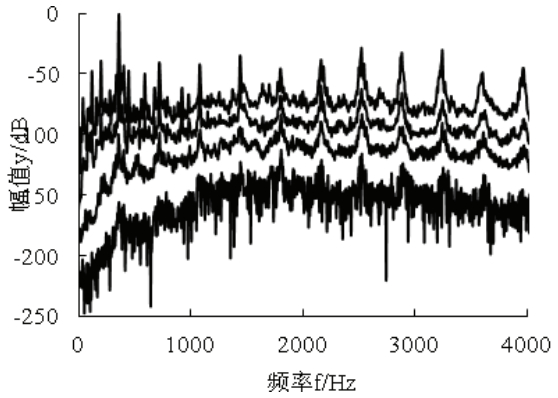


Fig. 4. Singular decomposition curves.

To distinguish between inherent modal and harmonic modal better, the Butterworth band-pass filter is designed in this paper, shown as Fig. 5. The frequency response curve which could be seen of the filter is the maximum flat within the pass-band, there are no ups and downs; the decay is slow; and which is enhancing continuously as the order of filter is increasing; of which amplitude attenuation speed of the resistance band is fast, which totally keeps the linearity and distortion of the filter. Take each peak frequency in Fig. 4 as the center frequency, half-power bandwidth as pass band width to take the response signal into Butterworth band-pass filter shown as Fig. 5 to filter. For the signal filtered are analyses by adopting the probability statistical characteristics analysis. The probability density curve of part of the modal frequency is shown in Fig. 6.

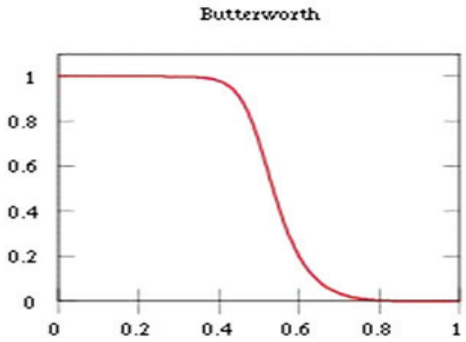
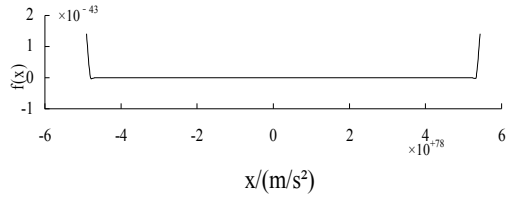
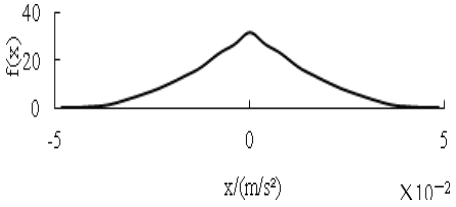


Fig. 5. Butterworth band-pass filter.

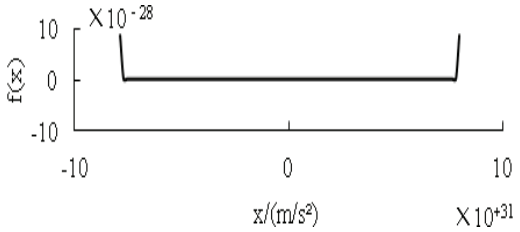
There are two peaks for modal probability distribution of 2 order modal, in the curves shown (Fig. 6 (a), Fig. 6 (c)), which could be seen from Fig. 6 and corresponds to the form of probability density curve harmonic response of inspiring the structure cyclically of the derivation above. Single peak exists in modal probability distribution of Fig. 6 (b), Fig. 6 (c), which approximates Gaussian distribution, could be confirmed as the form of probability density curve caused by the random response of structure caused by random incentive.



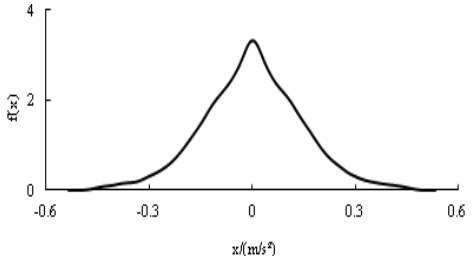
(a) 360 Hz.



(b) 448 Hz.



(c) 720 Hz.



(d) 980 Hz.

Fig. 6. Part of probability density curves of the random response and harmonic response.

On the other hand, the calculating formula for mesh frequency of the reducer gear is known as

$$f = \frac{z_1 n_1}{60} = \frac{z_2 n_2}{60}, \tag{12}$$

where in z_1, z_2 are the number of teeth of two gears respectively, n_1, n_2 are rotational speed of two gears respectively.

Equation (12) is applied to calculate transmission meshing frequency of the reducer at all levels. The first level 360 Hz, the second one is 303 Hz, the third level is 40 Hz, the fourth one is 40 Hz, and the fifth level is 40 Hz respectively. Periodical vibration is aroused when reducer gears are in the mesh per level. Therefore harmonic modal that is aroused by part of the periodical signal of system could be identified indirectly by gear-mesh frequency of this reducer per level.

It could be seen that two modal frequency 360 Hz and 720 Hz of the existence of double peak density curve obtained in Fig. 6 are just integer multiples of the mesh frequency of this reducer, which indicates that the two modal frequency is really harmonic modal produced by harmonic response that is inspired by cycle.

Singular value decomposition is done in power spectral density matrix gotten after calculating all points of measurement data and of which the probability density curve is solved. All harmonic modes could be identified that the reducer is inspired by harmonic excitation, and which is removed, and eventually real inherent modal parameters of eliminating harmonic modal of the reducer are gotten.

In order to verify the method above is effective further on, LMS Test. Lab commercial machine software is applied to analyses the operational modal of the reducer in this paper. (There is a certain authority for the commercial machine to analyses the modal, but is very expensive). Firstly a simplified model of entity of the same shape with the reducer is established in LMS software, then stationing of virtual sensors is done on the modal according to location coordinates of the same sensors in experiments above, in order to distribute vibration signal data of sensors to the corresponding virtual sensors in experiment. In this way, inherent modal parameters obtained from frequency domain decomposition technique and method studied in this paper could be compared with ones obtained from LMS Test. Lab commercial software. Situation being contrasted of inherent modal parameters obtained by two methods of identification of the first six orders shown by Table 1.

The data of comparison is shown from Table 1, inherent modal parameters obtained from commercial software LMS Test. Lab and inherent modal parameters obtained from frequency domain decomposition technique of which result is consistent. Because of Sampling frequency of FDD method of identification being restricted during the experiments (relation to the acquisition equipment), of which frequency resolution is 4 Hz. Thus, the results are integers; wherever the precision of LMS commercial software is high; the precision of resolution is also higher. The result of table 1 has proved the feasibility and effectiveness of frequency domain decomposition technique and path studied in this paper. Modal parameters obtained brings sufficient basis to correction of parameters of the reducer prototype.

Table. 1. Comparison between the results of frequency domain decomposition and LMS software.

Modal order	FDD (Hz)	LMS recognition (Hz)
1	338	336.6
2	448	447.1
3	536	542.5
4	739	741.4
5	921	911.8
6	980	979.9

5. Conclusions

1) After operational modal experiments is done on the mining machine reducer, frequency domain decomposition technique proposed in this paper is applied to calculate modal parameters to obtain significant peaks in spectrum, which provides sufficient and necessary conditions for the identification of modals in the next step.

2) Through filtering the signals at the frequency peak and analysis of the identifications of the probability density curve characteristics, the mathematical properties of each signal in inherent and harmonic modals can be identified, so as to separate inherent modals from the harmonic, which are both aroused by random and harmonic excitation coexisting in the structure. This can be applied in any equipment with the above mentioned conditions.

3) The research method provides a new path to identify modal parameter in operational modal analysis of machinery transmission systems, which establishes accurate basis for the correction of structural parameters. The same findings could be obtained without having to use expensive commercial machines in this process; hence, it is more credible and economical for any institute and individual researchers who will be engaged in the research of work modals.

Acknowledgements

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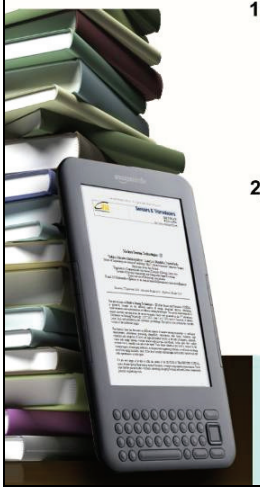

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Improving the Lorentz Force Amplitude of an EMAT Using Stacked Coil Configuration

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Abstract: This research presents a study of a novel method of improving the Lorentz force amplitude of a planer spiral coil EMAT configuration by stacking the coils. A 2D axisymmetric Finite Element Model was developed to study and compare the density of the Lorentz Force (LF) generated at the skin depth by the Stacked Spiral-EMAT (SS-EMAT) configuration on cs70 grades of pipe steel materials. The response of the novel EMAT configuration to variation of parameters such as the static magnetic field, internal diameter, excitation frequency, lift-off, coil stacking, the thickness of the insulation layer and number of cycles per tone burst were also investigated to ascertain the viability of the Stacked Spiral-EMAT. The results obtained showed that there is a significant improvement in the density of Lorentz Force generated by the novel EMAT when stacked. Hence a new EMAT configuration is proposed to enhance the performance of radially polarised bulk wave EMATs. *Copyright © 2013 IFSA.*

Keywords: Lorentz force, Electromagnetic acoustic transducer, Eddy current.

1. Introduction

Electromagnetic Acoustic Transducers (EMATs) can generate and detect ultrasonic wave without making contact with the test material. This feature makes it an attractive option in contactless measurement such as when carrying out inspection in high temperature environment or taking measurement in a moving object [1-3]. EMATs have also found usefulness in many non-destructive evaluation applications, such as the determination of physical parameters of conducting materials, flaw characterization, dimension and thickness measurement. EMATs consist of coils and permanent or electromagnet located in proximity to the surface of the material. The coils are excited with a high frequency tone burst pulse, this creates a dynamic magnetic field which further induces a high frequency eddy current on the conducting material [1, 4]. The interaction of the eddy current and the static magnetic field (bias field) gives rise to mechanical strain and hence produces ultrasonic waves on the material [1, 4, 5]. The outcome of researches during the last couple of years on the performance of EMAT on conducting materials (ferro-magnetic and non-ferromagnetic material) have been reported [6-9]. The reports have established that EMAT has low transduction efficiency which gives rise to reduced signal to noise ratio (SNR). In other to increase the signal to noise

ratio of EMAT, several attempts have been made to improve the performance of EMATs 10-18].

In [19] it was established that LF mechanism is the dominant transduction mechanism for normal biased EMATs configuration on ferro-magnetic materials. Table 1 shows the measured electromagnetic properties of the grade of steel (pipe steel) used in this study.

The research paper presents a 2 D axisymmetric finite element model of a Stacked spiral EMAT coil configuration and a novel stacked spiral EMAT (SS-EMAT) coil configuration to optimize the generation of ultrasonic wave on steel material based in LF mechanism. This is achieved by improving the strength of LF generated by the EMAT system. The SS-EMAT model is an improvement on the conventional planner circular spiral coil configurations.

Fig. 2: 2D Representation of SS-EMAT showing Eddy current, static field and effective Lorentz force; (a) single coil layer over pipe steel specimen; (b) stacked (two) coil layer over pipe steel specimen; (c) proposed stacked (three) coil layer over pipe steel specimen. All drawings are to scale.

Table 1. Measured Electromagnetic properties of CS70 grade of pipe steel material [6].

Material designation	Relative permeability μ_r	Electrical Conductivity (S/m)
CS70	59	3.77×10^6

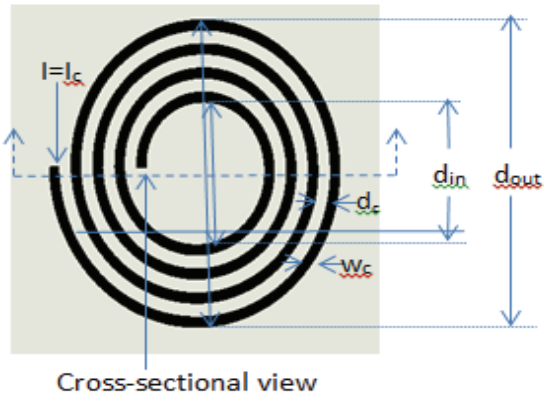
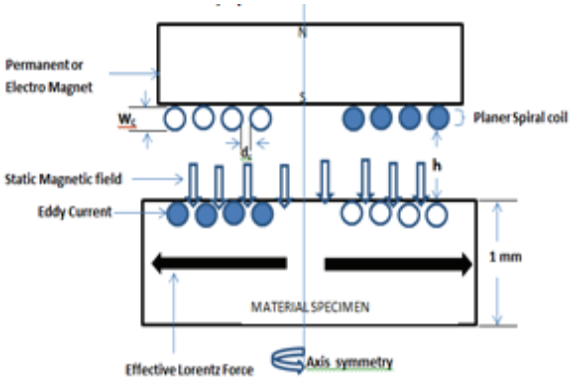
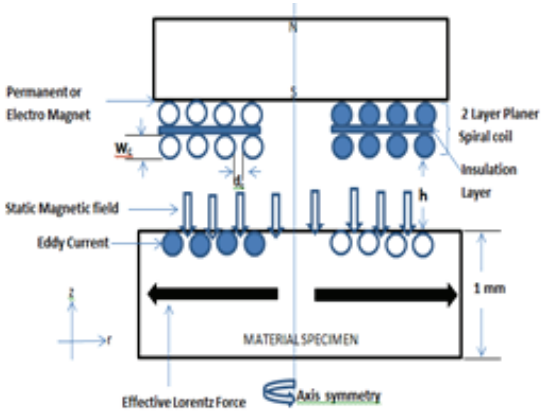


Fig. 1. The structural diagram of a circular planner spiral coil.



(a)



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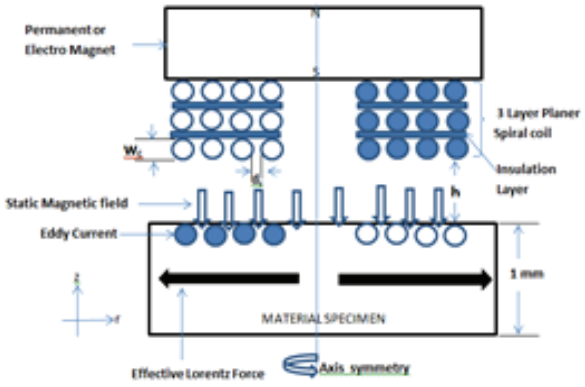

(c)

Fig. 2. 2D Representation of SS-EMAT showing Eddy current, static field and effective Lorentz force.

2. Theory

This analysis is based on the fact that in normal bias EMAT structure, LF has been proven to be the dominant transduction mechanism for both Ferro-magnetic and non ferro-magnetic materials [20].

Lorentz force is generated as a result of the interaction between the magnetic flux density and the eddy current density $\vec{J}_E$ induced by a coil excited with a sinusoidal current. The sinusoidal excitation current generate dynamic flux density $\vec{B}_d$ as well as Eddy current $\vec{J}_E$ in the steel plate. The Eddy current experiences a Lorentz force under the interaction of the dynamic magnetic field from the excitation current and the static magnetic field $\vec{B}_s$ from the permanent or electromagnet. The LF causes the vibration of the atomic structure of the material, which then leads to the generation of acoustic wave inside the conducting material. Generally, the direction and strength of the LF is governed by the following vector equations [21].

$$\nabla\times\vec{H}_d=\vec{J}_E, \tag{1}$$

$$\vec{B}_d=\mu_r\vec{H}_d, \tag{2}$$

$$\nabla\times\vec{E}_E=-\partial\vec{B}_d/\partial t, \tag{3}$$

$$\vec{J}_E=\sigma\vec{E}_E, \tag{4}$$

$$\vec{f}_d=\vec{J}_E\times\vec{B}_d, \tag{5}$$

$$\vec{f}_s=\vec{J}_E\times\vec{B}_s, \tag{6}$$

$$\vec{f}_L=\vec{f}_d+\vec{f}_s, \tag{7}$$

$$\vec{F}_L=\int\int\int_v\vec{f}_L\,dV, \tag{8}$$

where μ_r and σ, are the relative permeability and conductivity of CS70 grade of pipe steel; $\vec{E}_E$, $\vec{H}_d$ are the electric field intensity and dynamic magnetic intensity respectively; $\vec{f}_d$ and $\vec{f}_s$ are the Lorentz force densities due to the dynamic and static magnetic field respectively; $\vec{f}_L$ is the composite Lorentz force densities of $\vec{f}_d$ and $\vec{f}_s$; and $\vec{F}_L$ is the Lorentz force on a given volume, where ρ is mass volume density and τ and μ are the Lamé constants.

3. Finite Element Modelling

The structure of the EMAT considered in this research is shown in Fig. 2. In Fig. 2 (a), the structure consist of a rectangular meander-line coil, Fig. 2 (b) consist of a planer spiral coil while Fig. 2 (c) and Fig. 2 (d) consist of the novel single and double layer SS-EMAT coil respectively. Each structure consists of identical cross-sectional area of thickness t_c, width w_c, a permanent magnet and pipe steel specimen of 1mm thick. The lift off distance between the rectangular coils and the steel specimen is "h" and the distance between each coil turn is denoted as "d". In Fig. 2 (b) and Fig. 2 (c), the internal diameter of the coil is denoted as "d_{in}".

A 2D axis symmetrical finite element model was established using a commercial software called ComsolMultiphysics®, as shown in Fig. 5.

The coil was excited with a transient current [11]:

$$i_k(t)=\begin{cases}(-1)^k I_0\left[1-\cos\left(\frac{w_0 t}{n}\right)\right]\cos(w_0 t), \\ \quad for\ 0\le t\le(2n\pi)/w_0 \\ 0, \quad for \quad t\ge(2n\pi)/w_0\end{cases} \tag{9}$$

where I_0, w_0 and n are the amplitude of the current, the angular centre frequency and the number of cycles respectively.

$$w_0=2\pi f_0 \tag{10}$$

Fig. 3 shows the time history of external current density of the SS-EMAT at 30 A.

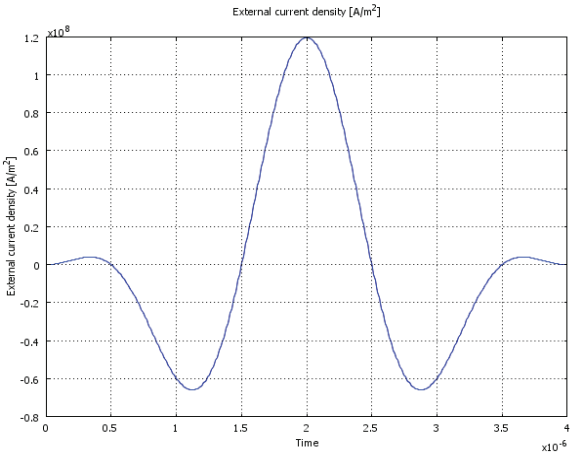

Fig. 3. Time history of the External current density of three layer SS-EMAT (I_0=30 A).

3.1. Modelling of Single Layer PS-EMAT

The parameters used for the simulation of the 1 layer SP-EMAT in Fig. 2 (a) are: f_0=500 kHz, n=2, I_0=30 A, B_0=1.2 T, W_c=0.8 mm, d_{in}=5 mm, d_c=0.1 mm and h=0.5 mm.

3.2. Modelling of Proposed Stacked SP-EMAT

The simulation was carried out using the same coil parameters as in 3.2 above but with adjustment in the number of layers of the coil to two in Fig. 2 (b) and three in Fig. 2 (c). An insulation layer thickness (t_{in}) of 0.1 mm was also included in the model. The insulation layer was modelled as a composite polymer (Z1258 anti rust silicone baffle paint) with performance index as presented in Table 2 [23].

Table 2. Performance index of Z1258 anti rust silicone baffle paint [23].

No.	Index name	Unit	Index
1.	Appearance		The film becomes bright after drying
2.	Viscosity	S	≥40
3.	Content of solid (10 g paint, (23±2^0 C/h)	%	55±5
4.	Drying time under room temperature	H	≤24
5.	Dielectric strength	MV/m	≥30
6.	Volume resistivity	Ω-m	≥ 1×10^8
7.	Fineness (scrapper fineness meter)	μm	≤45
8.	Adhesion coil	grade	1~2

3.3. Finite Element Mesh

The finite element model was obtained by discretising the physical model in to numerous second order quadratic triangular elements consisting of 6555 elements. The mesh element were refined at about 0.2 mm around the skin depth surface of the steel material and around the excitation coil with minimum element quality of 0.04897. This refinement of the model significantly improves its accuracy. Though the number of elements as wells as the solution time was greatly increased. This is achieved by continuously comparing the calculated result at different degree of refinement until when the result hardly changes with refinement as shown in Fig. 4.

To obtain a more precise result, the time steps, relative tolerance and the absolute tolerance of then finite element model were adjusted.

4. Simulation and Analisis

Applying the external source current in equation (9) and the static magnetic field in the finite element model, equation (8) is solved to obtain the Lorentz force density in the steel as shown in Fig. 5. Also to obtain an accurate record of the time variation of the Lorentz force from the simulation, the 2D axis symmetrical model was revolved about its axis, and a 3D surface plot was obtained. A line extrusion plot of the Lorentz force obtained at the skin depth of the steel material for both the single layer and multi-layer SS-EMAT as shown in Fig. 6, Fig. 7 and Fig. 8.

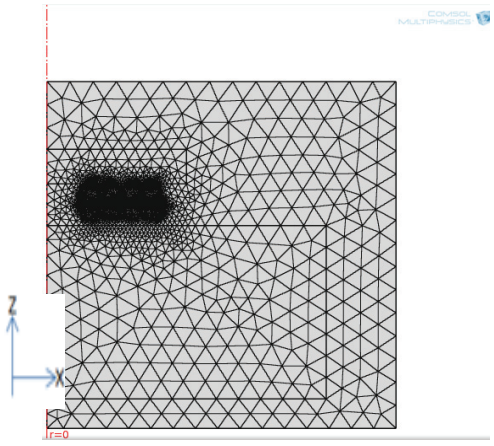

Fig. 4. Finite element mesh of the model used.

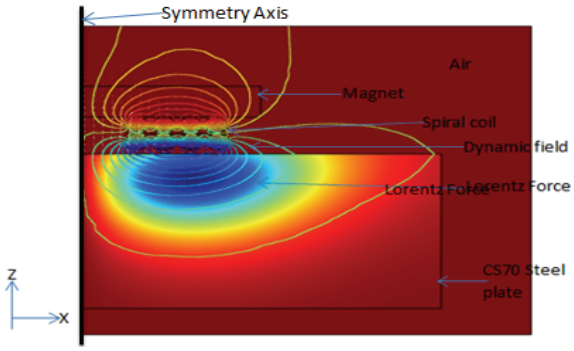

Fig. 5. Axi-symmetrical FE model of a stacked three layer SS-EMAT.

4.1. Influense of Coil Stacking on the Lorentz Force Density

As shown in equation (5), the dynamic magnetic field generated by the EMATs coil is a very important factor that influence the performance of EMATs in general. To improve the strength of the dynamic Lorentz force, the coils were stacked in layers and connected in series. The stacked coils were excited with a tone burst sinusoidal pulse as in equation (9). Fig. 6, Fig. 7 and Fig. 8 shows a 3D surface plot of a single, two and three layers SS-EMAT. The values of the density of LF were obtained by measuring the peak to peak value of the surface plot. This procedure was repeated to obtain the LF density for various number of coils in a stack as shown in Fig. 9. The values of the LF density obtained was 3.1×10^8 N/m^3, 1.9×10^9 N/m^3 and 3.5×10^9 N/m^3 for the single layer, two layers and three layers respectively.

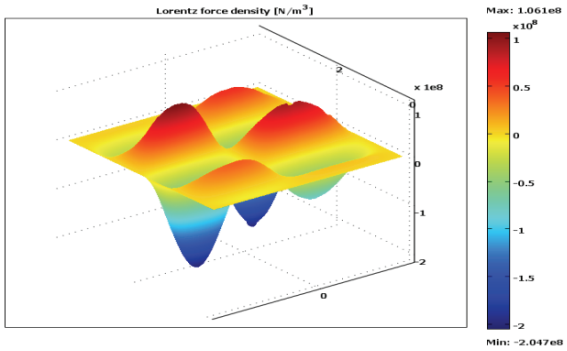

Fig. 6. 3D line extrusion surface plot of the Lorentz force generated by a single layer PS-EMAT.

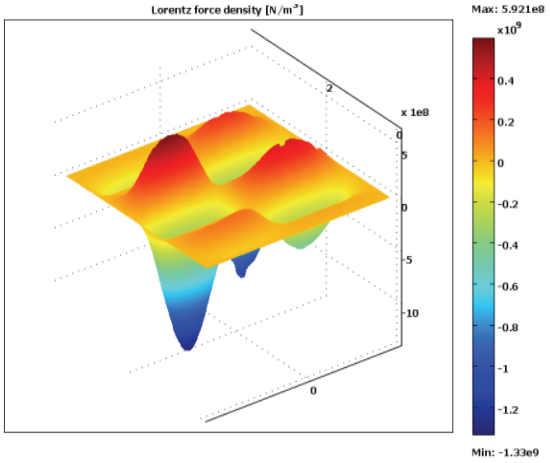

Fig. 7. 3D line extrusion surface plot of the Lorentz force generated by a Two layer SS-EMAT.

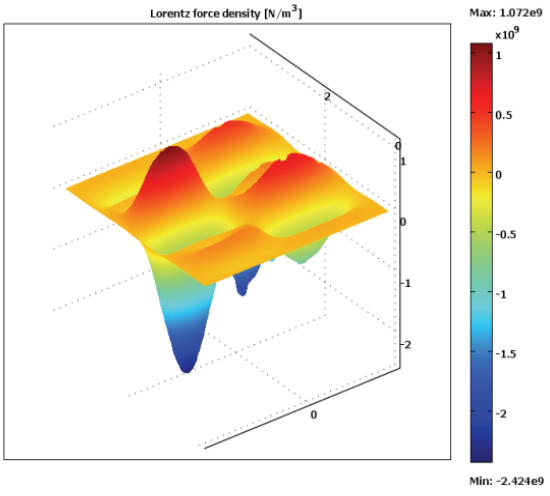

Fig. 8. 3D cross section linear extrusion surface plot of the Lorentz force generated by a Three layer SS-EMAT.

It was observed that the density of the LF in the materials increases exponentially with the increase in the number of coils in the stack/layers and saturates when the number of coil increases beyond eight coils per stack. Any further increment in the number of coil beyond this point yielded no significant increase in the Lorentz force density. This is expected since an increase in the number of coils in a stack, translates to an increase in the total dynamic magnetic flux. Beyond the Eighth coil, the distance between the coil and the specimen (lift off) will be enough as to obscure the effect of the additional coil, hence no significant increment was observed beyond nine coil per stack. Also, it worthy to note that in practice, an increment in the number of coil per stack does not lead to excessive heating due to the fact that the duration of the excitation current is typically at micro-second level [21]. Therefore, the SS-EMAT configuration performs optimally when the number of coil in a stack does not exceed 8.

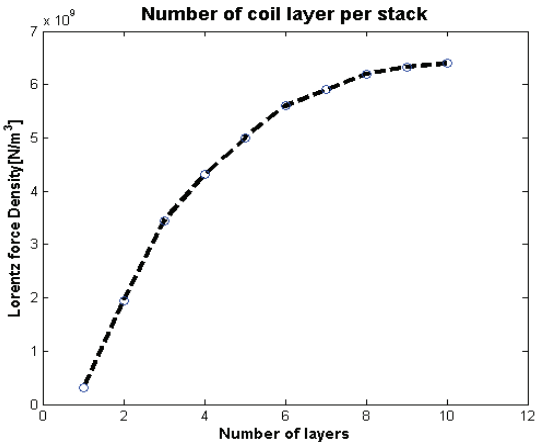

Fig. 9. LF strength under different number of coil layers.

4.2. Influence of Lift-off Distance on the Lorentz Force Density

The performance of the single, two and three layer stacked SS-EMAT configuration on CS70 grade of pipe steel was investigated, by varying the lift-off distance from 0.5 mm to 8 mm above the surface of the steel specimen. The measurement of LF density was taken within the skin depth (at about 0.1 mm below the surface) of the material specimen at the second turn of the coil. The results shown in Fig. 10 indicates that the LF density decreases exponentially with the lift-off distance with the three layer SS-EMAT configuration having the best lift off characteristics, followed by the two Layered configuration. This phenomenon also confirms the correlation between the coil layers, the dynamic magnetic field, and the LF and is in good agreement with findings of [22]. The results also indicates that for single and two layers SS-EMAT configuration, any increase beyond 4.0 mm and 4.5 mm respectively of lift off, the LF generated in the specimen is almost zero. While for three layer SS-EMAT, the same effect was observed beyond 5.5 mm of lift-off. This further indicates that there is a significant improvement in the lift off distance from 1 mm-3 mm obtained in most commercial EMAT [1] to 1 mm-5 mm.

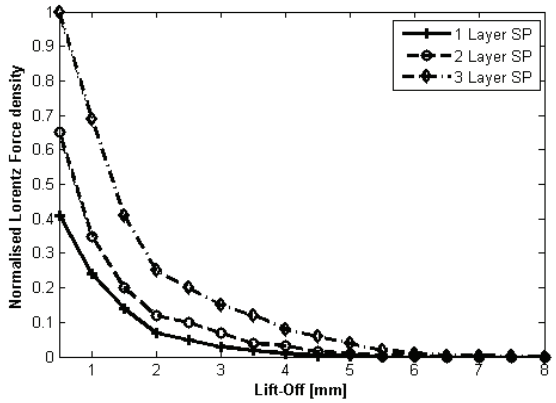

Fig. 10. Variation of Lift off distance with the number of coils per stack, on CS70 grade of pipe steel.

4.3. Influense of Coil Internal Diameter (d_{in}) on the Lorentz Force Density

Fig. 11 shows a correlation between the internal diameter of the three layer SS-EMAT coil and LF density generated on the surface of the pipe steel specimen. As the internal diameter is increased the LF also increases up to about 14 mm, at this point any further increment has no noticeable effect on the LF density. This phenomenon is because as the internal diameter is increased, the interaction between opposing dynamic magnetic field in the coil is reduced. This continues up to a point where there will be little or no effect of the opposing field. Therefore, for optimal performance of this stacked PS-EMAT configuration, the internal diameter must be greater than 14 mm.

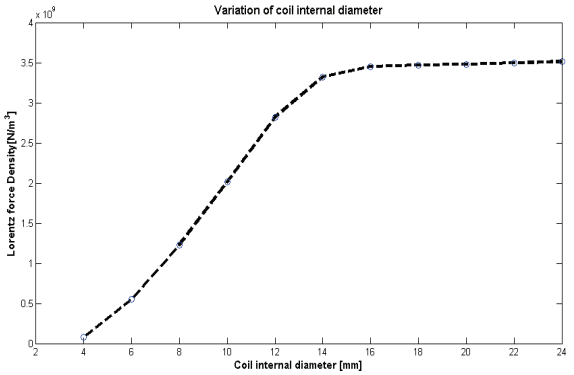

Fig. 11. LF strength under different internal diameter of SS-EMAT.

4.4. Effect of Variation of Static Flux Density (B_0) on the Lorentz Force Density

The static flux density is another important factor that affects the performance of EMATs, hence the need to investigate its effect on the LF generated by the three layer SS-EMAT, on the material specimen. Fig. 12 shows the result of the simulation obtained by varying the static magnetic field from 0.5 T to 2.0 T. The result shows that by increasing the static magnetic flux, the LF generated in the material specimen also increases and vice versa. At 0.5 T, the peak to peak value of the LF is 3.0×10^9 N/m^3 while the value of 2.0 T produced the highest LF of 4.7×10^9 N/m^3. The time variation of the LF was observed not be affected by the variation of the magnetic flux density. The result is in good agreement with published work of [9].

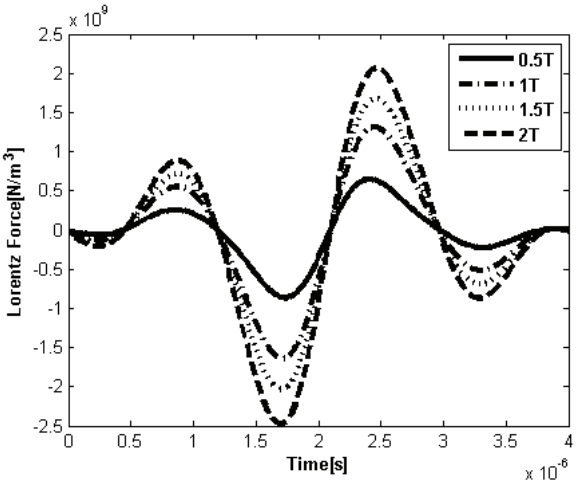

Fig. 12. Time variation of static flux density B_0 from (0.5 T to 2.0 T).

4.5. Effect of Variation of Insulation Layer Thickness (t_{in}) on the Lorentz Force Density

As observed in section 3.4, stacking coils in layers improves the performance of the SS-EMAT. To prevent short circuiting and binding the coils tightly together, they are coated with insulation material. The insulation material used for this simulation was Z1258 Anti-rust silicone baffle paint and has the following properties [23]:

(i) Good heat resistance
(ii) Firm adhesion
(iii) Short drying period
(iv) High dielectric strength of 30×10^6 V/m.

Fig. 13 shows the result of the simulation obtained by varying the insulation thickness from 0.1 mm to 2.5 mm. The result implies that, the LF strength decreases exponentially with increasing insulation thickness. The highest value of LF strength of 1.95×10^9 N/m^3 was obtained at insulation thickness of 0.1 mm while the least value of 1.1×10^9 N/m^3 was obtained at a thickness of 2.5 mm. Therefore, for optimal performance of the KT coil, the insulation thickness must not exceed 1 mm. The performance index of Z1258 anti rust silicone baffle paint is given in Table 2.

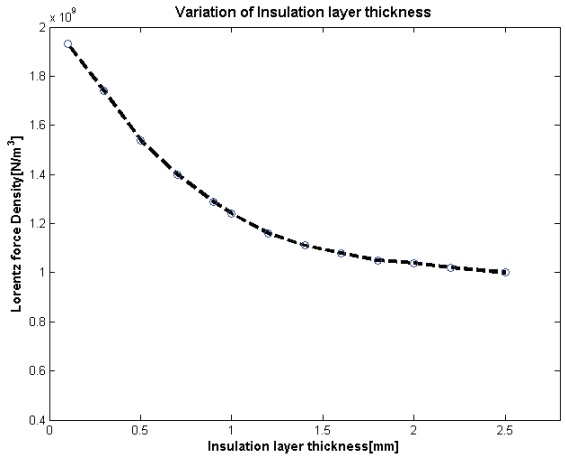

Fig. 13. Variation of insulation layer thickness from (0.1 mm to 10 mm).

4.6. Effect of Variation of Excitation Period on the Lorentz Force Density

It has been shown in [22] that the period number of a tone burst signal, determines the quantity of energy transferred from the power source to the coil, which has a direct effect on the LF strength and mechanical energy generated. This implies that more period number gives rise to more excitation energy. In the previous results (Fig. 5 to Fig. 13) the SS-EMATs were excited with a 2 period tone burst signal. To understand the effect of variation of the number of periods on the SS- EMAT under a given excitation current amplitude and frequency of 30 A and 500 kHz respectively, the number of period of the tone burst signal was varied in the following sequence (2 to 12) with all other parameters remaining unchanged. The result shows that an increase in the period, leads to an increase in the width of the LF wave packet (Fig. 14 (a) and Fig. 14 (b)). Analysing the result further, a linear relationship was established between the LF strength and the variation in the number of periods of the tone burst signal, see Fig. 14 (c).

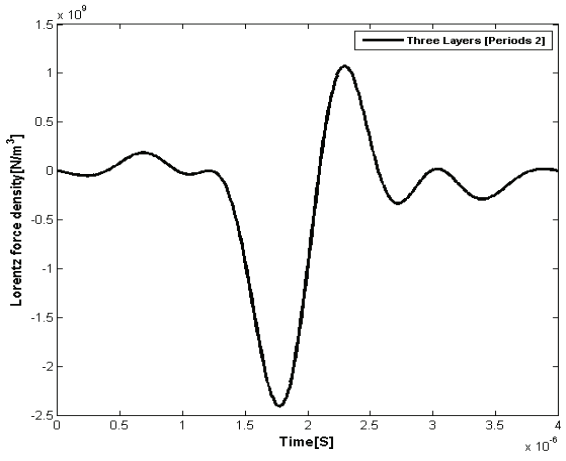

Fig. 14 (a). Lorentz force density for three layer SS-EMAT under different period number: 2 periods.

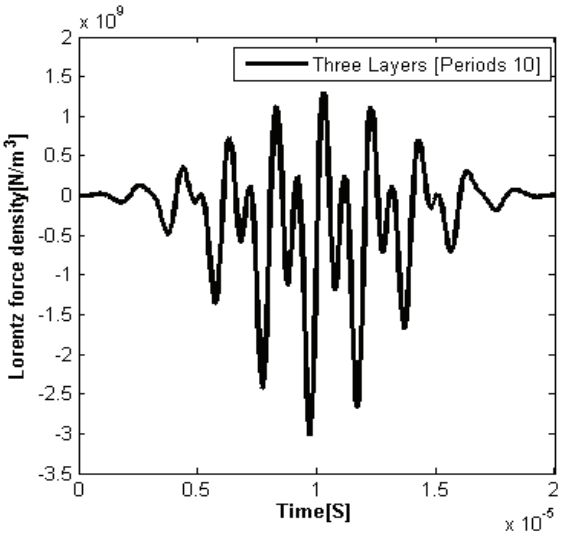

Fig. 14 (b). Lorentz force density for three layer SS-EMAT under different period number: 10 periods.

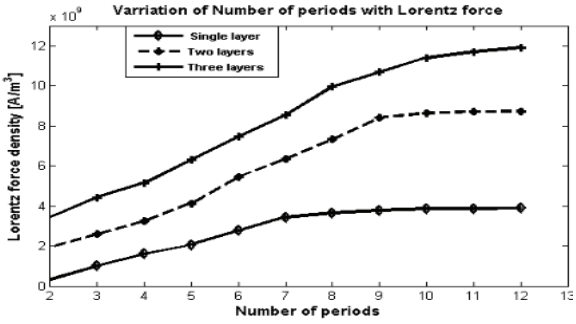

Fig. 14 (c). Variation of Lorentz force density with the number of periods from (2 to 12).

4.7. Effect of Variation of Excitation Frequency on the Density of Lorentz Force

To investigate the effect of variation of excitation frequency on the density of the LF generated by the proposed Three layer SS-EMAT, the static field and the excitation current were maintained at 0.7 T and 30 A respectively.

The excitation frequency was varied from 500 kHz to 8 MHz, and the corresponding change in LF generated in the material specimen was calculated. Fig. 15, shows that the LF density increases as the excitation frequency varies from 0.5 MHz and reached a peak value a at 4.5 MHz, thereafter it starts decreasing with a more pronounced effect seen in the three layer PS-EMAT. This implies that any increase in frequency beyond 4.5 MHz has a negative effect on the density of the LF generated in the material specimen. Hence for optimal performance of this EMAT, the right excitation frequency between 2.5 and 5 MHz should not be exceeded. This result is in good agreement with work published by [18].

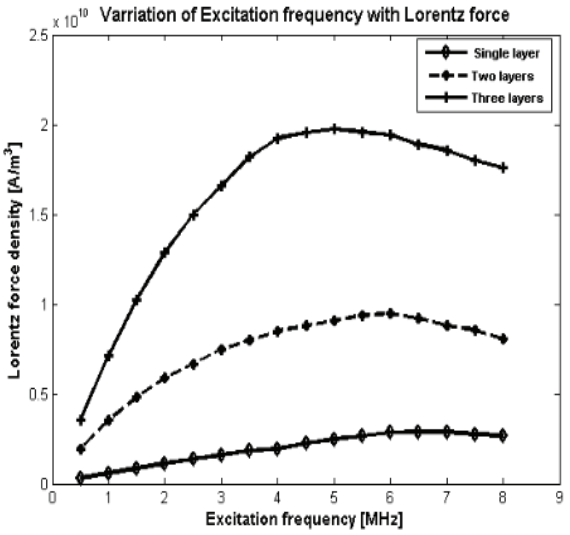

Fig. 15. Variation of LF density with excitation frequency.

5. Conclusion

In this research, a novel stacked PS-EMA transducer was presented and studied. Also, a 2D axis symmetrical finite element model in the time domain was successfully developed and implemented, we have used the model to study the effect of some parameters on the performance of a novel SS-EMAT configuration. The study carried out in the first stage includes calculating and comparing the LF density generated by the three SS-EMAT configuration (single, two and three layers) on CS70 grade of pipe steel material. The second stage of the study investigated the effect of variation of some EMAT parameters (such as coil internal diameter, lift off, coil width, excitation frequency, coil stacking, insulation layer thickness and excitation period) on the LF generated by the SS-EMAT configuration. The numerical results were in good agreement with previous work done by other researchers. According to the study, the proposed SS-EMAT is effective in improving the Lorentz force density, which is a very good parameter required for non destructive testing of steel materials. This is so because an increase in the Lorentz force density leads to a corresponding increase in the strength of the acoustic signal generated on the material specimen.

The result also shows that close attention should be paid to the these EMAT parameters in other to obtained accurate measurement using the SS-EMAT configuration.

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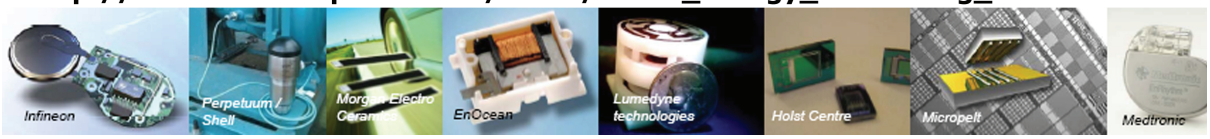

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Design of Automatic Production Line Training System Based on PLC

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Abstract: Automatic production line training system is used to simulate the actual process of industrial production. According to the actual needs of teaching instrument market at present, for the purpose of students or staff training, the author designs a practical automatic production line simulation training system. Using Siemens S7-200 PLC as the control core, setting sensors, buttons and lights, power supply module, motors, pneumatic components, mechanical structures for the integration, dividing into feeding, processing, assembly, sorting, robot units, and achieving interconnection between the unit controllers via PROFIBUS. It is used to simulate the various manufacturing aspects of workpieces from raw materials to finished products in the actual industrial production lines. The system covers mechanical principles, pneumatic control technology, sensors, electrical control technology, programmable logic controller and other related knowledge points, supports professional experimental courses of automation, mechanotronics and so on. A series of practical ideas are adopted in the subject. The entire system is fully open, students can have a comprehensive view of the internal wiring situation; freely detachable, can operate under single-station or overall system; supporting secondary development, can be re-build according to the innovation requirements. *Copyright © 2013 IFSA.*

Keywords: Automatic production line, PLC, Training system.

1. Introduction

Automatic production line training system is used to simulate the actual process of industrial production. It is used in training room of colleges and universities, to cultivate students' cognition, operation, then using and developing abilities. In order to cater to the educational philosophy and teaching need of today's colleges and universities, the training system is a comprehensive experimental teaching system which is connecting theoretical teaching with practical training. Through the demonstration experiments, assembly debugging and programming, students can stimulate learning interests, exercise practical abilities and thinking skills, consolidate expertise, improve professionalism and guide students to adapt to the demand for jobs in the future.

1.1. Research Status

Before the 1920s, the production line and semi-automatic production line appeared in the automotive industry; 1920s, Henry Ford founded the automotive industry assembly line; later, combination automatic line gradually emerged; the development of automatic control and computer technology,

prompted the completion of the automation workshops and factories [1]. Today, automatic production line is developing toward the direction of adjustable varieties, higher flexibility, faster and lower cost. With the automatic production line developing by leaps and bounds, the development of automatic production line simulation device has been greatly improved [2]. At present, many ordinances manufacturers have achieved the rapid development of automatic production line simulation ordinances. Most of today's automatic production simulators have achieved demonstration, debugging operations, programming and other basic functions, but there is considerable room for improvement [3]. Developing a removable, secondary developing, working sub-station or run-online training system is a far-reaching issue.

1.2. Issues to be Addressed

The system uses Siemens S7-200 PLC as the control core, setting sensors, buttons and lights, power supply module, motors, pneumatic components, mechanical structures for the integration, dividing into feeding, processing, assembly, sorting, robot units, can operate under single-station or overall system. It is used to simulate the actual process of industrial production. The entire system is fully open, freely detachable, supporting secondary development, can be re-build according to the innovation requirements.

2. The Overall System

Automatic production line training system consists of five units: feeding unit, processing unit, assembly unit, sorting unit and robot unit. The units are all installed in the aluminum alloy rail training table, the overall system block diagram is shown as Fig. 1.

Each unit of the system is an independent mechatronic system. Feeding, processing, assembly, sorting units basically depend on pneumatic actuator, robot unit takes the position controlling mode driven by the stepper motor. In terms of on-line operation, each unit is undertaken by a PLC, and achieving interconnection between the unit controllers via PROFIBUS. The entire system can simulate supplying, processing, assembly, sorting, transporting and other process of actual industrial production line.

Button and indicator modules including start, stop, reset, function buttons, manual / automatic, single / overall knob, the alarm indicator and emergency stop button. Start and stop buttons are used to control the start and stop operation of the unit; reset button can reset all the action element to initial position; function button is set for secondary development; spinning manual / automatic, single / online knob at the same time can select single-station or overall system situation.

Each unit of the system needs to be powered separately. The power supply module includes air switch and one-channel switching power, one-channel switching power can convert 220 VAC to 24 VDC and supply to each unit.

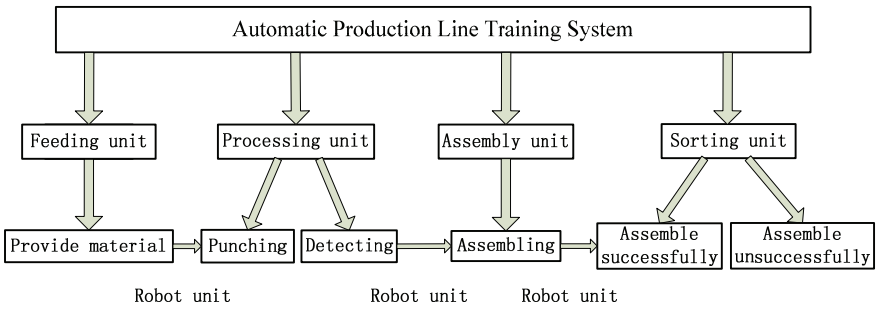

Fig. 1. The overall system block diagram.

3. Feeding Unit

The feeding unit is the first unit which plays a feeding role of supplying raw materials to the next unit. It can launch the material in the silo to the transfer stage automatically, so that the robot will crawl and transport it to other units. The configuration diagram of feeding unit is shown in Fig.2.

Feeding unit selects fiber optic sensor and amplifier, photoelectric sensor. Fiber optic sensor is mounted on the bottom side of the silo to detect the material; photoelectric sensor is used to detect whether the material is reaching.

Material transporting conveyor belt is driven by motor, the driving wheel is φ90 × 36 mm (wheel diameter × wheel thickness), the driven wheel is 40 × 36 mm (wheel diameter × wheel thickness). Micro-AC gear motor and the gearbox can regulate the actual speed between 9-135 rpm [4]. Lower speed can not only provide a greater torque, but also make the material smoothly fall on the conveyor belt, and obtained the similar speed with the conveyor belt within a shorter time and a smaller displacement, thus ensure the stability.

Choosing SMC mini pen-type cylinder, the cylinder push block pushes the material in the hopper to the material transfer stage. Connect SMC one-way

solenoid valve to the pen-type cylinder. One-way solenoid valve is a single electronic control components, it will control one-way cylinder, and return to its original position when lost power. Each pneumatic circuit consists of cylinder, one-way throttle valve, two-five way valve, manual valve, pneumatic FRL, pressure gauge, gas supply, muffler and other components. When the two-five way valve coil is energized, the cylinder piston driven stem extending and pushing the material; when the two-five way valve coil is de-energized, the valve stem move back to the initial position [5].

The flowchart of feeding unit is shown as Fig. 3.

4. Processing Unit

This unit has four workstations. Robot grabs the workpiece to the first workstation. When the workpiece is detected, the dial rotates, moving workpiece to the stamping punch workstation, then moving it to the stamping detection workstation to detect whether it is qualified, and then sending it to the fourth workstation, waiting for the robot to send it to the next unit. The configuration diagram of feeding unit is shown in Fig. 4.

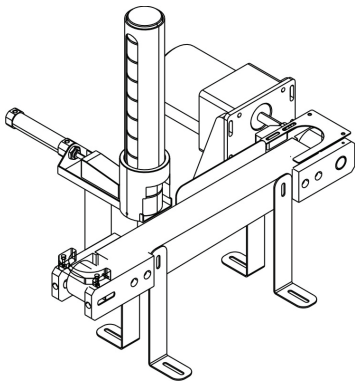

Fig. 2. The configuration diagram of feeding unit.

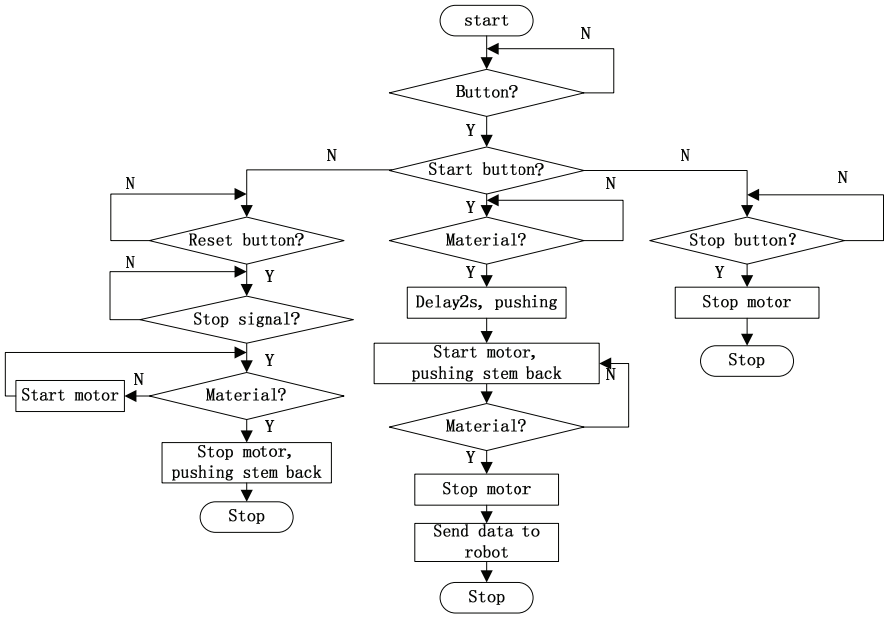

Fig. 3. The flowchart of feeding unit.

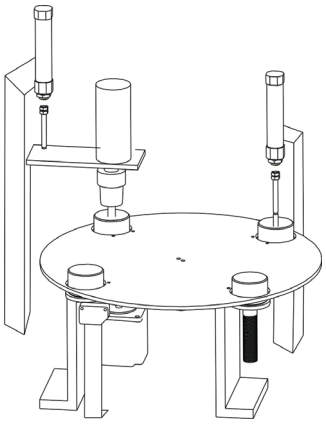

Fig. 4. The configuration diagram of processing unit.

Processing unit selects photoelectric sensor, proximity switch and two micro-switches. The photoelectric sensor is installed below the first workstation, using to detect whether the aluminum material is sent here; proximity switch is mounted on the bottom of the second workstation to detect whether the workpiece is arriving accurately; the two micro-switches are fixed on the first and fourth workstation of the robot run rails, using to judge the current running position of robot.

The dial controlling motor is hybrid stepping motor, which is installed on the bottom of the turntable and driving turntable rotating. Thin cylinder drives a DC gear deceleration motor rotating downward, to simulate the drilling movement.

SMC thin cylinder drives stamping punch motor move along the Y-axis direction; SMC mini pen cylinder simulates stamping detection operation. When the two-five way valve coil, which connected to thin cylinder is energized, stem drives drilling motor to punch; when the valve coil is de-energized, the valve stem moves back to the initial position [6]. Mini pen cylinder acts similarly.

Processing unit works as follows: The robot transports the workpiece to the first work station of the turntable, photoelectric sensor detects the presence of the metal parts, motor drives turntable rotating ninety degrees to the second workstation, DC gear motor moves along the Y-axis negative direction to simulation the operation of the stamping, and then the dial rotates to simulate punch detecting, finally the dial rotates to the fourth station, the proximity switch detects the metal workpiece and then sends the information to PLC, robot will send workpiece to the next unit.

5. Assembly Unit

The function of assembly unit is to embed the metal pin into the workpiece, and complete the assembly process. This unit focuses on pneumatic controlling, is the main unit of pneumatic control training. The configuration diagram of assembly unit is shown in Fig. 5.

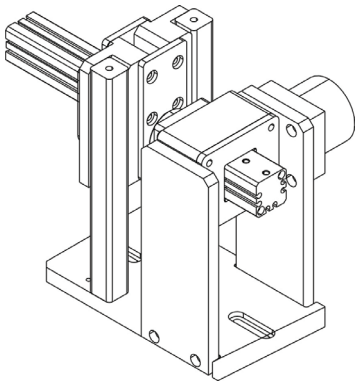

Fig. 5. The configuration diagram of assembly unit.

The unit contains a rotating cylinder and two square cylinders, selecting SMC swing cylinder to control the rotation of the assembly slot. Pushing cylinder is SMC thin cylinder, using for pushing out the metal pin. Ejecting cylinder is SMC standard cylinder, using to top out the assembly slot. The joint action of pushing and ejecting cylinder makes the workpiece complete the assembly process quickly and accurately [7]. Rotating cylinder and ejector cylinder connect to the two-way solenoid valve. The two-way solenoid valve is a double electronic controlling components, it can control two-stroke movement.

When the left coil of solenoid valve which is corresponding to rotating cylinder is energized, the left intake path and the right exhaust path switch on at the same time, rotating cylinder swings clockwise, and drives assembly platform to rotate 90 degrees, when the left coil is de-energized, rotating cylinder stops turning and keeps the original state; when the right coil is energized, the right intake path and the left exhaust path switch on at the same time, rotating cylinder swings counterclockwise, and drives assembly platform to rotate negative 90 degrees, assembly station returns to the initial position, when the right coil is de-energized, the bilateral gas path close at the same time, rotating cylinder stops.

When the left coil of solenoid valve which is corresponding to standard cylinder is energized, the ejector cylinder piston drives stem out and tops out the assembly slot, when the left coil is de-energized, the bilateral gas path close at the same time and keep the original state; when the right coil is energized, the ejector cylinder piston drives stem back to the original state, when the right coil is de-energized, the ejector cylinder piston stops.

When the coil of solenoid valve which is corresponding to the thin cylinder is energized, the pushing cylinder piston drives the stem out, pushes metal pin out to the hole of the workpiece; when the coil is de-energized, the stem of pushing cylinder returns.

The pneumatic picture of assembly unit is shown as Fig. 6.

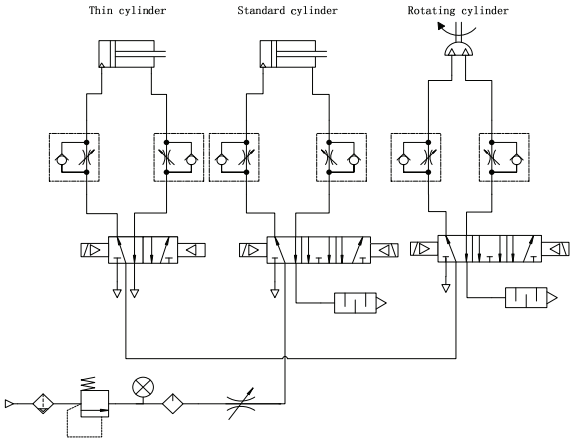

Fig. 6. The pneumatic picture of assembly unit.

The assembly unit works as follows: workpiece is placed on the assembly platform, rotating cylinder rotates ninety degrees, the ejecting cylinder tops out the assembly platform, then the pushing cylinder pushes the metal pin out to workpiece, completing assembly process. The ejecting cylinder returns, rotating assembly platform goes back to the initial position, robot sends the workpiece to the next unit.

6. Sorting Unit

This unit includes a fiber-optic sensor and its amplifier, proximity switch and beam photoelectric

switch. Fiber-optic sensor is installed on the sorting table to detecting the material. Proximity switch is suspended on the sorting platform, the distance between the metal pin that assemble successfully and the proximity switch is 4 mm, while the distance between the workpiece and the proximity switch is above 5 mm. Due to the detection range of the proximity switch is 3-5 mm [8], when workpiece arrives within the detection range of the sensor, proximity switch can detect its presence or not, that is whether the workpiece has been successfully assembled; beam photoelectric switches are installed in the middle of the trough to detect whether sorting operation is successful or not, then control the motor to stop.

DC reduction gear motor can drive conveyor belt to transport the material to sorting detection point.

Selecting small no-shaft-rotating pen cylinder to regulate the extending of baffle at the successfully assembled trough. In order to reduce the impact of the baffle caused by the movement of the workpiece, to make workpiece change direction when encountering baffle, this requires the contact surface between baffle and workpiece is arc, which requires the cylinder that controlling the damper must be no-shaft-rotating kind [9]. When the solenoid valve coil is energized, the cylinder piston drives the stem out, pushes baffle out; when the solenoid valve coil is de-energized, the stem of cylinder returns.

Sorting unit works as follows: fiber optic sensor detects the sorting materials, then controls motor to drive the transmission belt. Proximity switch detects the material when it reaches the centre of the belt, if it can detect the presence of the metal pin, indicating that the workpiece has been successfully assembled, cylinder pushes the baffle out, intercepts the workpiece, lets it slide into assembled slot; if proximity switch can not detect metal pin, indicating that the workpiece has not been successfully assembled, cylinder remain intact, lets it slide into no-assembled slot; when the workpiece slide into the slot, beam photoelectric switches detect the workpiece, baffle returns back, motor stops, sends completing signal to the robot unit, accepts the next workpiece sorting task.

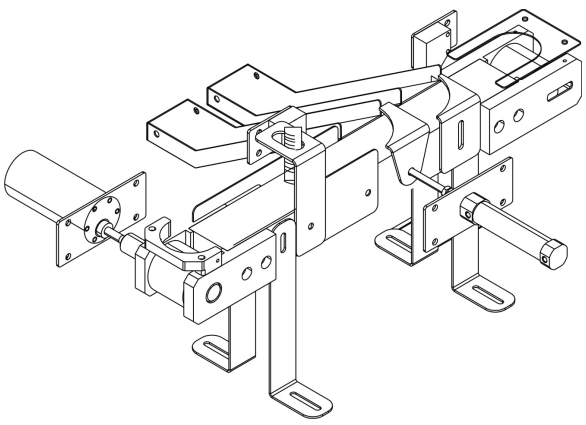

Fig. 7. The configuration diagram of sorting unit.

7. Robot Unit

The robot unit is the driving unit of the entire system, plays an important role of linking the remaining four units to complete the handling function among the other units. In this unit, the coordinates setting of robot's moving direction is shown as Fig. 8.

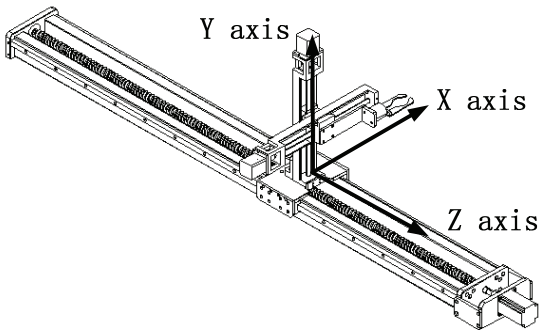

Fig. 8. The coordinate diagram of robot moving direction.

The robot unit has four micro-switches, the X and Y-axis rails each has a micro switch respectively, indicating the robot's moving limit position in X, Y direction; Z-axis rail has two micro switches, corresponding to the dwell position of the robot at the assembly unit and the sorting unit respectively.

X-axis selects 1500 mm thread screw and EG series linear guide; due to Y-axis and Z-axis having a shorter stroke (300 mm) and a smaller load, selects straight-line modules to realize the moving of gripper in this two-axis direction [10].

This unit has three stepper motors, which are used to control the movement of the manipulator in the three directions -X, Y and Z. Using two-claws sliding rail gripper to grip the workpiece. The towline can avoid the winding knot problems caused by the robot's movement [11]. When the solenoid valve coil is energized, the gripper clamps, grabs the workpiece; when the coil is de-energized, the gripper releases, puts down the workpiece. Fig. 9 is the electrical diagram of robot unit.

This unit works as follows: the robot can move along the X, Y, Z axis direction, rely on pulses and micro switches to locate accurately, indicate its duration position by micro switches, transport the workpiece to the corresponding position according to the needs and complete crawling and placing function.

8. Network Control and Overall System Running

8.1. Network Control

The system is running with the STEP7 programming software. In the network settings, the robot is driving unit, the other four are driven units.

Setting the communicate parameters in the system block, that is all the units are communicating with the robot unit. The transmission baud rate is set to 9600 bps with PPI communication protocol [12].

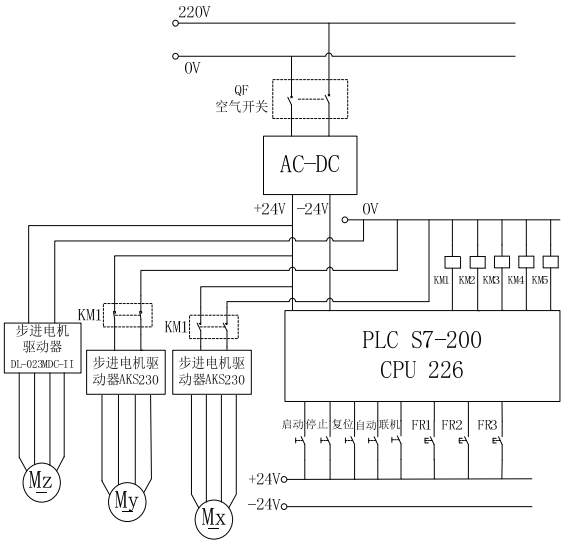

Fig. 9. The electrical diagram of robot unit.

In the PPI network, the robot unit must specify the driving property with the Port0 controlling byte SMB30 in the first scan cycle. After setting, the robot program will generate NET_EXE subroutine [13].

In the processing and robot program that selecting stepper or servo motor, the STEP7 wizard will generate the PTO (Pulse Train Output) subroutine automatically [14].

8.2. The Overall System Running

When the overall system is running, the running diagram is shown as Fig. 10. The thick line is PLC-PROFIBUS bus cable, it is used to connect PLC via Port 0, to realize communication of the whole system [15]. After programming and debugging, we can download the program to PLC via serial programming cable, in order to achieve the running purpose of the overall system without computer.

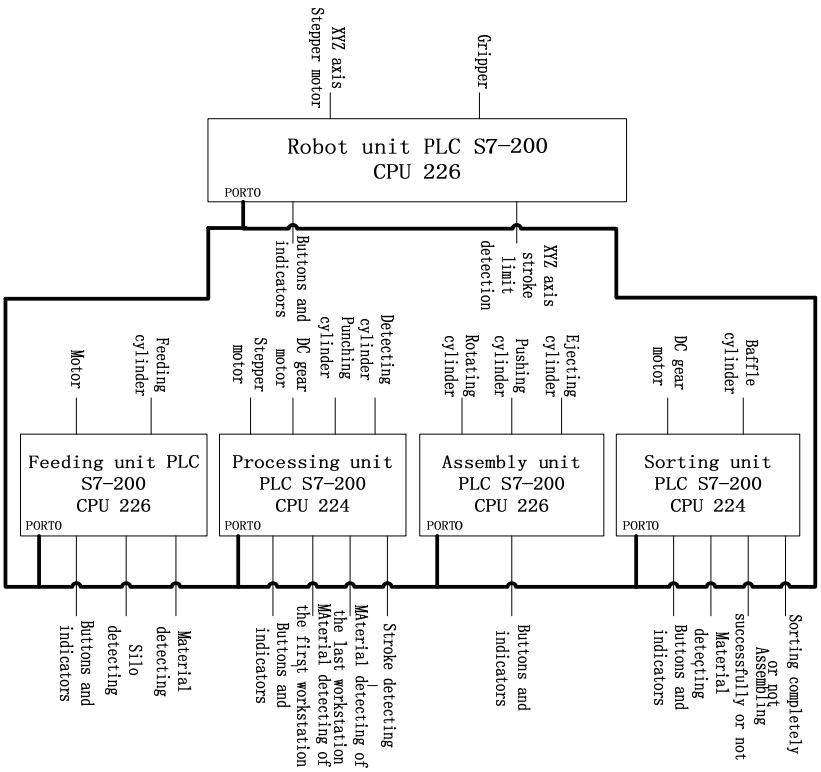

Fig. 10. The overall system running diagram.

9. Results and Discussion

The automatic production line training system can be applied in the training room of colleges and universities as the following three aspects: automatic production line simulation demonstration; according to the hardware design, compiling to achieve the overall function; reassembling the system according to the requirements and programming with the reserved bits properly, innovating more practical functions.

Compared with the other training systems at the teaching instrument market at present, the system has the following obvious advantages:

The System is fully open, especially the controlling cabinet, students can have a comprehensive view of the internal wiring, it is easy to apperceive and understand for students.

The system is freely detachable, it can operate under single-station or overall system. We can choose the type and quantity of the workstations according to the training requirements.

The controlling system has reserved some controlling bits, they can be used for developing and innovating.

10. Conclusions

The automatic production line training system covers electrical control technology, pneumatic control technology, sensors, mechanical principles and other related knowledge points. The entire system is fully open, freely detachable, can operate under single-station or overall system, can be re-build according to the innovation requirements. Experiments show that the system has simple hardware structure and reasonable programming, runs reliably and stably. It fits today's education philosophy of closely integration between theory and practice, has important and long-term significance of high-quality talents cultivation, the seamless connection of universities and enterprises.

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Design of Control System for the Transmitter of iFTEM Type Time-domain Fixed-wing Airborne Electromagnetic Survey System

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Abstract: To meet the need of a transmitter of domestic iFTEM type time-domain fixed-wing airborne electromagnetic survey system, we design a control system based on PC104 bus for iFTEM system. This system consists of a driving signal generator, synchronization control system, protection control system, voltage, current monitoring and transmitter waveform monitoring. In the design, we use WinDriver to develop a driver for the PC104 driving signal generator and design a practical control panel for the control system by using LabVIEW. The control system has advantages of small size, high reliability and being easy to operate. Experiments shows that our system is stable and reliable, achieves controlling over the transmitter of iFTEM system, and holds its reference value in research of domestic time-domain fixed-wing airborne electromagnetic method. *Copyright © 2013 IFSA.*

Keywords: Time-domain fixed-wing airborne electromagnetic survey system, Transmitter, Control system, PC104, WinDriver, LabVIEW.

1. Introduction

In 2007, Institute of Geophysical and Geochemical Exploration of CAGS began the research on iFTEM system; iFTEM system is a time-domain fixed-wing aircraft electromagnetic survey system. Time-domain fixed-wing aircraft electromagnetic method takes a flying fixed wing aircraft as the carrier, produces a large transmitting magnetic moment through the transmitter loop wired around the aircraft, thus to stimulate the earth to produce a secondary induction field, and then uses the electromagnetic field theory to analyze this secondary field response in order to get the geological information of hundreds of meters depth [1]. Advantages of this method include high speed, being free of ground conditions (such as sea, river, lake and desert), relatively uniform working accuracy, and being able to work in the case of complex terrain conditions. Now this method is gaining more and more attention in the geophysical field.

After half a century of development, foreign time-domain fixed-wing airborne electromagnetic method and the instrument system have become mature. Now systems widely used over the world and

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getting good results in practical applications are GEOTEM, MEGATEM and MEGATEMII system developed by Canadian Fugro [2, 3], TEMPEST system from the Joint Research Centre of the Australian mineral exploration [4] and the SPECTREM system from South African Anglo American Corporation (AAC) [5]. So far the method has found good application in mineral resources exploration [6], large area geological mapping [7], hydrogeology, engineering geology [8], expedition and environmental monitoring [9]. However Chinese domestic time-domain fixed-wing airborne electromagnetic method and its instrument industry have been left behind many other countries. In 1970s, Changchun College of Geology, Heilongjiang Geological Bureau and some other institutions did research and developed the first set time-domain airborne electromagnetic system M-1 induction pulse transient airborne electromagnetic instrument. In 1977, M-1 gained national recognition [10], but because of various reasons, the system had not been able to put into actual production. Since then our country has not carried out any related research. So far our country still does not have a time-domain fixed-wing airborne electromagnetic survey system in real production.

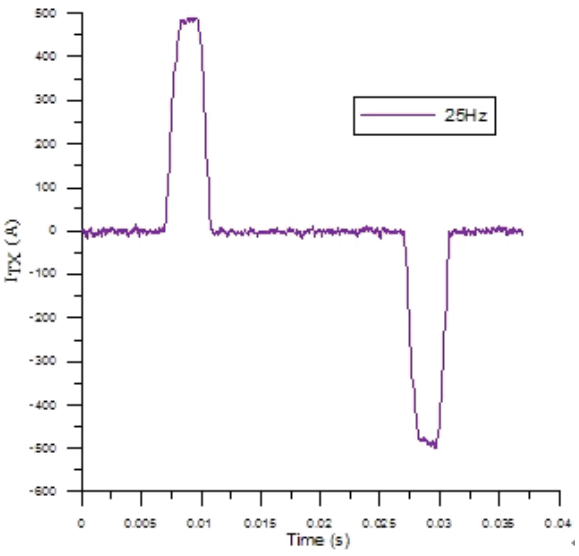
iFTEM type time-domain fixed-wing airborne electromagnetic exploration system described in this article, is funded by national high technology research and development Program, "Airborne geophysical survey system" sub-topic "Time-domain fixed-wing airborne electromagnetic exploration systems research and development". iFTEM systems are divided into transmitters and receivers. In the paper we mainly discuss the control system of a transmitter. The control system which is the core of a transmitter, must have the ability to control the transmitter waveform, to protect the transmitting system when transmitter errors appear, to keep receiver and transmitter in pace, to monitor the voltage, currents and the transmitter waveform in real-time. Our control system is based on PC104 bus, has advantages of small size, high reliability and operational easiness.

2. Introduction of the Transmitter

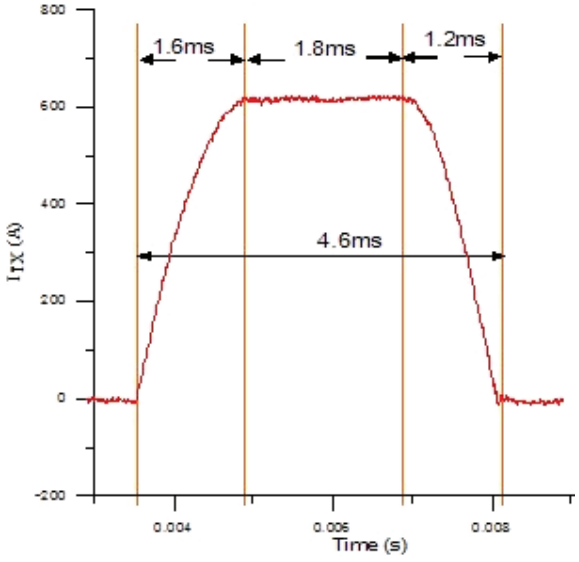
iFTEM's transmitter waveform is bipolar combination wave, base transmitter frequency being 12.5, 25 or 50 Hz. When the base frequency is 25 Hz, the transmitter waveform is shown in Fig. 1 (a); in Fig. 1 (b), we can find that the waveform is composed of rising edge, flat top, falling edge.

Fig. 2 shows transmitter architecture. The functions of each module are described below.

1) The core of the transmitter is the full-bridge inverter circuit. Full-bridge inverter circuit converts direct current into bipolar AC [11]; at the same time the full-bridge inverter circuit is also the power section of the transmitter. IGBT which is the power component switching converter is the core of the full-bridge inverter circuit, thus DC power is converted into bipolar pulse alternating current in order to achieve a large power output.



(a)



(b)

Fig. 1. iFTEM's transmitter waveform (25 Hz example).

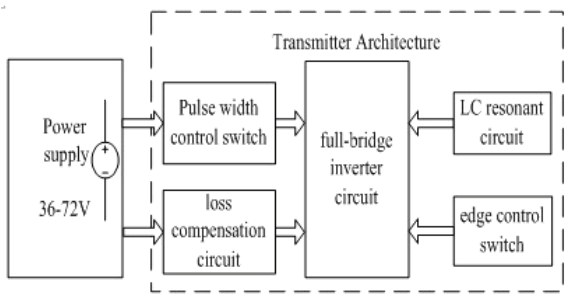


Fig. 2. Transmitter Architecture.

2) During the transmitter on-time, the high-voltage power stored in super capacitors is released into the transmitter loop; the current of the transmitter loop rapidly rises, and then forms the rising edge of transmitter waveform. An edge switch signal controls the rising edge time.

3) During the transmitter off-time, by a resonance circuit, the power of the inductance in the transmitter loop is stored into super capacitors, resulting in the falling edge of the transmitter waveform. We design a LC resonant circuit in iFTEM system.

4) When the transmitter current reaches the maximum value, in order to maintain a high current in the transmitter loop, we use the low-voltage power to supply continued flow for the transmitter loop, and then form the flat top of the transmitter waveform. The time of the flat top is controlled by a pulse width control switch signal.

5) Due to energy loss in the transmitter loop, in order to make the transmitter to achieve a large transmitter power, we must pre-charge resonant capacitors during the transmitter off-time. We design a loss compensation circuit to complete the task.

The driving signals used in the control system are shown in Fig. 3. Trigger is a synchronization signal of iFTEM system. Trigger can be provided by the transmitter or receiver, Trigger's frequency is 12.5, 25, 50 Hz optional. S1 and S2 are the switch control signals of the full-bridge inverter circuit, their phase being reverse, S1 and S2 controlling four IGBT of full-bridge inverter circuit to alternately turn it on. Then the bipolar waveform will appear in the transmitter loop. S3 controls the rising edge of combination wave and controls the rising time of transmitter waveform. S4 controls the pulse width of the combination wave and controls pulse width time of transmitter waveform. S5 is the switch control signal of loss compensation circuit, controlling the charging time of super capacitors. S3, S4 and S5 are all based on the rising edge of Trigger for the trigger edge, delay time and pulse width time are adjustable.

3. Control System of the Transmitter

The control system consists of driving signal generator, protection control system, synchronization control system, voltage and current monitoring and the transmitter waveform monitoring (Fig. 4). The driving signal generator, protection control system and synchronization control system work on the PC104 signal generator card, while the monitoring of voltages, currents and the transmitter waveform work on the PC104 data acquisition card.

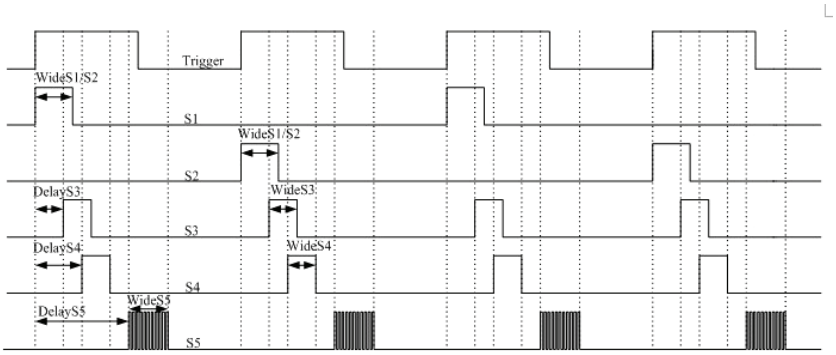


Fig. 3. The driving signals used in the control system.

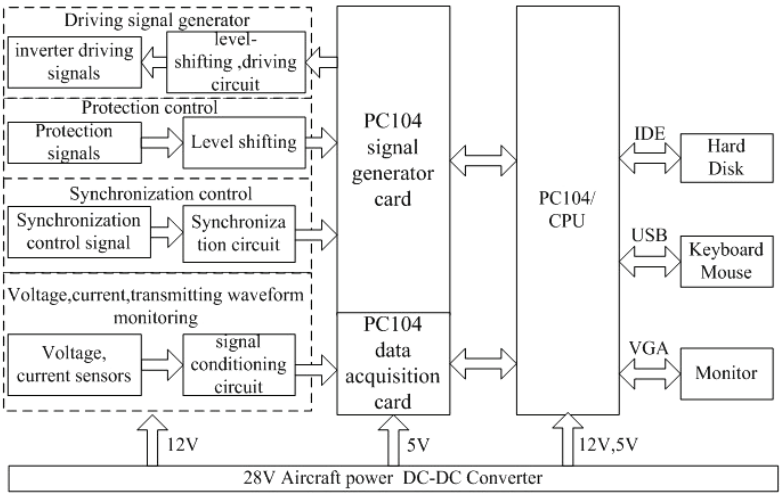


Fig. 4. Hardware structure of control system.

All modules required for the 12 V and 5 V power are supplied by aircraft power whose voltage is 28 V, which will be converted to 12 V or 5 V by DC-DC converter. The core of the control system is PC104/CPU module. When PC104/CPU module is connected to the hard disk, keyboard, mouse and monitor, a micro-computer is composed, and then users can operate the PC104 control system as easily as to operate their own PC, so users can quickly grasp the use of the PC104 control system. Signals generated by PC104 signal card are sent into the transmitter after passing level-shifting circuits and driver circuits, then become the inverter driving signals of the transmitter; external protection signals enter PC104 signal card after passing level-shifting circuits, then if the external protection signals are effective, the signal card will shut down the transmitter; the synchronization control is that the receiver produces a "Trigger" signal of signal generator card; voltage, current monitoring and the transmitter waveform monitoring are implemented by the PC104 data acquisition card. After passing the voltage and current sensor and signal conditioning circuit, voltages, currents and the transmitter waveform signals are sent into the PC104 data acquisition card. In addition, when the transmitter is running, the transient current will bring the electromagnetic interference to the control system, so each unit should be independent so Opto-isolation method is applied to weaken the interference of electromagnetic.

In the control system, the PC104/CPU module is Em104P-i8523 from Taiwan ARBOR Co., Ltd, and the PC104 data acquisition card is PM518 data acquisition card of Beijing ZTIC Co., Ltd. In the control system the driving signal generator is the most important part. Control system requires the driving signal generator not only to have the function of signal generator, but also to have the function of synchronization control and protection control, so users need to design dedicated signal generator for the control system. Below we will focus on the driving signal generator.

Design the driving signal generator for the control system is based on PC104 bus, the driving signal generator is designed as a PC104 signal generator card (Fig. 5).

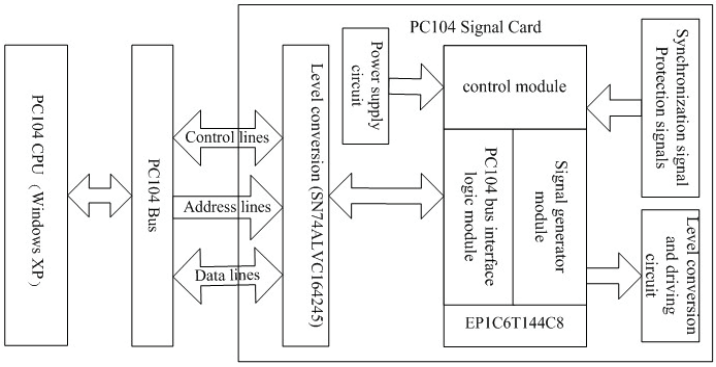


Fig. 5. Hardware structure of the signal generator.

PC104 signal card communicates with PC104 CPU by PC104 signal lines. The signal lines used in PC104 signal card are control lines, address lines and data lines. Signal lines must be level-shifting so as to communicate with FPGA chip of PC104 signal card. PC104 signal card's main task is completed by FPGA (EP1C6T144C8), including the PC104 bus interface logic module, signal generator module and control module. PC104 bus interface logic module completes the PC104 bus logic decoding and sequential circuits. FPGA's IO output level is 3.3 V, with pin's driving capability being only 24 mA (Altera Corporation 2008), but the transmitter requires the signal level to be 12 V, and driving capability not to be less than 200 mA, so the output signals need to go through level conversion and driving circuit before being sent into the transmitter. When external protection signals are effective, the signal card shuts the transmitter down. The signal card also controls the transmitter and receiver to keep pace.

Develop of PC104 signal card drive: On Windows XP, users cannot access hardware directly from the application level (user mode), where software development is usually done. Hardware can only be accessed from within the operating system itself, utilizing software modules called device drivers. In order to access PC104 signal card from the application level, we must develop a device driver for PC104 signal card. Often users use a certain driver development tool to develop device drivers, and then users' applications can access hardware. In a variety of device driver development tools, WinDriver from Jungo Co., Ltd is a rising star, which is well known for being simple and efficient, cross operating systems and good compatibility. You do not need to have any device driver knowledge, nor do you have to be familiar with operating system kernel programming, DDK, ETK or DDI/DKI. Using WinDriver only, you can develop a highly efficient device driver within a very short time.

Fig. 6 shows WinDriver architecture [12]. WinDriver provides two driver development modes: user mode and kernel mode. In user mode, users can directly use the WinDriver's driver template, without any deep understanding on the knowledge of hardware or Windows kernel. Neither understanding of development of device drivers nor operating system-related knowledge is necessary. For convenience I developed the PC104 signal card's driver in user mode.

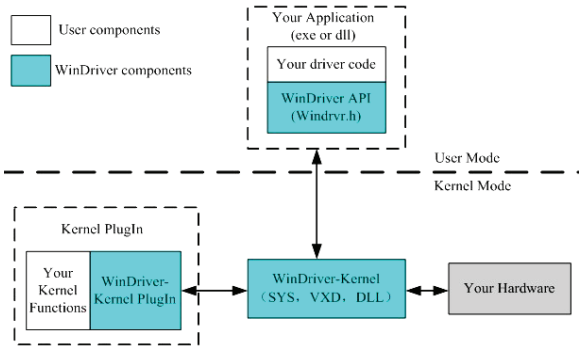


Fig. 6. WinDriver Architecture.

Fig. 7 shows running processes of the driver program. Firstly, the card calls OpenDevicePM( ) function to turn on the signal generator, and then calls initialization InitPM536( ) function and the following three interrupt routine functions:

PM536_IntIrq5Enable (hPM536, PM536_IntIrq 5HandlerRoutine)

PM536_IntIrq7Enable (hPM536, PM536_IntIrq 7HandlerRoutine)

PM536_IntIrq12Enable (hPM536, PM536_IntIr q12HandlerRoutine)

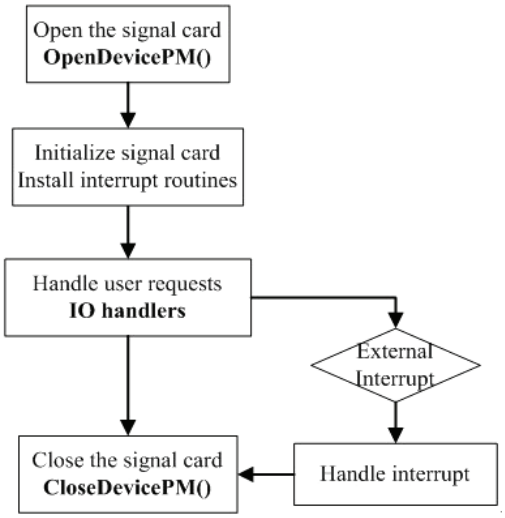


Fig. 7. Drivers running processes.

After completing the signal card initialization and interrupt routine installation, the program can handle users' requests to the signal card. User requests are always IO reading or writing through IO handlers. If the external interrupts (IRQ5, IRQ7 and IRQ12) are effective, they will be handled first, and finally function CloseDevicePM( ) is called to turn off the signal card.

4. Panel Design of Control System

The panel of control system implements the function of setting parameters of the signal generator, waveform display and other functions required by the control system. The control panel must be user friendly, operationally stable. We select the U.S. NI LabVIEW virtual instrument software to design the panel. LabVIEW (Laboratory Virtual Instrument Engineering Workbench) is a product of National Instruments, being one of the most important tools internationally and most widely used in data acquisition and control system development environment. LabVIEW is also used in instrument control, data acquisition, data analysis, data display system. Its programming language is graphical language (G language), and it is designed for test engineers rather than professional programmers. Using LabVIEW programming is very convenient, man-machine interface being intuitive and friendly, with powerful data visualization analysis and instrument control capabilities and other features.

The control panel consists of waveform display panel, power, rulers, coupling panel, voltage, current display panel, waveform recorder button, the transmitter switch, internal and external trigger buttons, parameter settings dialog box and other components (Fig. 8). Waveform storage path, waveform frequency, pulse width and edge width can be set at the preferences dialog. User can also set the PC104 data acquisition board, the number of acquisition channels, sampling frequency and sampling points. The parameters set by the user can be saved, so if the user wants the same setting next time, he or she only needs to import the saved parameter file.

Control panel running processes is shown in Fig. 9. When the system starts to boot, a login screen pops up. Only when the correct user name and password are entered, can the user login. After successful login, the initialization of the signal card and data acquisition card are completed. Before turning on the transmitter, users may need to reset the transmitter waveform parameters, data acquisition card parameters, internal/external triggering options, and determining synchronous control signal is provided by the transmitter or receiver. After completing parameter setting, users can turn on the transmitter. When the program is running, users can save the transmitter waveform at any time. Remember not to reset internal/external trigger when program is running. During program running process, if users need to reset the transmitter waveform or data acquisition parameters, they must make sure that the transmitter has been turned off.

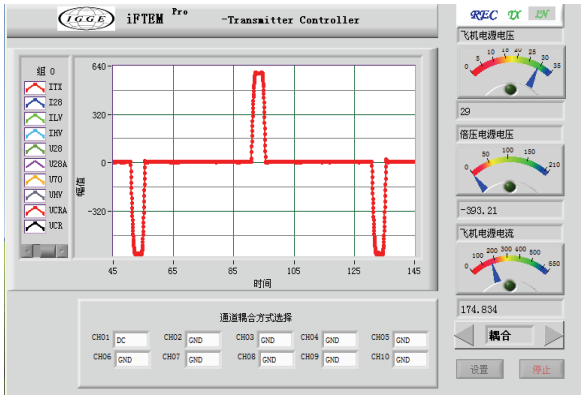


Fig. 8. The panel of control system.

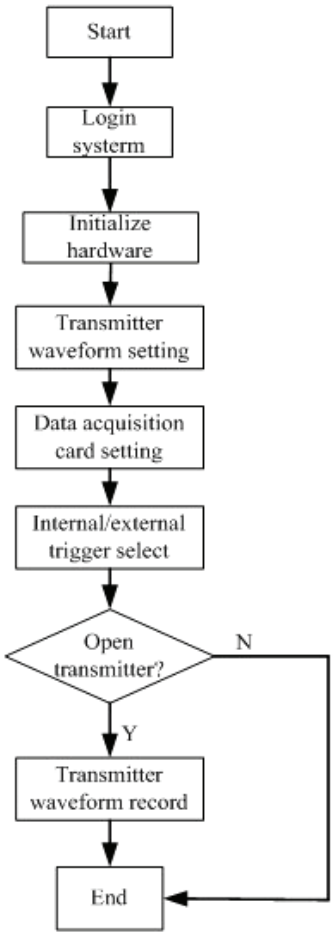


Fig. 9. Control panel running processes.

6. Conclusions

Experimental results show that through PC104 bus in the control system of iFTEM transmitter, waveform control, synchronization control, protection control, voltage, current and transmitter waveform monitoring are implemented well together, with small size, high reliability and scalability obtained at the same time. Our work contributes to the research of domestic time-domain fixed-wing airborne electromagnetic method.

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Study on a Pan-tilt Control Technique for the Airship Camera Orientation

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Abstract: A pan-tilt control technique for the camera orientation of the airship is proposed. This technique is constructed by combining the airship states and the error of the image centre as a feedback. For the ground target coordinates testing system installed in airship, a two dimension pan-tilt is used to drive the camera tracking the ground target in order to catch the image of the measured target reliably. The control method of a double adaptive varying velocity PID with dead zone is proposed. Based the movement characteristic of the airship, the relationship between the movement of the target tracking and the angular velocity of the pan-tilt is analyzed; the control algorithm of the angular velocity of the pan-tilt is designed. This method make the camera continue to monitor the testing region and get the original image of the target in time. The result of experimentation shows that the method can track the ground target rapidly and steadily.
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Keywords: Ground target position measurement, Airship sensors, Dual-PID self-adaptive control, Image process, Vision measurement.

1. Introduction

For the measurement of the ground target position in bigger region, the air-platform, such as hot-air balloon, airship and unmanned aircraft, sometimes are used to the to take the vision sensors as CCD cameras or TV theodolite [1] system to monitoring and track the peculiar point in the ground testing region. Although this kind of measurement technique have some advancement, for example, covering bigger test area, avoiding affection test object, etc., the limited by load capacity of the air-platform, Especially of the small airship, this platform only take light cameras, pan-tile, driver and controller of the pan-tile and the communication devices. So design a good performance of the controller used in airship is necessary. This controller can automatically calculate the controller output signal on the horizontal and vertical direction according to the deviation between the target and the ground point to be measured and the output of the angle sensors. Ref. [2] analyses the method of the measurement of the 2D and 3D coordinates of the target coordinates by using the TV measurement system, and analyses the output accuracy of the TV tracker. But in fact, real-time and accurately test the established goals of the premise and the key is to treat measuring target tracking area, thus to build a test system of tracking control algorithm is necessary. Due to the non stability of the aerial platform, often deviate from the

locked area, by rotation of pan-tilt control visual sensor tracking control for the ground wide range of areas, and become one of the most active research directions in this field [3-7].

During the process of tracking and controlling, most of the control is to track the moving ground target; the main task of the vision sensor is to estimate the movement states of the moving target, which is used as feedback information to form a closed loop control, and make the air-platform fly to track the target. Because the vision sensor move and rotate with the air-platform, in the video image, the background of the target is dynamically changing, which adds the difficultly to extract the target position from the image code. To solve this problem, the researchers put forward a lot of different solutions. Ref. [8] puts forward a method that based on the motion compensation to detect the dynamic target in the image code. The core idea of this method is to use a bilinear function to describe the camera movement which causes the movement of the background image, and estimates the parameter by the position relationship of the feature point in adjacent image, so as to realize the simulation of the background movement. In Ref. [9], Han Y *et al.* estimated the speed of the target move in the image plane, and the robot movement controller is designed by using this information and Jacobian matrix to achieve the track of the target. This method can be preferably utilized to avoid the loss of target caused by the image distortion. However, Ref. [10] proposes a visual probabilistic data association filter; it can test and track the specific target in complex background. In addition, according to the ground mobile robot, Ref. [11-14] design a variety of different control strategies to achieve vision-based dynamic target tracking, and these methods usually put the estimated target states to realize the control of movement of the mobile robot. This paper mainly tracks the stillness testing region. Because of the sway of the air-platform, it is necessary to control the track, make the testing region always stay in the view field of the vision sensor, measure the target which located in the testing region.

The movement of dynamic objects in the image plane mainly comes from two aspects: the movement of the target and the camera. While the movement of static targets in the image plane is only associated with the camera motion. So in order to obtain the target status relative to the world coordinate system, it is necessary to carry out a more accurate estimation of the image change caused by the movement of the camera itself. The object to be tested does not always appear in the field of view and the location is random, so it can't provide information to estimate the change of the image caused by camera movement. To solve this problem, it adopts a method of stationary reference target estimation. This method estimates the image change which is caused by camera motion, with the help of static reference target movement information, reduces the reference of the inertial sensor noise to the estimation of target state, and realizes the tracking of the measuring region.

This paper mainly aims at the measurement the position of the ground stationary area. Due to airship moves according to the scheduled routes, and it also have a swing caused by the wind, it make the field of the vision sensor off the area to be tested. This paper, on the basis of two-dimensional reference target measurement, analyze the characteristics of airship movement and inertia of the camera pan-tile, propose the double PID adaptive control tracking strategy with dead zone of the variable speed type, induce the relationship the camera rotation angular velocity and image shift, and design the trace parameters algorithm. By the simulation and experiment, verified the correctness of this method.

2. The Structure of Measurement System

The structure of measurement system is shown by Fig. 1. The system is divided into airship-load system and ground monitoring system. There are three kinds of sensors in the airship-load system: cameras parts is used to capture the image, GPS is used to get the altitude of the camera, and the angle sensor is used to output the angles degrees formed by the light axis of camera with the horizontal and vertical. Cameras parts include CCD camera, pan-tilt, the driver and controller of the pan-tilt. The camera captures the images of the ground test area and the controller analyze and processes the angular deflection and give the signal to driver so as to drive the pan-tilt and camera to track the ground target. And same time the image and other information are transformed to ground monitoring system by the model of wireless.

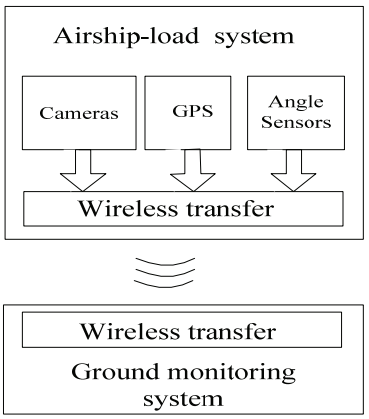


Fig. 1. The structure of measurement system.

3. The Controller Based on Airship

The ground target position measurement system uses the air-platform as the device carrier, and measures the target position with the photoelectric device. In the ground target region tracking system of

vision-based aerial platform, the low altitude flying aerial platform acquires visual information immediately on the ground by the clouds terrace control system which carries the camera. The whole clouds terrace is modulated automatically based on the acquired information to keep the ground target region in the visual field of the camera. Meanwhile, the visual information is also used for the estimation of the location that included the target to be measured, which achieves the measurement of the target.

To solver the target position in the testing region is a necessity, choose a fixed target on the ground as track reference target. The air-platform revolves this reference target. The track control system is working in two methods: one is manual search and another is automatic track. Firstly finding the reference target uses the manual search method. Secondly, the pan-tilt tracks the reference target with the automatic track control system. So the testing region is monitored by the vision measurement system.

3.1. The Conventional PID Control

The PID control is one kinds of the widely used process control, which is realized, to put it simply, by proportion, integral and derivative of deviation [15-18]. PID controller is a simple algorithm based on the information estimation of "past", "present" and "future". In the automatic control system, PID controller possesses the advantages including easy algorithm and structure, steady performance, definite physical meaning and good robustness, which dominates the automatic control system. Nowadays, although various kinds of advanced control are developing continuously, conventional PID control is the most frequently applied in practical manufacturing process, with the following reasons like this:

1) Different kinds of advanced control are not complete practically;

2) The practical requirements can be satisfied by conventional PID control for most control object;

3) Advanced control is hard to be mastered by technical staff in company.

Motor speed is directly proportional with the size of the applied armature voltage basically, which composes the foundation of PID modulation. As the computer can only identify digital quantity and cannot calculate the continuous control formula direct, it is necessary to apply the discretization algorithm design of control law in computer control system. The motor control system is the time discrete control system, and PID algorithm is applied to the motor control system.

The scheme of theory of PID control system is illustrated in Fig. 2.

As a kind of linear controllers, it controls the controlled object through composing the control deviation e from the setting value and actual output, and composing the control value u by linear combination of deviation proportional, integral and differential [19-23].

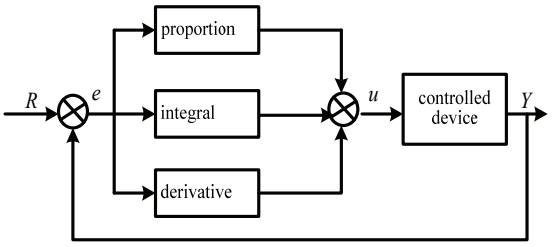


Fig. 2. The control system with PID regulator.

In the expression (1) of the digital PID speed control algorithm: U_n is the control parameter of the current system; k_p, k_d, k_i is the parameter of the proportion, differential and integration in the digital PID control algorithm; e_n and e_{n-1} are the system error of the current sample and last cycle sample.

$$U_n = k_p e_n + k_d (e_n - e_{n-1}) + k_i \sum_{i=1}^{n} e_i \quad (1)$$

The controlling of the proportion, differential and integration coefficients are correlated, which means that they can be adjusted individually or only one or two of them can be adjusted in the controlling scheme. Their value depends on the type of the object being controlled, and also needs to be regulated in the application in order to have a better performance. In simple words, the function of the different parts of the PID controller is as following [24-25]: the proportion part reflects the system error in a certain proportion, and once here are errors it will act to suppress them. The proportional part has a great effect which will shorten the regulation time and decrease the error, but a proportion coefficient that is too large will damage the stability of the system or even destroy it. The integration part mainly works to eliminate the static error of the system and increase indiscrimination degree. The integration part will keep working until the error becomes zero and its output is always constant. The effect of the integration depends on its time constant, and a smaller time constant means a better effect and vice versa. The integration will weaken the system stability and the response speed. It is often combined with the other regulation part, such as PI or PID controller. The differential part reflects the rate of change of the system error, which means it can predict the trend of the error, thus adjust the system before the error is produced, which improve the dynamic performance of the system. Under an appropriate differential time the overshoot and adjusting time can be effectively decreased. However, the differential part would also amplitude the system noise, which means that a better differential part will threaten the systems anti-noise

ability. Moreover, the differential part cannot be used without other parts, there could only be a PD or PID controller.

3.2. The Dual-PID Self-adaptive Control

As the air-platform hover, it is belongs the large disturbance environment. To track the reference target reliable, use the double adaptive PID with dead zone control method are based on the classical PID control algorithm. The block diagram of the rotational station control track shows as Fig. 3.

The whole control system is composed by two sections: manual search and automatic track. The manual search is K_1 on, K_2 off. The keyboard is controlled by operator, and the horizontal orientation $\Delta\Omega x$, pitch search signal $\Delta\Omega y$ are sent to the controller by the computer serial ports, control the reference target in the view field, and guarantee the optical axis point to the reference target. This can track the region of the target. The automatic track is K_2 on, K_1 off, the vision tracker get the image, and calculate the deviation angle $\Delta\Omega x$ and $\Delta\Omega y$ between target and center of the optical axis, send these to the main controller to form the control loop of the space position; The main controller calculates the motor speed ω_x and ω_y according to the current attitude of the air-platform which is got by the GPS, and the motor rate $\Delta\omega_x$ and $\Delta\omega_y$, sends these parameter to the main controller to form the control loop of speed rate, achieves the track control to the fixed reference target steadily.

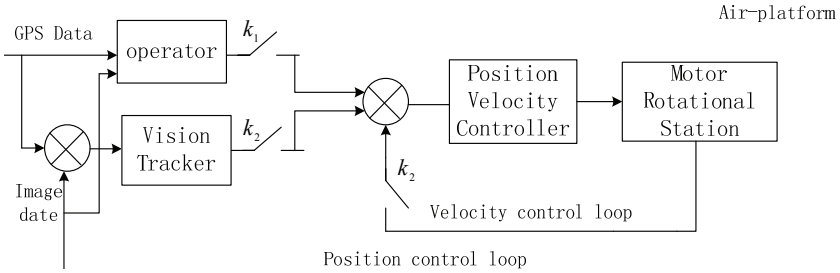


Fig. 3. The diagram of the pan-tilt control track.

In the track control condition, the digital controller uses digital PID control algorithm, and the period of the sampling is 20 ms. The signal of the TV field blanking is the signal for the system synchronous.

In the control loop of the position, need to calculate the deviation angle $\Delta\Omega x$ and $\Delta\Omega y$ between target and center of the optical axis. When the fixed reference target drift near the center of the optical axis, the control of the direction will change with unstop. This makes the motor shake. The position loop uses the PID controller with 10 % of the central angle dead zone to improve the track stability.

In the ordinary PID control algorithm, integral increment keeps invariant in the whole control process because the integral coefficient k_i is a constant. However, in the ground target location test system, the air platform has larger motion amplitude and it requires integral action to be weakened or even disappear when the system deviation is large and be enhanced when the system deviation is small.

Square up the tilt angle γ between the optical axis of the camera and the ground, this cause the size showed by the unit pixels of the camera image plane different from the real, and the smaller of the angle, the bigger of the real size showed by the unit pixels. The rotational station rotates should follow a principle: the larger γ, the faster of the angular velocity, the smaller γ, the slower of the angular velocity. When the set value of motor speed suddenly changes or the motor speed abruptly changes, it causes the step of the deviation and the PID output increases or decreases dramatically so much so that it exceeds the upper and lower limits of control variable. The actual control variable is only limited at the maximum now. The motor speed is rising, but its growth rate slows down because the control variable is restricted. Deviation lasts longer than normal circumstances in a larger deviation. So that the integral term in the PID equation continues to be cumulative. When motor speed exceeds the set value, a negative deviation appears. But due to the integral term has a considerable cumulative value, the control variable disengages the saturated zone after a considerable period of time. This is forward integral saturation, and the reverse integral saturation is similar to this. Solutions: Shorten the sampling period of the PID and tuning the PID parameters; use gearshift integral PID algorithm to improve the PID algorithm. To suffice the stability and the quick of the system, the PID track control with the shift speed and 10 % of the central dead zone to control the speed can be used.

The basic idea of the changing integration PID algorithm is to make the accumulating rate of integral term with the deviation corresponds to the size by changing it; the greater of the deviation, the slower it integral, and goes faster in the opposite case. Then the non-linear function can be defined as:

$$f(e_n)=\begin{cases}1 & |e_n|\le B\\ \dfrac{A-|e_n|+B}{A} & B<|e_n|\le A+B\\ 0 & |e_n|>A+B\end{cases} \quad (2)$$

In this case PID algorithm can be improved as:

$$U_n = k_p e_n + k_d(e_n - e_{n-1}) + k_i\sum_{i=1}^{n}[f(e_i)e_i] \quad (3)$$

While the value of f is in the range from 0 to 1. It can be proved the algorithm has entered the saturation region when the deviation is bigger than $(A+B)$, in this case, it is no longer accumulation of the integral term; when

$$B<|e_n|\le A+B, \quad (4)$$

the value of f is increased with the decreasing of deviation, the speed of accumulation goes faster until the deviation is less than B, while the speed of accumulation is up to the maximum which is 1. It is not required accurately of the algorithm to the parameters A and B, when the greater of A and B value the weaker the changing integration to the saturated inhibition, and goes stronger in the opposite case. Generally take $A=30\%[e_n]_{\max}$, $B=20\%[e_n]_{\max}$ in appropriation.

3.3. The Double PID Parameter Design

PID parameter tuning is setting and adjusting the PID parameters, makes the transition process of the control system achieve a satisfactory control quality. Parameter tuning has the direct relationship with quality control of the system. It is determined by process characteristics that the proportional coefficient of PID controller, integral time and derivative time. There are lots of methods to adjust the parameter of PID, summarized by two categories: one is theoretical calculation tuning method, it depends on the math model of system, theoretical calculation determines the parameter of the controller. Another is practical tuning method, it depends on the project experience, control the system directly during the experiment, and it is a simple easy way to master, and widely adopted in the engineering practice. The methods of the practical tuning mainly include critical ratio method and response curve method. The critical ratio method is a kind of cut and tries in essence. Its operation tedious and low efficiency and it may damage the device because of the improper handling. Another method is response curve method, it determines the approximate transfer function of the process by measured the step response curve of the controlled object, then use the parameters setting algorithm to tuning. The two methods have their own characteristics. The same point is experiment, then according to the project experience formula tuning the parameter of the controller. But no matter the parameter get in which way, it needs to adjust and improve in the practical operation.

4. The Control Parameter Design

According to the block diagram of the rotational station track control, to get the quick and accurate track and control, need calculate the angle of the fixed reference point offset the optical axis and the angular velocity of the reference point offset position according to the image data and GPS measure data, this can control the motor effectively through the controller.

4.1. The Parameter of the Reference Point Offset Angle

For calculate the offset angle of the fixed reference point, the position of camera and fixed reference point is showed as Fig. 4 (a), point *M* is the camera position, *h* is the height of the camera, *MA* is the main optical axis, *L* is the length of the *MA*, γ is the angle between the main optical axis and the vertical direction, B is the reference point, and need to move to point *A*, the pixel coordinates of *A* is $A'(X_A, Y_A)$, the pixel coordinates of B is $B'(X_B, Y_B)$, where

$$\begin{aligned} X_A &= W/2\\ Y_A &= H/2 \end{aligned} \quad (5)$$

The position of image plane is shown as Fig. 4 (b). The image of point A and point B in the image plane is showed as Fig. 5, ΔnX is the number of pixel need to move toward X direction, ΔnY is the number of pixel need to move toward Y direction.

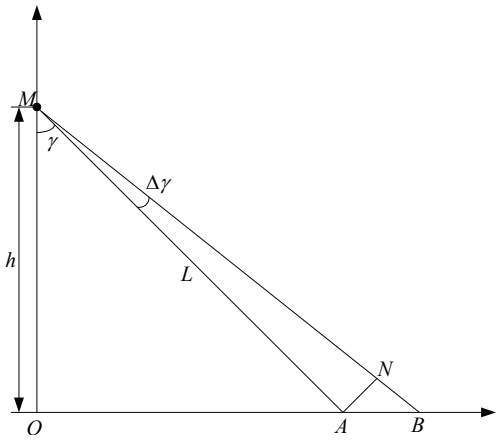


Fig. 4 (a). The relative position of camera and reference point.

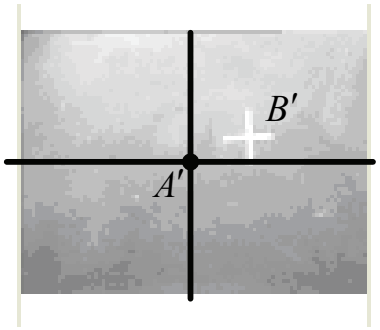


Fig. 4 (b). The image plane position of camera and reference point.

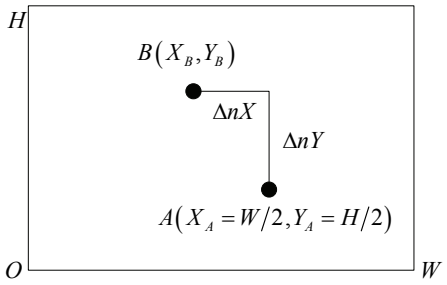


Fig. 5. Pixel coordinates.

$$\begin{cases} \Delta nX = X_B - X_A \\ \Delta nY = Y_B - Y_A \end{cases} \tag{6}$$

$$\rho = \frac{L}{f} = \frac{h/\cos\gamma}{f}, \tag{7}$$

where ρ is the magnification of camera. Suppose ΔX is the real size needed to move toward X direction, ΔY is the real size needed to move toward Y direction, μ is the pixel size.

$$\begin{cases} \Delta X = \Delta nX * \mu * \rho \\ \Delta Y = \Delta nY * \mu * \rho \end{cases} \tag{8}$$

The angle needed to rotate of rotational station is pitch angle $\Delta\Omega y$, it is $\Delta\gamma$ and horizontal angle $\Delta\Omega x$.

$$\begin{cases} \Delta\Omega x = \arctan\dfrac{\Delta X}{L} = \arctan\dfrac{\mu(X_B - X_A)}{f} \\ \Delta\Omega y = \arctan\dfrac{\Delta Y}{L} = \arctan\dfrac{\mu(Y_B - Y_A)}{f} \end{cases} \tag{9}$$

These parameters are related to the focal length of the optical system, In the track process, the change of the optical focal length will affect the gain of track systems. The focal length determine, can use camera to get the image and measure the focal length and the focal length voltage, adopt the piecewise linearization and polynomial fitting to get the relationship of focal length and the focal length voltage, according to this relationship, can get the value of the focal length real-time.

4.2. The Angle Velocity Parameter of the Reference Point Offset Position

Assume the T is the time needed to rotate elevation angle $\Delta\Omega y$ and level angle $\Delta\Omega x$, the rotational station rotate angular velocity is pitching angular velocity ω_y and level angular velocity ω_x. Because of the bigger of γ, the camera inclined, the lager displacement of the optical axis center caused by the rotational station rotate the same angle. It should satisfy:

$$\begin{cases} T_X = f_X(\gamma)T \\ T_Y = f_Y(\gamma)T \end{cases}$$

The AN is the projection of the AB in the camera image plane in Fig. 5.

$$AN = \Delta Y = (Y_B - Y_A) * u * \frac{h}{f\cos\gamma}, \tag{11}$$

$$\begin{aligned} AB &= OB - OA \\ &= \tan(\gamma + \arctan\frac{\mu(Y_B - Y_A)}{f}) * h - \tan\gamma * h \end{aligned}, \tag{12}$$

$$\begin{aligned} k_Y &= \frac{AN}{AB} \\ &= \frac{(Y_B - Y_A) * \mu * f}{\cos\gamma(\tan(\gamma + \arctan\frac{\mu(Y_B - Y_A)}{f}) - \tan\gamma)} \end{aligned} \tag{13}$$

In Fig. 6, O is camera, rectangular $ABCD$ is the ground, F is the intersection point of the optical axis of the camera and ground when the camera vertical, E is the intersection point of the optical axis and ground when the angle between camera and vertical is γ.

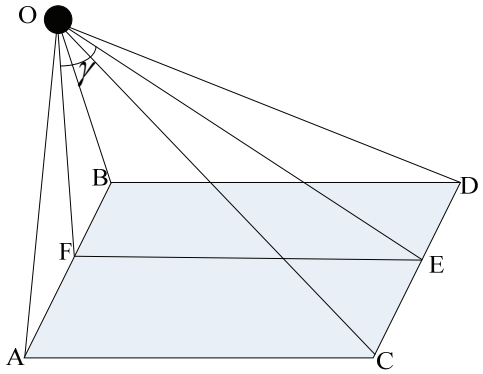


Fig. 6. The camera coordinates.

Assume the length of AB and CD is m, $OF = h$, so the length of AB and CD is $A'B'$ and $C'D'$ in the pixel coordinate:

$$\begin{cases} A'B' = \dfrac{AB*f}{OF} = \dfrac{m*f}{h} \\ C'D' = \dfrac{CD*f}{OE} = \dfrac{m*f}{h/\cos\gamma} \end{cases} \tag{14}$$

Let be:

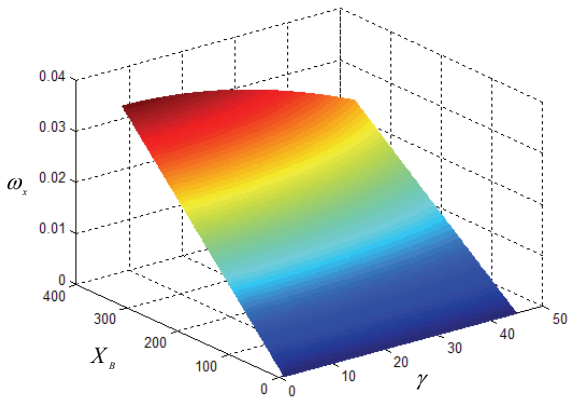
$$K_X = \frac{A'B'}{C'D'} = \cos\gamma, \tag{15}$$

$$\begin{cases} f_x(\gamma) = \dfrac{1}{k_X} = \dfrac{1}{\cos\gamma} \\ f = \dfrac{1}{k_Y} \\ = \dfrac{\cos\gamma\left(\tan\left(\gamma + \arctan\dfrac{\mu(Y_B - Y_A)}{f}\right) - \tan\gamma\right)}{(Y_B - Y_A)*u*f} \end{cases} \tag{16}$$

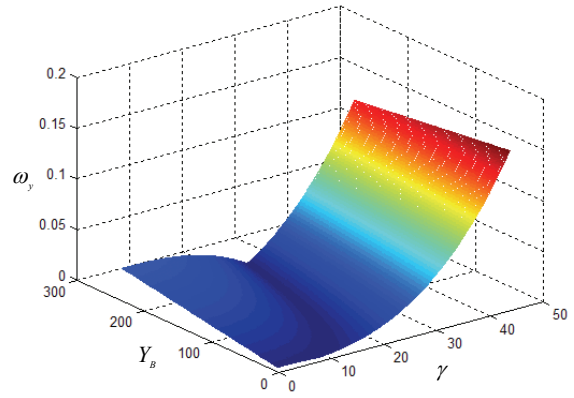
The angular velocity of the rotational station is pitching angular velocity ω_y and level angular velocity ω_x:

$$\begin{aligned} \omega_y &= \frac{\Delta\Omega y}{T_Y} \\ &= \frac{f\arctan(\frac{u(Y_B - Y_A)}{f})\cos\gamma}{(Y_B - Y_A)*u*T} \\ &* \frac{(\tan(\gamma + \arctan(\frac{u(Y_B - Y_A)}{f})) - \tan\gamma)}{(Y_B - Y_A)*u*T} \end{aligned} \tag{17}$$

$$\omega_x = \frac{\Delta\Omega x}{T_x} = \cos\gamma * \arctan\frac{\mu(X_B - X_A)}{f} / T \tag{18}$$



(a) The position relation of level angular ω_x and tilt angle γ.



(b) The position relation of pitching angular velocity ω_y and tilt angle γ.

Fig. 7. The position relation of speed control angular velocity ω_x, ω_y and tilt angle γ.

5. Simulation

According to the calculate of equation (15) and (16), the control parameter of the pan-tilt speed can be getten, and the testing region can be tracked timely. From the expression of the speed control, when the testing system is fixed, speed is related to tilt angle γ and the position of the target. Assume the change range of tilt angle γ is 0°~45°, the resolution of the camera is 640*480, the focal length is 50 mm, the coordinate of optical center is $(320, 240)$, the size of pixel is 0.006 mm, the relationship of the speed control ω_x, ω_y and the position of the target tilt angle γ is showed as Fig. 7.

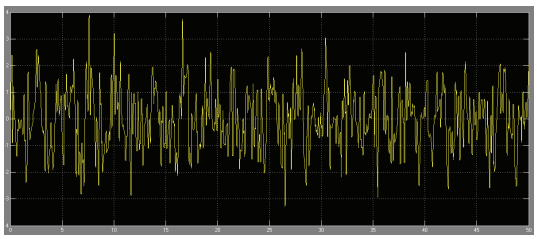
From the Fig. 7 can get the position relation of level angular ω_x, pitching angular velocity ω_y and tilt angle γ is not the linear relation, and near the optical, the small angular velocity ω_x, ω_y needed. This control relation satisfies the requirement of speed control.

According to the control algorithm, assume the sloshing of the air-platform is the Gaussian white noise with max value 4 degree of the slip angle. As Fig. 8 (a) shows, through the simulation, we can get the track result of the view field center as Fig. 8 (b) shows, the amplitude range is [-1.6,1.8]. Through the comparison, the angle deviation gets the effective control, which can satisfy the requirement of testing region track.

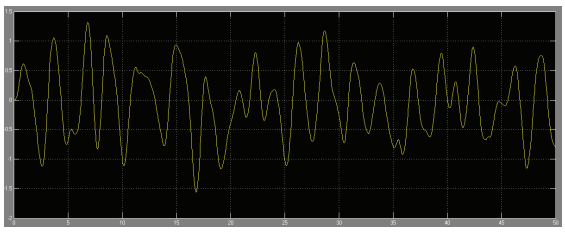
According to the relation between angle and angular velocity, the wave pattern of angular velocity is shown as Fig. 9 (a), the amplitude range is[-25,32], the result wave pattern of the angular velocity control is shown as Fig. 9 (b), the amplitude range is [-3,4]. Through the comparison, the angular velocity can get the effective control, satisfy the requirement of angular velocity track.

In the actual environment, airship random swing will bring deviation interference. In order to measure the time response and steady-state characteristics of the designed controller, experiment simulation is

carried out with impulse signal as interference, the respond is shown as Fig. 10, the angle deviation of position loop reached steady state within the fixed time, which is shown as Fig. 10 (a) and correspond angular velocity of velocity loop also decreases rapidly to zero, which is shown as Fig. 10 (b). Finally, the design control system can automatic track the region of reference target.

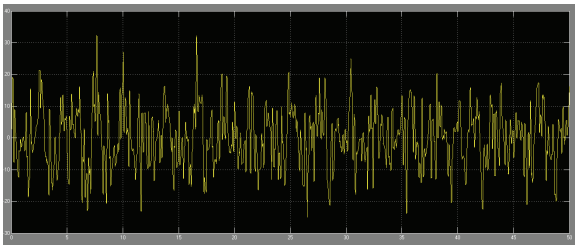


(a) The wave pattern of the angle deviation.

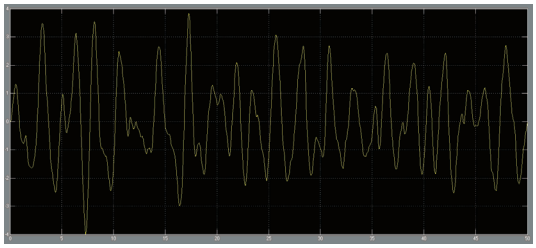


(b) The result wave pattern of the angle deviation control.

Fig. 8. The comparison of the angle deviation control.

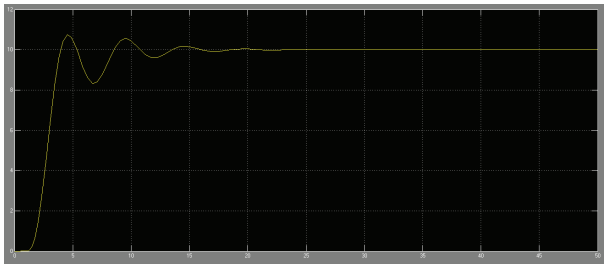


(a) The angular velocity wave pattern calculated by the angle deviation.

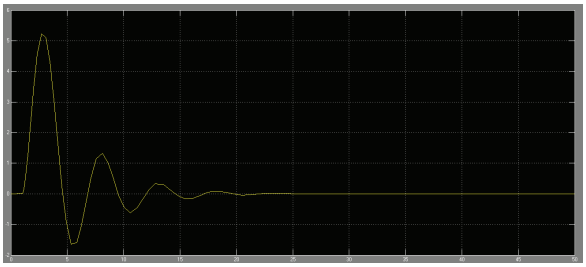


(b) The result wave pattern of the angular velocity control.

Fig. 9. The original wave pattern and the control wave pattern of the angular velocity.



(a) The angle deviation control respond of position loop.



(b) The angle velocity control respond of velocity loop.

Fig. 10. The impulse respond of the controller.

6. Conclusions

The airship ground target position testing system in this paper is researched by ourself. It is limited by the airship-load, so this condition needs all devices must light and efficient. In the condition of the movement of the airship, to guarantee to catch the image of the target, it must track the target region. By designing the ground fixed reference target, using 2D pan-tilt to load testing device, rotating the camera to track imaging the ground fixed reference target, according to the control method of the pan-tilt and the control strategy, to drive the pan-tile reliably and steadily, It can ensure to get the original image of the measured target timely, and let the camera can monitor the testing region of the measured target continuously.

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Signal Simulation Electric Systems of Admittance Monitoring

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Abstract: Admittance method is proposed to evolve the mathematical model of electric systems of monitoring the nonelectric nature object quality. The analysis of response models for water salt-solutions due the results of admittance monitoring has been conducted, to detect the peculiar properties of response surfaces. The possibility of applying the models of reactive conductivity component dependence on salt solution concentration and frequency of a test signal is suggested for maintaining and calibrating a measuring device.
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Keywords: Leave mathematical model, Admittance monitoring, Nonelectric nature objects, Salt solution, Capacitance sensor, Reactive conductivity component.

1. Introduction

The research of the system "primary transducer – control object" (PT – CO) in the real scale is a highly laborious task. Its complexity is caused by a number of influence-factors, and considerable interlacing of their mutual connections as well as those with system elements. One of the ways of those systems' study is a design of a mathematical model that should include the immanent characteristics of a real system. The investigation of a mathematical model in its turn allows us to sift the essential characteristics of the researched system [1-3].

The systems of monitoring the nonelectric nature objects' quality by dint of the admittance method constitute a subject of the proposed mathematical simulation. The admittance method is grounded on measuring the active G and reactive B components of complex conductivity [4]. It is worth noting that till recent time the admittance method was involved in the research of electric nature objects only, namely the elements of electric circuits: resistors, condensers, induction coils and etc. However the objects of nonelectric nature are much more frequent in our environment, e. g., water, soils, food products, wares of light industry and so on. Those objects are rather nontraditional for admittance monitoring [5-7].

Thus the necessity of studying the peculiarities of admittance method application in the estimation of quality indices of nonelectric nature objects is of topicality nowadays. For this purpose we should substantiate the choice of immanent parameters of the system "primary transducer – control object". According to the experimental values of these peculiar parameters, the mathematical model covering also the effects of controlled medium parameters on the nature of PT-CO system responses at the simultaneous alteration of those parameters

should be evolved. The objectives of the proposed work include the study into the peculiarities of the admittance monitoring method applied to the objects of nonelectric nature, namely water salt-solutions.

2. Results of Investigations

2.1. Equipment and Conditions of Research

The experiments were carried out with the usage of an immittance-meter with a primary transducer of capacitance type. The range of the researched frequencies made: 25 Hz - 1 MHz, voltage level of a test signal: 0.01 V - 2 V, electrode material: platinum, distance between electrodes: 5 mm, and length of a transducer working part: 6 mm. The stability of electrodes was assured by their soldering into the chemically stable glass tubes.

2.2. Analysis of the Researched System Parameters

The admittance method of object research is based on measuring and stiffing the dependence of admittance on the frequency F [8, 9]. Different objects and processes are characterized by different frequency dependences of active and reactive admittance components, which allows us to solve the reverse problem, i.e. to extract information on those objects by dint of analyzing their response frequency characteristics in the form of the reactive admittance component B. Thus the immittance method whose informative parameter in a concrete case is admittance enables assessing the quality of nonelectric nature objects through their electric parameters. Hereby, the chemical properties of nonelectric nature objects could be represented in the form of the electric models described by mathematical expressions.

The structure of the researched system is notified in the Fig. 1.

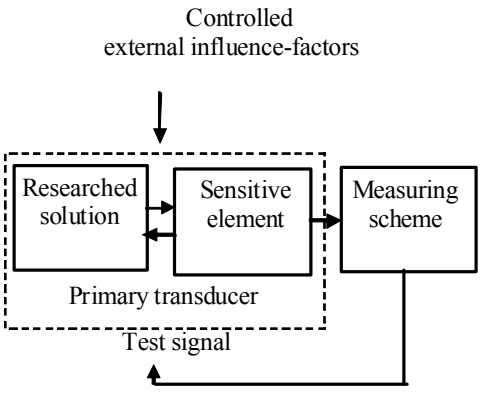


Fig. 1. Structure of the researched system Immittance PT-CO.

The peculiar parameter of the researched object, water salt-solution, is the index of its chemical composition – the salt concentration in the solution C. Since the object of our research is water solution, i.e. the conductor of the second type, one of the external factors of influence on the PT-CO system is the temperature T, whose influence nature requires further specification.

The application of the admittance method consists in the usage of some test signal of the voltage U, whose responses are under consideration. The parameters of a capacitance primary transducer could be represented by its geometric sizes, namely by the constant K that is determined by the ratio of a distance between electrodes L to the area of those electrodes S.

Thus the set of PT-CO system parameters for water salt-solution includes the concentration of salt-solution C, voltage value of test signals U, their frequency F, solution temperature T and also the mentioned before constant K.

2.3. Development and Analysis of Mathematical Model of a Signal in the Researched System

According to the expounded before analysis, the research on water salt-solutions with the usage of the admittance method is characterized by such parameters:

- Concentration of salt-solution C;
- Value of test signals' voltage U;
- Frequency of test signals F;
- Solution temperature T;
- Constant K characterizing the geometric sizes of electrodes.

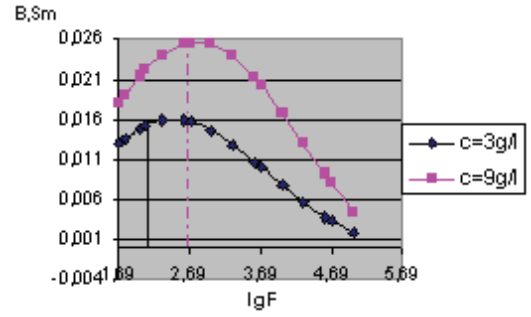
It means that the response signal Y could be described by some function:

$$Y = f(C, U, F, T, K) \tag{1}$$

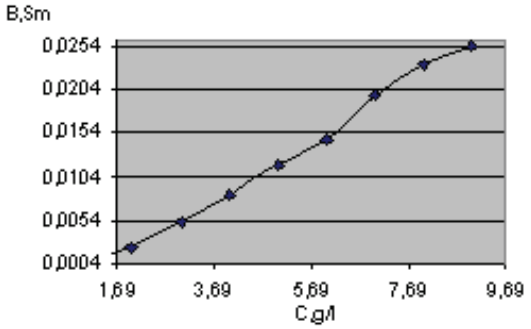
In general case the dependence (1) is quite complicated. To simplify it, we should investigate the dependence of the response signal Y on the less number of parameters at the certain fixed values of others [10-13]. This approach enables undertaking the separate research on the dependences of a response signal on certain parameters, and graphically visualizes their results.

The conducted experimental research has proved that the alteration in a response signal does not depend considerably on the change in test voltage values U. Apart from, the research is normally being conducted using the same primary transducer, to wit, the size-dependent constant K does not change within the experiment. Thus while applying the admittance method in the research on water salt-solution we could restrict the number of influence-parameters to three: salt-solution concentration, test signal frequency and temperature.

The experimental research has been carried out for different value combinations of the indicated parameters particularly the concentration has been changed within the range from 1 g/l to 9 g/l, frequency – from 0.05 kHz to 100 kHz, and temperature – from 20 °C to 50 °C. The additional analysis of the dependence of a response signal at the fixed values of salt-solution concentration and temperature has also been performed. The premise of this analysis was made by the research in the dependences of system responses on one changing parameter, under the condition of steadiness of others. Particularly the Fig. 2 (a) represents the dependence of a reactive conductivity component on the frequency of a test signal for various fixed values of salt-solution concentration. The dependence of a reactive conductivity component on the concentration of salt-solution at the fixed value of test signal frequency (F = 1 kHz) is notified in the Fig. 2 (b). The value of test signal frequency was logarithmized (lg$F$) for the convenience of experiment result processing.



(a)



(b)

Fig. 2. The dependences of a reactive conductivity component at (a) the fixed values of salt-solution concentration; (b) the fixed value of test signal frequency.

According to the results of the experiments conducted at the fixed temperature value, the dependence of a response signal on the concentration of NaCl solution and frequency of a test signal has been derived:

$$B = f_T(C, F), \tag{2}$$

where B is the reactive conductivity component.

To describe this dependence after the method of the smallest squares, the following expression has been got:

$$\begin{aligned}\hat{B}(\lg F, C) &= 0.00868 - 0{,}0091 \cdot \lg F - \\ &- 0{,}0054 \cdot C + 0{,}0075 \cdot \lg F \cdot C + \\ &+ 0{,}00261 \cdot (\lg F)^2 + 0{,}00022 \cdot (\lg F)^3 \cdot C - \\ &- 0{,}00243 \cdot (\lg F)^2 \cdot C + 0{,}00008 \cdot (\lg F) \cdot C^2 - \\ &- 0{,}00022 \cdot (\lg F)^3 - 0{,}0000027 \cdot (\lg F)^3 \cdot C^2.\end{aligned} \tag{3}$$

The expression (3) was calculated on the basis of 144 experimental values, gained for 9 different values of salt-solution concentration and 15 different values of test signal frequency. The observed values of a reactive conductivity component were changed from 0.0000703 Sm to 0.0256358 Sm. Since the values of reactive conductivity could change almost 1000 times as much, the method of the smallest squares with a relative error was applied to the derivation of the expression (3). The average relative error of reactive conductivity component reproducing by the model (3) makes 8 %, and the determination coefficient makes R^2 = 0.9. The largest value of a relative error, near 30 %, was fixed only in one point that corresponds to the maximal concentration of salt-solution 9 g/l and minimal signal frequency 0.05 kHz, and the relative error of more than 13 % was observed in four points.

The research on the mathematical model (3) has made it possible to characterize the system peculiarities in the different sub bandwidths of signal frequency and temperature alteration. So at the frequency alteration within the range of values due to which the model (3) was built, a response surface acquires a characteristic sag, whose maximum tends to shift with the rise in frequency and salt-solution concentration C (Fig. 2 (a), Fig. 3 (a)). The dependence of a reactive conductivity component at the fixed values of salt-solution concentration is notified in the Fig. 3 (b), Fig. 4 (a), Fig. 3 (d) and Fig. 4 (c). The system response surface for frequencies F, exceeding the bandwidth that serves as a basis for the model 3, thus constituting the predictable values of a response surface, is represented in the Fig. 3 (d), Fig. 4 (c).

The possibility of applying the model (3) to the frequency values of test signals outside the underlying range of the model (3) has been proved experimentally. Particularly, the representations in the Fig. 3 (c), Fig. 4 (b) prove the possibility of applying the model (3) to the higher values of test signal frequencies. It is worth mentioning that in the case of predictable values of signal frequency alteration F within the bandwidth from 10 kHz to 1000 kHz (look Fig. 4 (c)), a response surface also has a typical sag – minimal values of response signal are displacing with the rise in signal frequency and salt-solution concentration C which also was confirmed by the results of observation (Fig. 4 (b)).

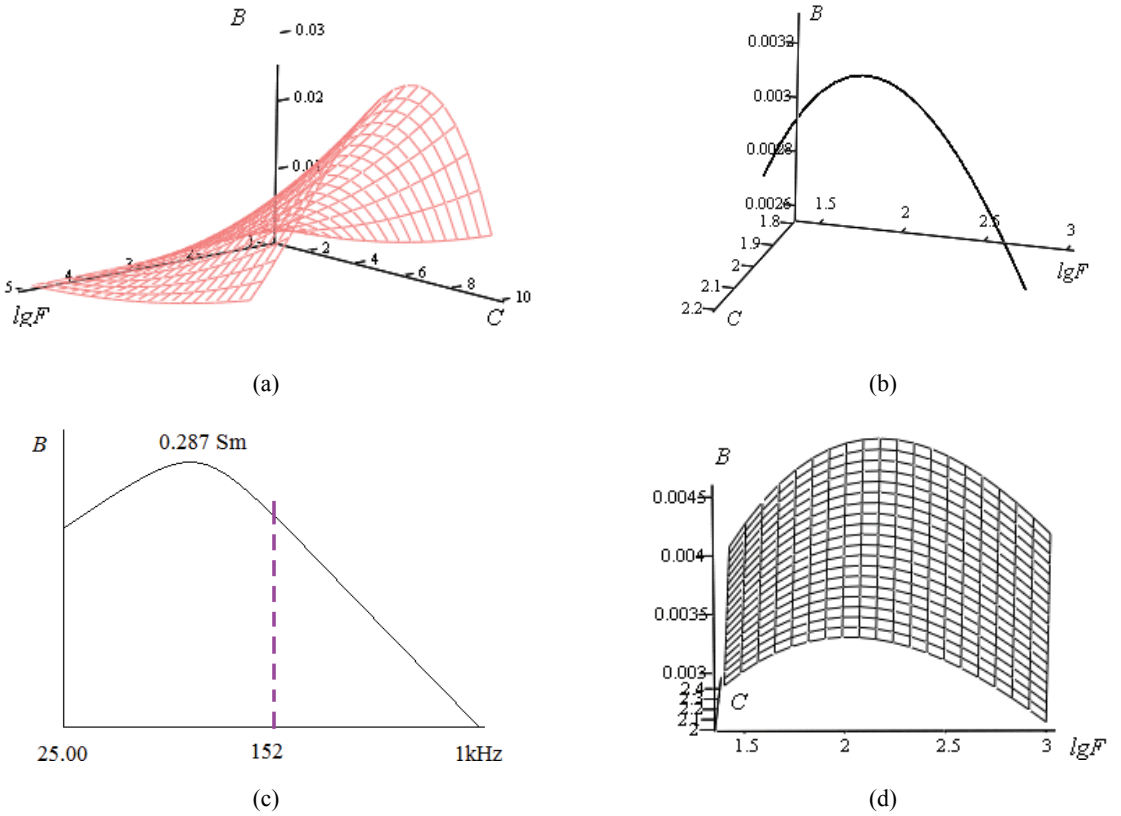


Fig. 3. The reflection of a system response surface: (a) within the range of parameters supporting the mathematical model (3); (b) and (c) at the fixed solution concentration C=1 g/l and within the bandwidth F = (0.025÷1) kHz (constructed according to an experiment model); (d) within the range C = (1÷3) g/l, F = (0.025÷1) kHz, constructed on a basis of the model.

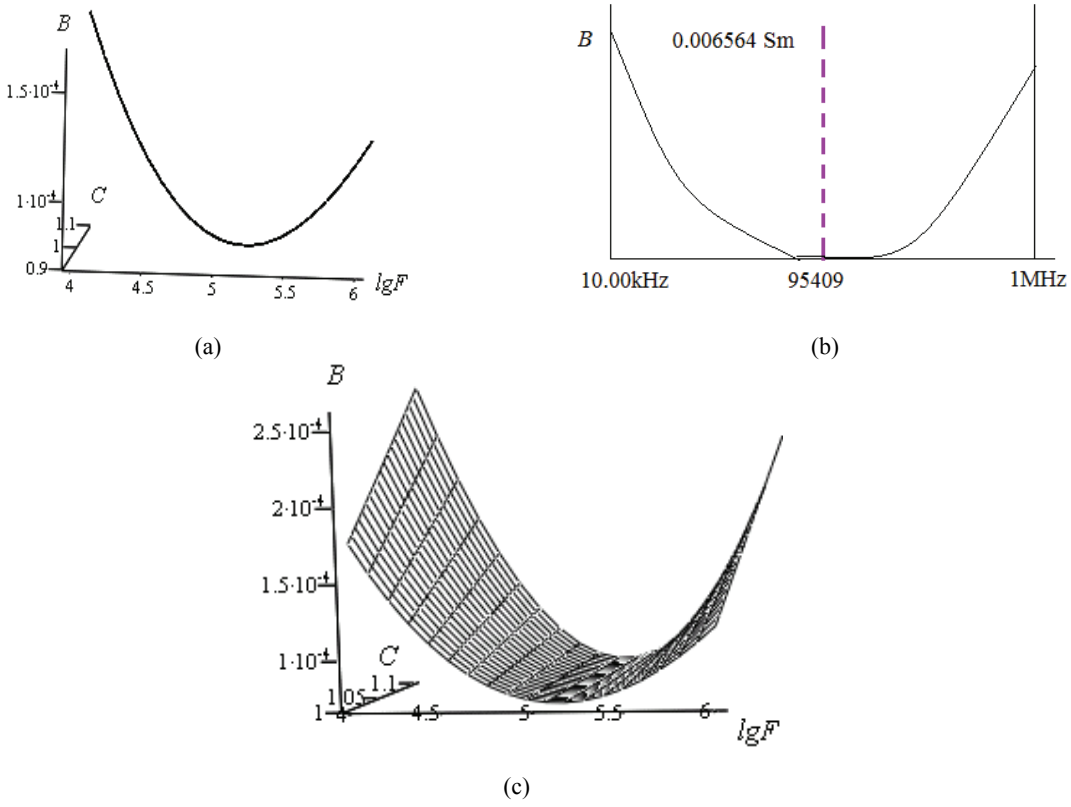


Fig. 4. The reflection of a system response surface: (a), (b) at the fixed solution concentration C = 1g/l and within the bandwidth F = (10÷1000) kHz, correspondently constructed on a basis of the model and experiment; (c) within the range C = (1÷3) g/l, F = (10÷1000) kHz, constructed on a basis of the model.

To wit, following the results of model analysis (3), we could observe the characteristic sags of a system response surface. The maximal values of reactive conductivity, reached while increasing the test signal frequency, are fixed at the largest solution concentrations.

At the fixed values of salt-solution concentration, the dependence of system responses on signal frequency and solution temperature has been researched:

$$B = f_C(F,T) \tag{4}$$

In accordance with the results of observing the dependence of reactive conductivity component of NaCl water solution on signal frequency and solution temperature, the following expression has been derived:

$$\begin{aligned}\hat{B}(\lg F,T) = &-0{,}01223 + 0{,}015574\cdot\lg F + \\ &+ 0{,}00007303\cdot T - 0{,}000027\cdot\lg F\cdot T - \\ &- 0{,}006158\cdot(\lg F)^2 + 0{,}0000026\cdot(\lg F)^2\cdot T + \\ &+ 0{,}0008492\cdot(\lg F)^3 - 0{,}0000057\cdot(\lg F)^5.\end{aligned} \tag{5}$$

The expression (5) has been calculated after the method of the smallest squares with an appropriate relative error on the ground of 80 observations through a reactive system conductivity component for the signal frequency within the bandwidth from 0.05 kHz to 100 kHz and solution temperature - from 33 °C to 50 °C. Since the observed values of a reactive system conductivity component have been altered within the range [0.0000372 Sm; 0.002482 Sm], the model (5) has been calculated under the condition of a minimum of relative error squares' sum. The average value of relative error of reactive conductivity component reproducing by the model (5) makes 0.03 %, and the determination coefficient makes $R^2 = 1.1$. The largest value of the relative error of the model (5), near 16 %, has been observed only in one point that corresponds to the temperature 33 °C and signal frequency 20 kHz; the considerably lower values of the relative error have pertained to other observation points.

The dependence of reactive system conductivity component on temperature at the fixed concentration of salt solution is practically linear (Fig. 5).

The research on the mathematical model (5) has enabled characterizing the system peculiarities for different alteration ranges of test signal frequency and temperature. So with altering the frequency within the range of model (5) underlying values, a response surface has acquired a typical sag – its maximum is not shifting with the rise in frequency and temperature at the given fixed solution concentration, but augmenting monotonously with temperature rise (Fig. 6 (a)).

The system responses within the range of temperature alteration values from 0 °C to 20 °C, exceeding the values of the model (5), are reflected in the Fig. 6 (b). The response surface representation in the Fig. 6 (b) shows that the nature of the dependence of a reactive NaCl solution conductivity component on signal frequency at the decrease in temperature is stable – a fall in the response value corresponds exactly with that of temperature.

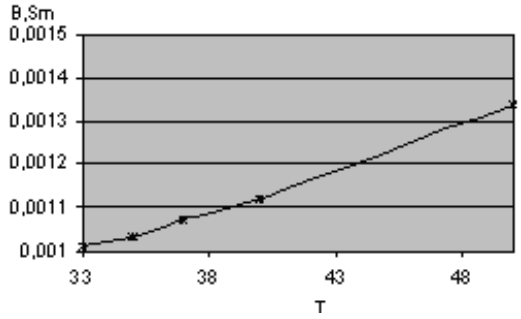


Fig. 5. Dependence of reactive system conductivity component on temperature at the fixed solution concentration.

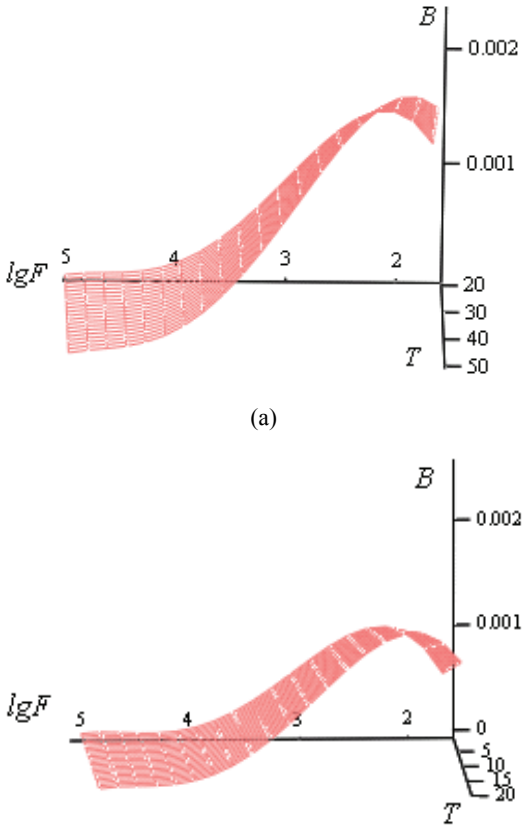


Fig. 6. Dependence of probability of the revealed fluctuations on the time of disequilibrium system observation.

Thus according to the results of analyzing the models (3) and (5), we might observe the typical sags of a system response surface. The maximal values of reactive conductivity at the increase in test signal frequency are observed for higher solution

concentrations. With the rise in solution temperature, those maximal values are increasing proportionally.

3. Conclusions

The conducted research supports the possibility of applying the admittance method to the assessment of nonelectric nature objects' quality. The gained models of the dependence of reactive response signal conductivity on the concentration of water salt-solution, its temperature and test signal frequency prove the expediency of involving them in the research into the nature of reactive conductivity dependence on these parameters.

Due to the analysis of these models the availability of reactive conductivity maximal values at the certain values of test signal frequency and salt-solution concentration has been revealed. With the rise in solution temperature those maximal values are increasing proportionally. This peculiarity of reactive conductivity dependence on solution concentration and test signal frequency could be employed during the maintaining and graduating of measuring devices.

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Development of Noise Measurements. Part 6. Opposing Method in Stepless Shifting Regulation

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Abstract: It is considered the hydraulic technique and studied nanoobject stepless shifting means along the Z-axis with the working liquid monomolecular layer thickness order error. Systematic multiplicative error elimination is obtained by means of the opposing method usage when calibrating. *Copyright © 2013 IFSA.*

Keywords: Nanoshift, Shift stepless control along the *Z*-axis, Relative error, Opposing method of error minimization, Calibration.

1. Introduction

Accuracy and stability problems of *3-D* positioning are considered to be principal in the nanotechnology. Authors [1] employ a combination of simulation techniques to model three different physical properties of the dopant atom, accessible in an experiment, and then use this information to triangulate the dopant position to ±2.5 nm in two directions and ± 15 nm in the third direction.

Today there are *3-D* positioning precise means of nanomachines, shift mechanism in atomic-force microscopes (later AFM) and familiar to them means. To reduce the scanner positioning error along the axes *X* and *Y* the AFM is preliminarily calibrated by the lateral standard. The calibration consists in standard surface measurement with the help of AFM and further usage of obtained length unit as a standard [2].

It is more complicated to calibrate along the axis *Z*. Paper [3] focuses on the on the design and calibration of an elastically guided vertical axis that will be applied in a small high precision 3D Coordinate Measuring Machine aiming a volumetric uncertainty of 25 nm. In the metrology part of this paper the calibration methods to determine the linearity as well as motion straightness and axis rotation errors were discussed. Finally first calibration results of *Z*-axis show nanometer repeatability of the probing point over the 4 mm stroke of this axis. The causes of the short-term variations with a bandwidth of about ± 10 nm are under investigation. Error compensation may reduce the residual error of the probing point to the nanometer level.

So a step standard [4] is used for calibration with next calculation – by the calibration results – calibration coefficient C_z. To detect scanner nonlinearities the calibration measurements are carried out by the means of different height step standards set, in other words by the multivalued measure producing of which is very complicated. At the same time own calibration coefficient $C_z(h)$ is calculated for every standard, and calibration

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coefficients for interim height standards will be calculated by the linear interpolation. For instance, the key features of standards TGZ set of NT-MDT firm are given in the Table 1 [5].

Table 1. Standard TGZ features as multivalued height measure.

Material	Silicium
Standard size	5×5×0.5 mm^3
Certificated area size	Central part 3×3 mm^2
Structure type	One-dimensional right-angled structure
Period	3.0 μm
Height	TGZ1 19±1 nm TGZ2 104±1.5 nm TGZ3 540±2 nm

A number of additional requirements appear during the preparation to the calibration on the multivalued step standard:

- Standard lateral positioning – in horizontal plane the structure must be located in the centre of controlled range;
- The structure must be located at angle of 90^0 to the measurement direction;
- Measurement speed is chosen so to provide the minimal distortion of structure fronts;
- Gauge range must exceed in 4 times the step width (to provide the measurement data sufficient quantity).

After calibration measurements providing the next stage is the step height estimation, for instance, in accordance with the method of ISO 5436 [4]. Here, the step height calibration artefact is a machined block of steel. It automatically detects the steps and calculates the step height or groove depth according to the mentioned standard. In this case the average step height is evaluated to 572 nm. All samples values of which exceed the threshold level are accepted as high level, and others – as low one. In this case the noise problem is neglected under this method usage that is not always acceptable, for instance, at step height small values. The step height h is calculated as the difference between high level and basic plane level.

To create Z-shift realization means the mentioned method is not enough as well as the thickness special measure usage: monomolecular layer coating, multilayer film etc. The main shortcomings of the mentioned means are gradation positioning, some measure quantity necessity, calibration process complication and its significant duration and cost.

2. Work Goal

The work goal is stepless highly precise nanoshift hydraulic technique development and positioning means along the axis Z-elaboration (within 2 nm) of the nanomachine or other device, for instance AFM contact assembly.

3. Theoretical and Practical Basis of Hydraulic Unit Creation for Nanosized Objects Shift Along the Z-axis

During the creation of hydraulic positioning unit along the axis Z of the nanomachine with providing stepless shift and position control possibility along this axis it is proposed to use hydraulic potential. To this purpose U-shaped hydraulic construction is proposed with ends of larger (D = 200 mm) and smaller (d = 0.3 mm) diameter. Diameters ratio is equal: $\frac{D}{d} = 20.0\,mm / 0.3\,mm = 66.66$. In the case of liquid level shift control in the narrow end with error ± 2.5 μm it is appeared the possibility to provide stepless and quite accurate level displacement in the wide end. This is the object of this study.

Hydraulic positioning is based on the laws of liquid motion and balance. The main part of hydraulic unit consists of two hermetic cylinders of different diameters connected by the interim diameter tube. The space in cylinders is partly filled of liquid (generally with distilled water, alcohol etc.). The height set levels of liquid column in every cylinder are fixed as zero levels. Spontaneous or enforced liquid level shift, for instance, in the small diameter tube or end, is detected by means of micrometer head (Fig. 1).

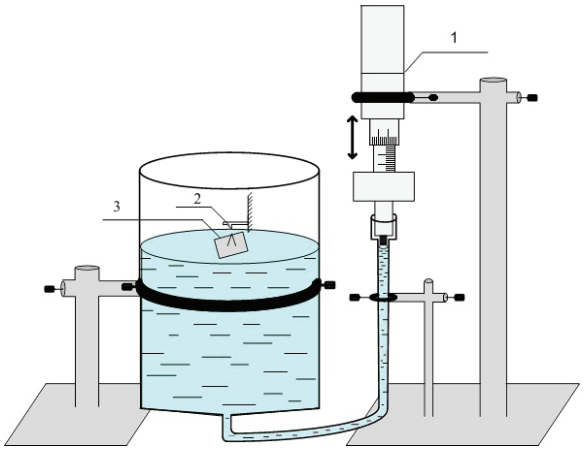

Fig. 1. The unit of nanosized objects hydraulic positioning: 1 - shift micrometer head; 2 – cantilever; 3 – float with mounted research nanoobject.

3.1. Measurement Operation

In the motionless fixed tube its pivot sinks by the shift micrometer head pressing hydraulic liquid to the

wide cylinder. Consequently the liquid level is increased in this end – on Δh_1. It results to weak but appreciable liquid level increase in the wide end. In accordance with joined vessels law the level increases on ΔH_1 in the wide end. Floating plate mounted in this wide end lifts the same as the studied nanoobject mounted on it.

In this case the condition of liquid quantity invariability under pouring from one end to another one can be described by the next formula: $\frac{\pi}{4} D^2 \Delta H = \frac{\pi}{4} d^2 \Delta h$. If we use $A = \frac{D^2}{d^2}$ as constant for this unit construction then with mentioned above ends diameters $A = \frac{D^2}{d^2} = \frac{400}{0.09} = 4444.4$. So we have got the formula specifying the level drops changes in wide (ΔH_1) and narrow (Δh_1) device ends.

$$\Delta H_1 = \frac{\Delta h_1}{A} \tag{1}$$

Accordingly with error approach principles [6] the relative errors of liquid level change in both device ends are linked between each other by the formula getting from (1):

$$\delta(\Delta H_1) = \delta(\Delta h_1) + \delta A\,, \tag{2}$$

where $\delta(\Delta h_1)$ is the device relative error specified by the inaccuracy of the drop level measurement by the micrometer head; δA is the device relative error specified by the inaccuracy of constant value A which can reach significant values and, even, change during the experiment. Let's consider consequently these two error components and abilities to decrease or/and eliminate them.

The first error component is determined by the following way. So far as liquid level measurement drop in the narrow end is 5 μm then the device measurement step is 1 ΔH determined by the formula (1) in the wide end is 5.0 μm / 4444.4 = 1.1 nm. Micrometer head absolute error in accordance with passport is ± 2.5 μm. Its value included to the result of the level shift in the wide end is ± 0.55 nm. Hereby the absolute measurement error of the liquid shift level in the wide end is specified by the micrometer head inaccuracy.

The step measurement refinement result with value error is equal to 1.10 + 0.55 nm. Otherwise the relative measurement error of hydraulic shift on 1.1 nm specified by the micrometer head capabilities is estimated as + 50 %. In the case of liquid level shift control in the narrow end with error ± 1.0 μm it seems possible to reach the relative measurement error of hydraulic shift ±20 %.

Hereinafter the unknown second component of the device error – the relative error of constant A value determination.

3.2. Metrological Experience of the Error Measurement Systematic Component Minimization

The error correction possibilities analysis enables to divide the absolute aggregate measurement error into three components: a) an additive error Δ_α which is independent from the value of measured quantity X and is also called zero error; b) the error Δ_{non} from the characteristic nonlinearity; c) the multiplicative error $\Delta_m = \delta_m X$ which is proportional to X and is called the sensitivity error.

The additive component can be easily detected if X=0. To detect nonlinear error component it should be used multivalued measures or scale converters. The multiplicative component can be detected under presence of the measure or the scale converter.

Relative multiplicative error δ_m can be eliminated by the means of the single-channel unregulated measure by the way of the measurement device verification if the X_0 value is known (in our case - Δh_0) and next division and multiplication.

For measurement accuracy improvement and multiplicative component minimization can be used the opposing method [7]. The method point consists in that the measured quantity H_X is compared twice with regulated measure - $H_{N1}; H_{N2}$; besides that before the second measurement it is rearranged with the measure. Consequently this quantity value with the eliminated measurement error multiplicative component can be gained:

$$H_X = \sqrt{H_{N1} H_{N2}}$$

3.3. Hydraulic Device Calibration

There is a real operation realizing before or after the measurement operation and is assigned to eliminate mentioned above multiplicative error component. In the current case it is assigned to decrease or/and eliminate the second component (in the formula "2") of the shifted liquid level relative error determination in the wide device end, in other words to set accurately Z-coordinate in the nanomachine.

The calibration operation (Fig. 2) is on principle inverse to the measurement operation.

The liquid level in the wide end is changed with the help of plunger connected at this time to micrometer head. After the liquid balance setting in two ends the liquid level shift in the narrow end is measured by the means of the Vernier-caliper gauge. Simultaneously the micrometer head is moved to the wide device end. It provides the plunger shift and liquid level on value that meets the level shift in the narrow end during the measurement. So $\Delta h_1 = \Delta H_2 = 5$ μm.

In this case the mentioned shift when the constant A value is equal to 4444.4 specifies the

liquid level shift in the narrow device end on $\Delta h_2 = A\,\Delta H_2 = 4444.4 \times 5 = 22.2$ mm. Under this, device constant A is determined from:

$$A = \frac{\Delta h_2}{\Delta H_2} \tag{4}$$

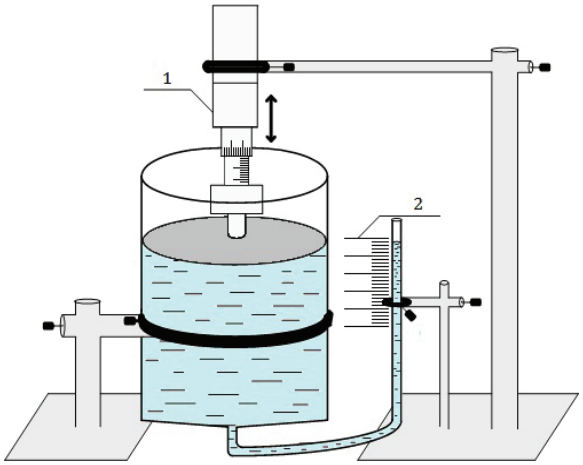

Fig. 2. Unit calibration for hydraulic positioning: 1- micrometer head; 2 – control linear scale of the liquid level shift.

Having placed (4) into (1) liquid column height desired change in the wider end can be found:

$$\Delta H_X = \Delta H_1 = \frac{\Delta h_1}{\Delta h_2}\Delta H_2 = \frac{\left(\Delta h_1\right)^2}{\Delta h_2} \tag{5}$$

Column height change ΔH_x numerical value assessment gives 5^2 μm^2/22.2 mm = 0.0011 μm = 1.1 nm. The error component is absent here that is specified by the device constant A unknown value. But completely known components of relative measurement error of the liquid level displacement appear. The next formula describes it. Aggregate relative error of the liquid level change measurement in the wide device end can be obtained by the differentiating (5):

$$\delta(\Delta H_X) = \delta\left[\frac{\left(\Delta h_1\right)^2}{\Delta h_2}\right] = 2\delta\left(\Delta h_1\right) + \delta\left(\Delta h_2\right) \tag{6}$$

Under the condition of the micrometer head double usage when level measurement is conducted with relative error ± 50 %, the aggregate relative error of before specified new liquid level in the wide end exceeds slightly ± 100 % (± 40 % in the case of liquid level shift control in the narrow end with error ± 1.0 μm). By the way the second component in (6) – insignificant (substantial level shift can be measured quite precisely with relative error not more than ± 2 %).

In the conditions of the studied nanoobject height single fixing so far as the unknown ideal fixing result it is more correctly to work not within the error approach but on the basis of the uncertainty approach. Accordingly with last one [9], [6] the height fixing result is more correctly to write in the form of readout result with established expended standard uncertainty of type A that is in our case $\pm\ 102/\sqrt{3}\ (42/\sqrt{3}) = \pm\ 58.89\ (24.25)$ %.

The calibration obtained results enable to reach efficiency in Z-axis shift control in the nanomachine. To this purpose the calibrated unit with concrete determined ration of two ends areas is used by appointment as described in the Fig. 1. If the average atoms size of liquid is approximately equal to ≈ 3Å = 0,3 nm then current device enables to set the liquid level and nanoobject mounted on the floating platform with absolute error which is slightly more than atoms size.

At the same time the work [8] gives close values of water film estimated thicknesses pretermitting the accuracy issue of their regulation and setting. For instance, thickness of a liquid layer seems to be of order 10 Å at the temperatures up to 3-4 K below the melting point of ice. To determine the thickness at temperatures closer to the melting point, larger simulation cells and more accurate estimates of the melting point are needed. The thickness of a liquid layers determined in this work seems to be of the same order of magnitude as determined by Bluhm *et al.* [10] from photoelectron microscopy.

Thanks to the wide device end and layer-by-layer filling of the liquid this hydraulic positioning method provides the required contact of cantilever and studied nanoobject mounted on floating platform.

During the study of cantilever Z-axis regulating capabilities or shift of the floating platform relative to it by the height with the help of mentioned above it is able to find the next one. Under accurate settings of very small Z-axis values the movable means elements (hydraulic liquid molecules) determine the noise device features. Perhaps, in future it should be taken into account the viscosity fluctuating deviations. They may be specified, for instance, by the temperature deviations from optimal regime, the diversions from the horizontality, the non-discrete character of the every layer filling and some other factors. They determine the noise threshold and limit the device sensitivity.

4. Conclusions

1. Nanotechnology further progress is impossible without methods and means development of nanosamples and sensitive elements stepless reciprocal shift along the Z-axis, for instance, probes and cantilevers that guarantee of tunnel and other phenomena studies that demand exceptional shift accuracy.

2. Hydraulic method and reciprocal shift means of the nanoobject and cantilever enable with the error

of working liquid monomolecular layer thickness order to provide their stepless shift along the *Z*-axis.

3. Simultaneously the systematic multiplicative component of aggregate error is eliminated by the opposing method of error minimization usage in the course of the calibration – standard method of the current error component elimination in the metrology. Therefore noise threshold is displaced in the direction of nanosizes and, as result, *Z*-axis accuracy setting is improved.

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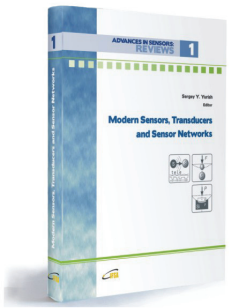
ADVANCES IN SENSORS: REVIEWS 1

Modern Sensors, Transducers and Sensor Networks

Sergey Y. Yurish, Editor



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Modern Sensors, Transducers and Sensor Networks is intended for anyone who wants to cover a comprehensive range of topics in the field of sensors paradigms and developments. It provides guidance for technology solution developers from academia, research institutions, and industry, providing them with a broader perspective of sensor science and industry.

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ADVANCES IN SENSORS: REVIEWS 2

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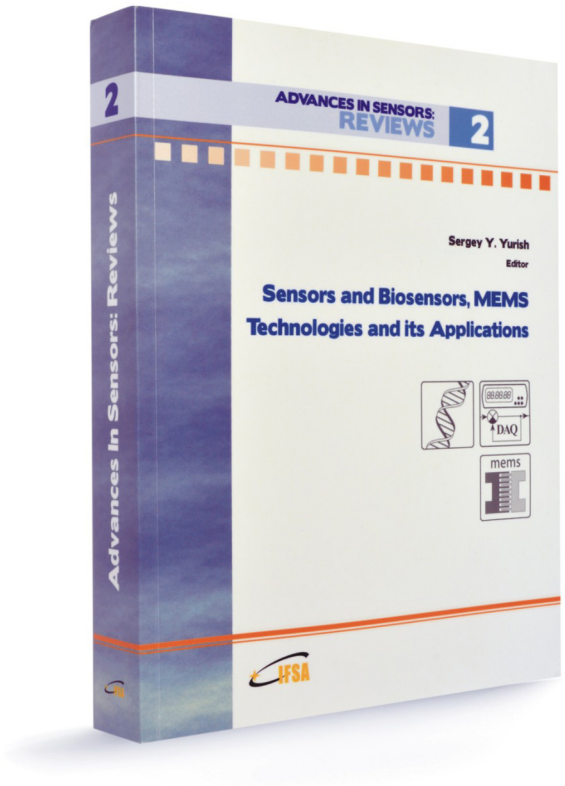
Sensors and Biosensors, MEMS Technologies and its Applications



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